GEOMETRY · COMMON CORE HIGH SCHOOL MATH

Answer Key

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

105 question sets • 3160 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

CONTENTS

Set 1	**HSG-CO.A.1**	Precise definition of a circle	9
Set 97	**HSG-CO.A.1**	Precise definition of an angle	40
Set 98	**HSG-CO.A.1**	Precise definition of a line segment	71
Set 2	**HSG-CO.A.1**	Precise definition of perpendicular lines	102
Set 99	**HSG-CO.A.1**	Precise definition of parallel lines	133
Set 3	**HSG-CO.A.2**	Transformations as functions: inputs, outputs, and preimages	170
Set 4	**HSG-CO.A.2**	Compare transformations that preserve distance and angle with those that do not	207
Set 5	**HSG-CO.A.3**	Rotations and reflections that carry a rectangle, parallelogram, or trapezoid onto it	238
Set 6	**HSG-CO.A.3**	Rotations and reflections that carry a regular polygon onto itself	275
Set 7	**HSG-CO.A.4**	Definition of a reflection	312
Set 100	**HSG-CO.A.4**	Definition of a translation	349
Set 8	**HSG-CO.A.4**	Definition of a rotation	380
Set 9	**HSG-CO.A.5**	Draw the image of a figure under a rotation, reflection, or translation	417
Set 10	**HSG-CO.A.5**	Specify a sequence of transformations that carries one figure onto another	454
Set 11	**HSG-CO.B.6**	Predict the effect of a rigid motion on a figure	491
Set 12	**HSG-CO.B.6**	Decide whether two figures are congruent using rigid motions	522
Set 13	**HSG-CO.B.7**	Congruent triangles: a rigid motion exactly when all corresponding parts are congr	553

CONTENTS (continued)

Set 14	**HSG-CO.B.8**	Explain SAS congruence with rigid motions	590
Set 15	**HSG-CO.B.8**	Explain ASA congruence with rigid motions	627
Set 16	**HSG-CO.B.8**	Explain SSS congruence with rigid motions	664
Set 17	**HSG-CO.C.9**	Prove that vertical angles are congruent	701
Set 18	**HSG-CO.C.9**	Prove the angle theorems for parallel lines cut by a transversal	738
Set 19	**HSG-CO.C.9**	Prove the perpendicular bisector theorem and its converse	775
Set 20	**HSG-CO.C.10**	Prove that the angles of a triangle add to 180°	812
Set 21	**HSG-CO.C.10**	Prove that the base angles of an isosceles triangle are congruent	849
Set 22	**HSG-CO.C.10**	Prove the midsegment theorem	886
Set 23	**HSG-CO.C.10**	Prove that the medians of a triangle meet at one point	929
Set 24	**HSG-CO.C.11**	Prove that opposite sides of a parallelogram are congruent	972
Set 101	**HSG-CO.C.11**	Prove that opposite angles of a parallelogram are congruent	1009
Set 25	**HSG-CO.C.11**	Prove that the diagonals of a parallelogram bisect each other	1046
Set 26	**HSG-CO.C.11**	Prove the converses: conditions that make a quadrilateral a parallelogram	1083
Set 27	**HSG-CO.C.11**	Prove that rectangles are parallelograms with congruent diagonals	1120
Set 28	**HSG-CO.D.12**	Construct a copy of a segment	1157
Set 102	**HSG-CO.D.12**	Make formal constructions with a variety of tools and methods	1194

CONTENTS (continued)

Set 29	**HSG-CO.D.12**	Construct a copy of an angle	1225
Set 30	**HSG-CO.D.12**	Construct the perpendicular bisector of a segment (bisect a segment)	1262
Set 31	**HSG-CO.D.12**	Construct the bisector of an angle	1299
Set 32	**HSG-CO.D.12**	Construct a perpendicular to a line through a point on or off the line	1336
Set 33	**HSG-CO.D.12**	Construct a line parallel to a given line through a point not on it	1373
Set 34	**HSG-CO.D.13**	Construct an equilateral triangle inscribed in a circle	1410
Set 35	**HSG-CO.D.13**	Construct a square inscribed in a circle	1447
Set 36	**HSG-CO.D.13**	Construct a regular hexagon inscribed in a circle	1484
Set 37	**HSG-SRT.A.1.a**	Verify that a dilation maps a line to a parallel line (or to itself through the center)	1521
Set 38	**HSG-SRT.A.1.b**	Verify that a dilation multiplies lengths by the scale factor	1558
Set 39	**HSG-SRT.A.2**	Decide whether two figures are similar using similarity transformations	1595
Set 40	**HSG-SRT.A.2**	Explain similarity of triangles as equal angles and proportional sides	1632
Set 41	**HSG-SRT.A.3**	Establish the AA criterion with similarity transformations	1669
Set 42	**HSG-SRT.B.4**	Prove the triangle proportionality theorem	1706
Set 43	**HSG-SRT.B.4**	Prove the converse of the triangle proportionality theorem	1743
Set 44	**HSG-SRT.B.4**	Prove the Pythagorean Theorem using similar triangles	1780
Set 45	**HSG-SRT.B.5**	Use similarity criteria to solve problems in geometric figures	1817

CONTENTS (continued)

Set 46	**HSG-SRT.B.5**	Use congruence criteria to prove relationships in figures	1854
Set 47	**HSG-SRT.B.5**	Use similarity criteria to prove relationships in figures	1891
Set 48	**HSG-SRT.C.6**	Understand trigonometric ratios as properties of an angle, by similarity	1928
Set 49	**HSG-SRT.C.7**	Explain and use the relationship between the sine and cosine of complementary	1965
Set 50	**HSG-SRT.C.8**	Find unknown sides of right triangles with trigonometric ratios in applied problems	2002
Set 51	**HSG-SRT.C.8**	Find unknown angles with inverse trigonometric ratios in applied problems	2039
Set 52	**HSG-SRT.C.8**	Solve right triangles in applied problems with trigonometry and the Pythagorean T	2076
Set 53	**HSG-C.A.1**	Prove that all circles are similar	2113
Set 54	**HSG-C.A.2**	Relate inscribed angles and central angles	2150
Set 55	**HSG-C.A.2**	Inscribed angles on a diameter are right angles	2187
Set 56	**HSG-C.A.2**	A radius is perpendicular to the tangent at the point of tangency	2224
Set 57	**HSG-C.A.2**	Relate circumscribed angles and central angles	2261
Set 58	**HSG-C.A.2**	Relationships among radii and chords	2298
Set 59	**HSG-C.A.3**	Construct the inscribed circle of a triangle	2335
Set 60	**HSG-C.A.3**	Construct the circumscribed circle of a triangle	2372
Set 61	**HSG-C.A.3**	Prove that opposite angles of an inscribed quadrilateral are supplementary	2409
Set 62	**HSG-C.B.5**	Derive arc length from similarity and define radian measure	2446

CONTENTS (continued)

Set 63	**HSG-C.B.5**	Derive and use the formula for the area of a sector	2483
Set 64	**HSG-GPE.A.1**	Derive the equation of a circle	2520
Set 65	**HSG-GPE.A.1**	Complete the square to find the center and radius of a circle	2557
Set 66	**HSG-GPE.A.2**	Derive the equation of a parabola from a focus and directrix	2594
Set 67	**HSG-GPE.B.4**	Prove or disprove the type of a quadrilateral with coordinates	2637
Set 68	**HSG-GPE.B.4**	Prove or disprove that a point lies on a circle, and other coordinate proofs	2674
Set 69	**HSG-GPE.B.5**	Prove the slope criteria for parallel and perpendicular lines	2711
Set 70	**HSG-GPE.B.5**	Write equations of parallel and perpendicular lines	2748
Set 71	**HSG-GPE.B.5**	Solve geometric problems with the slope criteria	2791
Set 72	**HSG-GPE.B.6**	Partition a directed segment in a given ratio	2834
Set 73	**HSG-GPE.B.7**	Compute perimeters of polygons with coordinates	2871
Set 74	**HSG-GPE.B.7**	Compute areas of triangles and rectangles with coordinates	2908
Set 75	**HSG-GMD.A.1**	Informal arguments for the circumference of a circle	2945
Set 103	**HSG-GMD.A.1**	Informal arguments for the area of a circle	2982
Set 76	**HSG-GMD.A.1**	Informal argument for the volume of a cylinder	3019
Set 77	**HSG-GMD.A.1**	Informal arguments for the volume of a pyramid	3056
Set 104	**HSG-GMD.A.1**	Informal arguments for the volume of a cone	3093

CONTENTS (continued)

Set 78	**HSG-GMD.A.3**	Use volume formulas for cylinders, pyramids, and cones	3130
Set 79	**HSG-GMD.A.3**	Use the volume formula for spheres, including composite solids	3173
Set 80	**HSG-GMD.A.3**	Solve for an unknown dimension from a volume	3216
Set 81	**HSG-GMD.B.4**	Identify cross-sections of three-dimensional objects	3253
Set 82	**HSG-GMD.B.4**	Identify three-dimensional objects generated by rotations of two-dimensional obj	3290
Set 83	**HSG-MG.A.1**	Model objects with geometric shapes	3327
Set 84	**HSG-MG.A.2**	Apply density based on area	3364
Set 85	**HSG-MG.A.2**	Apply density based on volume	3401
Set 86	**HSG-MG.A.3**	Design to satisfy physical constraints	3438
Set 87	**HSG-MG.A.3**	Design to minimize cost	3481
Set 105	**HSG-MG.A.3**	Design typographic grid systems	3518
Set 88	**HSG-SRT.D.9**	Derive and use the formula A = ½ab sin C	3555
Set 89	**HSG-SRT.D.10**	Prove the Law of Sines	3592
Set 90	**HSG-SRT.D.10**	Prove the Law of Cosines	3629
Set 91	**HSG-SRT.D.11**	Apply the Law of Sines, including the ambiguous case	3672
Set 92	**HSG-SRT.D.11**	Apply the Law of Cosines, including resultant forces	3715
Set 93	**HSG-C.A.4**	Construct a tangent line from a point outside a circle	3752

CONTENTS (continued)

Set 94	**HSG-GPE.A.3**	Derive the equation of an ellipse from its foci	3789
Set 95	**HSG-GPE.A.3**	Derive the equation of a hyperbola from its foci	3826
Set 96	**HSG-GMD.A.2**	Use Cavalieri's principle to justify the volume of a sphere	3863

Precise definition of a circle

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 3	Use the center, radius, and diameter of a circle	**7.G.B.4**
BACKWARD 4	Find the distance from a center to points on a grid	**8.G.B.8**
FORWARD 1	Test definitions of circle vocabulary with counterexamples	**HSG-CO.A.1**
FORWARD 2	Write precise definitions of circle parts from the definition of a circ…	**HSG-CO.A.1**

SET 1 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of a circle. Base it on the notions of point, plane, and distance (you may use ordinary words such as "set", "all", and "given"). Name the center and the radius in your definition.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q1/5

11

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

A circle is the set of all points in a plane that are a given positive distance (the radius) from a given point in that plane (the center).

GRADING NOTE Any equivalent wording. Essential: (1) ALL points, (2) in one plane, (3) at one fixed positive distance, (4) from one fixed point, the center. Not accepted: "a round shape" (not precise); leaving out the plane (that describes a sphere); "points within a distance" (that is a disk).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q1/5

Kai writes: "A circle is a round curve with a center."

Explain what makes this imprecise, and rewrite it as a precise definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q2/5

12

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

"Round" and "curve" are not defined, and the center is not tied to the points. Precise: a circle is the set of all points in a plane at a given positive distance (the radius) from a given point in that plane (the center).

GRADING NOTE Any equivalent answer. Essential: names an undefined or vague term ("round", "curve") or the missing equal-distance condition, and gives a definition with all points, one plane, one fixed positive distance, and a center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q2/5

Lee writes: "The circle with center O and radius 4 is all the points of the plane that are within 4 units of O."

What figure does Lee's sentence describe? Rewrite it so that it defines the circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q3/5

13

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

It describes a disk (the circle and its inside). Precise: the circle with center O and radius 4 is the set of all points in the plane that are exactly 4 units from O.

GRADING NOTE Any equivalent answer. Essential: identifies the disk (points inside are included) and replaces "within" by "exactly" (equal to).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q3/5

Mia writes: "A circle is all the points that are 5 cm from point O."

What must she add to make the definition precise? Explain why.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q4/5

14

QUESTION ID

S1-M-Q4

TYPE

Short answer

CORRECT ANSWER

She must add "in a plane" (a plane that contains O). Without it, all points 5 cm from O form a sphere, not a circle.

GRADING NOTE Any equivalent wording that adds the plane condition and explains that the points in space would form a sphere.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q4/5

Ana writes: "A circle is the set of points in a plane at a distance from a point O in that plane."

Which condition is missing? Rewrite the definition so that it is precise.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q5/5

15

QUESTION ID

S1-M-Q5

TYPE

Short answer

CORRECT ANSWER

The distance must be one fixed (given, positive) distance; "at a distance" allows every point of the plane. Precise: the set of all points in the plane at a given positive distance r from O.

GRADING NOTE Any equivalent answer. Essential: the same fixed distance for every point, and a corrected definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSG-CO.A.1

BACKWARD 3

Use the center, radius, and diameter of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 7 cm. Point P is in the plane of the circle, 7 cm from O. Is P on the circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q1/5

17

QUESTION ID

S1-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Every point of the plane that is 7 cm from O is on the circle.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q1/5

A circle has a diameter of 18 in. What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q2/5

18

QUESTION ID

S1-B3-Q2

TYPE

Short answer

CORRECT ANSWER

9 in.

GRADING NOTE Any equivalent form. The radius is half the diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q2/5

A circle has center C and radius 4. Point D is in the same plane, 3 units from C. Where is D?

A. Inside the circle

B. Outside the circle

C. On the circle

D. There is not enough information

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q3/5

19

QUESTION ID

S1-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Inside the circle

GRADING NOTE D is closer to the center than the radius, so it is inside the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q3/5

True or false?

Every radius of a circle has the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q4/5

20

QUESTION ID

S1-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All points of a circle are the same distance from its center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q4/5

True or false?

A point 10 units from the center of a circle of radius 5 lies on the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q5/5

21

QUESTION ID

S1-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Points on the circle are exactly 5 units from the center; this point is outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSG-CO.A.1

BACKWARD 4

Find the distance from a center to points on a grid • Grade 8 • Geometry • 8.G.B.8

Find the distance from O(0, 0) to P(3, 4).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B4-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q1/5

23

QUESTION ID

S1-B4-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $\sqrt{(3^2 + 4^2)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B4-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q1/5

Find the distance from C(1, 1) to Q(1, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B4-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q2/5

24

QUESTION ID

S1-B4-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. Same x, so |6 − 1| = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B4-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q2/5

Which point is 5 units from O(0, 0)?

A. (5, 5)

B. (2, 3)

C. (4, 4)

D. (−4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B4-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q3/5

25

QUESTION ID

S1-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−4, 3)

GRADING NOTE $\sqrt{(16 + 9)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B4-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q3/5

True or false?

The points (5, 0) and (0, −5) are the same distance from the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B4-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q4/5

26

QUESTION ID

S1-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both are 5 units from (0, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B4-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q4/5

True or false?

The point (3, 3) is 3 units from the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B4-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q5/5

27

QUESTION ID

S1-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(9 + 9)} = \sqrt{18} \approx 4.24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B4-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q5/5

SET 1 • HSG-CO.A.1

FORWARD 1

Test definitions of circle vocabulary with counterexamples • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Proposed definition: "A circle is a set of points that are all the same distance from a point O."

Give a set that fits this wording but is NOT a circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q1/5

29

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

For example, two points that are each 3 cm from O (or an arc of a circle, or a sphere). The definition must say ALL points at that distance, in one plane.

GRADING NOTE Any set that fits the wording but is not a whole circle: finitely many points, an arc, or a sphere.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q1/5

Proposed definition: "A radius of a circle is a segment from the center."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q2/5

30

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A segment from the center that stops inside the circle (or goes past it). Repair: a radius is a segment from the center to a point of the circle.

GRADING NOTE Any counterexample with the other endpoint not on the circle, plus a repair that puts it on the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q2/5

Which set fits the description "all points 2 units from point O" but is NOT a circle?

A. A circle of radius 2 centered at O in a plane

B. A sphere of radius 2 centered at O

C. A disk of radius 2

D. A segment of length 2 from O

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q3/5

31

QUESTION ID

S1-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A sphere of radius 2 centered at O

GRADING NOTE In space, the points 2 units from O form a sphere. A disk also contains points closer than 2 units, so it does not fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q3/5

Proposed definition: "A chord is a segment inside a circle."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q4/5

32

QUESTION ID

S1-F1-Q4

TYPE

Short answer

CORRECT ANSWER

A short segment near the center with neither endpoint on the circle. Repair: a chord is a segment whose endpoints are both on the circle.

GRADING NOTE Any counterexample whose endpoints are not both on the circle, plus a repair requiring both endpoints on the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Proposed definition: "A diameter is a segment that passes through the center of a circle."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q5/5

33

QUESTION ID

S1-F1-Q5

TYPE

Short answer

CORRECT ANSWER

A short segment through the center whose endpoints are not on the circle. Repair: a diameter is a segment through the center whose endpoints are both on the circle.

GRADING NOTE Any counterexample plus a repair that requires both endpoints on the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSG-CO.A.1

FORWARD 2

Write precise definitions of circle parts from the definition of a circle • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of a chord of a circle.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q1/5

35

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A chord of a circle is a segment whose endpoints both lie on the circle.

GRADING NOTE Any equivalent wording. Both endpoints on the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q1/5

Write a precise definition of a diameter of a circle with center O.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q2/5

36

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A diameter is a chord that contains the center O (a segment through O with both endpoints on the circle).

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q2/5

Segment PQ is a diameter of a circle with center O. Write a precise definition of a semicircle with endpoints P and Q.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q3/5

37

QUESTION ID

S1-F2-Q3

TYPE

Short answer

CORRECT ANSWER

A semicircle is P, Q, and all points of the circle that lie on one side of line PQ.

GRADING NOTE Any equivalent wording. Essential: the endpoints of a diameter and the points of the circle on one side of the diameter line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q3/5

Which is a precise definition of the interior of a circle with center O and radius r?

A. All points of the plane whose distance from O is at most r

B. All points of the circle

C. All points whose distance from O is less than r

D. All points of the plane of the circle whose distance from O is less than r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q4/5

38

QUESTION ID

S1-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. All points of the plane of the circle whose distance from O is less than r

GRADING NOTE The interior is in the plane of the circle and excludes the circle itself (distance less than r). The second choice includes points in space.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q4/5

Two circles in the same plane have the same center. Write a precise definition of this pair ("concentric circles") and explain why two different concentric circles never meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q5/5

39

QUESTION ID

S1-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Concentric circles are circles in one plane with the same center. If they had a common point P, then OP would equal both radii, so the radii would be equal and the circles would be the same.

GRADING NOTE Any equivalent definition and explanation that uses the fixed distance from the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q5/5

Precise definition of an angle

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	See an angle as two rays with a common endpoint, measured by…	**4.MD.C.5.a**
BACKWARD 2	Identify rays and their endpoints	**4.G.A.1**
FORWARD 1	Define angle relationships from the definition of an angle	**HSG-CO.A.1**
FORWARD 2	Prove angle facts from the angle definitions	**HSG-CO.C.9**

SET 97 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of an angle. Name its vertex and its sides in your definition.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q1/5

42

QUESTION ID

S97-M-Q1

TYPE

Short answer

CORRECT ANSWER

An angle is the figure formed by two rays that have the same endpoint. The common endpoint is the vertex, and the two rays are the sides.

GRADING NOTE Any equivalent wording. Essential: two rays (not lines or segments), one common endpoint, that endpoint named as the vertex. Not accepted: "the space between two lines", "a turn", or "two lines that cross" (two crossing lines form four angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q1/5

Sam writes: "An angle is the space between two lines."

Explain two things that are wrong, and rewrite it as a precise definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q2/5

43

QUESTION ID

S97-M-Q2

TYPE

Short answer

CORRECT ANSWER

Two crossing lines make four angles, so "two lines" does not name one angle; "space between" is not defined (an angle is the rays, not a region). Precise: an angle is the union of two rays with a common endpoint (the vertex).

GRADING NOTE Any equivalent answer. Essential: two errors (lines instead of rays or no common endpoint; a region instead of the figure) and a correct definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q2/5

Jo writes: "An angle is two rays."

Give a pair of rays that fits Jo's wording but is not an angle, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q3/5

44

QUESTION ID

S97-M-Q3

TYPE

Short answer

CORRECT ANSWER

For example, two rays with different endpoints that never meet (such as two parallel rays). Repair: an angle is two rays with the same endpoint.

GRADING NOTE Any counterexample with rays that do not share an endpoint, plus the repaired definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q3/5

A 1° angle is defined using distance around a circular arc. Complete the definition, and explain why the measure of an angle does not depend on the radius of the circle used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q4/5

45

QUESTION ID

S97-M-Q4

TYPE

Short answer

CORRECT ANSWER

A 1° angle, with its vertex at the center of a circle, cuts off an arc that is 1/360 of the circle. Every circle centered at the vertex is cut in the same fraction, so the measure is the same for every radius.

GRADING NOTE Any equivalent answer. Essential: vertex at the center, 1/360 of the circle, and the fraction (not the length) sets the measure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q4/5

Rays QP and QR have the common endpoint Q, and ray RS starts at R. Using the definition, explain which pair of rays forms an angle and name the angle, and which pair does not.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q5/5

46

QUESTION ID

S97-M-Q5

TYPE

Short answer

CORRECT ANSWER

Rays QP and QR form ∠PQR (common endpoint Q, the vertex). Rays QR and RS have different endpoints, so they do not form an angle.

GRADING NOTE Any equivalent answer. Essential: the common-endpoint test applied to both pairs, and the angle named with its vertex in the middle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q5/5

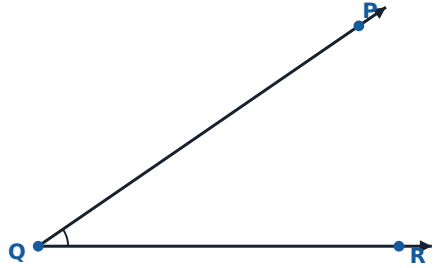
SET 97 • HSG-CO.A.1

BACKWARD 1

See an angle as two rays with a common endpoint, measured by an arc • Grade 4 • Measurement & Data • 4.MD.C.5.a

Name the vertex and the two sides of ∠PQR.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q1/5

48

QUESTION ID

S97-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Vertex Q; sides ray QP and ray QR.

GRADING NOTE Any equivalent form. The middle letter names the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q1/5

An angle has its vertex at the center of a circle and cuts off $\frac{1}{4}$ of the circle. What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q2/5

49

QUESTION ID

S97-B1-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form. 1/4 of 360° is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q2/5

An angle measures 60°. What fraction of a circle centered at its vertex does it cut off?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q3/5

50

QUESTION ID

S97-B1-Q3

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{6}$

GRADING NOTE Any equivalent form. 60/360 = 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q3/5

True or false?

The measure of an angle depends on how long its sides are drawn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q4/5

51

QUESTION ID

S97-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sides are rays; drawing them longer does not change the opening between them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q4/5

True or false?

A one-degree angle turns through $\frac{1}{360}$ of a full circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q5/5

52

QUESTION ID

S97-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is the meaning of one degree.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q5/5

SET 97 • HSG-CO.A.1

BACKWARD 2

Identify rays and their endpoints • Grade 4 • Geometry • 4.G.A.1

What is shown in the figure?

A. Segment AB

B. Ray BA

C. Line AB

D. Ray AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q1/5

54

QUESTION ID

S97-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Ray AB

GRADING NOTE It starts at A and goes through B without end, so it is ray AB (endpoint named first).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q1/5

How many endpoints does a ray have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q2/5

55

QUESTION ID

S97-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q2/5

Which figure goes on without end in exactly one direction?

A. A line segment

B. A point

C. A ray

D. A line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q3/5

56

QUESTION ID

S97-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A ray

GRADING NOTE A ray has one endpoint and continues in one direction; a line continues in both directions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q3/5

True or false?

Ray AB and ray BA are the same ray.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q4/5

57

QUESTION ID

S97-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Ray AB starts at A and goes through B; ray BA starts at B and goes through A, in the opposite direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q4/5

True or false?

Rays QA and QB share the endpoint Q.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q5/5

58

QUESTION ID

S97-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The first letter of a ray names its endpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q5/5

SET 97 • HSG-CO.A.1

FORWARD 1

Define angle relationships from the definition of an angle • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of adjacent angles.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q1/5

60

QUESTION ID

S97-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Two angles in a plane that share a vertex and a side, and whose interiors do not overlap.

GRADING NOTE Any equivalent wording. Essential: common vertex, common side, no common interior points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q1/5

Write a precise definition of vertical angles.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q2/5

61

QUESTION ID

S97-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Two angles whose sides form two pairs of opposite rays (the nonadjacent angles formed by two intersecting lines).

GRADING NOTE Any equivalent wording. Essential: opposite rays, or nonadjacent angles of two crossing lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q2/5

Write a precise definition of a linear pair of angles.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q3/5

62

QUESTION ID

S97-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Two adjacent angles whose noncommon sides are opposite rays.

GRADING NOTE Any equivalent wording. Not accepted: "two angles that add to 180°" (that defines supplementary angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q3/5

Proposed definition: "Two angles are vertical angles if they have the same vertex and are not adjacent."

Give a counterexample and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q4/5

63

QUESTION ID

S97-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Three rays from one point can form two nonadjacent angles that are not vertical (for example ∠AOB and ∠COD with four rays at 0°, 30°, 100°, 150°). Repair: their sides form two pairs of opposite rays.

GRADING NOTE Any valid counterexample (nonadjacent angles at one vertex without opposite rays) plus the repair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q4/5

Which is a precise definition of the bisector of ∠ABC?

A. Any line through B

B. A ray from B that lies outside ∠ABC

C. Ray BD, where D is in the interior of ∠ABC and m∠ABD = m∠DBC

D. The segment that joins A and C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q5/5

64

QUESTION ID

S97-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Ray BD, where D is in the interior of ∠ABC and m∠ABD = m∠DBC

GRADING NOTE The bisector is the ray from the vertex that splits the angle into two angles of equal measure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q5/5

SET 97 • HSG-CO.A.1

FORWARD 2

Prove angle facts from the angle definitions • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

∠1 and ∠2 form a linear pair, and ∠2 and ∠3 form a linear pair, where ∠1 and ∠3 are vertical angles. Prove that m∠1 = m∠3.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q1/5

66

QUESTION ID

S97-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠1 = m∠3

GRADING NOTE REQUIRED WORK: Use the linear pair postulate (a linear pair is supplementary) twice, then subtract.

Any valid proof. Essential: m∠1 + m∠2 = 180° and m∠2 + m∠3 = 180° (linear pairs); subtract or substitute to get m∠1 = m∠3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q1/5

Ray BD bisects ∠ABC and m∠ABD = (2x + 10)°, m∠DBC = (4x − 20)°. Find m∠ABC.

m∠ABC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q2/5

67

QUESTION ID

S97-F2-Q2

TYPE

Short answer

CORRECT ANSWER

80°

GRADING NOTE Any equivalent form. A bisector makes equal halves: 2x + 10 = 4x − 20, so x = 15; each half is 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q2/5

∠PQR and ∠RQS form a linear pair, and m∠PQR is 4 times m∠RQS. Find both angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q3/5

68

QUESTION ID

S97-F2-Q3

TYPE

Short answer

CORRECT ANSWER

144° and 36°

GRADING NOTE Form required: both angles. x + 4x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q3/5

Rays OA and OB are opposite rays, and ray OC is not on line AB. Ray OD bisects ∠AOC and ray OE bisects ∠COB. Prove that m∠DOE = 90°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q4/5

69

QUESTION ID

S97-F2-Q4

TYPE

Answer with required work

CORRECT ANSWER

m∠DOE = 90°

GRADING NOTE REQUIRED WORK: Use the linear pair and the definition of an angle bisector.

Any valid proof. Essential: m∠AOC + m∠COB = 180° (linear pair); m∠DOC = ½m∠AOC and m∠COE = ½m∠COB (bisectors); m∠DOE = m∠DOC + m∠COE = ½(180°) = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q4/5

True or false?

If two angles are adjacent and supplementary, their noncommon sides are opposite rays.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q5/5

70

QUESTION ID

S97-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Adjacent angles whose measures add to 180° have outer sides that make a straight angle, so they are opposite rays.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S97-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q5/5

SET 98 • HSG-CO.A.1 • CONGRUENCE: DEFINITIONS AND TRANSFORMATIONS

Precise definition of a line segment

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Identify points, lines, rays, and segments	**4.G.A.1**
BACKWARD 2	Find distances between points on a number line	**6.NS.C.8**
FORWARD 1	Define figures built from segments	**HSG-CO.A.1**
FORWARD 2	Test segment-based definitions with counterexamples	**HSG-CO.A.1**

SET 98 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Points A and B lie on line ℓ. Which is a precise definition of segment AB?

A. Point A and every point of line ℓ on the same side of A as B

B. Every point of the plane that is closer to A than to B

C. Every point of line ℓ

D. Points A and B together with every point C of line ℓ for which AC + CB = AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q1/5

73

QUESTION ID

S98-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Points A and B together with every point C of line ℓ for which AC + CB = AB

GRADING NOTE A segment is its two endpoints and every point between them; "C is between A and B" means C is on the line and AC + CB = AB (distance along a line). The other choices describe line ℓ, ray AB, and a half-plane.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q1/5

Write a precise definition of segment AB using the word "between".

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q2/5

74

QUESTION ID

S98-M-Q2

TYPE

Short answer

CORRECT ANSWER

Segment AB is the set made up of points A and B and all points of line AB that are between A and B.

GRADING NOTE Any equivalent wording. Essential: both endpoints, the points between them, and all on line AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q2/5

Ravi writes: "A segment is part of a line."

Give a part of a line that is not a segment, and rewrite the definition precisely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q3/5

75

QUESTION ID

S98-M-Q3

TYPE

Short answer

CORRECT ANSWER

A ray (or a line with one point removed) is part of a line but not a segment. Precise: segment AB is A, B, and all points of line AB between A and B.

GRADING NOTE Any valid counterexample plus a definition with two endpoints and the points between them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q3/5

Points A, B, and C are on one line with AC = 3, CB = 5, and AB = 8. Is C on segment AB? Use the definition to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q4/5

76

QUESTION ID

S98-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes.
AC + CB = 3 + 5 = 8 = AB, so C is between A and B, and points between A and B belong to segment AB.

GRADING NOTE Any equivalent explanation using AC + CB = AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q4/5

Points A, B, and D are on one line with AD = 3, DB = 11, and AB = 8. Is D on segment AB? Use the definition to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q5/5

77

QUESTION ID

S98-M-Q5

TYPE

Short answer

CORRECT ANSWER

No. AD + DB = 14 ≠ 8 = AB, so D is not between A and B (A is between D and B).

GRADING NOTE Any equivalent explanation using the betweenness test.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q5/5

SET 98 • HSG-CO.A.1

BACKWARD 1

Identify points, lines, rays, and segments • Grade 4 • Geometry • 4.G.A.1

How many endpoints does a line segment have? How many does a ray have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q1/5

79

QUESTION ID

S98-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A segment has 2 endpoints; a ray has 1.

GRADING NOTE Any equivalent form. Both numbers are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q1/5

True or false?

A line segment has exactly two endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q2/5

80

QUESTION ID

S98-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A segment is two endpoints and the points between them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q2/5

Which figure has two endpoints?

A. A ray

B. A line

C. An angle

D. A segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q3/5

81

QUESTION ID

S98-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A segment

GRADING NOTE Only a segment has two endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q3/5

True or false?

A line has two endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q4/5

82

QUESTION ID

S98-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A line goes on without end in both directions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q4/5

Which name describes the part of line AB from A to B, including A and B?

A. Ray BA

B. Line AB

C. Segment AB

D. Ray AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q5/5

83

QUESTION ID

S98-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Segment AB

GRADING NOTE The part between two endpoints is a segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q5/5

SET 98 • HSG-CO.A.1

BACKWARD 2

Find distances between points on a number line • Grade 6 • The Number System • 6.NS.C.8

On a number line, A is at 2 and B is at 9. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q1/5

85

QUESTION ID

S98-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. |9 − 2| = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

On a number line, A is at −3 and C is at 4. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q2/5

86

QUESTION ID

S98-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q2/5

On a number line, A is at 1, C is at 4, B is at 10. Which is true?

A. AB < AC

B. AC + CB > AB

C. AC + CB = AB

D. AC = CB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q3/5

87

QUESTION ID

S98-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. AC + CB = AB

GRADING NOTE 3 + 6 = 9 = AB, so C is between A and B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q3/5

True or false?

On a number line, the point at 5 is between the points at 2 and 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q4/5

88

QUESTION ID

S98-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2 < 5 < 8$; also $3 + 3 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q4/5

True or false?

On a number line, the point at 11 is between the points at 2 and 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q5/5

89

QUESTION ID

S98-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 11 is outside: 9 + 3 ≠ 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q5/5

SET 98 • HSG-CO.A.1

FORWARD 1

Define figures built from segments • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of ray AB. Use the word "between".

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q1/5

91

QUESTION ID

S98-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Ray AB is point A, point B, every point between A and B, and every point P of line AB such that B is between A and P.

GRADING NOTE Any equivalent definition built from A, B, line AB, and betweenness.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q1/5

Write a precise definition of the midpoint of segment AB.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q2/5

92

QUESTION ID
S98-F1-Q2

TYPE
Short answer

CORRECT ANSWER
The midpoint of segment AB is the point M on segment AB such that AM = MB.

GRADING NOTE Any equivalent wording. M must be on the segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q2/5

Write a precise definition of a triangle using segments.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q3/5

93

QUESTION ID

S98-F1-Q3

TYPE

Short answer

CORRECT ANSWER

For three points A, B, C not on one line, triangle ABC is the union of segments AB, BC, and CA.

GRADING NOTE Any equivalent wording. Essential: three noncollinear points and the three segments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q3/5

Write a precise definition of congruent segments.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q4/5

94

QUESTION ID

S98-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Two segments are congruent if they have the same length.

GRADING NOTE Any equivalent wording. Accept "if a rigid motion carries one onto the other".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q4/5

Write a precise definition of a segment bisector.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q5/5

95

QUESTION ID

S98-F1-Q5

TYPE

Short answer

CORRECT ANSWER

A segment bisector of AB is a point, segment, ray, line, or plane that passes through the midpoint of AB.

GRADING NOTE Any equivalent wording that requires passing through the midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q5/5

SET 98 • HSG-CO.A.1

FORWARD 2

Test segment-based definitions with counterexamples • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Proposed definition: "Point M is the midpoint of segment AB if AM = MB."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q1/5

97

QUESTION ID

S98-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The top vertex of an isosceles triangle with base AB has AM = MB but is not on AB. Repair: M is the point ON segment AB with AM = MB.

GRADING NOTE Any counterexample with M off segment AB and AM = MB, plus a repair that puts M on the segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q1/5

Proposed definition: "A triangle is three segments."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q2/5

98

QUESTION ID

S98-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Three segments that do not meet end to end (for example three parallel segments). Repair: for noncollinear A, B, C, the segments AB, BC, and CA.

GRADING NOTE Any valid counterexample plus a repair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q2/5

Proposed definition: "C is between A and B if AC + CB = AB."

Is this enough when A, B, and C are any three points of a plane? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q3/5

99

QUESTION ID

S98-F2-Q3

TYPE

Short answer

CORRECT ANSWER

Yes for the plane: AC + CB = AB forces C onto segment AB (otherwise the triangle inequality gives AC + CB > AB). Many texts still add "C on line AB" for clarity.

GRADING NOTE Any correct explanation using the triangle inequality. Accept "add C on line AB" as a valid repair if the student says why it is safe.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q3/5

True or false?

"The midpoint of AB is a point M with AM = MB" is precise enough, because every such point lies on segment AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q4/5

100

QUESTION ID

S98-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every point on the perpendicular bisector of AB satisfies AM = MB; only one of them is on the segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q4/5

Proposed definition: "A segment bisector is a line that cuts a segment into two parts."

Give a counterexample and repair it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q5/5

101

QUESTION ID

S98-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Any line crossing AB at a point other than the midpoint cuts it into two unequal parts. Repair: it passes through the midpoint.

GRADING NOTE Any valid counterexample plus a repair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S98-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q5/5

Precise definition of perpendicular lines

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Recognize perpendicular lines and right angles	**4.G.A.1**
BACKWARD 2	Find the other angles when two lines cross	**7.G.B.5**
FORWARD 1	Prove consequences of the definition of perpendicular lines	**HSG-CO.C.9**
FORWARD 2	Reason about perpendicular lines with a third line, in a plane and…	**HSG-CO.C.9**

SET 2 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Ben writes: "Perpendicular lines are lines that make a T."

Explain what is imprecise, and write a precise definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q1/5

104

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

"Make a T" is not defined (a T can be tilted or have unequal angles, and lines do not stop). Precise: two lines are perpendicular if they intersect and form a right angle.

GRADING NOTE Any equivalent answer. Essential: names the vague term and gives a definition with both parts (intersect; right angle). Accept "intersect to form congruent adjacent angles".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q1/5

Write a precise definition of perpendicular lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q2/5

105

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

Two lines are perpendicular if they intersect and form a right angle.

GRADING NOTE Any equivalent wording, such as "intersect to form four right angles" or "intersect to form congruent adjacent angles". Not accepted: "lines that cross" or "lines that make a T" (not precise).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q2/5

Lines a and b intersect at P. Two angles at P form a linear pair and have equal measures.

Use the definition to decide whether a and b are perpendicular. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q3/5

106

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes. A linear pair adds to 180°, so equal angles are 90° each. The lines intersect and form a right angle, so they are perpendicular.

GRADING NOTE Any equivalent explanation that finds a 90° angle and then cites both parts of the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q3/5

Line m passes through (0, 0) and (2, 1). Line n passes through (0, 0) and (−1, 2). Use the definition (intersect and form a right angle) to decide whether m and n are perpendicular. Show how you know the angle at (0, 0) is a right angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q4/5

107

QUESTION ID

S2-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes. They meet at (0, 0). With A(2, 1) and B(−1, 2): $OA^2 = 5$, $OB^2 = 5$, $AB^2 = 9 + 1 = 10 = OA^2 + OB^2$, so by the converse of the Pythagorean Theorem the angle at O is a right angle.

GRADING NOTE Any valid method that shows a right angle at the intersection (Pythagorean converse, or slopes ½ and −2 with a reason). Essential: the intersection and the right angle.

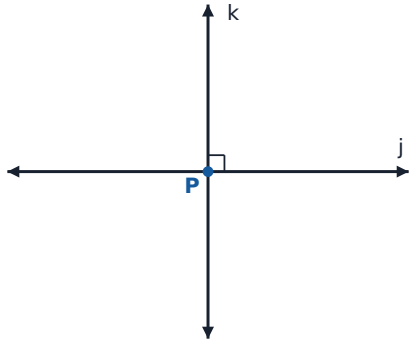
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q4/5

Lines j and k meet at point P, and one of the angles they form is marked as a right angle.

Are j and k perpendicular? Use the definition to explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q5/5

108

QUESTION ID

S2-M-Q5

TYPE

Short answer

CORRECT ANSWER

Yes. The lines intersect and form a right angle, which is the definition of perpendicular lines.

GRADING NOTE Any equivalent explanation that names both parts of the definition: they intersect, and an angle is a right angle (given by the mark).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q5/5

SET 2 • HSG-CO.A.1

BACKWARD 1

Recognize perpendicular lines and right angles • Grade 4 • Geometry • 4.G.A.1

True or false?

Lines that meet at a 90° angle are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q1/5

110

QUESTION ID

S2-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is the definition of perpendicular lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q1/5

True or false?

Any two lines that cross are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q2/5

111

QUESTION ID

S2-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Lines can cross at angles other than 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q2/5

Two perpendicular lines cross. How many right angles do they form?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q3/5

112

QUESTION ID

S2-B1-Q3

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q3/5

What is true of two sides of a rectangle that share a vertex?

A. They are perpendicular

B. They are parallel

C. They never meet

D. They form a 60° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q4/5

113

QUESTION ID

S2-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. They are perpendicular

GRADING NOTE Adjacent sides of a rectangle meet at a right angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q4/5

Which pair of streets is most like perpendicular lines?

A. Two straight streets that never meet

B. Two streets that curve around a park

C. Two streets that meet at a sharp point

D. Two streets that cross at a square corner

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q5/5

114

QUESTION ID

S2-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Two streets that cross at a square corner

GRADING NOTE Perpendicular lines meet at a right (square) angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

BACKWARD 2

Find the other angles when two lines cross • Grade 7 • Geometry • 7.G.B.5

Two lines cross. One of the angles measures 90°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q1/5

116

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

90°, 90°, 90°

GRADING NOTE Any order. The adjacent angles are 180° − 90°; the vertical angle is equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q1/5

Two lines cross. One of the angles measures 70°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q2/5

117

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

110°, 70°, 110°

GRADING NOTE Any order. Adjacent angles: 180° − 70° = 110°; the vertical angle is 70°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q2/5

True or false?

Vertical angles formed by two crossing lines are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q3/5

118

QUESTION ID

S2-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Vertical angles have equal measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q3/5

Two angles form a linear pair. One measures 90°. What does the other measure?

A. 180°

B. 45°

C. 90°

D. 270°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q4/5

119

QUESTION ID

S2-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 90°

GRADING NOTE A linear pair adds to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q4/5

True or false?

Two lines cross and one angle measures 100°.
The angle next to it also measures 100°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q5/5

120

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The angle next to it forms a linear pair with it, so it measures 80°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSG-CO.A.1

FORWARD 1

Prove consequences of the definition of perpendicular lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are perpendicular and meet at point P.
Prove that all four angles at P are right angles.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q1/5

122

QUESTION ID

S2-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

All four angles measure 90°.

GRADING NOTE REQUIRED WORK: Start from the definition (one angle at P is a right angle). Use the linear pair (supplementary) to get the adjacent angles and vertical angles (or another linear pair) to get the fourth, with a reason for each step.

Any valid proof. Essential: (1) definition gives one 90° angle; (2) each adjacent angle forms a linear pair with it, so it is 180° − 90° = 90°; (3) the fourth angle is vertical to the first (or forms a linear pair with a 90° angle). Invalid: "they look like right angles".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q1/5

Two lines meet at point P and form a linear pair of congruent angles. Prove that the lines are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q2/5

123

QUESTION ID

S2-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

Each angle of the pair measures 90°, so the lines are perpendicular.

GRADING NOTE REQUIRED WORK: Linear pair angles add to 180°; congruent means equal measures, so each is 90°; then apply the definition.

Any valid proof. Essential: x + x = 180° from the linear pair, so x = 90°; the lines meet and form a right angle, which is the definition of perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q2/5

Lines a and b intersect, and one of the angles they form measures (3x + 15)°. For what value of x are a and b perpendicular? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q3/5

124

QUESTION ID

S2-F1-Q3

TYPE

Short answer

CORRECT ANSWER

x = 25, because perpendicular lines form a 90° angle: 3x + 15 = 90.

GRADING NOTE Form required: the value with the reason from the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q3/5

Two intersecting lines form a pair of vertical angles that are supplementary. Prove that the lines are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q4/5

125

QUESTION ID

S2-F1-Q4

TYPE

Answer with required work

CORRECT ANSWER

Each vertical angle is 90°, so the lines are perpendicular.

GRADING NOTE REQUIRED WORK: Vertical angles are congruent; use the supplementary condition to find each angle, then apply the definition.

Any valid proof. Essential: the vertical angles are equal (x = y), x + y = 180°, so x = 90°; the lines intersect and form a right angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q4/5

Lines p and q meet at O. Ana says: "Two adjacent angles at O add to 180°, so p ⊥ q." Explain why her reason does not prove perpendicularity, and state a fact that does.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q5/5

126

QUESTION ID

S2-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Every pair of crossing lines has adjacent angles that add to 180° (linear pairs), so that is always true. A right angle at O (or a linear pair of congruent angles) proves p ⊥ q.

GRADING NOTE Any equivalent explanation that identifies the reason as always true and names a sufficient condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

FORWARD 2

Reason about perpendicular lines with a third line, in a plane and in space • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

In a plane, lines a and b are both perpendicular to line t. Prove that a and b are parallel.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q1/5

128

QUESTION ID

S2-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

a ∥ b

GRADING NOTE REQUIRED WORK: Use the corresponding angles that a and b make with t (both right angles, so congruent) and the converse of the corresponding angles postulate; or argue by contradiction with the triangle angle sum.

Any valid proof. Essential: both corresponding angles at t are 90°, so they are congruent; congruent corresponding angles make the lines parallel. Alternative: if a and b met, they would form a triangle with t having two 90° angles, so its angle sum would exceed 180°. Invalid: "they look parallel".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q1/5

True or false?

In three-dimensional space, two lines that are each perpendicular to the same line must be parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q2/5

129

QUESTION ID

S2-F2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Counterexample: in a corner of a room, the two floor edges are both perpendicular to the vertical edge, but they meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q2/5

In a plane, line a is parallel to line b, and line t is perpendicular to line a. What is true about t and b?

A. t and b do not meet

B. t is perpendicular to b

C. t is parallel to b

D. The angle between t and b cannot be known

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q3/5

130

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. t is perpendicular to b

GRADING NOTE t crosses a, so it crosses b. The corresponding angle at b equals the right angle at a, so t ⊥ b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q3/5

In a coordinate plane, line t is the x-axis. Line a is perpendicular to t at (0, 0), and line b is perpendicular to t and passes through (5, 2). Write an equation of line b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q4/5

131

QUESTION ID

S2-F2-Q4

TYPE

Short answer

CORRECT ANSWER

x = 5

GRADING NOTE Form required: an equation of the line. A line perpendicular to the x-axis is vertical, so every point has x = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q4/5

True or false?

In a plane, through a point on a line there is exactly one line perpendicular to that line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q5/5

132

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The perpendicular makes a 90° angle with the line at the point, and there is only one ray making a 90° angle on each side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q5/5

Precise definition of parallel lines

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Recognize parallel lines	**4.G.A.1**
BACKWARD 2	Decide whether two lines in the coordinate plane intersect	**8.EE.C.8**
BACKWARD 3	Visualize lines and planes in a box	**7.G.A.3**
FORWARD 1	Prove that lines are parallel from angle facts	**HSG-CO.C.9**
FORWARD 2	Use the parallel postulate	**HSG-CO.C.9**

SET 99 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of parallel lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q1/5

135

QUESTION ID

S99-M-Q1

TYPE

Short answer

CORRECT ANSWER

Two lines are parallel if they lie in the same plane and do not intersect.

GRADING NOTE Any equivalent wording. Essential: in the same plane, and no point in common. Not accepted: "lines that never meet" without the plane (skew lines in space never meet), or "lines that point the same way" (not precise).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q1/5

Lines ℓ and m do not intersect. Which additional fact makes them parallel?

A. They lie in the same plane

B. They both look horizontal in a drawing

C. They are drawn the same length

D. They are both perpendicular to one segment in space

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q2/5

136

QUESTION ID

S99-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. They lie in the same plane

GRADING NOTE Parallel lines are coplanar lines that do not meet. Lines in space can miss each other without being parallel (skew lines), even if both are perpendicular to one segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q2/5

Eli writes: "Parallel lines are lines that never meet."

Give an example of two lines that fit Eli's sentence but are not parallel, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q3/5

137

QUESTION ID

S99-M-Q3

TYPE

Short answer

CORRECT ANSWER

Two skew lines, such as a front top edge and a back side edge of a box that point in different directions. Repair: parallel lines lie in the same plane and never meet.

GRADING NOTE Any valid pair of skew lines plus the repaired definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q3/5

Box ABCD-EFGH has top face ABCD directly above bottom face EFGH (A above E, B above F, C above G, D above H). Using the definition, decide whether line AB is parallel to line HG, and whether line AB is parallel to line FG. Explain each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q4/5

138

QUESTION ID

S99-M-Q4

TYPE

Short answer

CORRECT ANSWER

AB ∥ HG: they lie in one plane (the slanted plane through A, B, G, H) and do not meet. AB and FG are not parallel: they never meet but do not lie in one plane (they are skew).

GRADING NOTE Form required: both decisions, each with both parts of the definition checked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q4/5

Line p is y = 2x + 1 and line q is y = 2x − 3. Use the definition to decide whether p and q are parallel. Show that they do not intersect.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q5/5

139

QUESTION ID

S99-M-Q5

TYPE

Short answer

CORRECT ANSWER

Yes. Both lie in the coordinate plane. If they met, 2x + 1 = 2x − 3 would be true for some x, which gives 1 = −3, so there is no intersection point.

GRADING NOTE Any equivalent argument that checks both parts: same plane and no common point (an equation with no solution).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q5/5

SET 99 • HSG-CO.A.1

BACKWARD 1

Recognize parallel lines • Grade 4 • Geometry • 4.G.A.1

True or false?

Two lines in a plane that never meet are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q1/5

141

QUESTION ID

S99-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is the definition (the plane is given).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q1/5

What is true of the opposite sides of a rectangle?

A. They are parallel

B. They are perpendicular

C. They meet at one point

D. They are neither parallel nor perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q2/5

142

QUESTION ID

S99-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. They are parallel

GRADING NOTE Opposite sides of a rectangle never meet and lie in one plane, so they are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q2/5

True or false?

Any two lines that look slanted the same amount in a drawing are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q3/5

143

QUESTION ID

S99-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A drawing can mislead; parallel lines must lie in one plane and never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q3/5

Which pair of streets is most like parallel lines?

A. Two streets that meet at a corner

B. Two streets that cross

C. A street and a bridge above it going another way

D. Two straight streets on flat ground that never meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q4/5

144

QUESTION ID

S99-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Two straight streets on flat ground that never meet

GRADING NOTE Parallel lines are coplanar and never meet; a bridge above going another way is like a skew line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q4/5

How many times do two parallel lines intersect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q5/5

145

QUESTION ID

S99-B1-Q5

TYPE

Short answer

CORRECT ANSWER

0 (never)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q5/5

SET 99 • HSG-CO.A.1

BACKWARD 2

Decide whether two lines in the coordinate plane intersect • Grade 8 • Expressions & Equations • 8.EE.C.8

Do the lines y = x + 2 and y = x − 1 intersect? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q1/5

147

QUESTION ID

S99-B2-Q1

TYPE

Short answer

CORRECT ANSWER

No. x + 2 = x − 1 has no solution (2 ≠ −1).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q1/5

Do the lines y = 2x and y = −x + 6 intersect? If so, where?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q2/5

148

QUESTION ID

S99-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes, at (2, 4).

GRADING NOTE Form required: the point. 2x = −x + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q2/5

Which line never meets y = 3x + 1?

A. y = x + 1

B. y = 3

C. y = −3x + 1

D. y = 3x − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q3/5

149

QUESTION ID

S99-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 3x − 4

GRADING NOTE Same slope, different intercept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q3/5

True or false?

Two different lines with the same slope never intersect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q4/5

150

QUESTION ID

S99-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Setting them equal gives a false statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q4/5

True or false?

The lines y = 2x + 5 and y = 5 never meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q5/5

151

QUESTION ID

S99-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They meet at (0, 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q5/5

SET 99 • HSG-CO.A.1

BACKWARD 3

Visualize lines and planes in a box • Grade 7 • Geometry • 7.G.A.3

True or false?

Any two edges of a box that never meet are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B3-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q1/5

153

QUESTION ID

S99-B3-Q1

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Skew edges of a box never meet and are not parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B3-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q1/5

In a box, the top front edge and the bottom back edge (running the same direction) are

A. perpendicular

B. skew

C. parallel

D. the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q2/5

154

QUESTION ID

S99-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. parallel

GRADING NOTE They lie in one slanted plane and never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How many faces of a box are there?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q3/5

155

QUESTION ID

S99-B3-Q3

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q3/5

Which pair of box edges are skew?

A. A top front edge and the bottom front edge below it

B. A top front edge and a bottom side edge that runs front to back

C. Two top edges that meet at a corner

D. Two vertical edges

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q4/5

156

QUESTION ID

S99-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A top front edge and a bottom side edge that runs front to back

GRADING NOTE They point in different directions and do not meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q4/5

True or false?

Two vertical edges of a box are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q5/5

157

QUESTION ID

S99-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They lie in one plane and never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q5/5

SET 99 • HSG-CO.A.1

FORWARD 1

Prove that lines are parallel from angle facts • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by transversal t, and a pair of alternate interior angles are congruent. Explain why ℓ and m cannot meet (so they are parallel).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q1/5

159

QUESTION ID

S99-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

$\ell \parallel m$

GRADING NOTE REQUIRED WORK: Argue by contradiction: if they met, they would form a triangle with an exterior angle equal to a remote interior angle.

Any valid proof. Essential: assume ℓ and m meet at P; then t, ℓ, m form a triangle in which one of the congruent angles is an exterior angle and the other a remote interior angle; the exterior angle theorem (or the angle sum) says the exterior angle is larger, a contradiction. Alternative: cite the alternate interior angles converse with the parallel postulate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q1/5

Lines ℓ and m are cut by a transversal. A pair of corresponding angles measure $(5x - 10)°$ and $(3x + 20)°$. For what x are ℓ and m parallel?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q2/5

160

QUESTION ID

S99-F1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. Congruent corresponding angles: $5x - 10 = 3x + 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q2/5

Same-side interior angles formed by lines a, b, and a transversal measure 115° and 65°. Are a and b parallel? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q3/5

161

QUESTION ID

S99-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Yes. Same-side interior angles that are supplementary (115° + 65° = 180°) make the lines parallel.

GRADING NOTE Any equivalent answer citing the converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q3/5

In a plane, line a ∥ line b and line b ∥ line c, where a and c are different lines. Prove that a ∥ c.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q4/5

162

QUESTION ID

S99-F1-Q4

TYPE

Answer with required work

CORRECT ANSWER

a ∥ c

GRADING NOTE

REQUIRED WORK: Argue by contradiction with the parallel postulate.

Any valid proof. Essential: if a and c met at P, there would be two lines through P parallel to b, contradicting the parallel postulate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q4/5

Which fact proves that two coplanar lines cut by a transversal are parallel?

A. A linear pair adds to 180°

B. Two angles are acute

C. A pair of alternate interior angles are congruent

D. A pair of vertical angles are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q5/5

163

QUESTION ID

S99-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. A pair of alternate interior angles are congruent

GRADING NOTE Vertical angles and linear pairs occur for every pair of crossing lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q5/5

SET 99 • HSG-CO.A.1

FORWARD 2

Use the parallel postulate • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

In a plane, through a point not on a line there is exactly one line parallel to the given line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q1/5

165

QUESTION ID

S99-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the parallel postulate of Euclidean geometry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q1/5

Point P is not on line ℓ. Line m passes through P and is parallel to ℓ. Line n also passes through P and is parallel to ℓ. What can you conclude about m and n? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q2/5

166

QUESTION ID

S99-F2-Q2

TYPE

Short answer

CORRECT ANSWER

m and n are the same line, by the parallel postulate (only one parallel to ℓ passes through P).

GRADING NOTE Any equivalent answer naming the postulate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q2/5

In the coordinate plane, find an equation of the line through (2, 5) parallel to y = −3x + 1, and explain why it is the only one.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q3/5

167

QUESTION ID

S99-F2-Q3

TYPE

Short answer

CORRECT ANSWER

y = −3x + 11. By the parallel postulate there is exactly one such line.

GRADING NOTE Form required: the equation and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q3/5

Which statement is the parallel postulate?

A. Vertical angles are congruent

B. All right angles are congruent

C. Through any two points there is exactly one line

D. Through a point not on a line there is exactly one parallel to the line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q4/5

168

QUESTION ID

S99-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Through a point not on a line there is exactly one parallel to the line

GRADING NOTE The others are different facts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q4/5

Why does the parallel postulate say "a point not on the line"?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q5/5

169

QUESTION ID

S99-F2-Q5

TYPE

Short answer

CORRECT ANSWER

A line through a point on ℓ either is ℓ or crosses ℓ at that point, so it cannot be a different line parallel to ℓ.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S99-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q5/5

Transformations as functions: inputs, outputs, and preimages

MAIN	Geometry main question	**HSG-CO.A.2**
BACKWARD 1	Evaluate a rule for a given input	**8.F.A.1**
BACKWARD 2	Apply coordinate rules for translations, reflections, and rotations	**8.G.A.3**
BACKWARD 3	Read and locate ordered pairs in all four quadrants	**6.NS.C.6.c**
FORWARD 1	Compose two transformations and write one rule	**HSG-CO.A.2**
FORWARD 2	Find the inverse of a transformation	**HSG-CO.A.2**

SET 3 • HSG-CO.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

The transformation T is defined by T(x, y) = (x + 4, y − 3). Find T(2, 5).

T(2, 5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q1/5

172

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

(6, 2)

GRADING NOTE Form required: an ordered pair. Input (2, 5): x + 4 = 6 and y − 3 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q1/5

The transformation R is defined by R(x, y) = (−y, x).
Find R(3, −1).

R(3, −1) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q2/5

173

QUESTION ID
S3-M-Q2

TYPE
Short answer

CORRECT ANSWER
(1, 3)

GRADING NOTE Form required: an ordered pair. −y = −(−1) = 1 and x = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q2/5

The transformation F takes each point (x, y) to (x, −y). The output point is F(P) = (4, 7). Find the input point P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q3/5

174

QUESTION ID

S3-M-Q3

TYPE

Short answer

CORRECT ANSWER

(4, −7)

GRADING NOTE Form required: an ordered pair. F(4, −7) = (4, 7).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q3/5

Every point of the plane is moved 5 units right and 2 units up. Which function describes this transformation?

A. T(x, y) = (x − 5, y − 2)

B. T(x, y) = (5x, 2y)

C. T(x, y) = (x + 2, y + 5)

D. T(x, y) = (x + 5, y + 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q4/5

175

QUESTION ID

S3-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. T(x, y) = (x + 5, y + 2)

GRADING NOTE Right adds to x and up adds to y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q4/5

The transformation D is defined by D(x, y) = (2x, 2y). Find every input point P whose output is itself (D(P) = P). Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q5/5

176

QUESTION ID

S3-M-Q5

TYPE

Short answer

CORRECT ANSWER

Only (0, 0). D(x, y) = (x, y) means 2x = x and 2y = y, so x = 0 and y = 0.

GRADING NOTE Form required: the point with the equations that show it is the only one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 1

Evaluate a rule for a given input • Grade 8 • Functions • 8.F.A.1

f(x) = 3x − 2. Find f(4).

f(4) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q1/5

178

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. 3(4) − 2 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q1/5

A rule takes a number, doubles it, and adds 1. What is the output for the input 6?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q2/5

179

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q2/5

g(x) = x + 5. Which input gives the output 12?

Input: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q3/5

180

QUESTION ID

S3-B1-Q3

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. 7 + 5 = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

A function assigns exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q4/5

181

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is the definition of a function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

The rule "the input 3 gives the outputs 4 and 5" describes a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q5/5

182

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A function gives exactly one output for each input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 2

Apply coordinate rules for translations, reflections, and rotations • Grade 8 • Geometry • 8.G.A.3

Translate the point (2, −1) 3 units left and 4 units up. Where does it go?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q1/5

184

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−1, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q1/5

Reflect the point (5, 2) across the y-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q2/5

185

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−5, 2)

GRADING NOTE Form required: an ordered pair. (x, y) → (−x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q2/5

Rotate the point (4, 1) by 180° about the origin.

A. (1, 4)

B. (−1, 4)

C. (4, −1)

D. (−4, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q3/5

186

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−4, −1)

GRADING NOTE A 180° rotation about the origin sends (x, y) to (−x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q3/5

True or false?

Reflecting the point (3, 6) across the x-axis gives (3, −6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q4/5

187

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (x, y) → (x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q4/5

True or false?

Rotating the point (0, 5) by 90° counterclockwise about the origin gives (5, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q5/5

188

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (x, y) → (−y, x) gives (−5, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSG-CO.A.2

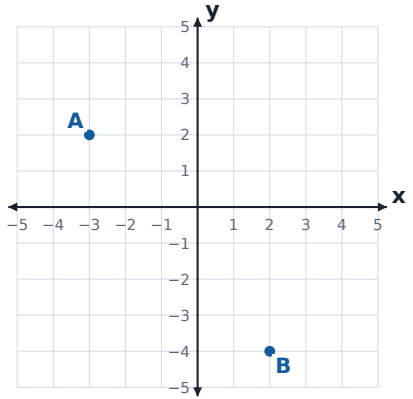
BACKWARD 3

Read and locate ordered pairs in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

ANSWER KEY Reduced copy of the question page, with its answer

What are the coordinates of point A?

A = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q1/5

190

QUESTION ID

S3-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, 2)

GRADING NOTE Form required: an ordered pair (x, y).

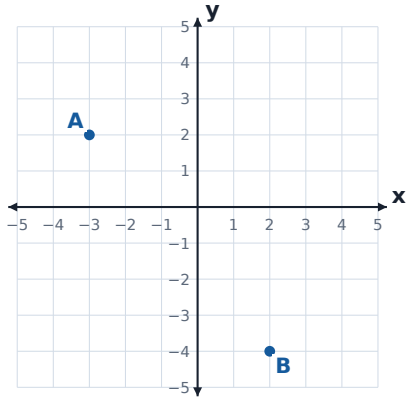
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What are the coordinates of point B?

B = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q2/5

191

QUESTION ID

S3-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(2, −4)

GRADING NOTE Form required: an ordered pair (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q2/5

Which point is in Quadrant III?

A. (−2, −5)

B. (−2, 5)

C. (2, 5)

D. (2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q3/5

192

QUESTION ID

S3-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (−2, −5)

GRADING NOTE Quadrant III has $x < 0$ and $y < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q3/5

True or false?

The point (0, −3) lies on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q4/5

193

QUESTION ID

S3-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its x-coordinate is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q4/5

True or false?

The point (−2, 5) is 2 units to the right of the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q5/5

194

QUESTION ID

S3-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x = −2 is 2 units to the LEFT of the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSG-CO.A.2

FORWARD 1

Compose two transformations and write one rule • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

T(x, y) = (x + 2, y) and R(x, y) = (−x, y). Find R(T(1, 4)).

R(T(1, 4)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q1/5

196

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, 4)

GRADING NOTE Form required: an ordered pair. T(1, 4) = (3, 4); R(3, 4) = (−3, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q1/5

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Write a single rule for $R(T(x, y))$.

$R(T(x, y)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q2/5

197

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(−x − 2, y)

GRADING NOTE Any equivalent form, such as $(-(x + 2), y)$. Apply T first, then R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q2/5

T(x, y) = (x + 2, y) and R(x, y) = (−x, y). Find T(R(1, 4)).

T(R(1, 4)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q3/5

198

QUESTION ID

S3-F1-Q3

TYPE

Short answer

CORRECT ANSWER

(1, 4)

GRADING NOTE Form required: an ordered pair. R(1, 4) = (−1, 4); T(−1, 4) = (1, 4). The order changes the answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q3/5

True or false?

For any two transformations, doing them in either order gives the same result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q4/5

199

QUESTION ID

S3-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A translation and a reflection give different results in different orders (for example, R(T(1, 4)) ≠ T(R(1, 4)) above).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q4/5

$T(x, y) = (x + 3, y - 1)$ and $S(x, y) = (x - 1, y + 5)$.
Which rule is $S(T(x, y))$?

A. $(x + 2, y + 4)$

B. $(x - 3, y - 5)$

C. $(x + 2, y - 4)$

D. $(x + 4, y - 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q5/5

200

QUESTION ID
S3-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. (x + 2, y + 4)

GRADING NOTE $S(x + 3, y - 1) = (x + 3 - 1, y - 1 + 5) = (x + 2, y + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

FORWARD 2

Find the inverse of a transformation • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Find a rule for the inverse of T(x, y) = (x − 6, y + 2).

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q1/5

202

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 6, y − 2)

GRADING NOTE Any equivalent form. The inverse undoes T: T(x + 6, y − 2) = (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q1/5

R(x, y) = (−y, x) is a 90° counterclockwise rotation about the origin. Find a rule for its inverse.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q2/5

203

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(y, −x)

GRADING NOTE Any equivalent form: a 90° clockwise rotation. Check: R(y, −x) = (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q2/5

True or false?

Every rotation is its own inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q3/5

204

QUESTION ID

S3-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A 90° rotation is undone by a 270° (or −90°) rotation; only a 180° rotation undoes itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q3/5

D(x, y) = (3x, 3y). Which rule undoes D?

A. (x − 3, y − 3)

B. (3y, 3x)

C. (−3x, −3y)

D. $(\frac{x}{3}, \frac{y}{3})$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q4/5

205

QUESTION ID

S3-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (x/3, y/3)

GRADING NOTE Dividing each coordinate by 3 undoes multiplying by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q4/5

True or false?

The transformation P(x, y) = (x, 0), which moves every point straight up or down to the x-axis, has an inverse transformation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q5/5

206

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE P is not one-to-one: (2, 1) and (2, 5) have the same output (2, 0), so no rule can undo it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q5/5

Compare transformations that preserve distance and angle with those that do not

MAIN	Geometry main question	**HSG-CO.A.2**
BACKWARD 1	Find the distance between two points	**8.G.B.8**
BACKWARD 2	Know that rigid motions keep lengths and angle measures	**8.G.A.1**
FORWARD 1	Classify transformations as rigid motions, similarity transformatio…	**HSG-SRT.A.2**
FORWARD 2	Prove with variables that a transformation preserves or scales ev…	**HSG-SRT.A.2**

SET 4 • HSG-CO.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Which transformation does NOT preserve distance?

A. R(x, y) = (−x, y)

B. T(x, y) = (x + 4, y − 1)

C. H(x, y) = (2x, y)

D. Q(x, y) = (−y, x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q1/5

209

QUESTION ID

S4-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. H(x, y) = (2x, y)

GRADING NOTE H stretches horizontally: (0, 0) and (1, 0) are 1 unit apart, but their images (0, 0) and (2, 0) are 2 units apart. The others are a translation, a reflection, and a rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q1/5

H(x, y) = (2x, y). Points A(0, 0) and B(3, 4) are 5 units apart.

Find H(A), H(B), and the distance between them. Does H preserve distance?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q2/5

210

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

H(A) = (0, 0), H(B) = (6, 4); the distance is √52 = 2√13 ≈ 7.21; no.

GRADING NOTE Form required: both images, the distance (exact or to 0.01), and the conclusion. $\sqrt{(6^2 + 4^2)} = \sqrt{52} \neq 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q2/5

The translation T(x, y) = (x − 2, y + 5) maps A(1, 1) and B(4, 5) to A′ and B′. Show that A′B′ = AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q3/5

211

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

A′(−1, 6), B′(2, 10). AB = √(3² + 4²) = 5 and A′B′ = √(3² + 4²) = 5, so A′B′ = AB.

GRADING NOTE Form required: both images and both distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q3/5

V(x, y) = (x, 3y). Does V preserve the measure of every angle? Use the right angle at O(0, 0) between the rays through (1, 1) and (1, −1) to decide.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q4/5

212

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. The images of (1, 1) and (1, −1) are (1, 3) and (1, −3); the angle between the rays from O through them is about 143°, not 90°.

GRADING NOTE Form required: the images, the new angle (or a reason it is not 90°, such as slopes 3 and −3 not multiplying to −1), and the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q4/5

S(x, y) = (3x, 3y). Does S preserve distance? Does S preserve angle measure? Explain each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q5/5

213

QUESTION ID

S4-M-Q5

TYPE

Short answer

CORRECT ANSWER

Distance: no; every distance is multiplied by 3. Angle measure: yes; S is a dilation, which keeps every angle measure.

GRADING NOTE Form required: both answers with reasons. Example: (0, 0) and (1, 0) are 1 apart; their images are 3 apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q5/5

SET 4 • HSG-CO.A.2

BACKWARD 1

Find the distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q1/5

215

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $\sqrt{(3^2 + 4^2)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q1/5

Find the distance between (−2, 3) and (4, 3).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q2/5

216

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. Same y, so |4 − (−2)| = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q2/5

Find the distance between (0, 0) and (5, 12).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q3/5

217

QUESTION ID

S4-B1-Q3

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form. √(25 + 144) = 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q3/5

True or false?

The distance from (0, 0) to (1, 1) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q4/5

218

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(1^2 + 1^2)} = \sqrt{2} \approx 1.41$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q4/5

Find the distance between (2, −1) and (5, 3).

A. 25

B. √7

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q5/5

219

QUESTION ID

S4-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 5

GRADING NOTE $\sqrt{(3^2 + 4^2)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

BACKWARD 2

Know that rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

True or false?

A rotation keeps every segment the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q1/5

221

QUESTION ID

S4-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rotations are rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q1/5

True or false?

A reflection can change the measure of an angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q2/5

222

QUESTION ID

S4-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Reflections keep angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q2/5

A triangle is translated. What changes?

A. Its angle measures

B. Its side lengths

C. The positions of its vertices

D. Its area

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q3/5

223

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The positions of its vertices

GRADING NOTE A translation moves the figure but keeps its lengths, angles, and area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q3/5

A segment 7 cm long is reflected across a line. How long is its image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q4/5

224

QUESTION ID

S4-B2-Q4

TYPE

Short answer

CORRECT ANSWER

7 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q4/5

A 35° angle is rotated 90° about a point. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q5/5

225

QUESTION ID

S4-B2-Q5

TYPE

Short answer

CORRECT ANSWER

35°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSG-CO.A.2

FORWARD 1

Classify transformations as rigid motions, similarity transformations, or neither • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Which transformation preserves angle measure but not distance?

A. (x, y) → (4x, y)

B. (x, y) → (x + 4, y)

C. (x, y) → (x, y + 4)

D. (x, y) → (4x, 4y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q1/5

227

QUESTION ID

S4-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (4x, 4y)

GRADING NOTE Multiplying both coordinates by 4 is a dilation: angles stay, distances are multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q1/5

Classify (x, y) → (−2x, −2y) as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q2/5

228

QUESTION ID
S4-F1-Q2

TYPE
Short answer

CORRECT ANSWER

A similarity transformation that is not rigid: every distance is doubled and angles are kept (a dilation by 2 followed by a 180° rotation).

GRADING NOTE Any equivalent explanation that shows the factor 2 for distances and that angles are kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q2/5

Classify (x, y) → (x + y, y) as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q3/5

229

QUESTION ID

S4-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Neither. (0, 0) and (0, 1) are 1 apart but their images (0, 0) and (1, 1) are √2 apart, while (0, 0) and (1, 0) stay 1 apart: distances are not all multiplied by one number, and the right angle at the origin becomes 45°.

GRADING NOTE Any correct counterexample showing that distances are not scaled by a single factor (or an angle changes).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q3/5

True or false?

Every similarity transformation is a rigid motion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q4/5

230

QUESTION ID

S4-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only similarity transformations with scale factor 1 are rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q4/5

True or false?

A transformation that multiplies every distance by the same number k > 0 always changes angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q5/5

231

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Multiplying all distances by k keeps triangles similar (SSS), so angle measures stay the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

FORWARD 2

Prove with variables that a transformation preserves or scales every distance • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Prove that T(x, y) = (x + a, y + b) preserves the distance between ANY two points (x_1, y_1) and (x_2, y_2).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q1/5

233

QUESTION ID

S4-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

The image distance equals $\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$, the original distance.

GRADING NOTE

REQUIRED WORK: Write both images, apply the distance formula to the images, and show that a and b cancel.

Any valid algebraic proof. Essential: images $(x_1 + a, y_1 + b)$ and $(x_2 + a, y_2 + b)$; the differences are $x_2 - x_1$ and $y_2 - y_1$, so the distance formula gives the same value. Invalid: checking one or two numerical examples only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q1/5

Prove that D(x, y) = (3x, 3y) multiplies the distance between ANY two points by 3.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q2/5

234

QUESTION ID

S4-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

The image distance is $3\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$.

GRADING NOTE REQUIRED WORK: Write both images, apply the distance formula, and factor out 9 under the square root.

Any valid proof. Essential: $\sqrt{((3x_2 - 3x_1)^2 + (3y_2 - 3y_1)^2)} = \sqrt{(9[(x_2 - x_1)^2 + (y_2 - y_1)^2])} = 3\cdot$(original distance).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q2/5

$H(x, y) = (2x, y)$. The distance between the images of (x_1, y_1) and (x_2, y_2) is $\sqrt{(4(x_2 - x_1)^2 + (y_2 - y_1)^2)}$.

For which pairs of points is the distance unchanged?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q3/5

235

QUESTION ID

S4-F2-Q3

TYPE

Short answer

CORRECT ANSWER

Exactly the pairs with $x_1 = x_2$ (points on the same vertical line).

GRADING NOTE Any equivalent condition. The distance is unchanged only if $4(x_2 - x_1)^2 = (x_2 - x_1)^2$, that is, $x_2 = x_1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q3/5

True or false?

One example in which a distance stays the same is enough to prove that a transformation preserves every distance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q4/5

236

QUESTION ID
S4-F2-Q4

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE H(x, y) = (2x, y) keeps vertical distances but changes others. A general claim needs a proof for all points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q4/5

True or false?

If a transformation preserves the distance between every pair of points, it also preserves every angle measure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q5/5

237

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The three distances of any triangle are kept, so each triangle maps to a congruent triangle (SSS), and its angles are kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q5/5

Rotations and reflections that carry a rectangle, parallelogram, or trapezoid onto itself

MAIN	Geometry main question	**HSG-CO.A.3**
BACKWARD 1	Identify lines of symmetry of figures	**4.G.A.3**
BACKWARD 2	Know the side and angle properties of parallelograms, rectangles,…	**5.G.B.3**
BACKWARD 3	Rotate points by 90° and 180° about the origin	**8.G.A.3**
FORWARD 1	Use a symmetry to deduce congruent parts of the figure	**HSG-CO.B.6**
FORWARD 2	Combine symmetries and identify a figure from its symmetries	**HSG-CO.A.5**

SET 5 • HSG-CO.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Rectangle ABCD is not a square. Describe every rotation (with angle greater than 0° and at most 360°) and every reflection that carries the rectangle onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q1/5

240

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

Rotations: 180° and 360° about the center O. Reflections: across the line through the midpoints of AB and CD, and across the line through the midpoints of BC and DA.

GRADING NOTE Form required: both rotations and both reflections, with the center and the lines named. The diagonals are NOT lines of symmetry of a non-square rectangle, and a 90° turn does not work because the sides have different lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q1/5

Parallelogram PQRS has no right angles, and its sides are not all the same length. Describe every rotation (angle greater than 0° and at most 360°, about any point) and every reflection that carries it onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q2/5

241

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

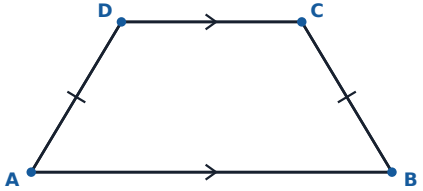
Rotations: 180° and 360° about the point where its diagonals meet. Reflections: none.

GRADING NOTE Form required: both rotations with the center, and the statement that no reflection works. The diagonals bisect each other, so a 180° turn about their intersection swaps P with R and Q with S; a parallelogram that is not a rhombus or a rectangle has no line of symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q2/5

Trapezoid ABCD has exactly one pair of parallel sides, the bases AB and DC (AB ≠ DC), and congruent legs AD = BC. Describe every rotation (angle greater than 0° and at most 360°, about any point) and every reflection that carries it onto itself.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q3/5

242

QUESTION ID

S5-M-Q3

TYPE

Short answer

CORRECT ANSWER

Only the reflection across the line through the midpoints of the two bases, and the 360° rotation.

GRADING NOTE Form required: the one reflection (named by its line) and the 360° rotation. No rotation of less than 360° works because the bases have different lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q3/5

Rectangle ABCD has A(1, 1), B(7, 1), C(7, 4), D(1, 4). Give the center of every rotation that carries it onto itself (angle greater than 0° and at most 360°), and the equation of every line of reflection that carries it onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q4/5

243

QUESTION ID

S5-M-Q4

TYPE

Short answer

CORRECT ANSWER

Rotations of 180° and 360° about (4, 2.5). Reflections across x = 4 and y = 2.5.

GRADING NOTE Form required: the center, both rotation angles, and both line equations. A 90° turn fails because AB = 6 and BC = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q4/5

Trapezoid EFGH has exactly one pair of parallel sides, EF ∥ HG, with EF = 4, HG = 8, and legs FG = 5 and EH = 3. Which rotations (angle greater than 0° and at most 360°, about any point) and which reflections carry it onto itself?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q5/5

244

QUESTION ID

S5-M-Q5

TYPE

Short answer

CORRECT ANSWER

Only the 360° rotation (the identity). It has no line of symmetry and no other rotation symmetry.

GRADING NOTE Form required: the conclusion with a reason. A symmetry would have to send the 3-unit leg to a congruent leg, but the only other leg is 5 units long, and the two bases have different lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 1

Identify lines of symmetry of figures • Grade 4 • Geometry • 4.G.A.3

How many lines of symmetry does a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q1/5

246

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. Two through opposite vertices and two through midpoints of opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q1/5

How many lines of symmetry does a rectangle that is not a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q2/5

247

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. Only the lines through midpoints of opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q2/5

True or false?

Each diagonal of a rectangle that is not a square is a line of symmetry of the rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q3/5

248

QUESTION ID

S5-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Folding along a diagonal does not match the sides of a non-square rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q3/5

Which figure has exactly one line of symmetry?

A. An isosceles triangle that is not equilateral

B. A square

C. A circle

D. A scalene triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q4/5

249

QUESTION ID

S5-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. An isosceles triangle that is not equilateral

GRADING NOTE The line through the vertex angle and the midpoint of the base is its only line of symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q4/5

True or false?

An isosceles trapezoid has a line of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q5/5

250

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The line through the midpoints of its two bases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 2

Know the side and angle properties of parallelograms, rectangles, and trapezoids • Grade 5 • Geometry • 5.G.B.3

True or false?

In every parallelogram, the diagonals have equal lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q1/5

252

QUESTION ID

S5-B2-Q1

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only rectangles (including squares) have equal diagonals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q1/5

True or false?

Every rectangle is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q2/5

253

QUESTION ID

S5-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both pairs of opposite sides of a rectangle are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q2/5

Which property do all rectangles have?

A. Diagonals that are perpendicular

B. Four sides of equal length

C. Four right angles

D. Exactly one pair of parallel sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q3/5

254

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Four right angles

GRADING NOTE A rectangle has four right angles; only a square has all sides equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q3/5

A parallelogram has one angle of 70°. Find its other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q4/5

255

QUESTION ID

S5-B2-Q4

TYPE

Short answer

CORRECT ANSWER

110°, 70°, 110°

GRADING NOTE Any order. Consecutive angles are supplementary; opposite angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q4/5

An isosceles trapezoid has a 65° angle at one end of its longer base. What is the angle at the other end of that base?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q5/5

256

QUESTION ID

S5-B2-Q5

TYPE

Short answer

CORRECT ANSWER

65°

GRADING NOTE Any equivalent form. Base angles of an isosceles trapezoid are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 3

Rotate points by 90° and 180° about the origin • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Rotate the point (3, 1) by 180° about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q1/5

258

QUESTION ID

S5-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, −1)

GRADING NOTE Form required: an ordered pair. (x, y) → (−x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q1/5

Rotate the point (2, 0) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q2/5

259

QUESTION ID

S5-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 2)

GRADING NOTE Form required: an ordered pair. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q2/5

True or false?

A 180° rotation about the origin sends (x, y) to (−x, −y).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q3/5

260

QUESTION ID

S5-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is the 180° rotation rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q3/5

Rotate the point (−4, 2) by 90° counterclockwise about the origin.

A. (4, −2)

B. (−4, −2)

C. (2, 4)

D. (−2, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q4/5

261

QUESTION ID

S5-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (−2, −4)

GRADING NOTE (x, y) → (−y, x) gives (−2, −4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q4/5

True or false?

A rotation of 360° about a point moves every point to a new position.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q5/5

262

QUESTION ID

S5-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A 360° rotation returns every point to where it started.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSG-CO.A.3

FORWARD 1

Use a symmetry to deduce congruent parts of the figure • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Isosceles trapezoid ABCD (AB ∥ DC) is carried onto itself by the reflection across the line through the midpoints of AB and DC. Which vertex does A go to? Use this to name a pair of congruent angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q1/5

264

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A goes to B (and D goes to C), so ∠DAB ≅ ∠CBA.

GRADING NOTE Any equivalent answer naming A → B and a pair of base angles (∠D ≅ ∠C also accepted). A reflection keeps angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q1/5

A 180° rotation about point O carries parallelogram PQRS onto itself, sending P to R and Q to S. Name two pairs of congruent sides that follow.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q2/5

265

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

PQ ≅ RS and QR ≅ SP.

GRADING NOTE Any equivalent form. The rotation sends segment PQ onto segment RS and QR onto SP.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q2/5

True or false?

Because a reflection carries an isosceles trapezoid onto itself, its two diagonals are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q3/5

266

QUESTION ID

S5-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The reflection that swaps A with B and D with C sends diagonal AC onto diagonal BD, and reflections keep lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q3/5

A reflection carries rectangle ABCD onto itself and swaps vertices A and B (and C and D). Which equality follows?

A. AB = BC

B. AC = AB

C. AD = AB

D. AD = BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q4/5

267

QUESTION ID

S5-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. AD = BC

GRADING NOTE The reflection sends side AD onto side BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q4/5

True or false?

If a 180° rotation about a point carries a quadrilateral onto itself, the quadrilateral must be a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q5/5

268

QUESTION ID

S5-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every parallelogram has 180° rotation symmetry; most are not rectangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

FORWARD 2

Combine symmetries and identify a figure from its symmetries • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Rectangle ABCD (not a square) has center O. Reflect it across the line through the midpoints of AB and CD, then across the line through the midpoints of BC and DA. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q1/5

270

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A 180° rotation about O.

GRADING NOTE Any equivalent form. Reflections across perpendicular lines through O combine to a 180° rotation about O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q1/5

A quadrilateral is carried onto itself by exactly two transformations: the 360° rotation and one reflection across a line through two of its vertices. Which could it be?

A. A parallelogram that is not a rectangle or a rhombus

B. A square

C. A rectangle that is not a square

D. A kite that is not a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q2/5

271

QUESTION ID

S5-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. A kite that is not a rhombus

GRADING NOTE A kite has one line of symmetry, through two opposite vertices. A rectangle has two lines and a 180° turn; a general parallelogram has a 180° turn and no line; a square has eight symmetries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q2/5

True or false?

A quadrilateral that is carried onto itself by a 180° rotation about a point must be a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q3/5

272

QUESTION ID

S5-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every parallelogram has 180° rotation symmetry about the crossing point of its diagonals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q3/5

A quadrilateral is carried onto itself by a 180° rotation but by no reflection. What is the most specific name for it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q4/5

273

QUESTION ID

S5-F2-Q4

TYPE

Short answer

CORRECT ANSWER

A parallelogram that is not a rectangle or a rhombus.

GRADING NOTE Any equivalent answer. A rectangle or a rhombus would also have reflection symmetries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q4/5

True or false?

Doing the same reflection twice gives the same result as the 360° rotation (the identity).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q5/5

274

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each point goes to its image and back again.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q5/5

Rotations and reflections that carry a regular polygon onto itself

MAIN	Geometry main question	**HSG-CO.A.3**
BACKWARD 1	Find the angle of a fraction of a full turn	**4.MD.C.5.b**
BACKWARD 2	Recognize regular polygons	**5.G.B.3**
BACKWARD 3	Find lines of symmetry of regular polygons	**4.G.A.3**
FORWARD 1	Work backward from a symmetry to the regular polygon	**HSG-CO.A.3**
FORWARD 2	Combine symmetries of a regular polygon	**HSG-CO.A.5**

SET 6 • HSG-CO.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Describe every rotation about its center, with angle greater than 0° and at most 360°, that carries a regular pentagon onto itself.

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q1/5

277

QUESTION ID

S6-M-Q1

TYPE

Short answer

CORRECT ANSWER

72°, 144°, 216°, 288°, and 360°

GRADING NOTE Form required: all five angles. Each is a multiple of 360° ÷ 5 = 72°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q1/5

How many reflections carry a regular hexagon onto itself? Describe their lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q2/5

278

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

6 reflections: 3 across lines through pairs of opposite vertices and 3 across lines through the midpoints of pairs of opposite sides.

GRADING NOTE Form required: the number and both kinds of lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q2/5

Describe every rotation about its center, with angle greater than 0° and at most 360°, that carries a regular octagon onto itself.

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q3/5

279

QUESTION ID
S6-M-Q3

TYPE
Short answer

CORRECT ANSWER
45°, 90°, 135°, 180°, 225°, 270°, 315°, and 360°

GRADING NOTE Form required: all eight angles. Each is a multiple of 360° ÷ 8 = 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q3/5

How many reflections carry a regular nonagon (9 sides) onto itself? Describe their lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q4/5

280

QUESTION ID

S6-M-Q4

TYPE

Short answer

CORRECT ANSWER

9 reflections, each across a line through one vertex and the midpoint of the opposite side.

GRADING NOTE Form required: the number and the kind of line. With an odd number of sides, no line passes through two vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q4/5

Describe every reflection that carries an equilateral triangle onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q5/5

281

QUESTION ID

S6-M-Q5

TYPE

Short answer

CORRECT ANSWER

3 reflections: across each line through a vertex and the midpoint of the opposite side.

GRADING NOTE Form required: the number and the lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 1

Find the angle of a fraction of a full turn • Grade 4 • Measurement & Data • 4.MD.C.5.b

How many degrees are in $\frac{1}{5}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q1/5

283

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

72°

GRADING NOTE Any equivalent form. 360 ÷ 5 = 72.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q1/5

How many degrees are in $\frac{1}{8}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q2/5

284

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form. 360 ÷ 8 = 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q2/5

How many degrees are in $\frac{3}{6}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q3/5

285

QUESTION ID

S6-B1-Q3

TYPE

Short answer

CORRECT ANSWER

180°

GRADING NOTE Any equivalent form. 3/6 = 1/2 and 1/2 of 360 is 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{12}$ of a full turn is 30°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q4/5

286

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 360 ÷ 12 = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q4/5

True or false?

$\frac{1}{9}$ of a full turn is 36°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q5/5

287

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 360 ÷ 9 = 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 2

Recognize regular polygons • Grade 5 • Geometry • 5.G.B.3

Which is a regular polygon?

A. A right triangle

B. A rhombus that is not a square

C. A rectangle that is not a square

D. An equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q1/5

289

QUESTION ID

S6-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. An equilateral triangle

GRADING NOTE A regular polygon has all sides equal and all angles equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q1/5

True or false?

A polygon with all sides equal must be regular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q2/5

290

QUESTION ID

S6-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A rhombus that is not a square has equal sides but unequal angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q2/5

How many sides does a regular decagon have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q3/5

291

QUESTION ID

S6-B2-Q3

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q3/5

What is the name of the regular polygon with 4 sides?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q4/5

292

QUESTION ID

S6-B2-Q4

TYPE

Short answer

CORRECT ANSWER

A square

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q4/5

True or false?

A regular hexagon has 6 equal angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q5/5

293

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Regular means all sides and all angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 3

Find lines of symmetry of regular polygons • Grade 4 • Geometry • 4.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

How many lines of symmetry does an equilateral triangle have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q1/5

295

QUESTION ID

S6-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q1/5

How many lines of symmetry does a regular pentagon have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q2/5

296

QUESTION ID

S6-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q2/5

True or false?

Every line of symmetry of a regular polygon passes through its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q3/5

297

QUESTION ID

S6-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A reflection that carries the polygon onto itself keeps its center fixed, so the center is on the line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q3/5

A line of symmetry of a square passes through

A. two adjacent vertices

B. exactly one vertex and no midpoint

C. no point of the square

D. two opposite vertices, or the midpoints of two opposite sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q4/5

298

QUESTION ID

S6-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. two opposite vertices, or the midpoints of two opposite sides

GRADING NOTE A square has 2 diagonal lines and 2 lines through midpoints of opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q4/5

True or false?

A regular hexagon has exactly 3 lines of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q5/5

299

QUESTION ID

S6-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 6: 3 through opposite vertices and 3 through midpoints of opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSG-CO.A.3

FORWARD 1

Work backward from a symmetry to the regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

The smallest positive rotation that carries a regular polygon onto itself is 24°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q1/5

301

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. 360 ÷ 24 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q1/5

A regular polygon has 20 lines of symmetry. What is the smallest positive rotation that carries it onto itself?

Rotation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q2/5

302

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

18°

GRADING NOTE Any equivalent form. A regular n-gon has n lines of symmetry, so n = 20 and 360 ÷ 20 = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A regular polygon can have 50° as its smallest positive rotation onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q3/5

303

QUESTION ID

S6-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 360 ÷ 50 = 7.2 is not a whole number of sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q3/5

Which can be the smallest positive rotation that carries a regular polygon onto itself?

A. 35°

B. 70°

C. 80°

D. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q4/5

304

QUESTION ID

S6-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 40°

GRADING NOTE The angle must be 360° ÷ n for a whole number n ≥ 3; 360 ÷ 40 = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q4/5

Counting the 360° rotation, how many rotations and reflections in all carry a regular 12-gon onto itself?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q5/5

305

QUESTION ID

S6-F1-Q5

TYPE

Short answer

CORRECT ANSWER

24

GRADING NOTE Any equivalent form. 12 rotations (30°, 60°, ..., 360°) and 12 reflections.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSG-CO.A.3

FORWARD 2

Combine symmetries of a regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Regular hexagon ABCDEF is labeled counterclockwise. It is rotated 120° counterclockwise about its center. Which vertex position does A move to?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q1/5

307

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The position of C

GRADING NOTE Any equivalent wording. Each 60° turn moves a vertex one place counterclockwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q1/5

Regular hexagon ABCDEF is labeled counterclockwise. Rotate it 60° counterclockwise about its center, then reflect it across line AD (through the original positions of A and D). Which original vertex position does vertex B end at?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q2/5

308

QUESTION ID

S6-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The position of E

GRADING NOTE Any equivalent wording. The rotation moves B to the position of C; the reflection across AD swaps C and E.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q2/5

True or false?

For a regular pentagon, a reflection followed by a different reflection that carries it onto itself is a rotation that carries it onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q3/5

309

QUESTION ID

S6-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two reflections across lines through the center combine into a rotation about the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q3/5

Square ABCD is labeled counterclockwise. Reflect it across line AC, then rotate it 90° counterclockwise about its center. Which original vertex position does vertex B end at?

A. The position of C

B. The position of A

C. The position of D

D. The position of B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q4/5

310

QUESTION ID

S6-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. The position of A

GRADING NOTE The reflection across AC sends B to the position of D; the 90° turn moves D to the position of A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q4/5

Two successive 72° rotations of a regular pentagon about its center equal which single transformation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q5/5

311

QUESTION ID

S6-F2-Q5

TYPE

Short answer

CORRECT ANSWER

A 144° rotation about the center

GRADING NOTE Any equivalent form. Rotation angles about the same center add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q5/5

Definition of a reflection

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Find the midpoint of a horizontal or vertical segment	**6.NS.C.8**
BACKWARD 2	Recognize perpendicular lines on a grid	**4.G.A.1**
BACKWARD 3	Reflect points with coordinate rules	**8.G.A.3**
FORWARD 1	Find lines of reflection and images in slanted, non-axis cases	**HSG-CO.A.4**
FORWARD 2	Prove from the definition that reflections keep lengths	**HSG-CO.A.4**

SET 7 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Complete a precise definition. A reflection across line ℓ maps each point P not on ℓ to the point P′ such that _____. Each point on ℓ maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q1/5

314

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

ℓ is the perpendicular bisector of segment PP′; itself.

GRADING NOTE Any equivalent wording. Both blanks are needed. "P′ is the same distance from ℓ" alone is not enough: it must be on the perpendicular through P, on the other side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q1/5

Noa writes: "A reflection across ℓ flips each point over ℓ."

Explain what is imprecise, and write a precise definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q2/5

315

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

"Flips over" does not say where the image is. Precise: the reflection across ℓ maps each point P on ℓ to itself, and each point P not on ℓ to the point P′ such that ℓ is the perpendicular bisector of PP′.

GRADING NOTE Any equivalent answer. Essential: both cases (on ℓ and not on ℓ) and the perpendicular bisector condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q2/5

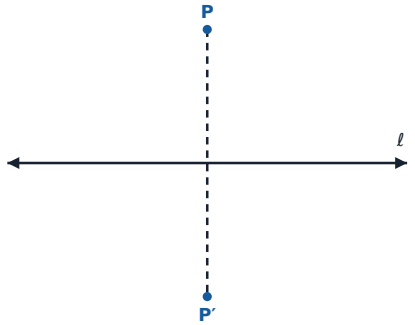
Point P′ is the image of point P under a reflection across line ℓ, and P is not on ℓ. Which must be true?

A. ℓ is perpendicular to PP′ and passes through its midpoint

B. ℓ passes through P

C. PP′ is parallel to ℓ

D. P and P′ are on the same side of ℓ



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q3/5

316

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ℓ is perpendicular to PP′ and passes through its midpoint

GRADING NOTE By the definition of a reflection, ℓ is the perpendicular bisector of PP′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ is y = 1. Use the definition to check that the reflection across ℓ maps P(2, 5) to P′(2, −3): test both conditions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q4/5

317

QUESTION ID

S7-M-Q4

TYPE

Short answer

CORRECT ANSWER

PP′ is vertical, so it is perpendicular to the horizontal line y = 1; its midpoint is (2, 1), which is on ℓ. So ℓ is the perpendicular bisector of PP′, and P′ is the image.

GRADING NOTE Form required: both conditions (perpendicular, and midpoint on ℓ) checked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q4/5

A reflection maps P(1, 2) to P′(5, 2). Use the definition to find the line of reflection.

Line: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q5/5

318

QUESTION ID

S7-M-Q5

TYPE

Short answer

CORRECT ANSWER

x = 3

GRADING NOTE Form required: an equation of the line. The line is the perpendicular bisector of PP′: PP′ is horizontal with midpoint (3, 2), so the line is the vertical line x = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 1

Find the midpoint of a horizontal or vertical segment • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the midpoint of the segment from (2, 4) to (8, 4).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q1/5

320

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 4)

GRADING NOTE Form required: an ordered pair. Halfway from x = 2 to x = 8 is x = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from (1, 1) to (1, 9).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q2/5

321

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 5)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q2/5

On a number line, what number is halfway between −3 and 7?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q3/5

322

QUESTION ID

S7-B1-Q3

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. (−3 + 7) ÷ 2 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q3/5

True or false?

The midpoint of the segment from (−4, 0) to (6, 0) is (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q4/5

323

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (−4 + 6) ÷ 2 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q4/5

True or false?

The midpoint of a segment is closer to one endpoint than to the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q5/5

324

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is the same distance from both endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 2

Recognize perpendicular lines on a grid • Grade 4 • Geometry • 4.G.A.1

True or false?

A horizontal line and a vertical line are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q1/5

326

QUESTION ID

S7-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Horizontal and vertical directions meet at right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q1/5

True or false?

Any two segments of the same length lie on parallel lines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q2/5

327

QUESTION ID

S7-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Length does not decide direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q2/5

Which pair of segments lie on perpendicular lines?

A. From (0, 0) to (4, 0) and from (2, −1) to (2, 3)

B. From (1, 1) to (2, 3) and from (1, 1) to (3, 2)

C. From (0, 0) to (3, 1) and from (0, 2) to (3, 3)

D. From (0, 0) to (3, 1) and from (0, 0) to (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q3/5

328

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. From (0, 0) to (4, 0) and from (2, −1) to (2, 3)

GRADING NOTE The first pair is horizontal and vertical; they cross at (2, 0) at a right angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q3/5

A segment goes from (2, 1) to (2, 6). Describe a segment through (5, 3) whose line is perpendicular to the line containing the first segment.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q4/5

329

QUESTION ID

S7-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Any horizontal segment through (5, 3), for example from (4, 3) to (8, 3).

GRADING NOTE Any horizontal segment containing (5, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q4/5

Which describes two lines that are both perpendicular to the x-axis?

A. They are parallel or the same line

B. They are perpendicular to each other

C. They meet at the origin

D. They always meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q5/5

330

QUESTION ID

S7-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. They are parallel or the same line

GRADING NOTE Both are vertical lines, so they are parallel (or the same line if they have the same x-value).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 3

Reflect points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Reflect the point (4, −3) across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q1/5

332

QUESTION ID

S7-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(4, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Reflect the point (−1, 5) across the y-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q2/5

333

QUESTION ID

S7-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 5)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q2/5

Reflect the point (2, 7) across the line y = x.

A. (7, 2)

B. (−7, −2)

C. (−2, 7)

D. (2, −7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q3/5

334

QUESTION ID

S7-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (7, 2)

GRADING NOTE Reflecting across y = x swaps the coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q3/5

True or false?

Reflecting the point (−6, 1) across the y-axis gives (6, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q4/5

335

QUESTION ID

S7-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (x, y) → (−x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q4/5

True or false?

Reflecting (3, 3) across the x-axis gives (−3, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q5/5

336

QUESTION ID

S7-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Across the x-axis, y changes sign: (3, −3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSG-CO.A.4

FORWARD 1

Find lines of reflection and images in slanted, non-axis cases • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A reflection maps P(0, 0) to P′(4, 2). Find an equation of the line of reflection.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q1/5

338

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$y = -2x + 5$

GRADING NOTE Any equivalent equation. The midpoint of PP′ is (2, 1); PP′ has slope 1/2, so the perpendicular bisector has slope −2: $y - 1 = -2(x - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q1/5

A reflection maps A(−2, 3) to A′(4, 3). Where does it map B(0, −1)?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q2/5

339

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(2, −1)

GRADING NOTE Form required: an ordered pair. The line of reflection is x = 1, so B is 1 unit left of it and its image is 1 unit right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q2/5

True or false?

A reflection maps P to P′ and Q to Q′, where P and Q are not on the line of reflection. Then PP′ is parallel to QQ′ (or on the same line).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q3/5

340

QUESTION ID
S7-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Both segments are perpendicular to the line of reflection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q3/5

A reflection maps A(1, 3) to A′(3, 1). Is the line of reflection y = x? Use the definition to decide.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q4/5

341

QUESTION ID

S7-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes. The midpoint (2, 2) is on y = x, and AA′ (slope −1) is perpendicular to y = x (slope 1).

GRADING NOTE Form required: the conclusion with both conditions of the definition checked (midpoint and perpendicularity).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q4/5

A reflection maps P(−1, 4) to P′(5, 4). Which point is on the line of reflection?

A. (−1, 0)

B. (5, 0)

C. (2, −3)

D. (4, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q5/5

342

QUESTION ID

S7-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. (2, −3)

GRADING NOTE The line is x = 2, the vertical perpendicular bisector of the horizontal segment PP′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

FORWARD 2

Prove from the definition that reflections keep lengths • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Points P and Q are on the same side of line ℓ, and line PQ is not perpendicular to ℓ. A reflection across ℓ maps them to P′ and Q′. Let PP′ meet ℓ at M and QQ′ meet ℓ at N. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q1/5

344

QUESTION ID

S7-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

PQ = P′Q′

GRADING NOTE REQUIRED WORK: Use the definition (ℓ is the perpendicular bisector of PP′ and of QQ′), then two pairs of congruent triangles (SAS): PMN ≅ P′MN, then PNQ ≅ P′NQ′.

Any valid proof. Essential: PM = P′M and QN = Q′N with right angles at M and N (definition); triangles PMN and P′MN are congruent by SAS, so PN = P′N and ∠PNM ≅ ∠P′NM; subtracting from the right angles gives ∠PNQ ≅ ∠P′NQ′; then triangles PNQ and P′NQ′ are congruent by SAS, so PQ = P′Q′. Alternative: coordinates with ℓ as the x-axis. Invalid: "reflections preserve distance" (that is what is being proved).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q1/5

Point P is on line ℓ and point Q is not. The reflection across ℓ maps Q to Q′. Prove that PQ = PQ′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q2/5

345

QUESTION ID

S7-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

PQ = PQ′

GRADING NOTE REQUIRED WORK: Let QQ′ meet ℓ at N. Use the definition and SAS (or the perpendicular bisector theorem).

Any valid proof. Essential: ℓ is the perpendicular bisector of QQ′ (definition), so QN = Q′N and the angles at N are right; PN is shared; triangles PNQ and PNQ′ are congruent by SAS (if P = N the result is QN = Q′N directly).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q2/5

Using coordinates with ℓ as the x-axis, the reflection is $(x, y) \to (x, -y)$. Prove that it keeps the distance between any two points (x_1, y_1) and (x_2, y_2).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q3/5

346

QUESTION ID

S7-F2-Q3

TYPE

Answer with required work

CORRECT ANSWER

The distance is unchanged.

GRADING NOTE REQUIRED WORK: Apply the distance formula before and after.

Any valid algebra. Essential: the image distance is $\sqrt{((x_2 - x_1)^2 + (-y_2 + y_1)^2)} = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q3/5

True or false?

Under a reflection, a point not on the line of reflection and its image are on the same side of the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q4/5

347

QUESTION ID

S7-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are on opposite sides; the line is the perpendicular bisector of the segment joining them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q4/5

In a proof that a reflection across ℓ keeps the length of segment PQ, which fact is used first?

A. ℓ is the perpendicular bisector of PP′ and of QQ′

B. ℓ bisects ∠PQP′

C. ℓ is parallel to PQ

D. ℓ passes through P

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q5/5

348

QUESTION ID
S7-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. ℓ is the perpendicular bisector of PP′ and of QQ′

GRADING NOTE The proof starts from the definition of a reflection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q5/5

Definition of a translation

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Recognize parallel segments on a grid	**4.G.A.1**
BACKWARD 2	Translate points with coordinate rules	**8.G.A.3**
FORWARD 1	Use the definition with points and their images	**HSG-CO.A.4**
FORWARD 2	Prove from the definition that translations keep lengths	**HSG-CO.A.4**

SET 100 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Points A and B are different. Write a precise definition of the translation that maps A to B.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q1/5

351

QUESTION ID

S100-M-Q1

TYPE

Short answer

CORRECT ANSWER

It maps each point P to the point P′ such that PP′ has the same length as AB, is parallel to AB (or on line AB), and points in the same direction.

GRADING NOTE Any equivalent wording. Essential: same length, parallel (or collinear), same direction. Accept "ABP′P is a parallelogram" for points P not on line AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q1/5

Tia writes: "A translation slides a figure."

Explain what is imprecise, and write a precise definition of the translation that maps A to B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q2/5

352

QUESTION ID

S100-M-Q2

TYPE

Short answer

CORRECT ANSWER

"Slides" does not say how far or in which direction each point moves. Precise: each point P goes to P′ with PP′ the same length as AB, parallel to AB (or on line AB), and in the same direction.

GRADING NOTE Any equivalent answer naming the missing length and direction and giving the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The translation that maps A(1, 2) to B(4, 6) maps P(−2, 0) to P′. Use the definition (same length, parallel, same direction) to find P′, and check the length.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q3/5

353

QUESTION ID

S100-M-Q3

TYPE

Short answer

CORRECT ANSWER

P′(1, 4). PP′ goes 3 right and 4 up, like AB, so it is parallel with the same direction; both have length 5.

GRADING NOTE Form required: P′ and the check of length and direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q3/5

A translation maps A to A′ and B to B′. Explain, using the definition, why AA′ and BB′ are parallel (or on one line) and have the same length.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q4/5

354

QUESTION ID

S100-M-Q4

TYPE

Short answer

CORRECT ANSWER

The definition says every segment from a point to its image has the same length and direction as the one that defines the translation; so AA′ and BB′ both match it, hence each other.

GRADING NOTE Any equivalent explanation based on the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q4/5

Is (x, y) → (x + 3, y − 4) a translation? Use the definition to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q5/5

355

QUESTION ID

S100-M-Q5

TYPE

Short answer

CORRECT ANSWER

Yes. Every point P moves to P′ with PP′ going 3 right and 4 down: the same length (5) and the same direction for every point.

GRADING NOTE Any equivalent explanation that checks same length and same direction for all points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q5/5

SET 100 • HSG-CO.A.4

BACKWARD 1

Recognize parallel segments on a grid • Grade 4 • Geometry • 4.G.A.1

True or false?

Two segments that each go 3 right and 1 up lie on parallel lines (or on one line).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q1/5

357

QUESTION ID

S100-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They have the same direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q1/5

True or false?

Any two segments of the same length are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q2/5

358

QUESTION ID

S100-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Length does not decide direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q2/5

Which pair of segments is parallel?

A. From (0, 0) to (3, 1) and from (0, 2) to (3, 3)

B. From (0, 0) to (4, 0) and from (0, 0) to (0, 4)

C. From (1, 1) to (2, 3) and from (1, 1) to (3, 2)

D. From (0, 0) to (3, 1) and from (0, 0) to (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q3/5

359

QUESTION ID

S100-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. From (0, 0) to (3, 1) and from (0, 2) to (3, 3)

GRADING NOTE Both segments in the first pair go 3 right and 1 up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q3/5

A segment goes from (1, 1) to (4, 3). Give the other endpoint of a parallel segment of the same length that starts at (0, 5).

Endpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q4/5

360

QUESTION ID

S100-B1-Q4

TYPE

Short answer

CORRECT ANSWER

(3, 7)

GRADING NOTE Form required: an ordered pair; (−3, 3) is also accepted (opposite direction, same length, parallel).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q4/5

Which describes two lines that are both perpendicular to the x-axis?

A. They are parallel or the same line

B. They are perpendicular to each other

C. They meet at the origin

D. They always meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q5/5

361

QUESTION ID

S100-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. They are parallel or the same line

GRADING NOTE Both are vertical lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q5/5

SET 100 • HSG-CO.A.4

BACKWARD 2

Translate points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Translate the point (−1, 5) 6 units right and 2 units down.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q1/5

363

QUESTION ID

S100-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Translate the point (3, −2) 4 units left.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q2/5

364

QUESTION ID

S100-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−1, −2)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q2/5

A translation moves (0, 0) to (2, −3). Where does it move (1, 1)?

A. (3, 4)

B. (2, −3)

C. (−1, 4)

D. (3, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q3/5

365

QUESTION ID

S100-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (3, −2)

GRADING NOTE Add (2, −3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q3/5

True or false?

Translating (3, 3) by 2 units left gives (1, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q4/5

366

QUESTION ID

S100-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Left changes x: 3 − 2 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q4/5

True or false?

Translating (3, 3) by 2 units left gives (3, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q5/5

367

QUESTION ID

S100-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Left changes x, not y: (1, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q5/5

SET 100 • HSG-CO.A.4

FORWARD 1

Use the definition with points and their images • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A translation maps A(2, −1) to A′(−3, 4). Where does it map B(0, 6)?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q1/5

369

QUESTION ID

S100-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(−5, 11)

GRADING NOTE Form required: an ordered pair. The translation moves every point 5 left and 5 up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q1/5

A translation maps P to P′(4, 1) and moves every point 2 left and 3 up. Find P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q2/5

370

QUESTION ID

S100-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(6, −2)

GRADING NOTE Form required: an ordered pair. Undo the move.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q2/5

True or false?

Under a translation, all the segments that join points to their images are parallel (or on one line) and have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q3/5

371

QUESTION ID

S100-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is what the definition of a translation says about every point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q3/5

A translation maps triangle ABC with A(0, 0), B(2, 0), C(0, 3) so that C goes to (5, 7). Find the images of A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q4/5

372

QUESTION ID

S100-F1-Q4

TYPE

Short answer

CORRECT ANSWER

A′(5, 4) and B′(7, 4).

GRADING NOTE Form required: both points. The move is 5 right and 4 up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q4/5

A translation maps (1, 2) to (4, 0). Which point maps to (0, 0)?

A. (−4, 0)

B. (−3, 2)

C. (3, −2)

D. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q5/5

373

QUESTION ID

S100-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. (−3, 2)

GRADING NOTE Undo 3 right, 2 down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q5/5

SET 100 • HSG-CO.A.4

FORWARD 2

Prove from the definition that translations keep lengths • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A translation maps P to P′ and Q to Q′, where Q is not on line PP′. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q1/5

375

QUESTION ID

S100-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

PQ = P′Q′

GRADING NOTE REQUIRED WORK: Use the definition (PP′ and QQ′ are parallel, the same length, and the same direction) to show PP′Q′Q is a parallelogram.

Any valid proof. Essential: PP′ ∥ QQ′ and PP′ = QQ′ in the same direction, so PP′Q′Q is a parallelogram (one pair of opposite sides parallel and congruent); opposite sides of a parallelogram are congruent, so PQ = P′Q′. Invalid: assuming translations preserve length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q1/5

True or false?

A translation maps P to P′ and Q to Q′. Then PP′Q′Q is always a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q2/5

376

QUESTION ID

S100-F2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE If Q is on line PP′, all four points lie on one line, so there is no parallelogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q2/5

Prove that T(x, y) = (x + a, y + b) maps every nonvertical line y = mx + k to a line with the same slope m (so the image is parallel to the line or is the line itself).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q3/5

377

QUESTION ID

S100-F2-Q3

TYPE

Answer with required work

CORRECT ANSWER

The image of (x, mx + k) is (X, Y) = (x + a, mx + k + b), and Y = m(X − a) + k + b = mX + (k + b − ma): slope m.

GRADING NOTE REQUIRED WORK: Take any point (x, mx + k), find its image, and show the image satisfies an equation with slope m.

Any valid algebraic proof. Essential: general point; image; substitution x = X − a; slope m; same line exactly when b = ma. Invalid: checking one example line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q3/5

A translation maps P to P′ and Q to Q′, where Q IS on line PP′. Prove that PQ = P′Q′ in this case too.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q4/5

378

QUESTION ID

S100-F2-Q4

TYPE

Answer with required work

CORRECT ANSWER

PQ = P′Q′

GRADING NOTE REQUIRED WORK: Work on the line: every point moves the same distance d in the same direction.

Any valid argument. Essential: on the line, coordinates shift by the same d, so the difference between the two points is unchanged.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q4/5

In the proof that a translation keeps the length of PQ (Q not on line PP′), which figure is used?

A. Rectangle PQRS

B. A circle centered at P

C. Parallelogram PP′Q′Q

D. Triangle PQQ′ only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q5/5

379

QUESTION ID
S100-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. Parallelogram PP′Q′Q

GRADING NOTE The definition gives one pair of opposite sides parallel and congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q5/5

Definition of a rotation

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Describe turns in degrees, clockwise and counterclockwise	**4.MD.C.5.b**
BACKWARD 2	Points on a circle are the same distance from its center	**7.G.B.4**
BACKWARD 3	Rotate points 90°, 180°, and 270° about the origin	**8.G.A.3**
FORWARD 1	Rotate about a center that is not the origin	**HSG-CO.A.4**
FORWARD 2	Find the center and angle of a rotation from points and their imag..	**HSG-CO.A.4**

SET 8 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Complete a precise definition. A counterclockwise rotation about point O through angle θ maps each point $P \neq O$ to the point P' such that _____ and _____. Point O maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q1/5

382

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

$OP' = OP$ (P' is on the circle with center O through P); $m\angle POP' = \theta$, measured counterclockwise from ray OP; itself.

GRADING NOTE Any equivalent wording. All three blanks are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q1/5

A rotation about point O maps point P to P′ (P ≠ P′). Which must be true?

A. O is the midpoint of PP′

B. O is on the perpendicular bisector of PP′

C. OP′ = 2 · OP

D. PP′ is perpendicular to OP

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q2/5

383

QUESTION ID

S8-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. O is on the perpendicular bisector of PP′

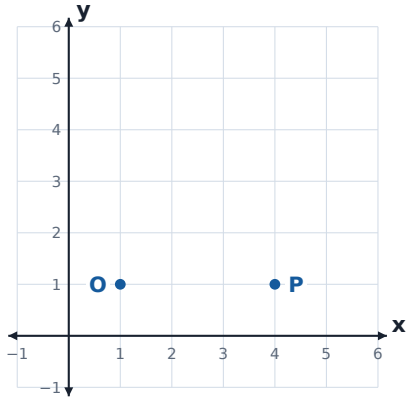
GRADING NOTE OP = OP′, so O is equidistant from P and P′ and lies on the perpendicular bisector of PP′. O is the midpoint only for a 180° rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q2/5

A rotation of 90° counterclockwise about O(1, 1) maps P(4, 1) to P′. Use the definition to find P′.

P′ = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q3/5

384

QUESTION ID

S8-M-Q3

TYPE

Short answer

CORRECT ANSWER

(1, 4)

GRADING NOTE Form required: an ordered pair. P is 3 units right of O; after a 90° counterclockwise turn it is 3 units above O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q3/5

A rotation about point O through 70° maps a point P ≠ O to P′. Use the definition to explain why P′ lies on the circle with center O through P, and why O is the only point that the rotation leaves where it is.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q4/5

385

QUESTION ID

S8-M-Q4

TYPE

Short answer

CORRECT ANSWER

The definition gives OP′ = OP, so P′ is on that circle. Every point P ≠ O is moved to a point 70° around its circle, so it changes; the definition sends O to itself.

GRADING NOTE Any equivalent explanation using OP′ = OP and the angle condition (for an angle strictly between 0° and 360°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q4/5

A rotation maps P(2, 0) to P′(0, 2). Explain why its center must lie on the line y = x.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q5/5

386

QUESTION ID

S8-M-Q5

TYPE

Short answer

CORRECT ANSWER

The center is the same distance from P and from P′, so it is on the perpendicular bisector of PP′. That bisector passes through (1, 1) with slope 1, which is the line y = x.

GRADING NOTE Any equivalent explanation that uses equal distances from the center (the definition) and the perpendicular bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 1

Describe turns in degrees, clockwise and counterclockwise • Grade 4 • Measurement & Data • 4.MD.C.5.b

A ray points right. It turns 90° counterclockwise. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q1/5

388

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Up

GRADING NOTE Any equivalent wording (north).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q1/5

A ray points up. It turns 180°. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q2/5

389

QUESTION ID
S8-B1-Q2

TYPE
Short answer

CORRECT ANSWER
Down

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q2/5

True or false?

A clockwise turn of 270° ends in the same direction as a counterclockwise turn of 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q3/5

390

QUESTION ID

S8-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 270° clockwise and 90° counterclockwise add to a full 360° turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q3/5

A ray points up. It turns 90° clockwise. Which way does it point now?

A. Down

B. Left

C. Right

D. Up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q4/5

391

QUESTION ID

S8-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Right

GRADING NOTE Clockwise from up is toward the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q4/5

How many degrees are in $\frac{3}{4}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q5/5

392

QUESTION ID

S8-B1-Q5

TYPE

Short answer

CORRECT ANSWER

270°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 2

Points on a circle are the same distance from its center • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 6. Point P is on the circle. What is OP?

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q1/5

394

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q1/5

True or false?

If OA = OB, then A and B lie on one circle with center O.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q2/5

395

QUESTION ID

S8-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both are on the circle with center O and radius OA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q2/5

Which point is on the circle with center (0, 0) and radius 5?

A. (4, 4)

B. (3, 4)

C. (2, 3)

D. (5, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q3/5

396

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (3, 4)

GRADING NOTE $\sqrt{(3^2 + 4^2)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q3/5

A circle has center O and diameter 10 cm. How far is every point of the circle from O?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q4/5

397

QUESTION ID

S8-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q4/5

True or false?

Two points the same distance from O are always the same point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q5/5

398

QUESTION ID

S8-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Many points lie on the circle of that radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 3

Rotate points 90°, 180°, and 270° about the origin • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (5, 2) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q1/5

400

QUESTION ID

S8-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(−2, 5)

GRADING NOTE Form required: an ordered pair. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q1/5

Rotate (−3, 4) by 180° about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q2/5

401

QUESTION ID

S8-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(3, −4)

GRADING NOTE Form required: an ordered pair. (x, y) → (−x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q2/5

Rotate (1, 6) by 90° clockwise about the origin.

A. (−6, 1)

B. (1, −6)

C. (−1, −6)

D. (6, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q3/5

402

QUESTION ID

S8-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (6, −1)

GRADING NOTE (x, y) → (y, −x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q3/5

True or false?

Rotating (2, 3) by 270° counterclockwise about the origin gives (3, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q4/5

403

QUESTION ID

S8-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 270° counterclockwise = 90° clockwise: (x, y) → (y, −x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q4/5

True or false?

Rotating (4, 0) by 90° counterclockwise about the origin gives (0, −4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q5/5

404

QUESTION ID

S8-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives (0, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSG-CO.A.4

FORWARD 1

Rotate about a center that is not the origin • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Rotate P(5, 4) by 90° clockwise about C(2, 3).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q1/5

406

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 0)

GRADING NOTE Form required: an ordered pair. P is (3, 1) from C; a 90° clockwise turn gives (1, −3) from C, which is (3, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q1/5

Rotate P(1, 5) by 180° about C(−1, 2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q2/5

407

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(−3, −1)

GRADING NOTE Form required: an ordered pair. P is (2, 3) from C; the image is (−2, −3) from C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q2/5

Rotate P(3, −2) by 90° counterclockwise about C(0, −2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q3/5

408

QUESTION ID

S8-F1-Q3

TYPE

Short answer

CORRECT ANSWER

(0, 1)

GRADING NOTE Form required: an ordered pair. P is (3, 0) from C; the image is (0, 3) from C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q3/5

Rotate P(1, 4) by 270° counterclockwise about C(1, 1).

A. (4, 4)

B. (4, 1)

C. (1, −2)

D. (−2, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q4/5

409

QUESTION ID

S8-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (4, 1)

GRADING NOTE P is (0, 3) from C; 270° counterclockwise (= 90° clockwise) gives (3, 0) from C, which is (4, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q4/5

True or false?

Rotating P(4, 0) by 90° counterclockwise about C(2, 0) gives the same image as rotating P by 90° counterclockwise about the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q5/5

410

QUESTION ID

S8-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE About C the image is (2, 2); about the origin it is (0, 4). The center matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

FORWARD 2

Find the center and angle of a rotation from points and their images • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A rotation maps B(1, 0) to B′(0, 1) and A(3, 1) to A′(−1, 3). The center is where the perpendicular bisectors of BB′ and AA′ meet. Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q1/5

412

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Center (0, 0); 90° counterclockwise.

GRADING NOTE Form required: the center and the angle with direction. The bisector of BB′ is y = x; the bisector of AA′ passes through (1, 2) with slope 2, which is y = 2x. They meet at (0, 0), and (1, 0) → (0, 1) is a 90° counterclockwise turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q1/5

A rotation maps P(4, 1) to P′(2, 3) and Q(4, 3) to Q′(0, 3). Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q2/5

413

QUESTION ID

S8-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Center (2, 1); 90° counterclockwise.

GRADING NOTE Form required: the center and the angle. Check: P is (2, 0) from (2, 1) and P′ is (0, 2) from it; Q is (2, 2) from it and Q′ is (−2, 2) from it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q2/5

True or false?

The center of a nonzero rotation moves to a new position under the rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q3/5

414

QUESTION ID

S8-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The center is the one fixed point of the rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q3/5

A rotation about C maps P to P′. CP = 5 and PP′ = 5√2. Which could be the angle of rotation?

A. 60°

B. 90°

C. 45°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q4/5

415

QUESTION ID

S8-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 90°

GRADING NOTE CP = CP′ = 5 and $PP'^2 = 50 = 5^2 + 5^2$, so triangle PCP′ has a right angle at C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q4/5

A rotation about the origin maps (5, 0) to (3, 4). What is the counterclockwise angle of rotation, to the nearest degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q5/5

416

QUESTION ID

S8-F2-Q5

TYPE

Short answer

CORRECT ANSWER

53°

GRADING NOTE Rounding to the nearest degree required. The angle at the origin has tangent 4/3, so it is about 53.13°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q5/5

Draw the image of a figure under a rotation, reflection, or translation

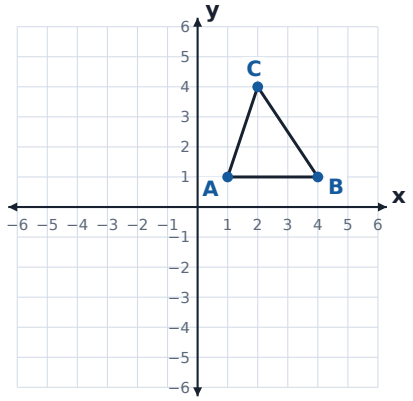
MAIN	Geometry main question	**HSG-CO.A.5**
BACKWARD 1	Plot and read points in all four quadrants	**6.NS.C.6.c**
BACKWARD 2	Find image points with coordinate rules	**8.G.A.3**
BACKWARD 3	Count the distance from a point to a horizontal or vertical line	**6.NS.C.8**
FORWARD 1	Draw the image under a sequence of two transformations	**HSG-CO.A.5**
FORWARD 2	Draw rotations and reflections off the grid with compass, protract…	**HSG-CO.A.5**

SET 9 • HSG-CO.A.5

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Draw the image of triangle ABC under the translation 5 units left and 2 units down. Label the image A′B′C′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q1/5

419

QUESTION ID

S9-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A′(−4, −1), B′(−1, −1), C′(−3, 2), joined to form triangle A′B′C′.

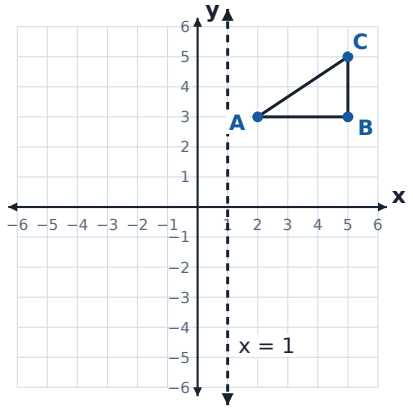
GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. (x, y) → (x − 5, y − 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q1/5

Draw the image of triangle ABC under the reflection across the line x = 1. Label the image A′B′C′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q2/5

420

QUESTION ID

S9-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

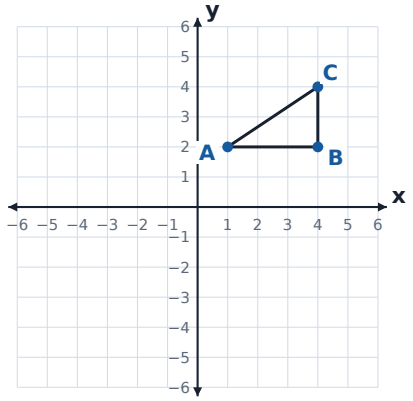
A′(0, 3), B′(−3, 3), C′(−3, 5).

GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. Each image is the same distance from x = 1 on the other side: x → 2 − x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q2/5

Draw the image of triangle ABC under the rotation of 90° counterclockwise about the origin. Label the image A′B′C′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q3/5

421

QUESTION ID

S9-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

A′(−2, 1), B′(−2, 4), C′(−4, 4).

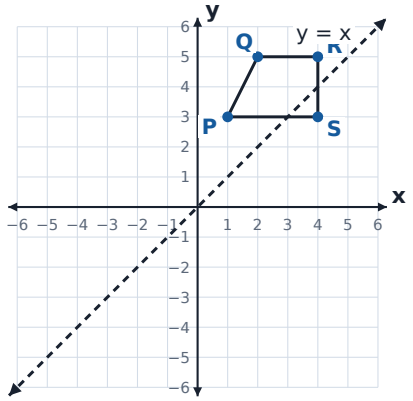
GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Draw the image of quadrilateral PQRS under the reflection across the line y = x. Label the image P′Q′R′S′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q4/5

422

QUESTION ID

S9-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

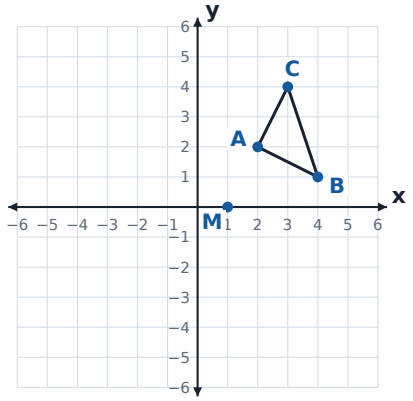
P′(3, 1), Q′(5, 2), R′(5, 4), S′(3, 4).

GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. Reflection across y = x swaps the coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q4/5

Draw the image of triangle ABC under the rotation of 180° about the point M(1, 0). Label the image A′B′C′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q5/5

423

QUESTION ID

S9-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A′(0, −2), B′(−2, −1), C′(−1, −4).

GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. Each image is directly opposite its point through M: (x, y) → (2 − x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

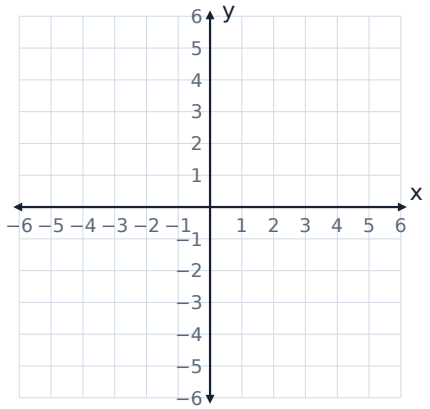
High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 1

Plot and read points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

Plot and label the points A(−3, 2), B(0, −4), and C(5, 1).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q1/5

425

QUESTION ID

S9-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A at (−3, 2), B at (0, −4), C at (5, 1).

GRADING NOTE Each point within about a quarter of a grid square of its location.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q1/5

A point is 4 units left of the y-axis and 1 unit below the x-axis. What are its coordinates?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q2/5

426

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(−4, −1)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q2/5

Which point is 2 units above the x-axis and on the y-axis?

A. (2, 2)

B. (0, −2)

C. (0, 2)

D. (2, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q3/5

427

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (0, 2)

GRADING NOTE On the y-axis, x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q3/5

True or false?

The point (3, −2) is in Quadrant IV.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q4/5

428

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x > 0$ and $y < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q4/5

True or false?

To plot (2, 5), start at the origin and move 2 units up and 5 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q5/5

429

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The x-coordinate (2) is the horizontal move.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 2

Find image points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Find the image of (3, −2) under a translation 4 units left and 1 unit up.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q1/5

431

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−1, −1)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q1/5

Find the image of (−2, 5) under a reflection across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q2/5

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, −5)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q2/5

Find the image of (3, 1) under a 90° clockwise rotation about the origin.

A. (−3, −1)

B. (3, −1)

C. (1, −3)

D. (−1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q3/5

433

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (1, −3)

GRADING NOTE (x, y) → (y, −x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q3/5

True or false?

The image of (4, −1) under a reflection across the y-axis is (−4, −1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q4/5

434

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (x, y) → (−x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q4/5

True or false?

The image of (2, 2) under a 180° rotation about the origin is (2, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q5/5

435

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is (−2, −2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 3

Count the distance from a point to a horizontal or vertical line • Grade 6 • The Number System • 6.NS.C.8

How far is the point (5, 3) from the vertical line x = 1?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q1/5

437

QUESTION ID

S9-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. |5 − 1| = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q1/5

How far is the point (−2, −1) from the horizontal line y = 3?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q2/5

438

QUESTION ID

S9-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. |−1 − 3| = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q2/5

Which point is 3 units from the line x = 2, on its left?

A. (−3, 4)

B. (−1, 4)

C. (2, 1)

D. (5, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q3/5

439

QUESTION ID

S9-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−1, 4)

GRADING NOTE 2 − 3 = −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q3/5

True or false?

The point (0, 4) is 4 units from the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q4/5

440

QUESTION ID

S9-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its distance from y = 0 is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q4/5

True or false?

The point (6, 2) is 6 units from the line x = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q5/5

441

QUESTION ID

S9-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is |6 − 2| = 4 units away.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q5

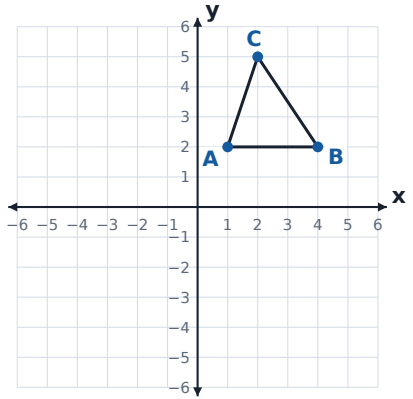
Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSG-CO.A.5

FORWARD 1

Draw the image under a sequence of two transformations • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Reflect triangle ABC across the y-axis, then translate the result 3 units down. Draw and label the final image A″B″C″.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q1/5

443

QUESTION ID

S9-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

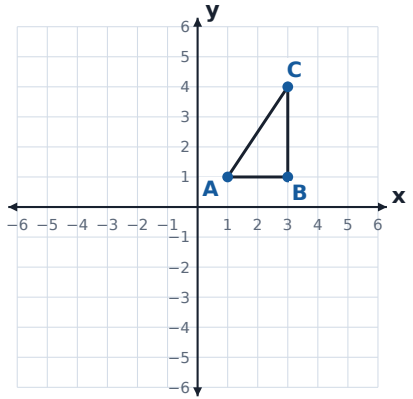
A″(−1, −1), B″(−4, −1), C″(−2, 2).

GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. (x, y) → (−x, y) → (−x, y − 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q1/5

Rotate triangle ABC by 90° counterclockwise about the origin, then reflect the result across the x-axis. Draw and label the final image A″B″C″.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S9-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q2/5

444

QUESTION ID

S9-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A″(−1, −1), B″(−1, −3), C″(−4, −3).

GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. (x, y) → (−y, x) → (−y, −x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S9-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q2/5

Triangle ABC has A(2, 1). It is translated 2 units right and then rotated 180° about the origin. Where does A end up?

A″ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q3/5

445

QUESTION ID

S9-F1-Q3

TYPE

Short answer

CORRECT ANSWER

(−4, −1)

GRADING NOTE Form required: an ordered pair. (2, 1) → (4, 1) → (−4, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q3/5

True or false?

Reflecting a figure across the x-axis and then across the y-axis gives the same image as reflecting it across the line y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q4/5

446

QUESTION ID

S9-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The two reflections give (x, y) → (−x, −y), a 180° rotation; reflecting across y = x gives (y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q4/5

Point P(1, 3) is translated 2 units left and then reflected across the line y = x. Where does it end up?

A. (3, 1)

B. (−1, 3)

C. (1, −3)

D. (3, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q5/5

447

QUESTION ID

S9-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. (3, −1)

GRADING NOTE (1, 3) → (−1, 3) → (3, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

FORWARD 2

Draw rotations and reflections off the grid with compass, protractor, and straightedge • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Use a compass and protractor. Rotate point P 60° counterclockwise about point O. Mark and label the image P′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1
High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q1/5

QUESTION ID
S9-F2-Q1

TYPE
Drawing / plotting

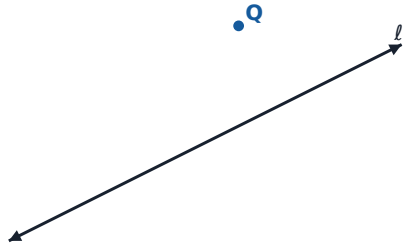
CORRECT ANSWER

P′ is on the circle with center O through P, with m∠POP′ = 60° measured counterclockwise from ray OP.

GRADING NOTE Accept within about 2° and 2 mm. Evidence: the arc (or equal compass distance) from O and the 60° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1
High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q1/5

Use a compass and straightedge. Reflect point Q across line ℓ. Mark and label the image Q′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q2/5

450

QUESTION ID

S9-F2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

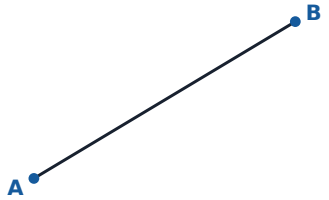
Q′ is on the perpendicular from Q to ℓ, on the other side of ℓ, the same distance from ℓ as Q (ℓ is the perpendicular bisector of QQ′).

GRADING NOTE Accept within about 2 mm. Evidence: the construction of the perpendicular from Q to ℓ and equal distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q2/5

Use a compass and protractor. Rotate segment AB 90° clockwise about point A. Draw and label the image AB′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q3/5

451

QUESTION ID

S9-F2-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

AB′ has the same length as AB, with m∠BAB′ = 90° measured clockwise from ray AB; A stays fixed.

GRADING NOTE Accept within about 2° and 2 mm. Evidence: equal lengths and the 90° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q3/5

Use a compass and straightedge. Rotate point P 180° about point O. Mark and label the image P′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q4/5

452

QUESTION ID

S9-F2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

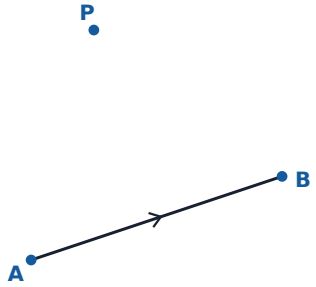
P′ is on ray opposite to ray OP (line PO extended past O) with OP′ = OP.

GRADING NOTE Accept within about 2 mm. Evidence: line PO drawn through O and the compass arc (radius OP) centered at O marking P′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q4/5

Use a compass and straightedge. The translation maps A to B. Draw the image P′ of point P (P is not on line AB).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q5/5

453

QUESTION ID

S9-F2-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

P′ is the fourth vertex of parallelogram ABP′P: PP′ = AB and BP′ = AP.

GRADING NOTE Accept within about 2 mm. Evidence: compass arcs of radius AB centered at P and radius AP centered at B crossing at P′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q5/5

Specify a sequence of transformations that carries one figure onto another

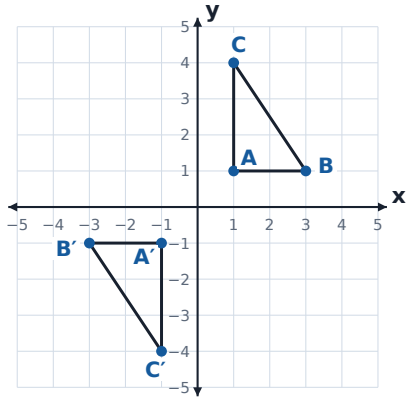
MAIN	Geometry main question	**HSG-CO.A.5**
BACKWARD 1	Understand congruence as the result of one rigid motion	**8.G.A.2**
BACKWARD 2	Find images with coordinate rules for sequences	**8.G.A.3**
BACKWARD 3	Read the orientation (clockwise or counterclockwise order) of a tr…	**8.G.A.1**
FORWARD 1	Use the fewest transformations and justify	**HSG-CO.A.5**
FORWARD 2	Specify a similarity sequence: a dilation and rigid motions	**HSG-SRT.A.2**

SET 10 • HSG-CO.A.5

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′ (A to A′, B to B′, C to C′).



Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q1/5

456

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

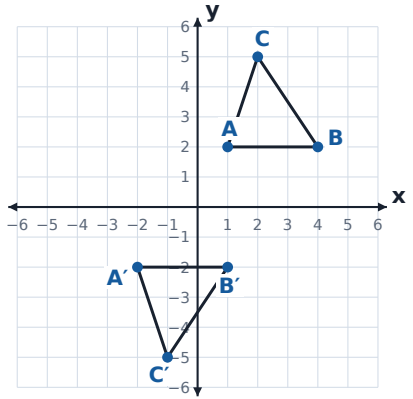
A 180° rotation about the origin.

GRADING NOTE Any valid sequence, for example a reflection across the x-axis followed by a reflection across the y-axis. Check: every vertex must land on its partner.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q1/5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′.



Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q2/5

457

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

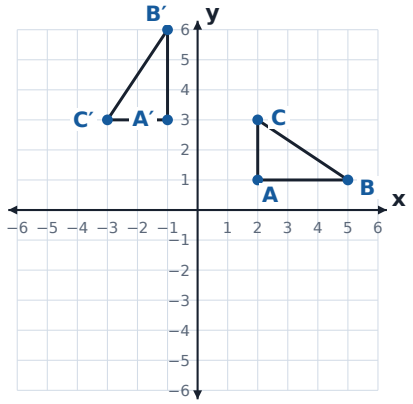
Reflect across the x-axis, then translate 3 units left.

GRADING NOTE Any valid sequence (for example, translate 3 units left, then reflect across the x-axis). The orientation is reversed, so the sequence must include a reflection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q2/5

On the grid, triangle A′B′C′ is congruent to triangle ABC. Describe a sequence of rigid transformations that carries ABC onto A′B′C′, with A → A′, B → B′, C → C′.



Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q3/5

458

QUESTION ID

S10-M-Q3

TYPE

Short answer

CORRECT ANSWER

Rotate 90° counterclockwise about the origin, then translate 1 unit up.

GRADING NOTE Any valid sequence. Check: (x, y) → (−y, x) → (−y, x + 1) sends A(2, 1) to (−1, 3), B to (−1, 6), C to (−3, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q3/5

Right triangle ABC has A(0, 0), B(2, 0), C(0, 3). Triangle A′B′C′ has A′(4, 0), B′(4, 2), C′(1, 0). Which sequence carries ABC onto A′B′C′?

A. Reflect across the y-axis, then translate 4 units right

B. Rotate 90° counterclockwise about the origin, then translate 4 units right

C. Rotate 90° clockwise about the origin, then translate 4 units right

D. Translate 4 units right, then rotate 90° counterclockwise about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q4/5

459

QUESTION ID

S10-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Rotate 90° counterclockwise about the origin, then translate 4 units right

GRADING NOTE The rotation sends B to (0, 2) and C to (−3, 0); the translation then gives B′(4, 2) and C′(1, 0). The other orders or turns send B or C elsewhere.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q4/5

Triangle PQR has P(1, 1), Q(4, 1), R(1, 3). Triangle P′Q′R′ has P′(1, −1), Q′(4, −1), R′(1, −3). Describe a sequence of rigid transformations that carries PQR onto P′Q′R′, and explain why it must include a reflection.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q5/5

460

QUESTION ID

S10-M-Q5

TYPE

Short answer

CORRECT ANSWER

Reflect across the x-axis. PQR is counterclockwise and P′Q′R′ is clockwise; rotations and translations keep orientation, so a reflection is needed.

GRADING NOTE Any valid sequence with a reflection, and the orientation reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 1

Understand congruence as the result of one rigid motion • Grade 8 • Geometry • 8.G.A.2

Triangle 2 is the image of triangle 1 after one rigid motion. Every point moved 3 units right. Which motion was it?

A. A reflection

B. A translation

C. A rotation

D. A dilation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q1/5

462

QUESTION ID

S10-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. A translation

GRADING NOTE Sliding every point the same way is a translation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q1/5

A triangle and its image face opposite ways, as in a mirror. Which kind of rigid motion was most likely used?

Motion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q2/5

463

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A reflection

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q2/5

True or false?

If one figure is the image of another under a sequence of rigid motions, the figures are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q3/5

464

QUESTION ID

S10-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rigid motions keep all lengths and angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q3/5

True or false?

A figure and its image under a rigid motion can have different areas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q4/5

465

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rigid motions keep area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q4/5

Segment AB goes from (0, 0) to (2, 0). Its image goes from (0, 0) to (0, 2). Which single rigid motion could do this?

A. A translation 2 units up

B. A reflection across the x-axis

C. A 90° counterclockwise rotation about the origin

D. A 180° rotation about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q5/5

466

QUESTION ID

S10-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. A 90° counterclockwise rotation about the origin

GRADING NOTE (2, 0) → (0, 2) under (x, y) → (−y, x), and the origin stays fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 2

Find images with coordinate rules for sequences • Grade 8 • Geometry • 8.G.A.3

Point (2, −3) is reflected across the y-axis. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q1/5

468

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−2, −3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q1/5

Point (−1, 4) is rotated 90° counterclockwise about the origin. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q2/5

469

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−4, −1)

GRADING NOTE Form required: an ordered pair. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q2/5

Point (5, 1) is translated 2 units left and 3 units down. Find its image.

A. (3, −2)

B. (7, 4)

C. (7, −2)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q3/5

470

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (3, −2)

GRADING NOTE (5 − 2, 1 − 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q3/5

True or false?

Point (0, 3) rotated 180° about the origin is (0, −3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q4/5

471

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (x, y) → (−x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q4/5

True or false?

Point (4, 1) reflected across the line y = x is (4, −1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q5/5

472

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is (1, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 3

Read the orientation (clockwise or counterclockwise order) of a triangle • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has A(0, 0), B(4, 0), C(0, 3). Going A → B → C, is the order clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q1/5

474

QUESTION ID

S10-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Counterclockwise

GRADING NOTE Any equivalent wording. From A go right to B, then up-left to C: a counterclockwise turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q1/5

True or false?

A reflection reverses the orientation of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q2/5

475

QUESTION ID

S10-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A mirror image runs the other way around.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q2/5

True or false?

A rotation reverses the orientation of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q3/5

476

QUESTION ID

S10-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rotations keep the order of the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q3/5

Triangle PQR is clockwise. After two reflections, its image is

A. counterclockwise

B. either, depending on the lines

C. clockwise

D. not a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q4/5

QUESTION ID

S10-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. clockwise

GRADING NOTE Each reflection reverses the order, so two reflections restore it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q4/5

Triangle ABC is counterclockwise. It is translated and then reflected. Is the final image clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q5/5

478

QUESTION ID

S10-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Clockwise

GRADING NOTE Any equivalent wording. The translation keeps the order; the reflection reverses it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSG-CO.A.5

FORWARD 1

Use the fewest transformations and justify • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Two reflections are done one after the other, across parallel lines that are 3 units apart. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q1/5

480

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A translation of 6 units, perpendicular to the lines (in the direction from the first line toward the second).

GRADING NOTE Any equivalent wording with the distance (twice the gap) and the direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q1/5

Two reflections are done one after the other, across lines that meet at point O and form a 40° angle. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q2/5

481

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A rotation of 80° about O (in the direction from the first line toward the second).

GRADING NOTE Any equivalent wording with the center and twice the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q2/5

Triangle ABC is clockwise and triangle A′B′C′ is congruent to it and counterclockwise. Explain why no single rotation or translation carries ABC onto A′B′C′.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q3/5

482

QUESTION ID

S10-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Rotations and translations keep orientation, but these triangles have opposite orientations, so a reflection is needed.

GRADING NOTE Any equivalent explanation based on orientation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q3/5

True or false?

Every rigid motion of the plane can be written as a sequence of at most three reflections.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q4/5

483

QUESTION ID

S10-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is a known theorem: a reflection fixes one vertex, a second fixes two, and a third (if needed) fixes the third vertex of a triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q4/5

Triangle A′B′C′ is the image of triangle ABC under some rigid motion, and both are counterclockwise. What is the fewest number of reflections that can always carry ABC onto A′B′C′?

A. 2

B. 4

C. 3

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q5/5

484

QUESTION ID

S10-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 2

GRADING NOTE The orientation is kept, so an even number is needed; two reflections give any rotation or translation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

FORWARD 2

Specify a similarity sequence: a dilation and rigid motions • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has A(0, 0), B(2, 0), C(0, 1). Triangle A′B′C′ has A′(0, 0), B′(0, 6), C′(−3, 0). Describe a sequence (a dilation and a rigid motion) that carries ABC onto A′B′C′.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q1/5

486

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Dilate by a factor of 3 about the origin, then rotate 90° counterclockwise about the origin.

GRADING NOTE Any valid sequence. Check: (2, 0) → (6, 0) → (0, 6) and (0, 1) → (0, 3) → (−3, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q1/5

Square S has vertices (0, 0), (1, 0), (1, 1), (0, 1). Square T has vertices (2, 2), (4, 2), (4, 4), (2, 4). Describe a similarity sequence carrying S onto T.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q2/5

487

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Dilate by 2 about the origin, then translate 2 units right and 2 units up.

GRADING NOTE Any valid sequence. (x, y) → (2x + 2, 2y + 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q2/5

True or false?

Any two figures that a sequence of rigid motions and a dilation carries one onto the other are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q3/5

488

QUESTION ID
S10-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They are similar; they are congruent only when the scale factor is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q3/5

Segment AB is from (0, 0) to (3, 0). Segment CD is from (1, 1) to (1, 7). Which sequence carries AB onto CD (A to C, B to D)?

A. Rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

B. Dilate by 2 about the origin, then translate 1 right and 1 up

C. Dilate by 3 about the origin, rotate 90° counterclockwise, then translate 1 right and 1 up

D. Dilate by 2 about the origin, rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q4/5

489

QUESTION ID
S10-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. Dilate by 2 about the origin, rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

GRADING NOTE (3, 0) → (6, 0) → (0, 6) → (1, 7), and (0, 0) → (1, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q4/5

Triangle PQR has sides 3, 4, 5. Triangle XYZ has sides 6, 8, 10. What scale factor must the dilation in a similarity sequence from PQR to XYZ have?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q5/5

490

QUESTION ID

S10-F2-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. Each side of XYZ is 2 times the matching side of PQR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q5/5

Predict the effect of a rigid motion on a figure

MAIN	Geometry main question	**HSG-CO.B.6**
BACKWARD 1	Know what rigid motions keep: lines, lengths, angles, and parallel l…	**8.G.A.1**
BACKWARD 2	Match the vertices and sides of a figure with its image	**8.G.A.2**
FORWARD 1	Predict the effect of a sequence of rigid motions	**HSG-CO.B.6**
FORWARD 2	Use a rigid motion to conclude that parts of a figure are congruent	**HSG-CO.B.7**

SET 11 • HSG-CO.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC has AB = 7, BC = 5, and m∠B = 48°. It is reflected across a line ℓ. What are A′B′ and m∠B′?

A. 14 and 48°

B. 7 and 132°

C. 7 and 48°

D. They depend on where ℓ is

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q1/5

493

QUESTION ID

S11-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. 7 and 48°

GRADING NOTE A reflection is a rigid motion: it keeps every length and every angle measure, wherever ℓ is.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q1/5

Quadrilateral PQRS has PQ ∥ SR. It is rotated 90° about a point O. What is true about P′Q′ and S′R′? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q2/5

494

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

P′Q′ ∥ S′R′. A rotation maps lines to lines and keeps them from meeting if they did not meet before, so parallel lines stay parallel.

GRADING NOTE Any equivalent explanation. Essential: the images are parallel because rigid motions take parallel lines to parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q2/5

Point A is 4 cm from line m. Triangle ABC is reflected across m. How far is A′ from m, and on which side of m is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q3/5

495

QUESTION ID
S11-M-Q3

TYPE
Short answer

CORRECT ANSWER
4 cm, on the other side of m.

GRADING NOTE Any equivalent wording. m is the perpendicular bisector of AA′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q3/5

Triangle ABC is labeled counterclockwise (A → B → C). It is translated 5 units right and then reflected across a line m. Is the final image A″B″C″ labeled clockwise or counterclockwise? Explain each step.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q4/5

496

QUESTION ID

S11-M-Q4

TYPE

Short answer

CORRECT ANSWER

Clockwise. The translation keeps the counterclockwise order (it slides every point the same way); the reflection reverses it.

GRADING NOTE Any equivalent explanation that predicts the effect of each motion on orientation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q4/5

Segment AB lies on line ℓ. A 180° rotation about a point O on ℓ maps AB to A′B′. Where does A′B′ lie? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q5/5

497

QUESTION ID
S11-M-Q5

TYPE
Short answer

CORRECT ANSWER
On ℓ. A 180° rotation about a point of ℓ sends every point of ℓ to the point of ℓ on the opposite side of O, so ℓ maps onto itself.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q5/5

SET 11 • HSG-CO.B.6

BACKWARD 1

Know what rigid motions keep: lines, lengths, angles, and parallel lines • Grade 8 • Geometry • 8.G.A.1

True or false?

Under a rotation, the image of a line is a line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q1/5

499

QUESTION ID

S11-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rigid motions map lines to lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q1/5

True or false?

A reflection can turn two parallel lines into two lines that meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q2/5

500

QUESTION ID

S11-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rigid motions keep parallel lines parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q2/5

A 52° angle is translated 6 units to the right. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q3/5

501

QUESTION ID

S11-B1-Q3

TYPE

Short answer

CORRECT ANSWER

52°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q3/5

A 9 cm segment is rotated 30° about one of its endpoints. How long is its image?

A. 30 cm

B. 4.5 cm

C. 39 cm

D. 9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q4/5

502

QUESTION ID

S11-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 9 cm

GRADING NOTE Rotations keep lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q4/5

Name two properties of a figure that a reflection does NOT change.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q5/5

503

QUESTION ID

S11-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Any two of: side lengths, angle measures, area, parallel sides, perimeter.

GRADING NOTE Any two correct properties.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

BACKWARD 2

Match the vertices and sides of a figure with its image • Grade 8 • Geometry • 8.G.A.2

Triangle ABC is rotated to triangle A′B′C′. Which side of the image corresponds to side BC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q1/5

505

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

B′C′

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q1/5

Quadrilateral PQRS is reflected to P′Q′R′S′. Which angle corresponds to ∠Q?

A. ∠R′

B. ∠P′

C. ∠Q′

D. ∠S′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q2/5

506

QUESTION ID

S11-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. ∠Q′

GRADING NOTE The prime marks the image of each vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q2/5

True or false?

If triangle ABC is translated to triangle A′B′C′, then AB = A′B′.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q3/5

507

QUESTION ID

S11-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Corresponding sides of an image under a rigid motion are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle DEF is reflected to D′E′F′. DE = 6 and EF = 9. Find D′E′ and E′F′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q4/5

508

QUESTION ID

S11-B2-Q4

TYPE

Short answer

CORRECT ANSWER

D′E′ = 6, E′F′ = 9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q4/5

True or false?

After a rotation, side A′B′ of the image always lies on top of side AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q5/5

509

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The image is usually in a new position; it only has the same length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSG-CO.B.6

FORWARD 1

Predict the effect of a sequence of rigid motions • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC is reflected across line ℓ and then across line m, where $\ell \parallel$ m. Is the final image clockwise or counterclockwise if ABC is counterclockwise? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q1/5

511

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Counterclockwise. Each reflection reverses the orientation, so two reflections restore it.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q1/5

A figure is rotated 70° counterclockwise about point O and then rotated 50° clockwise about the same point O. Which single rigid motion has the same effect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q2/5

512

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A rotation of 20° counterclockwise about O.

GRADING NOTE Form required: the angle, the direction, and the center. Directed angles add: +70° + (−50°) = +20°. (If both turns were in the same direction, the result would be 120°.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q2/5

True or false?

A translation that is not the identity has exactly one fixed point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q3/5

513

QUESTION ID

S11-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every point moves by the same nonzero amount, so no point is fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q3/5

Which rigid motion keeps exactly one point fixed?

A. A translation 3 units right

B. A reflection across line ℓ

C. A rotation of 90° about point O

D. The identity

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q4/5

514

QUESTION ID

S11-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. A rotation of 90° about point O

GRADING NOTE A 90° rotation fixes only its center; a reflection fixes every point of its line; a translation fixes none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q4/5

Line ℓ is reflected across line m, and ℓ is perpendicular to m. What is the image of ℓ? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q5/5

515

QUESTION ID

S11-F1-Q5

TYPE

Short answer

CORRECT ANSWER

ℓ itself. Each point of ℓ goes to the point of ℓ on the other side of m, the same distance away.

GRADING NOTE Any equivalent explanation (a line perpendicular to the mirror maps onto itself).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

FORWARD 2

Use a rigid motion to conclude that parts of a figure are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rotation about O maps triangle OAB onto triangle OCD, with A → C and B → D. OA = 5 and AB = 7. Find OC and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q1/5

517

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

OC = 5 and CD = 7.

GRADING NOTE Any equivalent form. The rotation sends OA to OC and AB to CD, keeping lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q1/5

A reflection across line ℓ maps triangle PQR onto triangle PSR (Q → S, and P and R are on ℓ). $m\angle PQR = 64°$. Find $m\angle PSR$.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q2/5

518

QUESTION ID
S11-F2-Q2

TYPE
Short answer

CORRECT ANSWER
64°

GRADING NOTE Any equivalent form. ∠PQR maps to ∠PSR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q2/5

True or false?

If a rigid motion maps segment AB onto segment CD, then AB could be longer than CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q3/5

519

QUESTION ID

S11-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rigid motions keep lengths, so AB = CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q3/5

A translation maps triangle KLM onto triangle K′L′M′. Which conclusion is NOT justified?

A. KL ∥ K′L′ or KL is on line K′L′

B. ∠L ≅ ∠L′

C. KL is on the same line as K′L′

D. KL = K′L′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q4/5

520

QUESTION ID

S11-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. KL is on the same line as K′L′

GRADING NOTE A translation keeps length and angle and keeps lines parallel to themselves, but the image is usually on a different line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q4/5

Explain why the image of an isosceles triangle under any rigid motion is also isosceles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q5/5

521

QUESTION ID

S11-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The two equal sides map to two sides of the same lengths, so the image has two equal sides.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q5/5

Decide whether two figures are congruent using rigid motions

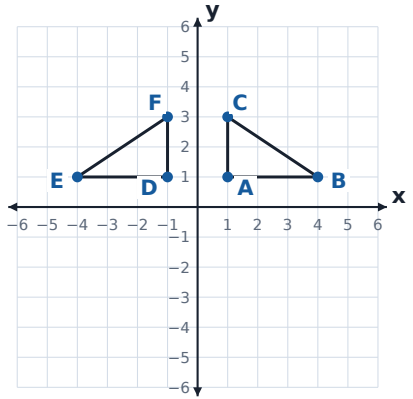
MAIN	Geometry main question	**HSG-CO.B.6**
BACKWARD 1	Recognize congruent figures as images under rigid motions	**8.G.A.2**
BACKWARD 2	Find lengths on a grid with the Pythagorean Theorem	**8.G.B.8**
FORWARD 1	Decide congruence for figures given by measurements, including…	**HSG-CO.B.6**
FORWARD 2	Use an established congruence to find unknown parts	**HSG-CO.B.7**

SET 12 • HSG-CO.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Is triangle ABC congruent to triangle DEF? If so, give a rigid motion that carries ABC onto DEF. If not, explain why.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q1/5

524

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

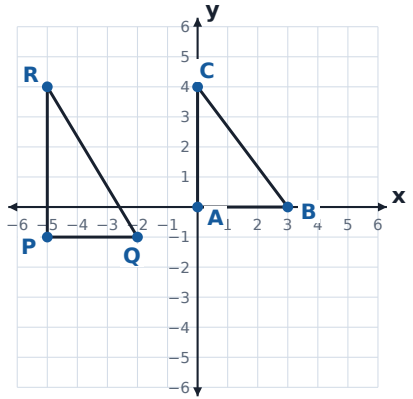
Yes. The reflection across the y-axis carries A to D, B to E, and C to F.

GRADING NOTE Any valid rigid motion or sequence. The conclusion and a motion that works are both required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q1/5

Is triangle ABC congruent to triangle PQR? If so, give a rigid motion that carries one onto the other. If not, explain why.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q2/5

525

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

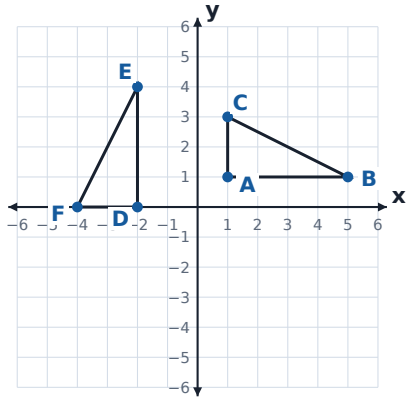
No. ABC has legs 3 and 4, but PQR has legs 3 and 5. A rigid motion keeps every length, so none can carry one onto the other.

GRADING NOTE Form required: "No" with a length (or area) that differs. Areas 6 and 7.5 also show it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q2/5

Is triangle ABC congruent to triangle DEF? If so, give a rigid motion (or sequence) that carries ABC onto DEF. If not, explain why.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q3/5

526

QUESTION ID

S12-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes. Rotate 90° counterclockwise about the origin (A → (−1, 1), B → (−1, 5), C → (−3, 1)), then translate 1 unit left and 1 unit down.

GRADING NOTE Any valid sequence that sends A to D, B to E, and C to F; the conclusion and a working motion are both required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q3/5

Quadrilateral Q has sides 3, 4, 3, 4 and four right angles. Quadrilateral R has sides 3, 4, 3, 4 and two 70° angles. Is Q congruent to R? Explain using rigid motions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q4/5

527

QUESTION ID

S12-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. Rigid motions keep angle measures, so no rigid motion can carry a 90° angle of Q onto a 70° angle, and every vertex of Q has a 90° angle.

GRADING NOTE Form required: "No" with a preserved measure (angle) that differs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q4/5

Rectangle R is 4 cm wide and 6 cm tall. Rectangle S is 6 cm wide and 4 cm tall. Are they congruent? Give a rigid motion or explain why not.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q5/5

528

QUESTION ID

S12-M-Q5

TYPE

Short answer

CORRECT ANSWER

Yes. A 90° rotation (followed by a translation if needed) carries R onto S.

GRADING NOTE Any valid rigid motion. Both are 4 by 6 rectangles; turning one a quarter turn matches the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q5/5

SET 12 • HSG-CO.B.6

BACKWARD 1

Recognize congruent figures as images under rigid motions • Grade 8 • Geometry • 8.G.A.2

True or false?

If one figure is a translation of another, the figures are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q1/5

530

QUESTION ID

S12-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A translation is a rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q1/5

True or false?

Two figures with the same shape but different sizes are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q2/5

531

QUESTION ID

S12-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are similar, not congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q2/5

Which pair of figures must be congruent?

A. A square and a larger square

B. A triangle and its reflection

C. A circle of radius 2 and a circle of radius 3

D. A triangle and a dilation of it by 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q3/5

532

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A triangle and its reflection

GRADING NOTE A reflection keeps size and shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q3/5

A triangle has sides 5, 6, and 7. It is rotated 90°. What are the side lengths of its image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q4/5

533

QUESTION ID

S12-B1-Q4

TYPE

Short answer

CORRECT ANSWER

5, 6, and 7

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q4/5

Is a figure congruent to its image after a reflection followed by a translation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q5/5

534

QUESTION ID

S12-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Any equivalent answer: both are rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

BACKWARD 2

Find lengths on a grid with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the length of the segment from (1, 2) to (5, 2).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q1/5

536

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q1/5

Find the length of the segment from (0, 0) to (3, 4).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q2/5

537

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q2/5

Find the length of the segment from (−2, 1) to (1, 5).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q3/5

538

QUESTION ID

S12-B2-Q3

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $\sqrt{(3^2 + 4^2)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q3/5

True or false?

The segments from (0, 0) to (2, 1) and from (0, 0) to (1, 2) have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q4/5

539

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both are √5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q4/5

True or false?

The segment from (0, 0) to (3, 3) has length 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q5/5

540

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its length is $\sqrt{18} = 3\sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSG-CO.B.6

FORWARD 1

Decide congruence for figures given by measurements, including curved figures • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Circle P has radius 3 and circle Q has radius 3. Explain, using a rigid motion, why they are congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q1/5

542

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Translate P's center to Q's center. Every point 3 units from P's center goes to a point 3 units from Q's center, so circle P lands on circle Q.

GRADING NOTE Any equivalent explanation using a translation that matches the centers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q1/5

Two quadrilaterals each have sides 5, 5, 5, 5. One is a square and the other has a 60° angle. Are they congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q2/5

543

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. A rigid motion keeps angle measures, but one has 90° angles and the other a 60° angle.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q2/5

True or false?

Two triangles with the same perimeter must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q3/5

544

QUESTION ID

S12-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A 3-4-5 triangle and a 4-4-4 triangle both have perimeter 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q3/5

Sector P has radius 4 and central angle 50°. Sector Q has radius 4 and central angle 50°, drawn in another place and turned. Are they congruent? Explain with a rigid motion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q4/5

545

QUESTION ID

S12-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes. Translate P's center onto Q's center, rotate so one bounding radius of P lies along one of Q's, and reflect across it if the arcs are on opposite sides. Equal radii and equal 50° angles make the arcs match.

GRADING NOTE Any valid motion with a reason using the equal radius and equal angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q4/5

Angle 1 measures 35° and angle 2 measures 35°. Describe a rigid motion that carries angle 1 onto angle 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q5/5

546

QUESTION ID

S12-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Translate the vertex of angle 1 to the vertex of angle 2, rotate so one side lies along a side of angle 2, and reflect across that side if the other side is on the wrong side.

GRADING NOTE Any valid description that matches vertex and both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

FORWARD 2

Use an established congruence to find unknown parts • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries quadrilateral ABCD onto quadrilateral EFGH with A → E, B → F, C → G, D → H. AB = 6, ∠C = 110°, and GH = 4. Find EF, ∠G, and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q1/5

548

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

EF = 6, m∠G = 110°, CD = 4.

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q1/5

Triangle RST is congruent to triangle XYZ by a rigid motion with R → X, S → Y, T → Z. RS = 2x + 3 and XY = 15. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q2/5

549

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. RS = XY, so 2x + 3 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q2/5

True or false?

If a rigid motion carries triangle ABC onto triangle DEF with A → D, B → E, C → F, then ∠B ≅ ∠F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q3/5

550

QUESTION ID

S12-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ∠B corresponds to ∠E.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q3/5

A reflection carries pentagon PQRST onto pentagon P′Q′R′S′T′. Which pair must be congruent?

A. ∠S and ∠S′

B. PQ and RS

C. QR and S′T′

D. ∠S and ∠T′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q4/5

551

QUESTION ID

S12-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. ∠S and ∠S′

GRADING NOTE Each part corresponds to its own image.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q4/5

Triangle ABC is congruent to triangle CDA by a rigid motion with A → C, B → D, C → A. m∠BAC = 3y° and m∠DCA = 42°. Find y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q5/5

552

QUESTION ID

S12-F2-Q5

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE Any equivalent form. ∠BAC corresponds to ∠DCA, so 3y = 42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q5/5

Congruent triangles: a rigid motion exactly when all corresponding parts are congruent

MAIN	Geometry main question	**HSG-CO.B.7**
BACKWARD 1	Read a congruence statement to match corresponding parts	**8.G.A.2**
BACKWARD 2	Rigid motions keep lengths and angle measures	**8.G.A.1**
BACKWARD 3	Move a point onto another and a ray onto another with rigid motio…	**8.G.A.3**
FORWARD 1	Find which three corresponding parts are enough	**HSG-CO.B.8**
FORWARD 2	Use congruent triangles to prove a relationship in a figure	**HSG-SRT.B.5**

SET 13 • HSG-CO.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries △ABC onto △DEF with A → D, B → E, C → F. Explain why AB = DE, BC = EF, CA = FD, ∠A ≅ ∠D, ∠B ≅ ∠E, and ∠C ≅ ∠F.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q1/5

555

QUESTION ID

S13-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Each side and angle of ABC is carried onto the corresponding side and angle of DEF, and rigid motions keep lengths and angle measures.

GRADING NOTE

REQUIRED WORK: Use the definition: the motion sends each vertex to its partner, so it sends each side and angle onto the corresponding one, and rigid motions keep length and angle measure.

Any valid explanation. Essential: (1) the motion maps segment AB onto segment DE (endpoints to endpoints), and similarly for the other sides and angles; (2) rigid motions preserve distance and angle measure. Invalid: "they look the same".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q1/5

In △ABC and △DEF, AB = DE, BC = EF, CA = FD, ∠A ≅ ∠D, ∠B ≅ ∠E, and ∠C ≅ ∠F. Explain how to build a rigid motion that carries △ABC onto △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q2/5

556

QUESTION ID

S13-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

A translation, a rotation, and (if needed) a reflection carry A to D, B to E, and C to F.

GRADING NOTE REQUIRED WORK: Translate A to D; rotate about D so ray AB lies on ray DE (B lands on E because AB = DE); if C is on the other side of line DE from F, reflect across DE. Explain why C then lands on F.

Any valid explanation. Essential: the translation and rotation match A and B; the reflection (if needed) puts C on F's side; C lands on F because ∠A ≅ ∠D puts C on ray DF and AC = DF fixes its distance from D. Accept using ∠B and ∠E instead. Invalid: stating the result without explaining why C lands on F.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q2/5

The rotation of 90° counterclockwise about the origin carries △ABC, with A(1, 0), B(4, 0), C(1, 2), onto △A′B′C′. Find A′, B′, C′, check that all six pairs of corresponding parts are congruent, and explain why the definition of congruence guarantees this for every rigid motion.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q3/5

557

QUESTION ID

S13-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

A′(0, 1), B′(0, 4), C′(−2, 1). AB = A′B′ = 3, AC = A′C′ = 2, BC = B′C′ = √13; the right angles at A and A′ match and the other angles match because the triangles have the same sides. In general a rigid motion sends each side and angle onto its partner and keeps length and angle measure.

GRADING NOTE REQUIRED WORK: Apply (x, y) → (−y, x); compare the three side lengths and the angles; then give the general reason.

Any valid explanation. Essential: correct images; the six comparisons (angles may be checked by slopes or by SSS); the general reason (the motion carries parts onto corresponding parts, and rigid motions preserve measures).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q3/5

△ABC has A(0, 0), B(3, 0), C(0, 4). △DEF has D(5, 1), E(5, 4), F(1, 1). Show that all six pairs of corresponding parts (A ↔ D, B ↔ E, C ↔ F) are congruent, and then give a rigid motion that carries △ABC onto △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q4/5

558

QUESTION ID

S13-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AB = DE = 3, AC = DF = 4, BC = EF = 5, and the right angles at A and D match. Translate 5 right and 1 up (A → D, B → (8, 1), C → (5, 5)), rotate 90° counterclockwise about D (B → (5, 4) = E, C → (1, 1) = F).

GRADING NOTE REQUIRED WORK: Compare the sides (and angles), then build the motion: move A onto D, turn ray AB onto ray DE, and reflect if needed.

Any valid motion. Essential: all six matches (angles via the right angles and equal sides), and a sequence that sends A, B, C to D, E, F.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q4/5

Explain why the converse direction needs the sides as well as the angles: give two triangles with all three pairs of angles congruent for which no rigid motion carries one onto the other, and say which part of the definition fails.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q5/5

559

QUESTION ID

S13-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

A 3-4-5 right triangle and a 6-8-10 right triangle have the same angles, but a rigid motion keeps lengths, so it cannot send a 3-unit side to a 6-unit side.

GRADING NOTE REQUIRED WORK: Use a counterexample and the fact that rigid motions preserve length.

Any valid counterexample with equal angles and unequal sides, and the reason (rigid motions preserve length).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 1

Read a congruence statement to match corresponding parts • Grade 8 • Geometry • 8.G.A.2

ANSWER KEY Reduced copy of the question page, with its answer

△ABC ≅ △XYZ. Which angle corresponds to ∠B?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

561

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

∠Y

GRADING NOTE Any equivalent form. The order of the letters gives the correspondence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

△ABC ≅ △XYZ. Which side corresponds to AC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

562

QUESTION ID

S13-B1-Q2

TYPE

Short answer

CORRECT ANSWER

XZ

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

△PQR ≅ △LMN. Which pair corresponds?

A. QR and MN

B. QR and LN

C. PQ and MN

D. PR and LM

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

563

QUESTION ID

S13-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. QR and MN

GRADING NOTE Q ↔ M and R ↔ N.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

True or false?

△DEF ≅ △GHI means that ∠F ≅ ∠I.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

564

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The third letters correspond.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

True or false?

△DEF ≅ △GHI means that DE = HI.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

565

QUESTION ID

S13-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE DE corresponds to GH.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 2

Rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

True or false?

A rotation keeps the measure of every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q1/5

567

QUESTION ID

S13-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rotations are rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q1/5

Segment PQ is 12 units long. It is reflected and then translated. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q2/5

568

QUESTION ID

S13-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12 units

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q2/5

A 71° angle is reflected across a line. Its image measures

A. 289°

B. 109°

C. 71°

D. 19°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q3/5

569

QUESTION ID

S13-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 71°

GRADING NOTE Reflections keep angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q3/5

True or false?

A translation can change the length of a segment if it moves the segment far enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q4/5

570

QUESTION ID

S13-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Translations keep every length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q4/5

Which three basic rigid motions keep lengths and angle measures?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q5/5

571

QUESTION ID

S13-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Translations, rotations, and reflections

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 3

Move a point onto another and a ray onto another with rigid motions • Grade 8 • Geometry • 8.G.A.3

Which translation moves (2, 3) onto (5, −1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q1/5

573

QUESTION ID

S13-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3 units right and 4 units down

GRADING NOTE Any equivalent form: (x, y) → (x + 3, y − 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q1/5

A ray starts at the origin and points along the positive x-axis. Which rotation about the origin makes it point along the positive y-axis?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q2/5

574

QUESTION ID

S13-B3-Q2

TYPE

Short answer

CORRECT ANSWER

90° counterclockwise

GRADING NOTE Any equivalent form (270° clockwise).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q2/5

After a rotation, point C is below line DE but should be above it, and D and E are on the line. Which motion moves C across without moving D or E?

A. A rotation about D

B. A translation along DE

C. A rotation about the midpoint of DE

D. A reflection across line DE

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q3/5

575

QUESTION ID

S13-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A reflection across line DE

GRADING NOTE A reflection across DE keeps every point of the line fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q3/5

True or false?

Reflecting across a line keeps every point of that line fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q4/5

576

QUESTION ID

S13-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Points on the mirror line are their own images.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q4/5

True or false?

Translating a triangle can keep exactly one of its vertices fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q5/5

577

QUESTION ID

S13-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A nonzero translation moves every point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q5/5

SET 13 • HSG-CO.B.7

FORWARD 1

Find which three corresponding parts are enough • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Must two triangles be congruent if all three pairs of corresponding sides are congruent? Use the result of the full six-part test.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q1/5

579

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Three congruent sides force the angles to be congruent too (SSS), so all six parts match and a rigid motion exists.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q1/5

Give an example of two triangles with two pairs of congruent sides and a pair of congruent angles that are NOT congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q2/5

580

QUESTION ID

S13-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Triangles with sides 6 and 8 and a 30° angle opposite the 6-unit side: the 8-unit side can make the third side either long or short (SSA).

GRADING NOTE Any valid SSA counterexample with the angle not between the sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q2/5

True or false?

Knowing that all three pairs of corresponding angles are congruent is enough to prove two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q3/5

581

QUESTION ID
S13-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE AAA gives similar triangles, not necessarily congruent ones.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q3/5

Which set of corresponding parts is NOT always enough for congruence?

A. Three sides

B. Two sides and an angle not between them

C. Two sides and the angle between them

D. Two angles and the side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q4/5

582

QUESTION ID

S13-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Two sides and an angle not between them

GRADING NOTE SSA can give two different triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q4/5

Two triangles have AB = DE, ∠A ≅ ∠D, and ∠B ≅ ∠E. Must ∠C ≅ ∠F? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q5/5

583

QUESTION ID

S13-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Yes. The angle sum is 180° in both triangles, so ∠C = 180° − ∠A − ∠B = ∠F.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

FORWARD 2

Use congruent triangles to prove a relationship in a figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB = CB and AD = CD. Prove that diagonal BD bisects ∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q1/5

585

QUESTION ID

S13-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

∠ABD ≅ ∠CBD, so BD bisects ∠ABC.

GRADING NOTE REQUIRED WORK: Prove △ABD ≅ △CBD (SSS, with BD shared), then use corresponding angles.

Any valid proof. Essential: AB = CB, AD = CD, BD = BD (reflexive); SSS gives the congruence; corresponding angles ∠ABD and ∠CBD are congruent, which is the definition of a bisector. Invalid: assuming the angles are equal from the picture.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q1/5

Segments AC and BD bisect each other at E. Prove that AB = CD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q2/5

586

QUESTION ID

S13-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

AB = CD

GRADING NOTE REQUIRED WORK: Use vertical angles and the bisected segments to prove △AEB ≅ △CED by SAS.

Any valid proof. Essential: AE = CE and BE = DE (definition of bisect); ∠AEB ≅ ∠CED (vertical angles); SAS; corresponding sides AB = CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q2/5

In △PQR and △STU, PQ = ST, QR = TU, and PR = SU. The triangles are congruent. What must be true of ∠R and ∠U? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q3/5

587

QUESTION ID

S13-F2-Q3

TYPE

Short answer

CORRECT ANSWER

∠R ≅ ∠U, because corresponding angles of congruent triangles are congruent.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q3/5

True or false?

Once two triangles are proved congruent, only the parts used in the proof are known to be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q4/5

588

QUESTION ID
S13-F2-Q4

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Congruence gives all six pairs of corresponding parts (CPCTC).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q4/5

To prove that point M is the midpoint of segment XY using congruent triangles, which pair of parts must you show is congruent?

A. ∠M and ∠X

B. ∠X and ∠Y

C. XM and YM

D. XY and XM

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q5/5

589

QUESTION ID
S13-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. XM and YM

GRADING NOTE Midpoint means XM = MY.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q5/5

Explain SAS congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Identify the included angle and included side	**7.G.A.2**
BACKWARD 2	Rigid motions map rays, lengths, and angles to equal ones	**8.G.A.1**
BACKWARD 3	Choose a translation, rotation, or reflection to move a point or ray…	**8.G.A.2**
FORWARD 1	Prove triangles congruent by SAS inside a larger figure	**HSG-SRT.B.5**
FORWARD 2	Decide whether given information is enough for SAS, and name a…	**HSG-SRT.B.5**

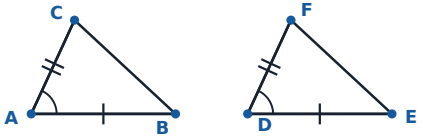
SET 14 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In △ABC and △DEF, AB = DE, AC = DF, and ∠A ≅ ∠D. Explain, using rigid motions, why △ABC ≅ △DEF.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q1/5

592

QUESTION ID

S14-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

A translation, a rotation, and possibly a reflection carry A to D, B to E, and C to F, so the triangles are congruent.

GRADING NOTE REQUIRED WORK: Translate A to D; rotate so ray AB lies on ray DE; reflect across DE if C is on the other side from F. Then explain why B lands on E and C lands on F.

Any valid explanation. Essential: (1) translate A onto D; (2) rotate about D so ray AB lies on ray DE; B lands on E because AB = DE; (3) reflect across DE if needed so C is on F's side; (4) ray AC lands on ray DF because ∠A ≅ ∠D; (5) C lands on F because AC = DF. Invalid: "the triangles look the same"; citing SAS to prove SAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q1/5

In △JKL and △MNO, JK = MN, JL = MO, and ∠J ≅ ∠M. △MNO is a mirror image of △JKL (its vertices run the opposite way around). Explain, using rigid motions, why △JKL ≅ △MNO, and say at which step a reflection is needed.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q2/5

593

QUESTION ID

S14-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

A translation, a rotation, and a reflection across line MN carry J to M, K to N, and L to O.

GRADING NOTE REQUIRED WORK: Translate J to M; rotate so ray JK lies on ray MN; then reflect across line MN because L is on the other side from O. Explain why K lands on N and L lands on O.

Any valid explanation. Essential: (1) J → M; (2) ray JK onto ray MN, K lands on N because JK = MN; (3) the reflection across MN is needed because the orientation is reversed, and it fixes M and N; (4) ray JL lands on ray MO because ∠J ≅ ∠M; (5) L lands on O because JL = MO. Invalid: citing SAS to prove SAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△ABC has A(0, 0), B(4, 0), C(1, 3). △DEF has D(2, −1), E(2, 3), F(−1, 0). AB = DE = 4, AC = DF = √10, and ∠A ≅ ∠D. Name a translation and a rotation that carry A to D and ray AB onto ray DE, and explain, using only the SAS facts, why C must then land on F.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q3/5

594

QUESTION ID

S14-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Translate 2 right and 1 down (A → D, B → (6, −1)); rotate 90° counterclockwise about D (B → E). Ray AC goes to the ray from D making ∠A with ray DE on F's side, which is ray DF (∠A ≅ ∠D); C is √10 from D on it, so C → F.

GRADING NOTE REQUIRED WORK: Find the motions, then use ∠A ≅ ∠D to place ray AC on ray DF and AC = DF to place C on F.

Any valid motions. Essential: the motions with the check B → E; the angle places the ray; the length places the point. A computation C → (−1, 0) is accepted as a check but the SAS reasoning is required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q3/5

In quadrilateral ABCD, AB = AD and diagonal AC bisects ∠BAD. Explain, using one reflection, why △ABC ≅ △ADC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q4/5

595

QUESTION ID

S14-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The reflection across line AC fixes A and C, sends ray AB onto ray AD, and sends B to D, so △ABC → △ADC.

GRADING NOTE REQUIRED WORK: Reflect across line AC. Explain where A, C, ray AB, and B go.

Any valid explanation. Essential: A and C are on the mirror line, so they are fixed; ∠BAC ≅ ∠DAC (bisector), with B and D on opposite sides of AC, so the reflection sends ray AB onto ray AD; AB = AD, so B lands on D. Invalid: "they look like mirror images".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Segments AC and BD bisect each other at E (A, E, C on one line and B, E, D on another). Explain, using a 180° rotation about E, why △AEB ≅ △CED.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q5/5

596

QUESTION ID

S14-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The 180° rotation about E sends A to C and B to D (and fixes E), so △AEB → △CED.

GRADING NOTE REQUIRED WORK: Use that a 180° rotation about E sends each ray from E to the opposite ray, and keeps distances from E.

Any valid explanation. Essential: ray EA goes to the opposite ray EC and EA = EC, so A → C; ray EB goes to ray ED and EB = ED, so B → D; E is fixed. (This is the SAS configuration with the vertical angles at E.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 1

Identify the included angle and included side • Grade 7 • Geometry • 7.G.A.2

In △ABC, which angle is included between sides AB and AC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q1/5

598

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

∠A

GRADING NOTE Any equivalent form. The included angle is at the shared vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q1/5

In △PQR, which side is included between ∠P and ∠Q?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q2/5

599

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

PQ

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In △XYZ, which angle is NOT included between sides XY and YZ?

A. Both ∠X and ∠Y

B. ∠X

C. ∠Y

D. None of the angles is excluded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q3/5

600

QUESTION ID

S14-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ∠X

GRADING NOTE Sides XY and YZ share vertex Y, so ∠Y is included; ∠X is not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q3/5

True or false?

Two sides of 5 cm and 7 cm with a 40° angle between them determine one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q4/5

601

QUESTION ID

S14-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two sides and the included angle fix the triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q4/5

True or false?

Two sides of 5 cm and 7 cm with a 40° angle that is not between them always determine one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q5/5

602

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE When the angle is not between the sides, there may be two triangles or none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 2

Rigid motions map rays, lengths, and angles to equal ones • Grade 8 • Geometry • 8.G.A.1

True or false?

Under a rigid motion, the image of a ray is a ray.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q1/5

604

QUESTION ID

S14-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rigid motions map rays to rays.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q1/5

A 30° angle is rotated about its vertex. What is the measure of the image angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q2/5

605

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q2/5

A point is 5 units from D along a ray. After a rotation about D, the image point is

A. closer to D

B. not on the image ray

C. 5 units from D on the image ray

D. farther from D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q3/5

606

QUESTION ID

S14-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 units from D on the image ray

GRADING NOTE A rotation about D keeps distances from D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q3/5

True or false?

A reflection across line DE can move point D.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q4/5

607

QUESTION ID

S14-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Points on the mirror line stay fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q4/5

Segment AB is translated so that A lands on D. AB = 9. How far is the image of B from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q5/5

608

QUESTION ID

S14-B2-Q5

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 3

Choose a translation, rotation, or reflection to move a point or ray onto a target • Grade 8 • Geometry • 8.G.A.2

Which kind of rigid motion moves point A onto point D without turning anything?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q1/5

610

QUESTION ID

S14-B3-Q1

TYPE

Short answer

CORRECT ANSWER

A translation

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q1/5

A and D are already together. Which kind of rigid motion turns ray AB onto ray DE without moving D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q2/5

611

QUESTION ID

S14-B3-Q2

TYPE

Short answer

CORRECT ANSWER

A rotation about D

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q2/5

D and E are fixed, and point C is on the wrong side of line DE. Which motion fixes D and E and moves C across the line?

A. A translation along DE

B. A rotation of 90° about D

C. A rotation of 180° about E

D. A reflection across line DE

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q3/5

612

QUESTION ID

S14-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A reflection across line DE

GRADING NOTE Only the reflection keeps both D and E fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q3/5

True or false?

A rotation about point D keeps D fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q4/5

613

QUESTION ID

S14-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The center of a rotation is its own image.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q4/5

True or false?

A translation that moves A onto D also keeps A fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q5/5

614

QUESTION ID

S14-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It moves A onto D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSG-CO.B.8

FORWARD 1

Prove triangles congruent by SAS inside a larger figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB ∥ DC and AB = DC. Diagonal AC is drawn. Prove that △ABC ≅ △CDA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q1/5

616

QUESTION ID

S14-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

△ABC ≅ △CDA by SAS.

GRADING NOTE REQUIRED WORK: Use alternate interior angles with the parallel sides and the shared diagonal to apply SAS.

Any valid proof. Essential: AB = CD (given); ∠BAC ≅ ∠DCA (alternate interior angles, AB ∥ DC); AC = CA (reflexive); SAS with the angle between the sides. Invalid: using ∠B ≅ ∠D without proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Point M is the midpoint of segment AB, and line ℓ is perpendicular to AB at M. Point P is on ℓ, and P ≠ M. Prove that △PMA ≅ △PMB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q2/5

617

QUESTION ID

S14-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

△PMA ≅ △PMB by SAS, so PA = PB.

GRADING NOTE REQUIRED WORK: Use the midpoint, the right angles, and the shared side PM with SAS.

Any valid proof. Essential: AM = BM (midpoint); ∠PMA ≅ ∠PMB (both right angles); PM = PM; SAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q2/5

Segments AC and BD cross at E, with AE = CE and BE = DE. Name the congruent triangles and the criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q3/5

618

QUESTION ID

S14-F1-Q3

TYPE

Short answer

CORRECT ANSWER

△AEB ≅ △CED by SAS (the vertical angles at E are the included angles).

GRADING NOTE Any equivalent correct pair (also △AED ≅ △CEB).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q3/5

True or false?

In an SAS proof, a side that the two triangles share can be one of the pairs of congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q4/5

619

QUESTION ID
S14-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The shared side is congruent to itself (reflexive property).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q4/5

Segments PQ and RS bisect each other at T. Which reason gives the included angles for SAS?

A. The angles are both right angles

B. Vertical angles are congruent

C. Base angles are congruent

D. Corresponding angles are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q5/5

620

QUESTION ID

S14-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Vertical angles are congruent

GRADING NOTE ∠PTR and ∠QTS (or the other pair) are vertical angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

FORWARD 2

Decide whether given information is enough for SAS, and name a missing condition • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC and △DEF, AB = DE and BC = EF. Name ONE more pair of congruent parts that would prove congruence by SAS.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q1/5

622

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

∠B ≅ ∠E

GRADING NOTE Form required: the included angle pair. (AC = DF would give SSS, not SAS.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △PQR and △XYZ, PQ = XY, PR = XZ, and ∠Q ≅ ∠Y. Is this enough to prove the triangles congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q2/5

623

QUESTION ID

S14-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No. The congruent angle is not between the congruent sides (SSA), which does not guarantee congruence.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q2/5

True or false?

In △ABC and △ADC (sharing AC), AB = AD and ∠BCA ≅ ∠DCA. These facts prove △ABC ≅ △ADC by SAS.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q3/5

624

QUESTION ID

S14-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The congruent angles at C are not included between AB and AC (this is SSA), so SAS does not apply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Kites ABCD and EFGH have AB = EF, BC = FG, and ∠B ≅ ∠F. Which triangles are congruent by SAS?

A. △BCD and △FGH

B. △ABC and △EFG

C. △ABD and △EFH

D. △ACD and △EGH

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q4/5

625

QUESTION ID

S14-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. △ABC and △EFG

GRADING NOTE The given parts are the two sides and the included angle of △ABC and △EFG.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q4/5

Two triangles have sides 7 and 9 with a 50° angle between them. A student says they need not be congruent because the third sides were not measured. Is the student right? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q5/5

626

QUESTION ID

S14-F2-Q5

TYPE

Short answer

CORRECT ANSWER

No. SAS guarantees congruence; the third side is then determined (and equal).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q5/5

Explain ASA congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Two angles and the included side determine a triangle	**7.G.A.2**
BACKWARD 2	Two different lines meet in at most one point	**8.EE.C.8.a**
BACKWARD 3	Rigid motions keep rays and angle measures	**8.G.A.1**
FORWARD 1	Explain AAS from ASA and the triangle angle sum	**HSG-CO.C.10**
FORWARD 2	Prove congruence with ASA or AAS in overlapping triangles	**HSG-SRT.B.5**

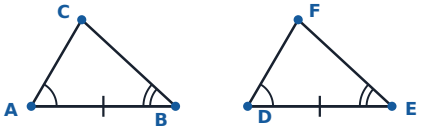
SET 15 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In △ABC and △DEF, ∠A ≅ ∠D, AB = DE, and ∠B ≅ ∠E. Explain, using rigid motions, why △ABC ≅ △DEF.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q1/5

629

QUESTION ID

S15-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

A translation, a rotation, and possibly a reflection carry A to D, B to E, and C to F.

GRADING NOTE REQUIRED WORK: Translate A to D; rotate so ray AB lies on ray DE (B lands on E); reflect across DE if needed; then use the two angles to place rays AC and BC on rays DF and EF.

Any valid explanation. Essential: (1) A → D and B → E using AB = DE; (2) C on F's side of DE (reflect if needed); (3) ray AC lands on ray DF because ∠A ≅ ∠D; (4) ray BC lands on ray EF because ∠B ≅ ∠E; (5) C is where rays AC and BC meet, so it lands where DF and EF meet, which is F. Invalid: using side lengths AC or BC (not given); citing ASA to prove ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q1/5

In △PQR and △STU, ∠P ≅ ∠S, PQ = ST, and ∠Q ≅ ∠T. △STU is a mirror image of △PQR (its vertices run the opposite way around). Explain, using rigid motions, why △PQR ≅ △STU, including where the reflection is used and why R lands on U.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q2/5

630

QUESTION ID

S15-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

A translation, a rotation, and a reflection across ST carry P → S, Q → T, and R → U.

GRADING NOTE REQUIRED WORK: Translate P to S; rotate so ray PQ lies on ray ST; reflect across line ST; then use the two angles to place two rays and their intersection.

Any valid explanation. Essential: P → S and Q → T (PQ = ST); the reflection (fixing S and T) puts R on U's side; ray PR lands on ray SU (∠P ≅ ∠S) and ray QR lands on ray TU (∠Q ≅ ∠T); two different lines meet in at most one point, so R lands on U.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△ABC has A(0, 0), B(4, 0), C(1, 3). △DEF has D(5, 5), E(1, 5), F(4, 2). ∠A ≅ ∠D, AB = DE, and ∠B ≅ ∠E. The 180° rotation about (2.5, 2.5) sends A to D and B to E. Explain, using only the ASA facts (not the coordinates of C), why it must send C to F.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q3/5

631

QUESTION ID

S15-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The image of ray AC is the ray from D making ∠A with ray DE on F's side, which is ray DF; likewise ray BC goes to ray EF; C, the intersection, goes to F.

GRADING NOTE REQUIRED WORK: Use the angles to place the images of rays AC and BC, then the intersection.

Any valid explanation. Essential: the rotation keeps angle measures and sends the side of line AB containing C to the side of DE containing F; the two angles place the two rays; the unique intersection. A coordinate check alone is not enough.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q3/5

In △ABC, ∠B ≅ ∠C. Let ℓ be the perpendicular bisector of BC. Explain, using the reflection across ℓ and the ASA argument, why the reflection carries △ABC onto △ACB, and conclude that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q4/5

632

QUESTION ID

S15-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The reflection swaps B and C, sends ray BA onto ray CA and ray CA onto ray BA, so it fixes A; hence segment AB maps onto AC and AB = AC.

GRADING NOTE REQUIRED WORK: The reflection swaps B and C. Use ∠B ≅ ∠C to find the images of rays BA and CA, then the image of A.

Any valid explanation. Essential: ℓ is the perpendicular bisector of BC, so B ↔ C; the image of ray BA is the ray from C making ∠B with CB on A's side, which is ray CA because ∠B ≅ ∠C (and vice versa); A is the intersection of the two rays, so A → A; lengths are kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q4/5

Segments AD and BC cross at E, which is the midpoint of AD, and AB ∥ CD (B and C on opposite sides of AD). Explain, using a 180° rotation about E, why △ABE ≅ △DCE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q5/5

633

QUESTION ID

S15-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The rotation sends A to D, line AB to line DC, and ray EB to ray EC, so B goes to C and △ABE → △DCE.

GRADING NOTE REQUIRED WORK: Find the images of A, line AB, and ray EB, then the image of B as an intersection.

Any valid explanation. Essential: EA = ED with opposite rays, so A → D; a 180° rotation sends line AB to a parallel line through D, which is line DC (parallel postulate); ray EB goes to the opposite ray EC; B = AB ∩ EB goes to DC ∩ EC = C (an ASA-type argument).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 1

Two angles and the included side determine a triangle • Grade 7 • Geometry • 7.G.A.2

True or false?

Two angles of 40° and 60° with a 5 cm side between them determine exactly one triangle (up to position).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q1/5

635

QUESTION ID

S15-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The side and the two angles at its ends fix where the other two sides meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q1/5

True or false?

Any two angles with a side between them make a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q2/5

636

QUESTION ID

S15-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The two angles must add to less than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q2/5

A triangle has angles of 50° and 70° at the ends of a 6 cm side. What is the third angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q3/5

637

QUESTION ID

S15-B1-Q3

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q3/5

Which set of conditions determines exactly one triangle?

A. Angles 30° and 80° with a 4 cm side between them

B. One angle of 45° and nothing else

C. Angles 30°, 60°, and 90°

D. Angles 100° and 90° with a 3 cm side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q4/5

638

QUESTION ID

S15-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Angles 30° and 80° with a 4 cm side between them

GRADING NOTE Two angles with the included side fix the triangle; three angles alone fix only its shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q4/5

Two angles of a triangle are 95° and 90°. Can the triangle exist? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q5/5

639

QUESTION ID

S15-B1-Q5

TYPE

Short answer

CORRECT ANSWER

No. 95° + 90° = 185° is more than 180°.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 2

Two different lines meet in at most one point • Grade 8 • Expressions & Equations • 8.EE.C.8.a

How many points do the lines y = x + 1 and y = −x + 3 have in common?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q1/5

641

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. They meet at (1, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q1/5

True or false?

Two different lines can meet in exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q2/5

642

QUESTION ID

S15-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two different lines meet in at most one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q2/5

Lines $y = 2x + 1$ and $y = 2x - 4$

A. never meet

B. are the same line

C. meet at one point

D. meet at two points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q3/5

643

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. never meet

GRADING NOTE Same slope, different intercepts: parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q3/5

True or false?

If two lines share two points, they are the same line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q4/5

644

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two points determine one line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q4/5

Where do the lines x = 2 and y = −3 meet?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q5/5

645

QUESTION ID

S15-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(2, −3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 3

Rigid motions keep rays and angle measures • Grade 8 • Geometry • 8.G.A.1

A 64° angle is translated and then reflected. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q1/5

647

QUESTION ID

S15-B3-Q1

TYPE

Short answer

CORRECT ANSWER

64°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q1/5

True or false?

A rigid motion maps an angle to an angle with the same measure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q2/5

QUESTION ID

S15-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rigid motions keep angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q2/5

A ray from E makes a 35° angle with ray ED above line DE. How many such rays are there?

A. Exactly one

B. Two

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q3/5

649

QUESTION ID

S15-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Exactly one

GRADING NOTE On one side of a line there is one ray at a given angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q3/5

True or false?

After a rotation, an angle can become larger if its sides are longer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q4/5

650

QUESTION ID

S15-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Angle measure does not depend on side length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q4/5

Segment AB = 4 cm is rotated so that A lands on D. Where is B, relative to D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q5/5

651

QUESTION ID

S15-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4 cm from D (on the image ray).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSG-CO.B.8

FORWARD 1

Explain AAS from ASA and the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC and △DEF, ∠A ≅ ∠D, ∠B ≅ ∠E, and BC = EF (the side is NOT between the angles). Prove △ABC ≅ △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q1/5

653

QUESTION ID

S15-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

△ABC ≅ △DEF by ASA after showing ∠C ≅ ∠F.

GRADING NOTE REQUIRED WORK: Use the triangle angle sum to show ∠C ≅ ∠F, then apply ASA with ∠B, BC, ∠C.

Any valid proof. Essential: m∠C = 180° − m∠A − m∠B = 180° − m∠D − m∠E = m∠F; then ∠B, side BC, ∠C match ∠E, EF, ∠F (ASA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q1/5

Two triangles have angles 40° and 75° and a 9 cm side opposite the 75° angle. Must they be congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q2/5

654

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes, by AAS: the third angles are both 65°, so ASA applies.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q2/5

In the argument above, how does AAS follow from ASA? Name the theorem used to get the third pair of angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q3/5

655

QUESTION ID

S15-F1-Q3

TYPE

Short answer

CORRECT ANSWER

The triangle angle sum theorem gives ∠C ≅ ∠F (180° minus equal sums), and then ∠B, side BC, ∠C satisfy ASA.

GRADING NOTE Any equivalent answer. Other valid derivations of AAS exist; this question asks about this argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q3/5

In △PQR and △XYZ, ∠P ≅ ∠X, ∠Q ≅ ∠Y, and PR = XZ. Which criterion proves congruence?

A. ASA

B. AAS

C. SAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q4/5

656

QUESTION ID

S15-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. AAS

GRADING NOTE PR is not between ∠P and ∠Q, so it is AAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q4/5

Which corresponding side would make AAS apply if ∠A ≅ ∠D and ∠C ≅ ∠F?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q5/5

657

QUESTION ID

S15-F1-Q5

TYPE

Short answer

CORRECT ANSWER

AB = DE or BC = EF

GRADING NOTE Any side not between the two angles (AC = DF would be ASA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

FORWARD 2

Prove congruence with ASA or AAS in overlapping triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Segments AD and BC cross at E. AB ∥ CD and E is the midpoint of AD. Prove that △ABE ≅ △DCE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q1/5

659

QUESTION ID

S15-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

△ABE ≅ △DCE by ASA.

GRADING NOTE REQUIRED WORK: Use alternate interior angles, the vertical angles at E, and AE = DE with ASA.

Any valid proof. Essential: ∠BAE ≅ ∠CDE (alternate interior, AB ∥ CD); AE = DE (midpoint); ∠AEB ≅ ∠DEC (vertical); ASA. AAS with ∠B ≅ ∠C also accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, the bisector of ∠A is perpendicular to BC at D. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q2/5

660

QUESTION ID

S15-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

AB = AC

GRADING NOTE REQUIRED WORK: Prove △ABD ≅ △ACD by ASA using the bisected angle, the shared side AD, and the right angles.

Any valid proof. Essential: ∠BAD ≅ ∠CAD (bisector); AD = AD; ∠ADB ≅ ∠ADC (right angles); ASA; corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q2/5

In overlapping triangles ABE and ACD, ∠A is shared, AB = AC, and ∠B ≅ ∠C. Which criterion proves △ABE ≅ △ACD?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q3/5

661

QUESTION ID

S15-F2-Q3

TYPE

Short answer

CORRECT ANSWER

ASA (∠A, AB = AC, ∠B ≅ ∠C)

GRADING NOTE Form required: the criterion with the matching parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q3/5

True or false?

An angle shared by two overlapping triangles cannot be used in a congruence proof, because it is the same angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q4/5

662

QUESTION ID

S15-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE By the reflexive property, the shared angle is congruent to itself and can be used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q4/5

In △JKL and △JML, ∠KJL ≅ ∠MJL and ∠JLK ≅ ∠JLM. Which side completes ASA?

A. JL = JL

B. JK = JM

C. KL = ML

D. JK = KL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q5/5

663

QUESTION ID

S15-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. JL = JL

GRADING NOTE JL is between the two pairs of angles and is shared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q5/5

Explain SSS congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Three side lengths determine a triangle (triangle inequality)	**7.G.A.2**
BACKWARD 2	Find the points at given distances from two points (two circles)	**7.G.B.4**
BACKWARD 3	Rigid motions keep every length	**8.G.A.1**
FORWARD 1	Explain the Hypotenuse-Leg criterion for right triangles	**HSG-SRT.B.5**
FORWARD 2	Decide which criteria guarantee congruence, with counterexampl…	**HSG-SRT.B.5**

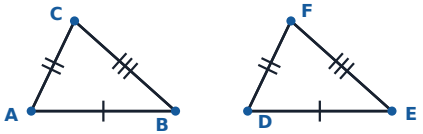
SET 16 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In △ABC and △DEF, AB = DE, BC = EF, and CA = FD. Explain, using rigid motions, why △ABC ≅ △DEF.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q1/5

666

QUESTION ID

S16-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

A translation and a rotation carry A to D and B to E; if needed, the reflection across DE carries C′ to F.

GRADING NOTE REQUIRED WORK: Move A to D and B to E (translation and rotation, using AB = DE). If the image C′ of C is not F, show that line DE is the perpendicular bisector of C′F, so the reflection across DE carries C′ to F.

Any valid explanation. Essential: (1) A → D, B → E (AB = DE); (2) DC′ = DF and EC′ = EF (rigid motions keep lengths); (3) if C′ ≠ F, D and E are each equidistant from C′ and F, so each lies on the perpendicular bisector of C′F, and line DE is that bisector; (4) the reflection across DE fixes D and E and maps C′ to F. Step 3 must not rest on SSS: justify it with SAS (the bisector of ∠C′DF meets C′F at M; △DMC′ ≅ △DMF by SAS, so MC′ = MF and the angles at M are a congruent linear pair, 90° each) or with coordinates (set 19). Invalid: using angles of the triangles (none are given); citing SSS, or the converse of the perpendicular bisector theorem proved with SSS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q1/5

In △ABC and △DEF, AB = DE, BC = EF, and CA = FD. A translation and a rotation have moved A onto D and B onto E, and the image C′ of C is on the opposite side of line DE from F, with DC′ = DF and EC′ = EF. Explain why the reflection across line DE carries C′ onto F. Justify each step without using SSS (it is what is being explained).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q2/5

667

QUESTION ID

S16-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

D and E are both on the perpendicular bisector of C′F, so line DE is that bisector, and the reflection across it maps C′ to F.

GRADING NOTE REQUIRED WORK: Show that D and E each lie on the perpendicular bisector of C′F using SAS, then use the definition of a reflection.

Any valid explanation. Essential: the bisector of ∠C′DF meets C′F at M; △DMC′ ≅ △DMF by SAS (DC′ = DF, equal angles, DM shared), so MC′ = MF and ∠DMC′ ≅ ∠DMF, a linear pair, so each is 90°: D is on the perpendicular bisector. The same for E. Two points determine the line, so DE is the perpendicular bisector, and by definition the reflection maps C′ to F. Invalid: citing SSS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△ABC has A(0, 0), B(5, 0), C(1, 3). △DEF has D(1, 1), E(1, 6), F(−2, 2). Show that the three pairs of corresponding sides are congruent, and explain, using rigid motions, why the triangles are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q3/5

668

QUESTION ID

S16-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

AB = DE = 5, BC = EF = 5, CA = FD = √10. Translate 1 right and 1 up, then rotate 90° counterclockwise about D: A → D, B → E, and C → F.

GRADING NOTE REQUIRED WORK: Compute the six lengths; then give motions that carry A to D and B to E and explain where C goes.

Any valid explanation. Essential: all six lengths; motions that match A and B; why C lands on F (the computed image (−2, 2), or the SSS argument that C′ is DF from D and EF from E on F's side).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD has AB = AD and CB = CD (a kite). Explain, using the reflection across line AC, why △ABC ≅ △ADC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q4/5

669

QUESTION ID

S16-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

A and C are each equidistant from B and D, so line AC is the perpendicular bisector of BD; the reflection across AC fixes A and C and swaps B and D.

GRADING NOTE REQUIRED WORK: Show that line AC is the perpendicular bisector of BD, then use the reflection.

Any valid explanation. Essential: equidistance of A and of C from B and D, justified without SSS (SAS with the angle bisector, or the perpendicular bisector converse proved independently); the reflection fixes A and C and sends B to D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q4/5

A triangle has sides 4, 5, and 6. Its 6-unit side is placed on segment DE (DE = 6). Explain why the third vertex can be in exactly two places, one on each side of line DE, and why the reflection across DE carries one place onto the other.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q5/5

670

QUESTION ID

S16-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The third vertex is 4 from D and 5 from E: the two circles meet in one point on each side of DE. The reflection across DE keeps distances from D and E, so it sends one intersection point to the other.

GRADING NOTE

REQUIRED WORK: Use the circles of radius 4 about D and radius 5 about E, and what a reflection across DE does to distances from D and E.

Any valid explanation. Essential: the two circles (4 + 5 > 6, so they meet); one point on each side; the reflection fixes D and E and keeps distances, so it swaps the two points. This is why SSS needs at most one reflection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 1

Three side lengths determine a triangle (triangle inequality) • Grade 7 • Geometry • 7.G.A.2

True or false?

Side lengths 3, 4, and 8 can form a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q1/5

672

QUESTION ID

S16-B1-Q1

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3 + 4 = 7 is less than 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q1/5

True or false?

Side lengths 5, 6, and 10 form exactly one triangle (up to position).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q2/5

673

QUESTION ID

S16-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 5 + 6 > 10, and three sides fix the triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q2/5

Which three lengths can form a triangle?

A. 5, 5, 11

B. 4, 5, 7

C. 2, 3, 6

D. 1, 1, 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q3/5

674

QUESTION ID

S16-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4, 5, 7

GRADING NOTE Each pair must add to more than the third side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two sides of a triangle are 6 and 10. Give the range of possible lengths for the third side.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q4/5

675

QUESTION ID

S16-B1-Q4

TYPE

Short answer

CORRECT ANSWER

Between 4 and 16

(4 < x < 16)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q4/5

Can two different-shaped triangles both have sides 7, 8, and 9? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q5/5

676

QUESTION ID

S16-B1-Q5

TYPE

Short answer

CORRECT ANSWER

No. Three side lengths determine one triangle shape and size.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 2

Find the points at given distances from two points (two circles) • Grade 7 • Geometry • 7.G.B.4

Point P is 5 units from A and 5 units from B, and A and B are 6 units apart. How many such points P are there in the plane?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q1/5

678

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. The circles of radius 5 about A and B meet twice, once on each side of AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q1/5

True or false?

In a given plane containing A, all points 3 cm from A form a circle with center A and radius 3 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q2/5

679

QUESTION ID

S16-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is the definition of a circle (in space, the points would form a sphere).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q2/5

Circles of radius 4 centered at A and B meet at two points. AB = 6. Where are the two points?

A. Both on line AB

B. One on each side of line AB

C. At A and B

D. Both on the same side of AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q3/5

680

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. One on each side of line AB

GRADING NOTE The picture is symmetric across line AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q3/5

True or false?

Two circles always meet in exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q4/5

681

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They can meet in 0, 1, or 2 points (or be the same circle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q4/5

Points A and B are 10 apart. Is there a point 3 units from A and 4 units from B? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q5/5

682

QUESTION ID

S16-B2-Q5

TYPE

Short answer

CORRECT ANSWER

No. 3 + 4 = 7 < 10, so the circles do not meet.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 3

Rigid motions keep every length • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has sides 3, 5, and 7. After a rotation and a reflection, what are the side lengths of the image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q1/5

684

QUESTION ID

S16-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3, 5, and 7

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q1/5

True or false?

Under a rigid motion, the distance from the image of A to the image of C equals AC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q2/5

685

QUESTION ID

S16-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rigid motions keep all distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which can a rigid motion change?

A. The perimeter

B. The length of a side

C. The area

D. The position of a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q3/5

686

QUESTION ID

S16-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The position of a triangle

GRADING NOTE Only the position changes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q3/5

True or false?

A reflection doubles the distance from a point to the line of reflection.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q4/5

687

QUESTION ID

S16-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The image is the same distance from the line, on the other side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q4/5

A point C is 6 units from D. After a reflection across a line through D, how far is the image of C from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q5/5

688

QUESTION ID

S16-B3-Q5

TYPE

Short answer

CORRECT ANSWER

6 units

GRADING NOTE Any equivalent form. D is on the line, so it is fixed, and distances are kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSG-CO.B.8

FORWARD 1

Explain the Hypotenuse-Leg criterion for right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangles ABC and DEF have right angles at C and F, hypotenuses AB = DE, and legs AC = DF. Prove that △ABC ≅ △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q1/5

690

QUESTION ID

S16-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

△ABC ≅ △DEF by SSS (BC = EF from the Pythagorean Theorem).

GRADING NOTE REQUIRED WORK: Use the Pythagorean Theorem to show BC = EF, then apply SSS.

Any valid proof. Essential: BC = √(AB² − AC²) = √(DE² − DF²) = EF; then SSS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q1/5

Two right triangles each have hypotenuse 13 and one leg 5. Are they congruent? What is the other leg?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q2/5

691

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes; the other leg is 12 in both.

GRADING NOTE Form required: the conclusion and the leg. $\sqrt{(169 - 25)} = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q2/5

True or false?

Hypotenuse-Leg is a special case of SSA that does guarantee congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q3/5

692

QUESTION ID
S16-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The right angle is opposite the hypotenuse; the Pythagorean Theorem fixes the third side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q3/5

Which information proves two right triangles congruent?

A. One pair of acute angles is equal

B. Their right angles are equal

C. Their hypotenuses are equal and one pair of legs is equal

D. Their hypotenuses are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q4/5

693

QUESTION ID

S16-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Their hypotenuses are equal and one pair of legs is equal

GRADING NOTE HL needs the hypotenuse and a leg.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangles PQR and XYZ (right angles at R and Z), PQ = XY = 10 and QR = YZ = 6. Find PR and XZ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q5/5

694

QUESTION ID

S16-F1-Q5

TYPE

Short answer

CORRECT ANSWER

PR = XZ = 8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

FORWARD 2

Decide which criteria guarantee congruence, with counterexamples • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Explain with a counterexample why AAA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q1/5

696

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Equilateral triangles with sides 2 and 5 have all angles 60° but are not congruent.

GRADING NOTE Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q1/5

Explain with a counterexample why SSA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q2/5

697

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

With sides 6 and 8 and a 30° angle opposite the 6, the third vertex can be in two places, giving two different triangles.

GRADING NOTE Any valid SSA counterexample (two non-congruent triangles with the same SSA data).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q2/5

Which list contains only valid triangle congruence criteria?

A. SSS, SAS, ASA, AAS

B. SSS, SSA, ASA, AAS

C. SSA, AAA, SAS, HL

D. SAS, AAA, ASA, SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q3/5

698

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. SSS, SAS, ASA, AAS

GRADING NOTE SSA and AAA are not criteria (HL is a special right-triangle case).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q3/5

True or false?

If two triangles have the same three side lengths, they have the same three angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q4/5

699

QUESTION ID

S16-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE SSS gives congruence, and corresponding angles of congruent triangles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q4/5

True or false?

Two triangles with two pairs of congruent sides and equal areas must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q5/5

700

QUESTION ID
S16-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The included angles could be θ and 180° − θ: equal areas (same sine), different third sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q5/5

Prove that vertical angles are congruent

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Angles in a linear pair add to 180°	**7.G.B.5**
BACKWARD 2	Use properties of equality as reasons	**HSA-REI.A.1**
BACKWARD 3	Identify vertical angles and linear pairs in a figure	**7.G.B.5**
FORWARD 1	Use the vertical angle theorem in multistep proofs	**HSG-CO.C.9**
FORWARD 2	Prove statements about the bisectors of angles formed by two lin…	**HSG-CO.C.9**

SET 17 • HSG-CO.C.9

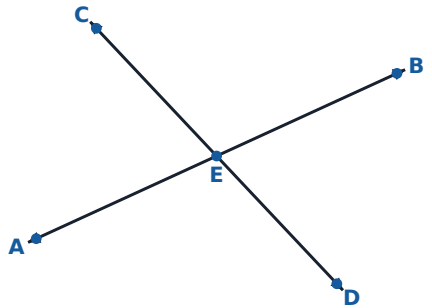
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Lines AB and CD intersect at E. Prove that ∠AEC ≅ ∠BED.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q1/5

703

QUESTION ID

S17-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

∠AEC ≅ ∠BED

GRADING NOTE REQUIRED WORK: Use the two linear pairs that share ∠CEB, and the subtraction property of equality.

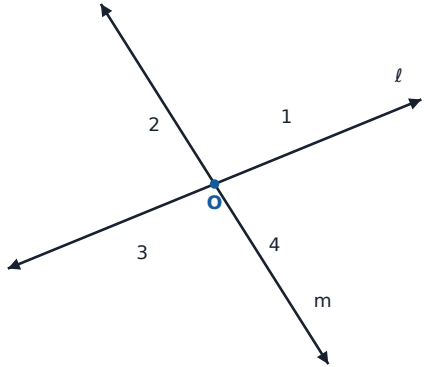
Any valid proof (paragraph, two-column, or flow). Essential: (1) m∠AEC + m∠CEB = 180° (linear pair: A, E, B are collinear); (2) m∠CEB + m∠BED = 180° (linear pair: C, E, D are collinear); (3) so m∠AEC + m∠CEB = m∠CEB + m∠BED (substitution/transitive); (4) subtract m∠CEB. Alternative: the 180° rotation about E maps ray EA to ray EB and ray EC to ray ED. Invalid: "they look equal", measuring, or citing the theorem itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q1/5

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, and $\angle 4$ in order around O. Prove that $\angle 2 \cong \angle 4$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q2/5

704

QUESTION ID

S17-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$\angle 2 \cong \angle 4$

GRADING NOTE REQUIRED WORK: Use the linear pairs $\angle 1$ and $\angle 2$, and $\angle 1$ and $\angle 4$, then the subtraction property of equality.

Any valid proof (paragraph, two-column, or flow). Essential: $m\angle 1 + m\angle 2 = 180°$ and $m\angle 1 + m\angle 4 = 180°$ (linear pairs, each with a reason); so $m\angle 2 = 180° - m\angle 1 = m\angle 4$. Using $\angle 3$ as the shared angle is also valid. Invalid: using $\angle 2 \cong \angle 4$ as a step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q2/5

Lines PQ and RS intersect at T (P, T, Q on one line and R, T, S on the other). Prove that ∠PTR ≅ ∠QTS.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q3/5

705

QUESTION ID

S17-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

∠PTR ≅ ∠QTS

GRADING NOTE REQUIRED WORK: Use the two linear pairs that share ∠RTQ, and the properties of equality.

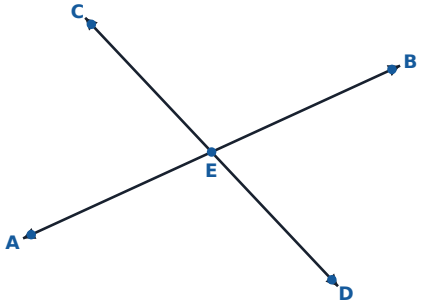
Any valid proof (paragraph, two-column, or flow). Essential: m∠PTR + m∠RTQ = 180° (linear pair: P, T, Q collinear); m∠RTQ + m∠QTS = 180° (linear pair: R, T, S collinear); substitution and subtraction give m∠PTR = m∠QTS. Invalid: citing the vertical angle theorem itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q3/5

Lines AB and CD intersect at E. Explain, using a rigid motion, why ∠AEC ≅ ∠BED.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q4/5

706

QUESTION ID

S17-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The 180° rotation about E maps ∠AEC onto ∠BED, so the angles are congruent.

GRADING NOTE

REQUIRED WORK: Use the 180° rotation about E and explain where it sends rays EA and EC.

Any valid explanation. Essential: a 180° rotation about E maps each line through E onto itself, sending ray EA to the opposite ray EB and ray EC to ray ED; so it maps ∠AEC onto ∠BED; rotations keep angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q4/5

Two lines intersect, forming vertical angles ∠1 and ∠3, and ∠2 is adjacent to both. Suppose m∠1 = x°. Prove that m∠3 = x° for every value of x, giving a reason for each step.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q5/5

707

QUESTION ID

S17-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

m∠3 = x°

GRADING NOTE REQUIRED WORK: Write m∠2 in terms of x using one linear pair, then m∠3 using the other.

Any valid proof (paragraph, two-column, or flow). Essential: m∠2 = 180° − x° (linear pair with ∠1); m∠3 = 180° − m∠2 = 180° − (180° − x°) = x° (linear pair with ∠2); the argument uses no particular value, so it holds for all vertical angles. Invalid: checking one value of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 1

Angles in a linear pair add to 180° • Grade 7 • Geometry • 7.G.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles form a linear pair. One measures 128°. Find the other.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q1/5

709

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

52°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q1/5

∠PQR and ∠RQS form a linear pair, and m∠PQR = 3 · m∠RQS. Find m∠RQS.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q2/5

710

QUESTION ID
S17-B1-Q2

TYPE
Short answer

CORRECT ANSWER
45°

GRADING NOTE Any equivalent form. 4x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q2/5

True or false?

The two angles of a linear pair are always supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q3/5

711

QUESTION ID

S17-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Together they form a straight angle, 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q3/5

Points A, B, C are on a line with B between A and C, and ray BD is drawn. Which angles form a linear pair?

A. ∠DBC and ∠ABC

B. ∠BAD and ∠BCD

C. ∠ABD and ∠DBC

D. ∠ABD and ∠ABC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q4/5

712

QUESTION ID

S17-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. ∠ABD and ∠DBC

GRADING NOTE They share ray BD and their other sides are opposite rays BA and BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q4/5

True or false?

Two angles that add to 180° must form a linear pair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q5/5

713

QUESTION ID

S17-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Supplementary angles need not be adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 2

Use properties of equality as reasons • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

From m∠1 + 40° = m∠2 + 40°, which property gives m∠1 = m∠2?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q1/5

715

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Subtraction property of equality

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q1/5

If x + y = 180 and y + z = 180, which property lets you write x + y = y + z?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q2/5

716

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Substitution (or the transitive property of equality)

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q2/5

Which property justifies: if a = b, then b = a?

A. Reflexive property

B. Symmetric property

C. Distributive property

D. Transitive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q3/5

717

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Symmetric property

GRADING NOTE Swapping the sides is the symmetric property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q3/5

True or false?

The reflexive property says that any quantity equals itself, such as m∠A = m∠A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q4/5

718

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE a = a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q4/5

True or false?

From 2x = 2y you may conclude x = y by the subtraction property of equality.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q5/5

719

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That step divides by 2 (division property).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 3

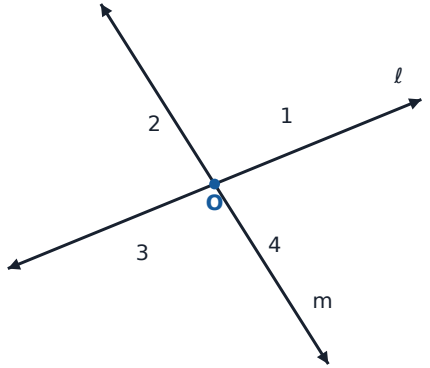
Identify vertical angles and linear pairs in a figure • Grade 7 • Geometry • 7.G.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Lines ℓ and m meet at O, forming ∠1, ∠2, ∠3, ∠4 in order. Which angle is vertical to ∠1?

A. ∠3 **B.** ∠1

C. ∠4 **D.** ∠2



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q1/5

721

QUESTION ID
S17-B3-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. ∠3

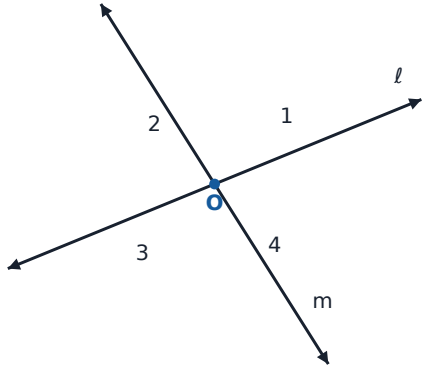
GRADING NOTE Vertical angles are opposite each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Lines ℓ and m meet at O, forming ∠1, ∠2, ∠3, ∠4 in order. Name the two angles that form linear pairs with ∠2.



Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q2/5

722

QUESTION ID

S17-B3-Q2

TYPE

Short answer

CORRECT ANSWER

∠1 and ∠3

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q2/5

True or false?

Vertical angles share a side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q3/5

723

QUESTION ID

S17-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They share only the vertex; their sides are opposite rays.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q3/5

True or false?

When two lines cross, they form two pairs of vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q4/5

724

QUESTION ID

S17-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ∠1 with ∠3, and ∠2 with ∠4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q4/5

Two lines cross. One angle is 63°. Find its vertical angle and the angles next to it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q5/5

725

QUESTION ID

S17-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Vertical: 63°; adjacent: 117° and 117°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • HSG-CO.C.9

FORWARD 1

Use the vertical angle theorem in multistep proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Lines AB and CD intersect at E, and ray EF bisects ∠AEC. Ray EG is the ray opposite ray EF. Prove that ray EG bisects ∠BED.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

727

QUESTION ID

S17-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠BEG = m∠GED, so EG bisects ∠BED.

GRADING NOTE REQUIRED WORK: Combine the bisector definition with vertical angles (∠AEF and ∠BEG; ∠FEC and ∠GED).

Any valid proof (paragraph, two-column, or flow). Essential: m∠AEF = m∠FEC (bisector); ∠AEF ≅ ∠BEG and ∠FEC ≅ ∠DEG (vertical angles, since EG is opposite EF, EB opposite EA, ED opposite EC); so m∠BEG = m∠GED.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

Three lines meet at point O, forming six angles. Two of the angles that are next to each other are 50° and 70°. Prove that the six angles are 50°, 70°, 60°, 50°, 70°, 60° in order.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

728

QUESTION ID

S17-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

The angles in order are 50°, 70°, 60°, 50°, 70°, 60°.

GRADING NOTE REQUIRED WORK: Use the straight angle (180°) to find the third angle on one side, then vertical angles for the other three.

Any valid proof (paragraph, two-column, or flow). Essential: three adjacent angles on one side of a line add to 180°, so the third is 60°; each of the other three is vertical to one of these.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

Two lines meet. One angle is (4x + 10)° and its vertical angle is (6x − 30)°. Find x and all four angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

729

QUESTION ID

S17-F1-Q3

TYPE

Short answer

CORRECT ANSWER

x = 20; angles 90°, 90°, 90°, 90°

GRADING NOTE Form required: x and the four angles. 4x + 10 = 6x − 30 gives x = 20, so the angle is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

True or false?

If one angle formed by two crossing lines is a right angle, all four angles are right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

730

QUESTION ID

S17-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its vertical angle is 90° and its neighbors are 180° − 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

Lines AB and CD meet at E with m∠AEC = 2x and m∠CEB = 3x. What is m∠BED?

A. 72°

B. 144°

C. 36°

D. 108°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

731

QUESTION ID
S17-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. 72°

GRADING NOTE 2x + 3x = 180, so x = 36 and m∠AEC = 72°; ∠BED is vertical to ∠AEC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

FORWARD 2

Prove statements about the bisectors of angles formed by two lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

∠ABD and ∠DBC form a linear pair. Ray BE bisects ∠ABD and ray BF bisects ∠DBC. Prove that ∠EBF is a right angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q1/5

733

QUESTION ID

S17-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠EBF = 90°

GRADING NOTE REQUIRED WORK: Express m∠EBF as half of m∠ABD plus half of m∠DBC.

Any valid proof (paragraph, two-column, or flow). Essential: m∠EBD = ½m∠ABD, m∠DBF = ½m∠DBC (bisectors); m∠EBF = ½(m∠ABD + m∠DBC) = ½(180°) = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q1/5

Is the converse of the vertical angle theorem true: "If two angles are congruent, they are vertical angles"? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q2/5

734

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No. Two 30° angles in different places in a figure are congruent but not vertical angles.

GRADING NOTE Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q2/5

Two lines meet at O. Prove that the bisectors of a pair of vertical angles lie on one line.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q3/5

735

QUESTION ID

S17-F2-Q3

TYPE

Answer with required work

CORRECT ANSWER

The bisector rays are opposite rays, so they lie on one line.

GRADING NOTE REQUIRED WORK: Show that the angle between the two bisector rays is 180°.

Any valid proof (paragraph, two-column, or flow). Essential: if the vertical angles each measure 2a and the other two each 180° − 2a, the angle from one bisector to the other is a + (180° − 2a) + a = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q3/5

True or false?

The bisectors of a pair of vertical angles are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q4/5

736

QUESTION ID

S17-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They form one straight line: each bisector is the extension of the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two lines meet, and the bisectors of two adjacent angles are drawn. What angle do the bisectors form?

A. 45°

B. It depends on the lines

C. 180°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q5/5

737

QUESTION ID

S17-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 90°

GRADING NOTE Adjacent angles formed by two lines are a linear pair, so their bisectors meet at 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q5/5

SET 18 • HSG-CO.C.9 • CONGRUENCE: PROVING THEOREMS

Prove the angle theorems for parallel lines cut by a transversal

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Name and use angles formed by parallel lines and a transversal	**8.G.A.5**
BACKWARD 2	Vertical angles and linear pairs as reasons	**HSG-CO.C.9**
BACKWARD 3	The parallel postulate and the definition of parallel lines	**HSG-CO.A.1**
FORWARD 1	Prove the converse: congruent alternate interior angles make line...	**HSG-CO.C.9**
FORWARD 2	Use parallel-line theorems to prove the triangle angle sum	**HSG-CO.C.10**

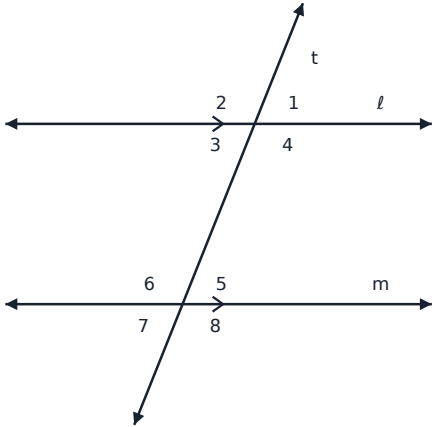
SET 18 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are parallel and are cut by transversal t, crossing ℓ at P and m at Q. Explain, using a translation, why corresponding angles $\angle 1$ and $\angle 5$ are congruent. (This argument uses only the parallel postulate and rigid motions, so the later items may use its result.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q1/5

740

QUESTION ID

S18-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

The translation that moves Q to P maps m onto ℓ and maps t onto itself, so it maps $\angle 5$ onto $\angle 1$; translations keep angle measure.

GRADING NOTE REQUIRED WORK: Translate along the transversal from Q to P. Explain why the image of m is ℓ and why the image of $\angle 5$ is $\angle 1$.

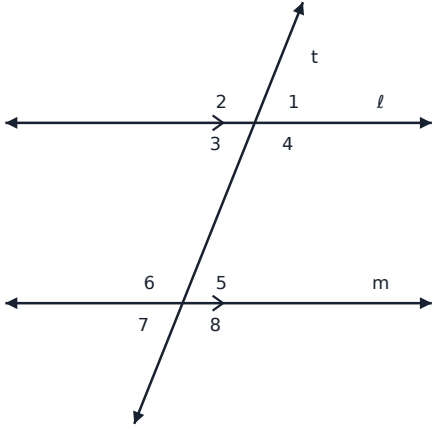
Any valid explanation. Essential: (1) the translation maps t onto itself (it moves along t); (2) it maps m to a line through P parallel to m; (3) by the parallel postulate that line is ℓ; (4) so the angle formed by t and m at Q maps to the corresponding angle at P; (5) angle measure is kept. Invalid: assuming corresponding angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q1/5

Lines ℓ and m are parallel and are cut by transversal t. Use the corresponding angles theorem (proved with a translation) to prove that alternate interior angles $\angle 3$ and $\angle 5$ are congruent.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q2/5

741

QUESTION ID

S18-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$\angle 3 \cong \angle 5$

GRADING NOTE REQUIRED WORK: Use the corresponding angle of $\angle 5$ at the top crossing and vertical angles.

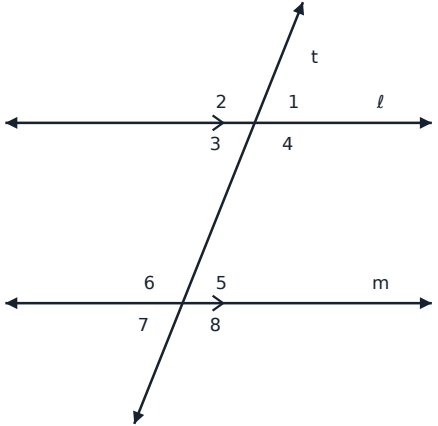
Any valid proof (paragraph, two-column, or flow). Essential: $\angle 1 \cong \angle 5$ (corresponding angles theorem, $\ell \parallel$ m); $\angle 1 \cong \angle 3$ (vertical angles); so $\angle 3 \cong \angle 5$ (transitive). Invalid: "they look equal"; assuming alternate interior angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q2/5

Lines ℓ and m are parallel and are cut by transversal t. Use the corresponding angles theorem to prove that alternate exterior angles $\angle 2$ and $\angle 8$ are congruent.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q3/5

742

QUESTION ID

S18-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$\angle 2 \cong \angle 8$

GRADING NOTE REQUIRED WORK: Use a corresponding angle and vertical angles.

Any valid proof (paragraph, two-column, or flow). Essential: $\angle 2 \cong \angle 6$ (corresponding angles, $\ell \parallel$ m); $\angle 6 \cong \angle 8$ (vertical angles); so $\angle 2 \cong \angle 8$ (transitive).

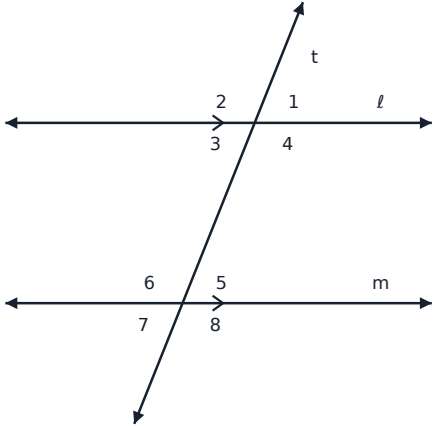
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Lines ℓ and m are parallel and are cut by transversal t. Prove that same-side interior angles $\angle 4$ and $\angle 5$ are supplementary.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q4/5

743

QUESTION ID

S18-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

$m\angle 4 + m\angle 5 = 180°$

GRADING NOTE REQUIRED WORK: Use a corresponding angle and a linear pair.

Any valid proof (paragraph, two-column, or flow). Essential: $\angle 4 \cong \angle 8$ (corresponding angles, $\ell \parallel$ m); $m\angle 8 + m\angle 5 = 180°$ (linear pair at Q); substitute: $m\angle 4 + m\angle 5 = 180°$.

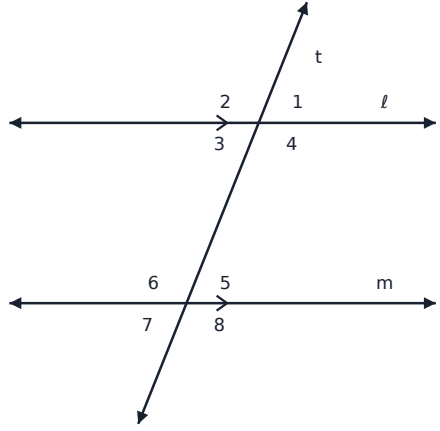
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Lines ℓ and m are parallel and are cut by transversal t at P (on ℓ) and Q (on m). Let M be the midpoint of PQ. Explain, using the 180° rotation about M, why alternate interior angles $\angle 3$ and $\angle 5$ are congruent.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q5/5

744

QUESTION ID

S18-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The rotation swaps P and Q, maps t onto itself, and maps ℓ to the line through Q parallel to ℓ, which is m; so $\angle 3$ maps onto $\angle 5$.

GRADING NOTE REQUIRED WORK: Find the images of P, Q, t, and ℓ under the rotation, then the image of $\angle 3$.

Any valid explanation. Essential: MP = MQ, so P ↔ Q; t passes through M, so it maps onto itself (each ray reversed); a 180° rotation maps ℓ to a parallel line through Q, which is m (parallel postulate); $\angle 3$ (between ray PQ and ℓ on one side) goes to the angle between ray QP and m on the other side, $\angle 5$; rotations keep angle measure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

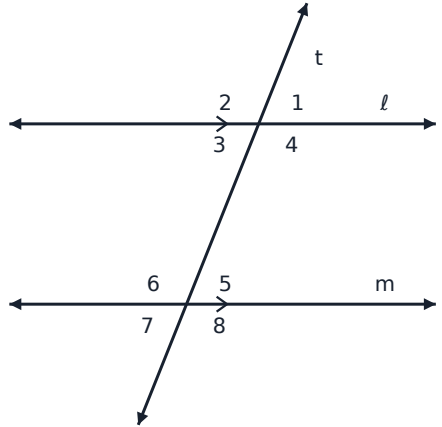
High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 1

Name and use angles formed by parallel lines and a transversal • Grade 8 • Geometry • 8.G.A.5

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ measures $70°$. Find the measure of $\angle 6$.



Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q1/5

746

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

70°

GRADING NOTE Any equivalent form. $\angle 4$ and $\angle 6$ are alternate interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q1/5

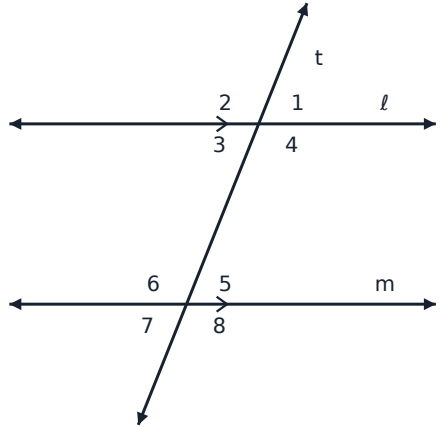
Lines $\ell \parallel m$ are cut by transversal t. Which angle corresponds to $\angle 2$?

A. $\angle 7$

B. $\angle 8$

C. $\angle 6$

D. $\angle 5$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q2/5

747

QUESTION ID
S18-B1-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. $\angle 6$

GRADING NOTE Both are in the upper-left position at their crossings.

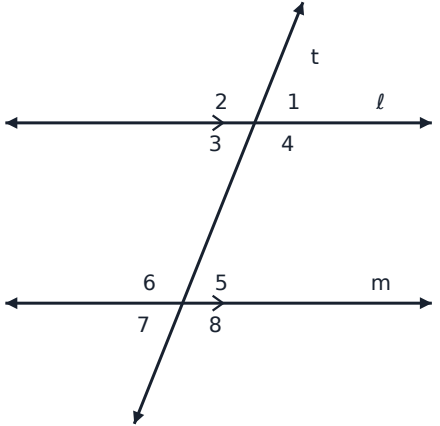
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q2/5

True or false?

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ and $\angle 5$ are same-side interior angles, so they add to 180°.

A. True **B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q3/5

748

QUESTION ID
S18-B1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE They are on the same side of t, between the lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q3/5

True or false?

When two lines are cut by a transversal, corresponding angles are congruent even if the lines are not parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q4/5

749

QUESTION ID

S18-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

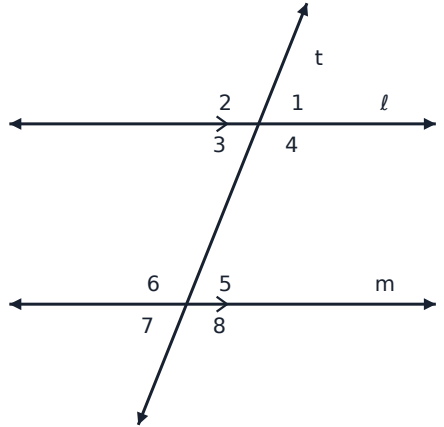
GRADING NOTE They are congruent only when the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Lines $\ell \parallel m$ are cut by transversal t. $\angle 8$ measures 112°. Find the measure of $\angle 2$.



Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q5/5

750

QUESTION ID
S18-B1-Q5

TYPE
Short answer

CORRECT ANSWER
112°

GRADING NOTE Any equivalent form. $\angle 8$ and $\angle 2$ are alternate exterior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 2

Vertical angles and linear pairs as reasons • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Two lines cross. Name the reason two opposite angles are congruent.

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q1/5

752

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Vertical angles are congruent

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q1/5

Name the reason two adjacent angles on a line add to 180°.

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q2/5

753

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

They form a linear pair (supplementary)

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q2/5

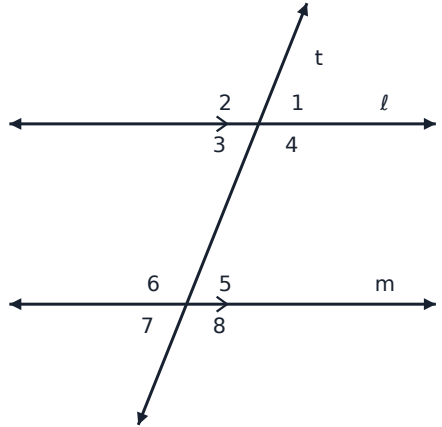
At the top crossing, ∠1 = 110°. What is ∠2 and why?

A. 110°, linear pair

B. 70°, linear pair

C. 110°, vertical angles

D. 70°, vertical angles



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q3/5

754

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 70°, linear pair

GRADING NOTE ∠1 and ∠2 are next to each other on line ℓ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q3/5

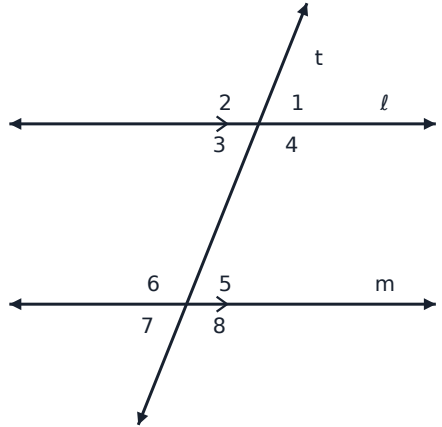
ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Angles 1 and 3 are vertical angles in the figure.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q4/5

755

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They are opposite each other at the top crossing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

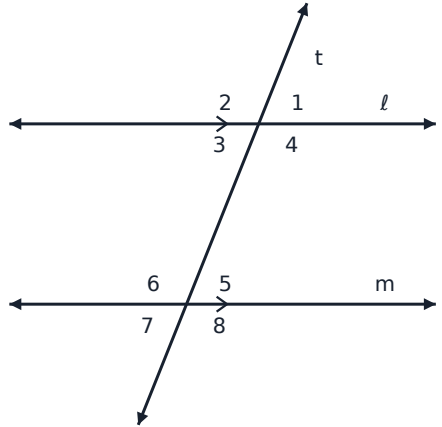
High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q4/5

True or false?

Angles 1 and 4 are vertical angles in the figure.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q5/5

756

QUESTION ID
S18-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They are adjacent: a linear pair along t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 3

The parallel postulate and the definition of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Through a point not on a line, there is exactly one line parallel to it (in a plane).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q1/5

758

QUESTION ID

S18-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The parallel postulate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ is parallel to line m. Line n passes through a point of ℓ and is also parallel to m. What can you conclude about n?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q2/5

759

QUESTION ID

S18-B3-Q2

TYPE

Short answer

CORRECT ANSWER

n is the same line as ℓ.

GRADING NOTE Any equivalent answer: by the parallel postulate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q2/5

Which pair of lines in a plane are parallel?

A. x = 2 and y = 2

B. y = x and y = −x

C. y = 2x + 1 and y = −2x + 1

D. y = 2x + 1 and y = 2x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q3/5

760

QUESTION ID

S18-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 2x + 1 and y = 2x − 5

GRADING NOTE Same slope, different intercepts: they never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q3/5

True or false?

Two lines in a plane that never meet are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q4/5

761

QUESTION ID
S18-B3-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition of parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q4/5

Line $\ell \parallel$ line m. A translation maps m to a line through a point of ℓ. Must that image line be parallel to m or the same as m?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q5/5

762

QUESTION ID

S18-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Parallel to m (a translation maps a line to a parallel line or itself); so it is ℓ.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q5/5

SET 18 • HSG-CO.C.9

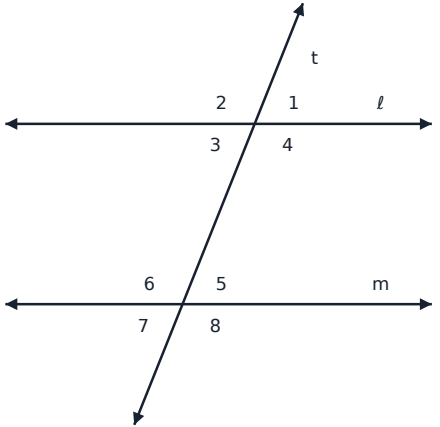
FORWARD 1

Prove the converse: congruent alternate interior angles make lines parallel • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Lines ℓ and m are cut by transversal t, and alternate interior angles $\angle 3$ and $\angle 5$ are congruent. Prove that $\ell \parallel$ m.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q1/5

764

QUESTION ID

S18-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

$\ell \parallel$ m

GRADING NOTE

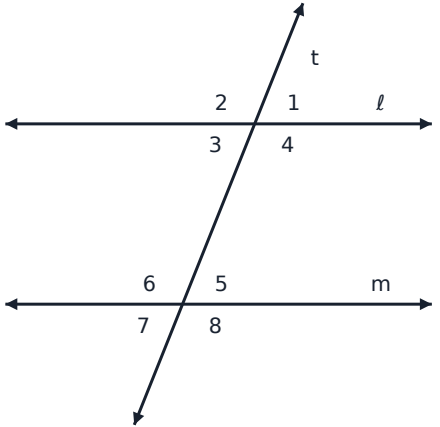
REQUIRED WORK: Indirect proof: suppose ℓ and m meet; they would form a triangle with t whose angles contradict a known result. Or use the 180° rotation about the midpoint of PQ.

Any valid proof (paragraph, two-column, or flow). Essential (indirect): if ℓ and m met at a point R, triangle PQR would have an exterior angle equal to a remote interior angle, which is impossible (an exterior angle is greater than each remote interior angle); so they do not meet. Alternative: the 180° rotation about the midpoint of PQ swaps P and Q and maps ℓ onto m, so if they met on one side they would also meet on the other, giving two common points. Invalid: assuming $\ell \parallel$ m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q1/5

Lines ℓ and m are cut by transversal t with $m\angle 4 = (5x - 20)°$ and $m\angle 6 = (3x + 30)°$. For what value of x is $\ell \parallel$ m? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q2/5

765

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 25, because then the alternate interior angles are congruent (105°).

GRADING NOTE Form required: x with the converse as the reason. $5x - 20 = 3x + 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q2/5

True or false?

If same-side interior angles are supplementary, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q3/5

766

QUESTION ID

S18-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then the alternate interior angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q3/5

Which fact proves that lines ℓ and m are parallel?

A. A pair of corresponding angles is congruent

B. A pair of vertical angles is congruent

C. A linear pair adds to 180°

D. Two angles at the same crossing are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q4/5

767

QUESTION ID

S18-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

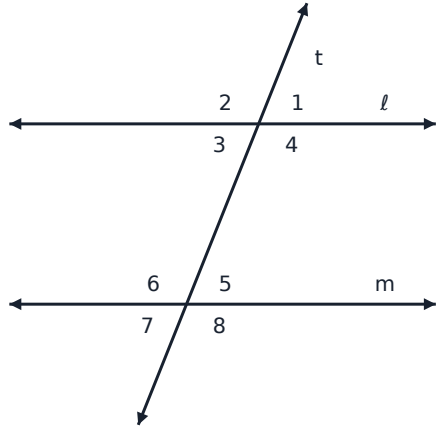
A. A pair of corresponding angles is congruent

GRADING NOTE Vertical angles and linear pairs hold for any crossing lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q4/5

Lines ℓ and m are cut by a transversal. $m\angle 1 = 72°$ and $m\angle 7 = 108°$. Are ℓ and m parallel? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q5/5

768

QUESTION ID

S18-F1-Q5

TYPE

Short answer

CORRECT ANSWER

No. ∠1 and ∠7 are alternate exterior angles; if $\ell \parallel$ m they would be congruent, but 72° ≠ 108°.

GRADING NOTE Form required: the conclusion with a reason. Equivalently, ∠5 is vertical to ∠7, so $m\angle 5 = 108°$, but its corresponding angle ∠1 is 72°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSG-CO.C.9

FORWARD 2

Use parallel-line theorems to prove the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that the angles of triangle ABC add to 180°. Begin by drawing the line through B parallel to AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q1/5

770

QUESTION ID

S18-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠A + m∠B + m∠C = 180°

GRADING NOTE REQUIRED WORK: Name the two new angles at B, use alternate interior angles, and the straight angle at B.

Any valid proof (paragraph, two-column, or flow). Essential: the line through B parallel to AC exists (parallel postulate); the angles it makes with BA and BC are congruent to ∠A and ∠C (alternate interior angles); those two angles and ∠ABC form a straight angle, 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q1/5

In triangle ABC, a line through B parallel to AC makes a 50° angle with BA (on A's side) and a 65° angle with BC (on C's side). Find all three angles of the triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q2/5

771

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

∠A = 50°, ∠C = 65°, ∠B = 65°

GRADING NOTE Form required: all three. Alternate interior angles give ∠A and ∠C; 180 − 50 − 65 = 65.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q2/5

True or false?

The proof of the triangle angle sum works without any fact about parallel lines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q3/5

772

QUESTION ID

S18-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It uses the parallel through one vertex and the alternate interior angle theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q3/5

Which theorem is used to show the new angles at B equal ∠A and ∠C?

A. Alternate interior angles theorem

B. Isosceles triangle theorem

C. Triangle inequality

D. Vertical angles theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q4/5

773

QUESTION ID

S18-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Alternate interior angles theorem

GRADING NOTE Sides BA and BC are transversals of the parallel lines (AC and the line through B).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q4/5

Explain why the exterior angle at C of triangle ABC equals m∠A + m∠B.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q5/5

774

QUESTION ID

S18-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The exterior angle forms a linear pair with ∠C, so it is 180° − m∠C, which equals m∠A + m∠B by the angle sum.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q5/5

SET 19 • HSG-CO.C.9 • CONGRUENCE: PROVING THEOREMS

Prove the perpendicular bisector theorem and its converse

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Use the definitions of midpoint and perpendicular bisector	**HSG-CO.A.1**
BACKWARD 2	Use SAS and SSS to prove triangles congruent	**HSG-CO.B.8**
BACKWARD 3	Corresponding parts of congruent triangles and right angles from…	**HSG-CO.B.7**
FORWARD 1	Use the perpendicular bisector to find points equidistant from two…	**HSG-GPE.B.5**
FORWARD 2	Prove that the perpendicular bisectors of a triangle meet at one p…	**HSG-C.A.3**

SET 19 • HSG-CO.C.9

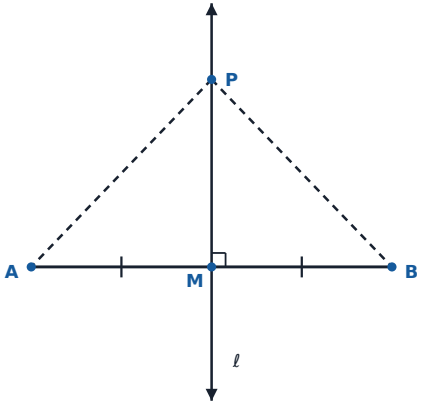
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ is the perpendicular bisector of segment AB, meeting AB at M. Point P is on ℓ. Prove that PA = PB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q1/5

777

QUESTION ID

S19-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

PA = PB

GRADING NOTE REQUIRED WORK: Use SAS on triangles PMA and PMB.

Any valid proof (paragraph, two-column, or flow). Essential: AM = BM (M is the midpoint); ∠PMA ≅ ∠PMB (both right angles, $\ell \perp$ AB); PM = PM (reflexive); △PMA ≅ △PMB (SAS); PA = PB (corresponding parts). If P = M, PA = MA = MB = PB directly. Alternative: the reflection across ℓ fixes P and swaps A and B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Point P is not on line AB, and PA = PB. Let M be the midpoint of AB. Prove that line PM is perpendicular to AB, so P is on the perpendicular bisector of AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q2/5

778

QUESTION ID

S19-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

PM ⊥ AB, so P is on the perpendicular bisector of AB.

GRADING NOTE REQUIRED WORK: Use SSS on triangles PMA and PMB, then the linear pair of congruent angles at M.

Any valid proof (paragraph, two-column, or flow). Essential: PA = PB (given), AM = BM (midpoint), PM = PM; △PMA ≅ △PMB (SSS); ∠PMA ≅ ∠PMB; they form a linear pair, so each is 90°; so PM is perpendicular to AB through its midpoint. SSS may be used here: in set 16 it was explained with SAS and a reflection, not with this converse, so the reasoning is not circular. Invalid: drawing the perpendicular bisector through P and assuming it passes through P (that is the claim).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Point P is not on line AB, and PA = PB. The bisector of ∠APB meets AB at M. Prove that line PM is the perpendicular bisector of AB (a second proof of the converse, using SAS).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q3/5

779

QUESTION ID

S19-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

AM = BM and PM ⊥ AB, so PM is the perpendicular bisector of AB.

GRADING NOTE REQUIRED WORK: Prove △PMA ≅ △PMB by SAS, then use corresponding parts.

Any valid proof (paragraph, two-column, or flow). Essential: PA = PB (given); ∠APM ≅ ∠BPM (bisector); PM = PM; SAS; AM = BM and ∠PMA ≅ ∠PMB; these angles form a linear pair, so each is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q3/5

In a coordinate plane, A(−a, 0) and B(a, 0) with a > 0. Prove that every point P(0, y) on the y-axis is equidistant from A and B, and that every point P(x, y) with PA = PB has x = 0.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q4/5

780

QUESTION ID

S19-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

PA = PB = √(a² + y²) for P(0, y); and PA = PB gives (x + a)² = (x − a)², so x = 0.

GRADING NOTE

REQUIRED WORK: Use the distance formula in both directions.

Any valid proof (paragraph, two-column, or flow). Essential: both directions. (1) √(a² + y²) both ways. (2) (x + a)² + y² = (x − a)² + y² simplifies to 4ax = 0, and a ≠ 0, so x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q4/5

A(1, 1) and B(5, 3). Prove that every point on the line $y = -2x + 8$ is equidistant from A and B, and that every point equidistant from A and B lies on $y = -2x + 8$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q5/5

781

QUESTION ID

S19-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

For P(x, −2x + 8), $PA^2 = PB^2 = 5x^2 - 30x + 50$; and PA = PB simplifies to $8x + 4y = 32$, that is, $y = -2x + 8$.

GRADING NOTE REQUIRED WORK: Use the distance formula (squared) in both directions.

Any valid proof (paragraph, two-column, or flow). Essential: both directions with algebra. (1) $PA^2 = (x - 1)^2 + (-2x + 7)^2$ and $PB^2 = (x - 5)^2 + (-2x + 5)^2$, both $5x^2 - 30x + 50$. (2) $(x - 1)^2 + (y - 1)^2 = (x - 5)^2 + (y - 3)^2$ gives $8x + 4y = 32$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 1

Use the definitions of midpoint and perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

M is the midpoint of AB and AB = 14. Find AM.

AM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q1/5

783

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector of AB passes through the midpoint of AB and is perpendicular to AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q2/5

784

QUESTION ID

S19-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is its definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q2/5

Line ℓ meets AB at M with AM = MB, but ℓ is not perpendicular to AB. What is ℓ?

A. Not related to AB

B. Parallel to AB

C. The perpendicular bisector of AB

D. A bisector of AB that is not the perpendicular bisector

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q3/5

785

QUESTION ID

S19-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A bisector of AB that is not the perpendicular bisector

GRADING NOTE It bisects the segment but is not perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q3/5

Line ℓ is perpendicular to AB at M, and AM = 3, MB = 5. Is ℓ the perpendicular bisector of AB? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q4/5

786

QUESTION ID

S19-B1-Q4

TYPE

Short answer

CORRECT ANSWER

No. M is not the midpoint (3 ≠ 5).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q4/5

True or false?

A segment has exactly one perpendicular bisector in a plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q5/5

787

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Exactly one line is perpendicular to AB at its midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 2

Use SAS and SSS to prove triangles congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In △PMA and △PMB, PM is shared, AM = BM, and ∠PMA ≅ ∠PMB. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q1/5

789

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

SAS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △PMA and △PMB, PA = PB, AM = BM, and PM is shared. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q2/5

790

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

SSS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q2/5

Two triangles share a side and have two other pairs of congruent sides. Which criterion applies?

A. SAS

B. ASA

C. SSS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q3/5

791

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. SSS

GRADING NOTE Three pairs of congruent sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

A side shared by two triangles counts as a pair of congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q4/5

792

QUESTION ID

S19-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

SSA is a valid congruence criterion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q5/5

793

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two sides and a non-included angle do not always determine a triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 3

Corresponding parts of congruent triangles and right angles from a linear pair • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

ANSWER KEY Reduced copy of the question page, with its answer

△PMA ≅ △PMB. Which angle corresponds to ∠PMA?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q1/5

795

QUESTION ID

S19-B3-Q1

TYPE

Short answer

CORRECT ANSWER

∠PMB

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q1/5

Two congruent angles form a linear pair. What is each angle's measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q2/5

796

QUESTION ID

S19-B3-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q2/5

True or false?

If $\triangle PMA \cong \triangle PMB$, then $PA = PB$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q3/5

797

QUESTION ID

S19-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q3/5

△XYZ ≅ △XWZ. Which pair of segments must be congruent?

A. YZ and XW

B. YZ and WZ

C. XY and XZ

D. XZ and YW

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q4/5

798

QUESTION ID

S19-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. YZ and WZ

GRADING NOTE Y ↔ W and Z ↔ Z.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q4/5

True or false?

Two angles in a linear pair are always congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q5/5

799

QUESTION ID

S19-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are supplementary; they are congruent only when both are 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • HSG-CO.C.9

FORWARD 1

Use the perpendicular bisector to find points equidistant from two points • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Find an equation of the perpendicular bisector of the segment from A(2, 1) to B(6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q1/5

801

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$y = -x + 7$

GRADING NOTE Any equivalent equation. Midpoint (4, 3); slope of AB is 1, so the bisector has slope −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q1/5

Find the point on the x-axis that is equidistant from A(2, 1) and B(6, 5).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q2/5

802

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(7, 0)

GRADING NOTE Form required: an ordered pair. It is where y = −x + 7 meets y = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q2/5

True or false?

The point (4, 4) is equidistant from (0, 0) and (8, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q3/5

803

QUESTION ID

S19-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Distances are $\sqrt{32} \approx 5.66$ and $\sqrt{20} \approx 4.47$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q3/5

Which point is equidistant from A(0, 0) and B(4, 0)?

A. (3, 1)

B. (4, 4)

C. (2, 7)

D. (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q4/5

804

QUESTION ID

S19-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 7)

GRADING NOTE Points on x = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q4/5

Points A(−1, 2) and B(3, 2). Describe all points equidistant from A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q5/5

805

QUESTION ID

S19-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The vertical line x = 1.

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

FORWARD 2

Prove that the perpendicular bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

Prove that the perpendicular bisectors of the sides of triangle ABC meet at one point O, and that O is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q1/5

807

QUESTION ID

S19-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

The three perpendicular bisectors meet at O, and OA = OB = OC.

GRADING NOTE REQUIRED WORK: Let the bisectors of AB and BC meet at O; use the theorem and its converse.

Any valid proof (paragraph, two-column, or flow). Essential: the bisectors of AB and BC are not parallel (AB and BC are not parallel), so they meet at O; OA = OB and OB = OC (theorem); so OA = OC; by the converse O is on the bisector of AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q1/5

Triangle ABC has A(0, 0), B(6, 0), C(0, 8). Find the point equidistant from all three vertices.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q2/5

808

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 4)

GRADING NOTE Form required: an ordered pair. Bisectors x = 3 and y = 4 meet at (3, 4); it is 5 from each vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q2/5

True or false?

The point where the perpendicular bisectors of a triangle meet is always inside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q3/5

809

QUESTION ID

S19-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For an obtuse triangle it is outside; for a right triangle it is on the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q3/5

Three towns that are not on one straight line will share a well equally distant from all three. Where should it go?

A. Where the perpendicular bisectors of the triangle's sides meet

B. Where the angle bisectors meet

C. At the midpoint of the longest side

D. Where the medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q4/5

QUESTION ID

S19-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Where the perpendicular bisectors of the triangle's sides meet

GRADING NOTE Points equidistant from two towns are on the perpendicular bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q4/5

Explain why showing that the perpendicular bisectors of AB and BC meet at O is enough to show that all three meet at O.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q5/5

811

QUESTION ID

S19-F2-Q5

TYPE

Short answer

CORRECT ANSWER

OA = OB and OB = OC give OA = OC, and by the converse O is on the perpendicular bisector of AC.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q5/5

Prove that the angles of a triangle add to 180°

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Use the angle sum and parallel-line angles informally	**8.G.A.5**
BACKWARD 2	Angles that form a straight angle add to 180°	**4.MD.C.7**
BACKWARD 3	Alternate interior angles of parallel lines are congruent	**HSG-CO.C.9**
FORWARD 1	Prove the exterior angle theorem	**HSG-CO.C.10**
FORWARD 2	Generalize the angle sum to polygons	**HSG-CO.C.10**

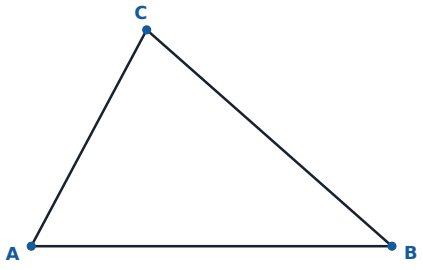
SET 20 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that
m∠A + m∠B + m∠C = 180° in triangle ABC. Draw the line through C parallel to AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q1/5

814

QUESTION ID

S20-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠A + m∠B + m∠C = 180°

GRADING NOTE REQUIRED WORK: Name the two new angles at C, use alternate interior angles with AB, and the straight angle at C.

Any valid proof (paragraph, two-column, or flow). Essential: (1) through C there is exactly one line parallel to AB (parallel postulate); (2) the angles it makes with CA and CB are congruent to ∠A and ∠B (alternate interior angles, transversals CA and CB); (3) those two angles and ∠ACB form a straight angle, 180°; (4) substitute. Invalid: measuring angles; citing the exterior angle theorem, because in this course that theorem is proved FROM the angle sum (set 20, FORWARD 1), so using it here would be circular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle PQR, draw the line through P parallel to QR. Prove that the angles of triangle PQR add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q2/5

815

QUESTION ID

S20-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

m∠P + m∠Q + m∠R = 180°

GRADING NOTE REQUIRED WORK: Same strategy with vertex P: alternate interior angles and a straight angle at P.

Any valid proof (paragraph, two-column, or flow). Essential: unique parallel through P; the two angles at P next to ∠QPR are congruent to ∠Q and ∠R (alternate interior); the three angles at P make 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q2/5

In triangle ABC, extend side BC past C to point D, and draw ray CE parallel to BA (E on the same side of line BD as A). Prove that the angles of triangle ABC add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q3/5

816

QUESTION ID

S20-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

m∠A + m∠B + m∠ACB = 180°

GRADING NOTE REQUIRED WORK: Use alternate interior angles (∠A and ∠ACE) and corresponding angles (∠B and ∠ECD), then the straight angle BCD.

Any valid proof (paragraph, two-column, or flow). Essential: ∠ACE ≅ ∠A (alternate interior, transversal AC); ∠ECD ≅ ∠B (corresponding, transversal BD); ∠ACB + ∠ACE + ∠ECD = 180° (straight angle at C).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q3/5

In triangle ABC, translate the triangle along ray AB so that A moves to B; C moves to C′ and B moves to D (on ray AB beyond B). Prove that the angles of triangle ABC add to 180°, using the three angles at B.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q4/5

817

QUESTION ID

S20-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

m∠A + m∠ABC + m∠C = 180°

GRADING NOTE REQUIRED WORK: Use the straight angle ABD at B, the translation (∠C′BD ≅ ∠A), and alternate interior angles (BC′ ∥ AC).

Any valid proof (paragraph, two-column, or flow). Essential: ∠ABC + ∠CBC′ + ∠C′BD = 180° (straight angle, A, B, D collinear); ∠C′BD ≅ ∠CAB (the translation maps ∠CAB onto it); the translation maps line AC to the parallel line BC′, so ∠CBC′ ≅ ∠ACB (alternate interior angles, transversal BC); substitute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q4/5

In triangle PQR, extend side PQ past Q to S, and draw ray QT parallel to PR (T on the same side of line PS as R). Prove that the angles of triangle PQR add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q5/5

818

QUESTION ID

S20-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

m∠P + m∠PQR + m∠R = 180°

GRADING NOTE REQUIRED WORK: Use alternate interior angles (∠R and ∠RQT) and corresponding angles (∠P and ∠TQS), then the straight angle PQS.

Any valid proof (paragraph, two-column, or flow). Essential: ∠RQT ≅ ∠R (alternate interior, transversal QR); ∠TQS ≅ ∠P (corresponding, transversal PS); ∠PQR + ∠RQT + ∠TQS = 180° (straight angle at Q); substitute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 1

Use the angle sum and parallel-line angles informally • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a triangle are 47° and 68°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q1/5

820

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

65°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q1/5

True or false?

Tearing off the three corners of a paper triangle and lining them up makes a straight angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q2/5

821

QUESTION ID

S20-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has angles x, 2x, and 3x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q3/5

822

QUESTION ID

S20-B1-Q3

TYPE

Short answer

CORRECT ANSWER

30

GRADING NOTE Any equivalent form. 6x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q3/5

Lines $\ell \parallel m$ are cut by a transversal. An angle at ℓ is 72°. Which angle at m must also be 72°?

A. Its alternate interior angle

B. Its same-side interior angle

C. Every angle at m

D. The angle next to it on the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q4/5

823

QUESTION ID

S20-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Its alternate interior angle

GRADING NOTE Alternate interior angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q4/5

True or false?

A triangle can have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q5/5

824

QUESTION ID

S20-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two angles over 90° would already add to more than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 2

Angles that form a straight angle add to 180° • Grade 4 • Measurement & Data • 4.MD.C.7

Three angles together form a straight angle. Two are 40° and 75°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q1/5

826

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

65°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q1/5

True or false?

The angles around a point on one side of a straight line add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q2/5

827

QUESTION ID

S20-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They make a straight angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q2/5

∠1, ∠2, and ∠3 form a straight angle, with ∠1 = ∠3 = 50°. Find ∠2.

A. 50°

B. 100°

C. 80°

D. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q3/5

828

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 80°

GRADING NOTE 180 − 100 = 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q3/5

Angle ABC is a straight angle. Ray BD makes ∠ABD = 3x and ∠DBC = 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q4/5

829

QUESTION ID

S20-B2-Q4

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form. 5x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q4/5

True or false?

Two angles that make up a straight angle must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q5/5

830

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 180° but can be any pair, such as 30° and 150°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 3

Alternate interior angles of parallel lines are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Lines $\ell \parallel m$ are cut by transversal t. An alternate interior angle of a 58° angle measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q1/5

832

QUESTION ID

S20-B3-Q1

TYPE

Short answer

CORRECT ANSWER

58°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q1/5

True or false?

Alternate interior angles formed by a transversal of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q2/5

833

QUESTION ID

S20-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the theorem proved in an earlier set.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q2/5

Triangle ABC has A at the left end of base AB, B at the right end, and C above AB. Line ℓ through C is parallel to AB; D is on ℓ to the left of C and E is on ℓ to the right of C. Which angle is congruent to ∠CAB?

A. ∠ACB

B. ∠ECA

C. ∠DCA

D. ∠ECB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q3/5

834

QUESTION ID

S20-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. ∠DCA

GRADING NOTE With transversal CA, ∠CAB and ∠DCA are alternate interior angles (on opposite sides of CA, between the parallel lines). ∠ECB is congruent to ∠CBA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q3/5

Why is it allowed to draw the line through C parallel to AB?

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q4/5

835

QUESTION ID

S20-B3-Q4

TYPE

Short answer

CORRECT ANSWER

The parallel postulate: through a point not on a line there is exactly one parallel line.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q4/5

True or false?

Alternate interior angles are congruent for any two lines cut by a transversal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q5/5

836

QUESTION ID

S20-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 1

Prove the exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, side BC is extended past C to D. Prove that m∠ACD = m∠A + m∠B.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q1/5

838

QUESTION ID

S20-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠ACD = m∠A + m∠B

GRADING NOTE REQUIRED WORK: Use the linear pair at C and the triangle angle sum.

Any valid proof (paragraph, two-column, or flow). Essential: m∠ACD + m∠ACB = 180° (linear pair); m∠A + m∠B + m∠ACB = 180° (angle sum); substitute and subtract m∠ACB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q1/5

An exterior angle of a triangle measures 125°. One remote interior angle is 48°. Find the other remote interior angle and the adjacent interior angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q2/5

839

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Remote interior: 77°; adjacent interior: 55°.

GRADING NOTE Form required: both angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q2/5

True or false?

An exterior angle of a triangle is always greater than 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q3/5

840

QUESTION ID

S20-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The exterior angle at an obtuse vertex is acute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q3/5

The exterior angle at C is (5x + 10)° and the remote interior angles are (2x + 15)° and (2x + 20)°. Find x.

A. 5

B. 35

C. 25

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q4/5

841

QUESTION ID

S20-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 25

GRADING NOTE 5x + 10 = 4x + 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q4/5

Prove that the three exterior angles of a triangle (one at each vertex) add to 360°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q5/5

842

QUESTION ID
S20-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Each exterior angle is 180° minus its interior angle, so the sum is 540° − 180° = 360°.

GRADING NOTE Any valid argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 2

Generalize the angle sum to polygons • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Prove that the interior angles of any convex polygon with n sides add to $(n - 2) \cdot 180°$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q1/5

844

QUESTION ID

S20-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

$(n - 2) \cdot 180°$

GRADING NOTE REQUIRED WORK: Draw the diagonals from one vertex to divide the polygon into triangles and use the triangle angle sum.

Any valid proof (paragraph, two-column, or flow). Essential: from one vertex there are n − 3 diagonals, making n − 2 triangles; their angles together make up exactly the polygon's angles; each triangle contributes 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q1/5

Find the measure of each interior angle of a regular octagon.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q2/5

845

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

135°

GRADING NOTE Any equivalent form. $(8 - 2) \cdot 180 \div 8 = 135$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q2/5

Each interior angle of a regular polygon is 156°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q3/5

846

QUESTION ID

S20-F2-Q3

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. Each exterior angle is 24°; 360 ÷ 24 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q3/5

True or false?

The interior angles of a hexagon add to 1,080°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q4/5

847

QUESTION ID

S20-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(6 - 2) \cdot 180 = 720°$; 1,080° is the sum for an octagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q4/5

A pentagon has angles 100°, 110°, 120°, 95°, and x°. Find x.

A. 125

B. 115

C. 135

D. 105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q5/5

848

QUESTION ID

S20-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 115

GRADING NOTE 540 − 425 = 115.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q5/5

SET 21 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove that the base angles of an isosceles triangle are congruent

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Prove triangles congruent with SAS or SSS	**HSG-CO.B.8**
BACKWARD 2	Use the definitions of angle bisector and midpoint	**HSG-CO.A.1**
BACKWARD 3	Classify triangles and name legs, base, and base angles	**4.G.A.2**
FORWARD 1	Prove the converse: congruent base angles give congruent sides	**HSG-CO.C.10**
FORWARD 2	Use isosceles triangle theorems in multistep problems and proofs	**HSG-CO.C.10**

SET 21 • HSG-CO.C.10

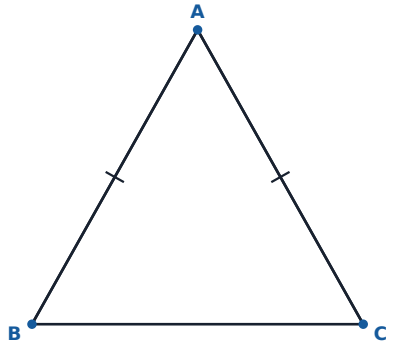
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C. (Draw the bisector of ∠A, meeting BC at D.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q1/5

851

QUESTION ID

S21-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

∠B ≅ ∠C

GRADING NOTE REQUIRED WORK: Prove △ABD ≅ △ACD by SAS, then use corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: AB = AC (given); ∠BAD ≅ ∠CAD (definition of angle bisector); AD = AD; △ABD ≅ △ACD (SAS); ∠B ≅ ∠C (corresponding angles). Invalid: "∠B and ∠C look equal"; using the fact that AD ⊥ BC (not yet known).

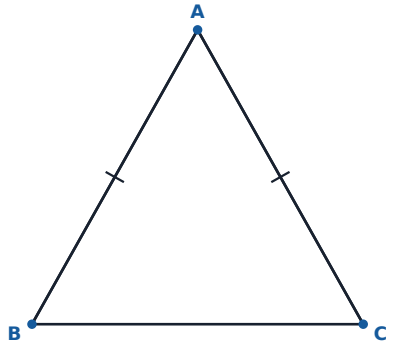
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using the midpoint M of BC.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q2/5

852

QUESTION ID

S21-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

∠B ≅ ∠C

GRADING NOTE

REQUIRED WORK: Prove △ABM ≅ △ACM by SSS.

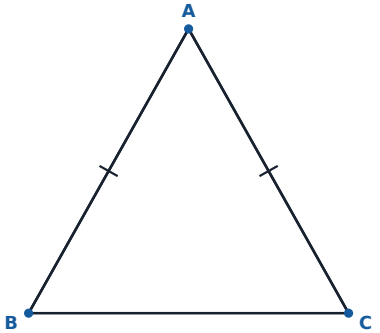
Any valid proof (paragraph, two-column, or flow). Essential: AB = AC; BM = CM (midpoint); AM = AM; SSS; corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q2/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using a rigid motion: the reflection across the bisector of ∠A.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q3/5

853

QUESTION ID

S21-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The reflection maps ∠ABC onto ∠ACB, so ∠B ≅ ∠C.

GRADING NOTE REQUIRED WORK: Explain where the reflection sends ray AB, point B, and point C.

Any valid explanation. Essential: the reflection across the bisector of ∠A swaps rays AB and AC (the bisector makes equal angles with them); B goes to the point of ray AC at distance AB = AC, which is C; so B ↔ C and the triangle maps onto itself with ∠B ↔ ∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q3/5

Prove that every equilateral triangle is equiangular (all three angles are congruent).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q4/5

854

QUESTION ID

S21-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

All three angles are congruent (each is 60°).

GRADING NOTE REQUIRED WORK: Apply the isosceles base angle theorem twice, with two different pairs of equal sides.

Any valid proof (paragraph, two-column, or flow). Essential: AB = AC gives ∠B ≅ ∠C; AB = BC gives ∠A ≅ ∠C; so all three are congruent (and each is 60° by the angle sum).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q4/5

Triangle ABC has A(0, h), B(−b, 0), and C(b, 0), with b > 0 and h > 0, so AB = AC. Prove that ∠B ≅ ∠C using the reflection (x, y) → (−x, y).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q5/5

855

QUESTION ID

S21-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The reflection fixes A and swaps B and C, so it maps ∠ABC onto ∠ACB; reflections keep angle measure.

GRADING NOTE

REQUIRED WORK: Find the images of A, B, and C, and explain why the reflection carries ∠ABC onto ∠ACB.

Any valid explanation. Essential: A(0, h) → A, B(−b, 0) → C(b, 0), C → B; so ray BA → ray CA and ray BC → ray CB; ∠ABC → ∠ACB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 1

Prove triangles congruent with SAS or SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share side AD, and AB = AC, ∠BAD ≅ ∠CAD. Which criterion shows △ABD ≅ △ACD?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q1/5

857

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

SAS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q1/5

Two triangles share side AM, and AB = AC, BM = CM.
Which criterion shows △ABM ≅ △ACM?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q2/5

858

QUESTION ID
S21-B1-Q2

TYPE
Short answer

CORRECT ANSWER
SSS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q2/5

True or false?

A shared side counts as a pair of congruent sides in a congruence proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q3/5

859

QUESTION ID

S21-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q3/5

Which information is NOT enough for congruence?

A. Three sides

B. Two sides and the included angle

C. Two angles and the included side

D. Two sides and a non-included angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q4/5

860

QUESTION ID

S21-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Two sides and a non-included angle

GRADING NOTE SSA is not a criterion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q4/5

True or false?

If △ABD ≅ △ACD, then ∠ABD ≅ ∠ACD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q5/5

861

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 2

Use the definitions of angle bisector and midpoint • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Ray AD bisects ∠BAC, and m∠BAC = 64°. Find m∠BAD.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q1/5

863

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

32°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q1/5

M is the midpoint of BC, and BC = 18. Find BM.

BM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q2/5

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q2/5

True or false?

An angle bisector divides an angle into two congruent angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q3/5

865

QUESTION ID

S21-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q3/5

Ray AD bisects ∠BAC. Which equation is true?

A. m∠BAD = m∠BAC

B. m∠BAD = m∠DAC

C. AD = AB

D. BD = DC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q4/5

866

QUESTION ID

S21-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. m∠BAD = m∠DAC

GRADING NOTE Equal angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q4/5

True or false?

The bisector of ∠A of triangle ABC always passes through the midpoint of BC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q5/5

867

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when AB = AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 3

Classify triangles and name legs, base, and base angles • Grade 4 • Geometry • 4.G.A.2

Triangle PQR has PQ = PR. Name the base and the base angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q1/5

869

QUESTION ID

S21-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Base QR; base angles ∠Q and ∠R

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q1/5

True or false?

An isosceles triangle has at least two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q2/5

870

QUESTION ID

S21-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition used in this course.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q2/5

Which triangle is scalene?

A. Sides 3, 3, 3

B. Sides 5, 5, 5

C. Sides 4, 5, 6

D. Sides 4, 4, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q3/5

871

QUESTION ID

S21-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Sides 4, 5, 6

GRADING NOTE No two sides are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q3/5

In isosceles triangle XYZ with XY = XZ, which angle is the vertex angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q4/5

872

QUESTION ID

S21-B3-Q4

TYPE

Short answer

CORRECT ANSWER

∠X

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q4/5

True or false?

Every equilateral triangle is isosceles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q5/5

873

QUESTION ID

S21-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It has at least two congruent sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 1

Prove the converse: congruent base angles give congruent sides • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, ∠B ≅ ∠C. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q1/5

875

QUESTION ID

S21-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AB = AC

GRADING NOTE REQUIRED WORK: Draw the bisector of ∠A meeting BC at D and prove △ABD ≅ △ACD by AAS.

Any valid proof (paragraph, two-column, or flow). Essential: ∠B ≅ ∠C (given); ∠BAD ≅ ∠CAD (bisector); AD = AD; AAS; AB = AC. Invalid: using AB = AC to choose the auxiliary line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q1/5

In triangle PQR, m∠Q = 70° and m∠R = 70°, and PQ = 3x + 2, PR = 5x − 6. Find x and PQ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q2/5

876

QUESTION ID

S21-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 4; PQ = 14

GRADING NOTE Form required: x and PQ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q2/5

True or false?

A triangle with two congruent angles must have two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q3/5

877

QUESTION ID

S21-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the converse of the isosceles triangle theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q3/5

Triangle ABC has m∠A = 50° and m∠B = 65°. Which two sides are congruent?

A. No two sides

B. AB and BC

C. AB and AC

D. AC and BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q4/5

878

QUESTION ID

S21-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. AB and AC

GRADING NOTE m∠C = 65° = m∠B, so the sides opposite them, AC and AB, are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q4/5

Prove that every equiangular triangle is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5
High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q5/5

879

QUESTION ID
S21-F1-Q5

TYPE
Answer with required work

CORRECT ANSWER
All three sides are congruent.

GRADING NOTE REQUIRED WORK: Apply the converse twice.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A ≅ ∠B gives BC = AC; ∠B ≅ ∠C gives AC = AB; so all three sides are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5
High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 2

Use isosceles triangle theorems in multistep problems and proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC and the vertex angle ∠A measures 38°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q1/5

881

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

71°

GRADING NOTE Any equivalent form. (180 − 38) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Point D is on side BC of triangle ABC with AD = BD = CD. Find m∠BAC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q2/5

882

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form. Triangles ABD and ACD are isosceles, so ∠BAD = ∠B and ∠CAD = ∠C; then 2(∠B + ∠C) = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q2/5

In triangle ABC, AB = AC, and D and E are points on BC with BD = CE. Prove that AD = AE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q3/5

883

QUESTION ID

S21-F2-Q3

TYPE

Answer with required work

CORRECT ANSWER

AD = AE

GRADING NOTE

REQUIRED WORK: Use the base angles and SAS on triangles ABD and ACE.

Any valid proof (paragraph, two-column, or flow). Essential: ∠B ≅ ∠C (base angles); AB = AC; BD = CE; △ABD ≅ △ACE (SAS); AD = AE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q3/5

True or false?

If the vertex angle of an isosceles triangle is 100°, each base angle is 80°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q4/5

884

QUESTION ID

S21-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (180 − 100) ÷ 2 = 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q4/5

In isosceles triangle PQR with PQ = PR, the exterior angle at Q (on the extension of RQ past Q) is 112°. Find m∠P.

A. 56°

B. 112°

C. 68°

D. 44°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q5/5

885

QUESTION ID

S21-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 44°

GRADING NOTE ∠PQR = 68° = ∠R, so ∠P = 180 − 136 = 44.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q5/5

SET 22 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove the midsegment theorem

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Find midpoints of segments in the coordinate plane	**6.G.A.3**
BACKWARD 2	Find and compare slopes to decide whether lines are parallel	**8.EE.B.6**
BACKWARD 3	Find lengths with the distance formula	**8.G.B.8**
BACKWARD 4	Prove lines parallel from congruent alternate interior angles	**HSG-CO.C.9**
FORWARD 1	Prove that the midpoints of a quadrilateral's sides form a parallelo	**HSG-CO.C.11**
FORWARD 2	Midsegment triangles: similarity, scale factor, and area	**HSG-SRT.A.2**

SET 22 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(2a, 0), and C(2b, 2c), with a > 0 and c ≠ 0. D is the midpoint of AC and E is the midpoint of BC. Prove that DE ∥ AB and DE = ½AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q1/5

888

QUESTION ID

S22-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

D(b, c), E(a + b, c); DE is horizontal like AB, and DE = a = ½(2a) = ½AB.

GRADING NOTE REQUIRED WORK: Find D and E with the midpoint formula, then compare slopes and lengths.

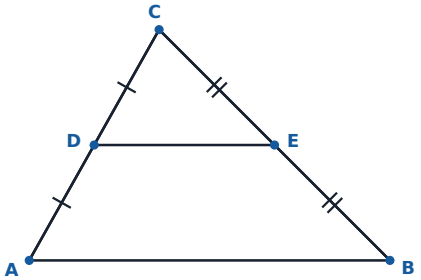
Any valid proof (paragraph, two-column, or flow). Essential: both midpoints; DE has slope 0 (same y) and AB has slope 0, so DE ∥ AB (and DE ≠ AB since c ≠ 0); DE = |a + b − b| = a and AB = 2a. Invalid: checking one numerical triangle only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q1/5

In triangle ABC, D is the midpoint of AC and E is the midpoint of BC. Extend DE past E to F so that EF = DE. Prove that DE ∥ AB and DE = ½AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q2/5

889

QUESTION ID

S22-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

DE ∥ AB and DE = ½AB

GRADING NOTE REQUIRED WORK: Prove △DEC ≅ △FEB (SAS), deduce BF = DC = AD and BF ∥ AD, so ABFD is a parallelogram; then use its opposite sides.

Any valid proof (paragraph, two-column, or flow). Essential: CE = BE, DE = FE, vertical angles at E, so △DEC ≅ △FEB (SAS); BF = DC = AD; ∠DCE ≅ ∠FBE gives BF ∥ AC (alternate interior converse); one pair of opposite sides of ABFD is parallel and congruent, so ABFD is a parallelogram; so DF ∥ AB and DF = AB; DE = ½DF = ½AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(2a, 0), and C(2b, 2c), with a > 0 and c ≠ 0. F is the midpoint of AB and E is the midpoint of AC. Prove that FE ∥ BC and FE = ½BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q3/5

890

QUESTION ID

S22-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

F(a, 0), E(b, c); FE goes from (a, 0) to (b, c), the same direction as BC from (2a, 0) to (2b, 2c), and FE = ½BC.

GRADING NOTE REQUIRED WORK: Find F and E, then compare slopes (or note both are vertical) and lengths.

Any valid proof (paragraph, two-column, or flow). Essential: both midpoints; FE = (b − a, c) and BC = (2b − 2a, 2c) = 2(b − a, c), so they are parallel (equal slopes c/(b − a), or both vertical when b = a) and FE = √((b − a)² + c²) = ½BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q3/5

In triangle ABC, F is the midpoint of AB and E is the midpoint of AC. Extend FE past E to G so that EG = FE. Prove that FE ∥ BC and FE = ½BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q4/5

891

QUESTION ID

S22-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

FE ∥ BC and FE = ½BC

GRADING NOTE REQUIRED WORK: Prove △AEF ≅ △CEG (SAS); then show FBCG is a parallelogram.

Any valid proof (paragraph, two-column, or flow). Essential: AE = CE, FE = GE, vertical angles at E, so △AEF ≅ △CEG (SAS); CG = AF = FB and ∠FAE ≅ ∠GCE, so CG ∥ AB (alternate interior converse); FBCG has a pair of opposite sides parallel and congruent, so it is a parallelogram; FG ∥ BC and FG = BC; FE = ½FG = ½BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(2p, 2q), B(2r, 2s), and C(2t, 2u), not on one line, with AB not vertical. D is the midpoint of AC and E is the midpoint of BC. Prove that DE ∥ AB and DE = ½AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q5/5

892

QUESTION ID

S22-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

D(p + t, q + u), E(r + t, s + u); DE = (r − p, s − q) and AB = (2r − 2p, 2s − 2q) = 2·DE, so DE ∥ AB and DE = ½AB.

GRADING NOTE REQUIRED WORK: Find D and E, then compare the coordinate differences of DE and AB.

Any valid proof (paragraph, two-column, or flow). Essential: both midpoints; slope DE = (s − q)/(r − p) = slope AB; DE = √((r − p)² + (s − q)²) = ½AB. DE and AB are different lines because C is not on AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 1

Find midpoints of segments in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (2, 6) to (8, 2).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q1/5

894

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 4)

GRADING NOTE Form required: an ordered pair. Average the coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from (−3, 1) to (5, −7).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q2/5

895

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(1, −3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q2/5

Find the midpoint of the segment from (0, 0) to (2a, 2c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q3/5

896

QUESTION ID

S22-B1-Q3

TYPE

Short answer

CORRECT ANSWER

(a, c)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q3/5

True or false?

The midpoint of the segment from (1, 1) to (7, 1) is (4, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q4/5

897

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ((1 + 7) ÷ 2, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q4/5

True or false?

The midpoint of the segment from (0, 4) to (6, 0) is (6, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q5/5

898

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is (3, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 2

Find and compare slopes to decide whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (1, 2) and (4, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

900

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. (8 − 2) ÷ (4 − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (b, c) and (a + b, c), where a ≠ 0.

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

901

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form. The y-coordinates are equal and the points are different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Two different nonvertical lines with the same slope are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

902

QUESTION ID

S22-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same steepness, different positions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

Which segment is parallel to the segment from (0, 0) to (4, 2)?

A. From (1, 3) to (5, 5)

B. From (2, 2) to (2, 6)

C. From (1, 3) to (3, 7)

D. From (0, 0) to (2, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

903

QUESTION ID

S22-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. From (1, 3) to (5, 5)

GRADING NOTE Both have slope 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

A horizontal segment and a vertical segment have the same slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

904

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Horizontal has slope 0; vertical is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 3

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

Find the distance between (b, c) and (a + b, c), with a > 0.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q1/5

906

QUESTION ID

S22-B3-Q1

TYPE

Short answer

CORRECT ANSWER

a (that is, |a|, with a > 0)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (6, 8).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q2/5

907

QUESTION ID

S22-B3-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q2/5

Find the length of the segment from (−2, 5) to (4, 5).

A. 3

B. 10

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q3/5

908

QUESTION ID

S22-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

GRADING NOTE Horizontal: |4 − (−2)| = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q3/5

True or false?

The distance from (1, 2) to (4, 6) is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q4/5

909

QUESTION ID

S22-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sqrt{(9 + 16)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q4/5

True or false?

The distance from (0, 0) to (2a, 0) is a, when a > 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q5/5

910

QUESTION ID

S22-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 4

Prove lines parallel from congruent alternate interior angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by a transversal, and alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q1/5

912

QUESTION ID

S22-B4-Q1

TYPE

Short answer

CORRECT ANSWER

$\ell \parallel m$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q1/5

True or false?

If two lines are cut by a transversal and alternate interior angles are congruent, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q2/5

913

QUESTION ID

S22-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse of the alternate interior angles theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q2/5

In △ABC, D is the midpoint of AC and E is the midpoint of BC. DE is extended to F so that EF = DE, and △DEC ≅ △FEB. Which fact shows BF ∥ AC?

A. DE = EF

B. ∠DCE ≅ ∠FBE are alternate interior angles

C. BF = DC

D. E is the midpoint of BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q3/5

914

QUESTION ID
S22-B4-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. ∠DCE ≅ ∠FBE are alternate interior angles

GRADING NOTE Congruent alternate interior angles (transversal BC) give parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q3/5

Two lines are cut by a transversal and corresponding angles measure 70° and 70°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q4/5

915

QUESTION ID

S22-B4-Q4

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q4/5

True or false?

Congruent vertical angles prove that two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q5/5

916

QUESTION ID

S22-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Vertical angles are congruent for any crossing lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 1

Prove that the midpoints of a quadrilateral's sides form a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is any quadrilateral, and P, Q, R, S are the midpoints of AB, BC, CD, DA. Prove that PQRS is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q1/5

QUESTION ID

S22-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

PQ ∥ AC ∥ SR and PQ = ½AC = SR, so PQRS is a parallelogram.

GRADING NOTE REQUIRED WORK: Draw diagonal AC and apply the midsegment theorem in triangles ABC and ADC.

Any valid proof (paragraph, two-column, or flow). Essential: in △ABC, PQ ∥ AC and PQ = ½AC; in △ADC, SR ∥ AC and SR = ½AC; so PQ ∥ SR and PQ = SR; a quadrilateral with one pair of opposite sides parallel and congruent is a parallelogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q1/5

Quadrilateral ABCD has diagonals AC = 10 and BD = 14. Find the perimeter of the quadrilateral formed by the midpoints of its sides.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q2/5

919

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

24

GRADING NOTE Any equivalent form. Its sides are ½AC, ½BD, ½AC, ½BD: 5 + 7 + 5 + 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q2/5

True or false?

The midpoint quadrilateral of any quadrilateral is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q3/5

920

QUESTION ID

S22-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is Varignon's theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q3/5

If the diagonals of ABCD are perpendicular, the midpoint quadrilateral is a

A. rectangle

B. trapezoid that is not a parallelogram

C. rhombus

D. kite that is not a parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q4/5

921

QUESTION ID

S22-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. rectangle

GRADING NOTE Its sides are parallel to the diagonals, so adjacent sides are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, D, E, F are the midpoints of AB, BC, CA. AB = 12, BC = 16, CA = 10. Find the perimeter of triangle DEF.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q5/5

922

QUESTION ID

S22-F1-Q5

TYPE

Short answer

CORRECT ANSWER

19

GRADING NOTE Any equivalent form. Each midsegment is half the parallel side: 6 + 8 + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 2

Midsegment triangles: similarity, scale factor, and area • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

D, E, F are the midpoints of the sides of triangle ABC. Explain why triangle DEF is similar to triangle ABC, and give the scale factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q1/5

924

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Each side of DEF is half of a side of ABC (midsegment theorem), so the triangles are similar by SSS similarity with scale factor ½.

GRADING NOTE Any equivalent explanation with the factor ½ (or 2 from DEF to ABC).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has area 48. Find the area of the triangle formed by the midpoints of its sides.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q2/5

925

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form. Scale factor ½ gives area factor ¼; also the 4 small triangles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q2/5

True or false?

The four triangles formed by drawing all three midsegments of a triangle are similar to it with scale factor ⅓.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q3/5

926

QUESTION ID
S22-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The scale factor is ½: each side is half a side of the original.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q3/5

The midsegment parallel to AB in triangle ABC is 9. A line segment connects the midpoints of that midsegment's triangle the same way. How long is the new segment parallel to AB?

A. 9

B. 18

C. 3

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q4/5

927

QUESTION ID

S22-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 4.5

GRADING NOTE Each repetition halves the length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q4/5

Triangle ABC has vertices A(0, 0), B(8, 0), C(2, 6). Find the vertices of its midsegment triangle and verify the ratio of areas.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q5/5

928

QUESTION ID

S22-F2-Q5

TYPE

Short answer

CORRECT ANSWER

(4, 0), (5, 3), (1, 3); area of ABC = 24 and area of the midsegment triangle = 6, a ratio of 1 : 4.

GRADING NOTE Form required: the three midpoints and both areas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q5/5

Prove that the medians of a triangle meet at one point

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Find midpoints, including with variables	**6.G.A.3**
BACKWARD 2	Use the midsegment theorem	**HSG-CO.C.10**
BACKWARD 3	Find the point a given fraction of the way along a segment	**7.RP.A.2**
BACKWARD 4	Write the equation of a line through two points	**8.F.B.4**
FORWARD 1	Locate the centroid and use the 2 : 1 ratio	**HSG-GPE.B.6**
FORWARD 2	Prove that the angle bisectors of a triangle meet at one point	**HSG-C.A.3**

SET 23 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(2b, 0), and C(2c, 2d), with b ≠ 0 and d ≠ 0. Prove that the point G((2b + 2c)/3, 2d/3) lies on all three medians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q1/5

931

QUESTION ID

S23-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

G is on the median from A (to (b + c, d)), from B (to (c, d)), and from C (to (b, 0)), so the medians meet at G.

GRADING NOTE REQUIRED WORK: Find the midpoint of each side, then show G is two-thirds of the way from each vertex to the opposite midpoint (or on each median line).

Any valid proof (paragraph, two-column, or flow). Essential: the three midpoints (b + c, d), (c, d), (b, 0); for each median, G = vertex + ⅔(midpoint − vertex): A + ⅔((b + c, d) − A) = G; B + ⅔((c, d) − (2b, 0)) = G; C + ⅔((b, 0) − (2c, 2d)) = G. Alternative: show G satisfies each median's equation. Invalid: checking one triangle with numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q1/5

In triangle ABC, medians BE and CF meet at G. Let H and K be the midpoints of BG and CG. Prove that BG = 2GE. Then explain why the third median also passes through G.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q2/5

932

QUESTION ID

S23-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

BG = 2GE; the same argument for medians AD and BE gives the point that is two-thirds along BE, which is G, so AD passes through G.

GRADING NOTE REQUIRED WORK: Use the midsegment theorem in triangles ABC and GBC to show FEKH is a parallelogram, then its diagonals.

Any valid proof (paragraph, two-column, or flow). Essential: FE ∥ BC and FE = ½BC (midsegment in △ABC); HK ∥ BC and HK = ½BC (midsegment in △GBC); so FEKH is a parallelogram; its diagonals bisect each other, so HG = GE; with BH = HG, BG = 2GE; any two medians meet at the point two-thirds along each, so all three share it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(−2a, 0), B(2a, 0), and C(0, 2b), with a > 0 and b > 0. Prove that the three medians meet at (0, 2b/3).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q3/5

933

QUESTION ID

S23-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Midpoints (a, b), (−a, b), (0, 0); the point (0, 2b/3) is ⅔ of the way along each median.

GRADING NOTE REQUIRED WORK: Find the three midpoints and show that (0, 2b/3) is two-thirds of the way from each vertex to the opposite midpoint.

Any valid proof (paragraph, two-column, or flow). Essential: midpoint of BC (a, b), of AC (−a, b), of AB (0, 0); A + ⅔((a, b) − A) = (0, 2b/3); B + ⅔((−a, b) − B) = (0, 2b/3); C + ⅔((0, 0) − C) = (0, 2b/3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle PQR has P(0, 0), Q(6, 0), and R(0, 6). Find the three medians' equations and prove that they meet at one point.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q4/5

934

QUESTION ID

S23-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The medians meet at (2, 2).

GRADING NOTE REQUIRED WORK: Find each median as a line through a vertex and a midpoint, find where two meet, and check that the third passes through that point.

Any valid proof (paragraph, two-column, or flow). Essential: midpoints (3, 3), (0, 3), (3, 0); medians y = x, y = −½x + 3 (from Q), y = −2x + 6 (from R); y = x and y = −2x + 6 meet at (2, 2); (2, 2) satisfies y = −½x + 3. (This proves the claim for this triangle; the general proof needs variables.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$, not on one line. Let $G = ((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3)$. Prove that G lies on all three medians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q5/5

935

QUESTION ID

S23-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

For the median from A: $A + \frac{2}{3}(((x_2 + x_3)/2, (y_2 + y_3)/2) - A) = G$; the same holds from B and C, so the medians meet at G.

GRADING NOTE REQUIRED WORK: For each median, show G = vertex + ⅔(opposite midpoint − vertex).

Any valid proof (paragraph, two-column, or flow). Essential: the computation for one median with the general coordinates and the symmetry (or computation) for the other two. Invalid: drawing examples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 1

Find midpoints, including with variables • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (2b, 0) to (2c, 2d).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q1/5

937

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(b + c, d)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the midpoint of the segment from (0, 6) to (6, 0).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q2/5

938

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q2/5

A median of a triangle joins a vertex to

A. the midpoint of the opposite side

B. another vertex

C. the foot of the perpendicular from that vertex

D. the midpoint of an adjacent side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q3/5

939

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the midpoint of the opposite side

GRADING NOTE Definition of a median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q3/5

True or false?

A triangle has exactly three medians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q4/5

940

QUESTION ID

S23-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE One from each vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q4/5

True or false?

A median always bisects the angle at its vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q5/5

941

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only in special cases (for example, the vertex angle of an isosceles triangle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 2

Use the midsegment theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, F and E are the midpoints of AB and AC. BC = 14. Find FE.

FE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q1/5

943

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q1/5

True or false?

The segment joining the midpoints of two sides of a triangle is parallel to the third side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q2/5

944

QUESTION ID

S23-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Midsegment theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q2/5

In triangle GBC, H and K are midpoints of GB and GC. Which statement is true?

A. HK ⊥ BC

B. HK ∥ GB

C. HK ∥ BC and HK = ½BC

D. HK = BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q3/5

945

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. HK ∥ BC and HK = ½BC

GRADING NOTE Midsegment theorem in triangle GBC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q3/5

A midsegment of a triangle is 11 cm. How long is the side parallel to it?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q4/5

946

QUESTION ID

S23-B2-Q4

TYPE

Short answer

CORRECT ANSWER

22 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q4/5

True or false?

A midsegment of a triangle is always half as long as each of the other two sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q5/5

947

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only half as long as the side it is parallel to.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 3

Find the point a given fraction of the way along a segment • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

On a number line, what point is ⅔ of the way from 0 to 9?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q1/5

949

QUESTION ID

S23-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

On a number line, what point is ⅔ of the way from 3 to 12?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q2/5

950

QUESTION ID

S23-B3-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. 3 + ⅔(9).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q2/5

A point is ⅔ of the way from A to D. What is the ratio of its distances from A and from D?

A. 2 : 3

B. 3 : 2

C. 1 : 2

D. 2 : 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q3/5

951

QUESTION ID

S23-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2 : 1

GRADING NOTE ⅔ and ⅓ of the length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q3/5

True or false?

The point ⅔ of the way from (0, 0) to (6, 3) is (4, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q4/5

952

QUESTION ID

S23-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ⅔ of each coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q4/5

True or false?

The point ⅔ of the way from (0, 0) to (3, 6) is (2, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q5/5

953

QUESTION ID

S23-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is (2, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 4

Write the equation of a line through two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation of the line through (0, 0) and (3, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q1

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q1/5

955

QUESTION ID

S23-B4-Q1

TYPE

Short answer

CORRECT ANSWER

y = x

GRADING NOTE Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q1

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation of the line through (6, 0) and (0, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q2

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q2/5

956

QUESTION ID

S23-B4-Q2

TYPE

Short answer

CORRECT ANSWER

$y = -\frac{1}{2}x + 3$

GRADING NOTE Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q2

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q2/5

Which point is on the line y = −2x + 6?

A. (1, 2)

B. (3, 3)

C. (0, 2)

D. (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q3

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q3/5

957

QUESTION ID

S23-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 2)

GRADING NOTE −2(2) + 6 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q3

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q3/5

True or false?

The line through (0, 6) and (3, 0) has slope −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q4

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q4/5

958

QUESTION ID

S23-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (0 − 6) ÷ (3 − 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q4

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q4/5

True or false?

The line y = x passes through (2, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q5

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q5/5

959

QUESTION ID

S23-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2 \neq 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q5

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q5/5

SET 23 • HSG-CO.C.10

FORWARD 1

Locate the centroid and use the 2 : 1 ratio • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the point where the medians of the triangle with vertices (1, 2), (7, 4), and (4, 9) meet.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q1/5

961

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(4, 5)

GRADING NOTE Form required: an ordered pair. Average the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q1/5

Median AD of a triangle is 18 cm long. How far is the centroid from A and from D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q2/5

962

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12 cm from A and 6 cm from D

GRADING NOTE Form required: both distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q2/5

True or false?

The centroid divides each median so that the part from the vertex is half the part from the midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q3/5

963

QUESTION ID

S23-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The part from the vertex is twice the other part (2 : 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q3/5

G is the centroid of triangle ABC, and GE = 5 on median BE. Find BE.

A. 20

B. 10

C. 7.5

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q4/5

964

QUESTION ID

S23-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 15

GRADING NOTE BG = 10, so BE = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F1-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q4/5

Explain why the centroid of a triangle is the average of its three vertices.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q5/5

965

QUESTION ID

S23-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The centroid is ⅔ of the way from A to the midpoint of BC: A + ⅔((B + C)/2 − A) = (A + B + C)/3.

GRADING NOTE Any equivalent derivation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSG-CO.C.10

FORWARD 2

Prove that the angle bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Prove that the bisectors of the three angles of triangle ABC meet at one point that is equidistant from the three sides.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q1/5

967

QUESTION ID

S23-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

The three angle bisectors meet at one point I, equidistant from all three sides.

GRADING NOTE REQUIRED WORK: Use the fact that a point on an angle bisector is equidistant from the angle's sides, and its converse.

Any valid proof (paragraph, two-column, or flow). Essential: the bisectors of ∠A and ∠B meet at I; I is equidistant from AB and AC, and from AB and BC; so it is equidistant from AC and BC; by the converse it lies on the bisector of ∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q1/5

Why must the bisectors of ∠A and ∠B of a triangle meet inside the triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q2/5

968

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Each bisector goes from its vertex into the triangle's interior and reaches the opposite side, so the two bisectors cross inside the triangle.

GRADING NOTE Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q2/5

True or false?

A point on the bisector of an angle is equidistant from the two sides of the angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q3/5

969

QUESTION ID

S23-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Proved with congruent right triangles (AAS).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q3/5

The point where the angle bisectors of a triangle meet is the center of the

A. circumscribed circle

B. nine-point circle

C. inscribed circle

D. largest square inside the triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q4/5

970

QUESTION ID

S23-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. inscribed circle

GRADING NOTE It is equidistant from the sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q4/5

True or false?

The angle bisectors and the medians of every triangle meet at the same point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q5/5

971

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only in an equilateral triangle in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q5/5

Prove that opposite sides of a parallelogram are congruent

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Alternate interior angles of parallel lines	**HSG-CO.C.9**
BACKWARD 2	Prove triangles congruent with ASA	**HSG-CO.B.8**
BACKWARD 3	Know the definition of a parallelogram	**5.G.B.3**
FORWARD 1	Prove that the diagonals of a rhombus are perpendicular	**HSG-CO.C.11**
FORWARD 2	Use opposite sides with coordinates and algebra	**HSG-GPE.B.4**

SET 24 • HSG-CO.C.11

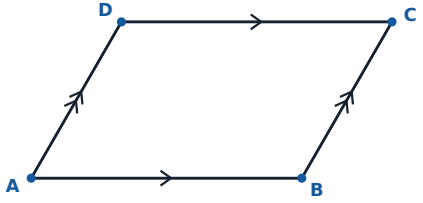
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram (AB ∥ DC and AD ∥ BC). Prove that AB = CD and AD = CB. (Draw diagonal AC.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q1/5

974

QUESTION ID

S24-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AB = CD and AD = CB

GRADING NOTE REQUIRED WORK: Prove △ABC ≅ △CDA by ASA, using alternate interior angles from both pairs of parallel sides and the shared diagonal.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BAC ≅ ∠DCA (alternate interior, AB ∥ DC); ∠BCA ≅ ∠DAC (alternate interior, AD ∥ BC); AC = CA (reflexive); △ABC ≅ △CDA (ASA); corresponding sides. Invalid: "opposite sides look equal"; using the theorem being proved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

PQRS is a parallelogram (PQ ∥ SR and PS ∥ QR). Prove that PQ = RS and QR = SP, using diagonal QS.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q2/5

975

QUESTION ID

S24-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

PQ = RS and QR = SP

GRADING NOTE REQUIRED WORK: Prove △PQS ≅ △RSQ by ASA with alternate interior angles and the shared diagonal QS.

Any valid proof (paragraph, two-column, or flow). Essential: ∠PQS ≅ ∠RSQ (alternate interior, PQ ∥ SR); ∠PSQ ≅ ∠RQS (alternate interior, PS ∥ QR); QS shared; ASA; corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q2/5

ABCD is a parallelogram and M is the midpoint of diagonal AC. Explain, using the 180° rotation about M, why AB = CD and AD = CB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q3/5

976

QUESTION ID

S24-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The rotation swaps A and C and sends B to D, so it maps AB onto CD and AD onto CB; rotations keep length.

GRADING NOTE REQUIRED WORK: Find the images of A, line AB, and line AD; then the image of B as an intersection of lines.

Any valid explanation. Essential: M is the midpoint of AC, so A ↔ C; a 180° rotation maps line AB to the parallel line through C, which is line CD (parallel postulate), and line CB to the parallel line through A, which is AD; B = AB ∩ CB goes to CD ∩ AD = D; lengths are kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram has vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c), with a > 0 and c > 0. Prove that AB = DC and AD = BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q4/5

977

QUESTION ID

S24-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AB = DC = a and AD = BC = √(b² + c²)

GRADING NOTE REQUIRED WORK: Use the distance formula on each pair of opposite sides.

Any valid proof (paragraph, two-column, or flow). Essential: AB = a; DC = |a + b − b| = a; AD = √(b² + c²); BC = √((a + b − a)² + c²) = √(b² + c²).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD and ABEF are parallelograms that share side AB, with C, D, E, F all different points. Prove that DC = FE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q5/5

978

QUESTION ID

S24-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

DC = AB = FE, so DC = FE.

GRADING NOTE REQUIRED WORK: Apply the theorem in each parallelogram and use the transitive property.

Any valid proof (paragraph, two-column, or flow). Essential: DC = AB (opposite sides of ABCD); FE = AB (opposite sides of ABEF); transitive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 1

Alternate interior angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

AB ∥ DC and diagonal AC is drawn. Which angle at C is congruent to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q1/5

980

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

∠DCA

GRADING NOTE Any equivalent form. Alternate interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q1/5

True or false?

Alternate interior angles formed by a transversal of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q2/5

981

QUESTION ID

S24-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem from set 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q2/5

In parallelogram ABCD with diagonal BD, which angle is congruent to ∠ABD?

A. ∠ABC

B. ∠ADB

C. ∠CBD

D. ∠CDB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q3/5

982

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. ∠CDB

GRADING NOTE AB ∥ DC with transversal BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q3/5

In parallelogram ABCD with diagonal AC, AD ∥ BC. Which angle is congruent to ∠DAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q4/5

983

QUESTION ID

S24-B1-Q4

TYPE

Short answer

CORRECT ANSWER

∠BCA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q4/5

True or false?

Alternate interior angles of parallel lines are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q5/5

984

QUESTION ID
S24-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 2

Prove triangles congruent with ASA • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Two triangles have two pairs of congruent angles and the congruent sides between them. Which criterion applies?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q1/5

986

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

ASA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q1/5

True or false?

A diagonal shared by two triangles can be the included side in ASA.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q2/5

987

QUESTION ID

S24-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q2/5

In △ABC and △CDA, ∠BAC ≅ ∠DCA, ∠BCA ≅ ∠DAC, and AC is shared. Which correspondence is correct?

A. A ↔ D, B ↔ C, C ↔ A

B. A ↔ A, B ↔ D, C ↔ C

C. A ↔ C, B ↔ D, C ↔ A

D. A ↔ C, B ↔ A, C ↔ D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q3/5

988

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A ↔ C, B ↔ D, C ↔ A

GRADING NOTE Match the equal angles: A ↔ C and C ↔ A, so B ↔ D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q3/5

After proving △ABC ≅ △CDA, which side corresponds to AB?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q4/5

989

QUESTION ID

S24-B2-Q4

TYPE

Short answer

CORRECT ANSWER

CD

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q4/5

True or false?

ASA requires the side to be between the two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q5/5

990

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Otherwise it is AAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 3

Know the definition of a parallelogram • Grade 5 • Geometry • 5.G.B.3

True or false?

A parallelogram is a quadrilateral with both pairs of opposite sides parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q1/5

992

QUESTION ID

S24-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q1/5

Which quadrilateral is always a parallelogram?

A. A trapezoid with exactly one pair of parallel sides

B. A kite that is not a rhombus

C. Any quadrilateral with one right angle

D. A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q2/5

993

QUESTION ID

S24-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. A rectangle

GRADING NOTE A rectangle has both pairs of opposite sides parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q2/5

Name the two pairs of opposite sides of parallelogram PQRS.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q3/5

994

QUESTION ID

S24-B3-Q3

TYPE

Short answer

CORRECT ANSWER

PQ and RS; QR and SP

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q3/5

True or false?

In parallelogram ABCD, sides AB and BC are opposite sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q4/5

995

QUESTION ID

S24-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are consecutive sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q4/5

Which sides of parallelogram WXYZ are parallel to WX and to XY?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q5/5

996

QUESTION ID

S24-B3-Q5

TYPE

Short answer

CORRECT ANSWER

ZY ∥ WX and WZ ∥ XY

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q5/5

SET 24 • HSG-CO.C.11

FORWARD 1

Prove that the diagonals of a rhombus are perpendicular • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a rhombus (a parallelogram with all four sides congruent), with diagonals meeting at E. Prove that AC ⊥ BD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q1/5

998

QUESTION ID

S24-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AC ⊥ BD

GRADING NOTE REQUIRED WORK: Use that the diagonals of a parallelogram bisect each other, then SSS on triangles AEB and AED (or the perpendicular bisector converse).

Any valid proof (paragraph, two-column, or flow). Essential: BE = ED (diagonals of a parallelogram bisect each other); AB = AD; AE = AE; △AEB ≅ △AED (SSS); ∠AEB ≅ ∠AED form a linear pair, so both are 90°. Alternative: A and C are each equidistant from B and D, so AC is the perpendicular bisector of BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q1/5

In rhombus ABCD, diagonals AC = 16 and BD = 12. Find the side length.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q2/5

999

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. Half-diagonals 8 and 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a rhombus. Prove that diagonal AC bisects ∠BAD and ∠BCD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q3/5

1000

QUESTION ID

S24-F1-Q3

TYPE

Answer with required work

CORRECT ANSWER

∠BAC ≅ ∠DAC and ∠BCA ≅ ∠DCA

GRADING NOTE REQUIRED WORK: Use SSS on triangles ABC and ADC.

Any valid proof (paragraph, two-column, or flow). Essential: AB = AD and CB = CD (all sides congruent); AC shared; SSS; corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Rhombus ABCD has perimeter 52 and diagonal AC = 24. Find diagonal BD.

BD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q4/5

1001

QUESTION ID

S24-F1-Q4

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. Side 13, half-diagonals 12 and 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q4/5

In rhombus PQRS, m∠PQR = 110°. Find m∠PQS, where QS is a diagonal, and give the reason.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q5/5

1002

QUESTION ID

S24-F1-Q5

TYPE

Short answer

CORRECT ANSWER

55°, because a diagonal of a rhombus bisects the angles at its ends.

GRADING NOTE Form required: the measure and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSG-CO.C.11

FORWARD 2

Use opposite sides with coordinates and algebra • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Parallelogram ABCD has A(1, 2), B(6, 2), and D(3, 5). Find C, and verify that AB = DC and AD = BC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q1/5

1004

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

C(8, 5); AB = DC = 5 and AD = BC = √13.

GRADING NOTE Form required: C and both pairs of lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q1/5

Explain why the vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c) describe a general parallelogram.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q2/5

1005

QUESTION ID

S24-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Any parallelogram can be moved (rigid motion) so that A is at the origin and AB lies on the positive x-axis; then D = (b, c) and C = B + (D − A) = (a + b, c).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram WXYZ, WX = 3x + 4 and YZ = 5x − 6. Find x and WX.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q3/5

1006

QUESTION ID

S24-F2-Q3

TYPE

Short answer

CORRECT ANSWER

x = 5; WX = 19

GRADING NOTE Form required: both. Opposite sides are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q3/5

For the parallelogram A(0, 0), B(6, 0), C(8, 4), D(2, 4), what is AD?

A. √20 + 2

B. 4

C. 2√5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q4/5

1007

QUESTION ID

S24-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2√5

GRADING NOTE √(4 + 16) = √20 = 2√5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Parallelogram ABCD has A(1, 1), B(5, 1), and D(2, 4). Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q5/5

1008

QUESTION ID
S24-F2-Q5

TYPE
Short answer

CORRECT ANSWER
(6, 4)

GRADING NOTE Form required: an ordered pair. C = B + (D − A).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q5/5

SET 101 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that opposite angles of a parallelogram are congruent

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Same-side interior angles of parallel lines	**HSG-CO.C.9**
BACKWARD 2	Name opposite and consecutive angles of a quadrilateral	**5.G.B.3**
BACKWARD 3	Prove triangles congruent with ASA and use corresponding angles	**HSG-CO.B.8**
FORWARD 1	Prove angle properties of special parallelograms	**HSG-CO.C.11**
FORWARD 2	Opposite angles and angles of rotation with coordinates	**HSG-CO.A.3**

SET 101 • HSG-CO.C.11

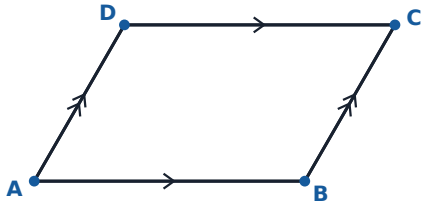
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram.
Prove that ∠B ≅ ∠D.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q1/5

1011

QUESTION ID
S101-M-Q1

TYPE
Answer with required work

CORRECT ANSWER
∠B ≅ ∠D

GRADING NOTE REQUIRED WORK: Use diagonal AC and the congruent triangles ABC and CDA (ASA), then corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BAC ≅ ∠DCA and ∠BCA ≅ ∠DAC (alternate interior); AC shared; △ABC ≅ △CDA (ASA); ∠B and ∠D are corresponding angles.

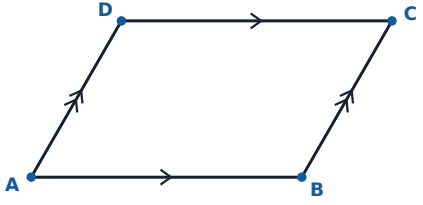
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram. Prove that ∠A ≅ ∠C using same-side interior angles (no diagonal).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q2/5

1012

QUESTION ID

S101-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

∠A ≅ ∠C

GRADING NOTE REQUIRED WORK: Show ∠A and ∠B are supplementary and ∠B and ∠C are supplementary, then subtract.

Any valid proof (paragraph, two-column, or flow). Essential: AD ∥ BC with transversal AB gives m∠A + m∠B = 180°; AB ∥ DC with transversal BC gives m∠B + m∠C = 180°; so m∠A = m∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

PQRS is a parallelogram. Prove that ∠P ≅ ∠R, using diagonal QS.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q3/5

1013

QUESTION ID

S101-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

∠P ≅ ∠R

GRADING NOTE REQUIRED WORK: Prove △PQS ≅ △RSQ by ASA, then use corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: ∠PQS ≅ ∠RSQ (alternate interior, PQ ∥ SR); ∠PSQ ≅ ∠RQS (alternate interior, PS ∥ QR); QS shared; ASA; ∠P and ∠R correspond.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram and M is the midpoint of diagonal AC. Explain, using the 180° rotation about M, why ∠DAB ≅ ∠BCD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q4/5

1014

QUESTION ID

S101-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The rotation maps ∠DAB onto ∠BCD, and rotations keep angle measure.

GRADING NOTE REQUIRED WORK: Use that the rotation swaps A and C and maps the sides at A onto the sides at C.

Any valid explanation. Essential: A ↔ C; line AB → the parallel line through C, which is CD; line AD → the parallel line through C, which is CB; ray AB goes to ray CD and ray AD to ray CB, so ∠DAB → ∠BCD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram with m∠A = x°. Prove that m∠C = x° and m∠B = m∠D = (180 − x)°, giving a reason for each step.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q5/5

1015

QUESTION ID

S101-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

m∠B = 180 − x,
m∠C = x,
m∠D = 180 − x

GRADING NOTE REQUIRED WORK: Use same-side interior angles for each pair of consecutive angles.

Any valid proof (paragraph, two-column, or flow). Essential: m∠A + m∠B = 180° (AD ∥ BC, transversal AB); m∠B + m∠C = 180° (AB ∥ DC); m∠C + m∠D = 180° (AD ∥ BC); solve each step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q5/5

SET 101 • HSG-CO.C.11

BACKWARD 1

Same-side interior angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Same-side interior angles formed by a transversal of parallel lines are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q1/5

1017

QUESTION ID

S101-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, side AB is a transversal of AD and BC. What is m∠A + m∠B?

A. It depends on the figure

B. 180°

C. 360°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q2/5

1018

QUESTION ID

S101-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 180°

GRADING NOTE Same-side interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, m∠A = 65°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q3/5

1019

QUESTION ID

S101-B1-Q3

TYPE

Short answer

CORRECT ANSWER

115°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Lines $\ell \parallel m$ are cut by a transversal; one same-side interior angle is 72°. Find the other.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q4/5

1020

QUESTION ID

S101-B1-Q4

TYPE

Short answer

CORRECT ANSWER

108°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Same-side interior angles of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q5/5

1021

QUESTION ID

S101-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q5/5

SET 101 • HSG-CO.C.11

BACKWARD 2

Name opposite and consecutive angles of a quadrilateral • Grade 5 • Geometry • 5.G.B.3

Name the two pairs of opposite angles of parallelogram WXYZ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q1/5

1023

QUESTION ID

S101-B2-Q1

TYPE

Short answer

CORRECT ANSWER

∠W and ∠Y; ∠X and ∠Z

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In quadrilateral ABCD, ∠A and ∠B are consecutive angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q2/5

1024

QUESTION ID
S101-B2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE They share side AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which angle is opposite ∠Q in quadrilateral PQRS?

A. ∠R

B. ∠P

C. ∠Q

D. ∠S

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q3/5

1025

QUESTION ID

S101-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. ∠S

GRADING NOTE Opposite vertices are not neighbors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q3/5

1025

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral KLMN: name the angles consecutive to ∠K.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q4/5

1026

QUESTION ID

S101-B2-Q4

TYPE

Short answer

CORRECT ANSWER

∠L and ∠N

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In quadrilateral ABCD, ∠A and ∠C share a side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q5/5

1027

QUESTION ID

S101-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are opposite angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q5/5

SET 101 • HSG-CO.C.11

BACKWARD 3

Prove triangles congruent with ASA and use corresponding angles • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

△ABC ≅ △CDA with A ↔ C, B ↔ D, C ↔ A. Which angle corresponds to ∠B?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q1/5

1029

QUESTION ID

S101-B3-Q1

TYPE

Short answer

CORRECT ANSWER

∠D

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q1/5

True or false?

After two triangles are proved congruent, their corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q2/5

1030

QUESTION ID

S101-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of congruence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q2/5

Which criterion uses two angles and the side between them?

A. ASA

B. SSS

C. SAS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q3/5

1031

QUESTION ID

S101-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ASA

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

△PQS ≅ △RSQ with P ↔ R. Which angle corresponds to ∠P?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q4/5

1032

QUESTION ID

S101-B3-Q4

TYPE

Short answer

CORRECT ANSWER

∠R

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q4/5

True or false?

ASA can use a shared diagonal as its side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q5/5

1033

QUESTION ID

S101-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q5/5

SET 101 • HSG-CO.C.11

FORWARD 1

Prove angle properties of special parallelograms • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

Prove that a parallelogram with one right angle has four right angles (so it is a rectangle).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q1/5

1035

QUESTION ID

S101-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

All four angles are 90°.

GRADING NOTE REQUIRED WORK: Use opposite angles congruent and consecutive angles supplementary.

Any valid proof (paragraph, two-column, or flow). Essential: the opposite angle is 90° (opposite angles); each consecutive angle is 180° − 90° = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, m∠A = (3x + 10)° and m∠C = (5x − 30)°. Find x and m∠B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q2/5

1036

QUESTION ID

S101-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 20; m∠B = 110°

GRADING NOTE Form required: both. 3x + 10 = 5x − 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Rhombus ABCD has m∠A = 60°. Prove that diagonal BD divides it into two equilateral triangles.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q3/5

1037

QUESTION ID

S101-F1-Q3

TYPE

Answer with required work

CORRECT ANSWER

△ABD and △CBD each have three 60° angles (and three equal sides).

GRADING NOTE REQUIRED WORK: Use the opposite angle, the equal sides, and the isosceles triangle angle facts.

Any valid proof (paragraph, two-column, or flow). Essential: AB = AD (rhombus), so △ABD is isosceles with vertex angle 60°, and its base angles are (180° − 60°) ÷ 2 = 60°: equilateral; m∠C = 60° (opposite angles) and CB = CD, so △CBD is equilateral too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q3/5

In parallelogram PQRS, m∠P is twice m∠Q. Find all four angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q4/5

1038

QUESTION ID

S101-F1-Q4

TYPE

Short answer

CORRECT ANSWER

∠P = ∠R = 120°,
∠Q = ∠S = 60°

GRADING NOTE Form required: all four. 2x + x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram has an angle of 47°. Which list gives all four angles in order?

A. 47°, 43°, 47°, 43°

B. 47°, 47°, 133°, 133°

C. 47°, 133°, 47°, 133°

D. 47°, 133°, 133°, 47°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q5/5

1039

QUESTION ID

S101-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 47°, 133°, 47°, 133°

GRADING NOTE Opposite angles are equal; consecutive angles add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q5/5

SET 101 • HSG-CO.C.11

FORWARD 2

Opposite angles and angles of rotation with coordinates • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Parallelogram ABCD has A(0, 0), B(4, 0), C(6, 3), D(2, 3). The 180° rotation about (3, 1.5) maps it onto itself. Which angle does ∠A map onto?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q1/5

1041

QUESTION ID

S101-F2-Q1

TYPE

Short answer

CORRECT ANSWER

∠C

GRADING NOTE Any equivalent form. A(0, 0) → (6, 3) = C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Parallelogram ABCD has A(0, 0), B(4, 0), C(6, 3), D(2, 3). Name the image of each vertex under the 180° rotation about (3, 1.5).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q2/5

1042

QUESTION ID

S101-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A → C, B → D, C → A, D → B

GRADING NOTE Form required: all four. (x, y) → (6 − x, 3 − y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q2/5

True or false?

A 180° rotation about the center of a parallelogram maps each angle onto the angle opposite it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q3/5

1043

QUESTION ID

S101-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It swaps opposite vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which transformation shows that opposite angles of every parallelogram are congruent?

A. A 90° rotation about a vertex

B. A translation along a side

C. A reflection across a diagonal

D. A 180° rotation about the intersection of the diagonals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q4/5

1044

QUESTION ID

S101-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A 180° rotation about the intersection of the diagonals

GRADING NOTE Reflections across diagonals work only for rhombuses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q4/5

Parallelogram PQRS has P(1, 1), Q(5, 1), R(7, 4), S(3, 4). Give the center and angle of a rotation that maps ∠P onto ∠R.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q5/5

1045

QUESTION ID

S101-F2-Q5

TYPE

Short answer

CORRECT ANSWER

180° about (4, 2.5)

GRADING NOTE Form required: the center and the angle. The center is the midpoint of PR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q5/5

SET 25 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that the diagonals of a parallelogram bisect each other

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Opposite sides of a parallelogram are congruent	**HSG-CO.C.11**
BACKWARD 2	Vertical and alternate interior angles at the crossing of the diago…	**HSG-CO.C.9**
BACKWARD 3	Find the midpoint of a segment and use the definition of bisect	**6.G.A.3**
FORWARD 1	Prove that a 180° rotation carries a parallelogram onto itself	**HSG-CO.A.3**
FORWARD 2	Use diagonal bisection with coordinates and algebra	**HSG-GPE.B.4**

SET 25 • HSG-CO.C.11

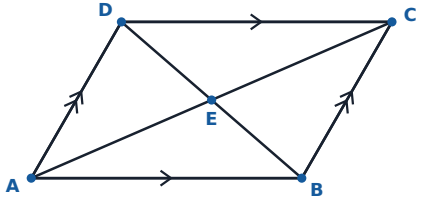
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram with diagonals AC and BD meeting at E. Prove that AE = CE and BE = DE.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q1/5

1048

QUESTION ID

S25-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AE = CE and BE = DE

GRADING NOTE REQUIRED WORK: Prove △ABE ≅ △CDE by ASA (or AAS), using alternate interior angles and AB = CD.

Any valid proof (paragraph, two-column, or flow). Essential: AB = CD (opposite sides of a parallelogram); ∠BAE ≅ ∠DCE and ∠ABE ≅ ∠CDE (alternate interior, AB ∥ DC); △ABE ≅ △CDE (ASA); corresponding sides. Vertical angles at E may replace one angle pair (AAS). Invalid: assuming E is a midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

PQRS is a parallelogram with diagonals PR and QS meeting at T. Prove that T is the midpoint of both diagonals.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q2/5

1049

QUESTION ID

S25-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

PT = TR and QT = TS, so T is the midpoint of both diagonals.

GRADING NOTE REQUIRED WORK: Prove △PTQ ≅ △RTS, using alternate interior angles and PQ = RS.

Any valid proof (paragraph, two-column, or flow). Essential: PQ = RS; two pairs of alternate interior angles (PQ ∥ SR); ASA; corresponding sides; definition of midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q2/5

A parallelogram has vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c). Prove that its diagonals bisect each other.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q3/5

1050

QUESTION ID

S25-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Both diagonals have midpoint ((a + b)/2, c/2).

GRADING NOTE REQUIRED WORK: Show the two diagonals have the same midpoint.

Any valid proof (paragraph, two-column, or flow). Essential: midpoint of AC = ((a + b)/2, c/2); midpoint of BD = ((a + b)/2, c/2); the same point lies on both diagonals and is the midpoint of each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram with diagonals meeting at E. Prove that AE = CE and BE = DE using triangles AED and CEB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q4/5

1051

QUESTION ID

S25-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AE = CE and BE = DE

GRADING NOTE REQUIRED WORK: Use AD = CB, alternate interior angles from AD ∥ BC, and ASA.

Any valid proof (paragraph, two-column, or flow). Essential: AD = CB (opposite sides); ∠DAE ≅ ∠BCE and ∠ADE ≅ ∠CBE (alternate interior, AD ∥ BC); △AED ≅ △CEB (ASA); corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

WXYZ is a parallelogram whose diagonals WY and XZ meet at V. Prove that V is the midpoint of WY and of XZ, using the 180° rotation about the midpoint M of WY (show that M is also the midpoint of XZ, so M = V).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q5/5

1052

QUESTION ID

S25-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The rotation swaps W and Y and maps X to Z, so M is the midpoint of XZ; the diagonals meet at M, so V = M.

GRADING NOTE REQUIRED WORK: Show that the rotation maps X to Z; then M is the midpoint of XZ too.

Any valid explanation. Essential: W ↔ Y; line WX → the parallel line through Y, which is YZ; line YX → the parallel line through W, which is WZ; X = WX ∩ YX → YZ ∩ WZ = Z; so M is the midpoint of XZ, and M lies on both diagonals, so it is their intersection V.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 1

Opposite sides of a parallelogram are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, AB = 9 and BC = 5. Find CD and DA.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q1/5

1054

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

CD = 9, DA = 5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q1/5

True or false?

Opposite sides of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q2/5

1055

QUESTION ID

S25-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Proved in the previous set.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q2/5

In parallelogram WXYZ, WX = 2x + 1 and YZ = x + 6. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q3/5

1056

QUESTION ID

S25-B1-Q3

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q3/5

In parallelogram PQRS, which side is congruent to QR?

A. PQ

B. PR

C. SP

D. RS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q4/5

1057

QUESTION ID

S25-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. SP

GRADING NOTE Opposite side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q4/5

True or false?

Consecutive sides of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q5/5

1058

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only in a rhombus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 2

Vertical and alternate interior angles at the crossing of the diagonals • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Diagonals of parallelogram ABCD meet at E. Which angle is vertical to ∠AEB?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q1/5

1060

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

∠CED

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q1/5

AB ∥ DC with transversal AC. Which angle is congruent to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q2/5

1061

QUESTION ID
S25-B2-Q2

TYPE
Short answer

CORRECT ANSWER
∠DCA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q2/5

True or false?

Vertical angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q3/5

1062

QUESTION ID

S25-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Proved earlier.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q3/5

AB ∥ DC with transversal BD. Which angle is congruent to ∠ABD?

A. ∠CDB

B. ∠ABC

C. ∠ADB

D. ∠CBD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q4/5

1063

QUESTION ID

S25-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. ∠CDB

GRADING NOTE Alternate interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q4/5

True or false?

∠AEB and ∠BEC are vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q5/5

1064

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They form a linear pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 3

Find the midpoint of a segment and use the definition of bisect • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (0, 0) to (a + b, c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q1/5

1066

QUESTION ID

S25-B3-Q1

TYPE

Short answer

CORRECT ANSWER

((a + b)/2, c/2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q1/5

Find the midpoint of the segment from (a, 0) to (b, c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q2/5

1067

QUESTION ID

S25-B3-Q2

TYPE

Short answer

CORRECT ANSWER

((a + b)/2, c/2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q2/5

True or false?

A segment bisects another segment if it passes through its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q3/5

1068

QUESTION ID

S25-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of bisect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Segment AC = 18 is bisected at E. Find AE.

AE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q4/5

1069

QUESTION ID

S25-B3-Q4

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q4/5

True or false?

If two segments have the same midpoint, each bisects the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q5/5

1070

QUESTION ID

S25-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Their common point is the midpoint of both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q5/5

SET 25 • HSG-CO.C.11

FORWARD 1

Prove that a 180° rotation carries a parallelogram onto itself • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

ABCD is a parallelogram whose diagonals meet at E. Prove that the 180° rotation about E maps ABCD onto itself.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q1/5

1072

QUESTION ID

S25-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

A ↔ C and B ↔ D, so the parallelogram maps onto itself.

GRADING NOTE REQUIRED WORK: Use the fact that E is the midpoint of both diagonals to find the images of the vertices.

Any valid proof (paragraph, two-column, or flow). Essential: a 180° rotation about E maps each point P to the point P′ with E the midpoint of PP′; E is the midpoint of AC and BD, so A → C, C → A, B → D, D → B; segments map to segments, so each side maps onto its opposite side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Parallelogram ABCD has diagonals meeting at E(2, 3), and A(−1, 1). Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q2/5

1073

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 5)

GRADING NOTE Form required: an ordered pair. E is the midpoint of AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD has diagonals that bisect each other at E. Explain why the 180° rotation about E carries ABCD onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q3/5

1074

QUESTION ID

S25-F1-Q3

TYPE

Short answer

CORRECT ANSWER

E is the midpoint of AC and of BD, so the rotation swaps A with C and B with D, and it maps each side onto the opposite side.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q3/5

The 180° rotation about the center of parallelogram PQRS maps side PQ onto

A. side SP

B. side RS

C. side QR

D. diagonal PR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q4/5

1075

QUESTION ID

S25-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. side RS

GRADING NOTE P → R and Q → S.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q4/5

A 180° rotation about the intersection of its diagonals carries a parallelogram onto itself. Explain why this implies that opposite angles of a parallelogram are congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q5/5

1076

QUESTION ID

S25-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The rotation maps each angle onto the opposite angle, and rotations keep angle measures.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSG-CO.C.11

FORWARD 2

Use diagonal bisection with coordinates and algebra • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Parallelogram ABCD has A(1, 2), B(7, 2), C(9, 6). Find D and the point where the diagonals meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q1/5

1078

QUESTION ID
S25-F2-Q1

TYPE
Short answer

CORRECT ANSWER
D(3, 6); diagonals meet at (5, 4)

GRADING NOTE Form required: both points. Midpoint of AC = (5, 4) = midpoint of BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q1/5

In parallelogram ABCD, diagonals meet at E.
AE = 3x − 2 and CE = x + 8. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q2/5

1079

QUESTION ID
S25-F2-Q2

TYPE
Short answer

CORRECT ANSWER
26

GRADING NOTE Any equivalent form. 3x − 2 = x + 8 gives x = 5; AE = 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q2/5

True or false?

The diagonals of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q3/5

1080

QUESTION ID

S25-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They bisect each other but are congruent only in a rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Three vertices of a parallelogram are (0, 0), (4, 0), (1, 3). Which could be the fourth vertex?

A. (4, 3)

B. (3, 3)

C. (5, 3)

D. (1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q4/5

1081

QUESTION ID

S25-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (5, 3)

GRADING NOTE (0, 0), (4, 0), (5, 3), (1, 3) is a parallelogram (others are possible: (3, −3), (−3, 3)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram PQRS, diagonals meet at T. PT = y + 4, TR = 2y − 1, QT = 3y. Find QS.

QS = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q5/5

1082

QUESTION ID
S25-F2-Q5

TYPE
Short answer

CORRECT ANSWER
30

GRADING NOTE Any equivalent form. y = 5, QT = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q5/5

SET 26 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove the converses: conditions that make a quadrilateral a parallelogram

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Converses of the parallel-line angle theorems	**HSG-CO.C.9**
BACKWARD 2	Prove triangles congruent with SSS and SAS	**HSG-CO.B.8**
BACKWARD 3	Angle sum of a quadrilateral	**8.G.A.5**
FORWARD 1	Prove that a quadrilateral is a parallelogram using coordinates	**HSG-GPE.B.4**
FORWARD 2	Decide which conditions are enough to guarantee a parallelogram	**HSG-CO.C.11**

SET 26 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

In quadrilateral ABCD, AB = CD and AD = CB. Prove that ABCD is a parallelogram. (Draw diagonal AC.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q1/5

1085

QUESTION ID

S26-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AB ∥ DC and AD ∥ BC, so ABCD is a parallelogram.

GRADING NOTE REQUIRED WORK: Prove △ABC ≅ △CDA by SSS, then use the converse of the alternate interior angles theorem twice.

Any valid proof (paragraph, two-column, or flow). Essential: SSS (AB = CD, BC = DA, AC shared); ∠BAC ≅ ∠DCA, so AB ∥ DC (converse); ∠BCA ≅ ∠DAC, so BC ∥ AD; definition. Invalid: using properties of parallelograms before proving it is one.

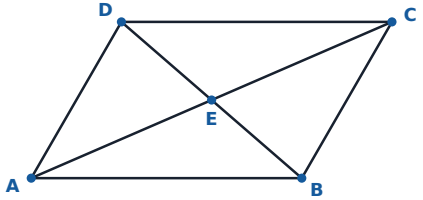
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrilateral ABCD, diagonals AC and BD bisect each other at E. Prove that ABCD is a parallelogram.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q2/5

1086

QUESTION ID

S26-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

AB ∥ DC and AD ∥ BC, so ABCD is a parallelogram.

GRADING NOTE REQUIRED WORK: Prove △AEB ≅ △CED and △AED ≅ △CEB by SAS (vertical angles), then use congruent alternate interior angles.

Any valid proof (paragraph, two-column, or flow). Essential: AE = CE, BE = DE (bisect); vertical angles; SAS twice; ∠EAB ≅ ∠ECD gives AB ∥ DC; ∠EAD ≅ ∠ECB gives AD ∥ BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q2/5

In quadrilateral ABCD, AB ∥ DC and AB = DC. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q3/5

1087

QUESTION ID

S26-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

AD ∥ BC, so with AB ∥ DC, ABCD is a parallelogram.

GRADING NOTE REQUIRED WORK: Draw diagonal AC; prove △ABC ≅ △CDA by SAS, then show AD ∥ BC.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BAC ≅ ∠DCA (alternate interior, AB ∥ DC); AB = CD; AC shared; SAS; ∠BCA ≅ ∠DAC; AD ∥ BC (converse).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrilateral ABCD, ∠A ≅ ∠C and ∠B ≅ ∠D. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q4/5

1088

QUESTION ID

S26-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AD ∥ BC and AB ∥ DC.

GRADING NOTE REQUIRED WORK: Use the quadrilateral angle sum to show consecutive angles are supplementary, then the converse of the same-side interior angles theorem.

Any valid proof (paragraph, two-column, or flow). Essential: 2m∠A + 2m∠B = 360°, so m∠A + m∠B = 180°; same-side interior angles supplementary gives AD ∥ BC; similarly m∠A + m∠D = 180° gives AB ∥ DC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrilateral ABCD, ∠A and ∠B are supplementary, and ∠B and ∠C are supplementary. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q5/5

1089

QUESTION ID

S26-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

AD ∥ BC and AB ∥ DC, so ABCD is a parallelogram.

GRADING NOTE REQUIRED WORK: Use the converse of the same-side interior angles theorem twice.

Any valid proof (paragraph, two-column, or flow). Essential: with transversal AB, same-side interior angles ∠A and ∠B are supplementary, so AD ∥ BC; with transversal BC, ∠B and ∠C are supplementary, so AB ∥ DC; definition of a parallelogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 1

Converses of the parallel-line angle theorems • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Two lines are cut by a transversal so that alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q1/5

1091

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The lines are parallel.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q1/5

True or false?

If same-side interior angles are supplementary, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q2/5

1092

QUESTION ID

S26-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q2/5

Which fact proves two lines are parallel?

A. Congruent corresponding angles

B. A linear pair

C. Two acute angles

D. Congruent vertical angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q3/5

1093

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Congruent corresponding angles

GRADING NOTE Corresponding angles converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q3/5

Same-side interior angles measure 104° and 76°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q4/5

1094

QUESTION ID

S26-B1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes (they add to 180°).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q4/5

True or false?

If alternate interior angles are 40° and 50°, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q5/5

1095

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They must be congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 2

Prove triangles congruent with SSS and SAS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

△ABC and △CDA share AC, and AB = CD, BC = DA.
Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q1/5

1097

QUESTION ID
S26-B2-Q1

TYPE
Short answer

CORRECT ANSWER
SSS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

△AEB and △CED: AE = CE, BE = DE, and ∠AEB ≅ ∠CED. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q2/5

1098

QUESTION ID
S26-B2-Q2

TYPE
Short answer

CORRECT ANSWER
SAS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q2/5

True or false?

Vertical angles can serve as the included angle in SAS.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q3/5

1099

QUESTION ID

S26-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

After △ABC ≅ △CDA, which angle is congruent to ∠BAC?

A. ∠BCA

B. ∠ABC

C. ∠DAC

D. ∠DCA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q4/5

1100

QUESTION ID

S26-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ∠DCA

GRADING NOTE A ↔ C, B ↔ D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

SSA proves two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q5/5

1101

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not a valid criterion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 3

Angle sum of a quadrilateral • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Three angles of a quadrilateral are 80°, 95°, and 110°. Find the fourth.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q1/5

1103

QUESTION ID

S26-B3-Q1

TYPE

Short answer

CORRECT ANSWER

75°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q1/5

True or false?

The angles of every quadrilateral add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q2/5

1104

QUESTION ID

S26-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A diagonal splits it into two triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A quadrilateral has angles x, x, 2x, 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q3/5

1105

QUESTION ID

S26-B3-Q3

TYPE

Short answer

CORRECT ANSWER

60

GRADING NOTE Any equivalent form. 6x = 360.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q3/5

Opposite angles of a quadrilateral are 70° and 70°, and another is 110°. Find the fourth.

A. 90°

B. 70°

C. 110°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q4/5

1106

QUESTION ID

S26-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 110°

GRADING NOTE 360 − 250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q4/5

True or false?

A quadrilateral can have four acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q5/5

1107

QUESTION ID

S26-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Four angles under 90° add to less than 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q5/5

SET 26 • HSG-CO.C.11

FORWARD 1

Prove that a quadrilateral is a parallelogram using coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove that A(1, 1), B(6, 2), C(8, 6), D(3, 5) form a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q1/5

1109

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

It is a parallelogram: the diagonals AC and BD have the same midpoint (4.5, 3.5), so they bisect each other.

GRADING NOTE Any valid method (slopes of opposite sides, or opposite side lengths, or diagonal midpoints) with the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q1/5

Prove or disprove that P(0, 0), Q(4, 1), R(5, 4), S(0, 3) form a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q2/5

1110

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Not a parallelogram: PQ has slope 1/4 but SR has slope 1/5, so PQ is not parallel to SR.

GRADING NOTE Any valid evidence and the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q2/5

True or false?

If the midpoints of both diagonals of a quadrilateral are the same point, the quadrilateral is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q3/5

1111

QUESTION ID
S26-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Diagonals bisect each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q3/5

Which test alone proves that ABCD is a parallelogram?

A. AC and BD have equal lengths

B. AB ⊥ BC

C. AB and DC have equal slopes and equal lengths

D. AB and DC have equal lengths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q4/5

1112

QUESTION ID

S26-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. AB and DC have equal slopes and equal lengths

GRADING NOTE One pair of opposite sides parallel and congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A(0, 0), B(5, 0), C(7, 3), D(x, 3) form a parallelogram ABCD. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q5/5

1113

QUESTION ID

S26-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. DC must equal AB = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSG-CO.C.11

FORWARD 2

Decide which conditions are enough to guarantee a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

Is a quadrilateral with one pair of opposite sides congruent and the other pair parallel always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q1/5

1115

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

No. An isosceles trapezoid is a counterexample.

GRADING NOTE Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q1/5

Is a quadrilateral with both pairs of opposite angles congruent always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q2/5

1116

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. The angle sum makes consecutive angles supplementary, so both pairs of opposite sides are parallel.

GRADING NOTE Any valid explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q2/5

True or false?

A quadrilateral with one diagonal that bisects the other must be a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q3/5

1117

QUESTION ID

S26-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A kite has one diagonal bisecting the other but is not a parallelogram in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q3/5

Which condition is NOT enough to prove a quadrilateral is a parallelogram?

A. One pair of opposite sides congruent

B. Both pairs of opposite sides congruent

C. Diagonals bisect each other

D. One pair of opposite sides parallel and congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q4/5

1118

QUESTION ID

S26-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. One pair of opposite sides congruent

GRADING NOTE One pair of congruent sides alone allows many shapes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q4/5

Is a quadrilateral with perpendicular diagonals always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q5/5

1119

QUESTION ID
S26-F2-Q5

TYPE
Short answer

CORRECT ANSWER
No. A kite (not a rhombus) has perpendicular diagonals but is not a parallelogram.

GRADING NOTE Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q5/5

SET 27 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that rectangles are parallelograms with congruent diagonals

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Parallelogram properties: opposite sides and consecutive angles	**HSG-CO.C.11**
BACKWARD 2	SAS and SSS with overlapping triangles	**HSG-CO.B.8**
BACKWARD 3	Use the Pythagorean Theorem for diagonals	**8.G.B.7**
FORWARD 1	Decide whether four points form a rectangle	**HSG-GPE.B.4**
FORWARD 2	Prove diagonal properties of rhombuses and squares	**HSG-CO.C.11**

SET 27 • HSG-CO.C.11

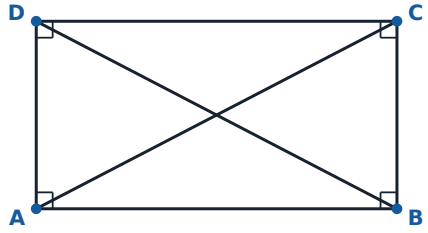
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a rectangle. Prove that AC = BD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q1/5

1122

QUESTION ID

S27-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AC = BD

GRADING NOTE REQUIRED WORK: Prove △ABC ≅ △BAD by SAS (right angles, shared side AB, and AD = BC).

Any valid proof (paragraph, two-column, or flow). Essential: a rectangle is a parallelogram, so AD = BC; ∠ABC ≅ ∠BAD (right angles); AB shared; SAS; corresponding sides AC = BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram with AC = BD. Prove that ABCD is a rectangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q2/5

1123

QUESTION ID

S27-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

All four angles are right angles, so ABCD is a rectangle.

GRADING NOTE REQUIRED WORK: Prove △ABC ≅ △BAD by SSS, show ∠A ≅ ∠B, then use that consecutive angles are supplementary.

Any valid proof (paragraph, two-column, or flow). Essential: BC = AD (opposite sides); AC = BD (given); AB shared; SSS; ∠DAB ≅ ∠CBA; they are supplementary (consecutive angles), so each is 90°; opposite angles then are 90° too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q2/5

Rectangle ABCD has A(0, 0), B(a, 0), C(a, b), D(0, b), with a, b > 0. Prove that its diagonals are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q3/5

1124

QUESTION ID

S27-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$AC = BD = \sqrt{(a^2 + b^2)}$

GRADING NOTE REQUIRED WORK: Use the distance formula on both diagonals.

Any valid proof (paragraph, two-column, or flow). Essential: $AC = \sqrt{(a^2 + b^2)}$ and $BD = \sqrt{(a^2 + b^2)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD has four right angles. Prove that ABCD is a parallelogram (so a rectangle is a parallelogram).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q4/5

1125

QUESTION ID

S27-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AB ∥ DC and AD ∥ BC.

GRADING NOTE REQUIRED WORK: Use the converse of the same-side interior angles theorem.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A + ∠B = 180° with transversal AB, so AD ∥ BC; ∠A + ∠D = 180° with transversal AD, so AB ∥ DC (or: both are perpendicular to the same line). Invalid: "it looks like a parallelogram".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The diagonals of quadrilateral ABCD bisect each other and are congruent. Prove that ABCD is a rectangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q5/5

1126

QUESTION ID

S27-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

ABCD is a parallelogram with congruent diagonals, so it is a rectangle.

GRADING NOTE REQUIRED WORK: First show ABCD is a parallelogram (diagonals bisect each other), then use the parallelogram-with-congruent-diagonals theorem or SSS.

Any valid proof (paragraph, two-column, or flow). Essential: diagonals that bisect each other make ABCD a parallelogram (set 26); then $\triangle ABC \cong \triangle BAD$ by SSS (BC = AD, AC = BD, AB shared), so $\angle A \cong \angle B$; consecutive angles are supplementary, so each is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 1

Parallelogram properties: opposite sides and consecutive angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, m∠A = 90°. Find the other three angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q1/5

1128

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

90°, 90°, 90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q1/5

True or false?

Consecutive angles of a parallelogram are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q2/5

1129

QUESTION ID

S27-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same-side interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram PQRS, m∠P = m∠Q. Find m∠P.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q3/5

1130

QUESTION ID

S27-B1-Q3

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form. They are also supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q3/5

In parallelogram ABCD, which side is congruent to AD?

A. AC

B. CD

C. AB

D. BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q4/5

1131

QUESTION ID

S27-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. BC

GRADING NOTE Opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q4/5

True or false?

In every parallelogram, all angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q5/5

1132

QUESTION ID

S27-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only in rectangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 2

SAS and SSS with overlapping triangles • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

△ABC and △BAD overlap and share AB. Which corresponding angle is at A in △BAD for ∠B in △ABC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q1/5

1134

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

∠BAD

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q1/5

True or false?

Overlapping triangles can share a side used in a congruence proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q2/5

1135

QUESTION ID

S27-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q2/5

△ABC and △BAD: AB shared, BC = AD, AC = BD. Which criterion?

A. SSS

B. ASA

C. SAS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q3/5

1136

QUESTION ID

S27-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. SSS

GRADING NOTE Three pairs of sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q3/5

△ABC and △BAD: AB shared, BC = AD, ∠ABC ≅ ∠BAD. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q4/5

1137

QUESTION ID

S27-B2-Q4

TYPE

Short answer

CORRECT ANSWER

SAS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q4/5

True or false?

In △ABC ≅ △BAD, vertex C corresponds to vertex C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q5/5

1138

QUESTION ID
S27-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE C ↔ D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 3

Use the Pythagorean Theorem for diagonals • Grade 8 • Geometry • 8.G.B.7

A rectangle is 6 by 8. Find the length of a diagonal.

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q1/5

1140

QUESTION ID

S27-B3-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q1/5

A rectangle is 5 by 12. Find the length of a diagonal.

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q2/5

1141

QUESTION ID

S27-B3-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q2/5

True or false?

A rectangle with sides a and b has diagonal $\sqrt{(a^2 + b^2)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q3/5

1142

QUESTION ID

S27-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q3/5

A square has side 4. Its diagonal is

A. 8

B. 4√2

C. 2√2

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q4/5

1143

QUESTION ID

S27-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4√2

GRADING NOTE √32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q4/5

True or false?

The two diagonals of a 3-by-7 rectangle have different lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q5/5

1144

QUESTION ID

S27-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both are √58.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • HSG-CO.C.11

FORWARD 1

Decide whether four points form a rectangle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove that A(1, 1), B(5, 3), C(4, 5), D(0, 3) form a rectangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q1/5

1146

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Rectangle: it is a parallelogram (AB ∥ DC, slope ½; AD ∥ BC, slope −2) and AB ⊥ AD (½ · −2 = −1).

GRADING NOTE Any valid method (or: diagonals bisect each other and are congruent, both 5 units long).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q1/5

Prove or disprove that P(0, 0), Q(4, 0), R(5, 3), S(1, 3) form a rectangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q2/5

1147

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Not a rectangle: it is a parallelogram, but its diagonals PR = √34 and QS = √18 are not congruent.

GRADING NOTE Any valid evidence and the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q2/5

True or false?

A quadrilateral whose diagonals are congruent and bisect each other is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q3/5

1148

QUESTION ID
S27-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Parallelogram with congruent diagonals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q3/5

Which set of facts proves ABCD is a rectangle?

A. Diagonals bisect each other and are congruent

B. Diagonals are congruent

C. One angle is a right angle

D. Opposite sides are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q4/5

1149

QUESTION ID

S27-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Diagonals bisect each other and are congruent

GRADING NOTE Congruent diagonals alone fit an isosceles trapezoid.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q4/5

A(0, 0), B(6, 0), and D(0, 4) are three vertices of rectangle ABCD. Find C and verify the diagonals are congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q5/5

1150

QUESTION ID

S27-F1-Q5

TYPE

Short answer

CORRECT ANSWER

C(6, 4);
AC = BD = √52
= 2√13.

GRADING NOTE Form required: C and both diagonal lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSG-CO.C.11

FORWARD 2

Prove diagonal properties of rhombuses and squares • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

Prove that a parallelogram whose diagonals are perpendicular is a rhombus.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q1/5

1152

QUESTION ID

S27-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

All four sides are congruent.

GRADING NOTE REQUIRED WORK: Use that the diagonals bisect each other, then SAS on two adjacent triangles.

Any valid proof (paragraph, two-column, or flow). Essential: diagonals bisect at E; right angles at E; shared segment; SAS gives AB = AD; opposite sides give the rest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q1/5

Which property must a quadrilateral have to be a square, using diagonals only?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q2/5

1153

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Its diagonals bisect each other, are congruent, and are perpendicular.

GRADING NOTE Any equivalent complete list.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q2/5

True or false?

A rectangle with congruent diagonals is a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q3/5

1154

QUESTION ID

S27-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every rectangle has congruent diagonals; a square needs perpendicular diagonals too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q3/5

A parallelogram has congruent, perpendicular diagonals. It is a

A. rectangle that is not a square

B. rhombus that is not a square

C. square

D. trapezoid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q4/5

1155

QUESTION ID

S27-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. square

GRADING NOTE Rectangle (congruent) and rhombus (perpendicular).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q4/5

A square has diagonal 10. Find its side length (exact) and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q5/5

1156

QUESTION ID

S27-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Side $5\sqrt{2}$; area 50

GRADING NOTE Form required: exact side and area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q5/5

Construct a copy of a segment

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Compare and measure lengths	**2.MD.A.1**
BACKWARD 2	A compass draws all points at one distance from its center	**7.G.B.4**
BACKWARD 3	Know rays, segments, and their endpoints	**HSG-CO.A.1**
FORWARD 1	Construct a triangle from three given segments	**HSG-CO.D.12**
FORWARD 2	Decide whether segments can form a triangle, and construct sums	**HSG-CO.D.12**

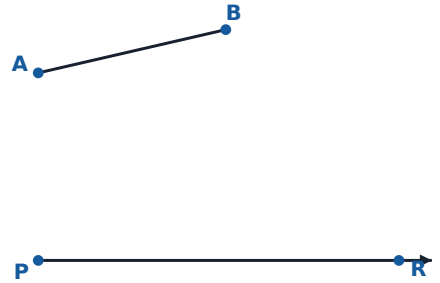
SET 28 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a point Q on ray PR so that PQ = AB.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q1/5

1159

QUESTION ID

S28-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A compass arc centered at P with radius AB crosses ray PR at Q; PQ = AB.

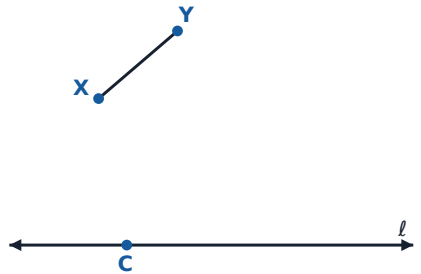
GRADING NOTE Grade the construction marks, not only the final figure: the compass set to AB (an arc or tick at A and B is allowed) and an arc centered at P meeting the ray at Q, which is labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

On line ℓ, starting at point C, construct a segment CD with CD = XY.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q2/5

1160

QUESTION ID

S28-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

An arc centered at C with radius XY crosses ℓ at D (either side of C is correct).

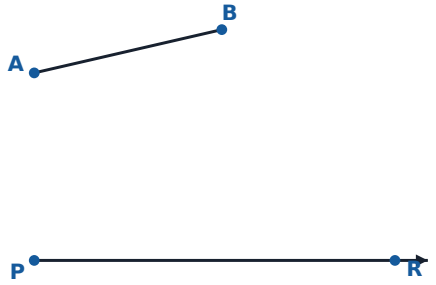
GRADING NOTE Grade the construction marks, not only the final figure: compass opening equal to XY, arc centered at C, D marked on ℓ. Either intersection point is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a point Q on ray PR so that PQ = 2 · AB.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q3/5

1161

QUESTION ID

S28-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Copy AB from P to get M on the ray, then copy AB again from M (away from P) to get Q; PQ = 2AB.

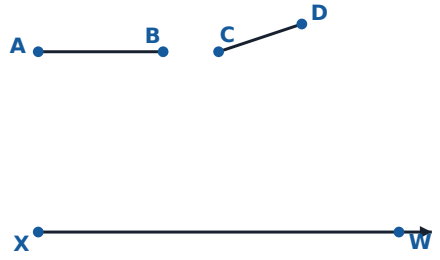
GRADING NOTE Grade the construction marks, not only the final figure: two arcs of radius AB, one centered at P and one centered at the first point, with Q labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

On ray XW, construct a point Z so that XZ = AB + CD.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q4/5

1162

QUESTION ID

S28-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Copy AB from X to get Y on the ray, then copy CD from Y (away from X) to get Z; XZ = AB + CD.

GRADING NOTE Grade the construction marks, not only the final figure: an arc of radius AB centered at X, an arc of radius CD centered at the first point, and Z labeled.

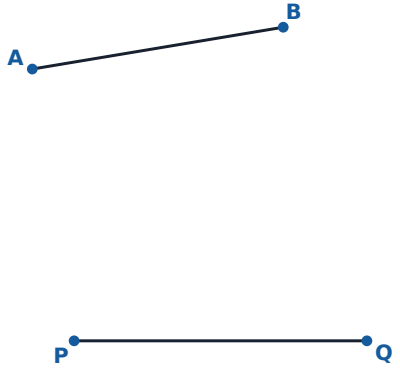
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct an isosceles triangle with base PQ whose two legs are each congruent to segment AB.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q5/5

1163

QUESTION ID

S28-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Arcs of radius AB centered at P and at Q meet at R; triangle PQR has PR = QR = AB.

GRADING NOTE Grade the construction marks, not only the final figure: two arcs with the compass set to AB, centered at P and Q, their crossing R, and the legs drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 1

Compare and measure lengths • Grade 2 • Measurement & Data • 2.MD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Segment AB is 6 cm and segment CD is 6 cm. Are they congruent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q1/5

1165

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Any equivalent answer: equal lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q1/5

True or false?

Two segments are congruent if they have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q2/5

1166

QUESTION ID

S28-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of congruent segments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q2/5

Which tool transfers a length without reading numbers?

A. A compass

B. A calculator

C. A protractor

D. A thermometer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q3/5

1167

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A compass

GRADING NOTE The compass keeps a fixed opening.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A segment 4.5 cm long is copied twice end to end on a line. How long is the result?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q4/5

1168

QUESTION ID

S28-B1-Q4

TYPE

Short answer

CORRECT ANSWER

9 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q4/5

True or false?

A segment drawn longer on the page than another is congruent to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q5/5

1169

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Congruent segments have equal lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 2

A compass draws all points at one distance from its center • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A compass is opened to 5 cm with its point on P. How far from P is every point on the arc it draws?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q1/5

1171

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q1/5

True or false?

Changing the compass opening between steps of a construction keeps the copy accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q2/5

1172

QUESTION ID

S28-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The opening must stay fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q2/5

An arc centered at P crosses a ray from P at Q. What is PQ?

A. It cannot be known

B. Half the opening

C. Twice the opening

D. The compass opening

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q3/5

1173

QUESTION ID

S28-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The compass opening

GRADING NOTE Q is on the arc, so it is one radius from P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q3/5

True or false?

Every point on a circle is the same distance from the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q4/5

1174

QUESTION ID

S28-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q4/5

An arc centered at C with radius 3 cm crosses a line through C. How many crossing points are there?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q5/5

1175

QUESTION ID

S28-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 3

Know rays, segments, and their endpoints • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Ray PR starts at P and continues through R without end.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q1/5

1177

QUESTION ID

S28-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of a ray.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q1/5

Q is on ray PR with PQ = 5. Must Q be between P and R? Describe all the cases.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q2/5

1178

QUESTION ID

S28-B3-Q2

TYPE

Short answer

CORRECT ANSWER

No. If PR > 5, Q is between P and R; if PR = 5, Q is R; if PR < 5, R is between P and Q.

GRADING NOTE Form required: all three cases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q2/5

Segment AB has endpoints

A. none

B. only B

C. only A

D. A and B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q3/5

1179

QUESTION ID

S28-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A and B

GRADING NOTE A segment has two endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q3/5

True or false?

A point on ray PR can be on the opposite side of P from R.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q4/5

1180

QUESTION ID

S28-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The ray goes only from P toward R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name the endpoint of ray PR.

Endpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q5/5

1181

QUESTION ID

S28-B3-Q5

TYPE

Short answer

CORRECT ANSWER

P

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q5/5

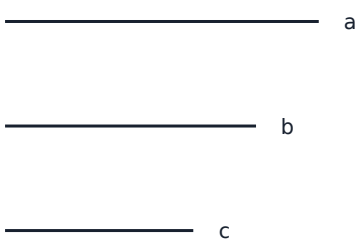
SET 28 • HSG-CO.D.12

FORWARD 1

Construct a triangle from three given segments • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a triangle whose sides are congruent to the three segments a, b, and c.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q1/5

1183

QUESTION ID

S28-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Copy a as base XY; arcs centered at X with radius b and at Y with radius c meet at Z; triangle XYZ has sides a, b, c.

GRADING NOTE Grade the construction marks, not only the final figure: the copied base, two arcs with radii b and c, and their intersection joined to the base endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q1/5

Explain why the two arcs in the triangle construction locate the third vertex.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q2/5

1184

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The third vertex must be b from one end of the base and c from the other, so it lies on both circles, where the arcs cross.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q2/5

True or false?

Any three segments can be used to construct a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q3/5

1185

QUESTION ID

S28-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The two arcs meet only if each length is less than the sum of the other two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q3/5

To construct a triangle from three given segments, arcs are drawn from the two ends of the base, and they cross at two points, one on each side of the base. The two triangles formed are

A. similar but not congruent

B. different in size

C. not triangles

D. congruent (mirror images)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q4/5

1186

QUESTION ID

S28-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. congruent (mirror images)

GRADING NOTE Reflection across the base swaps them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q4/5

Segments of length 3, 4, and 6 are given. Can a triangle be constructed? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q5/5

1187

QUESTION ID

S28-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Yes. 3 + 4 > 6, so the arcs meet.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSG-CO.D.12

FORWARD 2

Decide whether segments can form a triangle, and construct sums and differences • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Segments of length 2, 3, and 6 are given. Can a triangle be constructed? Explain what happens to the arcs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q1/5

1189

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

No. 2 + 3 < 6, so the arcs of radii 2 and 3 centered at the ends of the 6 base do not meet.

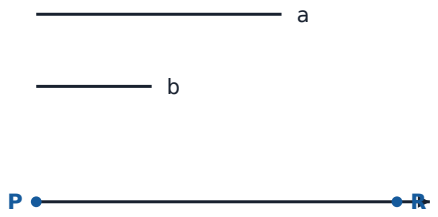
GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q1/5

Compass and straightedge only; leave all arcs.

On the ray from P, construct a segment of length a − b (a is the longer segment).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q2/5

1190

QUESTION ID

S28-F2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Copy a from P to Q on the ray; copy b from Q back toward P to S; PS = a − b.

GRADING NOTE Grade the construction marks, not only the final figure: two compass arcs (radius a at P, radius b at Q toward P), with the result labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q2/5

True or false?

If the arcs in a triangle construction just touch at one point on the base, the "triangle" is flat.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q3/5

1191

QUESTION ID

S28-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then one length equals the sum of the other two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q3/5

Which lengths CANNOT form a triangle?

A. 1, 2, 3

B. 3, 4, 5

C. 4, 4, 7

D. 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q4/5

1192

QUESTION ID

S28-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1, 2, 3

GRADING NOTE 1 + 2 = 3, so the figure is flat.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q4/5

Two sides of a constructed triangle are 5 and 8. For what third lengths will the arcs meet in two points?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q5/5

1193

QUESTION ID
S28-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Any length between 3 and 13 (3 < x < 13).

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q5/5

SET 102 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Make formal constructions with a variety of tools and methods

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Congruent segments have equal lengths	**2.MD.A.1**
BACKWARD 2	A reflection maps a point to its mirror image	**8.G.A.1**
FORWARD 1	Choose and justify a method for a construction	**HSG-CO.D.12**
FORWARD 2	Locate the circumcenter by folding	**HSG-C.A.3**

SET 102 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

You have no compass, only a straightedge and a strip of paper. Describe how to copy segment AB onto line ℓ starting at point C, and explain why the copy is exact.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q1/5

1196

QUESTION ID

S102-M-Q1

TYPE

Short answer

CORRECT ANSWER

Lay the strip along AB and mark the positions of A and B on its edge. Move the strip to ℓ, put the A-mark on C along ℓ, and mark the point at the B-mark: that is D. The strip carries the same length without measuring, so CD = AB.

GRADING NOTE Any valid method using a strip (or string) to transfer the length without reading a ruler, with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe how to construct the perpendicular bisector of segment AB by folding paper, and explain why the crease is the perpendicular bisector.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q2/5

1197

QUESTION ID

S102-M-Q2

TYPE

Short answer

CORRECT ANSWER

Fold so that A lands exactly on B and crease. The fold is a reflection that maps A to B, so by the definition of a reflection the crease is the perpendicular bisector of AB.

GRADING NOTE Any equivalent method and reason (the fold matches A onto B; a reflection line is the perpendicular bisector of PP′).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q2/5

Describe how to bisect an angle drawn on paper by folding, and explain why the crease bisects the angle.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q3/5

1198

QUESTION ID

S102-M-Q3

TYPE

Short answer

CORRECT ANSWER

Fold through the vertex so that one side lies exactly on the other side. The fold is a reflection that maps one side onto the other and fixes the vertex, so it makes two congruent angles.

GRADING NOTE Any equivalent method and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

You have a pencil, a pin, and a piece of string, but no compass. Describe how to construct an equilateral triangle on segment AB, and explain why it works.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q4/5

1199

QUESTION ID

S102-M-Q4

TYPE

Short answer

CORRECT ANSWER

Tie the string to the pin so the length from pin to pencil is AB. Pin it at A and draw an arc; pin it at B and draw an arc; the arcs cross at C. AC = BC = AB because the string length did not change.

GRADING NOTE Any valid string method that keeps one fixed length, with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In dynamic geometry software, describe how to construct the line through P parallel to line ℓ so that it stays parallel when P or ℓ is dragged, and explain why a line drawn "by eye" is not a construction.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q5/5

1200

QUESTION ID

S102-M-Q5

TYPE

Short answer

CORRECT ANSWER

Use the parallel-line tool (or construct a perpendicular to ℓ through P, then a perpendicular to that line at P). The software keeps the defining relationship when points move; a line placed by eye is not tied to ℓ and stops being parallel when anything is dragged.

GRADING NOTE Any valid software construction based on a defining relationship, and the drag-test reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q5/5

SET 102 • HSG-CO.D.12

BACKWARD 1

Congruent segments have equal lengths • Grade 2 • Measurement & Data • 2.MD.A.1

Segment PQ is 4.5 cm and segment RS is 45 mm. Are they congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q1/5

1202

QUESTION ID

S102-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: 45 mm = 4.5 cm, so the lengths are equal.

GRADING NOTE Any equivalent answer with the unit conversion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q1/5

True or false?

A strip of paper can carry a length from one place to another without a ruler.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q2/5

1203

QUESTION ID

S102-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Mark the two ends on the strip.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q2/5

Which of these can carry a length from one segment to another without a ruler?

A. A thermometer

B. A calculator

C. A folded paper strip marked at both ends

D. A protractor

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q3/5

1204

QUESTION ID

S102-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A folded paper strip marked at both ends

GRADING NOTE The marks keep a fixed length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A string is cut to the length of segment AB. It is laid along line ℓ from C. How long is the segment it marks?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q4/5

1205

QUESTION ID

S102-B1-Q4

TYPE

Short answer

CORRECT ANSWER

AB (the same length)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Two segments with different lengths can be congruent if they are drawn with the same tool.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q5/5

1206

QUESTION ID

S102-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Congruent segments have equal lengths, whatever tool is used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q5/5

SET 102 • HSG-CO.D.12

BACKWARD 2

A reflection maps a point to its mirror image • Grade 8 • Geometry • 8.G.A.1

True or false?

Folding paper along a line matches each point with its reflection across the fold.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q1/5

1208

QUESTION ID

S102-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The fold line is the mirror.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A paper is folded so that point A lands on point B. What is the fold line, in relation to segment AB?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q2/5

1209

QUESTION ID

S102-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A reflection keeps

A. neither

B. lengths and angle measures

C. only lengths

D. only angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q3/5

1210

QUESTION ID

S102-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. lengths and angle measures

GRADING NOTE Rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q3/5

True or false?

A point on the fold line moves when the paper is folded.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q4/5

1211

QUESTION ID

S102-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Points on the fold line stay fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Folding makes ray BA land on ray BC. What is true of ∠ABD and ∠DBC, where BD is the crease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q5/5

1212

QUESTION ID

S102-B2-Q5

TYPE

Short answer

CORRECT ANSWER

They are congruent.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q5/5

SET 102 • HSG-CO.D.12

FORWARD 1

Choose and justify a method for a construction • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which tool or method would you use to find the midpoint of a segment drawn on tracing paper, without a compass? Explain why it is exact.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q1/5

1214

QUESTION ID

S102-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Fold A onto B: the crease is the perpendicular bisector, so it crosses AB at the midpoint.

GRADING NOTE Any valid method with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A reflective device (a transparent mirror) is placed on line ℓ, and point P is seen in it. Explain how to use it to locate the reflection of P across ℓ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q2/5

1215

QUESTION ID

S102-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Look through the device placed along ℓ and mark the point where the image of P appears behind it; that point is P′, the reflection of P.

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why measuring a 60° angle with a protractor is not a formal construction, but stepping a compass around a circle is.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q3/5

1216

QUESTION ID

S102-F1-Q3

TYPE

Short answer

CORRECT ANSWER

A formal construction uses only the tools' exact operations (fixed lengths, lines through points), not readings of a scale; the protractor reading is an approximation.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe a way to construct the perpendicular from P to line ℓ by folding.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q4/5

1217

QUESTION ID

S102-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Fold the paper so that ℓ lands on itself and the crease passes through P; the crease is perpendicular to ℓ.

GRADING NOTE Any valid folding method with ℓ folded onto itself through P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q4/5

In dynamic geometry software, a student constructs a "square" by placing four points by eye. Describe a drag test that shows it is not a constructed square, and how to construct one properly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q5/5

1218

QUESTION ID

S102-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Drag a vertex: the figure stops being a square. Construct it from a segment with perpendiculars at both ends and circles of radius equal to the side, so it stays a square when dragged.

GRADING NOTE Any valid test and construction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q5/5

SET 102 • HSG-CO.D.12

FORWARD 2

Locate the circumcenter by folding • High School — Geometry • Geometry: Circles • HSG-C.A.3

Describe how to find the point equidistant from the three vertices of a paper triangle by folding.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q1/5

1220

QUESTION ID

S102-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Fold each of two vertices onto another (A onto B, then B onto C); the two creases are perpendicular bisectors and cross at the circumcenter.

GRADING NOTE Any valid folding method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q1/5

Why do two creases suffice, without folding the third pair of vertices?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q2/5

1221

QUESTION ID

S102-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The crossing point is equidistant from A and B and from B and C, so it is equidistant from A and C and lies on the third bisector too.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q2/5

True or false?

The point where the perpendicular-bisector creases of a triangle cross can lie outside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q3/5

1222

QUESTION ID

S102-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For an obtuse triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Folding vertex A onto vertex B of a triangle gives a crease that is

A. the perpendicular bisector of AB

B. the bisector of ∠C

C. the median from C

D. parallel to AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q4/5

1223

QUESTION ID

S102-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the perpendicular bisector of AB

GRADING NOTE The fold reflects A onto B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q4/5

After finding the circumcenter O by folding, how can you draw the circumscribed circle with a pin and string?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q5/5

1224

QUESTION ID

S102-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Pin the string at O and set its length to OA; draw the circle through A, which also passes through B and C.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q5/5

SET 29 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a copy of an angle

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Measure and draw angles with a protractor	**4.MD.C.6**
BACKWARD 2	Copy a segment with a compass	**HSG-CO.D.12**
BACKWARD 3	Arcs and circles in constructions	**7.G.B.4**
FORWARD 1	Justify why the angle-copy construction works	**HSG-CO.B.8**
FORWARD 2	Use angle copying to construct a parallel line	**HSG-CO.D.12**

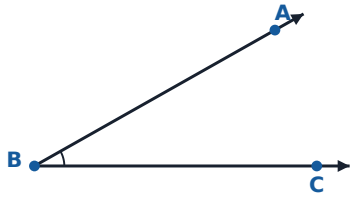
SET 29 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct an angle with vertex P and one side ray PR that is congruent to ∠ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q1/5

1227

QUESTION ID

S29-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Arc centered at B meets the sides at D and E; same arc centered at P meets ray PR at F; arc centered at F with radius DE meets the first arc at G; ray PG.

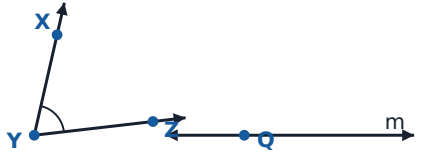
GRADING NOTE Grade the construction marks, not only the final figure: the arc at B, the same-radius arc at P, the arc with radius DE centered on ray PR, and ray PG drawn. The copy may open above or below ray PR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct an angle at point Q on line m congruent to ∠XYZ, opening upward.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q2/5

1228

QUESTION ID

S29-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same steps as copying an angle: arcs of the same radius at Y and Q, then the chord length transferred; the new ray from Q opens upward.

GRADING NOTE Grade the construction marks, not only the final figure: equal-radius arcs at Y and Q and the transferred chord arc.

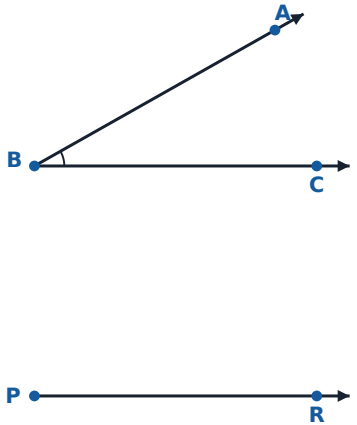
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct an angle with vertex P and one side ray PR that is congruent to ∠ABC, opening BELOW ray PR.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q3/5

1229

QUESTION ID

S29-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Equal-radius arcs at B and at P, the chord DE transferred from F on the lower side, and ray PG drawn below ray PR.

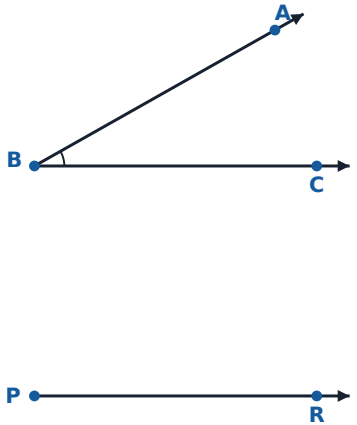
GRADING NOTE Grade the construction marks, not only the final figure: the arc at B, the same-radius arc at P, the chord arc centered on ray PR, and the new ray below PR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct an angle at P on ray PR whose measure is twice m∠ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q4/5

1230

QUESTION ID

S29-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Copy ∠ABC onto ray PR to get ray PG, then copy ∠ABC again onto ray PG (on the other side from R) to get ray PH; ∠RPH = 2∠ABC.

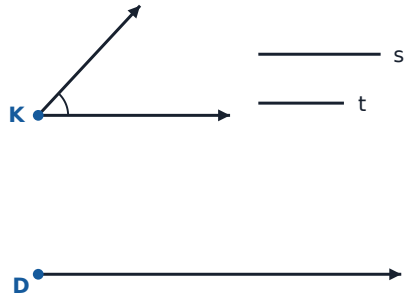
GRADING NOTE Grade the construction marks, not only the final figure: two complete angle copies sharing ray PG.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct triangle DEF with ∠D ≅ ∠K, DE = s, and DF = t.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q5/5

1231

QUESTION ID

S29-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Copy ∠K at D with one side on the ray; copy s along the ray from D to get E; copy t along the other side from D to get F; join E and F.

GRADING NOTE Grade the construction marks, not only the final figure: the complete angle copy at D and the two segment copies; triangle DEF drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 1

Measure and draw angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

ANSWER KEY Reduced copy of the question page, with its answer

A protractor shows a ray at 0° and the other ray at 65°. What is the angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q1/5

1233

QUESTION ID
S29-B1-Q1

TYPE
Short answer

CORRECT ANSWER
65°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q1/5

True or false?

An angle's measure does not change when its sides are drawn longer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q2/5

1234

QUESTION ID

S29-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Measure depends only on the opening.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which angle is obtuse?

A. 90°

B. 45°

C. 130°

D. 20°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q3/5

1235

QUESTION ID

S29-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 130°

GRADING NOTE Between 90° and 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q3/5

Two angles of 35° and 35° are placed side by side. What is the combined angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q4/5

1236

QUESTION ID

S29-B1-Q4

TYPE

Short answer

CORRECT ANSWER

70°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q4/5

True or false?

Two angles with sides of different lengths cannot be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q5/5

1237

QUESTION ID

S29-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only the measures matter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 2

Copy a segment with a compass • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

To copy ∠ABC to vertex P, an arc centered at B crosses BA at D and BC at E, and the same arc centered at P crosses ray PR at F. Which length is set on the compass next, to locate the second side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q1/5

1239

QUESTION ID
S29-B2-Q1

TYPE
Short answer

CORRECT ANSWER
DE, the chord between the two arc crossings

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q1/5

True or false?

To copy a length, set the compass to the length and do not change it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q2/5

1240

QUESTION ID

S29-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Fixed opening.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q2/5

An arc centered at F with radius DE crosses another arc at G. What is FG?

A. Half of DE

B. BD

C. DE

D. PF

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q3/5

1241

QUESTION ID

S29-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. DE

GRADING NOTE G is on the circle about F with radius DE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Arcs centered at B and at P are both drawn with radius 3 cm. How far is F (on the arc at P) from P?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q4/5

1242

QUESTION ID
S29-B2-Q4

TYPE
Short answer

CORRECT ANSWER
3 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q4/5

True or false?

A segment copy can be checked only by measuring with a ruler.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q5/5

1243

QUESTION ID

S29-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Compass marks are the evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 3

Arcs and circles in constructions • Grade 7 • Geometry • 7.G.B.4

True or false?

Two arcs centered at different points can cross at two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q1/5

1245

QUESTION ID

S29-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two circles can meet twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Points D and E are both on an arc centered at B with radius 4. Find BD and BE.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q2/5

1246

QUESTION ID

S29-B3-Q2

TYPE

Short answer

CORRECT ANSWER

BD = BE = 4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q2/5

Which describes an arc drawn with a compass?

A. Part of a circle

B. A ray

C. A straight segment

D. A polygon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q3/5

1247

QUESTION ID

S29-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Part of a circle

GRADING NOTE The compass draws circles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q3/5

True or false?

An arc centered at P passes through P.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q4/5

1248

QUESTION ID

S29-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Points on it are one radius away from P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q4/5

What does the point of the compass mark in a construction?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q5/5

1249

QUESTION ID

S29-B3-Q5

TYPE

Short answer

CORRECT ANSWER

The center of the arc

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSG-CO.D.12

FORWARD 1

Justify why the angle-copy construction works • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In copying ∠ABC to ∠FPG, BD = BE = PF = PG (same arc radius) and DE = FG (same compass setting). Prove that ∠DBE ≅ ∠FPG.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q1/5

1251

QUESTION ID

S29-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

∠DBE ≅ ∠FPG

GRADING NOTE REQUIRED WORK: Prove △DBE ≅ △FPG by SSS, then use corresponding angles.

Any valid proof. Essential: BD = PF, BE = PG, DE = FG (construction); SSS; corresponding angles ∠DBE and ∠FPG.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

∠XYZ is copied to vertex Q, opening downward: an arc of radius r centered at Y meets the sides at U and V, the same arc centered at Q meets the given ray at U′, and an arc of radius UV centered at U′ meets it at V′. Prove that ∠U′QV′ ≅ ∠XYZ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q2/5

1252

QUESTION ID

S29-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

∠U′QV′ ≅ ∠UYV, which is ∠XYZ.

GRADING NOTE

REQUIRED WORK: Name the three pairs of equal segments, then use SSS.

Any valid proof. Essential: YU = QU′ = r, YV = QV′ = r, UV = U′V′ (construction); △UYV ≅ △U′QV′ (SSS); corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

∠ABC is copied twice at P: first onto ray PR to give ray PG, then onto ray PG (on the side away from R) to give ray PH. Prove that m∠RPH = 2 · m∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q3/5

1253

QUESTION ID

S29-F1-Q3

TYPE

Answer with required work

CORRECT ANSWER

m∠RPH = m∠RPG + m∠GPH = 2m∠ABC

GRADING NOTE REQUIRED WORK: Use the justified copy twice and the angle addition postulate.

Any valid proof. Essential: each copy is congruent to ∠ABC (SSS justification); G is in the interior of ∠RPH, so the measures add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q3/5

In copying ∠ABC to P, a student keeps the arc radius but sets the compass to BD (a radius) instead of the chord DE for the last arc. Which step of the SSS justification fails, and what goes wrong with the copy?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q4/5

1254

QUESTION ID

S29-F1-Q4

TYPE

Short answer

CORRECT ANSWER

The third pair of sides is no longer DE = FG, so SSS does not apply; the new ray makes a 60° angle (an equilateral triangle) instead of ∠ABC unless ∠ABC is 60°.

GRADING NOTE Any equivalent explanation that names the missing side equality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q4/5

Explain why the angle-copy construction works using a rigid motion instead of SSS: there is a rigid motion taking B to P and ray BC onto ray PR.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q5/5

1255

QUESTION ID

S29-F1-Q5

TYPE

Answer with required work

CORRECT ANSWER

The motion sends E to F (same radius on ray PR) and D to the point on the arc at P that is DE from F on the chosen side, which is G; so ∠ABC → ∠FPG.

GRADING NOTE REQUIRED WORK: Follow D, E, F, G under the rigid motion and use the fact that two circles meet in at most one point on each side of a line.

Any valid explanation. Essential: rigid motions keep the distances BD, BE, DE; the image of D is on the arc at P and DE from F; on one side of line PR that point is G.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q5/5

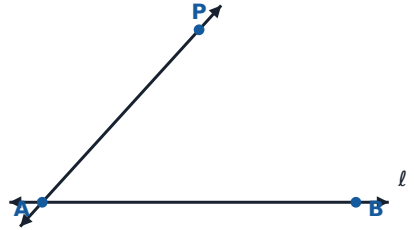
SET 29 • HSG-CO.D.12

FORWARD 2

Use angle copying to construct a parallel line • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ by copying an angle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q1/5

1257

QUESTION ID

S29-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Copy the angle that line AP makes with ℓ at A to vertex P, in the corresponding position; the new side through P is parallel to ℓ.

GRADING NOTE Grade the construction marks, not only the final figure: a complete angle copy at P (equal-radius arcs and chord) in the corresponding position, and the line drawn through P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q1/5

Why does copying a corresponding angle at P give a line parallel to ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q2/5

1258

QUESTION ID
S29-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Congruent corresponding angles make the lines parallel (converse of the corresponding angles theorem).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q2/5

True or false?

Copying the angle so the two angles are congruent same-side interior angles also produces a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q3/5

1259

QUESTION ID

S29-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Same-side interior angles must be supplementary, not congruent (unless both are 90°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q3/5

To construct the line through P parallel to ℓ by copying an angle, a transversal is drawn through P meeting ℓ. Which angle is copied?

A. The angle between ℓ and itself

B. The angle the transversal makes with ℓ

C. A right angle

D. Any 45° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q4/5

1260

QUESTION ID
S29-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. The angle the transversal makes with ℓ

GRADING NOTE The transversal through P and a point of ℓ creates the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q4/5

Describe a different way to construct a parallel through P using two perpendicular constructions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q5/5

1261

QUESTION ID

S29-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Construct the perpendicular from P to ℓ, then construct the perpendicular to that line at P; it is parallel to ℓ.

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q5/5

SET 30 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct the perpendicular bisector of a segment (bisect a segment)

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Points equidistant from A and B lie on the perpendicular bisector	**HSG-CO.C.9**
BACKWARD 2	Know midpoint and perpendicular bisector definitions	**HSG-CO.A.1**
BACKWARD 3	Arcs with one radius from two centers	**7.G.B.4**
FORWARD 1	Justify the perpendicular bisector construction	**HSG-CO.C.9**
FORWARD 2	Construct the circumcenter and circumscribed circle of a triangle	**HSG-C.A.3**

SET 30 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the perpendicular bisector of segment AB. Label the midpoint M.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q1/5

1264

QUESTION ID

S30-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Arcs of equal radius (more than half of AB) centered at A and at B meet at two points; the line through them is the perpendicular bisector, meeting AB at M.

GRADING NOTE Grade the construction marks, not only the final figure: two equal-radius arcs from A and B crossing on both sides of AB, the line through the crossings, and M labeled.

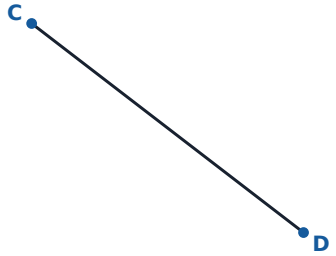
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the midpoint M of segment CD.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q2/5

1265

QUESTION ID

S30-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

The perpendicular bisector construction locates M where the line through the two arc crossings meets CD.

GRADING NOTE Grade the construction marks, not only the final figure: the two arc crossings and the line through them; M labeled on CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the point P on line ℓ that is the same distance from A as from B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q3/5

1266

QUESTION ID

S30-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular bisector of AB; the point P where it meets ℓ is equidistant from A and B.

GRADING NOTE Grade the construction marks, not only the final figure: the full perpendicular bisector construction for AB and its intersection with ℓ marked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the perpendicular bisector of the short vertical segment UV.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q4/5

1267

QUESTION ID

S30-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Equal arcs (radius more than half of UV) centered at U and V cross on the left and right; the horizontal line through the crossings bisects UV.

GRADING NOTE Grade the construction marks, not only the final figure: two equal-radius arcs from each endpoint, the two crossings, and the line through them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Divide segment AB into four congruent parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q5/5

1268

QUESTION ID

S30-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Bisect AB at M; then bisect AM and MB; the three points cut AB into four equal parts.

GRADING NOTE Grade the construction marks, not only the final figure: three perpendicular bisector constructions (for AB, AM, and MB).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 1

Points equidistant from A and B lie on the perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

A point equidistant from A and B lies on the perpendicular bisector of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q1/5

1270

QUESTION ID

S30-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse of the perpendicular bisector theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q1/5

Points P and Q are each equidistant from A and B. What line do P and Q determine?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q2/5

1271

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q2/5

In the perpendicular bisector construction for segment AB, the two arc crossings are equidistant from A and B because

A. the crossings are on AB

B. the arcs have the same radius

C. the compass is opened to AB

D. the arcs are drawn above AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q3/5

1272

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. the arcs have the same radius

GRADING NOTE Each crossing is one radius from A and one radius from B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q3/5

True or false?

Every point on the perpendicular bisector of AB is equidistant from A and B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q4/5

1273

QUESTION ID
S30-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q4/5

True or false?

Two points determine more than one line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q5/5

1274

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Exactly one line passes through two points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 2

Know midpoint and perpendicular bisector definitions • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

M is the midpoint of AB, and AM = 3.5. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q1/5

1276

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q1/5

True or false?

The perpendicular bisector of a segment passes through its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q2/5

1277

QUESTION ID

S30-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q2/5

A line crosses AB at its midpoint at a 60° angle. It is

A. the perpendicular bisector

B. not a bisector

C. parallel to AB

D. a bisector but not the perpendicular bisector

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q3/5

1278

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. a bisector but not the perpendicular bisector

GRADING NOTE It bisects but is not perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q3/5

Line k is perpendicular to AB and passes through its midpoint. What is k called?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q4/5

1279

QUESTION ID

S30-B2-Q4

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q4/5

True or false?

A segment has many perpendicular bisectors in one plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q5/5

1280

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Exactly one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 3

Arcs with one radius from two centers • Grade 7 • Geometry • 7.G.B.4

Two arcs of radius 5 are centered at A and B, and AB = 6. Do they cross?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q1/5

1282

QUESTION ID

S30-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Yes, at two points (5 + 5 > 6).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q1/5

True or false?

If the compass opening is less than half of AB, the arcs from A and B do not meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q2/5

1283

QUESTION ID

S30-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The two radii add to less than AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q2/5

Why must the opening be more than half of AB?

A. It does not matter

B. So the midpoint is on the arc

C. So the arcs from A and B cross

D. So the line is longer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q3/5

1284

QUESTION ID

S30-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. So the arcs from A and B cross

GRADING NOTE Equal radii must add to more than AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A point X is on both arcs (radius r) centered at A and B. Find XA and XB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q4/5

1285

QUESTION ID

S30-B3-Q4

TYPE

Short answer

CORRECT ANSWER

XA = XB = r

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q4/5

True or false?

The two arc crossings are on the same side of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q5/5

1286

QUESTION ID

S30-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE One is on each side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q5/5

SET 30 • HSG-CO.D.12

FORWARD 1

Justify the perpendicular bisector construction • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

To construct the perpendicular bisector of AB, arcs of radius r (more than half of AB) centered at A and B meet at P and Q. Prove that line PQ is the perpendicular bisector of AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q1/5

1288

QUESTION ID

S30-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

PQ is the perpendicular bisector of AB.

GRADING NOTE REQUIRED WORK: Use the converse of the perpendicular bisector theorem for P and for Q.

Any valid proof. Essential: PA = PB = r and QA = QB = r; each of P and Q is on the perpendicular bisector (converse); two points determine that line. Alternative: SSS gives congruent triangles APQ and BPQ, then SAS/linear pair at the midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q1/5

Explain why the perpendicular bisector construction for AB would still work with a larger compass opening.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q2/5

1289

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Any equal radius greater than half of AB gives two points equidistant from A and B.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q2/5

True or false?

The construction works only if the arcs are drawn with radius exactly AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q3/5

1290

QUESTION ID

S30-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any equal radius over half of AB works.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q3/5

Which theorem justifies that the arc crossings lie on the perpendicular bisector?

A. Midsegment theorem

B. Triangle angle sum

C. Converse of the perpendicular bisector theorem

D. Vertical angles theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q4/5

1291

QUESTION ID

S30-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Converse of the perpendicular bisector theorem

GRADING NOTE Equidistant points lie on the bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q4/5

Why does folding A onto B produce the perpendicular bisector?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q5/5

1292

QUESTION ID

S30-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The fold is a reflection swapping A and B, so the crease is the perpendicular bisector of AB (definition of reflection).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSG-CO.D.12

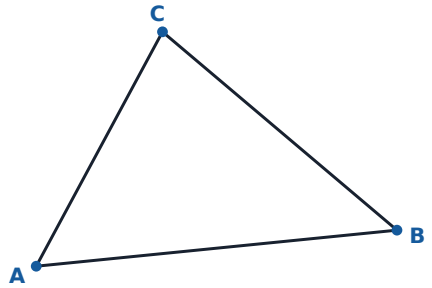
FORWARD 2

Construct the circumcenter and circumscribed circle of a triangle • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the point O equidistant from A, B, and C, and draw the circle through A, B, and C.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q1/5

1294

QUESTION ID

S30-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular bisectors of two sides; they meet at O; the circle centered at O through A also passes through B and C.

GRADING NOTE Grade the construction marks, not only the final figure: two complete perpendicular bisector constructions, O labeled, and the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q1/5

Why is it enough to construct only two perpendicular bisectors to find the circumcenter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2
High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q2/5

1295

QUESTION ID
S30-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Their crossing is equidistant from all three vertices, so it is on the third bisector too.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2
High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q2/5

True or false?

The circumcenter of an obtuse triangle lies on its longest side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q3/5

1296

QUESTION ID

S30-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It lies outside, beyond the longest side; it is on the longest side only for a right triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q3/5

Where is the circumcenter of a right triangle?

A. Inside, near the centroid

B. At the midpoint of the hypotenuse

C. Outside the triangle

D. At the right-angle vertex

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q4/5

1297

QUESTION ID

S30-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. At the midpoint of the hypotenuse

GRADING NOTE The hypotenuse is a diameter of the circumcircle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q4/5

A triangle has A(0, 0), B(8, 0), C(0, 6). Where is its circumcenter, and what is the radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q5/5

1298

QUESTION ID

S30-F2-Q5

TYPE

Short answer

CORRECT ANSWER

(4, 3); radius 5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q5/5

SET 31 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct the bisector of an angle

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Know the definition of an angle bisector	**HSG-CO.A.1**
BACKWARD 2	Prove triangles congruent with SSS	**HSG-CO.B.8**
BACKWARD 3	Measure angles with a protractor	**4.MD.C.6**
FORWARD 1	Justify the angle bisector construction	**HSG-CO.B.8**
FORWARD 2	Construct special angles and the incenter by bisecting	**HSG-CO.D.12**

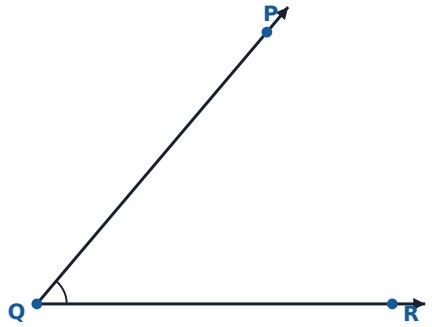
SET 31 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the bisector of ∠PQR.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q1/5

1301

QUESTION ID

S31-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Arc centered at Q crosses the sides at X and Y; equal arcs centered at X and Y meet at Z inside the angle; ray QZ is the bisector.

GRADING NOTE Grade the construction marks, not only the final figure: the arc at Q, two equal-radius arcs from X and Y crossing inside the angle, and ray QZ.

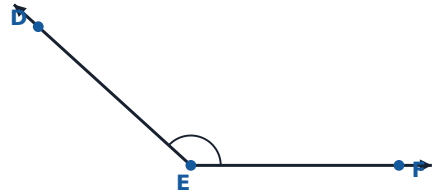
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the bisector of obtuse angle ∠DEF.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q2/5

1302

QUESTION ID

S31-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same steps: arc at E, equal arcs from the two crossing points, ray from E through their intersection.

GRADING NOTE Grade the construction marks, not only the final figure: all three arcs and the bisector ray.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a 45° angle with vertex A on line ℓ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q3/5

1303

QUESTION ID

S31-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular to ℓ at A (bisect the straight angle), then bisect one of the right angles.

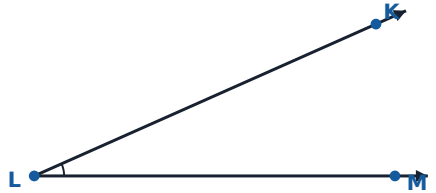
GRADING NOTE Grade the construction marks, not only the final figure: the perpendicular construction at A and the angle bisector construction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the bisector of the narrow angle ∠KLM.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q4/5

1304

QUESTION ID

S31-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Arc centered at L crosses both sides; equal arcs from those points cross inside the angle; ray from L through the crossing.

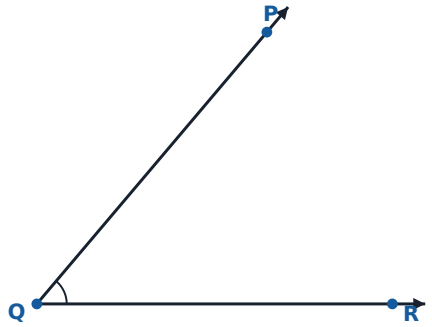
GRADING NOTE Grade the construction marks, not only the final figure: the arc at L, the two equal arcs, and the bisector ray.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct a ray QS inside ∠PQR so that m∠RQS = ¼ m∠PQR.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q5/5

1305

QUESTION ID

S31-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Bisect ∠PQR with ray QT, then bisect ∠TQR with ray QS.

GRADING NOTE Grade the construction marks, not only the final figure: two complete angle bisector constructions, the second inside the half next to ray QR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 1

Know the definition of an angle bisector • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Ray BD bisects ∠ABC, and m∠ABD = 27°. Find m∠ABC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q1/5

1307

QUESTION ID

S31-B1-Q1

TYPE

Short answer

CORRECT ANSWER

54°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q1/5

True or false?

An angle bisector divides the angle into two congruent angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q2/5

1308

QUESTION ID

S31-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q2/5

Ray BD bisects ∠ABC with m∠ABC = 3x + 6 and m∠ABD = 24. Find x.

A. 6

B. 10

C. 18

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q3/5

1309

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 14

GRADING NOTE 3x + 6 = 48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A 90° angle is bisected. What is each part?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q4/5

1310

QUESTION ID

S31-B1-Q4

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q4/5

True or false?

An angle bisector is a segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q5/5

1311

QUESTION ID

S31-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is a ray from the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 2

Prove triangles congruent with SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Two triangles share a side and have two more pairs of congruent sides. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q1/5

1313

QUESTION ID

S31-B2-Q1

TYPE

Short answer

CORRECT ANSWER

SSS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q1/5

True or false?

SSS guarantees that corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q2/5

1314

QUESTION ID

S31-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Congruent triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q2/5

To bisect ∠ABC, an arc centered at B crosses BA at X and BC at Y, and equal arcs centered at X and Y meet at Z. Which triangles are congruent?

A. △ABC and △XYZ

B. △BXZ and △BYZ

C. △BXY and △XYZ

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q3/5

1315

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. △BXZ and △BYZ

GRADING NOTE BX = BY, XZ = YZ, BZ shared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q3/5

In the angle-bisector construction for ∠ABC (arc at B meeting the sides at X and Y, equal arcs from X and Y meeting at Z), △BXZ ≅ △BYZ. Which angle is congruent to ∠XBZ?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q4/5

1316

QUESTION ID
S31-B2-Q4

TYPE
Short answer

CORRECT ANSWER
∠YBZ

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q4/5

True or false?

SSS requires knowing one angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q5/5

1317

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only three sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 3

Measure angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

ANSWER KEY Reduced copy of the question page, with its answer

A protractor reads 0° on one side and 112° on the other. Half of this angle is

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q1/5

1319

QUESTION ID

S31-B3-Q1

TYPE

Short answer

CORRECT ANSWER

56°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q1/5

True or false?

Measuring with a protractor is a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q2/5

1320

QUESTION ID

S31-B3-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Constructions use compass and straightedge.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

An angle of 150° is bisected. Each part is

A. 60°

B. 150°

C. 75°

D. 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q3/5

1321

QUESTION ID

S31-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 75°

GRADING NOTE 150 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

After bisecting, a student measures 38° and 38°. What was the original angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q4/5

1322

QUESTION ID

S31-B3-Q4

TYPE

Short answer

CORRECT ANSWER

76°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q4/5

True or false?

A protractor can check a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q5/5

1323

QUESTION ID

S31-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Measuring can check, though it does not replace the construction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q5/5

SET 31 • HSG-CO.D.12

FORWARD 1

Justify the angle bisector construction • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In bisecting ∠ABC, an arc centered at B crosses the sides at X and Y, and equal arcs centered at X and Y meet at Z. Prove that ray BZ bisects ∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q1/5

1325

QUESTION ID

S31-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

∠XBZ ≅ ∠YBZ

GRADING NOTE

REQUIRED WORK: Prove △BXZ ≅ △BYZ by SSS.

Any valid proof. Essential: BX = BY (same arc), XZ = YZ (equal arcs), BZ shared; SSS; corresponding angles; definition of bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q1/5

In the angle bisector construction, an arc centered at the vertex crosses the sides at X and Y, and then arcs from X and Y cross at Z. Why must the arcs from X and Y have the same radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q2/5

1326

QUESTION ID

S31-F1-Q2

TYPE

Short answer

CORRECT ANSWER

So that XZ = YZ, which SSS needs.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q2/5

True or false?

The bisector construction works for obtuse angles too.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q3/5

1327

QUESTION ID

S31-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The SSS argument does not depend on the size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q3/5

Which criterion justifies the angle bisector construction?

A. AAS

B. ASA

C. SSS

D. SAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q4/5

1328

QUESTION ID

S31-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. SSS

GRADING NOTE Three pairs of sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q4/5

Explain why folding one side onto the other gives the bisector.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q5/5

1329

QUESTION ID

S31-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The fold is a reflection that maps one side onto the other, so it makes two congruent angles at the vertex.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSG-CO.D.12

FORWARD 2

Construct special angles and the incenter by bisecting • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a 30° angle. (Hint: start from an equilateral triangle.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q1/5

1331

QUESTION ID

S31-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct an equilateral triangle on AB (60° angle at A), then bisect the angle at A to get 30°.

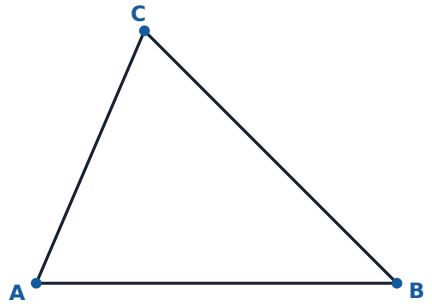
GRADING NOTE Grade the construction marks, not only the final figure: arcs of radius AB from A and B, the 60° angle, and its bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q1/5

Compass and straightedge only; leave all arcs.

Construct the point inside triangle ABC that is equidistant from its three sides.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q2/5

1332

QUESTION ID

S31-F2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Bisect two angles of the triangle; their intersection is the incenter.

GRADING NOTE Grade the construction marks, not only the final figure: two complete angle bisector constructions and the crossing labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q2/5

True or false?

Bisecting an angle and then bisecting each half divides the angle into three equal parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q3/5

1333

QUESTION ID

S31-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives four equal parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Starting from 60° and 90° angles, which angle CANNOT be constructed by bisecting angles and adding angles together?

A. 15°

B. 75°

C. 20°

D. 22.5°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q4/5

1334

QUESTION ID

S31-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 20°

GRADING NOTE Bisecting gives 30°, 15°, 45°, 22.5°, ... and 75° = 60° + 15°. 20° would require trisecting 60°, which these steps cannot do.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how to construct a 135° angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q5/5

1335

QUESTION ID

S31-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Construct a right angle, bisect it to get 45°, and add the 45° angle to the right angle (90° + 45°).

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q5/5

SET 32 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a perpendicular to a line through a point on or off the line

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Know the definition of perpendicular lines	**HSG-CO.A.1**
BACKWARD 2	Construct the perpendicular bisector of a segment	**HSG-CO.D.12**
BACKWARD 3	Recognize perpendicular lines in figures	**4.G.A.1**
FORWARD 1	Construct an altitude of a triangle and use it for area	**HSG-CO.D.12**
FORWARD 2	Construct a square on a given segment	**HSG-CO.D.12**

SET 32 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P perpendicular to line ℓ. (P is on ℓ.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q1/5

1338

QUESTION ID

S32-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Arcs centered at P mark A and B on ℓ equally far from P; then construct the perpendicular bisector of AB, which passes through P.

GRADING NOTE Grade the construction marks, not only the final figure: the two equal marks on ℓ and the perpendicular bisector arcs, with the line drawn through P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the line through P perpendicular to line ℓ. (P is not on ℓ.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q2/5

1339

QUESTION ID

S32-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

An arc centered at P crosses ℓ at A and B; equal arcs from A and B meet at Q below ℓ; line PQ is perpendicular to ℓ.

GRADING NOTE Grade the construction marks, not only the final figure: the arc from P meeting ℓ twice, equal arcs from A and B, and line PQ.

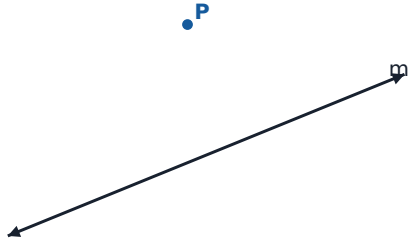
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the perpendicular from point P to the slanted line m.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q3/5

1340

QUESTION ID

S32-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Same method as for a point not on the line: arc from P meets m twice, equal arcs from those points meet, and the line through P and the crossing.

GRADING NOTE Grade the construction marks, not only the final figure: all arcs and the perpendicular line.

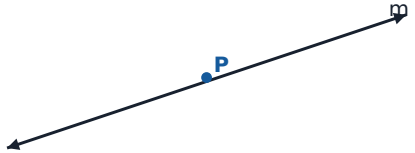
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q3/5

1340

Compass and straightedge only; leave all arcs.

Construct the line perpendicular to the slanted line m at the point P on m.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q4/5

1341

QUESTION ID

S32-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Arcs centered at P mark two points on m equally far from P; the perpendicular bisector of those two points passes through P.

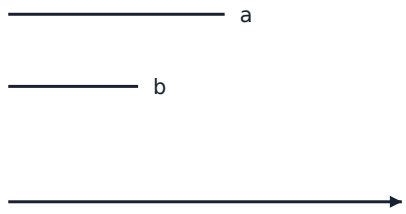
GRADING NOTE Grade the construction marks, not only the final figure: the two equal marks on m and the perpendicular bisector arcs, with the line drawn through P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct a right triangle whose legs are congruent to segments a and b.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q5/5

1342

QUESTION ID

S32-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Copy a onto the ray as XY; construct the perpendicular to the ray at X; copy b onto it as XZ; join Y and Z.

GRADING NOTE Grade the construction marks, not only the final figure: the copy of a, the perpendicular construction at X, the copy of b, and the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 1

Know the definition of perpendicular lines • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Perpendicular lines meet at a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q1/5

1344

QUESTION ID

S32-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q1/5

Line k ⊥ line ℓ. What is the measure of each angle they form?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q2/5

1345

QUESTION ID
S32-B1-Q2

TYPE
Short answer

CORRECT ANSWER
90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of lines is perpendicular?

A. y = 3 and y = −3

B. y = 2x and y = 2x + 1

C. y = x and y = 2x

D. y = 2x and y = −½x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q3/5

1346

QUESTION ID

S32-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 2x and y = −½x

GRADING NOTE Slopes multiply to −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q3/5

True or false?

Through a point, there are many lines perpendicular to a given line in a plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q4/5

1347

QUESTION ID

S32-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Exactly one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q4/5

The shortest path from a point to a line follows which kind of line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q5/5

1348

QUESTION ID

S32-B1-Q5

TYPE

Short answer

CORRECT ANSWER

The perpendicular from the point to the line

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 2

Construct the perpendicular bisector of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

After marking A and B on ℓ, which construction finishes the perpendicular?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q1/5

1350

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q1/5

True or false?

The perpendicular bisector construction needs arcs of equal radius from both endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q2/5

1351

QUESTION ID

S32-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equidistant crossings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q2/5

The arc crossings of the perpendicular bisector construction are

A. equidistant from A and B

B. on segment AB

C. on line ℓ

D. at A and B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q3/5

1352

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. equidistant from A and B

GRADING NOTE Same radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P is on ℓ, and A and B are marked on ℓ with PA = PB.
Where does the perpendicular bisector of AB cross ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q4/5

1353

QUESTION ID
S32-B2-Q4

TYPE
Short answer

CORRECT ANSWER
At P

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q4/5

True or false?

The perpendicular bisector of AB passes through A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q5/5

1354

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It passes through the midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 3

Recognize perpendicular lines in figures • Grade 4 • Geometry • 4.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A square's adjacent sides are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q1/5

1356

QUESTION ID

S32-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They meet at right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q1/5

Which letter is made of perpendicular segments?

A. V

B. T

C. O

D. S

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q2/5

1357

QUESTION ID

S32-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. T

GRADING NOTE T has a right angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q2/5

How many right angles do two perpendicular lines make?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q3/5

1358

QUESTION ID

S32-B3-Q3

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q3/5

True or false?

Two lines that look almost perpendicular must be perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q4/5

1359

QUESTION ID

S32-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Appearance is not proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name a pair of perpendicular edges of a rectangular door.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q5/5

1360

GRADING NOTE Any correct pair.

QUESTION ID

S32-B3-Q5

TYPE

Short answer

CORRECT ANSWER

A vertical side and the top (or bottom) edge

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q5/5

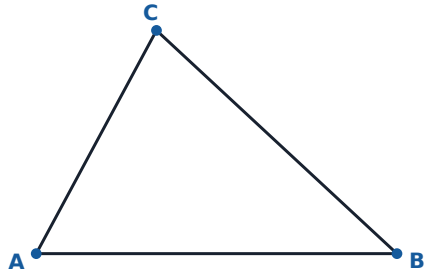
SET 32 • HSG-CO.D.12

FORWARD 1

Construct an altitude of a triangle and use it for area • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the altitude from C to side AB of triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q1/5

1362

QUESTION ID

S32-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular from C to line AB; the segment from C to the foot is the altitude.

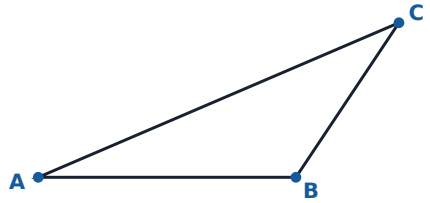
GRADING NOTE Grade the construction marks, not only the final figure: the perpendicular construction from C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q1/5

Compass and straightedge only; leave all arcs.

Construct the altitude from C in obtuse triangle ABC (the foot is outside AB). Extend AB as needed.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q2/5

1363

QUESTION ID

S32-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Extend line AB past B and construct the perpendicular from C to that line.

GRADING NOTE Grade the construction marks, not only the final figure: AB extended and the perpendicular construction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has base 10 and its constructed altitude measures 6. Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q3/5

1364

QUESTION ID

S32-F1-Q3

TYPE

Short answer

CORRECT ANSWER

30

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q3/5

True or false?

Every altitude of a triangle lies inside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q4/5

1365

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE An obtuse triangle has two altitudes outside it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q4/5

In a right triangle, the altitude from the right-angle vertex to the hypotenuse is

A. the hypotenuse

B. inside the triangle

C. outside the triangle

D. a leg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q5/5

1366

QUESTION ID

S32-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. inside the triangle

GRADING NOTE It meets the hypotenuse between its endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSG-CO.D.12

FORWARD 2

Construct a square on a given segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a square with side AB, above AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q1/5

1368

QUESTION ID

S32-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct perpendiculars to AB at A and at B; copy AB onto each, on the SAME side of AB (above it), to get D and C; join C and D.

GRADING NOTE Grade the construction marks, not only the final figure: two perpendicular constructions and two segment copies on the same side of AB. Copies on opposite sides give a crossed figure, not a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A square on segment AB is constructed by drawing perpendiculars to AB at A and at B and copying AB onto each, on the same side of line AB, to get D (from A) and C (from B). Explain why ABCD is a square, and why the two copies must be on the same side of AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q2/5

1369

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

AD ∥ BC (both ⊥ AB) and AD = BC, so ABCD is a parallelogram; it has a right angle at A, so it is a rectangle; AB = AD, so all sides are equal: a square. If D and C were on opposite sides, segment DC would cross AB and ABCD would not be a quadrilateral with these sides.

GRADING NOTE Any valid explanation with both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q2/5

True or false?

A quadrilateral with four congruent sides is always a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q3/5

1370

QUESTION ID

S32-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A rhombus need not have right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q3/5

To construct a rectangle 2AB long and AB wide, which additional step is needed?

A. Bisect AB

B. Construct a 60° angle

C. Draw a circle through A and B

D. Copy AB twice along the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q4/5

1371

QUESTION ID

S32-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Copy AB twice along the base

GRADING NOTE The base must be doubled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q4/5

A square is constructed on a segment of length 5. What is the length of its diagonal?

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q5/5

1372

QUESTION ID

S32-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$5\sqrt{2}$

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q5/5

SET 33 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a line parallel to a given line through a point not on it

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Copy an angle with compass and straightedge	**HSG-CO.D.12**
BACKWARD 2	Converse: congruent corresponding angles give parallel lines	**HSG-CO.C.9**
BACKWARD 3	Recognize parallel lines	**4.G.A.1**
FORWARD 1	Justify the parallel-line constructions	**HSG-CO.C.9**
FORWARD 2	Divide a segment into equal parts with parallel lines	**HSG-SRT.B.4**

SET 33 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q1/5

1375

QUESTION ID

S33-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Draw a transversal through P meeting ℓ at A; copy the angle it makes with ℓ at A to P in the corresponding position; the new side is parallel to ℓ.

GRADING NOTE Grade the construction marks, not only the final figure: a transversal, a complete angle copy at P (equal-radius arcs and transferred chord), and the parallel line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

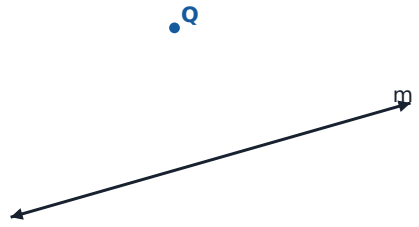
High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q1/5

1375

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the line through Q parallel to the slanted line m.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q2/5

1376

QUESTION ID

S33-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same method: a transversal through Q, then copy the corresponding (or alternate interior) angle at Q.

GRADING NOTE Grade the construction marks, not only the final figure: all arcs of the angle copy and the line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to ℓ using two perpendiculars.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q3/5

1377

QUESTION ID

S33-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular k from P to ℓ, then the perpendicular to k at P; it is parallel to ℓ.

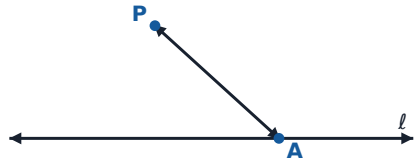
GRADING NOTE Grade the construction marks, not only the final figure: two complete perpendicular constructions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ by copying an ALTERNATE INTERIOR angle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q4/5

1378

QUESTION ID

S33-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Copy the angle between ℓ and line AP at A to vertex P, on the opposite side of line AP (alternate interior position); the new side through P is parallel to ℓ.

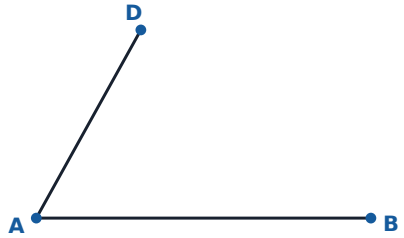
GRADING NOTE Grade the construction marks, not only the final figure: a complete angle copy at P in the alternate interior position and the line drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct parallelogram ABCD with sides AB and AD.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q5/5

1379

QUESTION ID

S33-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the line through D parallel to AB and the line through B parallel to AD; they meet at C.

GRADING NOTE Grade the construction marks, not only the final figure: two parallel-line constructions (angle copies or two perpendiculars) and the point C labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 1

Copy an angle with compass and straightedge • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

In copying an angle, which three lengths are transferred?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q1/5

1381

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The radius of the first arc (twice) and the chord between the side crossings

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q1/5

True or false?

An angle copy uses arcs of the same radius at both vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q2/5

1382

QUESTION ID

S33-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q2/5

Which step completes an angle copy?

A. Draw a circle

B. Measure with a protractor

C. Draw the ray from the new vertex through the last arc crossing

D. Bisect the angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q3/5

1383

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Draw the ray from the new vertex through the last arc crossing

GRADING NOTE The ray is the new side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q3/5

Why does the angle copy work?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q4/5

1384

QUESTION ID

S33-B1-Q4

TYPE

Short answer

CORRECT ANSWER

SSS makes the two triangles congruent.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q4/5

True or false?

An angle copy needs a protractor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q5/5

1385

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only compass and straightedge.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 2

Converse: congruent corresponding angles give parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Congruent corresponding angles mean the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q1/5

1387

QUESTION ID

S33-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q1/5

A transversal makes 62° angles with two lines in corresponding positions. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q2/5

1388

QUESTION ID

S33-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q2/5

Which also proves the lines parallel?

A. Congruent vertical angles

B. Supplementary linear pairs

C. Congruent alternate interior angles

D. Two acute angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q3/5

1389

QUESTION ID

S33-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Congruent alternate interior angles

GRADING NOTE Alternate interior converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q3/5

True or false?

Same-side interior angles of 70° and 70° prove the lines parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q4/5

1390

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They must add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Corresponding angles are (2x + 10)° and 50°. For which x are the lines parallel?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q5/5

1391

QUESTION ID

S33-B2-Q5

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 3

Recognize parallel lines • Grade 4 • Geometry • 4.G.A.1

True or false?

The opposite edges of a ruler are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q1/5

1393

QUESTION ID

S33-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q1/5

Which letter has parallel segments?

A. V

B. A

C. H

D. X

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q2/5

1394

QUESTION ID

S33-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. H

GRADING NOTE H's two uprights.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two railroad rails stay the same distance apart. What are they modeling?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q3/5

1395

QUESTION ID

S33-B3-Q3

TYPE

Short answer

CORRECT ANSWER

Parallel lines

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q3/5

True or false?

Lines that cross are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q4/5

1396

QUESTION ID

S33-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Parallel lines never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In a plane, how many lines through a point NOT on a given line are parallel to that line?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q5/5

1397

QUESTION ID

S33-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Parallel postulate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q5/5

SET 33 • HSG-CO.D.12

FORWARD 1

Justify the parallel-line constructions • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

To construct a line through P parallel to ℓ, a transversal through P meets ℓ at A, and the angle at A is copied at P in the corresponding position. Prove that the new line is parallel to ℓ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q1/5

1399

QUESTION ID

S33-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

The new line $\parallel \ell$.

GRADING NOTE REQUIRED WORK: Use the congruence of the copied angle (SSS) and the converse of the corresponding angles theorem.

Any valid proof. Essential: the copy is congruent (SSS from the construction); the angles are corresponding angles along AP; the converse gives parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the two-perpendicular construction, k ⊥ ℓ and n ⊥ k at P. Prove that n ∥ ℓ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q2/5

1400

QUESTION ID

S33-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

n ∥ ℓ

GRADING NOTE REQUIRED WORK: Both n and ℓ make right angles with k (corresponding angles congruent), or two lines perpendicular to the same line in a plane are parallel.

Any valid proof (paragraph, two-column, or flow). Essential: n ⊥ k and ℓ ⊥ k give a pair of congruent (90°) corresponding angles where k crosses n and ℓ; congruent corresponding angles give n ∥ ℓ. Alternative: two lines perpendicular to the same line are parallel (with that theorem cited). Invalid: "they look parallel".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q2/5

True or false?

Copying the alternate interior angle at P also gives a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q3/5

1401

QUESTION ID

S33-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Alternate interior converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q3/5

Which construction gives a line through P parallel to ℓ?

A. Copy, at P, the angle that a transversal through P makes with ℓ, in the corresponding position

B. Bisect the angle at P

C. Draw a circle centered at P

D. Draw the perpendicular bisector of the transversal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q4/5

1402

QUESTION ID

S33-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Copy, at P, the angle that a transversal through P makes with ℓ, in the corresponding position

GRADING NOTE Only the angle copy gives congruent corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q4/5

By the parallel postulate, how many lines could the construction produce?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q5/5

1403

QUESTION ID

S33-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Exactly one (the unique parallel through P).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSG-CO.D.12

FORWARD 2

Divide a segment into equal parts with parallel lines • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Divide segment AB into three congruent parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q1/5

1405

QUESTION ID

S33-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Draw a ray from A; mark three equal steps C, D, E on it with the compass; join E to B; construct parallels to EB through D and C; they cut AB into three equal parts.

GRADING NOTE Grade the construction marks, not only the final figure: three equal compass steps on the auxiliary ray and two parallel constructions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q1/5

To divide AB into three equal parts, three equal steps are marked on a ray from A, the last mark is joined to B, and parallels to that segment are drawn through the other marks. Why do the parallel lines divide AB into equal parts?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q2/5

1406

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Parallel lines cut the two sides of the triangle proportionally (triangle proportionality), so equal steps on the ray give equal parts of AB.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q2/5

True or false?

The same method can divide a segment into 5 equal parts only if its length is a multiple of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q3/5

1407

QUESTION ID

S33-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Equal compass steps on the auxiliary ray work for any segment; nothing is measured.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q3/5

To divide AB in the ratio 2 : 3, how many equal steps are marked on the auxiliary ray?

A. 2

B. 5

C. 3

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q4/5

1408

QUESTION ID

S33-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5

GRADING NOTE 2 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A segment of length 12 is divided into 4 parts this way. How long is each part?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q5/5

1409

QUESTION ID

S33-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q5/5

SET 34 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct an equilateral triangle inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Central angles of a regular polygon	**HSG-CO.A.3**
BACKWARD 2	Copy a length with a compass	**HSG-CO.D.12**
BACKWARD 3	Circle vocabulary: center, radius, chord, diameter	**7.G.B.4**
FORWARD 1	Justify that the inscribed triangle is equilateral	**HSG-C.A.2**
FORWARD 2	Construct a regular dodecagon by bisecting arcs	**HSG-CO.D.13**

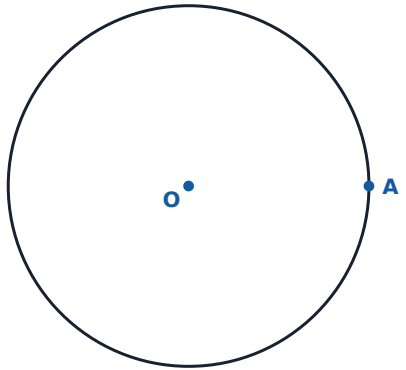
SET 34 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle, with one vertex at A.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q1/5

1412

QUESTION ID

S34-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

With the compass set to the radius, step around the circle from A marking six points; join every other point (A, the 2nd, the 4th).

GRADING NOTE Grade the construction marks, not only the final figure: radius-length arcs stepped around the circle (or the diameter-and-arcs method) and the triangle joining alternate points.

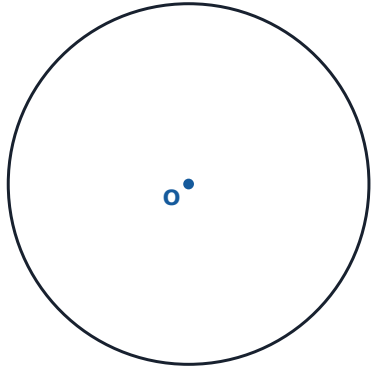
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle (choose any starting vertex).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q2/5

1413

QUESTION ID

S34-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Mark a point on the circle, step the radius around the circle to get six points, and join alternate points.

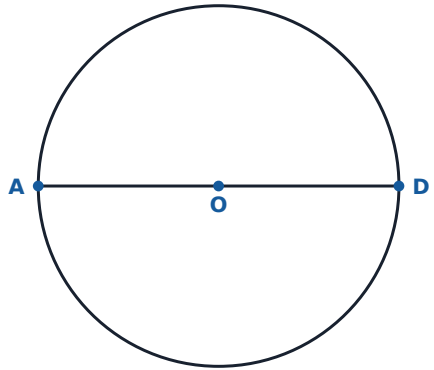
GRADING NOTE Grade the construction marks, not only the final figure: stepping arcs and the triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle using a diameter: draw diameter AD, then an arc centered at D with radius DO.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q3/5

1414

QUESTION ID

S34-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

The arc centered at D with radius DO meets the circle at B and C; triangle ABC is equilateral.

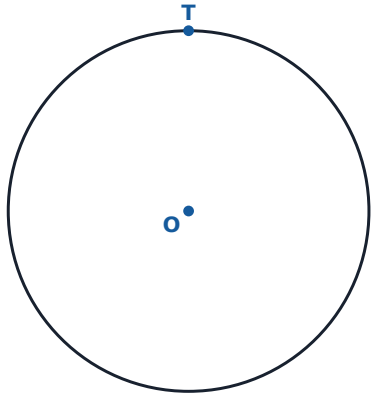
GRADING NOTE Grade the construction marks, not only the final figure: the arc at D and the triangle ABC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle with one vertex at the top point T.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q4/5

1415

QUESTION ID

S34-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Step the radius around the circle from T to mark six points; join T to the 2nd and 4th points.

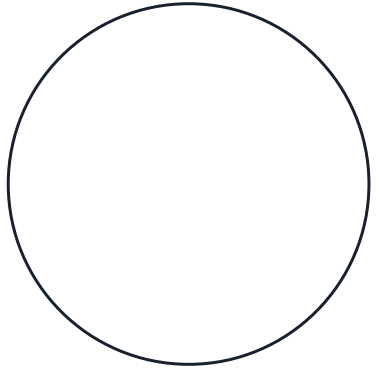
GRADING NOTE Grade the construction marks, not only the final figure: stepping arcs from T and the triangle joining alternate points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

The center of this circle is not marked. Construct an equilateral triangle inscribed in it.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q5/5

1416

QUESTION ID

S34-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Draw two chords and construct their perpendicular bisectors; they meet at the center O; then step the radius around the circle and join alternate points.

GRADING NOTE Grade the construction marks, not only the final figure: two perpendicular bisector constructions locating O, the compass set to the radius, and the triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 1

Central angles of a regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

What rotation carries an equilateral triangle onto itself, about its center (smallest positive)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q1/5

1418

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

120°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Into how many equal central angles does a regular hexagon divide 360°?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q2/5

1419

QUESTION ID

S34-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6 (each 60°)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q2/5

True or false?

The vertices of an inscribed equilateral triangle are 120° apart around the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q3/5

1420

QUESTION ID

S34-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 360 ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q3/5

Joining every other vertex of a regular hexagon gives

A. a square

B. a rectangle

C. an equilateral triangle

D. a pentagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q4/5

1421

QUESTION ID

S34-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. an equilateral triangle

GRADING NOTE Every other vertex: 120° apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q4/5

True or false?

The central angles of an inscribed equilateral triangle are 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q5/5

1422

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 2

Copy a length with a compass • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

The compass is set to the radius of a circle. What length does each step around the circle mark?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q1/5

1424

QUESTION ID

S34-B2-Q1

TYPE

Short answer

CORRECT ANSWER

A chord equal to the radius

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q1/5

True or false?

Keeping the compass opening fixed is required when stepping around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q2/5

1425

QUESTION ID

S34-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All chords must be equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q2/5

Each radius-length chord and the two radii to its ends form

A. a square

B. an equilateral triangle

C. a right triangle

D. an isosceles triangle with a 90° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q3/5

1426

QUESTION ID

S34-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. an equilateral triangle

GRADING NOTE All three sides equal the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q3/5

A circle has radius 4 cm. How long is each chord marked by stepping the radius?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q4/5

1427

QUESTION ID

S34-B2-Q4

TYPE

Short answer

CORRECT ANSWER

4 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q4/5

True or false?

Stepping a compass set to the diameter marks six points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q5/5

1428

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The diameter fits only twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 3

Circle vocabulary: center, radius, chord, diameter • Grade 7 • Geometry • 7.G.B.4

What is a chord of a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q1/5

1430

QUESTION ID

S34-B3-Q1

TYPE

Short answer

CORRECT ANSWER

A segment with both endpoints on the circle

GRADING NOTE Any equivalent definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q1/5

True or false?

A diameter is a chord through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q2/5

1431

QUESTION ID

S34-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q2/5

A circle has radius 6. Its diameter is

A. 3

B. 36

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q3/5

1432

QUESTION ID

S34-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12

GRADING NOTE Twice the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q3/5

What does "inscribed in a circle" mean for a polygon?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q4/5

1433

QUESTION ID

S34-B3-Q4

TYPE

Short answer

CORRECT ANSWER

All its vertices lie on the circle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q4/5

True or false?

An inscribed triangle can have a vertex inside the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q5/5

1434

QUESTION ID

S34-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE All vertices are on the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q5/5

SET 34 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed triangle is equilateral • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Points A, B, C, D, E, F are marked around a circle with center O by stepping the radius. Prove that triangle ACE is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q1/5

1436

QUESTION ID

S34-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AC = CE = EA

GRADING NOTE REQUIRED WORK: Show each central angle between consecutive marks is 60°, so AOC, COE, EOA are each 120°, then use congruent triangles (SAS).

Any valid proof. Essential: each small triangle (O, two consecutive marks) is equilateral, so each central angle is 60°; ∠AOC = ∠COE = ∠EOA = 120°; triangles AOC, COE, EOA are congruent by SAS (radii); so AC = CE = EA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center O and diameter AD. An arc centered at D with radius DO meets the circle at B and C. Prove that triangle ABC is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q2/5

1437

QUESTION ID

S34-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

AB = BC = CA

GRADING NOTE REQUIRED WORK: Show △DOB and △DOC are equilateral, find the central angles, and use SAS (or inscribed angles).

Any valid proof. Essential: DB = DO = OB, so △DOB is equilateral and ∠DOB = 60°; likewise ∠DOC = 60°; so ∠BOC = 120° and ∠AOB = ∠AOC = 180° − 60° = 120°; SAS with radii gives AB = BC = CA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q2/5

Explain why stepping a compass set to the radius around a circle returns exactly to the starting point after six steps.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q3/5

1438

QUESTION ID

S34-F1-Q3

TYPE

Answer with required work

CORRECT ANSWER

Each step makes a 60° central angle, and 6 × 60° = 360°.

GRADING NOTE REQUIRED WORK: Use the equilateral triangles formed with the center.

Any valid explanation. Essential: each chord equals the radius, so the triangle with the center is equilateral (60° at O); six steps fill 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the stepped construction, points A, B, C, D, E, F lie on a circle with center O, each 60° from the next. Prove that ∠CAE = 60°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q4/5

1439

QUESTION ID

S34-F1-Q4

TYPE

Answer with required work

CORRECT ANSWER

m∠CAE = 60°

GRADING NOTE REQUIRED WORK: Use the inscribed angle theorem (or the isosceles triangles AOC and AOE).

Any valid proof. Essential: arc CE (not containing A) measures 120°, and an inscribed angle is half its arc. Alternative: ∠OAC = ∠OAE = 30° from isosceles triangles with 120° apex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q4/5

Ana inscribes a triangle by marking points every 120° around the center with a protractor. Explain why her triangle is equilateral, and why her method is not a compass-and-straightedge construction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q5/5

1440

QUESTION ID

S34-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Equal central angles of 120° cut off equal chords (SAS with radii), so the sides are equal. Measuring with a protractor is not one of the allowed construction tools.

GRADING NOTE Any equivalent explanation with both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q5/5

SET 34 • HSG-CO.D.13

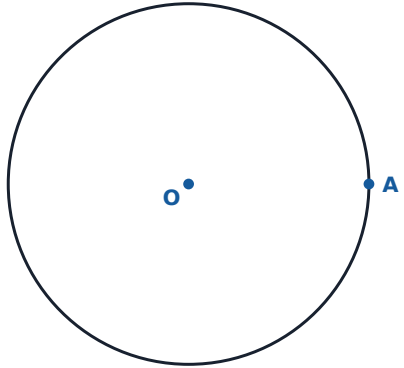
FORWARD 2

Construct a regular dodecagon by bisecting arcs • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a regular 12-sided polygon inscribed in the circle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q1/5

1442

QUESTION ID

S34-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Mark six points by stepping the radius, then bisect each of the six arcs (perpendicular bisector of each chord) to get six more; join the 12 points in order.

GRADING NOTE Grade the construction marks, not only the final figure: the six stepped points and the bisector constructions for the arcs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Each side of the inscribed regular hexagon subtends 60° at the center. What does each side of the dodecagon subtend?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q2/5

1443

QUESTION ID
S34-F2-Q2

TYPE
Short answer

CORRECT ANSWER
30°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Bisecting the arcs of an inscribed regular hexagon gives the vertices of a regular nonagon.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q3/5

1444

QUESTION ID

S34-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Doubling the vertices gives a regular 12-gon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q3/5

Starting from an inscribed square, bisecting arcs gives a regular

A. pentagon

B. hexagon

C. decagon

D. octagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q4/5

1445

QUESTION ID

S34-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. octagon

GRADING NOTE 4 → 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q4/5

How many sides does the polygon have after bisecting the arcs of an inscribed equilateral triangle twice?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q5/5

1446

QUESTION ID

S34-F2-Q5

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE 3 → 6 → 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q5/5

SET 35 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct a square inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Construct a perpendicular bisector	**HSG-CO.D.12**
BACKWARD 2	Diagonals of a square	**HSG-CO.C.11**
BACKWARD 3	Diameters and the center of a circle	**7.G.B.4**
FORWARD 1	Justify that the inscribed quadrilateral is a square	**HSG-CO.C.11**
FORWARD 2	Construct a regular octagon inscribed in a circle	**HSG-CO.D.13**

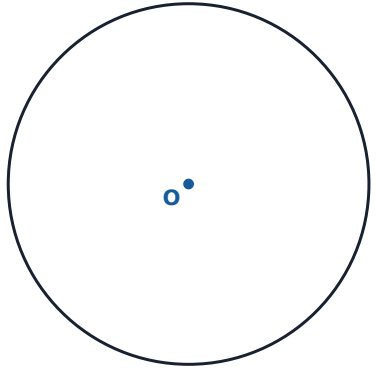
SET 35 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q1/5

1449

QUESTION ID

S35-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Draw a diameter AC through O; construct the perpendicular bisector of AC (a second diameter BD); join A, B, C, D.

GRADING NOTE Grade the construction marks, not only the final figure: a diameter, the perpendicular bisector construction for it (arcs), and the square joining the four endpoints.

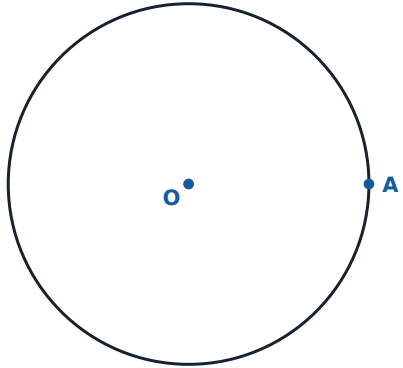
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle with one vertex at A.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q2/5

1450

QUESTION ID

S35-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Draw diameter AC through O; construct the perpendicular to AC at O, meeting the circle at B and D; join ABCD.

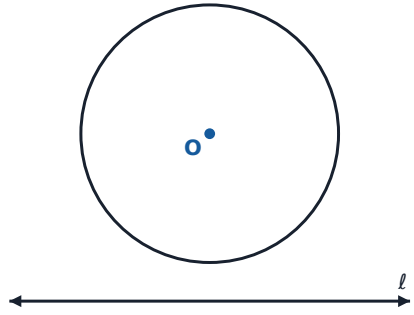
GRADING NOTE Grade the construction marks, not only the final figure: diameter through A and the perpendicular construction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle with one side parallel to the given line ℓ.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q3/5

1451

QUESTION ID

S35-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular from O to ℓ (one diameter); bisect the right angles at O to get diagonals at 45°; their endpoints form a square with sides parallel to ℓ.

GRADING NOTE Grade the construction marks, not only the final figure: the perpendicular, the angle bisectors, and the square. Accept any valid method.

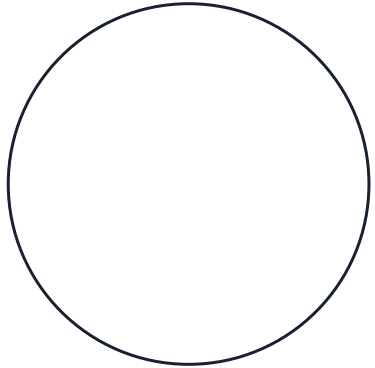
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

The center of this circle is not marked. Construct a square inscribed in it.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q4/5

1452

QUESTION ID

S35-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Locate the center O with the perpendicular bisectors of two chords; draw a diameter through O; construct the perpendicular diameter; join the four endpoints.

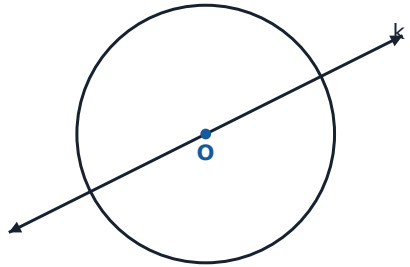
GRADING NOTE Grade the construction marks, not only the final figure: the center construction, two perpendicular diameters, and the square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle so that one of its diagonals lies along line k.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q5/5

1453

QUESTION ID

S35-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Line k meets the circle at A and C; construct the perpendicular to k at O, meeting the circle at B and D; join ABCD.

GRADING NOTE Grade the construction marks, not only the final figure: the perpendicular construction at O and the square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 1

Construct a perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction turns one diameter into two perpendicular diameters?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q1/5

1455

QUESTION ID

S35-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of the diameter

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector of a diameter passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q2/5

1456

QUESTION ID

S35-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The center is the diameter's midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q2/5

Two perpendicular diameters cut the circle into

A. six 60° arcs

B. four 45° arcs

C. four 90° arcs

D. two 90° arcs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q3/5

1457

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. four 90° arcs

GRADING NOTE 360 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q3/5

A diameter is 10 cm. Where does its perpendicular bisector cross it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q4/5

1458

QUESTION ID

S35-B1-Q4

TYPE

Short answer

CORRECT ANSWER

At the center, 5 cm from each end

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q4/5

True or false?

A perpendicular bisector can be drawn by measuring with a ruler only.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q5/5

1459

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A construction uses compass and straightedge.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 2

Diagonals of a square • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

The diagonals of a square are congruent and perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q1/5

1461

QUESTION ID

S35-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A square is a rectangle and a rhombus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q1/5

A quadrilateral's diagonals are congruent, perpendicular, and bisect each other. What is its most specific name?

A. rhombus that is not a square

B. square

C. trapezoid

D. kite that is not a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q2/5

1462

QUESTION ID

S35-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. square

GRADING NOTE All three properties together give a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A square has diagonal 8. Find its side.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q3/5

1463

QUESTION ID

S35-B2-Q3

TYPE

Short answer

CORRECT ANSWER

4√2

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q3/5

True or false?

A rectangle always has perpendicular diagonals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q4/5

1464

QUESTION ID

S35-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only a square does.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q4/5

The diagonals of a square meet at O. Name the angle between them.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q5/5

1465

QUESTION ID

S35-B2-Q5

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 3

Diameters and the center of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A circle has radius 3. How long is any diameter?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q1/5

1467

QUESTION ID

S35-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q1/5

True or false?

Every diameter passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q2/5

1468

QUESTION ID

S35-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q2/5

Where do two different diameters of a circle meet?

A. At the midpoint of a radius

B. On the circle

C. At the center

D. Nowhere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q3/5

1469

QUESTION ID

S35-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. At the center

GRADING NOTE Both pass through the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q3/5

What is the longest chord of a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q4/5

1470

QUESTION ID

S35-B3-Q4

TYPE

Short answer

CORRECT ANSWER

A diameter

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q4/5

True or false?

All diameters of a circle have different lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q5/5

1471

QUESTION ID

S35-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE All equal 2r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q5/5

SET 35 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed quadrilateral is a square • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Diameters AC and BD of a circle with center O are perpendicular. Prove that ABCD is a square.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q1/5

1473

QUESTION ID

S35-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

ABCD is a square.

GRADING NOTE REQUIRED WORK: Use that the diagonals are congruent (both diameters), bisect each other (radii), and are perpendicular; or SAS on the four triangles.

Any valid proof. Essential: OA = OB = OC = OD; four right angles at O; SAS gives four congruent sides; each triangle is isosceles right, so each vertex angle is 45° + 45° = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q1/5

A square is inscribed in a circle of radius 5. Find its side length and area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q2/5

1474

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Side 5√2; area 50

GRADING NOTE Form required: exact side and area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q2/5

True or false?

The side of a square inscribed in a circle of radius r is 2r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q3/5

1475

QUESTION ID

S35-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The side is r√2; 2r is its diagonal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q3/5

The area of a square inscribed in a circle of radius r is

A. $2r^2$

B. πr^2

C. $4r^2$

D. r^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q4/5

1476

QUESTION ID

S35-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $2r^2$

GRADING NOTE $(r\sqrt{2})^2 = 2r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q4/5

What fraction of the circle's area does the inscribed square cover? Give the exact value.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q5/5

1477

QUESTION ID

S35-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2/π

GRADING NOTE Form required: exact value. $2r^2 \div \pi r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSG-CO.D.13

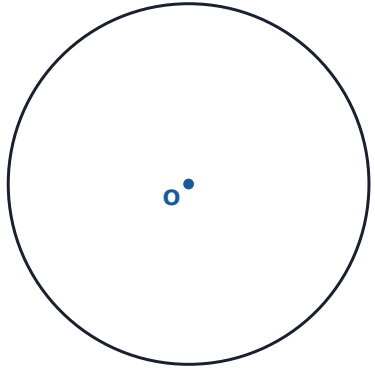
FORWARD 2

Construct a regular octagon inscribed in a circle • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a regular octagon inscribed in the circle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q1/5

1479

QUESTION ID

S35-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct two perpendicular diameters, then bisect two adjacent right angles at O (or the four arcs); the eight points on the circle are the vertices.

GRADING NOTE Grade the construction marks, not only the final figure: the perpendicular diameters and the bisector constructions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q1/5

What is the central angle for each side of a regular octagon?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q2/5

1480

QUESTION ID
S35-F2-Q2

TYPE
Short answer

CORRECT ANSWER
45°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q2/5

True or false?

A regular octagon has interior angles of 120°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q3/5

1481

QUESTION ID

S35-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (8 − 2) · 180 ÷ 8 = 135°; 120° is the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q3/5

Bisecting the arcs of an inscribed regular octagon gives a regular

A. hexagon

B. 16-gon

C. 12-gon

D. square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q4/5

1482

QUESTION ID

S35-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 16-gon

GRADING NOTE 8 → 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q4/5

A regular octagon is constructed in a circle by bisecting the four 90° central angles of an inscribed square. Explain why the eight points are equally spaced.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q5/5

1483

QUESTION ID

S35-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Each central angle is half of 90°, which is 45°, and equal central angles give equal chords.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct a regular hexagon inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Construct an inscribed equilateral triangle	**HSG-CO.D.13**
BACKWARD 2	An equilateral triangle has equal sides and 60° angles	**4.G.A.2**
BACKWARD 3	Circle facts: radius and chord	**7.G.B.4**
FORWARD 1	Justify that the inscribed hexagon is regular	**HSG-C.A.2**
FORWARD 2	Use inscribed hexagons to estimate π	**HSG-GMD.A.1**

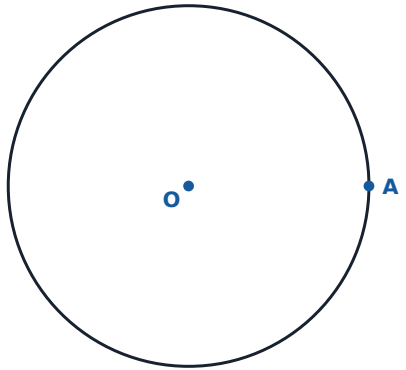
SET 36 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle, with one vertex at A.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q1/5

1486

QUESTION ID

S36-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Set the compass to the radius OA; from A step around the circle marking 5 more points; join the six points in order.

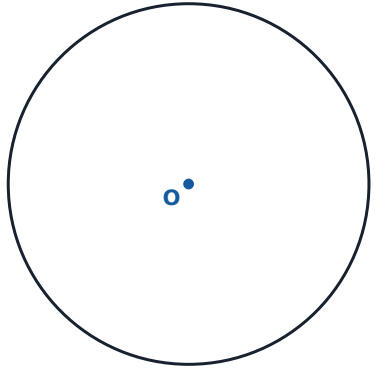
GRADING NOTE Grade the construction marks, not only the final figure: radius-length arcs around the circle and the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle (choose any starting point).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q2/5

1487

QUESTION ID

S36-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Mark a point, step the radius around the circle six times, and join consecutive points.

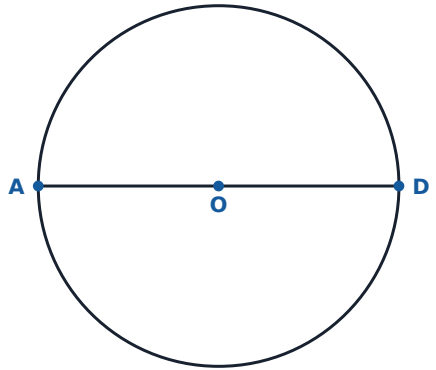
GRADING NOTE Grade the construction marks, not only the final figure: all stepping arcs and the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle using diameter AD and arcs centered at A and D with radius equal to the circle's radius.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q3/5

1488

QUESTION ID

S36-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

The arc at A meets the circle at B and F; the arc at D meets it at C and E; join A, B, C, D, E, F.

GRADING NOTE Grade the construction marks, not only the final figure: the two arcs and the hexagon.

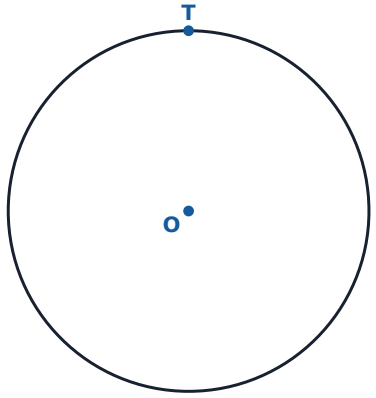
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle with one vertex at the top point T.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q4/5

1489

QUESTION ID

S36-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Keep the compass set to the radius and step around the circle from T; join the six points in order.

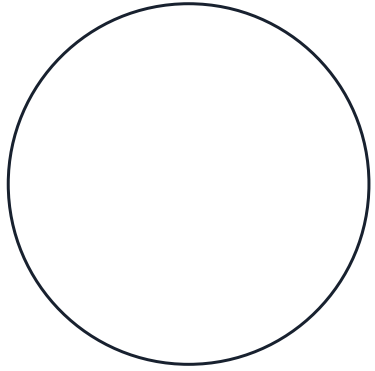
GRADING NOTE Grade the construction marks, not only the final figure: all stepping arcs from T and the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

The center of this circle is not marked. Construct a regular hexagon inscribed in it.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q5/5

1490

QUESTION ID

S36-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Locate the center with the perpendicular bisectors of two chords; set the compass to the radius; step around the circle and join consecutive points.

GRADING NOTE Grade the construction marks, not only the final figure: the center construction, the radius setting, and the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 1

Construct an inscribed equilateral triangle • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

How are the inscribed equilateral triangle and the inscribed regular hexagon related?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q1/5

1492

QUESTION ID

S36-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The triangle joins every other vertex of the hexagon.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q1/5

True or false?

Both constructions start by stepping the radius around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q2/5

1493

QUESTION ID

S36-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same six points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q2/5

How many points does stepping the radius mark?

A. 12

B. 6

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q3/5

1494

QUESTION ID

S36-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6

GRADING NOTE 60° each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An equilateral triangle is inscribed. How many more points must be added to make a regular hexagon?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q4/5

1495

QUESTION ID

S36-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q4/5

True or false?

The inscribed equilateral triangle and hexagon have the same side length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q5/5

1496

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Triangle side r√3; hexagon side r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 2

An equilateral triangle has equal sides and 60° angles • Grade 4 • Geometry • 4.G.A.2

True or false?

Each angle of an equilateral triangle measures 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q1/5

1498

QUESTION ID

S36-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 180 ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q1/5

A triangle has three sides equal to the radius of a circle. What kind of triangle is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q2/5

1499

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Equilateral

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Six equilateral triangles meeting at a point fill an angle of

A. 300°

B. 360°

C. 60°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q3/5

1500

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 360°

GRADING NOTE 6×60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q3/5

1500

A regular hexagon has side 5. How long are the segments from its center to its vertices?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q4/5

1501

QUESTION ID

S36-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q4/5

True or false?

A regular hexagon splits into four equilateral triangles from its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q5/5

1502

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Six.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 3

Circle facts: radius and chord • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A chord equals the radius. What central angle does it subtend?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q1/5

1504

QUESTION ID

S36-B3-Q1

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q1/5

True or false?

A chord equal to the radius forms an equilateral triangle with the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q2/5

1505

QUESTION ID

S36-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All sides equal r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q2/5

A circle has radius 7. A regular inscribed hexagon has perimeter

A. 21

B. 14

C. 42

D. 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q3/5

1506

QUESTION ID

S36-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 42

GRADING NOTE 6 × 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q3/5

A circle has circumference 31.4 (π ≈ 3.14). Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q4/5

1507

QUESTION ID

S36-B3-Q4

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Every chord of a circle is as long as the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q5/5

1508

QUESTION ID

S36-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Chords range up to the diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q5/5

SET 36 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed hexagon is regular • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Six points are marked around a circle with center O by stepping the radius. Prove that the hexagon they form is regular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q1/5

1510

QUESTION ID

S36-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

The hexagon is regular.

GRADING NOTE REQUIRED WORK: Show the six triangles from O are equilateral (all sides r), so all sides equal r and all angles equal 120°.

Any valid proof. Essential: each triangle O-consecutive points has three sides equal to r (equilateral, 60° at O); six 60° angles fill 360°, so the stepping closes; all sides equal r; each hexagon angle is 60° + 60° = 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A compass set to the radius of a circle is stepped around the circle from a starting point. Why does the sixth step land exactly back on the starting point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q2/5

1511

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Each step makes a 60° central angle, and 6 × 60° = 360°.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q2/5

True or false?

Each interior angle of a regular hexagon is 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q3/5

1512

QUESTION ID

S36-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each interior angle is 120°; 60° is the central angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q3/5

A regular hexagon inscribed in a circle of radius 4 has perimeter

A. 12

B. 48

C. 24

D. 8π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q4/5

1513

QUESTION ID

S36-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 24

GRADING NOTE 6 × 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q4/5

Find the area of a regular hexagon inscribed in a circle of radius 2.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q5/5

1514

QUESTION ID

S36-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$6\sqrt{3}$

GRADING NOTE Form required: exact. Six equilateral triangles of side 2: $6 \cdot \sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSG-CO.D.13

FORWARD 2

Use inscribed hexagons to estimate π • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A regular hexagon is inscribed in a circle of radius r. Compare its perimeter with the circumference. What estimate of π does this give?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q1/5

1516

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Perimeter

6r < circumference

2πr, so π > 3.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q1/5

A regular 12-gon inscribed in a circle of radius 1 has perimeter about 6.212. What estimate of π does it give, to the nearest hundredth?

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q2/5

1517

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

About 3.11
(π > 3.10)

GRADING NOTE Rounding to 0.01 required, with the explanation (half the perimeter: 6.212 ÷ 2 = 3.106).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q2/5

True or false?

Doubling the number of sides of an inscribed regular polygon makes its perimeter longer than the circumference.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q3/5

1518

QUESTION ID

S36-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The perimeter gets closer to the circumference but stays shorter (each side is a chord).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q3/5

Which polygon's perimeter is closest to the circumference of its circle?

A. Inscribed regular 96-gon

B. Inscribed regular hexagon

C. Inscribed square

D. Inscribed triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q4/5

1519

QUESTION ID

S36-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Inscribed regular 96-gon

GRADING NOTE More sides fit the circle more closely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q4/5

Explain why an inscribed polygon's perimeter is always less than the circumference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q5/5

1520

QUESTION ID

S36-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Each side is a chord, shorter than the arc it cuts off.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q5**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q5/5

Verify that a dilation maps a line to a parallel line (or to itself through the center)

MAIN	Geometry main question	**HSG-SRT.A.1.a**
BACKWARD 1	Dilate points with coordinates	**8.G.A.3**
BACKWARD 2	Compare slopes to test whether lines are parallel	**8.EE.B.6**
BACKWARD 3	Write the equation of a line through two points	**8.F.B.4**
FORWARD 1	Dilate lines about a center other than the origin	**HSG-SRT.A.1.a**
FORWARD 2	Prove algebraically that a dilation maps a line to a parallel line	**HSG-SRT.A.1.a**

SET 37 • HSG-SRT.A.1.a

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ is $y = 2x + 1$ (it does not pass through the origin). Dilate it about the origin with scale factor 3: find the images of (0, 1) and (1, 3), write an equation of the image line, and state what this example shows about dilations and lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q1/5

1523

QUESTION ID

S37-M-Q1

TYPE

Short answer

CORRECT ANSWER

(0, 3) and (3, 9); y = 2x + 3; the image is parallel to ℓ (both slopes 2). This is evidence (one example) that a line not through the center maps to a parallel line.

GRADING NOTE Form required: both images, the equation, the comparison, and a statement that the example supports (does not prove) the property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q1/5

Line m is y = −x, which passes through the origin. Dilate it about the origin with scale factor 2: find the images of (1, −1) and (3, −3). What is the image line, and what does this example suggest about lines through the center?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q2/5

1524

QUESTION ID
S37-M-Q2

TYPE
Short answer

CORRECT ANSWER
(2, −2) and (6, −6); the image is y = −x, the same line. It suggests that a line through the center maps onto itself.

GRADING NOTE Form required: the images, the image line, and the observation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The horizontal line y = 4 is dilated about the center C(0, 2) with scale factor ½. Find the images of (0, 4) and (6, 4), write an equation of the image line, and compare it with the original.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q3/5

1525

QUESTION ID

S37-M-Q3

TYPE

Short answer

CORRECT ANSWER

Images (0, 3) and (3, 3); image line y = 3; it is parallel to y = 4 (both horizontal), and y = 4 does not pass through C.

GRADING NOTE Form required: both images (center + ½ · (point − center)), the equation, and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Line n is y = −½x + 4. Dilate it about the center C(2, 1) with scale factor 3: find the images of (0, 4) and (4, 2), write an equation of the image line, and compare it with n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q4/5

1526

QUESTION ID

S37-M-Q4

TYPE

Short answer

CORRECT ANSWER

Images (−4, 10) and (8, 4); image line y = −½x + 8; it is parallel to n (same slope −½), and n does not pass through C.

GRADING NOTE Form required: both images (center + 3 · (point − center)), the equation, and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q4/5

Line p is y = 2x, which passes through C(1, 2). Dilate it about C with scale factor 4: find the images of (0, 0) and (2, 4) and the image line. What do you observe?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q5/5

1527

QUESTION ID

S37-M-Q5

TYPE

Short answer

CORRECT ANSWER

Images (−3, −6) and (5, 10); the image line is y = 2x, the same line, because p passes through the center.

GRADING NOTE Form required: both images, the image line, and the observation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-M-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 1

Dilate points with coordinates • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Dilate (2, −3) about the origin with scale factor 4.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

1529

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(8, −12)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Dilate (6, 10) about the origin with scale factor ½.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

1530

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 5)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

A dilation about the origin sends (3, 1) to (9, 3). What is the scale factor?

A. 6

B. 9

C. 3

D. ⅓

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

1531

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE Each coordinate is tripled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

True or false?

A dilation about the origin with scale factor 1 leaves every point where it is.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

1532

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (x, y) → (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

True or false?

A dilation with scale factor 2 moves the center to a new point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

1533

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The center is fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 2

Compare slopes to test whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (0, 3) and (3, 9).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q1/5

1535

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q1/5

True or false?

y = 2x + 1 and y = 2x + 3 are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q2/5

1536

QUESTION ID

S37-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same slope, different intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q2/5

Which line is parallel to y = −3x + 2?

A. y = ⅓x

B. y = −3

C. y = −3x − 5

D. y = 3x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q3/5

1537

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = −3x − 5

GRADING NOTE Slope −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Are y = 5 and y = 2 parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q4/5

1538

QUESTION ID

S37-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Yes (both horizontal)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Lines with slopes 2 and −2 are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q5/5

1539

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Different slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 3

Write the equation of a line through two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation of the line through (0, 3) and (3, 9).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q1

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q1/5

1541

QUESTION ID

S37-B3-Q1

TYPE

Short answer

CORRECT ANSWER

y = 2x + 3

GRADING NOTE Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q1

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation of the line through (2, −2) and (6, −6).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q2

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q2/5

1542

QUESTION ID

S37-B3-Q2

TYPE

Short answer

CORRECT ANSWER

y = −x

GRADING NOTE Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q2

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on y = 2x + 3?

A. (5, 1)

B. (2, 3)

C. (0, 2)

D. (1, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q3

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q3/5

1543

QUESTION ID

S37-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 5)

GRADING NOTE 2(1) + 3 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q3

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q3/5

True or false?

The line through (0, 0) and (4, 8) is y = 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q4

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q4/5

1544

QUESTION ID

S37-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Slope 2, intercept 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q4

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q4/5

True or false?

The line through (1, 1) and (2, 4) has slope 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q5

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q5/5

1545

QUESTION ID

S37-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(4 - 1) \div (2 - 1) = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q5

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q5/5

SET 37 • HSG-SRT.A.1.a

FORWARD 1

Dilate lines about a center other than the origin • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

Dilate the line y = x + 2 about the center (2, 0) with scale factor 2. Find the images of (0, 2) and (2, 4) and the image line.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q1/5

1547

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(−2, 4) and (2, 8); y = x + 6, parallel to y = x + 2.

GRADING NOTE Form required: images, equation, and conclusion. Image = center + 2(point − center).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q1/5

Does the line y = x − 2 pass through the center (2, 0)? What happens to it under any dilation about (2, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q2/5

1548

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes, 0 = 2 − 2; it maps onto itself.

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q2/5

True or false?

Under a dilation about (1, 1) with scale factor 3, the point (2, 1) goes to (4, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q3/5

1549

QUESTION ID

S37-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (1, 1) + 3(1, 0) = (4, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q3/5

A dilation about (0, 2) with scale factor ½ maps (4, 6) to

A. (4, 4)

B. (8, 10)

C. (2, 3)

D. (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q4/5

1550

QUESTION ID

S37-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 4)

GRADING NOTE (0, 2) + ½(4, 4) = (2, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q4/5

Line ℓ: $y = 3x$. A dilation about $(1, 0)$ with scale factor 2 maps ℓ to ℓ'. Find ℓ'.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q5/5

1551

QUESTION ID

S37-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$y = 3x + 3$

GRADING NOTE Any equivalent equation. $(0, 0) \to (-1, 0)$ and $(1, 3) \to (1, 6)$; slope 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSG-SRT.A.1.a

FORWARD 2

Prove algebraically that a dilation maps a line to a parallel line • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

Prove that the dilation about the origin with scale factor k ($k > 0$, $k \neq 1$) maps the line $y = mx + b$ to the line $y = mx + kb$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q1/5

1553

QUESTION ID

S37-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

The image of (x, mx + b) is (kx, kmx + kb), which satisfies y = mx + kb.

GRADING NOTE REQUIRED WORK: Take any point (x, mx + b) on the line, find its image, and show the image satisfies y = mx + kb (and every point of that line is an image).

Any valid proof (paragraph, two-column, or flow). Essential: image (X, Y) = (kx, k(mx + b)); Y = m(kx) + kb = mX + kb; conversely (X, mX + kb) is the image of (X/k, mX/k + b). Conclusion: same slope m, so parallel when b ≠ 0, the same line when b = 0. Invalid: checking only two points of one example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q1/5

From the general result, when does the image of $y = mx + b$ coincide with the original line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q2/5

1554

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

When b = 0 (the line passes through the center).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q2/5

True or false?

The dilation $(x, y) \to (kx, ky)$ maps the vertical line $x = c$ to the line $x = c + k$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q3/5

1555

QUESTION ID

S37-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It maps $x = c$ to $x = kc$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q3/5

Under the dilation about the origin with scale factor 4, the line $y = -2x + 5$ maps to

A. $y = -8x + 5$

B. $y = -2x + 20$

C. $y = -2x + 5$

D. $y = -8x + 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q4/5

1556

QUESTION ID

S37-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $y = -2x + 20$

GRADING NOTE $y = mx + kb$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q4/5

Explain why measuring a few dilated lines is not a proof of the property.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q5/5

1557

QUESTION ID

S37-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Examples cannot cover every line; a proof must work for all lines (with variables).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q5/5

Verify that a dilation multiplies lengths by the scale factor

MAIN	Geometry main question	**HSG-SRT.A.1.b**
BACKWARD 1	Find lengths with the distance formula	**8.G.B.8**
BACKWARD 2	Scale factor in scale drawings	**7.G.A.1**
BACKWARD 3	Dilate points about the origin	**8.G.A.3**
FORWARD 1	Find the center or scale factor from a segment and its image	**HSG-SRT.A.1.b**
FORWARD 2	Perimeter and area of dilated figures	**HSG-SRT.A.2**

SET 38 • HSG-SRT.A.1.b

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

Segment AB has A(1, 2) and B(4, 6). Dilate it about the origin with scale factor 3. Find A′B′ and AB, and compare.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q1/5

1560

QUESTION ID

S38-M-Q1

TYPE

Short answer

CORRECT ANSWER

A′(3, 6), B′(12, 18); AB = 5 and A′B′ = 15 = 3 · AB.

GRADING NOTE Form required: images, both lengths, and the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q1/5

Segment PQ has P(−4, 2) and Q(8, −6). Dilate it about the origin with scale factor ½. Find PQ and P′Q′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q2/5

1561

QUESTION ID
S38-M-Q2

TYPE
Short answer

CORRECT ANSWER
P′(−2, 1), Q′(4, −3); PQ = √208 = 4√13 ≈ 14.42 and P′Q′ = √52 = 2√13 ≈ 7.21, half as long.

GRADING NOTE Form required: both lengths (exact or to 0.01) and the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q2/5

Segment CD has C(2, 1) and D(5, 5). Dilate it about the center (2, 1) with scale factor 2. Find C′D′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q3/5

1562

QUESTION ID

S38-M-Q3

TYPE

Short answer

CORRECT ANSWER

C′ = (2, 1), D′ = (8, 9); C′D′ = 10 = 2 · CD (CD = 5).

GRADING NOTE Form required: images and lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-M-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(4, 0), C(0, 3). Dilate it about (1, 1) with scale factor 2. Find all three image side lengths, compare each with the original, and say what the comparison supports.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q4/5

1563

QUESTION ID

S38-M-Q4

TYPE

Short answer

CORRECT ANSWER

A′(−1, −1), B′(7, −1), C′(−1, 5); A′B′ = 8 = 2 · 4, A′C′ = 6 = 2 · 3, B′C′ = 10 = 2 · 5. Every length doubled: evidence that a dilation multiplies lengths by k.

GRADING NOTE Form required: images, three lengths with ratios, and the conclusion stated as evidence (not a general proof).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q4/5

Segment EF has E(−3, −1) and F(3, 7). Dilate it about the origin with scale factor 0.4. Find EF and E′F′, and compare.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q5/5

1564

QUESTION ID

S38-M-Q5

TYPE

Short answer

CORRECT ANSWER

E′(−1.2, −0.4), F′(1.2, 2.8); EF = 10 and E′F′ = 4 = 0.4 · 10.

GRADING NOTE Form required: images, both lengths, and the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-M-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 1

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (3, 6) and (12, 18).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q1/5

1566

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (−2, 1) and (4, −3).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q2/5

1567

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

√52 = 2√13

GRADING NOTE Any equivalent exact form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (5, 12).

A. √17

B. 7

C. 17

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q3/5

1568

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 13

GRADING NOTE √169.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q3/5

True or false?

The distance between (1, 2) and (4, 6) is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q4/5

1569

QUESTION ID

S38-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE √(9 + 16).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q4/5

True or false?

The distance between (0, 0) and (3, 3) is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q5/5

1570

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3\sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 2

Scale factor in scale drawings • Grade 7 • Geometry • 7.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A drawing uses scale factor 4. A 3 cm segment in the original is how long in the drawing?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q1/5

1572

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q1/5

A 15 m wall is drawn 5 cm long. What scale factor (drawing ÷ actual, in the same units) was used?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q2/5

1573

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5/1500 = 1/300

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q2/5

True or false?

A scale factor less than 1 makes a figure smaller.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q3/5

1574

QUESTION ID

S38-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Lengths shrink.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle 4 by 6 is enlarged by a scale factor of 1.5. Its new dimensions are

A. 5.5 by 7.5

B. 4 by 9

C. 8 by 12

D. 6 by 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q4/5

1575

QUESTION ID

S38-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 6 by 9

GRADING NOTE Multiply each side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q4/5

True or false?

Enlarging by a scale factor of 2 doubles the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q5/5

1576

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Area is multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 3

Dilate points about the origin • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Dilate (4, 6) about the origin with scale factor 3.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q1/5

1578

QUESTION ID

S38-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(12, 18)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Dilate (−4, 2) about the origin with scale factor ½.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q2/5

1579

QUESTION ID

S38-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, 1)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q2/5

Dilation about the origin sends (2, 5) to (5, 12.5). Scale factor?

A. 2.5

B. 3

C. 0.4

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q3/5

1580

QUESTION ID

S38-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2.5

GRADING NOTE 5 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q3/5

True or false?

Dilating about the origin multiplies both coordinates by the scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q4/5

1581

QUESTION ID

S38-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (kx, ky).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q4/5

True or false?

Dilating (3, 0) by 2 about the origin gives (5, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q5/5

1582

QUESTION ID

S38-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (6, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • HSG-SRT.A.1.b

FORWARD 1

Find the center or scale factor from a segment and its image • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

A dilation about the origin maps a segment of length 6 to a segment of length 15. What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q1/5

1584

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q1/5

A dilation maps A(1, 1) to A′(3, 3) and B(2, 1) to B′(5, 3). Find the scale factor and the center.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q2/5

1585

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Scale factor 2; center (−1, −1).

GRADING NOTE Form required: both. A′B′ = 2 = 2AB; lines AA′ and BB′ meet at (−1, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q2/5

True or false?

Under a dilation with scale factor k ≠ 1, the line through any point P other than the center and its image P′ passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q3/5

1586

QUESTION ID

S38-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE P′ is on ray CP from the center C, and P′ ≠ P when P ≠ C and k ≠ 1, so line PP′ is line CP.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A dilation maps P(4, 0) to P′(1, 0) and its center is the origin. Scale factor?

A. 4

B. −3

C. ¼

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q4/5

1587

QUESTION ID

S38-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. ¼

GRADING NOTE 1 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A segment of length 9 is dilated by an unknown factor k, and the image is 3 units shorter than the original. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q5/5

1588

QUESTION ID

S38-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{3}$

GRADING NOTE Any equivalent form. 9k = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F1-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSG-SRT.A.1.b

FORWARD 2

Perimeter and area of dilated figures • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

A triangle with perimeter 18 and area 12 is dilated with scale factor 3. Find the new perimeter and area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q1/5

1590

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Perimeter 54; area 108

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q1/5

A square of area 50 is dilated so its area becomes 8. Find the scale factor.

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q2/5

1591

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{5}$

GRADING NOTE Any equivalent form. $k^2 = 8/50 = 4/25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q2/5

True or false?

If lengths are multiplied by k, areas are multiplied by 2k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q3/5

1592

QUESTION ID
S38-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Areas are multiplied by k^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q3/5

A figure is dilated with scale factor ½. Its area is multiplied by

A. ½

B. 4

C. 2

D. ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q4/5

1593

QUESTION ID

S38-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ¼

GRADING NOTE $(½)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q4/5

Explain why a dilation with factor k multiplies the perimeter by k.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q5/5

1594

QUESTION ID

S38-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Every side is multiplied by k, and the perimeter is the sum of the sides.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q5/5

Decide whether two figures are similar using similarity transformations

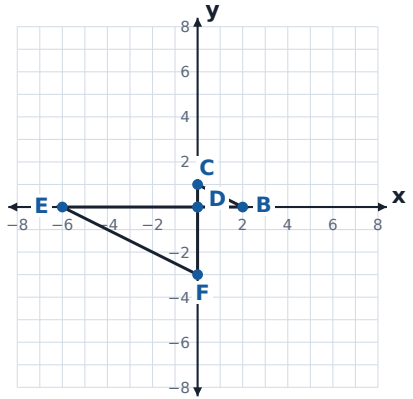
MAIN	Geometry main question	**HSG-SRT.A.2**
BACKWARD 1	Similar figures as a sequence of transformations including a dilati…	**8.G.A.4**
BACKWARD 2	Specify a sequence of rigid motions	**HSG-CO.A.5**
BACKWARD 3	Dilate figures about the origin	**8.G.A.3**
FORWARD 1	Find the scale factor and correspondence when the image is turne	**HSG-SRT.A.2**
FORWARD 2	Prove that all circles are similar	**HSG-C.A.1**

SET 39 • HSG-SRT.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Is triangle ABC similar to triangle DEF? If so, describe a similarity transformation that maps ABC onto DEF; if not, explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q1/5

1597

QUESTION ID

S39-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Dilate by 3 about the origin, then rotate 180° about the origin.

GRADING NOTE Any valid similarity transformation. Check: (2, 0) → (6, 0) → (−6, 0); (0, 1) → (0, 3) → (0, −3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q1/5

Is rectangle R (4 by 6) similar to rectangle S (6 by 8)? Explain using similarity transformations.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q2/5

1598

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. A dilation would need factor 6/4 = 1.5 for one side and 8/6 ≈ 1.33 for the other; no single scale factor works.

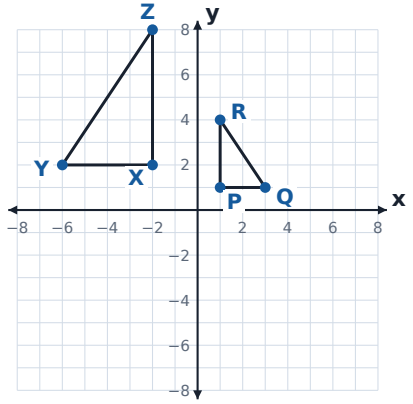
GRADING NOTE Any equivalent explanation comparing the ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Is triangle PQR similar to triangle XYZ? If so, describe a similarity transformation; if not, explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q3/5

1599

QUESTION ID

S39-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes. Dilate by 2 about the origin, then reflect across the y-axis.

GRADING NOTE Any valid sequence. (1, 1) → (2, 2) → (−2, 2); (3, 1) → (6, 2) → (−6, 2); (1, 4) → (2, 8) → (−2, 8).

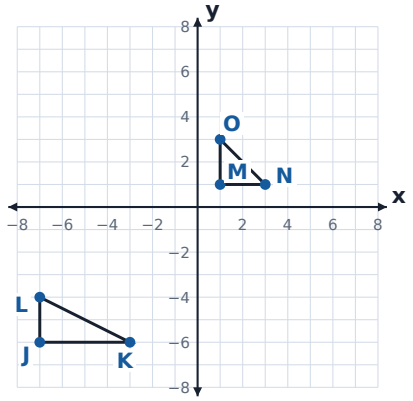
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Is triangle JKL similar to triangle MNO? If so, describe a similarity transformation; if not, explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q4/5

1600

QUESTION ID

S39-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. JKL has legs 4 and 2 (ratio 2 : 1) while MNO has legs 2 and 2 (ratio 1 : 1); a dilation multiplies both legs by the same factor, so no similarity transformation maps one onto the other.

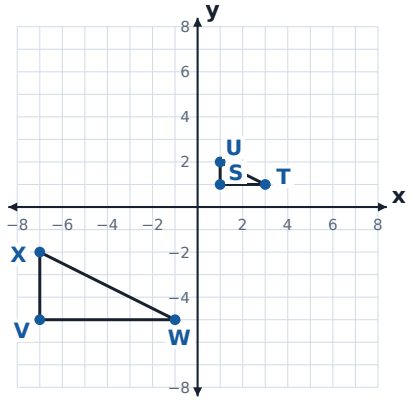
GRADING NOTE Any equivalent explanation comparing ratios of corresponding sides (or the unequal acute angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is triangle STU similar to triangle VWX? If so, describe a similarity transformation; if not, explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q5/5

1601

QUESTION ID

S39-M-Q5

TYPE

Short answer

CORRECT ANSWER

Yes. Dilate by 3 about S(1, 1), then translate 8 left and 6 down.

GRADING NOTE Any valid sequence. About S: T → (7, 1), U → (1, 4); then (x − 8, y − 6) gives V(−7, −5), W(−1, −5), X(−7, −2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q5/5

SET 39 • HSG-SRT.A.2

BACKWARD 1

Similar figures as a sequence of transformations including a dilation • Grade 8 • Geometry • 8.G.A.4

True or false?

A figure and its image after a dilation and a translation are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q1/5

1603

QUESTION ID

S39-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A similarity sequence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two figures are similar. What may be different?

A. Their shape

B. Their size

C. Their angle measures

D. The ratios of their sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q2/5

1604

QUESTION ID

S39-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Their size

GRADING NOTE Size may change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q2/5

A triangle is dilated by 2 and then rotated. Its sides were 3, 4, 5. What are the image's sides?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q3/5

1605

QUESTION ID

S39-B1-Q3

TYPE

Short answer

CORRECT ANSWER

6, 8, 10

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q3/5

True or false?

Two figures with the same angles but different side ratios are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q4/5

1606

QUESTION ID

S39-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Side ratios must all be equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name the transformation that changes size.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q5/5

1607

QUESTION ID

S39-B1-Q5

TYPE

Short answer

CORRECT ANSWER

A dilation

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSG-SRT.A.2

BACKWARD 2

Specify a sequence of rigid motions • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Which rigid motion maps (x, y) to (−x, y)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q1/5

1609

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Reflection across the y-axis

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q1/5

Which rigid motion maps (x, y) to (−x, −y)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q2/5

1610

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Rotation of 180° about the origin

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q2/5

Which rigid motion maps (2, 3) to (−3, 2)?

A. Translation 5 left

B. Rotation 90° counterclockwise about the origin

C. Rotation 180°

D. Reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q3/5

1611

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Rotation 90° counterclockwise about the origin

GRADING NOTE (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q3/5

True or false?

A translation then a reflection is a valid sequence of rigid motions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q4/5

1612

QUESTION ID

S39-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any composition of rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q4/5

True or false?

A dilation is a rigid motion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q5/5

1613

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It changes lengths unless k = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSG-SRT.A.2

BACKWARD 3

Dilate figures about the origin • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Dilate triangle (0, 0), (2, 0), (0, 1) by 3 about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q1/5

1615

QUESTION ID

S39-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(0, 0), (6, 0), (0, 3)

GRADING NOTE Form required: the three vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q1/5

True or false?

A dilation about the origin with factor 3 triples each coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q2/5

1616

QUESTION ID

S39-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (3x, 3y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q2/5

(4, −2) dilated by ½ about the origin is

A. (8, −4)

B. (2, −1)

C. (4.5, −1.5)

D. (2, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q3/5

1617

QUESTION ID

S39-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (2, −1)

GRADING NOTE Halve each coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q3/5

A dilation about the origin maps (5, 5) to (2, 2). What is the factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q4/5

1618

QUESTION ID

S39-B3-Q4

TYPE

Short answer

CORRECT ANSWER

0.4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q4/5

True or false?

A dilation can change the angles of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q5/5

1619

QUESTION ID

S39-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Angles are preserved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q5/5

SET 39 • HSG-SRT.A.2

FORWARD 1

Find the scale factor and correspondence when the image is turned or flipped • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has sides AB = 4, BC = 6, CA = 8. Triangle PQR has PQ = 9, QR = 12, RP = 6. Are they similar? If so, give the correspondence and the scale factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q1/5

1621

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: A ↔ R, B ↔ P, C ↔ Q (AB ↔ RP = 6, BC ↔ PQ = 9, CA ↔ QR = 12); scale factor 1.5.

GRADING NOTE Form required: the correspondence and the factor. Match shortest to shortest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q1/5

A 3-4-5 triangle and a triangle with sides 7.5, 10, 12.5: similar? Scale factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q2/5

1622

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes; 2.5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q2/5

True or false?

Matching the shortest side with the shortest side, middle with middle, and longest with longest gives the correspondence for similar triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q3/5

1623

QUESTION ID
S39-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Similarity preserves the order of side lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q3/5

Triangles with sides 5, 6, 7 and 10, 12, 15: are they similar?

A. Cannot tell

B. Yes, factor 2.5

C. Yes, factor 2

D. No, 15/7 ≠ 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q4/5

1624

QUESTION ID

S39-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. No, 15/7 ≠ 2

GRADING NOTE Ratios 2, 2, 15/7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q4/5

Triangle ABC ~ triangle DEF with a factor of 4 from ABC to DEF. DE = 10. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q5/5

1625

QUESTION ID

S39-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSG-SRT.A.2

FORWARD 2

Prove that all circles are similar • High School — Geometry • Geometry: Circles • HSG-C.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Circle C_1 has center A and radius r; circle C_2 has center B and radius s. Prove that C_1 and C_2 are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q1/5

1627

QUESTION ID

S39-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

A translation and a dilation map C_1 onto C_2.

GRADING NOTE REQUIRED WORK: Translate A to B, then dilate about B with scale factor s/r; show the image of C_1 is C_2.

Any valid proof (paragraph, two-column, or flow). Essential: the translation maps C_1 to the circle with center B and radius r; the dilation about B with factor s/r sends each point at distance r from B to a point at distance s, so the image is C_2 (and every point of C_2 is reached).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q1/5

Circle P has radius 4 and circle Q has radius 10. What scale factor maps P onto Q (after a translation)?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q2/5

1628

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q2/5

True or false?

Any two rectangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q3/5

1629

QUESTION ID

S39-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A 1 × 2 and a 1 × 3 rectangle do not have proportional sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of figures is NOT always similar?

A. Two equilateral triangles

B. Two squares

C. Two circles

D. Two ellipses

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q4/5

1630

QUESTION ID

S39-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Two ellipses

GRADING NOTE Ellipses can have different shapes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q4/5

Explain why all equilateral triangles are similar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q5/5

1631

QUESTION ID

S39-F2-Q5

TYPE

Short answer

CORRECT ANSWER

All their angles are 60°, so AA applies (or all side ratios are equal).

GRADING NOTE Any valid explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q5/5

Explain similarity of triangles as equal angles and proportional sides

MAIN	Geometry main question	**HSG-SRT.A.2**
BACKWARD 1	Read correspondence from a similarity or congruence statement	**HSG-CO.B.7**
BACKWARD 2	Solve proportions	**7.RP.A.2**
BACKWARD 3	Scale factor between figures	**7.G.A.1**
FORWARD 1	Use the AA criterion to recognize similar triangles	**HSG-SRT.A.3**
FORWARD 2	Use similarity to find unknown lengths in problems	**HSG-SRT.B.5**

SET 40 • HSG-SRT.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A similarity transformation (a dilation with scale factor k followed by rigid motions) maps △ABC onto △DEF with A → D, B → E, C → F. Explain why ∠A ≅ ∠D, ∠B ≅ ∠E, ∠C ≅ ∠F and DE/AB = EF/BC = FD/CA = k.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q1/5

1634

QUESTION ID

S40-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angles are preserved and every length is multiplied by k.

GRADING NOTE REQUIRED WORK: Use what dilations and rigid motions do to angles and to lengths.

Any valid explanation. Essential: dilations and rigid motions preserve angle measure, so corresponding angles are congruent; the dilation multiplies every length by k and rigid motions keep lengths, so each image side is k times its original side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q1/5

△ABC ~ △DEF with A ↔ D, B ↔ E, C ↔ F. AB = 6, BC = 9, DE = 10. Find EF and the scale factor from ABC to DEF.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q2/5

1635

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

EF = 15; scale factor 5/3

GRADING NOTE Form required: both. 10/6 = 5/3; EF = 9 · 5/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△PQR ~ △STU with P ↔ S, Q ↔ T, R ↔ U. m∠P = 42° and m∠Q = 71°. Find m∠U.

m∠U = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q3/5

1636

QUESTION ID

S40-M-Q3

TYPE

Short answer

CORRECT ANSWER

67°

GRADING NOTE Any equivalent form. ∠U ≅ ∠R = 180 − 113.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q3/5

△ABC has AB = 4, BC = 6, CA = 8, and △XYZ has XY = 6, YZ = 9, ZX = 10. Using the meaning of similarity, decide whether △ABC ~ △XYZ with A ↔ X, B ↔ Y, C ↔ Z, and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q4/5

1637

QUESTION ID

S40-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. XY/AB = 1.5 and YZ/BC = 1.5 but ZX/CA = 1.25; similar triangles need all three ratios equal (one scale factor).

GRADING NOTE Form required: all three ratios and the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q4/5

A dilation with scale factor 2.5 followed by a rotation maps △JKL onto △MNO (J → M, K → N, L → O). m∠J = 50°, m∠K = 60°, JK = 4, KL = 6. Find m∠O, MN and NO, and explain each answer using what the transformations do.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q5/5

1638

QUESTION ID

S40-M-Q5

TYPE

Short answer

CORRECT ANSWER

m∠O = m∠L = 70° (angles are preserved); MN = 2.5 · 4 = 10 and NO = 2.5 · 6 = 15 (the dilation multiplies lengths by 2.5; the rotation keeps them).

GRADING NOTE Form required: three values, each with its reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q5/5

SET 40 • HSG-SRT.A.2

BACKWARD 1

Read correspondence from a similarity or congruence statement • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

ANSWER KEY Reduced copy of the question page, with its answer

△ABC ~ △XYZ. Which angle corresponds to ∠C?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q1/5

1640

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

∠Z

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q1/5

△ABC ~ △XYZ. Which side corresponds to BC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q2/5

1641

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

YZ

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△KLM ~ △RST. Which ratio equals KL/RS?

A. LM/RT

B. RS/LM

C. LM/ST

D. KM/ST

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q3/5

1642

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. LM/ST

GRADING NOTE Second pair of letters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

△DEF ~ △GHI means ∠E ≅ ∠H.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q4/5

1643

QUESTION ID

S40-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Middle letters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q4/5

True or false?

△DEF ~ △GHI means DF corresponds to GH.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q5/5

1644

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE DF ↔ GI.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSG-SRT.A.2

BACKWARD 2

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve 6/9 = 10/x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q1/5

1646

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve x/4 = 15/6.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q2/5

1647

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which ratio is equal to 3 : 5?

A. 9 : 12

B. 6 : 8

C. 12 : 20

D. 5 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q3/5

1648

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12 : 20

GRADING NOTE Multiply both by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

2/3 = 8/12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q4/5

1649

QUESTION ID

S40-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both reduce to 2/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

4/6 = 6/9 = 10/14

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q5/5

1650

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10/14 = 5/7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSG-SRT.A.2

BACKWARD 3

Scale factor between figures • Grade 7 • Geometry • 7.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A figure is enlarged so that a 4 cm side becomes 10 cm. Scale factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q1/5

1652

QUESTION ID

S40-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q1/5

True or false?

All lengths of a scale drawing are multiplied by the same number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q2/5

1653

QUESTION ID

S40-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The scale factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q2/5

With scale factor 3, a 7 cm side becomes

A. 7 cm

B. 10 cm

C. 73 cm

D. 21 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q3/5

1654

QUESTION ID

S40-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 21 cm

GRADING NOTE Multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A scale drawing's 12 cm side represents 60 cm. Scale factor (drawing to actual)?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q4/5

1655

QUESTION ID

S40-B3-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{5}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q4/5

True or false?

A scale drawing changes the angles of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q5/5

1656

QUESTION ID

S40-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Angles stay the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q5/5

SET 40 • HSG-SRT.A.2

FORWARD 1

Use the AA criterion to recognize similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

Two triangles have angles of 35° and 80°. Are they similar? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q1/5

1658

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes, by AA: two pairs of congruent angles (the third angles are both 65°).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has m∠A = 50°, m∠B = 60°. Triangle DEF has m∠D = 60°, m∠F = 70°. Are they similar? Give the correspondence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q2/5

1659

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: ∠C = 70°, ∠E = 50°; A ↔ E, B ↔ D, C ↔ F.

GRADING NOTE Form required: the decision and the correspondence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q2/5

True or false?

Two right triangles with one pair of congruent acute angles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q3/5

1660

QUESTION ID

S40-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two pairs of congruent angles (the right angles and the acute angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q3/5

Which pair is NOT necessarily similar?

A. Two triangles with angles 30° and 60°

B. Two right isosceles triangles

C. Two equilateral triangles

D. Two isosceles triangles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q4/5

1661

QUESTION ID

S40-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Two isosceles triangles

GRADING NOTE Isosceles triangles can have different vertex angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In a figure, DE ∥ BC with D on AB and E on AC. Explain why △ADE ~ △ABC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q5/5

1662

QUESTION ID

S40-F1-Q5

TYPE

Short answer

CORRECT ANSWER

∠A is shared and ∠ADE ≅ ∠ABC (corresponding angles), so AA.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSG-SRT.A.2

FORWARD 2

Use similarity to find unknown lengths in problems • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A 1.8 m person casts a 2.4 m shadow when a tree casts a 16 m shadow. How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q1/5

1664

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

12 m

GRADING NOTE Any equivalent form. 1.8/2.4 = h/16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC.
AD = 4, DB = 6, and DE = 5. Find BC.

BC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q2/5

1665

QUESTION ID
S40-F2-Q2

TYPE
Short answer

CORRECT ANSWER
12.5

GRADING NOTE Any equivalent form. AD/AB = DE/BC: 4/10 = 5/BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q2/5

True or false?

In similar triangles with scale factor 3, a 7 cm side corresponds to a 10 cm side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q3/5

1666

QUESTION ID

S40-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 7 × 3 = 21 cm; a scale factor multiplies, it does not add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A 6 ft pole casts a 4 ft shadow. At the same time, a building casts a 30 ft shadow. How tall is the building?

A. 36 ft

B. 45 ft

C. 20 ft

D. 24 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q4/5

1667

QUESTION ID

S40-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 45 ft

GRADING NOTE 6/4 = h/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two similar triangles have perimeters 24 and 36. A side of the smaller is 10. Find the corresponding side of the larger.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q5/5

1668

QUESTION ID

S40-F2-Q5

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. Factor 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q5/5

Establish the AA criterion with similarity transformations

MAIN	Geometry main question	**HSG-SRT.A.3**
BACKWARD 1	Prove triangles congruent with ASA	**HSG-CO.B.8**
BACKWARD 2	Dilations multiply lengths and keep angles	**HSG-SRT.A.1.b**
BACKWARD 3	Use the angle sum to find a third angle	**8.G.A.5**
FORWARD 1	Establish the SAS and SSS similarity criteria	**HSG-SRT.A.3**
FORWARD 2	Use AA to set up and solve problems	**HSG-SRT.B.5**

SET 41 • HSG-SRT.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC and △DEF, ∠A ≅ ∠D and ∠B ≅ ∠E. Prove that △ABC ~ △DEF using similarity transformations.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q1/5

1671

QUESTION ID

S41-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

△ABC ~ △DEF.

GRADING NOTE REQUIRED WORK: Dilate △ABC by k = DE/AB; show the image is congruent to △DEF by ASA; conclude a dilation followed by rigid motions maps ABC onto DEF.

Any valid proof (paragraph, two-column, or flow). Essential: (1) dilate by k = DE/AB to get △A′B′C′ with A′B′ = DE; (2) dilations keep angles, so ∠A′ ≅ ∠D and ∠B′ ≅ ∠E; (3) △A′B′C′ ≅ △DEF by ASA; (4) so rigid motions map A′B′C′ onto DEF, and the dilation followed by those motions maps ABC onto DEF. Invalid: assuming the sides are proportional.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-M-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △PQR and △STU, m∠P = m∠S = 40°, m∠Q = m∠T = 75°, PQ = 4 and ST = 10. Prove that △PQR ~ △STU by describing a similarity transformation and justifying each step.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q2/5

1672

QUESTION ID

S41-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

△PQR ~ △STU (scale factor 2.5)

GRADING NOTE REQUIRED WORK: Dilate △PQR by 10/4 = 2.5, show the image is congruent to △STU by ASA, and conclude.

Any valid proof (paragraph, two-column, or flow). Essential: dilation with factor 2.5 gives △P′Q′R′ with P′Q′ = 10 = ST; angles are preserved, so ∠P′ ≅ ∠S and ∠Q′ ≅ ∠T; ASA gives △P′Q′R′ ≅ △STU; the dilation followed by the rigid motions maps PQR onto STU.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that two right triangles with one pair of congruent acute angles are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q3/5

1673

QUESTION ID

S41-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The triangles are similar by AA.

GRADING NOTE REQUIRED WORK: Name the triangles, list the two pairs of congruent angles, and apply AA (stating why AA is valid).

Any valid proof (paragraph, two-column, or flow). Essential: the right angles are congruent; the given acute angles are congruent; two pairs of congruent angles; AA. Invalid: assuming the sides are proportional.

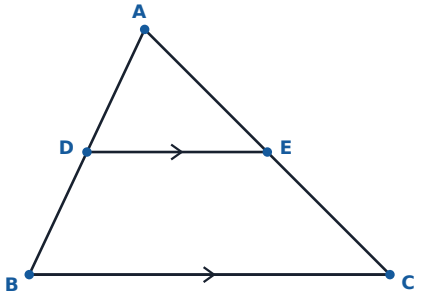
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-M-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. Prove that △ADE ~ △ABC.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q4/5

1674

QUESTION ID

S41-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

△ADE ~ △ABC

GRADING NOTE REQUIRED WORK: Find two pairs of congruent angles using the parallel lines, then apply AA.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A ≅ ∠A (shared); ∠ADE ≅ ∠ABC (corresponding angles, DE ∥ BC, transversal AB); AA. Invalid: using side ratios not given.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that two isosceles triangles with congruent vertex angles are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q5/5

1675

QUESTION ID

S41-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The triangles are similar by AA.

GRADING NOTE REQUIRED WORK: Let the vertex angles both measure v; find each base angle; apply AA.

Any valid proof (paragraph, two-column, or flow). Essential: base angles of an isosceles triangle are congruent, so each base angle measures (180° − v)/2 in both triangles; the vertex angles are congruent; two pairs of congruent angles; AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q5/5

SET 41 • HSG-SRT.A.3

BACKWARD 1

Prove triangles congruent with ASA • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Two triangles have ∠A ≅ ∠D, AB = DE, ∠B ≅ ∠E. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q1/5

1677

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

ASA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q1/5

True or false?

ASA needs the side between the two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q2/5

1678

QUESTION ID

S41-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Included side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q2/5

Which set of parts gives ASA?

A. Three angles

B. Three sides

C. Two sides and the angle between them

D. Two angles and the side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q3/5

1679

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Two angles and the side between them

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q3/5

After proving △PQR ≅ △XYZ, are their corresponding sides equal?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q4/5

1680

QUESTION ID

S41-B1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Corresponding parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q4/5

True or false?

ASA works only for right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q5/5

1681

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSG-SRT.A.3

BACKWARD 2

Dilations multiply lengths and keep angles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

A 5 cm segment is dilated by 1.6. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q1/5

1683

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-B2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q1/5

True or false?

A dilation preserves angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q2/5

1684

QUESTION ID

S41-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Similar shapes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q2/5

A dilation maps AB = 4 to A′B′ = 10. Factor?

A. 2.5

B. 6

C. 0.4

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q3/5

1685

QUESTION ID

S41-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2.5

GRADING NOTE 10 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What scale factor makes a 6 cm segment 6 cm long?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q4/5

1686

QUESTION ID

S41-B2-Q4

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-B2-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q4/5

True or false?

A dilation by 2 doubles the angles of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q5/5

1687

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Angles are unchanged.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSG-SRT.A.3

BACKWARD 3

Use the angle sum to find a third angle • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has angles 47° and 88°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q1/5

1689

QUESTION ID

S41-B3-Q1

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q1/5

True or false?

If two angles of one triangle equal two angles of another, the third angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q2/5

1690

QUESTION ID

S41-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Angle sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q2/5

Triangles with angles 30°, 70°, x and 70°, 80°, y. Which is true?

A. x = 80, y = 30; the triangles are similar

B. x = 80, y = 30; not similar

C. x = 70, y = 30

D. x = 30, y = 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q3/5

1691

QUESTION ID

S41-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 80, y = 30; the triangles are similar

GRADING NOTE 180 − 100 = 80; 180 − 150 = 30; same angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a triangle are equal and the third is 40°. Find the equal angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q4/5

1692

QUESTION ID

S41-B3-Q4

TYPE

Short answer

CORRECT ANSWER

70° each

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Knowing one angle of each triangle is enough for similarity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q5/5

1693

QUESTION ID

S41-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Need two pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q5/5

SET 41 • HSG-SRT.A.3

FORWARD 1

Establish the SAS and SSS similarity criteria • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC and △DEF, DE/AB = DF/AC = k and ∠A ≅ ∠D. Prove that △ABC ~ △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q1/5

1695

QUESTION ID

S41-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

△ABC ~ △DEF (SAS similarity).

GRADING NOTE REQUIRED WORK: Dilate △ABC by k, then use SAS congruence with △DEF.

Any valid proof (paragraph, two-column, or flow). Essential: dilate by k: A′B′ = DE, A′C′ = DF, ∠A′ ≅ ∠A ≅ ∠D; SAS congruence; dilation then rigid motions map ABC onto DEF.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q1/5

Triangles with sides 4, 6, 8 and 6, 9, 12. Similar? Which criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q2/5

1696

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes, SSS similarity (ratio 1.5).

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q2/5

True or false?

Two triangles with two pairs of proportional sides and congruent included angles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q3/5

1697

QUESTION ID

S41-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE SAS similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is NOT a similarity criterion?

A. SAS

B. SSA

C. SSS

D. AA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q4/5

1698

QUESTION ID

S41-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. SSA

GRADING NOTE SSA does not determine shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q4/5

△PQR: PQ = 5, PR = 7, ∠P = 40°. △XYZ: XY = 10, XZ = 14, ∠X = 40°. Similar? Factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q5/5

1699

QUESTION ID

S41-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Yes (SAS similarity); factor 2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSG-SRT.A.3

FORWARD 2

Use AA to set up and solve problems • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Two vertical poles are 3 m and 7 m tall. A wire from the top of each to the bottom of the other crosses at a point P. How high above the ground is P? (Use similar triangles.)

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q1/5

1701

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2.1 m

GRADING NOTE Any equivalent form. h = (3 · 7)/(3 + 7) = 2.1, from two pairs of AA-similar triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with ∠ADE ≅ ∠ACB. AD = 4, AE = 6, AB = 12. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q2/5

1702

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. △ADE ~ △ACB (AA): AD/AC = AE/AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q2/5

True or false?

A mirror on the ground can be used to find a building's height with similar triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q3/5

1703

QUESTION ID

S41-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal angles of incidence and reflection give AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Diagonals of trapezoid ABCD (AB ∥ DC) meet at E. AB = 12, DC = 4, AE = 9. Find EC.

A. 6

B. 27

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q4/5

1704

QUESTION ID

S41-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE △ABE ~ △CDE (AA): EC = 9 · 4/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A 2 m stick casts a 1.5 m shadow. A flagpole casts a 9 m shadow. How tall is the flagpole?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q5/5

1705

QUESTION ID
S41-F2-Q5

TYPE
Short answer

CORRECT ANSWER
12 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q5/5

Prove the triangle proportionality theorem

MAIN	Geometry main question	**HSG-SRT.B.4**
BACKWARD 1	Use AA to show triangles are similar	**HSG-SRT.A.3**
BACKWARD 2	Corresponding angles of parallel lines	**HSG-CO.C.9**
BACKWARD 3	Rewrite and solve proportions algebraically	**7.RP.A.2**
FORWARD 1	Prove that parallel lines cut transversals proportionally	**HSG-SRT.B.4**
FORWARD 2	Prove the angle bisector theorem	**HSG-SRT.B.5**

SET 42 • HSG-SRT.B.4

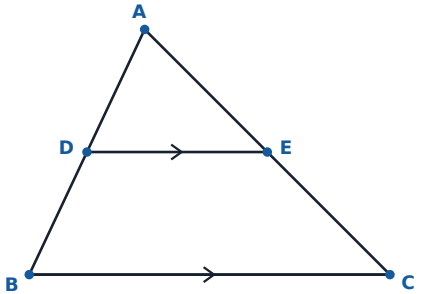
MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. Prove that AD/DB = AE/EC.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q1/5

1708

QUESTION ID

S42-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AD/DB = AE/EC

GRADING NOTE REQUIRED WORK: Prove △ADE ~ △ABC by AA, write AD/AB = AE/AC, then rewrite with AB = AD + DB and AC = AE + EC.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A shared; ∠ADE ≅ ∠ABC (corresponding angles, DE ∥ BC); AA; AD/AB = AE/AC; so AB/AD = AC/AE, (AD + DB)/AD = (AE + EC)/AE, 1 + DB/AD = 1 + EC/AE, DB/AD = EC/AE, and AD/DB = AE/EC.

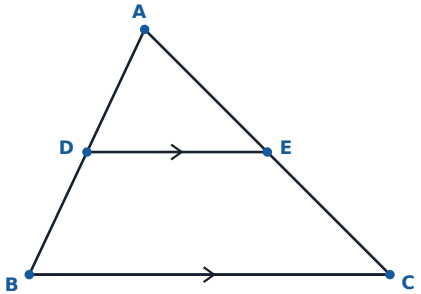
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. Prove that AD/AB = AE/AC = DE/BC.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q2/5

1709

QUESTION ID

S42-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

AD/AB = AE/AC = DE/BC

GRADING NOTE REQUIRED WORK: Prove △ADE ~ △ABC by AA, then use that corresponding sides of similar triangles are proportional.

Any valid proof (paragraph, two-column, or flow). Essential: shared ∠A; ∠ADE ≅ ∠ABC (corresponding angles); AA; corresponding sides proportional with AD ↔ AB, AE ↔ AC, DE ↔ BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In △PQR, a line parallel to QR meets PQ at S and PR at T. Prove that PS/SQ = PT/TR.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q3/5

1710

QUESTION ID

S42-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

PS/SQ = PT/TR

GRADING NOTE REQUIRED WORK: AA similarity of △PST and △PQR, then the same algebra.

Any valid proof (paragraph, two-column, or flow). Essential: shared ∠P and corresponding angles give △PST ~ △PQR; then the proportion algebra.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Coordinate proof. Place $\triangle ABC$ with $A(0, 0)$, $B(p, 0)$ and $C(q, r)$, where $p > 0$ and $r > 0$. $D(tp, 0)$ is on AB with $0 < t < 1$, and E is on AC with $DE \parallel BC$. Prove that $AD/DB = AE/EC$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q4/5

1711

QUESTION ID

S42-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

$AD/DB = AE/EC = t/(1 - t)$

GRADING NOTE REQUIRED WORK: Write $E = (sq, sr)$ with $0 < s < 1$, use equal slopes of DE and BC to show $s = t$, then compute both ratios.

Any valid proof (paragraph, two-column, or flow). Essential: slope $DE = sr/(sq - tp)$ and slope $BC = r/(q - p)$ (or both vertical); equal slopes give $s(q - p) = sq - tp$, so $sp = tp$ and $s = t$; $AD = tp$, $DB = (1 - t)p$; $AE = t \cdot AC$, $EC = (1 - t) \cdot AC$; both ratios are $t/(1 - t)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. Prove that AD/DB = AE/EC using the dilation about A with scale factor k = AB/AD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q5/5

1712

QUESTION ID

S42-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

AD/DB = AE/EC

GRADING NOTE REQUIRED WORK: Show the dilation maps D to B and line DE to line BC, so it maps E to C; compare lengths, then rewrite the proportion.

Any valid proof (paragraph, two-column, or flow). Essential: the dilation sends D to B (B is on ray AD with AB = k · AD); it sends line DE to the line through B parallel to DE, which is line BC; E is on ray AC, so its image is on ray AC and on line BC, which is C; hence AC = k · AE, so AD/AB = AE/AC = 1/k; then (AB − AD)/AD = (AC − AE)/AE gives DB/AD = EC/AE and AD/DB = AE/EC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q5/5

SET 42 • HSG-SRT.B.4

BACKWARD 1

Use AA to show triangles are similar • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Two pairs of congruent angles make two triangles similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q1/5

1714

QUESTION ID

S42-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. Name two pairs of congruent angles in △ADE and △ABC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q2/5

1715

QUESTION ID
S42-B1-Q2

TYPE
Short answer

CORRECT ANSWER
∠A ≅ ∠A and ∠ADE ≅ ∠ABC (or ∠AED ≅ ∠ACB)

GRADING NOTE Any two valid pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q2/5

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. Which ratio equals AD/AB?

A. DB/AB

B. AE/AC

C. DE/AB

D. AE/EC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q3/5

1716

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. AE/AC

GRADING NOTE Corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. AD = 3, AB = 9, AE = 4. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q4/5

1717

QUESTION ID

S42-B1-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q4/5

True or false?

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. Then AD/DB = DE/BC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q5/5

1718

QUESTION ID

S42-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE DB is not a side of either triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSG-SRT.B.4

BACKWARD 2

Corresponding angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

When a transversal crosses parallel lines, corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q1/5

1720

QUESTION ID

S42-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q1/5

In △ABC, D is on AB and E is on AC with DE ∥ BC, so AB is a transversal of DE and BC. Which angle corresponds to ∠ADE?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q2/5

1721

QUESTION ID
S42-B2-Q2

TYPE
Short answer

CORRECT ANSWER
∠ABC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. With AC as the transversal, ∠AED corresponds to

A. ∠BAC

B. ∠ABC

C. ∠DEC

D. ∠ACB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q3/5

1722

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. ∠ACB

GRADING NOTE Same position.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. m∠ABC = 65°. Find m∠ADE.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q4/5

1723

QUESTION ID

S42-B2-Q4

TYPE

Short answer

CORRECT ANSWER

65°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q4/5

True or false?

Corresponding angles are always supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q5/5

1724

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Congruent for parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSG-SRT.B.4

BACKWARD 3

Rewrite and solve proportions algebraically • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

If a/b = c/d, where a, b, c and d are all nonzero, write the equation you get by taking reciprocals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q1/5

1726

QUESTION ID

S42-B3-Q1

TYPE

Short answer

CORRECT ANSWER

b/a = d/c

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 4/(4 + x) = 6/15.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q2/5

1727

QUESTION ID

S42-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 60 = 24 + 6x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If (a + b)/a = (c + d)/c, then

A. b/a = d/c

B. b = d

C. a = c

D. a/b = c/d is false

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q3/5

1728

QUESTION ID

S42-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. b/a = d/c

GRADING NOTE Subtract 1 from both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

3/5 = 6/10 implies 5/3 = 10/6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q4/5

1729

QUESTION ID

S42-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reciprocals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(2 + 3)/2 = 5/2 implies 3/2 = 5/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q5/5

1730

QUESTION ID

S42-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Subtracting 1 gives 3/2 = 3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q5/5

SET 42 • HSG-SRT.B.4

FORWARD 1

Prove that parallel lines cut transversals proportionally • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Three parallel lines cut transversal t at A, B, C and transversal u at D, E, F (in order). Prove that AB/BC = DE/EF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q1/5

1732

QUESTION ID

S42-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AB/BC = DE/EF

GRADING NOTE

REQUIRED WORK: Draw segment AF; apply the triangle proportionality theorem in △ACF and in △FDA.

Any valid proof (paragraph, two-column, or flow). Essential: AF crosses the middle line at G; in △ACF, BG ∥ CF so AB/BC = AG/GF; in △FAD, GE ∥ AD so FG/GA = FE/ED, so AG/GF = DE/EF; combine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q1/5

Three parallel lines cut one transversal into segments 4 and 6 and another into segments 5 and x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q2/5

1733

QUESTION ID

S42-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q2/5

True or false?

Parallel lines that cut equal segments on one transversal cut equal segments on every transversal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q3/5

1734

QUESTION ID

S42-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Ratio 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. AD = 5, DB = 3, AE = 7.5. Find EC.

A. 12.5

B. 4.5

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q4/5

1735

QUESTION ID

S42-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4.5

GRADING NOTE 5/3 = 7.5/EC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Lots between parallel streets have frontages 30 m and 45 m on one avenue. On another avenue the first frontage is 40 m. Find the second.

Frontage: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q5/5

1736

QUESTION ID
S42-F1-Q5

TYPE
Short answer

CORRECT ANSWER
60 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSG-SRT.B.4

FORWARD 2

Prove the angle bisector theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, the bisector of ∠A meets BC at D. Prove that BD/DC = AB/AC. (Draw the line through C parallel to AD, meeting line AB at E.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q1/5

1738

QUESTION ID

S42-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

BD/DC = AB/AC

GRADING NOTE REQUIRED WORK: Show △ACE is isosceles (AE = AC), then apply the triangle proportionality theorem in △BCE.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BAD ≅ ∠AEC (corresponding) and ∠DAC ≅ ∠ACE (alternate interior), and ∠BAD ≅ ∠DAC, so ∠AEC ≅ ∠ACE and AE = AC; in △BCE, AD ∥ EC gives BD/DC = BA/AE = AB/AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, the bisector of ∠A meets BC at D. AB = 9, AC = 6, BC = 10. Find BD and DC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q2/5

1739

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

BD = 6, DC = 4

GRADING NOTE Form required: both. BD/DC = 9/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q2/5

True or false?

The bisector of an angle of a triangle always meets the opposite side at its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q3/5

1740

QUESTION ID

S42-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when the two sides are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q3/5

The bisector of ∠P in △PQR divides QR into 3 and 5. PQ = 6. Find PR.

A. 10

B. 15

C. 3.6

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q4/5

1741

QUESTION ID

S42-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE 3/5 = 6/PR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the angle bisector theorem is a relationship that similarity or proportionality, not congruence, proves.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q5/5

1742

QUESTION ID

S42-F2-Q5

TYPE

Short answer

CORRECT ANSWER

It states a ratio of lengths, which comes from proportional segments (parallel lines), not from equal lengths.

GRADING NOTE Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q5/5

Prove the converse of the triangle proportionality theorem

MAIN	Geometry main question	**HSG-SRT.B.4**
BACKWARD 1	Use the triangle proportionality theorem	**HSG-SRT.B.4**
BACKWARD 2	SAS similarity	**HSG-SRT.A.3**
BACKWARD 3	Converse of the corresponding angles theorem	**HSG-CO.C.9**
FORWARD 1	Use the converse to decide parallelism in figures	**HSG-SRT.B.4**
FORWARD 2	Prove the trapezoid midsegment theorem	**HSG-SRT.B.5**

SET 43 • HSG-SRT.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with AD/DB = AE/EC. Prove that DE ∥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q1/5

1745

QUESTION ID

S43-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

DE ∥ BC

GRADING NOTE REQUIRED WORK: Turn the given ratio into AD/AB = AE/AC, prove △ADE ~ △ABC by SAS similarity, then use congruent corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: AD/DB = AE/EC gives AD/AB = AE/AC (add 1 to the reciprocals); ∠A shared; SAS similarity; ∠ADE ≅ ∠ABC; congruent corresponding angles make DE ∥ BC (converse). Invalid: assuming DE ∥ BC to get the similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △PQR, S is on PQ and T is on PR with PS = 4, SQ = 6, PT = 6, TR = 9. Prove that ST ∥ QR.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q2/5

1746

QUESTION ID

S43-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

ST ∥ QR

GRADING NOTE REQUIRED WORK: Show PS/PQ = PT/PR, then SAS similarity and corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: PS/PQ = 4/10 = 0.4 and PT/PR = 6/15 = 0.4; ∠P shared; SAS similarity; corresponding angles congruent; converse of corresponding angles theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with AD/AB = AE/AC. Prove that DE ∥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q3/5

1747

QUESTION ID

S43-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

DE ∥ BC

GRADING NOTE REQUIRED WORK: SAS similarity using the shared angle, then congruent corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: AD/AB = AE/AC and ∠A shared; SAS similarity gives △ADE ~ △ABC; ∠ADE ≅ ∠ABC; congruent corresponding angles (transversal AB) make DE ∥ BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, M is the midpoint of AB and N is the midpoint of AC. Use the converse of the triangle proportionality theorem to prove that MN ∥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q4/5

1748

QUESTION ID

S43-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

MN ∥ BC

GRADING NOTE REQUIRED WORK: Show the two side ratios are equal, then apply the converse.

Any valid proof (paragraph, two-column, or flow). Essential: AM = MB and AN = NC, so AM/MB = 1 = AN/NC; by the converse, MN ∥ BC. Invalid: assuming MN ∥ BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with AD = 3, DB = 5, AE = 4.5, EC = 7.5, and BC = 16. Prove that DE ∥ BC, then find DE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q5/5

1749

QUESTION ID

S43-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

DE ∥ BC; DE = 6

GRADING NOTE REQUIRED WORK: Compare AD/DB and AE/EC, apply the converse, then use △ADE ~ △ABC.

Any valid proof (paragraph, two-column, or flow). Essential: 3/5 = 0.6 = 4.5/7.5, so DE ∥ BC by the converse; △ADE ~ △ABC with ratio AD/AB = 3/8; DE = 16 · 3/8 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q5/5

SET 43 • HSG-SRT.B.4

BACKWARD 1

Use the triangle proportionality theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC.
AD = 6, DB = 4, AE = 9. Find EC.

EC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q1/5

1751

QUESTION ID
S43-B1-Q1

TYPE
Short answer

CORRECT ANSWER
6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q1/5

True or false?

In △ABC, D is on AB and E is on AC. If DE ∥ BC, then AD/DB = AE/EC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q2/5

1752

QUESTION ID

S43-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. AD = 2, AB = 8, AC = 12. Find AE.

A. 9

B. 4

C. 6

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q3/5

1753

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE 2/8 = AE/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC. State the converse of "If DE ∥ BC, then AD/DB = AE/EC."

Converse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q4/5

1754

QUESTION ID

S43-B1-Q4

TYPE

Short answer

CORRECT ANSWER

If AD/DB = AE/EC, then DE ∥ BC.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q4/5

True or false?

A theorem and its converse are always both true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q5/5

1755

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Some converses are false; this one must be proved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSG-SRT.B.4

BACKWARD 2

SAS similarity • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Two pairs of proportional sides with congruent included angles make triangles similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q1/5

1757

QUESTION ID

S43-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE SAS similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC, so △ADE and △ABC share ∠A. AD/AB = AE/AC = 0.4. Are they similar? Criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q2/5

1758

QUESTION ID

S43-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes, SAS similarity

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q2/5

After △ADE ~ △ABC, which angle is congruent to ∠ADE?

A. ∠ACB

B. ∠BAC

C. ∠ABC

D. ∠DEA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q3/5

1759

QUESTION ID

S43-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. ∠ABC

GRADING NOTE D ↔ B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Ratios 3/6 and 4/8 with a shared included angle: similar? Factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q4/5

1760

QUESTION ID

S43-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Yes; factor 2 (or ½)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q4/5

1760

True or false?

SAS similarity needs the angle to be between the two pairs of sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q5/5

1761

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSG-SRT.B.4

BACKWARD 3

Converse of the corresponding angles theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Congruent corresponding angles make two lines parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q1/5

1763

QUESTION ID

S43-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC. ∠ADE ≅ ∠ABC, with AB as the transversal. What can you conclude about DE and BC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q2/5

1764

QUESTION ID

S43-B3-Q2

TYPE

Short answer

CORRECT ANSWER

DE ∥ BC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q2/5

In △ABC, D is on AB and E is on AC. Which angles, if congruent, prove DE ∥ BC?

A. ∠A and ∠B

B. ∠B and ∠C

C. ∠AED and ∠ACB

D. ∠ADE and ∠AED

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S43-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q3/5

1765

QUESTION ID

S43-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. ∠AED and ∠ACB

GRADING NOTE Corresponding angles along AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S43-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q3/5

Corresponding angles measure 58° and 62°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q4/5

1766

QUESTION ID

S43-B3-Q4

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q4/5

True or false?

Vertical angles prove lines parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q5/5

1767

QUESTION ID

S43-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are always congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • HSG-SRT.B.4

FORWARD 1

Use the converse to decide parallelism in figures • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB with AD = 5, DB = 10, and E is on AC with AE = 4. How long must EC be for DE ∥ BC?

EC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q1/5

1769

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △PQR, S on PQ, T on PR: PS = x, SQ = 6, PT = 4, TR = 8. For which x is ST ∥ QR?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q2/5

1770

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q2/5

True or false?

The midsegment theorem's parallel part is a special case of the converse (ratio 1 : 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q3/5

1771

QUESTION ID

S43-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE AD/DB = AE/EC = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q3/5

Points divide two sides in ratios 2 : 3 and 4 : 6. The segment joining them is

A. perpendicular to the third side

B. half the third side

C. parallel to the third side

D. not parallel to the third side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q4/5

1772

QUESTION ID

S43-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. parallel to the third side

GRADING NOTE 2 : 3 = 4 : 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB with AD : DB = 1 : 2 and DE ∥ BC with E on AC. Find DE : BC.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q5/5

1773

QUESTION ID

S43-F1-Q5

TYPE

Short answer

CORRECT ANSWER

1 : 3

GRADING NOTE Any equivalent form. AD : AB = 1 : 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSG-SRT.B.4

FORWARD 2

Prove the trapezoid midsegment theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Trapezoid ABCD has AB ∥ DC. M and N are the midpoints of legs AD and BC. Prove that MN ∥ AB and MN = ½(AB + DC).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

1775

QUESTION ID

S43-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

MN ∥ AB and MN = ½(AB + DC)

GRADING NOTE REQUIRED WORK: Draw diagonal AC; let P be the midpoint of AC; use the midsegment theorem (or the converse) in △ACD and △ABC.

Any valid proof (paragraph, two-column, or flow). Essential: in △ADC, MP ∥ DC and MP = ½DC; in △ABC, PN ∥ AB and PN = ½AB; AB ∥ DC so MP and PN are both parallel to AB through P, so M, P, N are collinear; MN = ½(AB + DC).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A trapezoid has bases 10 and 16. Find the length of the segment joining the midpoints of the legs.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

1776

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

True or false?

The midsegment of a trapezoid is half the length of the longer base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

1777

QUESTION ID

S43-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is half the sum of the bases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A trapezoid's midsegment is 12 and one base is 9. Find the other base.

A. 3

B. 15

C. 6

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

1778

QUESTION ID

S43-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 15

GRADING NOTE $(9 + b)/2 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

Trapezoid ABCD has AB ∥ DC; M and N are the midpoints of legs AD and BC, and P is the midpoint of diagonal AC. Explain why M, P and N lie on one line.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

1779

QUESTION ID

S43-F2-Q5

TYPE

Short answer

CORRECT ANSWER

MP and PN are both parallel to the bases through P, and only one line through P is parallel to them.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

Prove the Pythagorean Theorem using similar triangles

MAIN	Geometry main question	**HSG-SRT.B.4**
BACKWARD 1	Show triangles similar with AA	**HSG-SRT.A.3**
BACKWARD 2	Cross-multiply proportions	**7.RP.A.2**
BACKWARD 3	Explain a proof of the Pythagorean Theorem informally	**8.G.B.6**
FORWARD 1	Use the geometric mean relationships in a right triangle	**HSG-SRT.B.5**
FORWARD 2	The Law of Cosines as a generalization of the Pythagorean Theor	**HSG-SRT.D.10**

SET 44 • HSG-SRT.B.4

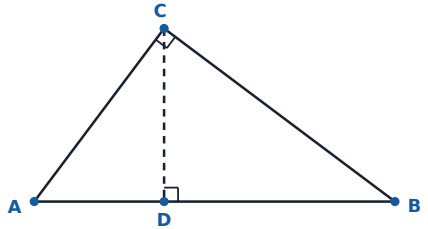
MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has the right angle at C, legs a = BC, b = AC, and hypotenuse c = AB. The altitude from C meets AB at D. Prove that $a^2 + b^2 = c^2$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q1/5

1782

QUESTION ID

S44-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$a^2 + b^2 = c^2$

GRADING NOTE REQUIRED WORK: Show △ACD ~ △ABC and △CBD ~ △ABC (AA), write two proportions, and add.

Any valid proof (paragraph, two-column, or flow). Essential: △ADC ~ △ACB (shared ∠A, right angles), so AD/AC = AC/AB and $b^2 = c \cdot AD$; △CDB ~ △ACB (shared ∠B), so BD/BC = BC/AB and $a^2 = c \cdot BD$; adding, $a^2 + b^2 = c(AD + BD) = c \cdot c = c^2$. Invalid: assuming $a^2 + b^2 = c^2$ anywhere; using the distance formula (it depends on the theorem).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle PQR has the right angle at R, legs p = QR and q = PR, and hypotenuse r = PQ. The altitude from R meets PQ at S. Prove that $p^2 + q^2 = r^2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q2/5

1783

QUESTION ID

S44-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$p^2 + q^2 = r^2$

GRADING NOTE REQUIRED WORK: Prove △PSR ~ △PRQ and △RSQ ~ △PRQ (AA), write the two proportions, and add.

Any valid proof (paragraph, two-column, or flow). Essential: shared ∠P and right angles give △PSR ~ △PRQ, so PS/PR = PR/PQ and $q^2 = r \cdot PS$; shared ∠Q gives △RSQ ~ △PRQ, so QS/QR = QR/QP and $p^2 = r \cdot QS$; adding, $p^2 + q^2 = r(PS + QS) = r^2$. Invalid: assuming the theorem or using the distance formula.

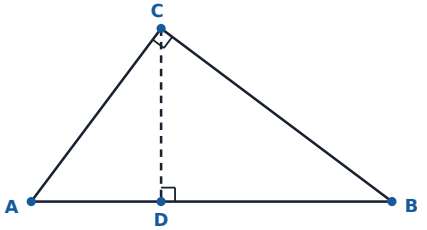
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Isosceles right triangle ABC has the right angle at C and legs AC = BC = a. The altitude from C meets AB at D. Using similar triangles (not the Pythagorean Theorem), prove that $AB^2 = 2a^2$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q3/5

1784

QUESTION ID

S44-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$AB^2 = 2a^2$

GRADING NOTE REQUIRED WORK: Use the similarity proportions $a^2 = AB \cdot AD$ and $a^2 = AB \cdot BD$, then add.

Any valid proof (paragraph, two-column, or flow). Essential: $\triangle ADC \sim \triangle ACB$ gives $a^2 = AB \cdot AD$; $\triangle CDB \sim \triangle ACB$ gives $a^2 = AB \cdot BD$; adding, $2a^2 = AB(AD + BD) = AB^2$.

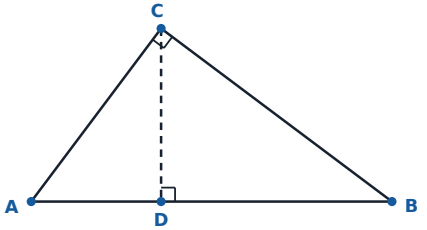
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has the right angle at C, AC = 6 and BC = 8. The altitude from C meets AB at D; let x = AD and y = DB. Use similar triangles (not the Pythagorean Theorem) to find x, y and AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q4/5

1785

QUESTION ID

S44-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AB = 10, x = 3.6, y = 6.4

GRADING NOTE REQUIRED WORK: Write $6^2 = AB \cdot x$ and $8^2 = AB \cdot y$ from the similar triangles, and use $x + y = AB$.

Any valid proof (paragraph, two-column, or flow). Essential: $\triangle ADC \sim \triangle ACB$ gives $36 = AB \cdot x$; $\triangle CDB \sim \triangle ACB$ gives $64 = AB \cdot y$; adding, $100 = AB(x + y) = AB^2$, so AB = 10, x = 3.6, y = 6.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs 5 and 12. The altitude to the hypotenuse divides it into parts x (next to the leg 5) and y. Use similar triangles (not the Pythagorean Theorem) to find the hypotenuse c, x and y.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q5/5

1786

QUESTION ID

S44-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

c = 13, x = 25/13, y = 144/13

GRADING NOTE

REQUIRED WORK: Write $5^2 = c \cdot x$ and $12^2 = c \cdot y$, add, and use $x + y = c$.

Any valid proof (paragraph, two-column, or flow). Essential: $25 = cx$ and $144 = cy$; adding, $169 = c(x + y) = c^2$, so $c = 13$; $x = 25/13 \approx 1.92$ and $y = 144/13 \approx 11.08$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q5/5

SET 44 • HSG-SRT.B.4

BACKWARD 1

Show triangles similar with AA • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Two right triangles that share an acute angle are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q1/5

1788

QUESTION ID
S44-B1-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has the right angle at C, and the altitude from C meets hypotenuse AB at D. Name two pairs of congruent angles in △ADC and △ACB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q2/5

1789

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

∠A ≅ ∠A; ∠ADC ≅ ∠ACB (right angles)

GRADING NOTE Any two valid pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q2/5

Right triangle ABC has the right angle at C, and the altitude from C meets hypotenuse AB at D. Then △ADC ~ △ACB. Which ratio equals AD/AC?

A. DC/AB

B. AD/AB

C. AC/AB

D. CB/AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q3/5

1790

QUESTION ID
S44-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. AC/AB

GRADING NOTE AD ↔ AC, AC ↔ AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q3/5

Right triangle ABC has the right angle at C, and the altitude from C meets hypotenuse AB at D. △ADC ~ △ACB, AD = 4 and AB = 9. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q4/5

1791

QUESTION ID

S44-B1-Q4

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. $AC^2 = AD \cdot AB = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q4/5

True or false?

Similar triangles must be the same size.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q5/5

1792

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only the shape matches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSG-SRT.B.4

BACKWARD 2

Cross-multiply proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

From AD/AC = AC/AB, write an equation without fractions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q1/5

1794

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$AC^2 = AD \cdot AB$

GRADING NOTE Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $4/x = x/9$ for $x > 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q2/5

1795

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If $p/q = q/r$, then

A. $q^2 = pr$

B. $p = r$

C. $r^2 = pq$

D. $p^2 = qr$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q3/5

1796

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $q^2 = pr$

GRADING NOTE Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

From $a/b = b/c$ we get $b^2 = ac$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q4/5

1797

QUESTION ID

S44-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

From x/5 = 5/x we get x = 5 only.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q5/5

1798

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 = 25$, so $x = \pm 5$ ($x = 5$ for a length).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSG-SRT.B.4

BACKWARD 3

Explain a proof of the Pythagorean Theorem informally • Grade 8 • Geometry • 8.G.B.6

True or false?

The Pythagorean Theorem says $a^2 + b^2 = c^2$ for every right triangle with legs a, b and hypotenuse c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q1/5

1800

QUESTION ID

S44-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Name a way to show the theorem by rearranging area pieces.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q2/5

1801

QUESTION ID

S44-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Arrange four copies of the triangle inside a square of side a + b (the leftover is c^2, or $a^2 + b^2$).

GRADING NOTE Any valid area argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q2/5

Which triangle satisfies the theorem?

A. 5, 12, 13

B. 6, 7, 9

C. 5, 6, 7

D. 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q3/5

1802

QUESTION ID

S44-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5, 12, 13

GRADING NOTE 25 + 144 = 169.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Legs 9 and 12. Hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q4/5

1803

QUESTION ID

S44-B3-Q4

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q4/5

True or false?

Checking the theorem with three triangles proves it for all right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q5/5

1804

QUESTION ID

S44-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Examples are not proofs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • HSG-SRT.B.4

FORWARD 1

Use the geometric mean relationships in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has the right angle at C, and altitude CD meets hypotenuse AB at D. AD = 4 and DB = 9. Find CD.

CD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q1/5

1806

QUESTION ID
S44-F1-Q1

TYPE
Short answer

CORRECT ANSWER
6

GRADING NOTE Any equivalent form. $CD^2 = AD \cdot DB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC with the right angle at C, altitude CD meets hypotenuse AB at D. Prove that $CD^2 = AD \cdot DB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q2/5

1807

QUESTION ID

S44-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

$CD^2 = AD \cdot DB$

GRADING NOTE REQUIRED WORK: Show △ACD ~ △CBD and write a proportion.

Any valid proof (paragraph, two-column, or flow). Essential: △ADC ~ △CDB (AA: right angles at D, and ∠A ≅ ∠BCD since both are complementary to ∠B); AD/CD = CD/DB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q2/5

True or false?

The altitude to the hypotenuse is the geometric mean of the two parts of the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q3/5

1808

QUESTION ID

S44-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $CD^2 = AD \cdot DB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has its right angle at C, and the altitude from C meets hypotenuse AB at D. AD = 2 and AB = 8. Find AC.

A. 4

B. $2\sqrt{2}$

C. 6

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q4/5

1809

QUESTION ID
S44-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 4

GRADING NOTE $AC^2 = AD \cdot AB = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle's altitude to the hypotenuse is 12 and one part of the hypotenuse is 9. Find the other part.

Part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q5/5

1810

QUESTION ID
S44-F1-Q5

TYPE
Short answer

CORRECT ANSWER
16

GRADING NOTE Any equivalent form. 144 = 9x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSG-SRT.B.4

FORWARD 2

The Law of Cosines as a generalization of the Pythagorean Theorem • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

For a triangle with sides a, b, c and angle C opposite c, the Law of Cosines says $c^2 = a^2 + b^2 - 2ab \cos C$. What does it become when $C = 90°$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q1/5

1812

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$c^2 = a^2 + b^2$ (the Pythagorean Theorem), since $\cos 90° = 0$.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has sides 5 and 8 with a 60° angle between them. Find the third side.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q2/5

1813

QUESTION ID
S44-F2-Q2

TYPE
Short answer

CORRECT ANSWER
7

GRADING NOTE Any equivalent form. 25 + 64 − 40 = 49.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q2/5

True or false?

If $a^2 + b^2 > c^2$, the angle opposite c is obtuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S44-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q3/5

1814

QUESTION ID

S44-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is acute; the angle is obtuse when $a^2 + b^2 < c^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q3/5

A triangle has sides 4, 5, 7. The largest angle is

A. right

B. impossible to tell

C. obtuse

D. acute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q4/5

1815

QUESTION ID

S44-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. obtuse

GRADING NOTE $16 + 25 = 41 < 49$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q4/5

Use the converse idea: is a triangle with sides 7, 24, 25 right, acute, or obtuse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q5/5

1816

QUESTION ID

S44-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Right

(49 + 576 = 625)

GRADING NOTE Form required: the classification with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q5/5

Use similarity criteria to solve problems in geometric figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Similarity criteria: AA, SAS, SSS	**HSG-SRT.A.3**
BACKWARD 2	Solve proportions	**7.RP.A.2**
BACKWARD 3	Identify corresponding sides in overlapping triangles	**HSG-SRT.A.2**
FORWARD 1	Solve multistep problems with two or more pairs of similar triangl…	**HSG-SRT.B.5**
FORWARD 2	Solve similarity problems with algebra (unknowns on both sides)	**HSG-SRT.B.5**

SET 45 • HSG-SRT.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. AD = 6, DB = 9, DE = 8. Find BC. Name the criterion you used for similar triangles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q1/5

1819

QUESTION ID
S45-M-Q1

TYPE
Short answer

CORRECT ANSWER
BC = 20; △ADE ~ △ABC by AA.

GRADING NOTE Form required: the length and the criterion. 6/15 = 8/BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A lamppost lights a student who is 1.6 m tall and stands 3 m from the post. The student's shadow is 2 m long, measured away from the post. How tall is the lamppost? Name the similarity criterion you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q2/5

1820

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

4 m; AA (the shared angle at the tip of the shadow, and two right angles).

GRADING NOTE Form required: the height and the criterion. The tip of the shadow is 5 m from the post: 1.6/2 = h/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q2/5

Two triangles: △PQR with PQ = 8, QR = 12, ∠Q = 50°; △STU with ST = 6, TU = 9, ∠T = 50°. Are they similar? If so, find the ratio of their perimeters.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q3/5

1821

QUESTION ID

S45-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes, SAS similarity (8/6 = 12/9 = 4/3); perimeter ratio 4 : 3.

GRADING NOTE Form required: the criterion and the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To find the width w of a river, a surveyor marks points so that segments AD and BE cross at C, with ∠B and ∠E right angles. AB = 12 m, BC = 18 m and CE = 60 m; w = DE is across the river. Find w and name the similarity criterion you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q4/5

1822

QUESTION ID

S45-M-Q4

TYPE

Short answer

CORRECT ANSWER

w = 40 m; △ABC ~ △DEC by AA (right angles at B and E, vertical angles at C).

GRADING NOTE Form required: the width and the criterion. 12/18 = w/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

△KLM has KL = 6, LM = 8 and KM = 12, and m∠L ≈ 117°. △NOP has NO = 9, OP = 12 and NP = 18. Are the triangles similar? Name the criterion, give the scale factor, and find m∠O.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q5/5

1823

QUESTION ID

S45-M-Q5

TYPE

Short answer

CORRECT ANSWER

Yes, SSS similarity with scale factor 1.5 (9/6 = 12/8 = 18/12); ∠O corresponds to ∠L, so m∠O ≈ 117°.

GRADING NOTE Form required: the criterion, the factor, and the angle with the correspondence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 1

Similarity criteria: AA, SAS, SSS • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Which criterion uses two pairs of congruent angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q1/5

1825

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

AA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B1-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Sides 2, 3, 4 and 4, 6, 8. Criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q2/5

1826

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

SSS similarity

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q2/5

Which information is enough for similarity?

A. Two pairs of congruent angles

B. One pair of congruent angles

C. One pair of proportional sides

D. Two sides of each triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q3/5

1827

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Two pairs of congruent angles

GRADING NOTE AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q3/5

True or false?

SAS similarity requires the congruent angle to be between the proportional sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q4/5

1828

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q4/5

True or false?

Any two right triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q5/5

1829

QUESTION ID

S45-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They need another congruent angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 2

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve 6/15 = 8/x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q1/5

1831

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 1.6/2 = h/5.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q2/5

1832

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 12/40 = 18/x.

A. 26.7

B. 54

C. 46

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q3/5

1833

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60

GRADING NOTE $x = 18 \cdot 40/12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

3/4 = 9/12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q4/5

1834

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

5/8 = 10/14

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q5/5

1835

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10/16 would be equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 3

Identify corresponding sides in overlapping triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. Which side of △ABC corresponds to DE?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q1/5

1837

QUESTION ID

S45-B3-Q1

TYPE

Short answer

CORRECT ANSWER

BC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B3-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q1/5

True or false?

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. Side AD corresponds to side AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q2/5

1838

QUESTION ID

S45-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Correspondence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q2/5

In △ABC, D is on AB and E is on AC with DE ∥ BC. In △ADE ~ △ABC, the side of the large triangle that corresponds to AD is

A. DB

B. AC

C. BC

D. AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q3/5

1839

QUESTION ID

S45-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. AB

GRADING NOTE Whole side, not the piece DB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q3/5

D is on segment AB with AD = 6 and DB = 9. What is AB?

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q4/5

1840

QUESTION ID

S45-B3-Q4

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q4/5

True or false?

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. DB corresponds to DE.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q5/5

1841

QUESTION ID
S45-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE DB is not a side of △ADE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q5/5

SET 45 • HSG-SRT.B.5

FORWARD 1

Solve multistep problems with two or more pairs of similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Trapezoid ABCD has AB ∥ DC, AB = 15, DC = 10. The diagonals meet at E, and AC = 20. Find AE and EC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q1/5

1843

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

AE = 12, EC = 8

GRADING NOTE Form required: both. AE/EC = 15/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q1/5

A pinhole camera: an object 2 m tall is 8 m from the pinhole; the image is 5 cm from the pinhole on the screen. How tall is the image?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q2/5

1844

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1.25 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q2/5

True or false?

In a figure with two pairs of similar triangles, one ratio found in the first pair can be reused in the second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q3/5

1845

QUESTION ID

S45-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Shared sides link them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D and F are on AB and E and G are on AC, with DE ∥ BC and FG ∥ BC. AD = 2, DB = 6, AF = 5 and BC = 16. Find FG.

A. 10

B. 12

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q4/5

1846

QUESTION ID

S45-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE FG/BC = AF/AB = 5/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Using a mirror on the ground, Ana (eyes 1.5 m high) stands 2 m from the mirror and sees the top of a tree 12 m beyond the mirror. How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q5/5

1847

QUESTION ID

S45-F1-Q5

TYPE

Short answer

CORRECT ANSWER

9 m

GRADING NOTE Any equivalent form. 1.5/2 = h/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSG-SRT.B.5

FORWARD 2

Solve similarity problems with algebra (unknowns on both sides) • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In △ABC, D is on AB and E is on AC with DE ∥ BC.
AD = x, DB = 6, AE = 4, EC = x + 2. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q1/5

1849

QUESTION ID
S45-F2-Q1

TYPE
Short answer

CORRECT ANSWER
4

GRADING NOTE Any equivalent form. x(x + 2) = 24, x > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Similar triangles have corresponding sides x + 1 and 2x − 3 with scale factor 1.5 (larger to smaller is 2x − 3 to x + 1). Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q2/5

1850

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. 2x − 3 = 1.5(x + 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q2/5

True or false?

A proportion with the unknown in two places can lead to a quadratic equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q3/5

1851

QUESTION ID

S45-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cross-multiplying gives products.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC with the right angle at C, the altitude CD meets hypotenuse AB at D. AD = x, DB = x + 5, and CD = 6. Find x.

A. 4

B. 3

C. 6

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q4/5

1852

QUESTION ID

S45-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE x(x + 5) = 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two similar rectangles: the first is 4 by x, the second is x by 9. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q5/5

1853

QUESTION ID

S45-F2-Q5

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 4/x = x/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q5/5

Use congruence criteria to prove relationships in figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Choose and apply SSS, SAS, ASA, AAS	**HSG-CO.B.8**
BACKWARD 2	Find hidden parts: shared sides, vertical angles, right angles	**HSG-CO.C.9**
BACKWARD 3	Use corresponding parts to finish a proof	**HSG-CO.B.7**
FORWARD 1	Prove relationships that need two congruences in a row	**HSG-SRT.B.5**
FORWARD 2	Decide which given data are sufficient for a proof, and supply a m…	**HSG-SRT.B.5**

SET 46 • HSG-SRT.B.5

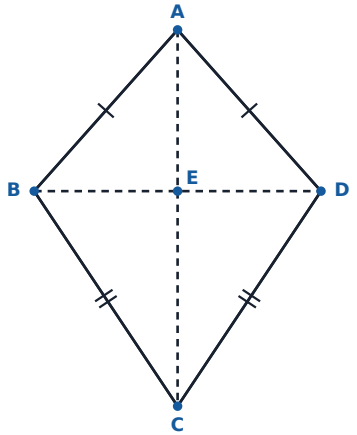
MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Kite ABCD has AB = AD and CB = CD. Its diagonals meet at E. Prove that AC ⊥ BD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q1/5

1856

QUESTION ID

S46-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AC ⊥ BD

GRADING NOTE REQUIRED WORK: Prove △ABC ≅ △ADC (SSS) to get ∠BAE ≅ ∠DAE; then △ABE ≅ △ADE (SAS); then the angles at E form a congruent linear pair.

Any valid proof (paragraph, two-column, or flow). Essential: SSS (AB = AD, CB = CD, AC shared); ∠BAC ≅ ∠DAC; SAS on △ABE and △ADE (AE shared); ∠AEB ≅ ∠AED, a linear pair, so 90°. Alternative: A and C are each equidistant from B and D, so AC is the perpendicular bisector of BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q1/5

Segments AB and CD bisect each other at M. Prove that AC ∥ DB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q2/5

1857

QUESTION ID

S46-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

AC ∥ DB

GRADING NOTE REQUIRED WORK: Prove △AMC ≅ △BMD (SAS with vertical angles), then use congruent alternate interior angles.

Any valid proof (paragraph, two-column, or flow). Essential: AM = BM, CM = DM (bisect), ∠AMC ≅ ∠BMD (vertical); SAS; ∠CAM ≅ ∠DBM; these are alternate interior angles for AC and DB with transversal AB; converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Point P is on the bisector of ∠ABC. PD ⊥ BA at D and PE ⊥ BC at E. Prove that PD = PE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q3/5

1858

QUESTION ID

S46-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

PD = PE

GRADING NOTE REQUIRED WORK: Prove △PBD ≅ △PBE by AAS.

Any valid proof (paragraph, two-column, or flow). Essential: ∠PBD ≅ ∠PBE (bisector), ∠PDB ≅ ∠PEB (right angles), PB shared; AAS; PD = PE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, AB = AC. M is the midpoint of BC. Prove that AM bisects ∠BAC and AM ⊥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q4/5

1859

QUESTION ID

S46-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

∠BAM ≅ ∠CAM and AM ⊥ BC

GRADING NOTE REQUIRED WORK: Prove △ABM ≅ △ACM (SSS), then use corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: SSS; ∠BAM ≅ ∠CAM (bisects); ∠AMB ≅ ∠AMC form a linear pair, so each is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrilateral ABCD, AB ∥ DC and AB = DC. The diagonals AC and BD meet at E. Prove that the diagonals bisect each other.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q5/5

1860

QUESTION ID

S46-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

AE = CE and BE = DE

GRADING NOTE REQUIRED WORK: Prove △ABE ≅ △CDE by ASA using alternate interior angles, then use corresponding parts.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BAE ≅ ∠DCE and ∠ABE ≅ ∠CDE (alternate interior angles, AB ∥ DC); AB = CD; ASA; AE = CE and BE = DE (corresponding parts).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 1

Choose and apply SSS, SAS, ASA, AAS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Two triangles share a side and have two more pairs of congruent sides. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q1/5

1862

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

SSS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q1/5

Two angles and a non-included side match. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q2/5

1863

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

AAS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q2/5

Which pair of facts plus a shared side gives SAS?

A. A pair of congruent sides and the congruent included angles

B. Two pairs of congruent sides not including the angle

C. Nothing more is needed

D. Two pairs of congruent angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q3/5

1864

QUESTION ID
S46-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. A pair of congruent sides and the congruent included angles

GRADING NOTE Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q3/5

True or false?

A shared side is a valid pair for any criterion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q4/5

1865

QUESTION ID

S46-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q4/5

True or false?

AAA proves congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q5/5

1866

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 2

Find hidden parts: shared sides, vertical angles, right angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Segments AB and CD cross at M. Which angles are vertical to ∠AMC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q1/5

1868

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

∠BMD

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q1/5

True or false?

Perpendicular lines give congruent right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q2/5

1869

QUESTION ID

S46-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q2/5

Which is a hidden fact in figures where two triangles share a side?

A. All angles are equal

B. The shared side is congruent to itself

C. The triangles are right

D. Nothing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q3/5

1870

GRADING NOTE Reflexive property.

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The shared side is congruent to itself

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q3/5

A bisector divides ∠ABC into which two congruent angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q4/5

1871

QUESTION ID

S46-B2-Q4

TYPE

Short answer

CORRECT ANSWER

∠ABP and ∠PBC (for bisector ray BP)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Vertical angles are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q5/5

1872

QUESTION ID

S46-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 3

Use corresponding parts to finish a proof • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

ANSWER KEY Reduced copy of the question page, with its answer

△ABC ≅ △ADC. Which angle corresponds to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q1/5

1874

QUESTION ID

S46-B3-Q1

TYPE

Short answer

CORRECT ANSWER

∠DAC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q1/5

True or false?

Once triangles are congruent, all corresponding parts are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q2/5

1875

QUESTION ID

S46-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q2/5

△PBD ≅ △PBE. Which conclusion follows?

A. PD = PE

B. PB = PD

C. BD = PE

D. ∠D ≅ ∠B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q3/5

1876

QUESTION ID

S46-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. PD = PE

GRADING NOTE Corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two congruent angles form a linear pair. Each measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q4/5

1877

QUESTION ID

S46-B3-Q4

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q4/5

True or false?

Corresponding parts can be used before the triangles are proved congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q5/5

1878

QUESTION ID

S46-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only after.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q5/5

SET 46 • HSG-SRT.B.5

FORWARD 1

Prove relationships that need two congruences in a row • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrilateral ABCD, AB = CD and AD = CB. Diagonals AC and BD meet at E. Prove that E is the midpoint of AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q1/5

1880

QUESTION ID

S46-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AE = CE

GRADING NOTE REQUIRED WORK: Prove △ABC ≅ △CDA (SSS) to get parallel sides, then △ABE ≅ △CDE (ASA).

Any valid proof (paragraph, two-column, or flow). Essential: SSS gives alternate interior angles congruent (AB ∥ DC); then ASA (or AAS) for △ABE and △CDE gives AE = CE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, AB = AC. D is on AB and E is on AC with BD = CE. Prove that BE = CD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q2/5

1881

QUESTION ID

S46-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

BE = CD

GRADING NOTE REQUIRED WORK: Show AD = AE, then prove △ABE ≅ △ACD by SAS (shared ∠A).

Any valid proof (paragraph, two-column, or flow). Essential: AD = AB − BD = AC − CE = AE; ∠A shared; AB = AC; SAS; corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q2/5

True or false?

Some proofs need one congruence to find a part used in a second congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q3/5

1882

QUESTION ID

S46-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Chained congruences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q3/5

In a two-step proof, the first congruence gives $\angle 1 \cong \angle 2$. This is used to

A. measure

B. end the proof

C. supply a pair of parts for the second congruence

D. restart

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q4/5

1883

QUESTION ID

S46-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. supply a pair of parts for the second congruence

GRADING NOTE Chaining.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q4/5

Explain why you cannot use corresponding parts of the second pair of triangles before proving them congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q5/5

1884

QUESTION ID
S46-F1-Q5

TYPE
Short answer

CORRECT ANSWER

That would assume what must be proved (circular reasoning).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSG-SRT.B.5

FORWARD 2

Decide which given data are sufficient for a proof, and supply a missing condition • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In △PQR and △PSR (sharing PR), PQ = PS. What one more fact would prove ∠QPR ≅ ∠SPR? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q1/5

1886

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

QR = SR (then SSS gives the congruence and the angles correspond).

GRADING NOTE Any sufficient condition with explanation (∠PRQ ≅ ∠PRS would give SSA, which is not enough).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q1/5

A student claims a kite's diagonals bisect each other. Prove or disprove.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q2/5

1887

GRADING NOTE Any valid counterexample or explanation.

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

False in general: only the diagonal BD is bisected (by AC); AC is bisected only if the kite is a rhombus. Example: A(0, 3), B(−1, 0), C(0, −1), D(1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q2/5

True or false?

Given AB = DE, BC = EF, and ∠C ≅ ∠F, the triangles must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q3/5

1888

QUESTION ID

S46-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE SSA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q3/5

Given ∠A ≅ ∠D and AB = DE in △ABC and △DEF, which extra fact gives ASA?

A. ∠C ≅ ∠F

B. BC = EF

C. AC = DF

D. ∠B ≅ ∠E

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q4/5

1889

QUESTION ID

S46-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ∠B ≅ ∠E

GRADING NOTE The angle at the other end of AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Given AB = DE and AC = DF, name an angle condition that guarantees congruence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q5/5

1890

QUESTION ID
S46-F2-Q5

TYPE
Short answer

CORRECT ANSWER
∠A ≅ ∠D (the included angle)

GRADING NOTE Form required: the included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q5/5

Use similarity criteria to prove relationships in figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Prove triangles similar with AA	**HSG-SRT.A.3**
BACKWARD 2	Turn a proportion into a product	**7.RP.A.2**
BACKWARD 3	Match corresponding sides of similar triangles	**HSG-SRT.A.2**
FORWARD 1	Use proved product relationships to compute	**HSG-SRT.B.5**
FORWARD 2	Prove relationships in circles with similar triangles	**HSG-C.A.2**

SET 47 • HSG-SRT.B.5

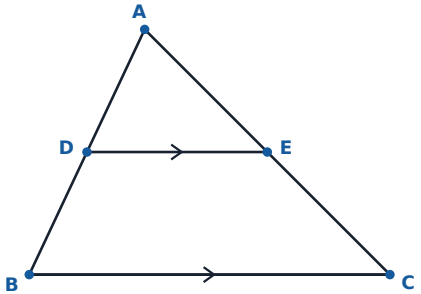
MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, DE ∥ BC with D on AB and E on AC. Prove that AD · AC = AE · AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q1/5

1893

QUESTION ID

S47-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AD · AC = AE · AB

GRADING NOTE REQUIRED WORK: Prove △ADE ~ △ABC (AA), write the proportion, and cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: AA (shared ∠A; corresponding angles); AD/AB = AE/AC; cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Chords AB and CD of a circle meet at P, and ∠BAD ≅ ∠BCD (given). Prove that PA · PB = PC · PD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q2/5

1894

QUESTION ID

S47-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

PA · PB = PC · PD

GRADING NOTE REQUIRED WORK: Prove △PAD ~ △PCB (AA: the given angles and vertical angles), then cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: ∠PAD ≅ ∠PCB (given), ∠APD ≅ ∠CPB (vertical); AA; PA/PC = PD/PB; cross-multiply.

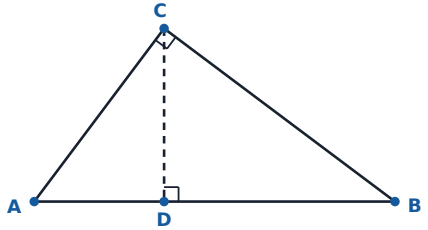
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC (right angle at C), CD is the altitude to AB. Prove that AC · BC = AB · CD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q3/5

1895

QUESTION ID

S47-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

AC · BC = AB · CD

GRADING NOTE REQUIRED WORK: Prove △ACB ~ △ADC (or △CDB) and use the ratio of corresponding sides.

Any valid proof (paragraph, two-column, or flow). Essential: △ADC ~ △ACB (AA); DC/CB = AC/AB; cross-multiply. Alternative: both products equal twice the area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In trapezoid ABCD with AB ∥ DC, the diagonals AC and BD meet at E. Prove that AE/EC = AB/DC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q4/5

1896

QUESTION ID

S47-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AE/EC = AB/DC

GRADING NOTE REQUIRED WORK: Prove △ABE ~ △CDE by AA (alternate interior angles or vertical angles).

Any valid proof (paragraph, two-column, or flow). Essential: ∠EAB ≅ ∠ECD and ∠EBA ≅ ∠EDC (alternate interior, AB ∥ DC); AA; corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q4/5

In acute triangle ABC, the altitudes from A and B meet BC at D and AC at E. Prove that CD · CB = CE · CA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q5/5

1897

QUESTION ID

S47-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

CD · CB = CE · CA

GRADING NOTE REQUIRED WORK: Prove △CDA ~ △CEB (AA), write the proportion, and cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: ∠C is shared; ∠CDA and ∠CEB are right angles; AA; CD/CE = CA/CB; cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 1

Prove triangles similar with AA • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Vertical angles can be one pair of congruent angles in an AA proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q1/5

1899

QUESTION ID

S47-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two triangles share ∠A and have another pair of congruent angles. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q2/5

1900

QUESTION ID
S47-B1-Q2

TYPE
Short answer

CORRECT ANSWER
AA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Parallel lines in a figure usually supply

A. equal side lengths

B. a scale factor

C. right angles

D. congruent corresponding or alternate interior angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q3/5

1901

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. congruent corresponding or alternate interior angles

GRADING NOTE Angle pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-B1-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

△PAD ~ △PCB. Which side of △PCB corresponds to PA?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q4/5

1902

QUESTION ID

S47-B1-Q4

TYPE

Short answer

CORRECT ANSWER

PC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-B1-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Similarity requires all three pairs of sides to be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q5/5

1903

QUESTION ID
S47-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Proportional, not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 2

Turn a proportion into a product • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

From a/b = c/d, write the product relationship.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q1/5

1905

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

ad = bc

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

From PA/PC = PD/PB, write the product relationship.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q2/5

1906

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

PA · PB = PC · PD

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If x/3 = 4/6, then

A. 6x = 12

B. x = 8

C. 3x = 24

D. 4x = 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q3/5

1907

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6x = 12

GRADING NOTE Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

a/b = c/d implies ab = cd.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q4/5

1908

QUESTION ID

S47-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It implies ad = bc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

2/5 = 6/15 implies 2 · 15 = 5 · 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q5/5

1909

QUESTION ID

S47-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 30 = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 3

Match corresponding sides of similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC, and △ADE ~ △ABC. Write the proportion with AD, AB, AE, AC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q1/5

1911

QUESTION ID

S47-B3-Q1

TYPE

Short answer

CORRECT ANSWER

AD/AB = AE/AC

GRADING NOTE Any equivalent proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q1/5

True or false?

In △ABE ~ △CDE, AB corresponds to CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q2/5

1912

QUESTION ID
S47-B3-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A ↔ C, B ↔ D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q2/5

△ACB ~ △ADC. Which side corresponds to CB?

A. DC

B. AD

C. AB

D. AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q3/5

1913

QUESTION ID

S47-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. DC

GRADING NOTE A ↔ A, C ↔ D, B ↔ C, so CB ↔ DC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-B3-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

△XYZ ~ △UVW. Write all three equal ratios.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q4/5

1914

QUESTION ID

S47-B3-Q4

TYPE

Short answer

CORRECT ANSWER

XY/UV = YZ/VW = XZ/UW

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q4/5

True or false?

Any two sides of similar triangles are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q5/5

1915

QUESTION ID

S47-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSG-SRT.B.5

FORWARD 1

Use proved product relationships to compute • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Chords AB and CD meet at P with PA = 4, PB = 9, PC = 6. Find PD.

PD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q1/5

1917

QUESTION ID
S47-F1-Q1

TYPE
Short answer

CORRECT ANSWER
6

GRADING NOTE Any equivalent form. 4 · 9 = 6 · PD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC (right angle at C), AC = 6, BC = 8, AB = 10. Find the altitude CD to AB.

CD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q2/5

1918

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4.8

GRADING NOTE Any equivalent form. 6 · 8 = 10 · CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q2/5

True or false?

If two chords meet inside a circle, the products of their segments are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q3/5

1919

QUESTION ID
S47-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Proved by similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Trapezoid diagonals: AB = 18, DC = 12, AE = 9. Find EC.

A. 8

B. 4

C. 13.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q4/5

1920

QUESTION ID

S47-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 6

GRADING NOTE AE/EC = 18/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC.
AD = 3, AB = 9, AC = 12. Find AE.

AE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q5/5

1921

QUESTION ID
S47-F1-Q5

TYPE
Short answer

CORRECT ANSWER
4

GRADING NOTE Any equivalent form. AD · AC = AE · AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSG-SRT.B.5

FORWARD 2

Prove relationships in circles with similar triangles • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

From point P outside a circle, a tangent touches the circle at T and a secant meets it at A and B (A between P and B). Given ∠PTA ≅ ∠PBT, prove that $PT^2 = PA \cdot PB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q1/5

1923

QUESTION ID

S47-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

$PT^2 = PA \cdot PB$

GRADING NOTE REQUIRED WORK: Prove △PTA ~ △PBT (AA), then cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: shared ∠P and the given angles; AA; PT/PB = PA/PT; cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q1/5

A tangent segment from P is 8 and the secant from P has outer part 4. Find the whole secant length.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q2/5

1924

QUESTION ID
S47-F2-Q2

TYPE
Short answer

CORRECT ANSWER
16

GRADING NOTE Any equivalent form. 64 = 4 · PB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q2/5

True or false?

Two secants from an outside point P satisfy PA + PB = PC + PD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q3/5

1925

QUESTION ID

S47-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The products are equal: PA · PB = PC · PD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

From P, secants meet a circle at A, B and at C, D with PA = 3, PB = 12, PC = 4. Find PD.

A. 7

B. 1

C. 16

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q4/5

1926

QUESTION ID

S47-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 9

GRADING NOTE $36 = 4 \cdot PD$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q4/5

Explain why the angle facts used in these circle proofs depend on inscribed angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q5/5

1927

QUESTION ID
S47-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Angles that intercept the same arc are congruent (inscribed angle theorem), which supplies the AA pairs.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q5/5

Understand trigonometric ratios as properties of an angle, by similarity

MAIN	Geometry main question	**HSG-SRT.C.6**
BACKWARD 1	Similar triangles have equal side ratios within each triangle	**HSG-SRT.A.2**
BACKWARD 2	Name the opposite side, adjacent side, and hypotenuse for an ang	**HSG-SRT.A.3**
BACKWARD 3	Write and simplify ratios	**6.RP.A.1**
FORWARD 1	Find trig ratios from one ratio, using a similar right triangle	**HSG-SRT.C.6**
FORWARD 2	The unit circle extends the ratios beyond acute angles	**HSF-TF.A.2**

SET 48 • HSG-SRT.C.6

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Right triangles ABC and DEF have right angles at C and F and ∠A ≅ ∠D. Explain why BC/AB = EF/DE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q1/5

1930

QUESTION ID

S48-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

BC/AB = EF/DE

GRADING NOTE REQUIRED WORK: Use AA similarity, then the proportion of corresponding sides, rearranged.

Any valid proof (paragraph, two-column, or flow). Essential: right angles and ∠A ≅ ∠D give AA similarity; BC/EF = AB/DE; rearranging gives BC/AB = EF/DE. So the ratio depends only on the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two right triangles each have a 30° angle. In one, the side opposite 30° is 4 and the hypotenuse is 8. In the other, the hypotenuse is 15. Find the side opposite 30°, and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q2/5

1931

QUESTION ID
S48-M-Q2

TYPE
Short answer

CORRECT ANSWER
7.5; the triangles are similar (AA), so opposite/hypotenuse = 4/8 = ½ in both.

GRADING NOTE Form required: the length and the similarity reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle PQR, the right angle is at R, PR = 5, QR = 12, PQ = 13. Write sin P, cos P, and tan P.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q3/5

1932

QUESTION ID

S48-M-Q3

TYPE

Short answer

CORRECT ANSWER

sin P = 12/13, cos P = 5/13, tan P = 12/5

GRADING NOTE Form required: all three ratios (any equivalent fractions).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has the right angle at C, AC = 4 and BC = 3. Right triangle DEF has the right angle at F, ∠D ≅ ∠A and DE = 20. Find EF and DF, and explain why tan A = tan D.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q4/5

1933

QUESTION ID

S48-M-Q4

TYPE

Short answer

CORRECT ANSWER

AB = 5 and the triangles are similar (AA) with factor 4, so EF = 12 and DF = 16; tan A = 3/4 and tan D = 12/16 = 3/4, equal because corresponding sides are proportional.

GRADING NOTE Form required: both lengths and the similarity explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A student says cos 50° depends on how big the right triangle is. Use two right triangles with a 50° angle, one with hypotenuse 1 and one with hypotenuse 10, to explain why the student is wrong, and give cos 50° to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q5/5

1934

QUESTION ID

S48-M-Q5

TYPE

Short answer

CORRECT ANSWER

The triangles are similar (AA), so the side adjacent to 50° is 10 times as long when the hypotenuse is 10 times as long; adjacent/hypotenuse is the same, cos 50° ≈ 0.643.

GRADING NOTE Form required: the similarity explanation and the value to the nearest thousandth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 1

Similar triangles have equal side ratios within each triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Triangles with sides 3, 4, 5 and 6, 8, 10 are similar.
Compare 3/5 and 6/10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q1/5

1936

QUESTION ID
S48-B1-Q1

TYPE
Short answer

CORRECT ANSWER
Both equal 0.6.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B1-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q1/5

True or false?

If △ABC ~ △DEF, then AB/AC = DE/DF.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q2/5

1937

QUESTION ID

S48-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Ratios within each triangle match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

△ABC ~ △DEF with BC/AB = 0.8. Find EF/DE.

A. Cannot tell

B. 0.8

C. 1.25

D. 0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q3/5

1938

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.8

GRADING NOTE Same ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q3/5

A triangle has legs 2 and 3. A similar one has shorter leg 8. Find the longer leg.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q4/5

1939

QUESTION ID

S48-B1-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B1-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Similar triangles have equal side lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q5/5

1940

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Proportional, not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 2

Name the opposite side, adjacent side, and hypotenuse for an angle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC with right angle at C, which side is opposite ∠A?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q1/5

1942

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

BC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC with right angle at C, which side is adjacent to ∠A (not the hypotenuse)?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q2/5

1943

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

AC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q2/5

In a right triangle, the hypotenuse is always

A. opposite the right angle

B. next to the right angle

C. the shortest side

D. opposite the smallest angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q3/5

1944

QUESTION ID

S48-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. opposite the right angle

GRADING NOTE Longest side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q3/5

True or false?

In right triangle ABC with the right angle at C, the side opposite ∠A is adjacent to ∠B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q4/5

1945

QUESTION ID

S48-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE BC is next to ∠B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q4/5

True or false?

The opposite side of an angle depends on the triangle's size.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q5/5

1946

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It depends on position.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 3

Write and simplify ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write 12 : 13 as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q1/5

1948

QUESTION ID

S48-B3-Q1

TYPE

Short answer

CORRECT ANSWER

12/13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify 8/10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q2/5

1949

QUESTION ID

S48-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4/5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equals 0.75?

A. 7/5

B. 3/7

C. 3/4

D. 4/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q3/5

1950

QUESTION ID

S48-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/4

GRADING NOTE 3 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

5/13 ≈ 0.385

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q4/5

1951

QUESTION ID

S48-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Division.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A ratio changes when both quantities are tripled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q5/5

1952

QUESTION ID

S48-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • HSG-SRT.C.6

FORWARD 1

Find trig ratios from one ratio, using a similar right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

For an acute angle θ, sin θ = 3/5. Find cos θ and tan θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q1/5

1954

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

cos θ = 4/5, tan θ = 3/4

GRADING NOTE Form required: both. A 3-4-5 triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

tan θ = 5/12 for an acute angle θ. Find sin θ.

sin θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q2/5

1955

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5/13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q2/5

True or false?

For any acute angle, $(\sin \theta)^2 + (\cos \theta)^2 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q3/5

1956

QUESTION ID

S48-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pythagorean Theorem divided by the hypotenuse squared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos θ = 0.6. Which is sin θ (θ acute)?

A. 0.8

B. 1.6

C. 0.4

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q4/5

1957

QUESTION ID

S48-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.8

GRADING NOTE √(1 − 0.36).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $\sin\theta < 1$ for every acute angle θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q5/5

1958

QUESTION ID

S48-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The opposite leg is shorter than the hypotenuse.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSG-SRT.C.6

FORWARD 2

The unit circle extends the ratios beyond acute angles • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

On a unit circle, the point for angle θ is (cos θ, sin θ).
Find the point for 90°.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q1/5

1960

QUESTION ID
S48-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(0, 1)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q1/5

Find the coordinates of the point on the unit circle at 30°.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q2/5

1961

QUESTION ID

S48-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(√3/2, ½)

GRADING NOTE Form required: exact values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q2/5

True or false?

For acute angles, the unit-circle definition of sine gives a different value from the right-triangle definition.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q3/5

1962

QUESTION ID
S48-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They agree: the hypotenuse is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

sin 150° on the unit circle equals

A. −½

B. −√3/2

C. √3/2

D. ½

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q4/5

1963

QUESTION ID

S48-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ½

GRADING NOTE Point (−√3/2, ½).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the unit circle lets us define sine for 120°, when no right triangle has a 120° acute angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q5/5

1964

QUESTION ID

S48-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The point on the unit circle at 120° has a y-coordinate, which defines sin 120°.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q5/5

Explain and use the relationship between the sine and cosine of complementary angles

MAIN	Geometry main question	**HSG-SRT.C.7**
BACKWARD 1	Complementary angles add to 90°	**7.G.B.5**
BACKWARD 2	Opposite and adjacent sides depend on the chosen angle	**HSG-SRT.C.6**
BACKWARD 3	Angle sum in a right triangle	**8.G.A.5**
FORWARD 1	Relationships between tangents of complementary angles	**HSG-SRT.C.7**
FORWARD 2	The Pythagorean identity	**HSF-TF.C.8**

SET 49 • HSG-SRT.C.7

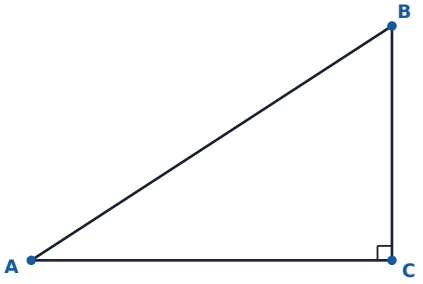
MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has its right angle at C. Explain why sin A = cos B.

Work space



Conclusion: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q1/5

1967

QUESTION ID

S49-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

sin A = BC/AB = cos B, and A and B are complementary.

GRADING NOTE REQUIRED WORK: Name the side opposite ∠A and the side adjacent to ∠B, and use the angle sum.

Any valid proof (paragraph, two-column, or flow). Essential: sin A = BC/AB (opposite over hypotenuse); cos B = BC/AB (BC is adjacent to ∠B); so they are equal; ∠A + ∠B = 90°, so sin A = cos(90° − A).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sin 25° = \cos x°$, where $0 < x < 90$. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q2/5

1968

QUESTION ID

S49-M-Q2

TYPE

Short answer

CORRECT ANSWER

65

GRADING NOTE Any equivalent form. Complementary angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

θ is acute and cos θ = 0.6. Find sin(90° − θ).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q3/5

1969

QUESTION ID

S49-M-Q3

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Both angles are acute and sin(2x + 10)° = cos(3x)°.
Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q4/5

1970

QUESTION ID
S49-M-Q4

TYPE
Short answer

CORRECT ANSWER
16

GRADING NOTE Any equivalent form. 2x + 10 + 3x = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A ladder leans against a vertical wall on level ground and makes an acute angle θ with the ground, with sin θ = 0.8. Find cos(90° − θ), say which angle in the picture measures 90° − θ, and explain why the two values are equal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q5/5

1971

QUESTION ID

S49-M-Q5

TYPE

Short answer

CORRECT ANSWER

0.8. The angle between the ladder and the wall is 90° − θ; for it, the height on the wall is the adjacent side, and for θ it is the opposite side, so both ratios are height/ladder.

GRADING NOTE Form required: the value, the angle, and the explanation with the shared side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 1

Complementary angles add to 90° • Grade 7 • Geometry • 7.G.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find the complement of 37°.

Complement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q1/5

1973

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

53°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q1/5

True or false?

The two acute angles of a right triangle are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q2/5

1974

QUESTION ID

S49-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They add to 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Angles x and 2x are complementary. x =

A. 90°

B. 45°

C. 60°

D. 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q3/5

1975

QUESTION ID

S49-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 30°

GRADING NOTE 3x = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The complement of θ is 90° − θ. If θ = 64°, find it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q4/5

1976

QUESTION ID

S49-B1-Q4

TYPE

Short answer

CORRECT ANSWER

26°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Complementary angles must be adjacent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q5/5

1977

QUESTION ID

S49-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They need only add to 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 2

Opposite and adjacent sides depend on the chosen angle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC (right angle at C), name the side adjacent to ∠B.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q1/5

1979

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

BC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-B2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In right triangle ABC with the right angle at C, the side opposite ∠A is the side adjacent to ∠B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q2/5

1980

QUESTION ID

S49-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both are BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC with the right angle at C, cos B equals

A. AC/AB

B. BC/AB

C. AC/BC

D. BC/AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q3/5

1981

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. BC/AB

GRADING NOTE Adjacent over hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Write sin A using the sides of right triangle ABC (right angle at C).

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q4/5

1982

QUESTION ID

S49-B2-Q4

TYPE

Short answer

CORRECT ANSWER

BC/AB

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

sin A and cos A are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q5/5

1983

QUESTION ID

S49-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when A = 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 3

Angle sum in a right triangle • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has an acute angle of 28°. Find the other acute angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q1/5

1985

QUESTION ID

S49-B3-Q1

TYPE

Short answer

CORRECT ANSWER

62°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A right triangle can have two right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q2/5

1986

QUESTION ID

S49-B3-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sum would exceed 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a right triangle, the acute angles are x and x + 20. x =

A. 70°

B. 45°

C. 55°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q3/5

1987

QUESTION ID

S49-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 35°

GRADING NOTE 2x + 20 = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why do the acute angles of a right triangle add to 90°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q4/5

1988

QUESTION ID

S49-B3-Q4

TYPE

Short answer

CORRECT ANSWER

The angles add to 180° and the right angle uses 90°.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q4/5

True or false?

The two acute angles of an isosceles right triangle are 45° each.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q5/5

1989

QUESTION ID

S49-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal and complementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q5/5

SET 49 • HSG-SRT.C.7

FORWARD 1

Relationships between tangents of complementary angles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $\tan\theta \cdot \tan(90° - \theta) = 1$ for an acute angle θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q1/5

1991

QUESTION ID

S49-F1-Q1

TYPE

Short answer

CORRECT ANSWER

tan θ = opposite/adjacent and tan(90° − θ) = adjacent/opposite, which are reciprocals.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q1/5

tan 20° ≈ 0.364. Use the complementary relationship to estimate tan 70° to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q2/5

1992

QUESTION ID

S49-F1-Q2

TYPE

Short answer

CORRECT ANSWER

About 2.75

(1 ÷ 0.364 ≈ 2.747)

GRADING NOTE Accept 2.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q2/5

True or false?

sin θ = cos θ exactly when θ = 45° (for acute θ).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q3/5

1993

QUESTION ID

S49-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then 90° − θ = θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equals 1 for every acute θ?

A. $\tan\theta + \tan(90° - \theta)$

B. $\tan\theta \cdot \tan(90° - \theta)$

C. $\sin\theta \cdot \cos\theta$

D. $\sin\theta + \cos\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q4/5

1994

GRADING NOTE Reciprocals.

QUESTION ID

S49-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $\tan\theta \cdot \tan(90° - \theta)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that sin θ / cos(90° − θ) = 1 for acute θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q5/5

1995

QUESTION ID

S49-F1-Q5

TYPE

Short answer

CORRECT ANSWER

cos(90° − θ) = sin θ, so the quotient is sin θ / sin θ = 1.

GRADING NOTE Any equivalent argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSG-SRT.C.7

FORWARD 2

The Pythagorean identity • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Prove that $(\sin \theta)^2 + (\cos \theta)^2 = 1$ for every acute angle θ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q1/5

1997

QUESTION ID

S49-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

$(\sin \theta)^2 + (\cos \theta)^2 = 1$

GRADING NOTE REQUIRED WORK: Use a right triangle with legs a, b and hypotenuse c and the Pythagorean Theorem.

Any valid proof (paragraph, two-column, or flow). Essential: $\sin \theta = a/c$, $\cos \theta = b/c$; $a^2 + b^2 = c^2$; divide by c^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

sin θ = 0.28 for acute θ. Find cos θ.

cos θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q2/5

1998

QUESTION ID

S49-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.96

GRADING NOTE Any equivalent form. √(1 − 0.0784).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin 35° + \cos 35° = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q3/5

1999

QUESTION ID
S49-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The squares add to 1; sin 35° + cos 35° ≈ 1.39.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos θ = 12/13 (θ acute). sin θ =

A. 1/13

B. 5/13

C. 12/5

D. 13/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q4/5

2000

QUESTION ID

S49-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5/13

GRADING NOTE √(1 − 144/169).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use complementary angles to show $(\sin 20°)^2 + (\sin 70°)^2 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q5/5

2001

QUESTION ID

S49-F2-Q5

TYPE

Short answer

CORRECT ANSWER

sin 70° = cos 20°, so the sum is $(\sin 20°)^2 + (\cos 20°)^2 = 1$.

GRADING NOTE Any equivalent argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q5**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q5/5

SET 50 • HSG-SRT.C.8 • SIMILARITY, RIGHT TRIANGLES, AND TRIGONOMETRY

Find unknown sides of right triangles with trigonometric ratios in applied problems

MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Choose the trigonometric ratio that links the known and unknown.	**HSG-SRT.C.6**
BACKWARD 2	Solve one-variable equations with a multiplication or division	**HSA-REI.B.3**
BACKWARD 3	Round to a stated precision and keep track of units	**HSN-Q.A.3**
FORWARD 1	Find a height from two angle-of-elevation observations	**HSG-SRT.C.8**
FORWARD 2	Find a feasible location under a clearance constraint	**HSG-MG.A.3**

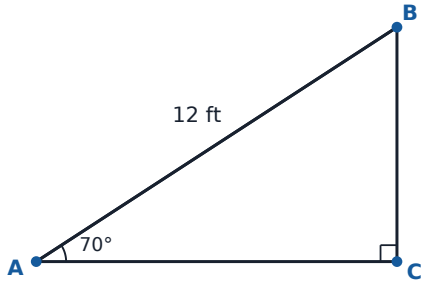
SET 50 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A 12 ft ladder leans against a wall, making a 70° angle with the level ground. How high up the wall does it reach, to the nearest tenth of a foot?



Not drawn to scale.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q1/5

2004

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

11.3 ft

GRADING NOTE Rounding to 0.1 required. 12 sin 70° = 11.2763.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A wheelchair ramp rises 2.5 ft and makes an 8° angle with the ground. How long is the ramp surface, to the nearest tenth of a foot?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q2/5

2005

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

18.0 ft

GRADING NOTE Rounding to 0.1 required. 2.5 ÷ sin 8° = 17.9632.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

From a point 25 m from the base of a tree, on level ground, the angle of elevation to the top is 38°. How tall is the tree, to the nearest tenth of a meter?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q3/5

2006

QUESTION ID

S50-M-Q3

TYPE

Short answer

CORRECT ANSWER

19.5 m

GRADING NOTE Rounding to 0.1 required. 25 tan 38° = 19.5321.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A kite string 60 m long makes a 50° angle with the level ground. About how high is the kite above the point where the string is held (nearest meter)?

A. 78 m

B. 46 m

C. 39 m

D. 72 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q4/5

2007

QUESTION ID

S50-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 46 m

GRADING NOTE Opposite = hypotenuse × sin 50° = 45.96.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 20 and a 32° angle. Find the leg adjacent to the 32° angle, to the nearest tenth.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q5/5

2008

QUESTION ID
S50-M-Q5

TYPE
Short answer

CORRECT ANSWER
17.0

GRADING NOTE Rounding to 0.1 required. 20 cos 32° = 16.9610.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 1

Choose the trigonometric ratio that links the known and unknown sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

You know the hypotenuse and an angle and want the opposite side. Which ratio?

A. none

B. sine

C. cosine

D. tangent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q1/5

2010

QUESTION ID

S50-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. sine

GRADING NOTE Opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

You know the adjacent side and an angle and want the opposite side. Which ratio?

A. none

B. tangent

C. cosine

D. sine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q2/5

2011

QUESTION ID

S50-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. tangent

GRADING NOTE Opposite/adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q2/5

2011

ANSWER KEY Reduced copy of the question page, with its answer

Which ratio uses the adjacent side and the hypotenuse?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q3/5

2012

QUESTION ID

S50-B1-Q3

TYPE

Short answer

CORRECT ANSWER

Cosine

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B1-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

To find the hypotenuse from the opposite side and the angle, use sine.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q4/5

2013

QUESTION ID

S50-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE sin A = opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Tangent uses the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q5/5

2014

QUESTION ID

S50-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Tangent uses the two legs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 2

Solve one-variable equations with a multiplication or division • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve x/12 = 0.94.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q1/5

2016

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

11.28

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B2-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 0.14 = 2.5/x. Round to the nearest hundredth.

x ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q2/5

2017

QUESTION ID

S50-B2-Q2

TYPE

Short answer

CORRECT ANSWER

17.86

GRADING NOTE Rounding to 0.01 required. 2.5 ÷ 0.14 = 17.857....

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B2-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q2/5

2017

ANSWER KEY Reduced copy of the question page, with its answer

If sin 30° = h/10, then h =

A. 10 sin 30°

B. 10 ÷ sin 30°

C. sin 30° ÷ 10

D. 30 ÷ 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q3/5

2018

QUESTION ID

S50-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10 sin 30°

GRADING NOTE Multiply both sides by 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

From 0.5 = 4/x we get x = 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q4/5

2019

QUESTION ID

S50-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE x = 4 ÷ 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

From tan 40° = 20/x we get x = 20 tan 40°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q5/5

2020

QUESTION ID

S50-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x = 20 ÷ tan 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q5/5

2020

SET 50 • HSG-SRT.C.8

BACKWARD 3

Round to a stated precision and keep track of units • High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Round 11.2763 to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q1/5

2022

QUESTION ID

S50-B3-Q1

TYPE

Short answer

CORRECT ANSWER

11.3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B3-Q1**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A calculation gives 19.532 m. Report it to the nearest tenth of a meter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q2/5

2023

QUESTION ID

S50-B3-Q2

TYPE

Short answer

CORRECT ANSWER

19.5 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B3-Q2**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q2/5

2023

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Rounding in the middle of a calculation can change the final answer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q3/5

2024

QUESTION ID
S50-B3-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Round at the end.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q3/5

2024

ANSWER KEY Reduced copy of the question page, with its answer

A height is measured in feet and the angle in degrees. The answer's unit is

A. degrees

B. feet per degree

C. feet

D. no unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q4/5

2025

QUESTION ID
S50-B3-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. feet

GRADING NOTE Lengths keep their units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

17.96 rounded to the nearest tenth is 17.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q5/5

2026

QUESTION ID

S50-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 18.0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q5/5

SET 50 • HSG-SRT.C.8

FORWARD 1

Find a height from two angle-of-elevation observations • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

From point A on level ground the angle of elevation to the top of a tower is 25°. From point B, 40 m closer, it is 40°. Find the tower's height to the nearest tenth of a meter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q1/5

2028

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

42.0 m

GRADING NOTE Rounding to 0.1 required. h = 40 ÷ (1/tan 25° − 1/tan 40°) = 41.984. Two equations: h = x tan 40° and h = (x + 40) tan 25°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

From point A on level ground the angle of elevation to the top of a tower is 25°. From point B, 40 m closer, it is 40°. How far is point B from the base of the tower, to the nearest tenth of a meter?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q2/5

2029

QUESTION ID

S50-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50.0 m

GRADING NOTE Rounding to 0.1 required. x = h ÷ tan 40° = 50.034.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q2/5

True or false?

With only one angle of elevation and no distance, the height cannot be found.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q3/5

2030

QUESTION ID

S50-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE One equation has two unknowns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equations describe two observations 40 m apart (angles 25° and 40°, distance x from the closer point)?

A. h = 40 tan 25°

B. h = x tan 25° and h = (x + 40) tan 40°

C. h = x tan 40° and h = (x + 40) tan 25°

D. h = x sin 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q4/5

2031

QUESTION ID

S50-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. h = x tan 40° and h = (x + 40) tan 25°

GRADING NOTE The closer point has the larger angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q4/5

Explain why the closer observation point has the larger angle of elevation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q5/5

2032

QUESTION ID

S50-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The same height over a shorter horizontal distance gives a larger tangent, so a larger angle.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSG-SRT.C.8

FORWARD 2

Find a feasible location under a clearance constraint • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A ramp must rise 3 ft, and its angle with the ground may be at most 4.8°. What is the shortest horizontal run, in whole feet, that meets the requirement?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q1/5

2034

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

36 ft

GRADING NOTE Form required: a whole number of feet that meets the constraint (round up). 3 ÷ tan 4.8° = 35.73.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A camera on a 6 m pole must see a point on the ground at an angle of depression of at least 20°. What is the greatest allowed distance from the pole, in tenths of a meter, that still meets the requirement?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q2/5

2035

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

16.4 m

GRADING NOTE Form required: a distance in tenths of a meter that meets the constraint (round DOWN). 6 ÷ tan 20° = 16.485; farther points need smaller angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q2/5

2035

True or false?

Rounding UP the run of a ramp with a maximum angle can make the ramp too steep.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q3/5

2036

QUESTION ID

S50-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A longer run makes the ramp less steep; rounding the run down is the risk.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A ladder must reach 10 ft up and make an angle of at most 75° with the ground. To the tenth of a foot, rounded so that the ladder still works, the shortest ladder is

A. 10.0 ft

B. 9.7 ft

C. 2.7 ft

D. 10.4 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q4/5

2037

QUESTION ID

S50-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 10.4 ft

GRADING NOTE At exactly 75°, length = 10 ÷ sin 75° = 10.35; a smaller angle needs a longer ladder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A building code allows stairs with an angle of at most 37°. A staircase must rise 9 ft. What is the shortest horizontal run, in tenths of a foot, that meets the code?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q5/5

2038

QUESTION ID

S50-F2-Q5

TYPE

Short answer

CORRECT ANSWER

12.0 ft

GRADING NOTE Form required: a run in tenths of a foot that meets the constraint (round UP; 11.9 ft would be too steep). 9 ÷ tan 37° = 11.943.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q5/5

Find unknown angles with inverse trigonometric ratios in applied problems

MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Inverse trig undoes a trig ratio	**HSG-SRT.C.6**
BACKWARD 2	Choose the ratio from the two known sides	**HSG-SRT.C.6**
BACKWARD 3	Use degree mode and round angles	**HSN-Q.A.3**
FORWARD 1	Solve a right triangle completely from two sides	**HSG-SRT.C.8**
FORWARD 2	Find directions (bearings) with inverse trigonometry	**HSG-SRT.C.8**

SET 51 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A ramp rises 1 m over a horizontal distance of 12 m. What angle does it make with the ground, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q1/5

2041

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

4.8°

GRADING NOTE Rounding to 0.1° required. $\tan^{-1}(1/12) = 4.764°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 6 m ladder reaches 5.5 m up a vertical wall. What angle does it make with the level ground, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q2/5

2042

QUESTION ID
S51-M-Q2

TYPE
Short answer

CORRECT ANSWER
66.4°

GRADING NOTE Rounding to 0.1° required. $\sin^{-1}(5.5/6) = 66.444°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A 10 m flagpole casts a 14 m shadow on level ground. Find the angle of elevation of the sun, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q3/5

2043

QUESTION ID
S51-M-Q3

TYPE
Short answer

CORRECT ANSWER
35.5°

GRADING NOTE Rounding to 0.1° required. $\tan^{-1}(10/14) = 35.538°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q3/5

A right triangle has legs 7 and 24. What is its smaller acute angle, to the nearest tenth of a degree?

A. 73.7°

B. 16.3°

C. 32.5°

D. 26.3°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q4/5

2044

QUESTION ID

S51-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 16.3°

GRADING NOTE $\tan^{-1}(7/24) = 16.260°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q4/5

A road rises 6 m for every 100 m of horizontal distance (a 6% grade). What angle does it make with the horizontal, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q5/5

2045

QUESTION ID

S51-M-Q5

TYPE

Short answer

CORRECT ANSWER

3.4°

GRADING NOTE Rounding to 0.1° required. $\tan^{-1}(0.06) = 3.434°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 1

Inverse trig undoes a trig ratio • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

sin θ = 0.5 for an acute angle θ. Find θ.

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q1/5

2047

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

tan θ = 1 for an acute angle θ. Find θ.

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q2/5

2048

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which gives an angle from a ratio?

A. cos

B. tan

C. sin

D. $\sin^{-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q3/5

2049

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\sin^{-1}$

GRADING NOTE Inverse functions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\cos^{-1}(0.5) = 60°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q4/5

2050

QUESTION ID

S51-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE cos 60° = 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin^{-1}(x)$ means $1 \div \sin(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q5/5

2051

QUESTION ID

S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is the angle whose sine is x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 2

Choose the ratio from the two known sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

You know the opposite side and the hypotenuse. Which inverse function finds the angle?

A. none

B. $\cos^{-1}$

C. $\sin^{-1}$

D. $\tan^{-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q1/5

2053

QUESTION ID
S51-B2-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. $\sin^{-1}$

GRADING NOTE Opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q1/5

You know both legs. Which inverse function finds an acute angle?

A. $\sin^{-1}$

B. $\tan^{-1}$

C. $\cos^{-1}$

D. none

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q2/5

2054

QUESTION ID

S51-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $\tan^{-1}$

GRADING NOTE Opposite/adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Rise 5 and run 12. Write the ratio for the angle with the ground.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q3/5

2055

QUESTION ID

S51-B2-Q3

TYPE

Short answer

CORRECT ANSWER

tan θ = 5/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B2-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q3/5

True or false?

Adjacent side and hypotenuse give the angle by $\cos^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q4/5

2056

QUESTION ID

S51-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cosine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q4/5

True or false?

Opposite side and hypotenuse give the angle by $\tan^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q5/5

2057

QUESTION ID

S51-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sin^{-1}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 3

Use degree mode and round angles • High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Round 4.7636° to the nearest tenth of a degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q1/5

2059

QUESTION ID

S51-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4.8°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B3-Q1**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A calculator in radian mode gives $\sin^{-1}(0.5) \approx 0.524$, not 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q2/5

2060

QUESTION ID
S51-B3-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

An acute angle of a right triangle must be

A. more than 90°

B. any angle

C. exactly 45°

D. between 0° and 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q3/5

2061

QUESTION ID

S51-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. between 0° and 90°

GRADING NOTE Acute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A computed angle is 112°. Can it be an acute angle of a right triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q4/5

2062

QUESTION ID
S51-B3-Q4

TYPE
Short answer

CORRECT ANSWER
No

GRADING NOTE Any equivalent answer; check the calculation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B3-Q4**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

35.537° to the nearest tenth is 35.6°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q5/5

2063

QUESTION ID

S51-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 35.5°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q5/5

SET 51 • HSG-SRT.C.8

FORWARD 1

Solve a right triangle completely from two sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs 9 and 14. Find the hypotenuse and both acute angles (nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q1/5

2065

QUESTION ID
S51-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Hypotenuse 16.6; angles 32.7° and 57.3°

GRADING NOTE Rounding to 0.1 required for all three; the angles add to 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 17 and one leg 8. Find the other leg and both acute angles (nearest tenth of a degree).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q2/5

2066

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Other leg 15; angles 28.1° and 61.9°

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q2/5

True or false?

Once one acute angle is known, the other is 90° minus it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q3/5

2067

QUESTION ID

S51-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Complementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q3/5

Legs 5 and 5. Angles?

A. 45° and 45°

B. 50° and 40°

C. 30° and 60°

D. 5° and 85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q4/5

2068

QUESTION ID

S51-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 45° and 45°

GRADING NOTE Isosceles right triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how to check a solved right triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q5/5

2069

QUESTION ID

S51-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The acute angles add to 90°, $a^2 + b^2 = c^2$, and each ratio matches its angle.

GRADING NOTE Any reasonable checks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSG-SRT.C.8

FORWARD 2

Find directions (bearings) with inverse trigonometry • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A boat travels 8 km east and then 5 km north. Find the angle north of east of its straight-line direction from the start, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q1/5

2071

QUESTION ID

S51-F2-Q1

TYPE

Short answer

CORRECT ANSWER

32.0°

GRADING NOTE Rounding to 0.1° required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A hiker walks 3 km north and 4 km west. What is the bearing angle west of north from the start, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q2/5

2072

QUESTION ID

S51-F2-Q2

TYPE

Short answer

CORRECT ANSWER

53.1°

GRADING NOTE Rounding to 0.1° required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q2/5

True or false?

Bearing problems become right-triangle problems when the motions are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q3/5

2073

QUESTION ID

S51-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE East-west and north-south legs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q3/5

A drone flies 120 m horizontally and rises 50 m. Its angle of climb is about

A. 22.6°

B. 24.6°

C. 67.4°

D. 50°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q4/5

2074

QUESTION ID

S51-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 22.6°

GRADING NOTE $\tan^{-1}(50/120)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A ship is 10 km east and 10 km south of port. Find the straight-line distance (to the nearest tenth of a km) and the angle south of east.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q5/5

2075

QUESTION ID

S51-F2-Q5

TYPE

Short answer

CORRECT ANSWER

14.1 km; 45°

GRADING NOTE Form required: both. $10\sqrt{2} \approx 14.1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q5/5

Solve right triangles in applied problems with trigonometry and the Pythagorean Theorem

MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Use the Pythagorean Theorem to find a side	**8.G.B.7**
BACKWARD 2	Interpret angles of elevation and depression	**HSG-SRT.C.6**
BACKWARD 3	Choose sine, cosine, or tangent	**HSG-SRT.C.6**
FORWARD 1	Solve problems that need two right triangles	**HSG-SRT.C.8**
FORWARD 2	Solve non-right triangles with the Law of Sines	**HSG-SRT.D.11**

SET 52 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

From the top of an 80 m cliff, the angle of depression to a boat is 25°. How far is the boat from the base of the cliff, to the nearest tenth of a meter? How far is it from the observer?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q1/5

2078

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

171.6 m from the base; 189.3 m from the observer

GRADING NOTE Rounding to 0.1 required for both. Horizontal = 80 ÷ tan 25°; line of sight = 80 ÷ sin 25°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane flying at an altitude of 2,000 m is seen at an angle of elevation of 14°. Find the horizontal ground distance and the line-of-sight distance to the plane, to the nearest meter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q2/5

2079

QUESTION ID
S52-M-Q2

TYPE
Short answer

CORRECT ANSWER
8022 m horizontal; 8267 m line of sight

GRADING NOTE Rounding to the nearest meter required for both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A ramp is 15 ft long and rises 2 ft. Find its horizontal run (nearest tenth of a foot) and its angle with the ground (nearest tenth of a degree).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q3/5

2080

QUESTION ID

S52-M-Q3

TYPE

Short answer

CORRECT ANSWER

Run 14.9 ft; angle 7.7°

GRADING NOTE Rounding required as stated. Run = $\sqrt{(15^2 - 2^2)}$; angle = $\sin^{-1}(2/15)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A 25 ft guy wire from the top of a pole meets the level ground 9 ft from the pole. How tall is the pole and what angle does the wire make with the ground (nearest tenth)?

A. 26.6 ft; 68.9°

B. 16.0 ft; 21.1°

C. 23.3 ft; 21.1°

D. 23.3 ft; 68.9°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q4/5

2081

QUESTION ID

S52-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 23.3 ft; 68.9°

GRADING NOTE Height $\sqrt{(25^2 - 9^2)}$; angle $\cos^{-1}(9/25)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 25 and a 40° angle. Find both legs and the other acute angle (legs to the nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q5/5

2082

QUESTION ID
S52-M-Q5

TYPE
Short answer

CORRECT ANSWER
Legs 16.1 and 19.2; other angle 50°

GRADING NOTE Form required: both legs and the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 1

Use the Pythagorean Theorem to find a side • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 25 and a leg 9. Find the other leg, to the nearest tenth.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q1/5

2084

QUESTION ID
S52-B1-Q1

TYPE
Short answer

CORRECT ANSWER
23.3

GRADING NOTE Any equivalent form. √(625 − 81).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Legs 8 and 15. Find the hypotenuse.

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q2/5

2085

QUESTION ID

S52-B1-Q2

TYPE

Short answer

CORRECT ANSWER

17

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q2/5

True or false?

In a right triangle, the hypotenuse is the longest side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q3/5

2086

QUESTION ID

S52-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is opposite the largest angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q3/5

Hypotenuse 13, leg 5. Other leg?

A. √194

B. 12

C. 8

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q4/5

2087

QUESTION ID

S52-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 12

GRADING NOTE √(169 − 25).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$a^2 + b^2 = c^2$ works for every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q5/5

2088

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 2

Interpret angles of elevation and depression • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

True or false?

An angle of depression is measured down from the horizontal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q1/5

2090

QUESTION ID
S52-B2-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The angle of depression from a cliff top to a boat is 25°. What is the angle of elevation from the boat to the cliff top?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q2/5

2091

QUESTION ID
S52-B2-Q2

TYPE
Short answer

CORRECT ANSWER
25°

GRADING NOTE Alternate interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-B2-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which side is horizontal in an angle-of-elevation triangle?

A. The hypotenuse

B. The leg along the ground

C. The vertical leg

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q3/5

2092

QUESTION ID

S52-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The leg along the ground

GRADING NOTE Level ground.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An observer looks up at 30° to a bird. Is that an angle of elevation or depression?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q4/5

2093

QUESTION ID

S52-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Elevation

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The angle of depression is measured from the vertical line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q5/5

2094

QUESTION ID
S52-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE From the horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 3

Choose sine, cosine, or tangent • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Known: opposite side and angle. Wanted: hypotenuse. Ratio?

A. none

B. tangent

C. sine

D. cosine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q1/5

2096

QUESTION ID

S52-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. sine

GRADING NOTE sin = opp/hyp.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Known: adjacent side and angle. Wanted: opposite side. Ratio?

A. cosine

B. tangent

C. none

D. sine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q2/5

2097

QUESTION ID

S52-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. tangent

GRADING NOTE tan = opp/adj.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Known: hypotenuse and angle. Wanted: adjacent side. Ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q3/5

2098

QUESTION ID

S52-B3-Q3

TYPE

Short answer

CORRECT ANSWER

Cosine

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q3/5

True or false?

Two different ratios can lead to the same correct side if used correctly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q4/5

2099

QUESTION ID

S52-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, finding the third side by trig or Pythagoras.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Tangent relates the hypotenuse and a leg.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q5/5

2100

QUESTION ID

S52-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Tangent relates the two legs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q5/5

SET 52 • HSG-SRT.C.8

FORWARD 1

Solve problems that need two right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

From a window 20 m up, the angle of elevation to the top of a building across the street is 30° and the angle of depression to its base is 40°. How tall is the building, to the nearest tenth of a meter?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q1/5

2102

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

33.8 m

GRADING NOTE Rounding to 0.1 required. Street width = 20 ÷ tan 40° = 23.835; height = 20 + width · tan 30°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two observers 100 m apart on opposite sides of a tower see its top at 35° and 50°. Find the tower's height, to the nearest tenth of a meter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q2/5

2103

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

44.1 m

GRADING NOTE Rounding to 0.1 required. h/tan 35° + h/tan 50° = 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q2/5

True or false?

A shared side links two right triangles in such problems.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q3/5

2104

QUESTION ID

S52-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, a common height or distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two observers 100 m apart on opposite sides of a tower of height h see its top at angles of elevation 35° and 50°. Which equation is correct?

A. h = 100 tan 85°

B. h/sin 35° + h/sin 50° = 100

C. h tan 35° + h tan 50° = 100

D. h/tan 35° + h/tan 50° = 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q4/5

2105

QUESTION ID
S52-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. h/tan 35° + h/tan 50° = 100

GRADING NOTE Horizontal distances add to 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two observers on opposite sides of a tower each measure an angle of elevation to its top, and the distance between them is known. Explain why the tower's height cannot be found from one of the two right triangles alone.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q5/5

2106

QUESTION ID

S52-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Each triangle alone has two unknowns (h and its distance); the shared height and total distance give the needed second equation.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSG-SRT.C.8

FORWARD 2

Solve non-right triangles with the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, A = 40°, B = 65°, and a = 12. Find b to the nearest tenth.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q1/5

2108

QUESTION ID
S52-F2-Q1

TYPE
Short answer

CORRECT ANSWER
16.9

GRADING NOTE Rounding to 0.1 required. b = a sin B ÷ sin A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, A = 52°, C = 70°, and c = 20. Find a to the nearest tenth.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q2/5

2109

QUESTION ID

S52-F2-Q2

TYPE

Short answer

CORRECT ANSWER

16.8

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

SOH-CAH-TOA applies directly to a triangle with no right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q3/5

2110

QUESTION ID

S52-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Those ratios are defined in right triangles; use the Law of Sines or Cosines, or draw an altitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q3/5

Which data allow the Law of Sines directly?

A. Two sides and the included angle

B. Two angles and any side

C. Three sides

D. Three angles only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q4/5

2111

QUESTION ID
S52-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. Two angles and any side

GRADING NOTE AAS or ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q4/5

Explain how dropping an altitude turns a non-right triangle into two right triangles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q5/5

2112

QUESTION ID
S52-F2-Q5

TYPE
Short answer

CORRECT ANSWER

The altitude creates two right triangles that share it, so h = b sin A = a sin B.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q5/5

SET 53 • HSG-C.A.1 • CIRCLES

Prove that all circles are similar

MAIN	Geometry main question	**HSG-C.A.1**
BACKWARD 1	A dilation multiplies distances by the scale factor	**HSG-SRT.A.1.b**
BACKWARD 2	Translate a point to another point	**8.G.A.3**
BACKWARD 3	A circle is all points at one distance from its center	**HSG-CO.A.1**
FORWARD 1	Find a single dilation that maps one circle onto another	**HSG-C.A.1**
FORWARD 2	Arcs of similar circles scale with the radius	**HSG-C.B.5**

SET 53 • HSG-C.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Circle C has center A and radius r, and circle D has center B and radius s. Prove that the two circles are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q1/5

2115

QUESTION ID

S53-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

A translation followed by a dilation maps C onto D, so they are similar.

GRADING NOTE REQUIRED WORK: Translate A to B, then dilate about B with scale factor s/r. Show the image of C is exactly D.

Any valid proof (paragraph, two-column, or flow). Essential: (1) the translation maps C onto the circle with center B and radius r (distances kept); (2) the dilation about B with factor s/r maps each point at distance r from B to a point at distance s on the same ray, so onto D; (3) every point of D is reached (the inverse dilation); a similarity transformation exists. Invalid: "circles all look alike"; only checking two specific circles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Circle C_1 has center (1, 2) and radius 3. Circle C_2 has center (−4, 0) and radius 6. Describe a similarity transformation that maps C_1 onto C_2 and explain why it works.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q2/5

2116

QUESTION ID

S53-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Translate 5 units left and 2 units down, then dilate about (−4, 0) with scale factor 2.

GRADING NOTE

REQUIRED WORK: Give the translation that moves the centers together and the dilation (center and factor).

Any valid proof (paragraph, two-column, or flow). Essential: the translation (x, y) → (x − 5, y − 2) moves the center to (−4, 0) keeping the radius 3; the dilation about (−4, 0) by 2 doubles every distance from the center, so radius 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Circle C_1 has center (0, 0) and radius 2. Circle C_2 has center (3, −1) and radius 5. Prove that C_1 and C_2 are similar by giving a similarity transformation, and show where it sends the point (2, 0) of C_1.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q3/5

2117

QUESTION ID

S53-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Translate by (x, y) → (x + 3, y − 1), then dilate about (3, −1) by 2.5; (2, 0) → (5, −1) → (8, −1), which is 5 from (3, −1).

GRADING NOTE REQUIRED WORK: Translate the center onto (3, −1), then dilate about (3, −1) with factor 5/2; follow (2, 0) through both steps and check its distance from (3, −1).

Any valid proof (paragraph, two-column, or flow). Essential: the translation keeps the radius 2 and moves the center to (3, −1); the dilation by 5/2 about the center makes every radius 5, so the image is C_2; the check point lands on C_2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q3/5

Explain why two circles with the same center but different radii, r and s, are similar using only ONE transformation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q4/5

2118

QUESTION ID

S53-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The dilation about the common center with factor s/r maps the first circle onto the second.

GRADING NOTE REQUIRED WORK: Name the transformation and its center and factor.

Any valid proof (paragraph, two-column, or flow). Essential: dilation about the shared center; factor s/r; each point at distance r goes to distance s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Circle C has center A and radius r, and circle D has center B and radius s. Prove that C and D are similar using a dilation FIRST and a translation SECOND.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q5/5

2119

QUESTION ID

S53-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The dilation followed by the translation maps C onto D, so they are similar.

GRADING NOTE REQUIRED WORK: Dilate about A with factor s/r, then translate A to B; show the image is exactly D.

Any valid proof (paragraph, two-column, or flow). Essential: the dilation about A by s/r maps C onto the circle with center A and radius s (every distance from A is multiplied by s/r); the translation A → B keeps distances, so it maps that circle onto the circle with center B and radius s, which is D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 1

A dilation multiplies distances by the scale factor • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

A dilation with factor 2.5 is applied to a segment of length 4. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q1/5

2121

QUESTION ID

S53-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q1/5

True or false?

A dilation about point O keeps O fixed and multiplies each distance from O by the factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q2/5

2122

QUESTION ID

S53-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q2/5

Which factor maps a circle of radius 4 about its center to one of radius 10?

A. 2.5

B. 0.4

C. 6

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q3/5

2123

QUESTION ID

S53-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2.5

GRADING NOTE 10 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle of radius 9 is dilated about its center by ⅓. New radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q4/5

2124

QUESTION ID
S53-B1-Q4

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q4/5

True or false?

A dilation can turn a circle into a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q5/5

2125

QUESTION ID

S53-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It keeps shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 2

Translate a point to another point • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Which translation moves (1, 2) onto (−4, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q1/5

2127

QUESTION ID

S53-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5 units left and 2 units down

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q1/5

True or false?

A translation keeps the radius of a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q2/5

2128

QUESTION ID

S53-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is a rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle centered at (3, 3) is translated by (x, y) → (x − 3, y + 1). New center?

A. (6, 2)

B. (0, 4)

C. (0, 3)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q3/5

2129

QUESTION ID

S53-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (0, 4)

GRADING NOTE Apply the rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A translation moves (2, −1) to (7, 4). Describe it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q4/5

2130

QUESTION ID

S53-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5 units right and 5 units up

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q4/5

True or false?

A translation changes the size of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q5/5

2131

QUESTION ID

S53-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 3

A circle is all points at one distance from its center • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Every point of a circle is the same distance from its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q1/5

2133

QUESTION ID

S53-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center (0, 0) and passes through (3, 4). Its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q2/5

2134

QUESTION ID

S53-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle is completely determined by

A. its color

B. one point on it

C. its center only

D. its center and radius

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q3/5

2135

QUESTION ID

S53-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. its center and radius

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two circles have the same center and the same radius. What can you say?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q4/5

2136

QUESTION ID

S53-B3-Q4

TYPE

Short answer

CORRECT ANSWER

They are the same circle.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q4/5

True or false?

A circle's radius can be different in different directions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q5/5

2137

QUESTION ID

S53-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The definition requires one distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • HSG-C.A.1

FORWARD 1

Find a single dilation that maps one circle onto another • High School — Geometry • Geometry: Circles • HSG-C.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Circle P: center (0, 0), radius 2. Circle Q: center (6, 0), radius 4. Find the center of the dilation with a positive factor that maps P onto Q.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q1/5

2139

QUESTION ID

S53-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(−6, 0)

GRADING NOTE Form required: an ordered pair. The center (c, 0) satisfies 6 = c + 2(0 − c), so c = −6; check: (0, 0) → −6 + 2(0 + 6) = 6, and the radius 2 → 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Circle P has center (0, 0) and radius 2. Circle Q has center (6, 0) and radius 4. What is the scale factor of a dilation that maps P onto Q?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q2/5

2140

QUESTION ID

S53-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 4 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q2/5

True or false?

Two circles with different radii can always be mapped onto each other by one dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q3/5

2141

QUESTION ID

S53-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Choose the center on the line of centers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q3/5

Circles with equal radii and different centers are related by

A. nothing

B. a translation (no dilation needed)

C. a dilation with factor 2

D. a reflection only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q4/5

2142

QUESTION ID

S53-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. a translation (no dilation needed)

GRADING NOTE Same size: congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q4/5

Circle A has radius 3 and circle B has radius 12, both centered at the origin. Which dilation maps A onto B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q5/5

2143

QUESTION ID

S53-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Center (0, 0), factor 4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSG-C.A.1

FORWARD 2

Arcs of similar circles scale with the radius • High School — Geometry • Geometry: Circles • HSG-C.B.5

Two circles have radii 3 and 7. A 60° central angle cuts off an arc in each. What is the ratio of the arc lengths?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q1/5

2145

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3 : 7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 90° arc of a circle with radius 4 has length 2π. How long is a 90° arc of a circle with radius 10?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q2/5

2146

QUESTION ID

S53-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q2/5

True or false?

For a fixed central angle, the arc length is the same in every circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q3/5

2147

QUESTION ID

S53-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Arc length is proportional to the radius; s/r is what stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An arc of length 6 in a circle of radius 2 corresponds, by similarity, to what arc length in a circle of radius 5 (same angle)?

A. 2.4

B. 15

C. 30

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q4/5

2148

QUESTION ID

S53-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 15

GRADING NOTE Multiply by 5/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q4/5

Explain how the similarity of circles leads to radian measure.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q5/5

2149

QUESTION ID

S53-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Since arc length is proportional to radius for a fixed angle, arc/radius depends only on the angle; that number is the angle in radians.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q5/5

SET 54 • HSG-C.A.2 • CIRCLES

Relate inscribed angles and central angles

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Base angles of an isosceles triangle are congruent	**HSG-CO.C.10**
BACKWARD 2	The exterior angle theorem	**HSG-CO.C.10**
BACKWARD 3	A central angle and its arc have the same measure	**4.MD.C.5.a**
FORWARD 1	Prove the inscribed angle theorem when the center is inside or out…	**HSG-C.A.2**
FORWARD 2	Inscribed angles on the same arc, and arc applications	**HSG-C.A.2**

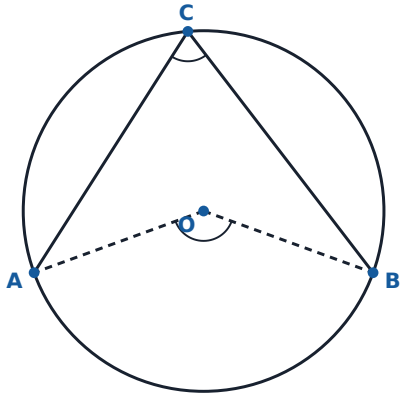
SET 54 • HSG-C.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, central angle AOB measures 110°. Inscribed angle ACB intercepts the same arc AB. Find m∠ACB.



m∠ACB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q1/5

2152

QUESTION ID

S54-M-Q1

TYPE

Short answer

CORRECT ANSWER

55°

GRADING NOTE Any equivalent form. An inscribed angle is half the central angle on the same arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q1/5

In circle O, inscribed angle PQR measures 38°. Find the measure of arc PR (the arc it intercepts) and of central angle POR.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q2/5

2153

QUESTION ID

S54-M-Q2

TYPE

Short answer

CORRECT ANSWER

Arc PR = 76°;
∠POR = 76°

GRADING NOTE Form required: both. The central angle equals its arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, AC is a diameter and B is another point on the circle. Prove that m∠BAC = ½ m∠BOC (the inscribed angle theorem when one side of the angle is a diameter).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q3/5

2154

QUESTION ID

S54-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

m∠BAC = ½ m∠BOC

GRADING NOTE REQUIRED WORK: Use isosceles triangle AOB (OA = OB) and the exterior angle theorem at O.

Any valid proof (paragraph, two-column, or flow). Essential: OA = OB (radii), so ∠OAB ≅ ∠OBA; ∠BOC is an exterior angle of △AOB, so m∠BOC = m∠OAB + m∠OBA = 2m∠BAC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed angle XYZ and central angle XOZ intercept the same arc XZ, and m∠XYZ = 47°. What is m∠XOZ?

A. 23.5°

B. 94°

C. 47°

D. 133°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q4/5

2155

QUESTION ID
S54-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 94°

GRADING NOTE Double the inscribed angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, points C and D are on the same side of chord AB, so inscribed angles ADB and ACB intercept the same arc AB. m∠ADB = (3x + 4)° and m∠ACB = (5x − 12)°. Find x, m∠ACB, and the central angle AOB that intercepts arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q5/5

2156

QUESTION ID

S54-M-Q5

TYPE

Short answer

CORRECT ANSWER

x = 8; m∠ACB = 28°; m∠AOB = 56°.

GRADING NOTE Form required: all three. Both inscribed angles are half of arc AB, so 3x + 4 = 5x − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 1

Base angles of an isosceles triangle are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

In △AOB, OA = OB and m∠OAB = 35°. Find m∠OBA and m∠AOB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q1/5

2158

QUESTION ID

S54-B1-Q1

TYPE

Short answer

CORRECT ANSWER

35° and 110°

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q1/5

True or false?

Two radii of a circle and the chord joining their ends form an isosceles triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q2/5

2159

QUESTION ID

S54-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The radii are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q2/5

Isosceles triangle with vertex angle 80°. Each base angle?

A. 100°

B. 50°

C. 80°

D. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q3/5

2160

QUESTION ID

S54-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 50°

GRADING NOTE (180 − 80)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In △OPQ with OP = OQ, m∠P = 2x and m∠Q = 50°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q4/5

2161

QUESTION ID

S54-B1-Q4

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q4/5

True or false?

In an isosceles triangle, the vertex angle always equals a base angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q5/5

2162

QUESTION ID

S54-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when equilateral.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 2

The exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has remote interior angles 30° and 30°. Find the exterior angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q1/5

2164

QUESTION ID

S54-B2-Q1

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q1/5

True or false?

An exterior angle equals the sum of the two remote interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q2/5

2165

QUESTION ID

S54-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Exterior angle 100°, one remote interior angle 45°. The other?

A. 45°

B. 80°

C. 55°

D. 135°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q3/5

2166

QUESTION ID

S54-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 55°

GRADING NOTE 100 − 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q3/5

If both remote interior angles equal x, the exterior angle is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q4/5

2167

QUESTION ID

S54-B2-Q4

TYPE

Short answer

CORRECT ANSWER

2x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q4/5

True or false?

An exterior angle equals the adjacent interior angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q5/5

2168

QUESTION ID
S54-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 3

A central angle and its arc have the same measure • Grade 4 • Measurement & Data • 4.MD.C.5.a

ANSWER KEY Reduced copy of the question page, with its answer

A central angle cuts off ¼ of a circle. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q1/5

2170

QUESTION ID

S54-B3-Q1

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q1/5

True or false?

The measure of a minor arc equals its central angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q2/5

2171

QUESTION ID

S54-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of arc measure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q2/5

An arc measures 70°. Its central angle measures

A. 70°

B. 140°

C. 290°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q3/5

2172

QUESTION ID

S54-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 70°

GRADING NOTE Equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A central angle of 150°. What fraction of the circle is its arc?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q4/5

2173

QUESTION ID

S54-B3-Q4

TYPE

Short answer

CORRECT ANSWER

5/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

All arcs of 60° have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q5/5

2174

QUESTION ID

S54-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Length depends on the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q5/5

SET 54 • HSG-C.A.2

FORWARD 1

Prove the inscribed angle theorem when the center is inside or outside the angle • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed angle ABC in circle O has the center O inside the angle. Prove that m∠ABC = ½ m(arc AC), where arc AC is the arc intercepted by ∠ABC (the arc not containing B). (Draw diameter BD.) If ∠ABC is obtuse, arc AC is a major arc and its central angle is a reflex angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q1/5

2176

QUESTION ID

S54-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠ABC = ½ m(arc AC)

GRADING NOTE REQUIRED WORK: Split ∠ABC along BD, apply the diameter case to each part, then add the arcs.

Any valid proof (paragraph, two-column, or flow). Essential: m∠ABD = ½ m(arc AD) and m∠DBC = ½ m(arc DC) (diameter case, with each arc measured by its central angle); D lies on arc AC, so m∠ABC = ½(m(arc AD) + m(arc DC)) = ½ m(arc AC). This also covers an obtuse ∠ABC, where arc AC is greater than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed angle ABC in circle O has the center O outside the angle. Prove that m∠ABC = ½ m(arc AC), where arc AC is the arc intercepted by ∠ABC (the arc not containing B). (Draw diameter BD.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q2/5

2177

QUESTION ID

S54-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

m∠ABC = ½ m(arc AC)

GRADING NOTE REQUIRED WORK: Use the diameter case twice and SUBTRACT the arcs.

Any valid proof (paragraph, two-column, or flow). Essential: label so that A lies on arc DC (A and C on the same side of BD); m∠DBC = ½ m(arc DC) and m∠DBA = ½ m(arc DA) (diameter case); subtract: m∠ABC = ½(m(arc DC) − m(arc DA)) = ½ m(arc AC). Here ∠ABC is acute and arc AC is a minor arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q2/5

True or false?

The inscribed angle theorem needs three cases for a complete proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q3/5

2178

QUESTION ID

S54-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Center on a side, inside, or outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q3/5

In the proof of the inscribed angle theorem for the case where the center lies inside the inscribed angle, a diameter splits the angle in two, and the two diameter-case results are

A. averaged

B. multiplied

C. subtracted

D. added

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q4/5

2179

QUESTION ID

S54-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. added

GRADING NOTE The angle is the sum of two pieces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q4/5

Why can't the proof of the inscribed angle theorem stop after the case in which one side of the angle is a diameter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q5/5

2180

QUESTION ID
S54-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Most inscribed angles do not have a diameter as a side; the other positions must be covered too.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSG-C.A.2

FORWARD 2

Inscribed angles on the same arc, and arc applications • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In a circle, inscribed angles ∠ADB and ∠ACB intercept the same arc AB, and m∠ADB = 41°. Find m∠ACB.

m∠ACB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q1/5

2182

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

41°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q1/5

In a circle, inscribed angles ADB and ACB intercept the same arc AB (C and D are on the same side of chord AB). Prove that ∠ADB ≅ ∠ACB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q2/5

2183

QUESTION ID

S54-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

They are congruent.

GRADING NOTE REQUIRED WORK: Each inscribed angle is half the measure of the same intercepted arc.

Any valid proof (paragraph, two-column, or flow). Essential: m∠ADB = ½ m(arc AB) and m∠ACB = ½ m(arc AB), where arc AB is the arc not containing C or D (it may be a major arc, with a reflex central angle); equal halves of the same arc, so the angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q2/5

True or false?

An inscribed angle whose intercepted arc is a 200° major arc measures 200°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q3/5

2184

QUESTION ID

S54-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is half the intercepted arc: 100°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q3/5

Points A, B, C, D are on a circle with arc AB = 80° and arc CD = 50° (arcs not containing the other points). Chords AC and BD meet at E inside. m∠AEB?

A. 30°

B. 65°

C. 15°

D. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q4/5

2185

QUESTION ID

S54-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 65°

GRADING NOTE Angle formed by two chords = half the sum of the intercepted arcs: (80 + 50)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

An inscribed angle measures 3x + 10 and its intercepted arc measures 110°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q5/5

2186

QUESTION ID

S54-F2-Q5

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. 3x + 10 = 55.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q5/5

SET 55 • HSG-C.A.2 • CIRCLES

Inscribed angles on a diameter are right angles

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Use the inscribed angle theorem	**HSG-C.A.2**
BACKWARD 2	Isosceles triangles from radii	**HSG-CO.C.10**
BACKWARD 3	Triangle angle sum	**8.G.A.5**
FORWARD 1	Use the converse to find a circle through a right triangle	**HSG-C.A.2**
FORWARD 2	Use the diameter property to construct tangents from an external…	**HSG-C.A.4**

SET 55 • HSG-C.A.2

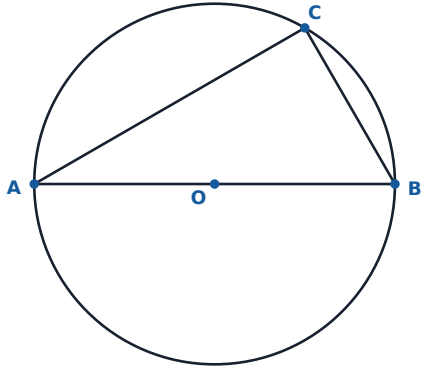
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

AB is a diameter of circle O and C is any other point on the circle. Prove that ∠ACB is a right angle.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q1/5

2189

QUESTION ID

S55-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠ACB = 90°

GRADING NOTE REQUIRED WORK: Use the inscribed angle theorem (the intercepted arc is 180°), or the two isosceles triangles AOC and BOC.

Any valid proof (paragraph, two-column, or flow). Essential (theorem): ∠ACB intercepts a semicircle, 180°, so it measures 90°. Alternative: OA = OC = OB, so △AOC and △BOC are isosceles; with base angles x and y, the angle sum of △ABC gives 2x + 2y = 180°, so x + y = 90° = m∠ACB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

AB is a diameter of a circle and C is on the circle. AC = 6 and BC = 8. Find AB and the radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q2/5

2190

QUESTION ID

S55-M-Q2

TYPE

Short answer

CORRECT ANSWER

AB = 10; radius 5

GRADING NOTE Form required: both. ∠C = 90°, so AB = √(36 + 64).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q2/5

AB is a diameter of circle O and C is on the circle with m∠CAB = 34°. Find m∠CBA.

m∠CBA = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q3/5

2191

QUESTION ID

S55-M-Q3

TYPE

Short answer

CORRECT ANSWER

56°

GRADING NOTE Any equivalent form. ∠C = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle PQR is inscribed in a circle with PR a diameter. m∠P = (3x)° and m∠R = (2x + 15)°. Find x and the measures of all three angles, and name the theorem you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q4/5

2192

QUESTION ID

S55-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = 15; m∠P = 45°, m∠R = 45°, m∠Q = 90° (an angle inscribed in a semicircle is a right angle).

GRADING NOTE Form required: x, three angles, and the theorem. 3x + 2x + 15 = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Points A(−5, 0), B(5, 0) and C(3, 4) are on the circle $x^2 + y^2 = 25$, and AB is a diameter. Use slopes to show that ∠ACB is a right angle, and explain how this agrees with the theorem.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q5/5

2193

QUESTION ID

S55-M-Q5

TYPE

Short answer

CORRECT ANSWER

Slope CA = 4/8 = ½ and slope CB = 4/(−2) = −2; the product is −1, so CA ⊥ CB. This matches the theorem: ∠ACB is inscribed in a semicircle.

GRADING NOTE Form required: both slopes, the perpendicularity conclusion, and the link to the theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 1

Use the inscribed angle theorem • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

An inscribed angle intercepts an arc of 140°. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q1/5

2195

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

70°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q1/5

True or false?

An inscribed angle is half its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q2/5

2196

QUESTION ID

S55-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q2/5

A semicircle measures

A. 360°

B. 180°

C. 90°

D. 120°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q3/5

2197

QUESTION ID

S55-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 180°

GRADING NOTE Half the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An inscribed angle measures 25°. Its arc?

Arc: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q4/5

2198

QUESTION ID

S55-B1-Q4

TYPE

Short answer

CORRECT ANSWER

50°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q4/5

True or false?

An inscribed angle equals its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q5/5

2199

QUESTION ID

S55-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 2

Isosceles triangles from radii • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

OA = OC (radii) and m∠OAC = 40°. Find m∠OCA.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q1/5

2201

QUESTION ID

S55-B2-Q1

TYPE

Short answer

CORRECT ANSWER

40°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q1/5

True or false?

Any two radii and the chord between them form an isosceles triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q2/5

2202

QUESTION ID

S55-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In △OBC with OB = OC, m∠BOC = 100°. m∠OBC =

A. 80°

B. 40°

C. 100°

D. 50°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q3/5

2203

QUESTION ID

S55-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 40°

GRADING NOTE (180 − 100)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why are ∠OAC and ∠OCA congruent when OA = OC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q4/5

2204

QUESTION ID

S55-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Base angles of an isosceles triangle are congruent.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A radius is always longer than a chord.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q5/5

2205

QUESTION ID

S55-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Chords can be longer (up to the diameter).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 3

Triangle angle sum • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a triangle are 90° and 34°. The third?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q1/5

2207

QUESTION ID

S55-B3-Q1

TYPE

Short answer

CORRECT ANSWER

56°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q1/5

True or false?

The acute angles of a right triangle add to 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q2/5

2208

QUESTION ID

S55-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 180 − 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$2x + 2y = 180$. Then $x + y =$

A. 45

B. 360

C. 90

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q3/5

2209

QUESTION ID

S55-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 90

GRADING NOTE Divide by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q3/5

2209

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has angles x, x, and 2x. Find 2x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q4/5

2210

QUESTION ID

S55-B3-Q4

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE 4x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q4/5

2210

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A triangle can have angles 90°, 60°, and 40°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q5/5

2211

QUESTION ID

S55-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 190°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q5/5

SET 55 • HSG-C.A.2

FORWARD 1

Use the converse to find a circle through a right triangle • High School — Geometry • Geometry: Circles • HSG-C.A.2

Right triangle ABC has its right angle at C, with A(0, 0) and B(8, 6). Where is the center of the circle through A, B, and C, and what is its radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q1/5

2213

QUESTION ID

S55-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Center (4, 3), the midpoint of AB; radius 5.

GRADING NOTE Form required: both. The hypotenuse is a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, m∠C = 90°. Prove that C lies on the circle with diameter AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q2/5

2214

QUESTION ID

S55-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

MC = MA = MB, so C is on the circle with center M and diameter AB.

GRADING NOTE REQUIRED WORK: Let M be the midpoint of AB and show MC = MA (for example, by completing a rectangle).

Any valid proof (paragraph, two-column, or flow). Essential: reflect C through M to D; ACBD is a parallelogram with a right angle, so a rectangle; its diagonals are congruent and bisect each other, so MC = MA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q2/5

True or false?

The hypotenuse of a right triangle is a diameter of its circumscribed circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q3/5

2215

QUESTION ID

S55-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q3/5

A right triangle has legs 5 and 12. Radius of its circumscribed circle?

A. 6.5

B. 8.5

C. 6

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q4/5

2216

QUESTION ID

S55-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6.5

GRADING NOTE Half the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q4/5

A carpenter's square (a right angle) is placed with its corner touching a circular board and its edges crossing the edge at P and Q. What is segment PQ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q5/5

2217

QUESTION ID
S55-F1-Q5

TYPE
Short answer

CORRECT ANSWER
A diameter of the board.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q5/5

SET 55 • HSG-C.A.2

FORWARD 2

Use the diameter property to construct tangents from an external point • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

Compass and straightedge only; leave all arcs.

Construct the two tangent lines from point P to circle O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q1/5

2219

QUESTION ID

S55-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the midpoint M of OP; draw the circle with center M through O and P; it meets circle O at T_1 and T_2; lines PT_1 and PT_2 are tangents.

GRADING NOTE Grade the construction marks, not only the final figure: the perpendicular bisector of OP, the circle on diameter OP, and both tangent lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

To construct tangents from P to circle O, a circle with diameter OP is drawn; it meets circle O at T_1 and T_2. Why are lines PT_1 and PT_2 tangent to circle O?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q2/5

2220

QUESTION ID

S55-F2-Q2

TYPE

Short answer

CORRECT ANSWER

T_1 is on the circle with diameter OP, so $\angle OT_1P = 90°$; a line perpendicular to a radius at its endpoint is tangent.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

From a point outside a circle there is exactly one tangent line to the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q3/5

2221

QUESTION ID

S55-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE There are two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q3/5

To construct a tangent from a point P outside circle O, a circle with diameter OP is drawn. That circle is used because

A. it passes through the center only

B. every angle inscribed in it on OP is a right angle

C. it has the same radius as circle O

D. it is tangent to circle O

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q4/5

2222

QUESTION ID

S55-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. every angle inscribed in it on OP is a right angle

GRADING NOTE Right angle at the tangent point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

O is the center, OP = 10, and the radius is 6. How long is each tangent segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q5/5

2223

QUESTION ID
S55-F2-Q5

TYPE
Short answer

CORRECT ANSWER
8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q5/5

SET 56 • HSG-C.A.2 • CIRCLES

A radius is perpendicular to the tangent at the point of tangency

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Use the Pythagorean Theorem and its converse	**8.G.B.7**
BACKWARD 2	The perpendicular is the shortest segment from a point to a line	**7.G.A.2**
BACKWARD 3	A tangent touches a circle at exactly one point	**HSG-CO.A.1**
FORWARD 1	Tangent segments from an external point are congruent	**HSG-C.A.2**
FORWARD 2	Write the equation of a tangent line in the coordinate plane	**HSG-GPE.B.5**

SET 56 • HSG-C.A.2

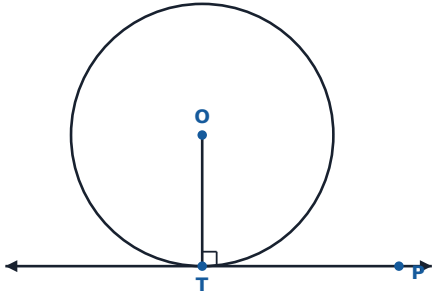
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ is tangent to circle O at T. Prove that OT $\perp \ell$. (Use an indirect argument.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q1/5

2226

QUESTION ID

S56-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

OT $\perp \ell$

GRADING NOTE REQUIRED WORK: Suppose OT is not perpendicular; drop the perpendicular from O to ℓ at Q ≠ T; show a second point of ℓ is on the circle or inside it.

Any valid proof (paragraph, two-column, or flow). Essential: if OQ $\perp \ell$ with Q ≠ T, then OQ < OT (the perpendicular is the shortest segment, as the hypotenuse OT is longest in right △OQT), so Q is inside the circle, and then ℓ would cross the circle twice, contradicting tangency. Invalid: "it looks perpendicular".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q1/5

2226

ANSWER KEY Reduced copy of the question page, with its answer

Line PT is tangent to circle O at T. OT = 5 and PT = 12. Find OP.

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q2/5

2227

QUESTION ID
S56-M-Q2

TYPE
Short answer

CORRECT ANSWER
13

GRADING NOTE Any equivalent form. ∠OTP = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A tangent from P touches circle O at T. OP = 17 and the radius is 8. Find PT.

PT = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q3/5

2228

QUESTION ID
S56-M-Q3

TYPE
Short answer

CORRECT ANSWER
15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

T lies on circle O, and P is outside it, with OT = 7, PT = 24 and OP = 26. Is line PT tangent to the circle at T? Justify your answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q4/5

2229

QUESTION ID

S56-M-Q4

TYPE

Short answer

CORRECT ANSWER

No, PT is not tangent: $7^2 + 24^2 = 625 \neq 676 = 26^2$, so $\angle OTP$ is not a right angle, and a tangent at T must be perpendicular to radius OT.

GRADING NOTE Form required: the computation, the conclusion "not tangent", and the reason (radius $\perp$ tangent).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

T is on circle O with radius 6. Line ℓ is perpendicular to OT at T, and Q is a point on ℓ with TQ = 8. Find OQ, and explain why Q is outside the circle. What does this show about line ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q5/5

2230

QUESTION ID

S56-M-Q5

TYPE

Short answer

CORRECT ANSWER

OQ = 10 > 6, so Q is outside. Every point of ℓ other than T is the end of a hypotenuse longer than OT, so ℓ meets the circle only at T: ℓ is tangent.

GRADING NOTE Form required: the length, the comparison, and the tangent conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 1

Use the Pythagorean Theorem and its converse • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Legs 5 and 12. Hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q1/5

2232

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse 17, leg 8. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q2/5

2233

QUESTION ID

S56-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A triangle with sides 7, 24, 25 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q3/5

2234

QUESTION ID

S56-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 49 + 576 = 625.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q3/5

Sides 6, 8, 11. Right triangle?

A. Yes

B. No

C. Only if 11 is a leg

D. Cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q4/5

2235

QUESTION ID

S56-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. No

GRADING NOTE 36 + 64 ≠ 121.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q4/5

True or false?

The Pythagorean Theorem applies to any triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q5/5

2236

QUESTION ID

S56-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Right triangles only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 2

The perpendicular is the shortest segment from a point to a line • Grade 7 • Geometry • 7.G.A.2

True or false?

The shortest segment from a point to a line is perpendicular to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

2238

QUESTION ID

S56-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any other segment is a hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a right triangle, which side is longest?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

2239

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The hypotenuse

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

O is 4 cm from line ℓ (perpendicular distance). Another segment from O to ℓ could be

A. 4 cm only

B. 5 cm

C. 2 cm

D. 3 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

2240

QUESTION ID

S56-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5 cm

GRADING NOTE At least 4 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

Why is the perpendicular segment shorter than any slanted one?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

2241

QUESTION ID

S56-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Each slanted segment is the hypotenuse of a right triangle with the perpendicular as a leg.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

True or false?

The distance from a point to a line is measured along any segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

2242

QUESTION ID

S56-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Along the perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 3

A tangent touches a circle at exactly one point • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

A tangent line meets the circle at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q1/5

2244

QUESTION ID
S56-B3-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q1/5

A line meets a circle at two points. It is a

A. secant

B. radius

C. diameter

D. tangent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q2/5

2245

QUESTION ID

S56-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. secant

GRADING NOTE Two points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q2/5

Point Q on a line is inside the circle. Can the line be tangent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q3/5

2246

QUESTION ID

S56-B3-Q3

TYPE

Short answer

CORRECT ANSWER

No; it would cross the circle twice.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the point where a tangent touches the circle called?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q4/5

2247

QUESTION ID

S56-B3-Q4

TYPE

Short answer

CORRECT ANSWER

The point of tangency

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q4/5

True or false?

A tangent can pass through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q5/5

2248

QUESTION ID

S56-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Then it would meet the circle twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q5/5

SET 56 • HSG-C.A.2

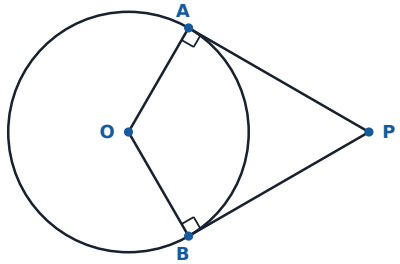
FORWARD 1

Tangent segments from an external point are congruent • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

From point P outside circle O, tangent segments PA and PB touch the circle at A and B. Prove that PA = PB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q1/5

2250

QUESTION ID

S56-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

PA = PB

GRADING NOTE REQUIRED WORK: Use the right angles at A and B, OA = OB, and the shared hypotenuse OP (HL, or the Pythagorean Theorem).

Any valid proof (paragraph, two-column, or flow). Essential: ∠OAP = ∠OBP = 90°; OA = OB; OP shared; HL (or PA = $\sqrt{(OP^2 - r^2)}$ = PB).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangent segments from P to a circle are 3x + 4 and 25. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q2/5

2251

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q2/5

True or false?

The segment from P to the center bisects the angle between the two tangents.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q3/5

2252

QUESTION ID
S56-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Congruent triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q3/5

A circle is inscribed in a triangle with sides tangent to it. From one vertex, the two tangent segments are

A. equal

B. parallel

C. unequal

D. perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q4/5

2253

QUESTION ID

S56-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. equal

GRADING NOTE Same external point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle is tangent to all sides of a quadrilateral ABCD. AB = 9, BC = 11, CD = 13. Find DA.

DA = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q5/5

2254

QUESTION ID

S56-F1-Q5

TYPE

Short answer

CORRECT ANSWER

11

GRADING NOTE Any equivalent form. AB + CD = BC + DA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSG-C.A.2

FORWARD 2

Write the equation of a tangent line in the coordinate plane • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find an equation of the line tangent to $x^2 + y^2 = 25$ at (3, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q1/5

2256

QUESTION ID

S56-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$y = -\frac{3}{4}x + 25/4$

GRADING NOTE Any equivalent form, such as 3x + 4y = 25. The radius has slope 4/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q1/5

Find an equation of the tangent to $x^2 + y^2 = 13$ at $(2, -3)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q2/5

2257

QUESTION ID

S56-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$2x - 3y = 13$

GRADING NOTE Any equivalent form. Radius slope $-3/2$; tangent slope $2/3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q2/5

True or false?

A tangent line has the same slope as the radius to its point of tangency.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q3/5

2258

QUESTION ID
S56-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its slope is the negative reciprocal (when neither line is vertical or horizontal).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The tangent to $x^2 + y^2 = 16$ at (0, 4) is

A. $x = 0$

B. $y = x + 4$

C. $x = 4$

D. $y = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q4/5

2259

QUESTION ID

S56-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. y = 4

GRADING NOTE The radius is vertical, so the tangent is horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is y = 7 tangent to the circle with center (2, 3) and radius 4? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q5/5

2260

QUESTION ID
S56-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Yes: the distance from (2, 3) to the line y = 7 is 4, the radius.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q5/5

SET 57 • HSG-C.A.2 • CIRCLES

Relate circumscribed angles and central angles

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	A tangent is perpendicular to the radius at the point of tangency	**HSG-C.A.2**
BACKWARD 2	The angles of a quadrilateral add to 360°	**8.G.A.5**
BACKWARD 3	Supplementary angles add to 180°	**7.G.B.5**
FORWARD 1	Use the circumscribed angle with arcs, including major arcs	**HSG-C.A.2**
FORWARD 2	Angles formed by secants and tangents outside a circle	**HSG-C.A.2**

SET 57 • HSG-C.A.2

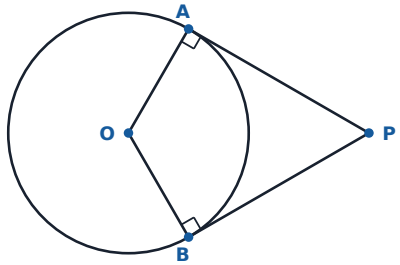
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. Prove that m∠APB + m∠AOB = 180°.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q1/5

2263

QUESTION ID

S57-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠APB + m∠AOB = 180°

GRADING NOTE REQUIRED WORK: Use the right angles at A and B and the angle sum of quadrilateral OAPB.

Any valid proof (paragraph, two-column, or flow). Essential: ∠OAP = ∠OBP = 90° (radius ⊥ tangent); the angles of quadrilateral OAPB add to 360°; so m∠APB + m∠AOB = 360° − 180° = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. The circumscribed angle APB measures 52°. Find the central angle AOB and the minor arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q2/5

2264

QUESTION ID

S57-M-Q2

TYPE

Short answer

CORRECT ANSWER

∠AOB = 128°; arc AB = 128°

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. Minor arc AB measures 100°. Find the circumscribed angle APB.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q3/5

2265

QUESTION ID
S57-M-Q3

TYPE
Short answer

CORRECT ANSWER
80°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. m∠APB = 90°. Find m∠AOB and classify quadrilateral OAPB as specifically as possible. Give reasons.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q4/5

2266

QUESTION ID

S57-M-Q4

TYPE

Short answer

CORRECT ANSWER

m∠AOB = 90°. OAPB is a square: all four angles are 90° (a rectangle), and OA = OB (radii) and PA = PB (tangent segments) with OA = PB (opposite sides), so all sides are equal.

GRADING NOTE Form required: the angle, "square", and the reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. $m\angle APB = (2x + 10)°$ and $m\angle AOB = (4x + 50)°$. Find x and both angle measures, and state the relationship you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q5/5

2267

QUESTION ID
S57-M-Q5

TYPE
Short answer

CORRECT ANSWER
x = 20; m∠APB = 50°, m∠AOB = 130°; a circumscribed angle and its central angle are supplementary.

GRADING NOTE Form required: x, both angles, and the relationship. 6x + 60 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 1

A tangent is perpendicular to the radius at the point of tangency • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

The radius to the point of tangency is perpendicular to the tangent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q1/5

2269

QUESTION ID

S57-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is m∠OAP when PA is tangent at A?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q2/5

2270

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q2/5

In quadrilateral OAPB with tangents PA and PB, how many right angles are certain?

A. 4

B. 0

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q3/5

2271

QUESTION ID

S57-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE At A and B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangent PA, radius OA = 9, PA = 12. OP?

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q4/5

2272

QUESTION ID

S57-B1-Q4

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The angle between a tangent and a radius can be 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q5/5

2273

QUESTION ID

S57-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Always 90° at the point of tangency.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 2

The angles of a quadrilateral add to 360° • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

A quadrilateral has angles 90°, 90°, and 70°. The fourth?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q1/5

2275

QUESTION ID

S57-B2-Q1

TYPE

Short answer

CORRECT ANSWER

110°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q1/5

True or false?

A diagonal splits a quadrilateral into two triangles, so its angles add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q2/5

2276

QUESTION ID

S57-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2 × 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q2/5

Two angles of a quadrilateral are right angles. The other two must

A. each be 90°

B. add to 360°

C. add to 180°

D. add to 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q3/5

2277

QUESTION ID

S57-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. add to 180°

GRADING NOTE 360 − 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Angles x, 90°, 90°, 2x in a quadrilateral. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q4/5

2278

QUESTION ID

S57-B2-Q4

TYPE

Short answer

CORRECT ANSWER

60

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q4/5

True or false?

The angles of a quadrilateral add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q5/5

2279

QUESTION ID

S57-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 3

Supplementary angles add to 180° • Grade 7 • Geometry • 7.G.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find the supplement of 52°.

Supplement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q1/5

2281

QUESTION ID

S57-B3-Q1

TYPE

Short answer

CORRECT ANSWER

128°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Supplementary angles add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q2/5

2282

QUESTION ID

S57-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Angles x and 3x are supplementary. x =

A. 60°

B. 90°

C. 45°

D. 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q3/5

2283

QUESTION ID

S57-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 45°

GRADING NOTE 4x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If two angles are supplementary and one is obtuse, the other is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q4/5

2284

QUESTION ID

S57-B3-Q4

TYPE

Short answer

CORRECT ANSWER

acute

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q4/5

True or false?

Supplementary angles must form a linear pair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q5/5

2285

QUESTION ID

S57-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They need not be adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • HSG-C.A.2

FORWARD 1

Use the circumscribed angle with arcs, including major arcs • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. The circumscribed angle APB is 40°. Find the minor arc AB and the major arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q1/5

2287

QUESTION ID
S57-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Minor arc 140°; major arc 220°

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. Show that m∠APB = ½(major arc AB − minor arc AB).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q2/5

2288

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

If the minor arc is m, the major arc is 360 − m and the angle is 180 − m = ½((360 − m) − m).

GRADING NOTE Any equivalent algebra.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q2/5

True or false?

A circumscribed angle can measure more than 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q3/5

2289

QUESTION ID
S57-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is 180° minus the minor arc, so it is less than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. m∠APB = 70°. An inscribed angle intercepting the minor arc AB measures

A. 110°

B. 70°

C. 55°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q4/5

2290

QUESTION ID
S57-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 55°

GRADING NOTE Minor arc = 110°, inscribed angle = 55°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q4/5

2290

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. Minor arc AB is x°, and the circumscribed angle APB is (x − 20)°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q5/5

2291

QUESTION ID

S57-F1-Q5

TYPE

Short answer

CORRECT ANSWER

100

GRADING NOTE Any equivalent form. x + x − 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q5/5

2291

SET 57 • HSG-C.A.2

FORWARD 2

Angles formed by secants and tangents outside a circle • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Two secants from P meet a circle at A, B and at C, D (A between P and B, C between P and D). Prove that m∠P = ½(arc BD − arc AC).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q1/5

2293

QUESTION ID

S57-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠P = ½(arc BD − arc AC)

GRADING NOTE REQUIRED WORK: Draw chord BC; use the exterior angle theorem in triangle PBC and the inscribed angle theorem.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BCD is exterior to △PBC, so m∠BCD = m∠P + m∠PBC; m∠BCD = ½ arc BD and m∠PBC = ½ arc AC; subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two secants from P intercept arcs of 110° and 30°. Find m∠P.

m∠P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q2/5

2294

QUESTION ID
S57-F2-Q2

TYPE
Short answer

CORRECT ANSWER
40°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q2/5

True or false?

The angle outside a circle formed by a tangent and a secant is half the sum of the intercepted arcs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q3/5

2295

QUESTION ID

S57-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is half the difference; half the sum is for two chords meeting inside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An angle outside a circle intercepts arcs of 160° and 50°. Its measure?

A. 110°

B. 55°

C. 210°

D. 105°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q4/5

2296

QUESTION ID

S57-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 55°

GRADING NOTE (160 − 50)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the formula for two tangents is a special case of the secant formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q5/5

2297

QUESTION ID
S57-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The tangent case is the limit where each secant's two intersection points merge; the arcs are the major and minor arcs.

GRADING NOTE Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q5/5

SET 58 • HSG-C.A.2 • CIRCLES

Relationships among radii and chords

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Right triangles: Hypotenuse-Leg and the Pythagorean Theorem	**HSG-SRT.B.5**
BACKWARD 2	The perpendicular bisector theorem	**HSG-CO.C.9**
BACKWARD 3	Radii of a circle are equal	**HSG-CO.A.1**
FORWARD 1	Locate the center of a circle from chords	**HSG-C.A.2**
FORWARD 2	Find chord lengths and central angles with trigonometry	**HSG-SRT.C.8**

SET 58 • HSG-C.A.2

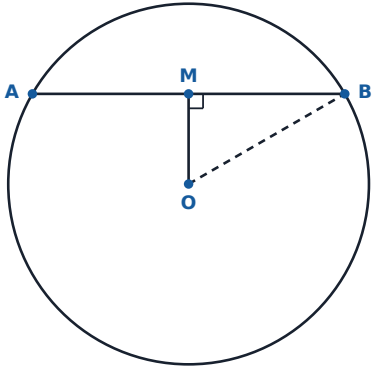
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, OM ⊥ chord AB at M. Prove that M is the midpoint of AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q1/5

2300

QUESTION ID

S58-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AM = MB

GRADING NOTE REQUIRED WORK: Prove △OMA ≅ △OMB (HL, using OA = OB).

Any valid proof (paragraph, two-column, or flow). Essential: OA = OB (radii); OM shared; right angles at M; HL (or the Pythagorean Theorem: AM = √(r² − OM²) = MB).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has radius 13. A chord is 5 units from the center. How long is the chord?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q2/5

2301

QUESTION ID
S58-M-Q2

TYPE
Short answer

CORRECT ANSWER
24

GRADING NOTE Any equivalent form. Half-chord $\sqrt{(169 - 25)} = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A chord of length 16 is in a circle of radius 10. How far is the chord from the center?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q3/5

2302

QUESTION ID
S58-M-Q3

TYPE
Short answer

CORRECT ANSWER
6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In circle O, chords AB and CD are congruent. OM ⊥ AB at M and ON ⊥ CD at N. Prove that OM = ON (congruent chords are equidistant from the center).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q4/5

2303

QUESTION ID

S58-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

OM = ON

GRADING NOTE REQUIRED WORK: Use the perpendicular-bisects-chord result, then HL on △OMA and △ONC.

Any valid proof (paragraph, two-column, or flow). Essential: OM ⊥ AB bisects AB and ON ⊥ CD bisects CD, so AM = ½AB = ½CD = CN; OA = OC (radii); right angles at M and N; HL gives △OMA ≅ △ONC, so OM = ON.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Circle O has radius 17. Chords AB and CD are each 30 units long. Find the distance from O to each chord, and explain why the distances are equal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q5/5

2304

QUESTION ID

S58-M-Q5

TYPE

Short answer

CORRECT ANSWER

Each distance is 8 (half-chord 15, √(289 − 225) = 8). They are equal because congruent chords are equidistant from the center.

GRADING NOTE Form required: the distance with work and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 1

Right triangles: Hypotenuse-Leg and the Pythagorean Theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangles with equal hypotenuses and one pair of equal legs are congruent by

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q1/5

2306

QUESTION ID

S58-B1-Q1

TYPE

Short answer

CORRECT ANSWER

HL

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse 10, leg 6. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q2/5

2307

QUESTION ID

S58-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q2/5

True or false?

In circle O, OM is perpendicular to chord AB at M. The radii OA and OB are congruent hypotenuses of right triangles OMA and OMB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q3/5

2308

QUESTION ID

S58-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE OA = OB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

OM ⊥ AB in circle O. △OMA and △OMB share

A. nothing

B. OM

C. AB

D. OA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q4/5

2309

QUESTION ID

S58-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. OM

GRADING NOTE Common leg.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-B1-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q4/5

True or false?

HL works for any triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q5/5

2310

QUESTION ID

S58-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Right triangles only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 2

The perpendicular bisector theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

A point equidistant from A and B lies on the perpendicular bisector of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q1/5

2312

QUESTION ID

S58-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q1/5

The center O of a circle is equidistant from the endpoints of chord AB. Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q2/5

2313

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

OA and OB are radii.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q2/5

The center of a circle is equidistant from the endpoints of each of its chords. So the center lies on

A. the perpendicular bisector of every chord

B. a tangent

C. the circle

D. every chord

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q3/5

2314

QUESTION ID

S58-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the perpendicular bisector of every chord

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q3/5

Two chords of a circle are not parallel, so their perpendicular bisectors are different lines; the bisectors meet at P. What is P?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q4/5

2315

QUESTION ID

S58-B2-Q4

TYPE

Short answer

CORRECT ANSWER

The center of the circle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q4/5

True or false?

The perpendicular bisector of a chord can miss the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q5/5

2316

QUESTION ID

S58-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It always passes through it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 3

Radii of a circle are equal • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center O. OA = 7. Find OB if B is on the circle.

OB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q1/5

2318

QUESTION ID

S58-B3-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q1/5

True or false?

All radii of one circle have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q2/5

2319

QUESTION ID

S58-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q2/5

AB is a chord of circle O, and OA = 5. OB =

A. unknown

B. 10

C. 2.5

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q3/5

2320

QUESTION ID

S58-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5

GRADING NOTE Radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the longest chord in a circle of radius 6?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q4/5

2321

QUESTION ID

S58-B3-Q4

TYPE

Short answer

CORRECT ANSWER

A diameter, 12

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q4/5

True or false?

A chord always passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q5/5

2322

QUESTION ID

S58-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q5/5

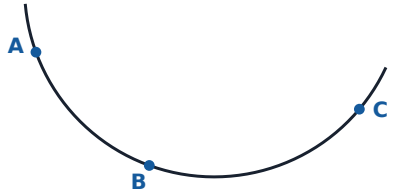
SET 58 • HSG-C.A.2

FORWARD 1

Locate the center of a circle from chords • High School — Geometry • Geometry: Circles • HSG-C.A.2

Compass and straightedge only; leave all arcs.

The arc below is part of a circle. Construct its center.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q1/5

2324

QUESTION ID

S58-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Draw chords AB and BC; construct their perpendicular bisectors; they meet at the center.

GRADING NOTE Grade the construction marks, not only the final figure: two chords and both perpendicular bisector constructions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle passes through (0, 0), (8, 0), and (0, 6). Find its center.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q2/5

2325

QUESTION ID

S58-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 3)

GRADING NOTE Form required: an ordered pair. Perpendicular bisectors x = 4 and y = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q2/5

True or false?

Three points not on one line determine exactly one circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q3/5

2326

QUESTION ID

S58-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The perpendicular bisectors meet in one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why can't three points on one line lie on a circle?

A. Three points on a line always lie on a circle

B. Their perpendicular bisectors are parallel, so no center exists

C. Only two points can lie on a circle

D. They are too close

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q4/5

2327

QUESTION ID
S58-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. Their perpendicular bisectors are parallel, so no center exists

GRADING NOTE Parallel bisectors never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q4/5

A circular plate is broken. Describe how to find its original radius from the piece.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q5/5

2328

QUESTION ID

S58-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Mark three points on the edge, construct the perpendicular bisectors of two chords to find the center, then measure to the edge.

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSG-C.A.2

FORWARD 2

Find chord lengths and central angles with trigonometry • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A chord subtends a central angle of 80° in a circle of radius 10. Find its length to the nearest tenth.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q1/5

2330

QUESTION ID

S58-F2-Q1

TYPE

Short answer

CORRECT ANSWER

12.9

GRADING NOTE Rounding to 0.1 required. Half the chord = 10 sin 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A chord of length 12 is in a circle of radius 10. Find the central angle to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q2/5

2331

QUESTION ID

S58-F2-Q2

TYPE

Short answer

CORRECT ANSWER

73.7°

GRADING NOTE Rounding to 0.1° required. Half-angle $\sin^{-1}(6/10)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q2/5

True or false?

A chord's length is 2r sin θ for central angle θ.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q3/5

2332

QUESTION ID

S58-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 2r sin(θ/2), from the right triangle with half the chord.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Central angle 60°, radius 9. Chord length?

A. 4.5

B. 18

C. 9

D. 9√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q4/5

2333

QUESTION ID

S58-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 9

GRADING NOTE $2 \cdot 9 \cdot \sin 30° = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A chord equals the radius. Find its central angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q5/5

2334

QUESTION ID

S58-F2-Q5

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form. Equilateral triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q5/5

SET 59 • HSG-C.A.3 • CIRCLES

Construct the inscribed circle of a triangle

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	Construct an angle bisector	**HSG-CO.D.12**
BACKWARD 2	Construct a perpendicular from a point to a line	**HSG-CO.D.12**
BACKWARD 3	Points on an angle bisector are equidistant from the sides	**HSG-SRT.B.5**
FORWARD 1	Prove that the incenter is equidistant from the sides	**HSG-C.A.3**
FORWARD 2	Locate a point equidistant from three lines in a design	**HSG-MG.A.3**

SET 59 • HSG-C.A.3

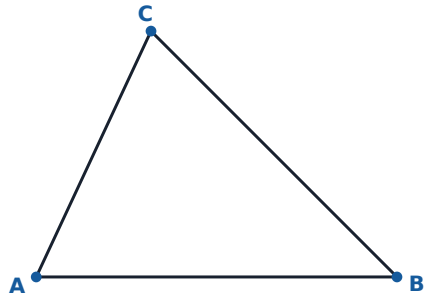
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q1/5

2337

QUESTION ID

S59-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Bisect two angles (for example ∠A and ∠B); they meet at the incenter I; construct the perpendicular from I to one side, meeting it at D; draw the circle with center I and radius ID.

GRADING NOTE Grade the construction marks, not only the final figure: two angle bisector constructions, the perpendicular from I to a side, and the circle touching all three sides.

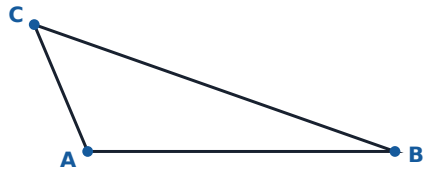
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of obtuse triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q2/5

2338

QUESTION ID

S59-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same steps: two angle bisectors meet at I (inside the triangle), perpendicular from I to a side gives the radius, then the circle.

GRADING NOTE Grade the construction marks, not only the final figure: all constructions and the circle.

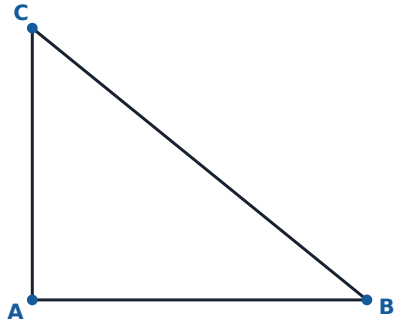
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of right triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q3/5

2339

QUESTION ID

S59-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Two angle bisectors, the perpendicular from their intersection to a side, and the circle.

GRADING NOTE Grade the construction marks, not only the final figure: all constructions and the circle.

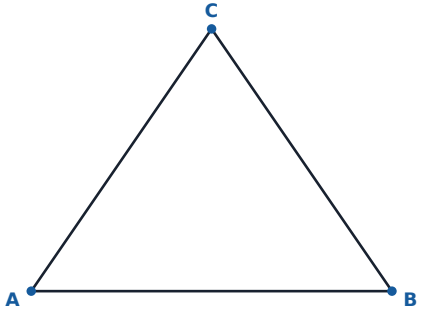
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of isosceles triangle ABC (AC = BC). Then, as evidence, construct the perpendiculars from the incenter to ALL THREE sides and mark the three points where the circle touches the sides.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q4/5

2340

QUESTION ID

S59-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Bisect two angles to get I; construct perpendiculars from I to AB, BC and CA (feet D, E, F); draw the circle with center I through D; it passes through E and F and touches each side once.

GRADING NOTE Grade the construction marks, not only the final figure: two angle bisectors, all three perpendiculars, and a circle through the three feet (evidence that the distances are equal).

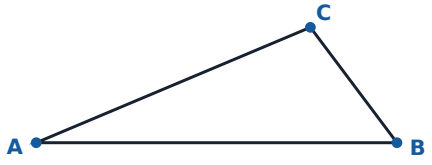
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of triangle ABC by bisecting ∠B and ∠C (not ∠A). Then construct the bisector of ∠A as a check.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q5/5

2341

QUESTION ID

S59-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

The bisectors of ∠B and ∠C meet at I; the perpendicular from I to a side gives the radius; the circle is drawn; the bisector of ∠A also passes through I (the check).

GRADING NOTE Grade the construction marks, not only the final figure: three angle bisectors meeting at one point, the perpendicular, and the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q5/5

SET 59 • HSG-C.A.3

BACKWARD 1

Construct an angle bisector • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

How many arcs are needed to bisect an angle with compass and straightedge?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q1/5

2343

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Three (one at the vertex, then two equal arcs from the crossing points)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q1/5

True or false?

An angle bisector construction ends with a ray from the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q2/5

2344

QUESTION ID

S59-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Through the crossing of the equal arcs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q2/5

The incenter is where the

A. medians meet

B. perpendicular bisectors meet

C. angle bisectors meet

D. altitudes meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q3/5

2345

QUESTION ID

S59-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. angle bisectors meet

GRADING NOTE Equidistant from sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q3/5

Why construct only two angle bisectors to find the incenter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q4/5

2346

QUESTION ID

S59-B1-Q4

TYPE

Short answer

CORRECT ANSWER

The third bisector passes through the same point.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Bisecting the sides locates the incenter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q5/5

2347

QUESTION ID

S59-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That locates the circumcenter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSG-C.A.3

BACKWARD 2

Construct a perpendicular from a point to a line • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction finds the point where the incircle touches a side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q1/5

2349

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

The perpendicular from the incenter to the side

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The perpendicular from a point to a line is the shortest segment to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q2/5

2350

QUESTION ID

S59-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q2/5

From point I, an arc meets side AB at two points. Next step?

A. Measure AB

B. Draw a circle at A

C. Bisect ∠I

D. Arcs of equal radius from those two points, crossing on the other side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q3/5

2351

QUESTION ID

S59-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Arcs of equal radius from those two points, crossing on the other side

GRADING NOTE Then draw the line through I and that crossing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q3/5

Why not measure from I to a vertex for the radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q4/5

2352

QUESTION ID

S59-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Vertices are farther than the sides; the circle must touch the sides, not pass through vertices.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q4/5

2352

True or false?

The incircle passes through the vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q5/5

2353

QUESTION ID

S59-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It touches the sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSG-C.A.3

BACKWARD 3

Points on an angle bisector are equidistant from the sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

True or false?

A point on the bisector of ∠A is equidistant from the two sides of ∠A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q1/5

2355

QUESTION ID

S59-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Congruent right triangles (AAS).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The incenter is 3 cm from side AB. How far is it from side BC?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q2/5

2356

QUESTION ID
S59-B3-Q2

TYPE
Short answer

CORRECT ANSWER
3 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q2/5

Distance from a point to a side is measured

A. to the nearest vertex

B. along the perpendicular

C. along the bisector

D. along any segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q3/5

2357

QUESTION ID

S59-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. along the perpendicular

GRADING NOTE Shortest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q3/5

Why is the incenter the same distance from all three sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q4/5

2358

QUESTION ID
S59-B3-Q4

TYPE
Short answer

CORRECT ANSWER
It is on all three angle bisectors, and each bisector point is equidistant from that angle's sides.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q4/5

True or false?

The incenter is equidistant from the three vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q5/5

2359

QUESTION ID
S59-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE That is the circumcenter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q5/5

SET 59 • HSG-C.A.3

FORWARD 1

Prove that the incenter is equidistant from the sides • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

The bisectors of ∠A and ∠B of △ABC meet at I. Prove that I is equidistant from all three sides and lies on the bisector of ∠C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q1/5

2361

QUESTION ID

S59-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

I is equidistant from AB, BC, CA and lies on the bisector of ∠C.

GRADING NOTE REQUIRED WORK: Use the angle bisector distance property twice, then its converse at C.

Any valid proof (paragraph, two-column, or flow). Essential: on the bisector of ∠A: d(I, AB) = d(I, AC); on the bisector of ∠B: d(I, AB) = d(I, BC); so d(I, AC) = d(I, BC); the converse puts I on the bisector of ∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q1/5

A triangle has area 30 and perimeter 30. Find the radius of its inscribed circle.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q2/5

2362

QUESTION ID

S59-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. Area = r · (perimeter ÷ 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q2/5

True or false?

The incenter of an obtuse triangle lies outside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q3/5

2363

QUESTION ID

S59-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The incenter is always inside; the circumcenter is the one that can be outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A 3-4-5 right triangle has inradius

A. 2.5

B. 1

C. 1.5

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q4/5

2364

QUESTION ID

S59-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1

GRADING NOTE Area 6 = r · 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why area = r · s, where s is half the perimeter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q5/5

2365

QUESTION ID

S59-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Joining I to the vertices splits the triangle into three triangles with heights r on bases a, b, c: area = ½r(a + b + c) = r · s.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSG-C.A.3

FORWARD 2

Locate a point equidistant from three lines in a design • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A circular fountain must touch three straight paths that form a triangle with sides 13, 14, and 15 m (area 84 m²). What is the largest possible radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q1/5

2367

QUESTION ID

S59-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 m

GRADING NOTE Any equivalent form. s = 21, r = 84/21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q1/5

Explain why the incircle gives the largest circle that fits inside a triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q2/5

2368

QUESTION ID

S59-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A larger circle would cross some side; the incircle touches all three sides.

GRADING NOTE Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q2/5

True or false?

A sprinkler equidistant from three straight fences forming a triangle should be at the circumcenter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q3/5

2369

QUESTION ID

S59-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use the incenter (equidistant from lines); the circumcenter is equidistant from the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q3/5

To be equidistant from three towns (points), use the

A. orthocenter

B. centroid

C. incenter

D. circumcenter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q4/5

2370

QUESTION ID

S59-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. circumcenter

GRADING NOTE Points, not lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

An equilateral triangle has side 6. Find its inradius exactly.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q5/5

2371

QUESTION ID
S59-F2-Q5

TYPE
Short answer

CORRECT ANSWER
√3

GRADING NOTE Form required: exact. Area 9√3, s = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q5/5

SET 60 • HSG-C.A.3 • CIRCLES

Construct the circumscribed circle of a triangle

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	Construct the perpendicular bisector of a segment	**HSG-CO.D.12**
BACKWARD 2	Points on the perpendicular bisector are equidistant from the end…	**HSG-CO.C.9**
BACKWARD 3	Triangle types: acute, right, obtuse	**4.G.A.2**
FORWARD 1	Find the circumcenter and radius with coordinates	**HSG-GPE.B.5**
FORWARD 2	Find the circumradius with trigonometry (extended Law of Sines)	**HSG-SRT.D.10**

SET 60 • HSG-C.A.3

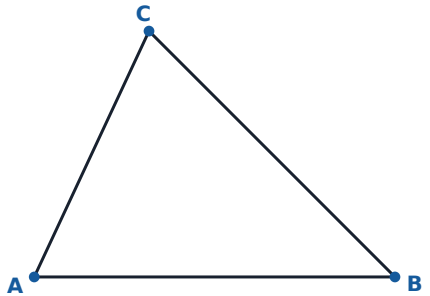
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q1/5

2374

QUESTION ID

S60-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular bisectors of two sides; they meet at O; draw the circle with center O through A (it also passes through B and C).

GRADING NOTE Grade the construction marks, not only the final figure: two perpendicular bisector constructions and the circle through all three vertices.

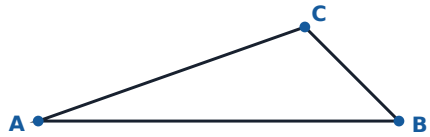
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of obtuse triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q2/5

2375

QUESTION ID

S60-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same steps; the circumcenter is outside the triangle.

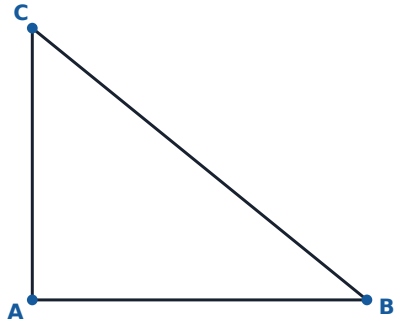
GRADING NOTE Grade the construction marks, not only the final figure: all constructions and the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of right triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q3/5

2376

QUESTION ID

S60-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

The perpendicular bisectors meet at the midpoint of hypotenuse BC; circle centered there through A.

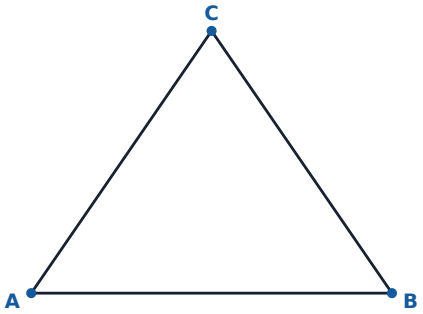
GRADING NOTE Grade the construction marks, not only the final figure: constructions and the circle; the center is on BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of isosceles triangle ABC (AC = BC). Then, as evidence, construct the perpendicular bisector of the third side and show that it passes through the same center.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q4/5

2377

QUESTION ID

S60-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Perpendicular bisectors of two sides meet at O; the third perpendicular bisector also passes through O; the circle with center O through A passes through B and C.

GRADING NOTE Grade the construction marks, not only the final figure: three perpendicular bisectors meeting at O and the circle through all three vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of triangle ABC using the perpendicular bisectors of AC and BC. Then check with the compass that the circle passes through all three vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q5/5

2378

QUESTION ID

S60-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

The perpendicular bisectors of AC and BC meet at O, outside the triangle below AB; the circle with center O through C also passes through A and B.

GRADING NOTE Grade the construction marks, not only the final figure: both perpendicular bisector constructions and the circle through A, B and C; the center is outside the triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 1

Construct the perpendicular bisector of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Which construction gives all points equidistant from A and B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q1/5

2380

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector construction uses equal arcs from both endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q2/5

2381

QUESTION ID

S60-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The circumcenter is where the

A. angle bisectors meet

B. sides meet

C. perpendicular bisectors of the sides meet

D. medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q3/5

2382

QUESTION ID

S60-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. perpendicular bisectors of the sides meet

GRADING NOTE Equidistant from vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

After constructing two perpendicular bisectors, why is the third unnecessary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q4/5

2383

QUESTION ID

S60-B1-Q4

TYPE

Short answer

CORRECT ANSWER

It passes through the same point.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The arcs in a perpendicular bisector construction can have any two different radii.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q5/5

2384

QUESTION ID

S60-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Equal radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 2

Points on the perpendicular bisector are equidistant from the endpoints • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Every point on the perpendicular bisector of AB is equidistant from A and B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q1/5

2386

QUESTION ID

S60-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

O is on the perpendicular bisectors of AB and BC.
Compare OA, OB, OC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q2/5

2387

QUESTION ID

S60-B2-Q2

TYPE

Short answer

CORRECT ANSWER

OA = OB = OC

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The circumcircle's radius is

A. the distance from O to any vertex

B. the longest side

C. the distance from O to a side

D. half a side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q3/5

2388

QUESTION ID

S60-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the distance from O to any vertex

GRADING NOTE Through the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q3/5

A circle passes through A, B, C. Where must its center lie relative to AB?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q4/5

2389

QUESTION ID

S60-B2-Q4

TYPE

Short answer

CORRECT ANSWER

On the perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q4/5

True or false?

The circumcenter is equidistant from the sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q5/5

2390

QUESTION ID

S60-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE From the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 3

Triangle types: acute, right, obtuse • Grade 4 • Geometry • 4.G.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has angles 30°, 45°, 105°. Classify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q1/5

2392

QUESTION ID

S60-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Obtuse

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q1/5

True or false?

A right triangle has one 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q2/5

2393

QUESTION ID

S60-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q2/5

Where is the circumcenter of a right triangle?

A. Inside

B. Outside

C. On the hypotenuse (its midpoint)

D. At the right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q3/5

2394

QUESTION ID

S60-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. On the hypotenuse (its midpoint)

GRADING NOTE Hypotenuse is a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has angles 50°, 60°, 70°. Classify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q4/5

2395

QUESTION ID

S60-B3-Q4

TYPE

Short answer

CORRECT ANSWER

Acute

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q4/5

True or false?

An acute triangle can have an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q5/5

2396

QUESTION ID

S60-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE All angles are acute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q5/5

SET 60 • HSG-C.A.3

FORWARD 1

Find the circumcenter and radius with coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Triangle with A(0, 0), B(6, 0), C(2, 4). Find the circumcenter and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q1/5

2398

QUESTION ID

S60-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 1); radius √10

GRADING NOTE Form required: both. Bisector of AB: x = 3; of AC: y − 2 = −½(x − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q1/5

Triangle with A(−2, 0), B(2, 0), C(0, 2). Find the circumcenter.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q2/5

2399

QUESTION ID
S60-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(0, 0)

GRADING NOTE Form required: an ordered pair. All three are 2 from the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q2/5

True or false?

The circumcenter is the intersection of two perpendicular bisector equations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q3/5

2400

QUESTION ID
S60-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Solve the system.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q3/5

A right triangle has vertices (0, 0), (8, 0), (0, 6). Circumradius?

A. 5

B. 10

C. 4

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q4/5

2401

QUESTION ID

S60-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 5

GRADING NOTE Half the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how to check a computed circumcenter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q5/5

2402

QUESTION ID

S60-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Verify it is the same distance from all three vertices.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSG-C.A.3

FORWARD 2

Find the circumradius with trigonometry (extended Law of Sines) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

In a triangle, a = 10 and A = 30°. Find the circumradius R using a/sin A = 2R.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q1/5

2404

QUESTION ID
S60-F2-Q1

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE Any equivalent form. 10 / 0.5 = 20 = 2R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Equilateral triangle with side 6. Find the circumradius exactly.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q2/5

2405

QUESTION ID
S60-F2-Q2

TYPE
Short answer

CORRECT ANSWER
2√3

GRADING NOTE Form required: exact. 6/sin 60° = 4√3 = 2R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q2/5

True or false?

In any triangle, a/sin A equals the radius of the circumcircle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q3/5

2406

QUESTION ID

S60-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It equals the diameter, 2R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q3/5

A triangle with a 90° angle opposite side 14 has circumradius

A. 3.5

B. 14

C. 28

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q4/5

2407

QUESTION ID

S60-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 7

GRADING NOTE 14/sin 90° = 14 = 2R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q4/5

Explain why a/sin A = 2R using an inscribed angle on a diameter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q5/5

2408

QUESTION ID

S60-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Draw the diameter from B; the inscribed angle at its far end equals A (same arc) and is in a right triangle with hypotenuse 2R, so sin A = a/(2R).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSG-C.A.3 • CIRCLES

Prove that opposite angles of an inscribed quadrilateral are supplementary

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	The inscribed angle theorem	**HSG-C.A.2**
BACKWARD 2	Supplementary angles	**7.G.B.5**
BACKWARD 3	Arcs of a circle add to 360°	**4.MD.C.7**
FORWARD 1	Use the converse: decide whether a quadrilateral can be inscribed	**HSG-C.A.3**
FORWARD 2	Exterior angle of a cyclic quadrilateral	**HSG-C.A.3**

SET 61 • HSG-C.A.3

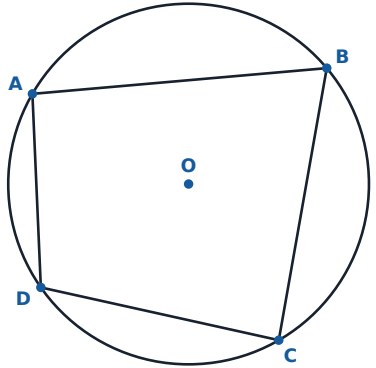
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD is inscribed in circle O. Prove that m∠A + m∠C = 180°.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q1/5

2411

QUESTION ID

S61-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠A + m∠C = 180°

GRADING NOTE REQUIRED WORK: Use the inscribed angle theorem: ∠A and ∠C intercept the two arcs BCD and BAD, which together make the whole circle.

Any valid proof (paragraph, two-column, or flow). Essential: m∠A = ½ arc BCD and m∠C = ½ arc BAD (inscribed angle theorem); the two arcs add to 360°; so the angles add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral PQRS is inscribed in a circle. Prove that ∠Q and ∠S are supplementary.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q2/5

2412

QUESTION ID

S61-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

m∠Q + m∠S = 180°

GRADING NOTE REQUIRED WORK: Use the inscribed angle theorem: ∠Q and ∠S intercept arcs PSR and PQR, which together make the whole circle.

Any valid proof (paragraph, two-column, or flow). Essential: each angle is half its intercepted arc, and the two intercepted arcs make up the whole circle (360°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD is inscribed in a circle with m∠A = 72° and m∠B = 105°. Find m∠C and m∠D, justifying each with the inscribed-quadrilateral theorem.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q3/5

2413

QUESTION ID

S61-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

m∠C = 108°,
m∠D = 75°

GRADING NOTE REQUIRED WORK: Use that opposite angles of an inscribed quadrilateral are supplementary.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A and ∠C are opposite, so m∠C = 180° − 72° = 108°; ∠B and ∠D are opposite, so m∠D = 180° − 105° = 75°; the theorem named for each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Parallelogram ABCD is inscribed in a circle. Prove that ABCD is a rectangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q4/5

2414

QUESTION ID

S61-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

ABCD is a rectangle.

GRADING NOTE REQUIRED WORK: Opposite angles of a parallelogram are congruent; opposite angles of an inscribed quadrilateral are supplementary.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A ≅ ∠C (parallelogram) and m∠A + m∠C = 180° (inscribed), so each is 90°; a parallelogram with a right angle is a rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q4/5

Trapezoid ABCD with AB ∥ DC is inscribed in a circle. Prove that ∠A ≅ ∠B (so the trapezoid is isosceles).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q5/5

2415

QUESTION ID

S61-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

∠A ≅ ∠B

GRADING NOTE REQUIRED WORK: Compare same-side interior angles (from the parallel sides) with opposite angles (from the circle).

Any valid proof (paragraph, two-column, or flow). Essential: m∠A + m∠D = 180° (same-side interior angles, AB ∥ DC); m∠B + m∠D = 180° (opposite angles of an inscribed quadrilateral); so m∠A = m∠B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 1

The inscribed angle theorem • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

An inscribed angle intercepts a 220° arc. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q1/5

2417

QUESTION ID

S61-B1-Q1

TYPE

Short answer

CORRECT ANSWER

110°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q1/5

True or false?

An inscribed angle can intercept an arc greater than 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q2/5

2418

QUESTION ID

S61-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then the angle is obtuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q2/5

An inscribed angle of 70° intercepts

A. 70°

B. 290°

C. 35°

D. 140°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q3/5

2419

QUESTION ID

S61-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 140°

GRADING NOTE Double.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two arcs together make a whole circle. Their sum?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q4/5

2420

QUESTION ID

S61-B1-Q4

TYPE

Short answer

CORRECT ANSWER

360°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q4/5

True or false?

An inscribed angle equals half the central angle only when it is acute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S61-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q5/5

2421

QUESTION ID

S61-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Always.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S61-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 2

Supplementary angles • Grade 7 • Geometry • 7.G.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Supplement of 72°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q1/5

2423

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

108°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q1/5

True or false?

Two angles whose measures add to 180° are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q2/5

2424

QUESTION ID

S61-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Angles 3x and 2x are supplementary. x =

A. 30°

B. 45°

C. 36°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q3/5

2425

QUESTION ID

S61-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 36°

GRADING NOTE 5x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If ∠A = ∠C and they are supplementary, find each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q4/5

2426

QUESTION ID

S61-B2-Q4

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Supplementary angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q5/5

2427

QUESTION ID

S61-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only if both are 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 3

Arcs of a circle add to 360° • Grade 4 • Measurement & Data • 4.MD.C.7

ANSWER KEY Reduced copy of the question page, with its answer

A circle is split into arcs of 100°, 80°, and x°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q1/5

2429

QUESTION ID

S61-B3-Q1

TYPE

Short answer

CORRECT ANSWER

180

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q1/5

True or false?

The arcs between four points on a circle add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q2/5

2430

QUESTION ID

S61-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Whole circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q2/5

Two arcs make a whole circle; one is 150°. The other is

A. 210°

B. 150°

C. 30°

D. 360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q3/5

2431

QUESTION ID

S61-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 210°

GRADING NOTE 360 − 150.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Half of 360° is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q4/5

2432

QUESTION ID

S61-B3-Q4

TYPE

Short answer

CORRECT ANSWER

180°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q4/5

True or false?

The arcs of a circle add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q5/5

2433

QUESTION ID

S61-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q5/5

SET 61 • HSG-C.A.3

FORWARD 1

Use the converse: decide whether a quadrilateral can be inscribed • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD has angles A = 85°, B = 70°, C = 95°, D = 110°. Can it be inscribed in a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q1/5

2435

QUESTION ID

S61-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: A + C = 180° and B + D = 180°.

GRADING NOTE Form required: the conclusion with the check. (The converse is true: such a quadrilateral is cyclic.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q1/5

An isosceles trapezoid has base angles of 64°. Can it be inscribed in a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q2/5

2436

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: its other angles are 116°, and 64° + 116° = 180° for each pair of opposite angles.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q2/5

True or false?

Every kite can be inscribed in a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q3/5

2437

QUESTION ID

S61-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only kites with two right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q3/5

Inscribed quadrilateral with angles (2x)°, (3x + 5)°, (2x + 20)°, and y°. If 2x and 2x + 20 are opposite, x =

A. 40

B. 50

C. 45

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q4/5

2438

QUESTION ID

S61-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 40

GRADING NOTE 4x + 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed quadrilateral: ∠A = (5y − 10)°,
∠C = (3y + 30)°, A and C opposite. Find y and ∠A.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q5/5

2439

QUESTION ID
S61-F1-Q5

TYPE
Short answer

CORRECT ANSWER
y = 20; ∠A = 90°

GRADING NOTE Form required: both. 8y + 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q5/5

SET 61 • HSG-C.A.3

FORWARD 2

Exterior angle of a cyclic quadrilateral • High School — Geometry • Geometry: Circles • HSG-C.A.3

ABCD is inscribed in a circle and side DC is extended past C to E. Prove that m∠BCE = m∠A.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q1/5

2441

QUESTION ID

S61-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠BCE = m∠A

GRADING NOTE REQUIRED WORK: Use the linear pair at C and the supplementary opposite angles.

Any valid proof (paragraph, two-column, or flow). Essential: m∠BCE = 180° − m∠BCD (linear pair); m∠A = 180° − m∠BCD (opposite angles); so equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In inscribed ABCD, ∠A = 77°. Side DC is extended past C to E. Find ∠BCE.

∠BCE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q2/5

2442

QUESTION ID

S61-F2-Q2

TYPE

Short answer

CORRECT ANSWER

77°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q2/5

True or false?

An exterior angle of a cyclic quadrilateral equals the interior angle next to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q3/5

2443

QUESTION ID

S61-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is supplementary to the adjacent interior angle and equals the opposite one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed ABCD with ∠B = 100°. The exterior angle at D (extending CD past D) is

A. 50°

B. 180°

C. 100°

D. 80°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q4/5

2444

QUESTION ID

S61-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 100°

GRADING NOTE Equals the opposite interior angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is inscribed in a circle, and side DC is extended past C. Why is the exterior angle at C equal to ∠A, not ∠B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q5/5

2445

QUESTION ID

S61-F2-Q5

TYPE

Short answer

CORRECT ANSWER

It is supplementary to ∠C, and the angle supplementary to ∠C is its opposite angle, ∠A.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q5/5

SET 62 • HSG-C.B.5 • CIRCLES

Derive arc length from similarity and define radian measure

MAIN	Geometry main question	**HSG-C.B.5**
BACKWARD 1	Similar figures: lengths scale by the scale factor	**HSG-SRT.A.2**
BACKWARD 2	Circumference of a circle	**7.G.B.4**
BACKWARD 3	Find a fraction of a whole	**7.RP.A.3**
FORWARD 1	Convert between degrees and radians	**HSG-C.B.5**
FORWARD 2	Radian measure for angles of rotation and the unit circle	**HSF-TF.A.1**

SET 62 • HSG-C.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A central angle θ cuts off arc length s_1 on a circle of radius r_1 and arc length s_2 on a circle of radius r_2 (same center). Explain, using similarity, why $s_1/r_1 = s_2/r_2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q1/5

2448

QUESTION ID

S62-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$s_2 = (r_2/r_1)s_1$, so $s_1/r_1 = s_2/r_2$.

GRADING NOTE REQUIRED WORK: Use the dilation about the center with factor r_2/r_1: it maps the first circle to the second and the first arc to the second arc.

Any valid proof (paragraph, two-column, or flow). Essential: the dilation about the center with factor $k = r_2/r_1$ maps the first circle onto the second and the arc of angle θ onto the arc of angle θ; it multiplies all lengths, including arc length, by k; so $s_2 = k\, s_1$ and s/r is the same for both. This constant is θ in radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q1/5

A central angle cuts off an arc of length 12 cm on a circle of radius 4 cm. What is the angle in radians?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q2/5

2449

QUESTION ID
S62-M-Q2

TYPE
Short answer

CORRECT ANSWER
3 radians

GRADING NOTE Any equivalent form. θ = s/r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the length of the arc cut off by a central angle of 2.5 radians in a circle of radius 6 m.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q3/5

2450

QUESTION ID
S62-M-Q3

TYPE
Short answer

CORRECT ANSWER
15 m

GRADING NOTE Any equivalent form. s = rθ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact length of the arc cut off by a 120° central angle in a circle of radius 9.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q4/5

2451

QUESTION ID
S62-M-Q4

TYPE
Short answer

CORRECT ANSWER
6π

GRADING NOTE Form required: exact value. 120° = 2π/3 radians; s = 9 · 2π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q4/5

Use s = rθ to explain why a full turn is 2π radians and a half turn is π radians, and convert 270° to radians.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q5/5

2452

QUESTION ID

S62-M-Q5

TYPE

Short answer

CORRECT ANSWER

A full turn cuts off the whole circumference 2πr, so θ = 2πr/r = 2π; half a turn cuts off πr, so θ = π; 270° is ¾ of a turn, 3π/2 radians.

GRADING NOTE Form required: both explanations and the conversion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 1

Similar figures: lengths scale by the scale factor • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

True or false?

Under a dilation with factor k, every length (straight or curved) is multiplied by k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q1/5

2454

QUESTION ID

S62-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two similar circles have radii 3 and 12. Scale factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q2/5

2455

QUESTION ID

S62-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-B1-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q2/5

An arc of length 5 on the small circle (radius 3) matches which arc length on the large one (radius 12)?

A. 20

B. 15

C. 8

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q3/5

2456

QUESTION ID

S62-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 20

GRADING NOTE × 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Are all circles similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q4/5

2457

QUESTION ID

S62-B1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q4/5

True or false?

Arc lengths do not change under dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q5/5

2458

QUESTION ID

S62-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 2

Circumference of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the circumference of a circle with radius 5 (exact).

Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q1/5

2460

QUESTION ID

S62-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q1/5

True or false?

$C = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q2/5

2461

QUESTION ID

S62-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q2/5

A circle has diameter 8. Its circumference is

A. 64π

B. 4π

C. 16π

D. 8π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q3/5

2462

QUESTION ID

S62-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8π

GRADING NOTE πd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has circumference 18π. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q4/5

2463

QUESTION ID

S62-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$C = \pi r^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q5/5

2464

QUESTION ID

S62-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 3

Find a fraction of a whole • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of a circle is a 90° arc?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q1/5

2466

QUESTION ID

S62-B3-Q1

TYPE

Short answer

CORRECT ANSWER

¼

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find ⅓ of 18π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q2/5

2467

QUESTION ID

S62-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q2/5

An arc of 45° is what fraction of the circumference?

A. 45/100

B. ⅛

C. 1/45

D. ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q3/5

2468

QUESTION ID

S62-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ⅛

GRADING NOTE 45/360.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q3/5

True or false?

A 120° arc is ⅓ of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q4/5

2469

QUESTION ID

S62-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 120/360.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A 60° arc is 1/60 of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q5/5

2470

QUESTION ID

S62-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q5/5

SET 62 • HSG-C.B.5

FORWARD 1

Convert between degrees and radians • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Convert 150° to radians (exact).

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q1/5

2472

QUESTION ID

S62-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5π/6

GRADING NOTE Form required: exact value. Multiply by π/180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Convert 3π/4 radians to degrees.

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q2/5

2473

QUESTION ID

S62-F1-Q2

TYPE

Short answer

CORRECT ANSWER

135°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

1 radian is about 3.14°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q3/5

2474

QUESTION ID

S62-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1 radian ≈ 57.3°; π radians = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q3/5

Which equals π/3 radians?

A. 60°

B. 120°

C. 30°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q4/5

2475

QUESTION ID

S62-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 60°

GRADING NOTE 180/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

An arc in a circle of radius 10 has length 5π. Find the central angle in degrees.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q5/5

2476

QUESTION ID

S62-F1-Q5

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form. θ = π/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSG-C.B.5

FORWARD 2

Radian measure for angles of rotation and the unit circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

On the unit circle, an angle of 2 radians cuts off an arc of what length?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q1/5

2478

QUESTION ID
S62-F2-Q1

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Any equivalent form. r = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q1/5

A wheel of radius 0.3 m rotates about its fixed center (axle) through 10 radians. How far does a point on its rim travel along its circular path?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q2/5

2479

QUESTION ID
S62-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An angle of 6 radians is more than one full turn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q3/5

2480

QUESTION ID
S62-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE One full turn is $2\pi \approx 6.28$ radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A point on the unit circle travels an arc of length π. The angle is

A. 1 radian

B. 2π radians

C. π°

D. π radians (180°)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q4/5

2481

QUESTION ID

S62-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. π radians (180°)

GRADING NOTE θ = s/r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the radian measure of an angle does not depend on the radius of the circle used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q5/5

2482

QUESTION ID
S62-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Arc length is proportional to radius, so s/r is the same for every circle.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSG-C.B.5 • CIRCLES

Derive and use the formula for the area of a sector

MAIN	Geometry main question	**HSG-C.B.5**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	The radius is half the diameter	**7.G.B.4**
BACKWARD 3	Fractions of a circle in degrees and radians	**HSG-C.B.5**
FORWARD 1	Solve inverse sector problems	**HSG-C.B.5**
FORWARD 2	Areas of segments and composite circular regions	**HSG-MG.A.1**

SET 63 • HSG-C.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Derive a formula for the area of a sector with central angle θ (in degrees) in a circle of radius r. Then rewrite it for θ in radians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q1/5

2485

QUESTION ID

S63-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

A = (θ/360)πr² for degrees; A = ½r²θ for radians.

GRADING NOTE REQUIRED WORK: Use the fraction of the whole circle that the sector is, times the circle's area.

Any valid proof (paragraph, two-column, or flow). Essential: the sector is θ/360 of the circle (central angles split the area in proportion); area = (θ/360)πr²; with θ radians, the fraction is θ/(2π), so A = (θ/(2π))πr² = ½r²θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q1/5

2485

ANSWER KEY Reduced copy of the question page, with its answer

Derive the formula A = ½rs for the area of a sector in terms of its radius r and arc length s. Then use it for a sector with r = 8 and s = 5.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q2/5

2486

QUESTION ID

S63-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

A = ½rs; A = 20

GRADING NOTE REQUIRED WORK: Use the fraction s/(2πr) of the circumference, which is the same fraction of the area.

Any valid proof (paragraph, two-column, or flow). Essential: the sector is s/(2πr) of the circle (arc length and area are in the same proportion to the whole circle); A = (s/(2πr)) · πr² = ½rs; ½ · 8 · 5 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q2/5

2486

ANSWER KEY Reduced copy of the question page, with its answer

Derive the sector area formula for θ in radians from the proportion $A/(\pi r^2) = \theta/(2\pi)$. Then find the area of a sector with central angle 1.2 radians in a circle of radius 5.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q3/5

2487

QUESTION ID

S63-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

A = ½r²θ; area 15

GRADING NOTE

REQUIRED WORK: Solve the proportion for A, then substitute.

Any valid proof (paragraph, two-column, or flow). Essential: the fraction of the full angle (2π) equals the fraction of the area; $A = \pi r^2 \cdot \theta/(2\pi) = \frac{1}{2}r^2\theta$; $\frac{1}{2} \cdot 25 \cdot 1.2 = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circular pizza of diameter d is cut into n equal slices. Derive a formula for the area of one slice in terms of d and n, then find the area for d = 14 in. and n = 8, to the nearest tenth of a square inch.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q4/5

2488

QUESTION ID

S63-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

A = πd²/(4n); 19.2 in²

GRADING NOTE REQUIRED WORK: Each slice is a sector with central angle 360°/n and radius d/2.

Any valid proof (paragraph, two-column, or flow). Essential: radius d/2 (not d); fraction 1/n; $A = (1/n)\pi(d/2)^2 = \pi d^2/(4n)$; π · 196/32 = 19.242. Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve the degree sector formula $A = (\theta/360)\pi r^2$ for θ, and use your formula to find the central angle of a sector with area 12π in a circle of radius 6.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q5/5

2489

QUESTION ID

S63-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$\theta = 360A/(\pi r^2)$;

$\theta = 120°$

GRADING NOTE

REQUIRED WORK: Isolate θ algebraically, then substitute.

Any valid proof (paragraph, two-column, or flow). Essential: multiply both sides by 360 and divide by πr^2; $\theta = 360 \cdot 12\pi/(36\pi) = 120°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact area of a circle with radius 6.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q1/5

2491

QUESTION ID

S63-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has diameter 10. Exact area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q2/5

2492

QUESTION ID

S63-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25π

GRADING NOTE Form required: exact value. Use the radius 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q2/5

True or false?

$A = \pi r^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q3/5

2493

QUESTION ID

S63-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q3/5

A circle has area 49π. Radius?

A. 14

B. 7

C. 24.5

D. 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q4/5

2494

QUESTION ID

S63-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 7

GRADING NOTE $r^2 = 49$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q4/5

True or false?

Doubling the radius doubles the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q5/5

2495

QUESTION ID

S63-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It multiplies the area by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 2

The radius is half the diameter • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A circle has diameter 14 in. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q1/5

2497

QUESTION ID

S63-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7 in.

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q1/5

True or false?

Using the diameter in $A = \pi r^2$ gives an area four times too large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q2/5

2498

QUESTION ID

S63-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(2r)^2 = 4r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A sector of a circle with diameter 20 uses r =

A. 10

B. 20

C. 5

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q3/5

2499

QUESTION ID

S63-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle's radius is 2.5 cm. Diameter?

Diameter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q4/5

2500

QUESTION ID

S63-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q4/5

2500

True or false?

The radius and diameter can be used interchangeably in area formulas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q5/5

2501

QUESTION ID

S63-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They differ by a factor of 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 3

Fractions of a circle in degrees and radians • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of a circle is a sector with central angle 45°?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q1/5

2503

QUESTION ID

S63-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{8}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of a circle is a sector with central angle π/2 radians?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q2/5

2504

QUESTION ID

S63-B3-Q2

TYPE

Short answer

CORRECT ANSWER

¼

GRADING NOTE Any equivalent form. (π/2)/(2π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q2/5

A full circle measures

A. 360° or 2π radians

B. 180° or π radians

C. 360 radians

D. 2π°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q3/5

2505

QUESTION ID

S63-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 360° or 2π radians

GRADING NOTE Both units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q3/5

True or false?

A central angle of 1 radian is 1/(2π) of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q4/5

2506

QUESTION ID

S63-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Fraction of 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A 90° sector is ⅓ of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q5/5

2507

QUESTION ID

S63-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ¼.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q5/5

SET 63 • HSG-C.B.5

FORWARD 1

Solve inverse sector problems • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A sector with a 40° angle has area 4π. Find the radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q1/5

2509

QUESTION ID
S63-F1-Q1

TYPE
Short answer

CORRECT ANSWER
6

GRADING NOTE Any equivalent form. $(40/360)\pi r^2 = 4\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q1/5

2509

ANSWER KEY Reduced copy of the question page, with its answer

A sector of a circle with radius 10 has area 30. Find the central angle in radians.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q2/5

2510

QUESTION ID
S63-F1-Q2

TYPE
Short answer

CORRECT ANSWER
0.6 radians

GRADING NOTE Any equivalent form. ½(100)θ = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q2/5

2510

True or false?

If the radius is tripled and the angle stays the same, the sector area is tripled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q3/5

2511

QUESTION ID

S63-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Area depends on r^2, so it is multiplied by 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q3/5

2511

ANSWER KEY Reduced copy of the question page, with its answer

A sector has area 50 and arc length 10. Radius?

A. 20

B. 5

C. 2.5

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q4/5

2512

QUESTION ID

S63-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 10

GRADING NOTE A = ½rs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q4/5

Show that the area of a sector is A = ½rs, where s is the arc length.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q5/5

2513

QUESTION ID

S63-F1-Q5

TYPE

Short answer

CORRECT ANSWER

A = ½r²θ and s = rθ, so A = ½r(rθ) = ½rs.

GRADING NOTE Any equivalent derivation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q5/5

SET 63 • HSG-C.B.5

FORWARD 2

Areas of segments and composite circular regions • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact area of the region between a 90° arc and its chord in a circle of radius 4 (a circular segment).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q1/5

2515

QUESTION ID
S63-F2-Q1

TYPE
Short answer

CORRECT ANSWER
4π − 8

GRADING NOTE Form required: exact value. Sector 4π minus right triangle 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q1/5

2515

ANSWER KEY Reduced copy of the question page, with its answer

A sprinkler waters a 120° sector of radius 9 m. Find the watered area to the nearest square meter.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q2/5

2516

QUESTION ID

S63-F2-Q2

TYPE

Short answer

CORRECT ANSWER

85 m²

GRADING NOTE Rounding to the nearest whole number required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q2/5

True or false?

The area of a circular segment (smaller than a semicircle) equals the sector area plus the area of the triangle formed by the two radii and the chord.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q3/5

2517

QUESTION ID

S63-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Subtract the triangle: segment = sector − triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q3/5

An annulus (ring) has inner radius 3 and outer radius 5. Its area is

A. 8π

B. 4π

C. 16π

D. 25π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q4/5

2518

QUESTION ID

S63-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 16π

GRADING NOTE 25π − 9π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A windshield wiper blade covers the part of its arm from 20 cm to 60 cm from the pivot and sweeps through 120°. Find the area it wipes to the nearest square centimeter.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q5/5

2519

QUESTION ID

S63-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3351 cm²

GRADING NOTE Rounding to the nearest whole number required. $\frac{1}{3}\pi(60^2 - 20^2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q5/5

Derive the equation of a circle

MAIN	Geometry main question	**HSG-GPE.A.1**
BACKWARD 1	Distance between two points with the Pythagorean Theorem	**8.G.B.8**
BACKWARD 2	Use the Pythagorean Theorem with expressions	**8.G.B.7**
BACKWARD 3	Shifting a graph: reading h and k	**HSF-BF.B.3**
FORWARD 1	Write circle equations from other given information	**HSG-GPE.A.1**
FORWARD 2	Prove or disprove that points lie on a circle	**HSG-GPE.B.4**

SET 64 • HSG-GPE.A.1

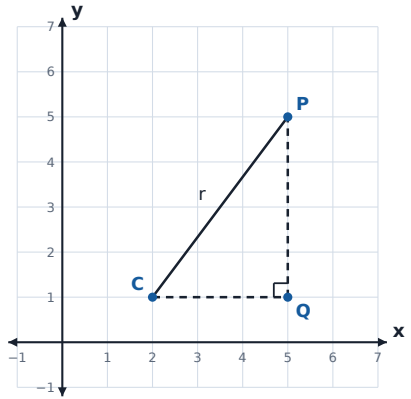
MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center (h, k) and radius r. Use the Pythagorean Theorem to derive the equation satisfied by every point (x, y) on the circle. The diagram shows one example, with center C(2, 1).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q1/5

2522

QUESTION ID

S64-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$(x - h)^2 + (y - k)^2 = r^2$

GRADING NOTE REQUIRED WORK: Draw the right triangle with legs |x − h| and |y − k| and hypotenuse r.

Any valid proof (paragraph, two-column, or flow). Essential: a right triangle with horizontal leg |x − h|, vertical leg |y − k|, hypotenuse r (the distance CP); the Pythagorean Theorem gives $(x - h)^2 + (y - k)^2 = r^2$; squaring removes the absolute values; the reasoning holds for every point on the circle (degenerate legs of length 0 included). Invalid: only checking the one example point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the Pythagorean Theorem (not a memorized formula) to derive the equation of the circle with center (3, −2) and radius 5. Show the right triangle for a general point (x, y) on the circle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q2/5

2523

QUESTION ID

S64-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$(x - 3)^2 + (y + 2)^2 = 25$

GRADING NOTE REQUIRED WORK: Use legs $|x - 3|$ and $|y + 2|$ and hypotenuse 5.

Any valid proof (paragraph, two-column, or flow). Essential: horizontal leg $|x - 3|$, vertical leg $|y - (-2)| = |y + 2|$, hypotenuse 5; $(x - 3)^2 + (y + 2)^2 = 25$ for every point on the circle. Invalid: writing the formula with no triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q2/5

A circle is centered at the origin and passes through (6, −8). Use the Pythagorean Theorem (not a memorized formula) to derive its equation, first finding the radius.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q3/5

2524

QUESTION ID

S64-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$r = 10$; $x^2 + y^2 = 100$

GRADING NOTE REQUIRED WORK: Find the radius as a hypotenuse, then use legs |x| and |y| for a general point.

Any valid proof (paragraph, two-column, or flow). Essential: $r^2 = 6^2 + 8^2 = 100$; for a general point the legs are |x| and |y|, so $x^2 + y^2 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center (−1, 4) and passes through (2, 8). Use the Pythagorean Theorem (not a memorized formula) to derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q4/5

2525

QUESTION ID

S64-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

$(x + 1)^2 + (y - 4)^2 = 25$

GRADING NOTE REQUIRED WORK: Find r from the right triangle with legs 3 and 4, then write the relationship for a general point (x, y).

Any valid proof (paragraph, two-column, or flow). Essential: legs 3 and 4 give r = 5; general legs |x + 1| and |y − 4|; $(x + 1)^2 + (y - 4)^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the Pythagorean Theorem (not a memorized formula) to derive the equation of the circle with center (−2, 5) and radius 3. Then use your equation to decide whether (−2, 8) is on the circle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q5/5

2526

QUESTION ID

S64-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$(x + 2)^2 + (y - 5)^2 = 9$; (−2, 8) is on it ($0 + 9 = 9$).

GRADING NOTE REQUIRED WORK: Use legs $|x + 2|$ and $|y - 5|$ and hypotenuse 3; substitute the point.

Any valid proof (paragraph, two-column, or flow). Essential: the right-triangle relationship with signs handled ($x - (-2) = x + 2$); $r^2 = 9$, not 3; the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q5/5

SET 64 • HSG-GPE.A.1

BACKWARD 1

Distance between two points with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q1/5

2528

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. Legs 3 and 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (5, 12).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q2/5

2529

QUESTION ID

S64-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q2/5

The legs of the right triangle joining (2, 1) and (7, 13) are

A. 2 and 1

B. 7 and 13

C. 9 and 14

D. 5 and 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q3/5

2530

QUESTION ID

S64-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5 and 12

GRADING NOTE Differences of coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q3/5

True or false?

The distance from $(-3, 0)$ to $(3, 0)$ is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q4/5

2531

QUESTION ID

S64-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Horizontal distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q4/5

True or false?

The distance from (0, 0) to (3, 4) is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q5/5

2532

QUESTION ID

S64-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 5; the legs are not added.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q5/5

2532

SET 64 • HSG-GPE.A.1

BACKWARD 2

Use the Pythagorean Theorem with expressions • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs 6 and 8. Find the hypotenuse.

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q1/5

2534

QUESTION ID

S64-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs a and b and hypotenuse c.
Write the Pythagorean relationship.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q2/5

2535

QUESTION ID

S64-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$a^2 + b^2 = c^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q2/5

A right triangle has legs $(x - 2)$ and $(y - 1)$ and hypotenuse 4. Which equation is true?

A. $(x - 2)^2 + (y - 1)^2 = 16$

B. $x^2 + y^2 = 16$

C. $(x - 2)^2 + (y - 1)^2 = 4$

D. $(x - 2) + (y - 1) = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q3/5

2536

QUESTION ID

S64-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(x - 2)^2 + (y - 1)^2 = 16$

GRADING NOTE Square the legs and the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q3/5

2536

True or false?

In a right triangle, the hypotenuse is opposite the right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q4/5

2537

QUESTION ID

S64-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$a^2 + b^2 = c^2$ holds for every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q5/5

2538

QUESTION ID

S64-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q5/5

2538

SET 64 • HSG-GPE.A.1

BACKWARD 3

Shifting a graph: reading h and k • High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $y = (x - 3)^2$ is the graph of $y = x^2$ shifted in which direction and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q1/5

2540

QUESTION ID

S64-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Right 3

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write x − (−4) without the double sign.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q2/5

2541

QUESTION ID

S64-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x + 4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In $(x + 2)^2$, the shift is

A. 2 down

B. 2 right

C. 2 left

D. 2 up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q3

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q3/5

2542

QUESTION ID
S64-B3-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. 2 left

GRADING NOTE $x - (-2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q3

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Replacing x with x − 5 shifts a graph 5 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q4/5

2543

QUESTION ID
S64-B3-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Horizontal translation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q4/5

True or false?

Replacing y with y + 1 shifts a graph 1 unit up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q5/5

2544

QUESTION ID
S64-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It shifts it 1 unit down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q5/5

SET 64 • HSG-GPE.A.1

FORWARD 1

Write circle equations from other given information • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

A diameter of a circle has endpoints (−2, 3) and (6, −3). Write the equation of the circle.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q1/5

2546

QUESTION ID

S64-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 2)^2 + y^2 = 25$

GRADING NOTE Form required: an equation. Center (2, 0) is the midpoint; radius 5 (half of 10).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center (4, −3) and is tangent to the x-axis. Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q2/5

2547

QUESTION ID

S64-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 4)^2 + (y + 3)^2 = 9$

GRADING NOTE Form required: an equation. Radius = distance to the x-axis = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q2/5

True or false?

The circle $(x - 1)^2 + (y - 1)^2 = 2$ passes through the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q3/5

2548

QUESTION ID

S64-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $1 + 1 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle with center (0, 5) passes through (0, −1). Its radius is

A. 4

B. 36

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q4/5

2549

QUESTION ID

S64-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 6

GRADING NOTE Vertical distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle is centered at (3, 3) and is tangent to both axes. Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q5/5

2550

QUESTION ID
S64-F1-Q5

TYPE
Short answer

CORRECT ANSWER
$(x - 3)^2 + (y - 3)^2 = 9$

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q5/5

SET 64 • HSG-GPE.A.1

FORWARD 2

Prove or disprove that points lie on a circle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Does (−4, 6) lie on the circle with center (2, −2) passing through (8, 6)? Show the check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q1/5

2552

QUESTION ID

S64-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: both points are 10 from the center (36 + 64 = 100).

GRADING NOTE Form required: the conclusion with the distance check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (7, 1) inside, on, or outside the circle $(x - 3)^2 + (y + 2)^2 = 25$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q2/5

2553

QUESTION ID

S64-F2-Q2

TYPE

Short answer

CORRECT ANSWER

On (16 + 9 = 25)

GRADING NOTE Form required: the conclusion with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q2/5

True or false?

A point is inside a circle when its distance from the center is greater than the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q3/5

2554

QUESTION ID

S64-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Greater than the radius means outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The point (1, 1) is ___ the circle $x^2 + y^2 = 4$.

A. on

B. outside

C. inside

D. the center of

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q4/5

2555

QUESTION ID

S64-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. inside

GRADING NOTE $1 + 1 = 2 < 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q4/5

A circle has the diameter with endpoints (−2, 3) and (6, −3). Does (5, 4) lie on it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q5/5

2556

QUESTION ID

S64-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Yes:

$(5 - 2)^2 + 4^2 = 25$.

GRADING NOTE Form required: the conclusion with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q5/5

Complete the square to find the center and radius of a circle

MAIN	Geometry main question	**HSG-GPE.A.1**
BACKWARD 1	Complete the square in one variable	**HSA-REI.B.4.a**
BACKWARD 2	Square a binomial	**HSA-APR.A.1**
BACKWARD 3	Keep an equation balanced	**HSA-REI.A.1**
FORWARD 1	Leading coefficients and equations that are not circles	**HSG-GPE.A.1**
FORWARD 2	Complete the square for an ellipse	**HSG-GPE.A.3**

SET 65 • HSG-GPE.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square to find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$. Show the standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q1/5

2559

QUESTION ID

S65-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 3)^2 + (y + 2)^2 = 25$; center $(3, -2)$; radius 5

GRADING NOTE Form required: both. $(x - 3)^2 + (y + 2)^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square to find the center and radius of the circle $x^2 + y^2 + 8x - 10y + 5 = 0$. Show the standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q2/5

2560

QUESTION ID
S65-M-Q2

TYPE
Short answer

CORRECT ANSWER
$(x + 4)^2 + (y - 5)^2 = 36$; center $(-4, 5)$; radius 6

GRADING NOTE Form required: both. $(x + 4)^2 + (y - 5)^2 = -5 + 16 + 25 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q2/5

2560

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square to write $x^2 + y^2 - 2x - 14y + 25 = 0$ in standard form. Then give the center and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q3/5

2561

QUESTION ID

S65-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$(x - 1)^2 + (y - 7)^2 = 25$; center (1, 7); radius 5

GRADING NOTE REQUIRED WORK: Group x and y terms, add $(b/2)^2$ for each to both sides, and factor.

Form required: the standard form, center, and radius. Added 1 and 49 to both sides: $-25 + 1 + 49 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q3/5

Complete the square to find the center and radius of the circle $x^2 + y^2 + 10y = 0$. Show the standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q4/5

2562

QUESTION ID

S65-M-Q4

TYPE

Short answer

CORRECT ANSWER

$x^2 + (y + 5)^2 = 25$; center $(0, -5)$; radius 5

GRADING NOTE Form required: both. $x^2 + (y + 5)^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square to write $x^2 + y^2 - 4x + 6y = 3$ in standard form. Then give the center and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q5/5

2563

QUESTION ID

S65-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$(x - 2)^2 + (y + 3)^2 = 16$; center $(2, -3)$; radius 4

GRADING NOTE REQUIRED WORK: Group x and y terms, add $(b/2)^2$ for each to both sides, and factor.

Form required: the standard form, center, and radius. Added 4 and 9 to both sides: 3 + 4 + 9 = 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q5/5

SET 65 • HSG-GPE.A.1

BACKWARD 1

Complete the square in one variable • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

What number completes the square: $x^2 - 6x + ___?$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q1/5

2565

QUESTION ID

S65-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $(6/2)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B1-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $x^2 + 10x + 25$ as a square.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q2/5

2566

QUESTION ID

S65-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 5)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B1-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 8x + 16$ equals

A. $(x + 4)^2$

B. $(x - 8)^2$

C. $(x - 4)^2$

D. $(x - 16)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q3/5

2567

QUESTION ID

S65-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(x - 4)^2$

GRADING NOTE Half of 8 is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B1-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

To complete $x^2 + bx$, add $(b/2)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q4/5

2568

QUESTION ID
S65-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 + 6x + 6$ is a perfect square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q5/5

2569

QUESTION ID
S65-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It needs + 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B1-Q5**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSG-GPE.A.1

BACKWARD 2

Square a binomial • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q1/5

2571

QUESTION ID
S65-B2-Q1

TYPE
Short answer

CORRECT ANSWER
$x^2 - 6x + 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(y + 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q2/5

2572

QUESTION ID
S65-B2-Q2

TYPE
Short answer

CORRECT ANSWER
$y^2 + 4y + 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x + 5)^2 =$

A. $x^2 + 10x + 25$

B. $x^2 + 25$

C. $x^2 + 10x + 10$

D. $x^2 + 5x + 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q3/5

2573

QUESTION ID
S65-B2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. $x^2 + 10x + 25$

GRADING NOTE Middle term 2 · 5x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a - b)^2 = a^2 - 2ab + b^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q4/5

2574

QUESTION ID
S65-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x - 3)^2 = x^2 - 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q5/5

2575

QUESTION ID
S65-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Missing the middle term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • HSG-GPE.A.1

BACKWARD 3

Keep an equation balanced • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

If 9 is added to the left side of an equation, what must be done to the right side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q1/5

2577

QUESTION ID
S65-B3-Q1

TYPE
Short answer

CORRECT ANSWER
Add 9

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B3-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 6x = 7$. Add 9 to both sides and factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q2/5

2578

QUESTION ID

S65-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 3)^2 = 16$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B3-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 + y^2 - 4x = 5$ becomes $(x - 2)^2 + y^2 =$

A. 4

B. 5

C. 1

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q3/5

2579

QUESTION ID
S65-B3-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 9

GRADING NOTE 5 + 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B3-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q3/5

True or false?

Adding the same number to both sides keeps the solutions the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q4/5

2580

QUESTION ID

S65-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equality property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B3-Q4**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q4/5

True or false?

A number added to one side need not be added to the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q5/5

2581

QUESTION ID
S65-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE That changes the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q5/5

SET 65 • HSG-GPE.A.1

FORWARD 1

Leading coefficients and equations that are not circles • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Find the center and radius of
$2x^2 + 2y^2 - 8x + 12y - 6 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q1/5

2583

QUESTION ID
S65-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Center (2, −3); radius 4

GRADING NOTE Form required: both. Divide by 2 first.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Does $x^2 + y^2 + 2x + 4y + 10 = 0$ represent a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q2/5

2584

QUESTION ID

S65-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No: it becomes $(x + 1)^2 + (y + 2)^2 = -5$, and a sum of squares cannot be negative.

GRADING NOTE Form required: the conclusion with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q2/5

True or false?

$x^2 + y^2 - 6x + 9 = 0$ represents a single point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q3/5

2585

QUESTION ID

S65-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x - 3)^2 + y^2 = 0$: only (3, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$3x^2 + 3y^2 - 12y = 0$ has radius

A. $\sqrt{12}$

B. 2

C. 12

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q4/5

2586

QUESTION ID

S65-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2

GRADING NOTE $x^2 + (y - 2)^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For what values of c is $x^2 + y^2 - 4x + c = 0$ a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q5/5

2587

QUESTION ID

S65-F1-Q5

TYPE

Short answer

CORRECT ANSWER

c < 4

GRADING NOTE Form required: an inequality. $r^2 = 4 - c > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q5/5

SET 65 • HSG-GPE.A.1

FORWARD 2

Complete the square for an ellipse • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Write $4x^2 + 9y^2 - 16x + 18y - 11 = 0$ in standard form and give the center.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q1/5

2589

QUESTION ID

S65-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 2)^2/9 + (y + 1)^2/4 = 1$; center $(2, -1)$

GRADING NOTE Form required: both. $4(x - 2)^2 + 9(y + 1)^2 = 11 + 16 + 9 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the center of $x^2 + 4y^2 + 6x - 8y + 9 = 0$.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q2/5

2590

QUESTION ID

S65-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(−3, 1)

GRADING NOTE Form required: an ordered pair. $(x + 3)^2 + 4(y - 1)^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q2/5

True or false?

After factoring 4 out of $4x^2 - 16x$, add 16 inside the parentheses to complete the square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q3/5

2591

QUESTION ID

S65-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Add 4 inside: $4(x^2 - 4x + 4)$, which adds 16 to the expression.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is NOT a circle?

A. $x^2 + 4y^2 = 4$

B. $2x^2 + 2y^2 = 8$

C. $x^2 + y^2 = 4$

D. $(x - 1)^2 + y^2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q4/5

2592

QUESTION ID

S65-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 4y^2 = 4$

GRADING NOTE Unequal coefficients give an ellipse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $x^2 + 4y^2 = 4$ is not a circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q5/5

2593

QUESTION ID

S65-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The x^2 and y^2 coefficients differ, so the "radius" differs by direction (2 across, 1 up).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q5/5

Derive the equation of a parabola from a focus and directrix

MAIN	Geometry main question	**HSG-GPE.A.2**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	Distance from a point to a horizontal line	**6.NS.C.8**
BACKWARD 3	Points equidistant from two objects	**HSG-CO.C.9**
BACKWARD 4	Expand and simplify squares	**HSA-APR.A.1**
FORWARD 1	Find the focus and directrix from an equation	**HSG-GPE.A.2**
FORWARD 2	Design with the reflective property of parabolas	**HSG-MG.A.3**

SET 66 • HSG-GPE.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Derive the equation of the parabola with focus (0, 2) and directrix y = −2.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q1/5

2596

QUESTION ID

S66-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x^2 = 8y$ ($y = x^2/8$)

GRADING NOTE REQUIRED WORK: Set the distance from (x, y) to the focus equal to its distance to the directrix, square, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{(x^2 + (y - 2)^2)} = |y + 2|$; square: $x^2 + y^2 - 4y + 4 = y^2 + 4y + 4$; simplify to $x^2 = 8y$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (equal distances to the focus and the directrix), the equation of the parabola with focus (0, −3) and directrix y = 3.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q2/5

2597

QUESTION ID

S66-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$x^2 = -12y$

GRADING NOTE REQUIRED WORK: Set $\sqrt{(x^2 + (y + 3)^2)} = |y - 3|$, square, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $x^2 + y^2 + 6y + 9 = y^2 - 6y + 9$; $x^2 = -12y$. Invalid: quoting $x^2 = 4py$ without the distance equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (equal distances to the focus and the directrix), the equation of the parabola with focus (2, 5) and directrix y = 1.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q3/5

2598

QUESTION ID

S66-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$(x - 2)^2 = 8(y - 3)$

GRADING NOTE REQUIRED WORK: Set $\sqrt{((x - 2)^2 + (y - 5)^2)} = |y - 1|$, square, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $(x - 2)^2 + y^2 - 10y + 25 = y^2 - 2y + 1$; $(x - 2)^2 = 8y - 24 = 8(y - 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (equal distances to the focus and the directrix), the equation of the parabola with focus (3, 0) and directrix $x = -3$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q4/5

2599

QUESTION ID

S66-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

$y^2 = 12x$

GRADING NOTE REQUIRED WORK: Set $\sqrt{((x-3)^2 + y^2)} = |x+3|$, square, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $x^2 - 6x + 9 + y^2 = x^2 + 6x + 9$; $y^2 = 12x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (equal distances to the focus and the directrix), the equation of the parabola with focus (1, 4) and directrix y = 0 (the x-axis). Then give its vertex.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q5/5

2600

QUESTION ID

S66-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$(x - 1)^2 = 8(y - 2)$; vertex $(1, 2)$

GRADING NOTE REQUIRED WORK: Set $\sqrt{((x - 1)^2 + (y - 4)^2)} = |y|$, square, simplify, and read the vertex.

Any valid proof (paragraph, two-column, or flow). Essential: $(x - 1)^2 + y^2 - 8y + 16 = y^2$; $(x - 1)^2 = 8y - 16 = 8(y - 2)$; the vertex (1, 2) is midway between the focus and the directrix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (3, 6) to (0, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q1/5

2602

QUESTION ID

S66-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the distance from (x, y) to (0, 2) as an expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q2/5

2603

QUESTION ID

S66-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$\sqrt{(x^2 + (y - 2)^2)}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The distance from (4, 1) to (1, 5) is

A. 7

B. 5

C. √7

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q3/5

2604

QUESTION ID

S66-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

GRADING NOTE Legs 3 and 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q3/5

True or false?

The distance formula comes from the Pythagorean Theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q4/5

2605

QUESTION ID

S66-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Legs are coordinate differences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{(9 + 16)} = 3 + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q5/5

2606

QUESTION ID

S66-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{25} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 2

Distance from a point to a horizontal line • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

How far is (5, 7) from the line y = 2?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q1/5

2608

QUESTION ID

S66-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How far is (1, −3) from the line y = 4?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q2/5

2609

QUESTION ID

S66-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The distance from (x, y) to y = −2 is

A. |x + 2|

B. y

C. |y − 2|

D. |y + 2|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q3/5

2610

QUESTION ID

S66-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. |y + 2|

GRADING NOTE Vertical difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q3/5

2610

True or false?

The distance from a point to a horizontal line is measured straight up or down.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q4/5

2611

QUESTION ID

S66-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q4/5

True or false?

The distance from (6, 1) to y = 4 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q5/5

2612

QUESTION ID

S66-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 3

Points equidistant from two objects • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

The set of points equidistant from two points is the perpendicular bisector of the segment joining them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q1/5

2614

QUESTION ID
S66-B3-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A parabola is the set of points equidistant from what two objects?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q2/5

2615

QUESTION ID

S66-B3-Q2

TYPE

Short answer

CORRECT ANSWER

A point (the focus) and a line (the directrix)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q2/5

The vertex of a parabola is

A. at the focus

B. twice as far from the focus

C. halfway between the focus and the directrix

D. on the directrix

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q3/5

2616

QUESTION ID

S66-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. halfway between the focus and the directrix

GRADING NOTE Equidistant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Focus (0, 6), directrix y = 0. Find the vertex.

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q4/5

2617

QUESTION ID

S66-B3-Q4

TYPE

Short answer

CORRECT ANSWER

(0, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q4/5

True or false?

Every point of a parabola is closer to the focus than to the directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q5/5

2618

QUESTION ID

S66-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Equal distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 4

Expand and simplify squares • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(y - 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q1/5

2620

QUESTION ID
S66-B4-Q1

TYPE
Short answer

CORRECT ANSWER
$y^2 - 4y + 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(y + 2)^2 - (y - 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q2/5

2621

QUESTION ID
S66-B4-Q2

TYPE
Short answer

CORRECT ANSWER
8y

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$y^2 + 6y + 9 - (y^2 - 6y + 9) =$

A. 18

B. $2y^2$

C. 0

D. 12y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q3/5

2622

QUESTION ID
S66-B4-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 12y

GRADING NOTE The squares cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(y + 3)^2 = y^2 + 6y + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q4/5

2623

QUESTION ID
S66-B4-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(y - 2)^2 - y^2 = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q5/5

2624

QUESTION ID
S66-B4-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $-4y + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q5/5

SET 66 • HSG-GPE.A.2

FORWARD 1

Find the focus and directrix from an equation • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Find the focus and directrix of $y = x^2/12$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q1/5

2626

QUESTION ID

S66-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Focus (0, 3); directrix y = −3

GRADING NOTE Form required: both. $x^2 = 12y = 4py$, $p = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q1/5

Find the vertex, focus, and directrix of $(x - 1)^2 = -8(y + 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q2/5

2627

QUESTION ID

S66-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Vertex (1, −2); focus (1, −4); directrix y = 0

GRADING NOTE Form required: all three. p = −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q2/5

True or false?

In $x^2 = 4py$ ($p \neq 0$), the focus is $|p|$ units from the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q3/5

2628

QUESTION ID

S66-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The focus is (0, p); its distance from the vertex (0, 0) is |p|, since p may be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q3/5

The parabola $y^2 = -20x$ opens

A. left

B. up

C. down

D. right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q4/5

2629

QUESTION ID

S66-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. left

GRADING NOTE Negative coefficient with y^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the focus of $y = 2x^2$.

Focus: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q5/5

2630

QUESTION ID

S66-F1-Q5

TYPE

Short answer

CORRECT ANSWER

(0, ⅛)

GRADING NOTE Form required: an ordered pair. $x^2 = \frac{1}{2}y$, $4p = \frac{1}{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q5/5

SET 66 • HSG-GPE.A.2

FORWARD 2

Design with the reflective property of parabolas • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A satellite dish is 4 ft across and 1 ft deep. How far from the vertex should the receiver (at the focus) be placed?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q1/5

2632

QUESTION ID

S66-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1 ft

GRADING NOTE Any equivalent form. $x^2 = 4py$ through (2, 1): $4 = 4p$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A headlight reflector has equation $x^2 = 12y$ (inches). Where should the bulb go?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q2/5

2633

QUESTION ID
S66-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3 in. above the vertex, at (0, 3)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q2/5

True or false?

Rays parallel to the axis of a parabolic mirror reflect through the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q3/5

2634

QUESTION ID
S66-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They reflect through the focus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q3/5

A dish has focus 2 m above its vertex. Its equation (vertex at the origin) is

A. $x^2 = 4y$

B. $x^2 = 8y$

C. $y = 8x^2$

D. $x^2 = 2y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q4/5

2635

QUESTION ID

S66-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $x^2 = 8y$

GRADING NOTE $4p = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q4/5

A dish has equation $x^2 = 16y$ (meters) and is 8 m across. How deep is it?

Depth: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q5/5

2636

QUESTION ID
S66-F2-Q5

TYPE
Short answer

CORRECT ANSWER
1 m

GRADING NOTE Any equivalent form. 16 = 16y at x = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q5/5

SET 67 • HSG-GPE.B.4 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Prove or disprove the type of a quadrilateral with coordinates

MAIN	Geometry main question	**HSG-GPE.B.4**
BACKWARD 1	Find the slope between two points	**8.F.B.4**
BACKWARD 2	Find the distance between two points	**8.G.B.8**
BACKWARD 3	Properties that define special quadrilaterals	**HSG-CO.C.11**
FORWARD 1	Coordinate proofs with variables	**HSG-GPE.B.4**
FORWARD 2	Use the classification to find area	**HSG-GPE.B.7**

SET 67 • HSG-GPE.B.4

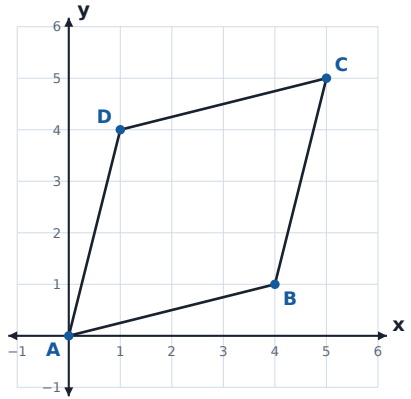
MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Prove that ABCD with A(0, 0), B(4, 1), C(5, 5), D(1, 4) is a rhombus.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q1/5

2639

QUESTION ID

S67-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AB = BC = CD = DA = √17, so ABCD is a rhombus.

GRADING NOTE REQUIRED WORK: Compute all four side lengths with the distance formula.

Any valid coordinate proof. Essential: all four distances computed correctly (√17 each) and the definition of a rhombus cited. Alternative: show it is a parallelogram (slopes) with perpendicular diagonals (slopes 1 and −1). Invalid: "it looks like a rhombus".

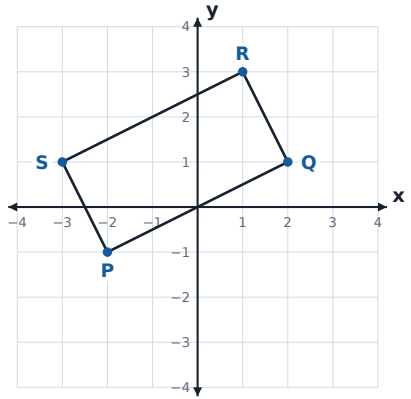
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that PQRS with P(−2, −1), Q(2, 1), R(1, 3), S(−3, 1) is a rectangle. Is it a square?

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q2/5

2640

QUESTION ID

S67-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Slopes PQ = RS = ½ and QR = SP = −2; adjacent products are −1, so all angles are right: a rectangle. Not a square: PQ = √20 but QR = √5.

GRADING NOTE REQUIRED WORK: Compute the slopes of all four sides; compare adjacent slopes.

Any valid coordinate proof. Essential: four slopes; opposite sides parallel; adjacent slopes multiply to −1 (right angles); side lengths √20 ≠ √5 for "not a square".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove: KLMN with K(0, 0), L(5, 0), M(6, 3), N(0, 3) is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q3/5

2641

QUESTION ID

S67-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Disproved: KL ∥ MN (slope 0), but LM has slope 3 while NK is vertical, so LM ∦ NK. KLMN is a trapezoid, not a parallelogram.

GRADING NOTE

REQUIRED WORK: Compare the slopes of both pairs of opposite sides.

Any valid coordinate proof. Essential: one pair of opposite sides shown NOT parallel (a single counterexample disproves). Alternative: KL = 5 but MN = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that the triangle with vertices A(1, 1), B(4, 5), C(8, 2) is an isosceles right triangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q4/5

2642

QUESTION ID

S67-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AB = BC = 5 and AC = √50; $5^2 + 5^2 = 50$, so ∠B is right. Isosceles right triangle.

GRADING NOTE REQUIRED WORK: Use distances for the equal sides and slopes (or the converse of the Pythagorean Theorem) for the right angle.

Any valid coordinate proof. Essential: three distances; two equal (isosceles); right angle at B by the converse of the Pythagorean Theorem or slopes 4/3 and −3/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove: EFGH with E(−1, 0), F(2, −1), G(3, 2), H(0, 3) is a square.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q5/5

2643

QUESTION ID

S67-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

Proved: all sides are √10; slopes EF = −⅓ and FG = 3 multiply to −1, so ∠F is right. A rhombus with a right angle is a square.

GRADING NOTE REQUIRED WORK: Show four equal sides and one right angle (or a rhombus with a right angle).

Any valid coordinate proof. Essential: four lengths √10; opposite slopes equal (−⅓ and 3); a right angle; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 1

Find the slope between two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (1, 2) and (5, 4).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q1/5

2645

QUESTION ID

S67-B1-Q1

TYPE

Short answer

CORRECT ANSWER

½

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (0, 3) and (2, −1).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q2/5

2646

QUESTION ID

S67-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The slope through (2, 5) and (6, 5) is

A. undefined

B. 0

C. 5

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q3/5

2647

QUESTION ID

S67-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0

GRADING NOTE Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A vertical segment has an undefined slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q4/5

2648

QUESTION ID

S67-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Run 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Slope is run divided by rise.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q5/5

2649

QUESTION ID

S67-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rise ÷ run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 2

Find the distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (4, 1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q1/5

2651

QUESTION ID

S67-B2-Q1

TYPE

Short answer

CORRECT ANSWER

√17

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q1/5

2651

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (1, 1) and (4, 5).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q2/5

2652

QUESTION ID

S67-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of points is 10 apart?

A. (2, 3) and (8, 3)

B. (0, 0) and (6, 8)

C. (0, 0) and (10, 10)

D. (1, 1) and (5, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q3/5

2653

GRADING NOTE 36 + 64 = 100.

QUESTION ID

S67-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (0, 0) and (6, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q3/5

True or false?

Two segments with the same length are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q4/5

2654

QUESTION ID

S67-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{10} + \sqrt{10} = \sqrt{20}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q5/5

2655

QUESTION ID

S67-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{10} + \sqrt{10} = 2\sqrt{10} = \sqrt{40}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 3

Properties that define special quadrilaterals • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Which quadrilateral has four congruent sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q1/5

2657

QUESTION ID

S67-B3-Q1

TYPE

Short answer

CORRECT ANSWER

A rhombus (a square is a special rhombus)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram with one right angle is a ___.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q2/5

2658

QUESTION ID

S67-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Rectangle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q2/5

Which shows a quadrilateral is a parallelogram?

A. One pair of sides is parallel

B. Two adjacent sides are equal

C. One right angle

D. Both pairs of opposite sides are parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q3/5

2659

QUESTION ID

S67-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Both pairs of opposite sides are parallel

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q3/5

True or false?

Every square is a rhombus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q4/5

2660

QUESTION ID

S67-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Four congruent sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Every rhombus is a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q5/5

2661

QUESTION ID

S67-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its angles need not be right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q5/5

SET 67 • HSG-GPE.B.4

FORWARD 1

Coordinate proofs with variables • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has vertices O(0, 0), A(a, 0), B(a, b), C(0, b). Prove that its diagonals are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q1/5

2663

QUESTION ID

S67-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

OB = AC = √(a² + b²)

GRADING NOTE

REQUIRED WORK: Compute OB and AC with the distance formula.

Any valid coordinate proof. Essential: OB = √(a² + b²), AC = √((0 − a)² + (b − 0)²) = √(a² + b²); equal for all a, b. Invalid: using numbers for a and b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle with vertices O(0, 0), A(2a, 0), B(2b, 2c), where a > 0 and c ≠ 0 (so the points are not collinear). M and N are the midpoints of OB and AB. Prove that MN ∥ OA and MN = ½OA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q2/5

2664

QUESTION ID

S67-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

M(b, c), N(a + b, c); MN is horizontal like OA, and MN = a = ½(2a).

GRADING NOTE REQUIRED WORK: Find M and N, then compare slope and length.

Any valid coordinate proof. Essential: correct midpoints; slope 0 for both; length a vs 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q2/5

True or false?

Placing a figure with a vertex at the origin and a side on the x-axis is allowed in a coordinate proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q3/5

2665

QUESTION ID

S67-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any figure can be moved there by a rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q3/5

Why do coordinates like (2a, 0) and (2b, 2c) make the midpoint proof easier?

A. The midpoints have no fractions

B. They make the triangle isosceles

C. They make a right angle

D. They make the slopes 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q4/5

2666

QUESTION ID

S67-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The midpoints have no fractions

GRADING NOTE Halves of even coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q4/5

Why is a proof using A(6, 0) and B(2, 4) not enough to prove the midsegment theorem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q5/5

2667

QUESTION ID
S67-F1-Q5

TYPE
Short answer

CORRECT ANSWER
It shows only one triangle; the theorem is about all triangles.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSG-GPE.B.4

FORWARD 2

Use the classification to find area • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ABCD with A(0, 0), B(4, 1), C(5, 5), D(1, 4) is a rhombus. Find its area using its diagonals.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q1/5

2669

QUESTION ID

S67-F2-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. AC = 5√2, BD = 3√2; ½ · 5√2 · 3√2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rectangle PQRS has P(−2, −1), Q(2, 1), R(1, 3), S(−3, 1). Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q2/5

2670

QUESTION ID
S67-F2-Q2

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE Any equivalent form. $\sqrt{20} \cdot \sqrt{5}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q2/5

True or false?

Once a quadrilateral is proved a rhombus, its area is the product of two adjacent side lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q3/5

2671

QUESTION ID

S67-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only for a square; use $\frac{1}{2}d_1d_2$ or base × height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q3/5

A square has vertices (0, 0), (3, 1), (2, 4), (−1, 3). Its area is

A. 16

B. √10

C. 10

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q4/5

2672

QUESTION ID

S67-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

GRADING NOTE Side √10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F2-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q4/5

Explain why proving the shape first matters before using a special area formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q5/5

2673

QUESTION ID
S67-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The formula (such as ½d_1d_2 or length × width) is valid only for that type of figure.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q5/5

Prove or disprove that a point lies on a circle, and other coordinate proofs

MAIN	Geometry main question	**HSG-GPE.B.4**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	The equation of a circle	**HSG-GPE.A.1**
BACKWARD 3	Substitute values into an expression	**6.EE.A.2**
FORWARD 1	Coordinate proofs about circles with variables	**HSG-GPE.B.4**
FORWARD 2	Find the tangent line to a circle at a point	**HSG-GPE.B.5**

SET 68 • HSG-GPE.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A circle is centered at (2, −1) and contains the point (2, 4). Prove or disprove that the point (5, 3) lies on the circle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q1/5

2676

QUESTION ID

S68-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Proved: the radius is 5 and the distance from (2, −1) to (5, 3) is √(9 + 16) = 5.

GRADING NOTE REQUIRED WORK: Find the radius, then compare the distance from the center to (5, 3).

Any valid coordinate proof. Essential: radius $|4 - (-1)| = 5$; distance $\sqrt{(3^2 + 4^2)} = 5$; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q1/5

Prove or disprove that (5, 2) lies on the circle with center (1, −1) and radius 5.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q2/5

2677

QUESTION ID

S68-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Proved:

$(5 - 1)^2 + (2 + 1)^2 = 16 + 9 = 25 = 5^2$.

GRADING NOTE

REQUIRED WORK: Compare the distance from the center to the point with the radius.

Any valid coordinate proof. Essential: the squared distance 25 computed from the coordinates; compared with $r^2 = 25$; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q2/5

Prove or disprove that $(-3, 6)$ lies on the circle $(x - 1)^2 + (y - 2)^2 = 30$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q3/5

2678

QUESTION ID

S68-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Disproved: 16 + 16 = 32 ≠ 30 (the point is outside).

GRADING NOTE REQUIRED WORK: Substitute the point into the equation, or compare its distance from the center with $\sqrt{30}$.

Any valid coordinate proof. Essential: substitution giving 32; 32 ≠ 30; conclusion (outside, since 32 > 30).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has A(0, 0), B(8, 0), C(0, 6). Prove that the midpoint M of hypotenuse BC is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q4/5

2679

QUESTION ID

S68-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

M(4, 3);

MA = MB = MC = 5

GRADING NOTE REQUIRED WORK: Find M, then compute MA, MB, MC.

Any valid coordinate proof. Essential: midpoint (4, 3); three distances of 5. Consequence: A, B, C lie on a circle centered at M.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that the point (4, −3) lies outside the circle $x^2 + y^2 = 20$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q5/5

2680

QUESTION ID

S68-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

Distance 5 > √20 ≈ 4.47, so the point is outside.

GRADING NOTE REQUIRED WORK: Compare the distance from the center to the point with the radius.

Any valid coordinate proof. Essential: center (0, 0), radius √20; distance √(16 + 9) = 5; 5 > √20 (25 > 20); conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q5/5

SET 68 • HSG-GPE.B.4

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (1, −1) to (5, 2)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q1/5

2682

QUESTION ID

S68-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (2, −1) to (5, 3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q2/5

2683

QUESTION ID

S68-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (0, 0) to (−3, 4) is

A. 5

B. 1

C. 7

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q3/5

2684

QUESTION ID

S68-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

GRADING NOTE 9 + 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(-3)^2 = 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q4/5

2685

QUESTION ID

S68-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Square of a negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q4/5

True or false?

The distance from (2, 2) to (2, 7) is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q5/5

2686

QUESTION ID

S68-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSG-GPE.B.4

BACKWARD 2

The equation of a circle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the circle with center (1, −1) and radius 5.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q1/5

2688

QUESTION ID

S68-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 1)^2 + (y + 1)^2 = 25$

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the radius of $(x - 1)^2 + (y - 2)^2 = 30$?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q2/5

2689

QUESTION ID

S68-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$\sqrt{30}$

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Center of $(x + 4)^2 + y^2 = 9$?

A. (4, 0)

B. (−4, 0)

C. (−4, 9)

D. (0, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q3/5

2690

QUESTION ID

S68-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−4, 0)

GRADING NOTE Opposite sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q3/5

True or false?

A point satisfies a circle's equation exactly when it is on the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q4/5

2691

QUESTION ID

S68-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Meaning of the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The radius of $x^2 + y^2 = 20$ is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q5/5

2692

QUESTION ID
S68-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $\sqrt{20}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSG-GPE.B.4

BACKWARD 3

Substitute values into an expression • Grade 6 • Expressions & Equations • 6.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(x - 1)^2 + (y - 2)^2$ when $x = -3$ and $y = 6$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q1/5

2694

QUESTION ID

S68-B3-Q1

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $x^2 + y^2$ when $x = 4$, $y = -3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q2/5

2695

QUESTION ID

S68-B3-Q2

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x + 1)^2$ when $x = -4$ is

A. -9

B. 9

C. 25

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q3/5

2696

QUESTION ID

S68-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9

GRADING NOTE $(-3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q3/5

True or false?

$(-5 - 1)^2 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q4/5

2697

QUESTION ID

S68-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-6)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(2 + 1)^2 = 2^2 + 1^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q5/5

2698

QUESTION ID

S68-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $9 \neq 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q5/5

SET 68 • HSG-GPE.B.4

FORWARD 1

Coordinate proofs about circles with variables • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A(−r, 0) and B(r, 0) are endpoints of a diameter of the circle $x^2 + y^2 = r^2$, and P(x, y) is any other point on the circle. Prove that ∠APB is a right angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q1/5

2700

QUESTION ID

S68-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

slope PA · slope PB = $y^2/(x^2 - r^2)$ = $y^2/(-y^2) = -1$, so PA ⊥ PB.

GRADING NOTE REQUIRED WORK: Compute the slopes of PA and PB and use $x^2 + y^2 = r^2$.

Any valid coordinate proof. Essential: slopes $y/(x + r)$ and $y/(x - r)$; product $y^2/(x^2 - r^2)$; substitute $x^2 - r^2 = -y^2$; product −1. (P is not A or B, so $y \neq 0$.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that if (a, b) is on $x^2 + y^2 = r^2$, then (b, a) and (−a, b) are too.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q2/5

2701

QUESTION ID

S68-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$b^2 + a^2 = r^2$ and $(-a)^2 + b^2 = r^2$, the same sum.

GRADING NOTE Any equivalent argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q2/5

True or false?

A coordinate proof with variables covers every case at once.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q3/5

2702

QUESTION ID
S68-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE General.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To prove four points lie on one circle, show they are

A. all the same distance from one point

B. the same distance from the origin only

C. vertices of a rectangle only

D. all on one line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q4/5

2703

QUESTION ID

S68-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. all the same distance from one point

GRADING NOTE Common center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove: A(1, 0), B(0, 1), C(−1, 0), D(0, −2) lie on one circle centered at the origin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q5/5

2704

QUESTION ID

S68-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Disproved: D is 2 from the origin, the others 1.

GRADING NOTE Form required: the conclusion with distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSG-GPE.B.4

FORWARD 2

Find the tangent line to a circle at a point • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find the equation of the line tangent to $x^2 + y^2 = 25$ at (3, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q1/5

2706

QUESTION ID

S68-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$y = -\frac{3}{4}x + 25/4$

GRADING NOTE Form required: an equation; equivalent forms (3x + 4y = 25) accepted. Radius slope 4/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the tangent to $(x - 1)^2 + (y - 1)^2 = 25$ at (4, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q2/5

2707

QUESTION ID

S68-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$y - 5 = -\frac{3}{4}(x - 4)$

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q2/5

True or false?

The tangent at P is parallel to the radius drawn to P.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S68-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q3/5

2708

QUESTION ID
S68-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is perpendicular to that radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S68-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q3/5

The radius to (0, 5) on $x^2 + y^2 = 25$ is vertical, so the tangent there is

A. $x = 0$

B. $x = 5$

C. $y = x + 5$

D. $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q4/5

2709

QUESTION ID

S68-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. y = 5

GRADING NOTE Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how the slope of the tangent at (a, b) on $x^2 + y^2 = r^2$ is found.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q5/5

2710

QUESTION ID
S68-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The radius has slope b/a, so the tangent has slope −a/b (negative reciprocal), for a, b ≠ 0.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q5/5

Prove the slope criteria for parallel and perpendicular lines

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Similar slope triangles give the same slope	**8.EE.B.6**
BACKWARD 2	Rotate a point 90° about the origin	**HSG-CO.A.5**
BACKWARD 3	Negative reciprocals	**7.NS.A.2**
FORWARD 1	Use the criteria to classify pairs of lines	**HSG-GPE.B.5**
FORWARD 2	Use the criteria inside a general coordinate proof	**HSG-GPE.B.4**

SET 69 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that two distinct nonvertical lines with the same slope m are parallel.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q1/5

2713

QUESTION ID

S69-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$mx + b_1 = mx + b_2$ gives $b_1 = b_2$, a contradiction; so the lines never meet and are parallel.

GRADING NOTE REQUIRED WORK: Write the lines as $y = mx + b_1$ and $y = mx + b_2$ with $b_1 \neq b_2$ and show there is no intersection.

Any valid proof (paragraph, two-column, or flow). Essential: equations with equal slopes and different intercepts; solving for an intersection leads to $b_1 = b_2$ (false); conclusion. Alternative: a translation maps one line to the other; or congruent slope triangles with corresponding angles equal.

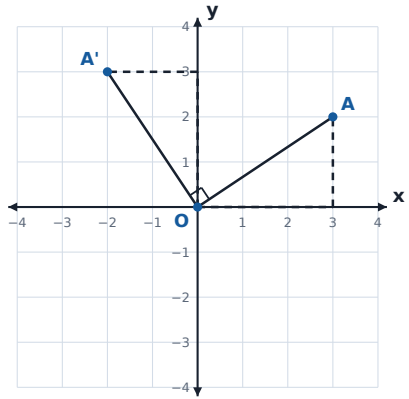
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Line OA passes through O(0, 0) and A(a, b) with a, b ≠ 0. Rotating A 90° counterclockwise about O gives A'(−b, a). Use this to prove that perpendicular lines have slopes whose product is −1. The diagram shows the case A(3, 2).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q2/5

2714

QUESTION ID

S69-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

slope OA = b/a, slope OA' = a/(−b) = −a/b, product −1.

GRADING NOTE REQUIRED WORK: Find the slopes of OA and OA', then their product.

Any valid proof (paragraph, two-column, or flow). Essential: a 90° rotation maps OA onto a perpendicular line; the image of (a, b) is (−b, a); slopes b/a and −a/b; product −1. Any pair of perpendicular nonvertical lines can be translated to pass through O, keeping slopes. Invalid: checking only the one example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that two distinct nonvertical lines that are parallel have equal slopes.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q3/5

2715

QUESTION ID

S69-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

If $m_1 \neq m_2$, then $m_1x + b_1 = m_2x + b_2$ has the solution $x = (b_2 - b_1)/(m_1 - m_2)$, so the lines meet, contradicting parallel. So $m_1 = m_2$.

GRADING NOTE REQUIRED WORK: Indirect: assume the slopes m_1 and m_2 are different and find an intersection point.

Any valid proof (paragraph, two-column, or flow). Essential: assumption of unequal slopes; solving for the intersection; contradiction with "parallel"; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that if two nonvertical lines have slopes m and $-1/m$ ($m \neq 0$), they are perpendicular. (Translate both lines so they pass through the origin first.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q4/5

2716

QUESTION ID

S69-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

(1, m) rotates to (−m, 1), on the line with slope 1/(−m) = −1/m; so the line with slope −1/m through O is the 90° image of y = mx, hence perpendicular.

GRADING NOTE REQUIRED WORK: Rotate the point (1, m) on y = mx by 90° about the origin and find the slope of its image line.

Any valid proof (paragraph, two-column, or flow). Essential: translations keep slopes; the rotation image of (1, m); its slope $-1/m$; a 90° rotation image is perpendicular; only one line through O has slope $-1/m$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Lines $y = mx$ and $y = -x/m$ ($m \neq 0$) pass through the origin O. Use the points P(1, m) and Q(−m, 1) and the converse of the Pythagorean Theorem to prove that the lines are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q5/5

2717

QUESTION ID

S69-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$OP^2 + OQ^2 = (1 + m^2) + (m^2 + 1) = 2 + 2m^2 = PQ^2$, so $\angle POQ = 90°$.

GRADING NOTE REQUIRED WORK: Compute OP^2, OQ^2, and PQ^2, and compare.

Any valid proof (paragraph, two-column, or flow). Essential: $OP^2 = 1 + m^2$; $OQ^2 = m^2 + 1$; $PQ^2 = (1 + m)^2 + (m - 1)^2 = 2 + 2m^2$; converse of the Pythagorean Theorem; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 1

Similar slope triangles give the same slope • Grade 8 • Expressions & Equations • 8.EE.B.6

True or false?

Any two slope triangles on the same line are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q1/5

2719

QUESTION ID

S69-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line passes through (0, 0) and (2, 3). Find the rise for a run of 4.

Rise: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q2/5

2720

QUESTION ID

S69-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope triangles with run 2, rise 3 and run 6, rise 9 give slopes

A. 6 and 18

B. 3/2 and 9/6 that differ

C. both 3/2

D. 2/3 and 6/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q3/5

2721

QUESTION ID
S69-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. both 3/2

GRADING NOTE 9/6 = 3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the slope of a line is the same between any two of its points.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q4/5

2722

QUESTION ID

S69-B1-Q4

TYPE

Short answer

CORRECT ANSWER

The slope triangles are similar, so rise/run is the same ratio.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slope between two points on a line depends on which points are chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q5/5

2723

QUESTION ID

S69-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 2

Rotate a point 90° about the origin • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (3, 2) 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q1/5

2725

QUESTION ID

S69-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−2, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (1, 4) 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q2/5

2726

QUESTION ID

S69-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−4, 1)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q2/5

The rule for a 90° counterclockwise rotation about the origin is

A. (x, y) → (−y, x)

B. (x, y) → (−x, −y)

C. (x, y) → (y, x)

D. (x, y) → (y, −x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q3/5

2727

QUESTION ID

S69-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (x, y) → (−y, x)

GRADING NOTE Rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q3/5

True or false?

A 90° rotation maps a line to a perpendicular line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q4/5

2728

QUESTION ID

S69-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Angle of rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q4/5

True or false?

A 90° rotation maps (2, 0) to (0, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q5/5

2729

QUESTION ID

S69-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Counterclockwise gives (0, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 3

Negative reciprocals • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of 2/3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q1/5

2731

QUESTION ID

S69-B3-Q1

TYPE

Short answer

CORRECT ANSWER

−3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(b/a)(-a/b)$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q2/5

2732

QUESTION ID

S69-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of −4?

A. ¼

B. −¼

C. 4

D. −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q3/5

2733

QUESTION ID

S69-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ¼

GRADING NOTE −1 ÷ (−4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A number times its negative reciprocal is −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q4/5

2734

QUESTION ID

S69-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The negative reciprocal of 1 is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q5/5

2735

QUESTION ID

S69-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q5/5

SET 69 • HSG-GPE.B.5

FORWARD 1

Use the criteria to classify pairs of lines • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Are y = 3x − 2 and 6x − 2y = 7 parallel, perpendicular, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q1/5

2737

QUESTION ID

S69-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Parallel (both slopes 3)

GRADING NOTE Form required: the classification with the slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Are 2x + 5y = 1 and 5x − 2y = 4 parallel, perpendicular, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q2/5

2738

QUESTION ID

S69-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Perpendicular (slopes −2/5 and 5/2)

GRADING NOTE Form required: the classification with the slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q2/5

True or false?

A horizontal line and a vertical line are perpendicular even though the slope product rule cannot be used.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q3/5

2739

QUESTION ID

S69-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Vertical slope is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q3/5

$y = \frac{1}{2}x + 1$ and $y = 2x - 1$ are

A. neither

B. the same line

C. perpendicular

D. parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q4/5

2740

QUESTION ID

S69-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. neither

GRADING NOTE Product 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find k so that y = kx + 1 is perpendicular to y = −⅓x.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q5/5

2741

QUESTION ID

S69-F1-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q5/5

SET 69 • HSG-GPE.B.5

FORWARD 2

Use the criteria inside a general coordinate proof • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Rhombus OABC has O(0, 0), A(a, 0), B(a + b, c), C(b, c), with $b^2 + c^2 = a^2$. Prove that its diagonals are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q1/5

2743

QUESTION ID

S69-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

slope OB · slope AC = c/(a + b) · c/(b − a) = c²/(b² − a²) = c²/(−c²) = −1

GRADING NOTE REQUIRED WORK: Find the slopes of OB and AC and multiply.

Any valid coordinate proof. Essential: both slopes; product simplified with $b^2 + c^2 = a^2$; conclusion. (Assume $b \neq \pm a$, so neither diagonal is vertical.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q1/5

A coordinate proof places rhombus OABC at O(0, 0), A(a, 0), B(a + b, c), C(b, c). Why must the proof assume $b^2 + c^2 = a^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q2/5

2744

QUESTION ID

S69-F2-Q2

TYPE

Short answer

CORRECT ANSWER

It says OC = a = OA, the equal sides that make OABC a rhombus.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q2/5

True or false?

The slope criterion turns "perpendicular" into an algebra statement.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q3/5

2745

QUESTION ID
S69-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Product −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q3/5

In a coordinate proof that lines are parallel, you show

A. equal slopes

B. equal intercepts

C. slope product −1

D. equal lengths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q4/5

2746

QUESTION ID

S69-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. equal slopes

GRADING NOTE Criterion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q4/5

Prove or disprove with slopes: A(0, 0), B(4, 2), C(3, 4) form a right angle at B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q5/5

2747

QUESTION ID

S69-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Proved: slope AB = ½ and slope BC = −2; the product is −1, so AB ⊥ BC.

GRADING NOTE Form required: the conclusion with slopes. Slope BC = (4 − 2)/(3 − 4) = −2; product −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q5/5

Write equations of parallel and perpendicular lines

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Read the slope of a line	**8.F.B.4**
BACKWARD 2	Rewrite an equation in slope-intercept form	**HSA-CED.A.4**
BACKWARD 3	Write a line from a point and a slope	**HSF-LE.A.2**
BACKWARD 4	Opposite reciprocals	**7.NS.A.2**
FORWARD 1	Write perpendicular bisectors and altitudes	**HSG-GPE.B.5**
FORWARD 2	Find the distance from a point to a line	**HSG-GPE.B.4**

SET 70 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line parallel to y = 3x − 5 that passes through (2, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q1/5

2750

QUESTION ID

S70-M-Q1

TYPE

Short answer

CORRECT ANSWER

y = 3x − 2

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line perpendicular to $y = \frac{2}{3}x + 1$ that passes through $(4, -1)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q2/5

2751

QUESTION ID

S70-M-Q2

TYPE

Short answer

CORRECT ANSWER

$y = -3/2\ x + 5$

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line parallel to 2x + 5y = 10 that passes through (5, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q3/5

2752

QUESTION ID
S70-M-Q3

TYPE
Short answer

CORRECT ANSWER
y = −2/5 x + 5

GRADING NOTE Form required: an equation; equivalent forms (2x + 5y = 25) accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line perpendicular to x = 4 that passes through (1, −3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q4/5

2753

QUESTION ID

S70-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = −3

GRADING NOTE Form required: an equation. x = 4 is vertical, so the line is horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q4/5

Which line is perpendicular to y = −4x + 1 and passes through the origin?

A. y = −4x

B. y = −¼x

C. y = ¼x

D. y = 4x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q5/5

2754

QUESTION ID

S70-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. y = ¼x

GRADING NOTE Negative reciprocal of −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 1

Read the slope of a line • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Slope of $y = -2x + 7$?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q1/5

2756

QUESTION ID

S70-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope of y = 5?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q2/5

2757

QUESTION ID

S70-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope of $y = \frac{2}{3}x - 4$?

A. 3/2

B. ⅔

C. −4

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q3/5

2758

QUESTION ID

S70-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ⅔

GRADING NOTE Coefficient of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In y = mx + b, m is the slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q4/5

2759

QUESTION ID

S70-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Slope-intercept form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slope of x = 3 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q5/5

2760

QUESTION ID

S70-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 2

Rewrite an equation in slope-intercept form • High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 5y = 10 for y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q1/5

2762

QUESTION ID

S70-B2-Q1

TYPE

Short answer

CORRECT ANSWER

y = −2/5 x + 2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3x − y = 4 for y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q2/5

2763

QUESTION ID

S70-B2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3x − 4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

6x + 3y = 9 in slope-intercept form is

A. y = 2x + 3

B. y = −6x + 9

C. y = −2x + 9

D. y = −2x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q3/5

2764

QUESTION ID

S70-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = −2x + 3

GRADING NOTE Divide by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x - 2y = 6$ has slope ½.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q4/5

2765

QUESTION ID

S70-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $y = ½x - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

4x + 2y = 8 has slope 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q5/5

2766

QUESTION ID

S70-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 3

Write a line from a point and a slope • High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the line with slope 3 through (2, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q1/5

2768

QUESTION ID

S70-B3-Q1

TYPE

Short answer

CORRECT ANSWER

y = 3x − 2

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B3-Q1**

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the line with slope −1 through (0, 6).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q2/5

2769

QUESTION ID

S70-B3-Q2

TYPE

Short answer

CORRECT ANSWER

y = −x + 6

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope 2 through (1, 5): b =

A. 5

B. 7

C. −3

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q3

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q3/5

2770

QUESTION ID

S70-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE 5 = 2 + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B3-Q3**

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q3/5

True or false?

$y - 4 = 3(x - 2)$ is the line with slope 3 through (2, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q4/5

2771

QUESTION ID

S70-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Point-slope form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q4/5

True or false?

A line through (0, 6) has y-intercept 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q5/5

2772

QUESTION ID

S70-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 4

Opposite reciprocals • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of ⅔?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q1/5

2774

QUESTION ID

S70-B4-Q1

TYPE

Short answer

CORRECT ANSWER

−3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of −5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q2/5

2775

QUESTION ID

S70-B4-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{5}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q2/5

Negative reciprocal of −2/5?

A. 5/2

B. 2/5

C. −2/5

D. −5/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q3/5

2776

QUESTION ID

S70-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5/2

GRADING NOTE Flip and change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q3/5

True or false?

The negative reciprocal of ¼ is −4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q4/5

2777

QUESTION ID

S70-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Flip and change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The negative reciprocal of 3 is −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q5/5

2778

QUESTION ID

S70-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −⅓.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q5/5

SET 70 • HSG-GPE.B.5

FORWARD 1

Write perpendicular bisectors and altitudes • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the perpendicular bisector of the segment from (2, 1) to (6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q1/5

2780

QUESTION ID

S70-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = −x + 7

GRADING NOTE Form required: an equation. Midpoint (4, 3), slope −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q1/5

Triangle ABC has A(0, 0), B(6, 0), C(2, 4). Write the equation of the altitude from A to BC.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q2/5

2781

QUESTION ID

S70-F1-Q2

TYPE

Short answer

CORRECT ANSWER

y = x

GRADING NOTE Form required: an equation. BC slope −1, so the altitude has slope 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q2/5

True or false?

A perpendicular bisector passes through the midpoint of the segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q3/5

2782

QUESTION ID

S70-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The perpendicular bisector of the segment from (0, 0) to (0, 8) is

A. x = 0

B. y = 4

C. x = 4

D. y = 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q4/5

2783

QUESTION ID

S70-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 4

GRADING NOTE Horizontal through (0, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the perpendicular bisector of the segment from (−3, 2) to (5, 2).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q5/5

2784

QUESTION ID

S70-F1-Q5

TYPE

Short answer

CORRECT ANSWER

x = 1

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSG-GPE.B.5

FORWARD 2

Find the distance from a point to a line • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from P(5, 5) to the line y = −x + 2.
Give the exact value.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q1/5

2786

QUESTION ID

S70-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

4√2 (about 5.66); the foot is (1, 1)

GRADING NOTE REQUIRED WORK: Write the perpendicular through P, find where it meets the line, then use the distance formula. Form required: exact value (a decimal to 0.01 is also accepted). Perpendicular y = x meets y = −x + 2 at (1, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (0, 0) to the line y = 4.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q2/5

2787

QUESTION ID

S70-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q2/5

True or false?

The distance from a point to a line can be measured along any segment from the point to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q3/5

2788

QUESTION ID
S70-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is measured along the perpendicular, the shortest segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q3/5

The foot of the perpendicular from (3, 0) to y = x is

A. (0, 3)

B. (0, 0)

C. (3, 3)

D. (1.5, 1.5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q4/5

2789

QUESTION ID

S70-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (1.5, 1.5)

GRADING NOTE y = −x + 3 meets y = x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (6, 1) to the line y = 2x − 1. Give the exact value.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q5/5

2790

QUESTION ID

S70-F2-Q5

TYPE

Short answer

CORRECT ANSWER

2√5 (about 4.47)

GRADING NOTE Form required: exact value (a decimal to 0.01 is also accepted). Foot (2, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q5/5

SET 71 • HSG-GPE.B.5 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Solve geometric problems with the slope criteria

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Slope from two points	**8.F.B.4**
BACKWARD 2	Solve a linear equation for an unknown	**HSA-REI.B.3**
BACKWARD 3	Equations of parallel and perpendicular lines	**HSG-GPE.B.5**
BACKWARD 4	Opposite sides of a parallelogram are parallel	**HSG-CO.C.11**
FORWARD 1	Find the orthocenter of a triangle	**HSG-GPE.B.5**
FORWARD 2	Shortest path to a straight line in a design	**HSG-MG.A.3**

SET 71 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

The line through (1, 2) and (4, k) is parallel to $y = 2x + 7$. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q1/5

2793

QUESTION ID

S71-M-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. $(k - 2)/3 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q1/5

ABCD is a parallelogram with A(0, 0), B(5, 1), C(7, 4). Find D.

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q2/5

2794

QUESTION ID

S71-M-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 3)

GRADING NOTE Form required: an ordered pair. AD ∥ BC and DC ∥ AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q2/5

The segment from (2, 1) to (5, k) is perpendicular to the line y = 3x. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q3/5

2795

QUESTION ID
S71-M-Q3

TYPE
Short answer

CORRECT ANSWER
0

GRADING NOTE Any equivalent form. (k − 1)/3 = −⅓.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(8, 4), C(4, 6). Find the foot of the altitude from C to AB.

Foot: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q4/5

2796

QUESTION ID

S71-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

(5.6, 2.8)

GRADING NOTE REQUIRED WORK: Write line AB and the perpendicular through C, then solve the system.

Form required: an ordered pair. AB: y = ½x; altitude: y = −2x + 14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q4/5

Triangle ABC has A(1, 1), B(4, 2), C(3, 5). At which vertex is the right angle?

A. C

B. B

C. A

D. There is no right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q5/5

2797

QUESTION ID

S71-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. B

GRADING NOTE Slope AB = ⅓, slope BC = −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 1

Slope from two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Slope through (1, 2) and (4, 8)?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q1/5

2799

QUESTION ID

S71-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope through (4, 2) and (3, 5)?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q2/5

2800

QUESTION ID

S71-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope through (0, 0) and (8, 4)?

A. ½

B. 8

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q3/5

2801

QUESTION ID

S71-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ½

GRADING NOTE 4/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q3/5

True or false?

The slope from A to B equals the slope from B to A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q4/5

2802

QUESTION ID
S71-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Same ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slope through (1, 1) and (1, 5) is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q5/5

2803

QUESTION ID

S71-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 2

Solve a linear equation for an unknown • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(k - 2)/3 = 2$.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q1/5

2805

QUESTION ID

S71-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-B2-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ½x = −2x + 14.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q2/5

2806

QUESTION ID

S71-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-B2-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(k - 1)/3 = -\frac{1}{3}$.

A. -2

B. 1

C. 2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q3/5

2807

QUESTION ID

S71-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE $k - 1 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q3/5

True or false?

Multiplying both sides by 3 is a valid first step in $(k - 2)/3 = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S71-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q4/5

2808

QUESTION ID

S71-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Inverse operation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S71-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The solution of $2x + 1 = 7$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q5/5

2809

QUESTION ID

S71-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 3

Equations of parallel and perpendicular lines • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Line perpendicular to y = ½x through (4, 6)?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q1/5

2811

QUESTION ID

S71-B3-Q1

TYPE

Short answer

CORRECT ANSWER

y = −2x + 14

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Line parallel to y = −x through (0, 3)?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q2/5

2812

QUESTION ID

S71-B3-Q2

TYPE

Short answer

CORRECT ANSWER

y = −x + 3

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Perpendicular slope to ⅓?

A. 3

B. −3

C. −⅓

D. ⅓

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q3/5

2813

QUESTION ID

S71-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −3

GRADING NOTE Negative reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q3/5

True or false?

An altitude is perpendicular to the side it meets.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q4/5

2814

QUESTION ID

S71-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q4/5

True or false?

An altitude passes through the midpoint of the opposite side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q5/5

2815

QUESTION ID

S71-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is a median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 4

Opposite sides of a parallelogram are parallel • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

In a parallelogram, both pairs of opposite sides are parallel and congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q1/5

2817

QUESTION ID

S71-B4-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram has A(0, 0), B(3, 0), C(4, 2). Find D (in order ABCD).

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q2/5

2818

QUESTION ID

S71-B4-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 2)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, side AB is parallel to

A. AC

B. BC

C. AD

D. DC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q3/5

2819

QUESTION ID

S71-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. DC

GRADING NOTE Opposite side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, going from B to C is "right 2, up 3". Going from A to D is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q4/5

2820

QUESTION ID

S71-B4-Q4

TYPE

Short answer

CORRECT ANSWER

Right 2, up 3

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A parallelogram's adjacent sides are always perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q5/5

2821

QUESTION ID

S71-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only rectangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q5/5

SET 71 • HSG-GPE.B.5

FORWARD 1

Find the orthocenter of a triangle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(6, 0), C(2, 4). Find the orthocenter (the intersection of the altitudes).

Orthocenter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q1/5

2823

QUESTION ID

S71-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

(2, 2)

GRADING NOTE REQUIRED WORK: Write two altitude equations and solve the system.

Form required: an ordered pair. Altitude from C: x = 2; from A: y = x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q1/5

Triangle with A(0, 0), B(4, 0), C(0, 3). Where is the orthocenter?

Orthocenter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q2/5

2824

QUESTION ID

S71-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 0)

GRADING NOTE Form required: an ordered pair. Right angle at A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q2/5

True or false?

The orthocenter of a right triangle is at the right-angle vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q3/5

2825

QUESTION ID
S71-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Two legs are altitudes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To find the orthocenter, you need at least

A. three medians

B. the circumcenter

C. one altitude

D. two altitudes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q4/5

2826

QUESTION ID

S71-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. two altitudes

GRADING NOTE Two lines meet in a point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Check that the third altitude of the triangle A(0, 0), B(6, 0), C(2, 4) also passes through (2, 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q5/5

2827

QUESTION ID
S71-F1-Q5

TYPE
Short answer

CORRECT ANSWER
AC has slope 2, so the altitude from B has slope −½: y = −½(x − 6); at x = 2, y = 2. Yes.

GRADING NOTE Any equivalent check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSG-GPE.B.5

FORWARD 2

Shortest path to a straight line in a design • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A well is at W(2, 9). A straight canal runs along $y = 2x - 1$ (units: 100 m). Where should a pipe from the well meet the canal so the pipe is as short as possible, and how long is it (to the nearest hundredth of a unit)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q1/5

2829

QUESTION ID

S71-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

At (4.4, 7.8); about 2.68 units (268 m)

GRADING NOTE REQUIRED WORK: Use the perpendicular from W to the canal.

Form required: the point and the length (rounding to 0.01 unit or 1 m). Perpendicular $y = -\frac{1}{2}x + 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pipe will connect a well to a straight canal. Why is the shortest possible pipe perpendicular to the canal?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q2/5

2830

QUESTION ID
S71-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The perpendicular segment is the shortest distance from a point to a line.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q2/5

True or false?

A pipe meeting the canal at a 60° angle is shorter than one meeting it at 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q3/5

2831

QUESTION ID

S71-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The perpendicular is shortest; the slanted pipe is a hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A path from (0, 5) to the road y = 0 is shortest at

A. (0, 0)

B. (5, 0)

C. (−5, 0)

D. (0, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q4/5

2832

QUESTION ID

S71-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (0, 0)

GRADING NOTE Straight down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A sensor at (3, 4) must connect to the line y = x. Where should the wire meet the line?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q5/5

2833

QUESTION ID

S71-F2-Q5

TYPE

Short answer

CORRECT ANSWER

(3.5, 3.5)

GRADING NOTE Form required: an ordered pair. y = −x + 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q5/5

2833

Partition a directed segment in a given ratio

MAIN	Geometry main question	**HSG-GPE.B.6**
BACKWARD 1	A ratio as a fraction of the whole	**6.RP.A.1**
BACKWARD 2	Horizontal and vertical change	**6.NS.C.8**
BACKWARD 3	Find a fraction of a number	**6.NS.A.1**
FORWARD 1	Work backward: find an endpoint or the ratio	**HSG-GPE.B.6**
FORWARD 2	The centroid as a partition point	**HSG-GPE.B.4**

SET 72 • HSG-GPE.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the point P on the directed segment from A(1, 2) to B(13, 8) that partitions it in the ratio 1 : 2.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q1/5

2836

QUESTION ID

S72-M-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 4)

GRADING NOTE Form required: an ordered pair. ⅓ of the way from A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q1/5

Find the point P on the directed segment from A(−4, 6) to B(6, 1) that partitions it in the ratio 3 : 2.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q2/5

2837

QUESTION ID

S72-M-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 3)

GRADING NOTE Form required: an ordered pair. 3/5 of the way from A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q2/5

Find the point P on the directed segment from A(2, −3) to B(−6, 9) that partitions it in the ratio 1 : 3.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q3/5

2838

QUESTION ID
S72-M-Q3

TYPE
Short answer

CORRECT ANSWER
(0, 0)

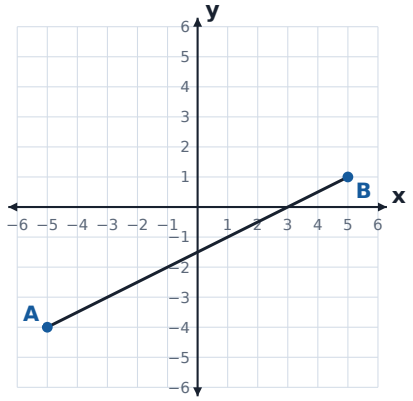
GRADING NOTE Form required: an ordered pair. ¼ of the way from A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the point P on the directed segment from A to B that partitions it in the ratio 2 : 3.



P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q4/5

2839

QUESTION ID

S72-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−1, −2)

GRADING NOTE Form required: an ordered pair. A(−5, −4), B(5, 1); 2/5 of (10, 5) is (4, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Point P partitions the directed segment from B(10, 5) to A(0, 0) in the ratio 1 : 4. P is

A. (8, 4)

B. (2, 1)

C. (5, 2.5)

D. (4, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q5/5

2840

QUESTION ID

S72-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. (8, 4)

GRADING NOTE Start at B: ⅕ of the way toward A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q5/5

2840

SET 72 • HSG-GPE.B.6

BACKWARD 1

A ratio as a fraction of the whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A ratio of 2 : 3 splits a whole into how many equal parts?

Parts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q1/5

2842

QUESTION ID

S72-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a 1 : 2 partition, the first part is what fraction of the whole?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q2/5

2843

QUESTION ID

S72-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{3}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a 3 : 2 partition, the first part is

A. 3/2 of the whole

B. 2/3 of the whole

C. 3/5 of the whole

D. 2/5 of the whole

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q3/5

2844

QUESTION ID

S72-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/5 of the whole

GRADING NOTE 3 of 5 parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q3/5

2844

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A ratio of 1 : 3 means the point is ¼ of the way along.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q4/5

2845

QUESTION ID

S72-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1 of 4 parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q4/5

True or false?

A ratio of 1 : 2 means the point is halfway.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q5/5

2846

QUESTION ID

S72-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\frac{1}{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 2

Horizontal and vertical change • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

From (1, 2) to (13, 8), what is the change in x?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q1/5

2848

QUESTION ID

S72-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

From (−4, 6) to (6, 1), what is the change in y?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q2/5

2849

QUESTION ID

S72-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−5

GRADING NOTE Any equivalent form (a decrease of 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

From (2, −3) to (−6, 9), the change in x is

A. −8

B. −4

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q3/5

2850

QUESTION ID

S72-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −8

GRADING NOTE −6 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q3/5

2850

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Change in x is the ending x minus the starting x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q4/5

2851

QUESTION ID

S72-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Direction matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q4/5

True or false?

The change in y from (0, 5) to (0, 1) is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q5/5

2852

QUESTION ID

S72-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 3

Find a fraction of a number • Grade 6 • The Number System • 6.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find ⅓ of 12.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q1/5

2854

QUESTION ID

S72-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 3/5 of −5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q2/5

2855

QUESTION ID

S72-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

2/5 of 10 is

A. 5

B. 25

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q3/5

2856

QUESTION ID

S72-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE $10 \div 5 \times 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q3/5

2856

True or false?

¼ of −8 is −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q4/5

2857

QUESTION ID

S72-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q4/5

True or false?

⅔ of 9 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q5/5

2858

QUESTION ID

S72-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q5/5

SET 72 • HSG-GPE.B.6

FORWARD 1

Work backward: find an endpoint or the ratio • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

P(3, 4) partitions the directed segment from A(1, 2) to B in the ratio 1 : 2. Find B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q1/5

2860

QUESTION ID

S72-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(7, 8)

GRADING NOTE Form required: an ordered pair. AP is ⅓ of AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(4, 3) is on the segment from A(0, 1) to B(10, 6). In what ratio does P partition AB?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q2/5

2861

QUESTION ID
S72-F1-Q2

TYPE
Short answer

CORRECT ANSWER
2 : 3

GRADING NOTE Form required: a ratio. 4/10 = 2/5 of the way.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q2/5

True or false?

The midpoint partitions a segment in the ratio $1 : 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q3/5

2862

QUESTION ID
S72-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Halfway.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P partitions AB in 3 : 1. Then P is closer to

A. B

B. neither

C. A

D. the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q4/5

2863

QUESTION ID

S72-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. B

GRADING NOTE ¾ of the way from A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

M(2, 5) is the midpoint of AB with A(−1, 1). Find B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q5/5

2864

QUESTION ID

S72-F1-Q5

TYPE

Short answer

CORRECT ANSWER

(5, 9)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSG-GPE.B.6

FORWARD 2

The centroid as a partition point • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Triangle with A(0, 0), B(6, 0), C(3, 9). Find the point ⅔ of the way from A to the midpoint of BC.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q1/5

2866

QUESTION ID
S72-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(3, 3)

GRADING NOTE Form required: an ordered pair. Midpoint (4.5, 4.5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that the centroid (the point 2 : 1 along a median from the vertex) of the triangle with vertices (x_1, y_1), (x_2, y_2), (x_3, y_3) is the average of the vertices.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q2/5

2867

QUESTION ID

S72-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

$((x_1 + x_2 + x_3)/3,$
$(y_1 + y_2 + y_3)/3)$

GRADING NOTE REQUIRED WORK: Partition the median from (x_1, y_1) to the midpoint of the opposite side in the ratio 2 : 1.

Any valid coordinate proof. Essential: midpoint $((x_2 + x_3)/2, (y_2 + y_3)/2)$; $x_1 + \frac{2}{3}((x_2 + x_3)/2 - x_1) = (x_1 + x_2 + x_3)/3$; the same for y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q2/5

True or false?

The centroid of the triangle (0, 0), (6, 0), (3, 9) is (3, 4.5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q3/5

2868

QUESTION ID
S72-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is (3, 3), the average of the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The centroid of (1, 2), (4, 8), (7, −1) is

A. (4, 3)

B. (4, 9)

C. (3, 3)

D. (12, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q4/5

2869

QUESTION ID

S72-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (4, 3)

GRADING NOTE (12/3, 9/3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the centroid of the triangle with vertices (−2, 5), (4, 1), (7, 0).

Centroid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q5/5

2870

QUESTION ID
S72-F2-Q5

TYPE
Short answer

CORRECT ANSWER
(3, 2)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q5/5

SET 73 • HSG-GPE.B.7 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Compute perimeters of polygons with coordinates

MAIN	Geometry main question	**HSG-GPE.B.7**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	Lengths of horizontal and vertical sides	**6.G.A.3**
BACKWARD 3	Simplify square roots	**HSN-RN.A.2**
FORWARD 1	Find a missing coordinate from a perimeter	**HSG-GPE.B.7**
FORWARD 2	Fencing and cost from a coordinate plan	**HSG-MG.A.3**

SET 73 • HSG-GPE.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of the triangle with vertices (0, 0), (3, 4), and (3, 0).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q1/5

2873

QUESTION ID
S73-M-Q1

TYPE
Short answer

CORRECT ANSWER
12 units

GRADING NOTE Any equivalent form. 5 + 4 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of the quadrilateral with vertices A(1, 1), B(4, 5), C(10, −3), D(7, −7).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q2/5

2874

QUESTION ID
S73-M-Q2

TYPE
Short answer

CORRECT ANSWER
30 units

GRADING NOTE Any equivalent form. 5 + 10 + 5 + 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of the triangle with vertices (0, 0), (4, 0), (2, 4). Give the exact value and a decimal to the nearest tenth.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q3/5

2875

QUESTION ID
S73-M-Q3

TYPE
Short answer

CORRECT ANSWER
$4 + 4\sqrt{5} \approx 12.9$ units

GRADING NOTE Form required: both. Two sides $\sqrt{20} = 2\sqrt{5}$.

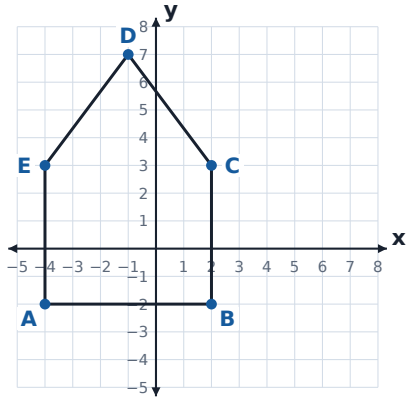
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of pentagon ABCDE.

Perimeter: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q4/5

2876

QUESTION ID

S73-M-Q4

TYPE

Short answer

CORRECT ANSWER

26 units

GRADING NOTE Any equivalent form. 6 + 5 + 5 + 5 + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q4/5

A square has vertices (0, 0), (3, 1), (2, 4), (−1, 3). Its perimeter is

A. 4√10

B. 10

C. 40

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q5/5

2877

QUESTION ID

S73-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 4√10

GRADING NOTE Side √10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (2, 3) to (−1, 7)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q1/5

2879

QUESTION ID

S73-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (4, 5) to (10, −3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q2/5

2880

QUESTION ID

S73-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (0, 0) to (2, 4)?

A. 20

B. 2√5

C. 6

D. √6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q3/5

2881

QUESTION ID

S73-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2√5

GRADING NOTE √20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q3/5

True or false?

The distance from (0, 0) to (3, 1) is √10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q4/5

2882

QUESTION ID

S73-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 9 + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q4/5

True or false?

Distances can be negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q5/5

2883

QUESTION ID

S73-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Never.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 2

Lengths of horizontal and vertical sides • Grade 6 • Geometry • 6.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Length of the side from (−4, −2) to (2, −2)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q1/5

2885

QUESTION ID

S73-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Length of the side from (2, −2) to (2, 3)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q2/5

2886

QUESTION ID

S73-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q2/5

Length from (−3, 4) to (−3, −1)?

A. 7

B. 3

C. 5

D. −5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q3/5

2887

QUESTION ID

S73-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5

GRADING NOTE $|4 - (-1)|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q3/5

True or false?

A horizontal side's length is the absolute value of the difference of the x-coordinates of its endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S73-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q4/5

2888

QUESTION ID

S73-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $|x_2 - x_1|$; a length is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S73-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q4/5

True or false?

The side from (1, 1) to (1, 9) has length 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q5/5

2889

QUESTION ID

S73-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 3

Simplify square roots • High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Simplify √20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q1/5

2891

QUESTION ID
S73-B3-Q1

TYPE
Short answer

CORRECT ANSWER
2√5

GRADING NOTE Form required: simplest radical form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-B3-Q1**

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify √50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q2/5

2892

QUESTION ID
S73-B3-Q2

TYPE
Short answer

CORRECT ANSWER
5√2

GRADING NOTE Form required: simplest radical form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-B3-Q2**

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$2\sqrt{5} + 2\sqrt{5} =$

A. $2\sqrt{10}$

B. $8\sqrt{5}$

C. $4\sqrt{10}$

D. $4\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q3/5

2893

QUESTION ID
S73-B3-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. $4\sqrt{5}$

GRADING NOTE Like terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{10} \approx 3.16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q4/5

2894

QUESTION ID

S73-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Calculator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{9} + \sqrt{16} = \sqrt{25}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q5/5

2895

QUESTION ID

S73-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3 + 4 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q5/5

SET 73 • HSG-GPE.B.7

FORWARD 1

Find a missing coordinate from a perimeter • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has vertices (0, 0), (k, 0), (k, 3), (0, 3) with k > 0 and perimeter 22. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q1/5

2897

QUESTION ID

S73-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. 2k + 6 = 22.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle with vertices (0, 0), (6, 0), (3, h), h > 0, has perimeter 16. Find h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q2/5

2898

QUESTION ID

S73-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. Each slanted side is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The triangle with vertices (0, 0), (6, 0), (3, h) has the same perimeter for h = 4 and for h = −4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q3/5

2899

QUESTION ID

S73-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both give slanted sides of 5 (perimeter 16); a condition such as h > 0 is needed to pick one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A square has vertices (0, 0), (a, 0), (a, a), (0, a) with a > 0 and perimeter 18. Find a.

A. 4

B. 3

C. 9

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q4/5

2900

QUESTION ID

S73-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 4.5

GRADING NOTE 4a = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has vertices (0, 0), (8, 0), (8, k) with k > 0 and perimeter 24. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q5/5

2901

QUESTION ID

S73-F1-Q5

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. $8 + k + \sqrt{(64 + k^2)} = 24$, so $\sqrt{(64 + k^2)} = 16 - k$; $64 + k^2 = 256 - 32k + k^2$; $k = 6$ (check: $8 + 6 + 10 = 24$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSG-GPE.B.7

FORWARD 2

Fencing and cost from a coordinate plan • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A garden plan has corners (0, 0), (8, 0), (8, 6), (0, 6), in meters. Fencing costs $12 per meter. Find the cost to fence it.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q1/5

2903

QUESTION ID
S73-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$336

GRADING NOTE Any equivalent form. 28 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangular lot has corners (0, 0), (30, 0), (30, 40) in feet. Edging costs $2.50 per foot. Find the total cost.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q2/5

2904

QUESTION ID
S73-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$300

GRADING NOTE Any equivalent form. 120 ft.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q2/5

True or false?

On a map with 1 unit = 5 m, an area of 12 square units is 60 m².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q3/5

2905

QUESTION ID

S73-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Areas scale by 5²: 12 × 25 = 300 m².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A path has corners (0, 0), (3, 4), (9, 12) (units: 10 m). Its length is

A. 150 m

B. 210 m

C. 15 m

D. 21 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q4/5

2906

QUESTION ID

S73-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 150 m

GRADING NOTE 5 + 10 = 15 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A run around a lot with corners (0, 0), (5, 12), (5, 0) in units of 100 m: how many meters?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q5/5

2907

QUESTION ID
S73-F2-Q5

TYPE
Short answer

CORRECT ANSWER
3000 m

GRADING NOTE Any equivalent form. 13 + 12 + 5 = 30 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q5/5

SET 74 • HSG-GPE.B.7 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Compute areas of triangles and rectangles with coordinates

MAIN	Geometry main question	**HSG-GPE.B.7**
BACKWARD 1	Area of a triangle from base and height	**6.G.A.1**
BACKWARD 2	Lengths on the coordinate plane	**6.G.A.3**
BACKWARD 3	Distance between two points	**8.G.B.8**
FORWARD 1	Find a missing vertex from an area	**HSG-GPE.B.7**
FORWARD 2	Use area from coordinates with density	**HSG-MG.A.2**

SET 74 • HSG-GPE.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of the triangle with vertices (0, 0), (8, 0), and (3, 5).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q1/5

2910

QUESTION ID
S74-M-Q1

TYPE
Short answer

CORRECT ANSWER
20 square units

GRADING NOTE Any equivalent form. Base 8, height 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rectangle ABCD has A(1, 1), B(5, 3), C(4, 5), D(0, 3). Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q2/5

2911

QUESTION ID
S74-M-Q2

TYPE
Short answer

CORRECT ANSWER
10 square units

GRADING NOTE Any equivalent form. AB = √20, BC = √5.

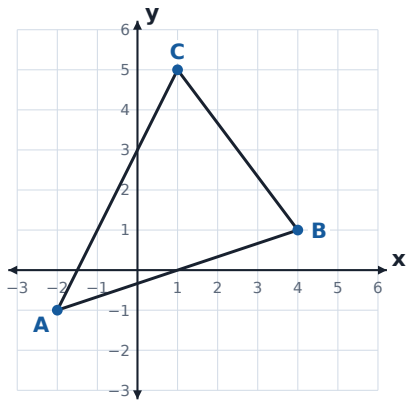
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of triangle ABC.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q3/5

2912

QUESTION ID
S74-M-Q3

TYPE
Short answer

CORRECT ANSWER
15 square units

GRADING NOTE Any equivalent form. Bounding rectangle 36 minus 6, 6, and 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of the triangle with vertices A(0, 0), B(6, 2), C(2, 6).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q4/5

2913

QUESTION ID

S74-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

16 square units

GRADING NOTE REQUIRED WORK: Enclose the triangle in a rectangle and subtract the three right triangles.

Any equivalent form. 36 − 6 − 8 − 6. Alternative methods (shoelace) accepted with work.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q4/5

A triangle has vertices (0, 0), (10, 0), and (x, 4). What is its area?

A. 20, for any x

B. It depends on x

C. 40

D. 20 only if x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q5/5

2914

QUESTION ID

S74-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 20, for any x

GRADING NOTE Base 10, height 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q5/5

SET 74 • HSG-GPE.B.7

BACKWARD 1

Area of a triangle from base and height • Grade 6 • Geometry • 6.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Area of a triangle with base 8 and height 5?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q1/5

2916

QUESTION ID

S74-B1-Q1

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Area of a right triangle with legs 6 and 2?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q2/5

2917

QUESTION ID

S74-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A =

A. ½bh

B. b + h

C. 2bh

D. bh

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q3/5

2918

QUESTION ID

S74-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ½bh

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q3/5

True or false?

The height of a triangle is perpendicular to the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q4/5

2919

QUESTION ID
S74-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The height must be one of the sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q5/5

2920

QUESTION ID

S74-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only in right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q5/5

SET 74 • HSG-GPE.B.7

BACKWARD 2

Lengths on the coordinate plane • Grade 6 • Geometry • 6.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Length from (−2, −1) to (4, −1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q1/5

2922

QUESTION ID

S74-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Height of the point (3, 5) above the x-axis?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q2/5

2923

QUESTION ID

S74-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q2/5

Width of the rectangle enclosing x from −2 to 4?

A. 2

B. 8

C. 4

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q3/5

2924

QUESTION ID

S74-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6

GRADING NOTE 4 − (−2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q3/5

True or false?

A point with y-coordinate −3 is 3 units from the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q4/5

2925

QUESTION ID

S74-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Absolute value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q4/5

True or false?

The length from (0, 2) to (0, 9) is 11.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q5/5

2926

QUESTION ID

S74-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSG-GPE.B.7

BACKWARD 3

Distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (1, 1) to (5, 3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q1/5

2928

QUESTION ID

S74-B3-Q1

TYPE

Short answer

CORRECT ANSWER

√20

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (5, 3) to (4, 5)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q2/5

2929

QUESTION ID

S74-B3-Q2

TYPE

Short answer

CORRECT ANSWER

√5

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sqrt{20} \cdot \sqrt{5} =$

A. 10

B. 25

C. $\sqrt{25}$

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q3/5

2930

QUESTION ID

S74-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE $\sqrt{100}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q3/5

True or false?

Slanted rectangle sides need the distance formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q4/5

2931

QUESTION ID

S74-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Not grid-aligned.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{20} \cdot \sqrt{5} = 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q5/5

2932

QUESTION ID

S74-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q5/5

SET 74 • HSG-GPE.B.7

FORWARD 1

Find a missing vertex from an area • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Triangle with vertices (0, 0), (6, 0), (2, k) has area 18.
Find all possible k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q1/5

2934

QUESTION ID
S74-F1-Q1

TYPE
Short answer

CORRECT ANSWER
6 or −6

GRADING NOTE Form required: both values. ½ · 6 · |k| = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has vertices (0, 0), (a, 0), (a, 5), (0, 5), $a > 0$, and area 35. Find a.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q2/5

2935

QUESTION ID

S74-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q2/5

True or false?

Moving the top vertex of a triangle parallel to its base does not change its area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q3/5

2936

QUESTION ID

S74-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q3/5

Triangle (0, 0), (4, 0), (x, 6). For which x is the area 12?

A. Any x

B. No x

C. x = 4 only

D. x = 2 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q4/5

2937

QUESTION ID

S74-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Any x

GRADING NOTE ½ · 4 · 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle (1, 1), (7, 1), (4, k) has area 21 and k > 1. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q5/5

2938

QUESTION ID

S74-F1-Q5

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. Base 6, height 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q5/5

SET 74 • HSG-GPE.B.7

FORWARD 2

Use area from coordinates with density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A park on a map has corners (0, 0), (4, 0), (4, 3), (0, 3), with 1 unit = 1 km. 1,800 people live there. Find the population density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1
High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q1/5

2940

QUESTION ID
S74-F2-Q1

TYPE
Short answer

CORRECT ANSWER
150 people per km²

GRADING NOTE Any equivalent form. Area 12 km².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1
High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangular field has vertices (0, 0), (200, 0), (0, 150) in meters. Seeds are planted at 4 per m². How many seeds?

Seeds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q2/5

2941

QUESTION ID
S74-F2-Q2

TYPE
Short answer

CORRECT ANSWER
60,000

GRADING NOTE Any equivalent form. Area 15,000 m².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q2/5

True or false?

Density based on area is an area divided by a quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q3/5

2942

QUESTION ID

S74-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is a quantity divided by an area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A lot with corners (0, 0), (6, 0), (6, 5), (0, 5) (meters) holds 60 plants. Density?

A. 2 plants per m²

B. 30 plants per m²

C. 11 plants per m²

D. 0.5 plants per m²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q4/5

2943

QUESTION ID

S74-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2 plants per m²

GRADING NOTE 60/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle with corners (0, 0), (3, 0), (3, 2), (0, 2) (miles) has 4,200 residents. Find the density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q5/5

2944

QUESTION ID
S74-F2-Q5

TYPE
Short answer

CORRECT ANSWER
700 people per square mile

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q5/5

SET 75 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the circumference of a circle

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Circumference formulas	**7.G.B.4**
BACKWARD 4	Perimeters of regular polygons	**3.MD.D.8**
BACKWARD 3	All circles are similar	**HSG-C.A.1**
FORWARD 1	Trap π between inscribed and circumscribed polygons	**HSG-GMD.A.1**
FORWARD 2	Extend the circumference argument to arcs and curved boundari…	**HSG-GMD.A.1**

SET 75 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Regular hexagons are inscribed in and circumscribed about a circle of radius 1. Their perimeters are 6 and $4\sqrt{3} \approx 6.93$. Explain why this shows that $3 < \pi < 2\sqrt{3} \approx 3.46$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q1/5

2947

QUESTION ID

S75-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

The circle lies between the two hexagons, so $6 < C < 4\sqrt{3}$; with $C = 2\pi$ this gives $3 < \pi < 2\sqrt{3} \approx 3.46$.

GRADING NOTE REQUIRED WORK: Compare the circumference 2π with the two perimeters, then divide by 2.

Any valid explanation. Essential: the inscribed hexagon is six equilateral triangles of side 1 (perimeter 6); the circumference is between the inner and outer perimeters; $C = 2\pi \cdot 1$; divide by 2. Invalid: quoting $\pi \approx 3.14$ as the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the ratio of circumference to diameter is the same number (π) for every circle, so that $C = \pi d = 2\pi r$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q2/5

2948

QUESTION ID

S75-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Every circle is a dilation of every other, so C and d scale by the same factor and C/d is constant; that constant is called π.

GRADING NOTE

REQUIRED WORK: Use the fact that all circles are similar.

Any valid explanation. Essential: circles are similar (dilation); lengths scale by the same factor k; $C_2/d_2 = kC_1/kd_1 = C_1/d_1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A wheel with diameter 70 cm rolls one full turn along level ground without slipping, and a student measures the distance traveled as 219.9 cm. Explain why that distance is the circumference, use the measurement to estimate π, and say why a measurement supports the formula but does not prove it.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q3/5

2949

QUESTION ID

S75-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Each point of the rim touches the ground once in one turn, so the distance equals the circumference; 219.9 ÷ 70 ≈ 3.141, close to π. A measurement has error and covers one wheel only.

GRADING NOTE REQUIRED WORK: Relate each point of the rim to the ground it touches; divide distance by diameter.

Any valid explanation. Essential: rolling without slipping lays the rim along the ground; C/d ≈ 3.141; evidence versus proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A regular hexagon is inscribed in a circle of radius 5. Find the hexagon's perimeter, compare it with the circle's circumference, and explain what happens as the number of sides increases.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q4/5

2950

QUESTION ID

S75-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

Perimeter 30 < 10π ≈ 31.4; with more sides the perimeter gets closer to the circumference.

GRADING NOTE REQUIRED WORK: Split the hexagon into six triangles from the center; compare 30 with 10π; describe the trend.

Any valid explanation. Essential: six equilateral triangles of side 5 (central angles 60°, two radii), so perimeter 30; 30 < 10π; more sides bring the perimeter closer to the circumference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Regular polygons with 6, 12 and 24 sides are inscribed in a circle of radius 1. Their perimeters are 6, 6.212 and 6.265. Explain what these values suggest about the circumference and how the formula C = 2πr follows for a circle of any radius r.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q5/5

2951

QUESTION ID

S75-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The perimeters increase toward 2π ≈ 6.283, the circumference of the unit circle; a circle of radius r is the unit circle dilated by r, so its circumference is 2π · r.

GRADING NOTE REQUIRED WORK: Describe the trend of the perimeters, then use similarity (scaling by r).

Any valid explanation. Essential: the inscribed perimeters approach the circumference from below; the limit for radius 1 is 2π; dilation by r multiplies lengths by r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 1

Circumference formulas • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact circumference of a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q1/5

2953

QUESTION ID

S75-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact circumference of a circle of diameter 9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q2/5

2954

QUESTION ID

S75-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9π

GRADING NOTE Form required: exact value. C = πd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Half the circumference of a circle of radius r is

A. πr^2

B. πr

C. $2\pi r$

D. r/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q3/5

2955

QUESTION ID

S75-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. πr

GRADING NOTE $\frac{1}{2} \cdot 2\pi r$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$C = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q4/5

2956

QUESTION ID

S75-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q4/5

True or false?

$A = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q5/5

2957

QUESTION ID

S75-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $A = \pi r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 4

Perimeters of regular polygons • Grade 3 • Measurement & Data • 3.MD.D.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of a regular hexagon with side 5.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q1/5

2959

QUESTION ID

S75-B4-Q1

TYPE

Short answer

CORRECT ANSWER

30

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of a regular octagon with side 2.5 cm.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q2/5

2960

QUESTION ID

S75-B4-Q2

TYPE

Short answer

CORRECT ANSWER

20 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A regular 12-gon has perimeter 36. Each side is

A. 432

B. 3

C. 12

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q3/5

2961

QUESTION ID

S75-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3

GRADING NOTE 36 ÷ 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q3/5

True or false?

A regular polygon with n sides of length s has perimeter ns.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q4/5

2962

QUESTION ID

S75-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q4/5

True or false?

A regular pentagon with side 4 has perimeter 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q5/5

2963

QUESTION ID

S75-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5 \times 4 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 3

All circles are similar • High School — Geometry • Geometry: Circles • HSG-C.A.1

True or false?

Any circle can be mapped to any other circle by a translation and a dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q1/5

2965

QUESTION ID

S75-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle of radius 2 is dilated to radius 6. By what factor is the circumference multiplied?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q2/5

2966

QUESTION ID

S75-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Circles of radii 3 and 12 have circumferences in the ratio

A. 1 : 16

B. 3 : 4

C. 1 : 4

D. 1 : 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q3/5

2967

QUESTION ID

S75-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1 : 4

GRADING NOTE Scale factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why does C/d not depend on the size of the circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q4/5

2968

QUESTION ID

S75-B3-Q4

TYPE

Short answer

CORRECT ANSWER

Both C and d are multiplied by the same scale factor.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A larger circle has a larger value of π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q5/5

2969

QUESTION ID

S75-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE π is constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q5/5

SET 75 • HSG-GMD.A.1

FORWARD 1

Trap π between inscribed and circumscribed polygons • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A circle has radius 1. Find the perimeters of the inscribed square and the circumscribed square, and use them to write bounds for π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q1/5

2971

QUESTION ID

S75-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Inscribed $4\sqrt{2} \approx 5.66$, circumscribed 8; $5.66 < 2\pi < 8$, so $2.83 < \pi < 4$.

GRADING NOTE Form required: both perimeters and the bounds. Inscribed side √2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed and circumscribed regular 12-gons of a circle of radius 1 have perimeters about 6.212 and 6.431. What bounds for π do they give (to the nearest hundredth)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q2/5

2972

QUESTION ID

S75-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$3.11 < \pi < 3.22$

GRADING NOTE Form required: both bounds. Divide each perimeter by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q2/5

True or false?

Using polygons with more sides gives closer bounds for π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q3/5

2973

QUESTION ID

S75-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both perimeters approach the circumference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q3/5

Why is the circumscribed polygon's perimeter larger than the circumference?

A. It has fewer sides

B. The circle lies inside it, and a curve inside a convex shape is shorter than the shape's boundary

C. Its radius is smaller

D. Its sides are curved

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q4/5

2974

QUESTION ID

S75-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. The circle lies inside it, and a curve inside a convex shape is shorter than the shape's boundary

GRADING NOTE The circle is enclosed by the polygon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q4/5

Archimedes used 96-gons and found $3\ 10/71 < \pi < 3\ 1/7$. Write these bounds as decimals to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q5/5

2975

QUESTION ID

S75-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$3.141 < \pi < 3.143$

GRADING NOTE Form required: both decimals. $3\ 10/71 \approx 3.1408$ and $3\ 1/7 \approx 3.1429$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q5/5

SET 75 • HSG-GMD.A.1

FORWARD 2

Extend the circumference argument to arcs and curved boundaries • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Use the circumference formula to explain why an arc with central angle 90° in a circle of radius 6 has length 3π, and find its length to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q1/5

2977

QUESTION ID

S75-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The arc is 90/360 = ¼ of the circle, so its length is ¼ · 2π · 6 = 3π ≈ 9.4.

GRADING NOTE Form required: the fraction-of-the-circle reasoning and the value (rounding to 0.1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the length of an arc is proportional to its central angle, so that an arc of θ° has length (θ/360) · 2πr.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q2/5

2978

QUESTION ID

S75-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Equal central angles cut off congruent arcs (a rotation about the center maps one onto another), so the 360° of the circle split the circumference 2πr in proportion to the angle.

GRADING NOTE Any equivalent explanation using equal angles ↔ equal arcs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q2/5

True or false?

A semicircular region of radius 5 has perimeter 5π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q3/5

2979

QUESTION ID

S75-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The perimeter also includes the diameter: 5π + 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q3/5

2979

ANSWER KEY Reduced copy of the question page, with its answer

A running track has two straight sides of 100 m and two semicircular ends of radius 30 m. Its length is about

A. 294 m

B. 200 m

C. 388π m

D. 388 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q4/5

2980

QUESTION ID
S75-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 388 m

GRADING NOTE 200 + 60π ≈ 388.5 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The circumference argument used inscribed polygons whose perimeters approach 2πr. Explain why inscribed broken lines (chords) can be used the same way to measure the length of an arc.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q5/5

2981

QUESTION ID

S75-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Chords that split the arc into smaller and smaller pieces form broken lines whose lengths increase toward the arc length, just as the polygon perimeters approach the circumference.

GRADING NOTE Any equivalent explanation of the limit of chord lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q5/5

SET 103 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the area of a circle

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	Area of a parallelogram	**6.G.A.1**
BACKWARD 3	Scale factors and area	**7.G.A.1**
FORWARD 1	Area of a circle as a limit of regular polygons	**HSG-GMD.A.1**
FORWARD 2	From area to volume: stacking disks	**HSG-GMD.A.1**

SET 103 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A circle of radius r is cut into many equal sectors, which are rearranged as shown. Explain why this shows that the area of the circle is πr^2.

Work space

Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q1/5

2984

QUESTION ID

S103-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Area = (half the circumference) × r = $\pi r \cdot r = \pi r^2$.

GRADING NOTE REQUIRED WORK: Describe the base and height of the rearranged shape and what happens as the number of sectors grows.

Any valid explanation. Essential: the shape approaches a parallelogram (rectangle); its height approaches r; its base is half the circumference, πr; area = $\pi r \cdot r = \pi r^2$. Invalid: "the formula says so" or measuring one drawing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle of radius r is cut into many thin rings, like the layers of an onion. Each ring is cut open and straightened into a thin strip, and the strips are stacked from shortest (at the top) to longest (at the bottom). Explain why the stack approaches a triangle and why its area is πr^2.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q2/5

2985

QUESTION ID

S103-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

The strips have lengths 2π · (distance from the center), from about 0 to 2πr, and the stack is r tall, so it approaches a triangle with base 2πr and height r: ½ · 2πr · r = πr².

GRADING NOTE REQUIRED WORK: Find the lengths of the shortest and longest strips and the height of the stack.

Any valid explanation. Essential: strip length grows in proportion to the distance from the center; outer strip 2πr; height r; triangle area ½bh = πr².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use similarity to explain why the area of a circle of radius r equals r^2 times the area of a circle of radius 1. Then use the fact that the unit circle has area π to write the formula.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q3/5

2986

QUESTION ID

S103-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

A dilation by r multiplies every area by r^2, so A = r^2 · (area of the unit circle) = πr^2.

GRADING NOTE REQUIRED WORK: A circle of radius r is a dilation of the unit circle by factor r; describe what a dilation does to areas. Any valid explanation. Essential: circles are similar; areas scale by k^2; the unit circle's area is π; A = πr^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A quarter circle of radius 10 is drawn on a grid of unit squares. Counting full squares and estimating the partial ones gives about 78.5 squares. Explain how this supports $A = \pi r^2$ and why counting is evidence rather than proof.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q4/5

2987

QUESTION ID

S103-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The whole circle covers about $4 \times 78.5 = 314$ squares, and $\pi \cdot 10^2 \approx 314.2$, which agrees. Counting is approximate and checks one circle only, so it supports but does not prove the formula.

GRADING NOTE REQUIRED WORK: Compare 4×78.5 with πr^2 for $r = 10$.

Any valid explanation. Essential: the scaled-up count (about 314); comparison with 100π; evidence versus proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the area of a circle of radius r is more than $2r^2$ and less than $4r^2$, using an inscribed square and a circumscribed square, and check that πr^2 fits these bounds.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q5/5

2988

QUESTION ID

S103-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The circumscribed square has side 2r (area $4r^2$); the inscribed square has diagonal 2r (area $2r^2$); the circle lies between them, and $2r^2 < \pi r^2 \approx 3.14r^2 < 4r^2$.

GRADING NOTE

REQUIRED WORK: Find the area of each square in terms of r.

Any valid explanation. Essential: both square areas from r (diagonal 2r gives area $\frac{1}{2}(2r)^2$); the circle is between them; $2 < \pi < 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q5/5

SET 103 • HSG-GMD.A.1

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact area of a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q1/5

2990

QUESTION ID

S103-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q1/5

2990

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact area of a circle of diameter 12.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q2/5

2991

QUESTION ID

S103-B1-Q2

TYPE

Short answer

CORRECT ANSWER

36π

GRADING NOTE Form required: exact value. r = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has area 25π. Its radius is

A. 5

B. 12.5

C. 25

D. √5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q3/5

2992

QUESTION ID

S103-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

GRADING NOTE $r^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q3/5

True or false?

A circle of radius 3 has area 9π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q4/5

2993

QUESTION ID

S103-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\pi \cdot 3^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q4/5

True or false?

Tripling the radius triples the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q5/5

2994

QUESTION ID

S103-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The area is multiplied by 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q5/5

SET 103 • HSG-GMD.A.1

BACKWARD 2

Area of a parallelogram • Grade 6 • Geometry • 6.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Area of a parallelogram with base 9 and height 4?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q1/5

2996

QUESTION ID

S103-B2-Q1

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q1/5

True or false?

A parallelogram has the same area as a rectangle with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q2/5

2997

QUESTION ID

S103-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cut and slide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram's area is

A. base × slanted side

B. ½ base × height

C. base × height

D. base + height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q3/5

2998

QUESTION ID

S103-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. base × height

GRADING NOTE Perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram has base πr and height r. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q4/5

2999

QUESTION ID

S103-B2-Q4

TYPE

Short answer

CORRECT ANSWER

πr^2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slanted side is the height of a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q5/5

3000

QUESTION ID

S103-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q5/5

SET 103 • HSG-GMD.A.1

BACKWARD 3

Scale factors and area • Grade 7 • Geometry • 7.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A scale drawing uses factor 3. A region of 5 cm² on the original covers how much on the drawing?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q1/5

3002

QUESTION ID

S103-B3-Q1

TYPE

Short answer

CORRECT ANSWER

45 cm²

GRADING NOTE Any equivalent form. $3^2 \cdot 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A square of side 2 is enlarged by factor 4. New area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q2/5

3003

QUESTION ID

S103-B3-Q2

TYPE

Short answer

CORRECT ANSWER

64

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Enlarging a figure by factor k multiplies its area by

A. k^3

B. k

C. 2k

D. k^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q3/5

3004

QUESTION ID

S103-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. k^2

GRADING NOTE Both dimensions scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q3/5

True or false?

Halving every length of a figure multiplies its area by ¼.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q4/5

3005

QUESTION ID

S103-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(½)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q4/5

True or false?

Enlarging by factor 3 triples the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B3-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q5/5

3006

QUESTION ID

S103-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9 times.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B3-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q5/5

SET 103 • HSG-GMD.A.1

FORWARD 1

Area of a circle as a limit of regular polygons • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A regular n-gon is inscribed in a circle of radius r. Its area is ½ · perimeter · apothem. Explain why this approaches πr² as n increases.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q1/5

3008

QUESTION ID

S103-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

The perimeter approaches 2πr and the apothem approaches r, so the area approaches ½ · 2πr · r = πr².

GRADING NOTE REQUIRED WORK: Describe what the perimeter and apothem approach.

Any valid explanation. Essential: perimeter → circumference; apothem → radius; substitution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of a regular hexagon with side 2 and apothem √3. Compare with the area of its circumscribed circle (radius 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q2/5

3009

QUESTION ID

S103-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$6\sqrt{3} \approx 10.4 < 4\pi \approx 12.6$

GRADING NOTE Form required: both numbers. ½ · 12 · √3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S103-F1-Q2**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q2/5

True or false?

An inscribed polygon always has less area than its circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q3/5

3010

QUESTION ID

S103-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It lies inside the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q3/5

3010

ANSWER KEY Reduced copy of the question page, with its answer

As n grows, the apothem of an inscribed regular n-gon approaches

A. 0

B. 2r

C. r

D. πr

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q4/5

3011

QUESTION ID

S103-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. r

GRADING NOTE The sides approach the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why do mathematicians call this an "informal" argument?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q5/5

3012

QUESTION ID

S103-F1-Q5

TYPE

Short answer

CORRECT ANSWER

It relies on what the shapes approach (a limit) without a formal proof of the limit.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q5/5

SET 103 • HSG-GMD.A.1

FORWARD 2

From area to volume: stacking disks • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A stack of 40 identical coins, each a disk of radius 1 cm and thickness 0.2 cm, forms a cylinder. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q1/5

3014

QUESTION ID

S103-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8π cm^3

GRADING NOTE Form required: exact value. $\pi \cdot 1^2 \cdot 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q1/5

True or false?

A cylinder can be thought of as a stack of congruent disks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q2/5

3015

QUESTION ID

S103-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Layers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q2/5

The volume of a stack of disks of area πr^2 and total height h is

A. $\pi r^2 h$

B. $2\pi rh$

C. πr^2

D. $\frac{1}{3}\pi r^2 h$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q3/5

3016

QUESTION ID

S103-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $\pi r^2 h$

GRADING NOTE Base area × height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how the area of a circle is used to find the volume of a cylinder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q4/5

3017

QUESTION ID

S103-F2-Q4

TYPE

Short answer

CORRECT ANSWER

The volume is the base area πr^2 times the height h.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Each disk in a stack has area 12 cm² and the stack is 5 cm high. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q5/5

3018

QUESTION ID
S103-F2-Q5

TYPE
Short answer

CORRECT ANSWER
60 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q5/5

SET 76 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal argument for the volume of a cylinder

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Volume of a rectangular prism as layers	**5.MD.C.5**
BACKWARD 2	Area of a circle	**7.G.B.4**
BACKWARD 3	Cross-sections parallel to the base	**7.G.A.3**
FORWARD 1	Cavalieri's principle with different shapes	**HSG-GMD.A.1**
FORWARD 2	Use the cylinder volume formula to solve problems	**HSG-GMD.A.3**

SET 76 • HSG-GMD.A.1

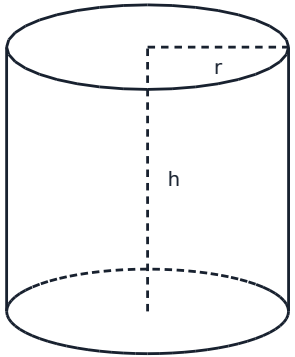
MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the volume of the cylinder shown is $V = \pi r^2 h$.

Work space



Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q1/5

3021

QUESTION ID

S76-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Each layer is a disk of area πr^2; stacking layers to height h gives volume (base area) × (height) = $\pi r^2 h$.

GRADING NOTE

REQUIRED WORK: Think of the cylinder as a stack of thin congruent disks.

Any valid explanation. Essential: congruent circular cross-sections of area πr^2; volume = base area × height, as for a prism (layers of unit thickness); result $\pi r^2 h$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A right cylinder and an oblique cylinder have the same radius r and the same height h. Explain why they have the same volume.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q2/5

3022

QUESTION ID

S76-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

At every height both cross-sections are disks of area πr², so by Cavalieri's principle their volumes are equal: πr²h.

GRADING NOTE

REQUIRED WORK: Use Cavalieri's principle.

Any valid explanation. Essential: equal heights; equal cross-sectional areas at every level; Cavalieri's principle cited. Invalid: "they look the same size".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder has radius r and height h. A square prism has base area πr^2 and the same height h. Use Cavalieri's principle to explain why the two solids have the same volume, and conclude the cylinder formula.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q3/5

3023

QUESTION ID

S76-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Every horizontal cross-section of the cylinder is a disk of area πr^2, and every cross-section of the prism has area πr^2; equal heights and equal cross-section areas give equal volumes, so $V = (\pi r^2)h$.

GRADING NOTE REQUIRED WORK: Compare the cross-section areas at every height.

Any valid explanation. Essential: both cross-section areas named and equal at every level; equal heights; Cavalieri's principle; prism volume Bh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why V = Bh works for both prisms and cylinders.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q4/5

3024

QUESTION ID

S76-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

Both have congruent cross-sections of area B at every height, so the volume is B times the height.

GRADING NOTE

REQUIRED WORK: Describe the cross-sections of each solid and how the volume is built from them.

Any valid explanation. Essential: congruent cross-sections parallel to the base; layers of thickness 1 (or thin layers) each of volume B; h layers give Bh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder of radius r and height h is approximated by prisms whose bases are regular polygons inscribed in its base circle. Explain why the prism volumes approach $\pi r^2 h$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q5/5

3025

QUESTION ID

S76-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

Each prism has volume (polygon area) × h; as the number of sides grows, the polygon area approaches πr^2, so the volumes approach $\pi r^2 h$.

GRADING NOTE REQUIRED WORK: Describe what the polygon base areas approach as the number of sides grows.

Any valid explanation. Essential: prism volume Bh; inscribed polygon areas approach the circle area; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 1

Volume of a rectangular prism as layers • Grade 5 • Measurement & Data • 5.MD.C.5

ANSWER KEY Reduced copy of the question page, with its answer

A prism has a 4 × 3 base and height 5. How many unit cubes are in one layer, and what is the volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q1/5

3027

QUESTION ID

S76-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 per layer; 60

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q1/5

True or false?

V = Bh, where B is the area of the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q2/5

3028

QUESTION ID

S76-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Layers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A prism has base area 20 and height 7. Volume?

A. 27

B. 140

C. 200

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q3/5

3029

QUESTION ID

S76-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 140

GRADING NOTE B × h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q3/5

3029

ANSWER KEY Reduced copy of the question page, with its answer

A prism has 6 layers of 10 cubes each. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q4/5

3030

QUESTION ID

S76-B1-Q4

TYPE

Short answer

CORRECT ANSWER

60 cubic units

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q4/5

True or false?

Volume is measured in square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q5/5

3031

QUESTION ID

S76-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Cubic units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 2

Area of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle of radius 3?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q1/5

3033

QUESTION ID

S76-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle of diameter 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q2/5

3034

QUESTION ID

S76-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A =

A. $2\pi r$

B. πr

C. πr^2

D. πd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q3/5

3035

QUESTION ID

S76-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. πr^2

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q3/5

True or false?

The base of a cylinder is a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q4/5

3036

QUESTION ID

S76-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q4/5

True or false?

The area of a circle of radius 2 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q5/5

3037

QUESTION ID

S76-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 3

Cross-sections parallel to the base • Grade 7 • Geometry • 7.G.A.3

True or false?

A slice of a right prism parallel to its base is congruent to the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q1/5

3039

QUESTION ID

S76-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same shape and size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder is sliced parallel to its base. Describe the cross-section.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q2/5

3040

QUESTION ID

S76-B3-Q2

TYPE

Short answer

CORRECT ANSWER

A circle congruent to the base

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q2/5

A rectangular prism sliced parallel to its 3 × 4 base gives a

A. smaller rectangle

B. 3 × 4 rectangle

C. triangle

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q3/5

3041

QUESTION ID

S76-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3 × 4 rectangle

GRADING NOTE Congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why do all horizontal slices of a cylinder have the same area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q4/5

3042

QUESTION ID

S76-B3-Q4

TYPE

Short answer

CORRECT ANSWER

They are all congruent to the base.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q4/5

True or false?

Horizontal slices of a cylinder get smaller toward the top.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q5/5

3043

QUESTION ID

S76-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is a cone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q5/5

SET 76 • HSG-GMD.A.1

FORWARD 1

Cavalieri's principle with different shapes • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder with base area 50 cm² and a square prism with base area 50 cm² both have height 8 cm. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q1/5

3045

QUESTION ID

S76-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Equal: 400 cm³ each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An oblique cylinder has radius 3 and vertical height 7 (its slanted side is 9). Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q2/5

3046

QUESTION ID

S76-F1-Q2

TYPE

Short answer

CORRECT ANSWER

63π

GRADING NOTE Form required: exact value. Use the vertical height, not the slant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q2/5

True or false?

Cavalieri's principle requires equal cross-sectional areas at every height and equal heights.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q3/5

3047

QUESTION ID

S76-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two stacks of 30 identical cards, one straight and one slanted, have

A. different volumes

B. equal volumes

C. equal surface areas

D. no volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q4/5

3048

QUESTION ID

S76-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. equal volumes

GRADING NOTE Same cross-sections.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

State Cavalieri's principle in your own words.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q5/5

3049

QUESTION ID

S76-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Two solids with the same height and equal cross-sectional areas at every level have the same volume.

GRADING NOTE Any equivalent statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSG-GMD.A.1

FORWARD 2

Use the cylinder volume formula to solve problems • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A can has radius 4 cm and height 12 cm. Find its volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q1/5

3051

QUESTION ID
S76-F2-Q1

TYPE
Short answer

CORRECT ANSWER
603 cm³

GRADING NOTE Rounding to the nearest whole number required. 192π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pipe 2 m long has inner radius 0.1 m. Find the volume of water it holds, to the nearest thousandth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q2/5

3052

QUESTION ID

S76-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.063 m³

GRADING NOTE Rounding to 0.001 required. 0.02π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q2/5

True or false?

Doubling the height of a cylinder multiplies its volume by 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q3/5

3053

QUESTION ID

S76-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volume is linear in h, so it doubles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q3/5

Doubling the radius of a cylinder multiplies its volume by

A. 4

B. 8

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q4/5

3054

QUESTION ID

S76-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE r^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q4/5

3054

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical tank has diameter 6 ft and height 10 ft. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q5/5

3055

QUESTION ID
S76-F2-Q5

TYPE
Short answer

CORRECT ANSWER
90π ft^3

GRADING NOTE Form required: exact value. r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q5/5

SET 77 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the volume of a pyramid

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Volume of a prism	**7.G.B.6**
BACKWARD 2	Base area of a pyramid	**6.G.A.1**
BACKWARD 4	Split a volume into equal parts	**5.MD.C.5**
FORWARD 1	Cavalieri's principle for pyramids	**HSG-GMD.A.1**
FORWARD 2	Use the pyramid volume formula	**HSG-GMD.A.3**

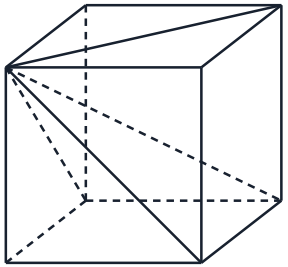
SET 77 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A cube with edge s is split into three congruent pyramids that share one vertex of the cube as their apex, as shown. Use this, together with scaling and Cavalieri's principle, to explain why EVERY pyramid with base area B and height h has volume ⅓Bh.



Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q1/5

3058

QUESTION ID

S77-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Each of the three pyramids has volume ⅓s³ = ⅓(s²)(s). Stretching vertically by h/s multiplies the volume by h/s, giving ⅓s²h. Any pyramid with base area B and height h has cross-sections B((h − y)/h)² at height y, the same as a square pyramid with base area B and height h, so by Cavalieri's principle its volume is ⅓Bh.

GRADING NOTE REQUIRED WORK: (1) Find the volume of one of the three pyramids. (2) Stretch it vertically to any height h. (3) Use Cavalieri's principle to compare any pyramid with a square pyramid of the same base area and height.

Any valid explanation. Essential: (1) the three congruent pyramids fill the cube, each with base s² and height s, so ⅓s³; (2) a vertical stretch extends this to any height; (3) cross-sections are scaled copies of the base with area B((h − y)/h)², so Cavalieri's principle extends it to any base shape and apex position. The cube case alone is NOT sufficient for full credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangular prism with base area B and height h is cut into three pyramids (tetrahedra), as Euclid did. Pairs of these pyramids have congruent bases and equal heights. Explain why each pyramid has volume ⅓Bh.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q2/5

3059

QUESTION ID

S77-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Pyramid 1 and pyramid 2 have equal bases and heights, and so do pyramids 2 and 3, so all three have equal volume. They fill the prism (volume Bh), so each has ⅓Bh; pyramid 1 has the prism's base and height.

GRADING NOTE REQUIRED WORK: Use the fact that pyramids with equal base areas and equal heights have equal volumes (Cavalieri's principle), applied to two pairs.

Any valid explanation. Essential: two equal-volume pairs give three equal volumes; their sum is Bh; one of them has base B and height h, so V = ⅓Bh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a class experiment, a hollow square pyramid and a hollow square prism have the same base and the same height. It takes three pyramids full of sand to fill the prism. Explain what this shows about the pyramid formula and why it is evidence, not a proof.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q3/5

3060

QUESTION ID

S77-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The pyramid holds about ⅓ of the prism, Bh, which supports V = ⅓Bh. The experiment has measurement error and tests one pyramid only, so it does not prove the formula for all pyramids.

GRADING NOTE REQUIRED WORK: Relate the prism volume Bh to the three pyramid fillings, then discuss measurement.

Any valid explanation. Essential: the ⅓ relationship with the prism; evidence versus proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cube with edge 6 is split into three congruent pyramids that share one vertex of the cube as their apex. Find the volume of each pyramid, then show that ⅓Bh gives the same value for one pyramid, and explain why this agreement is a check, not a proof for all pyramids.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q4/5

3061

QUESTION ID

S77-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

Each is 216 ÷ 3 = 72; one pyramid has base 6 × 6 = 36 and height 6, and ⅓ · 36 · 6 = 72. One numerical case cannot cover every pyramid.

GRADING NOTE

REQUIRED WORK: Divide the cube's volume by 3; then compute ⅓Bh for one pyramid.

Any valid explanation. Essential: 216 ÷ 3 = 72; B = 36 and h = 6 for one pyramid; ⅓Bh = 72; a single case is evidence only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid has base side s and height h. Imagine it built from n stacked square slabs, each of thickness h/n, where the slab k steps from the top has side ks/n. The total slab volume is $(s^2h/n^3) \times$ (the sum of the squares 1^2, 2^2, 3^2, and so on up to n^2). For n = 10 and n = 100 this equals $0.385s^2h$ and $0.338s^2h$. Explain how this supports $V = \frac{1}{3}s^2h$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q5/5

3062

QUESTION ID

S77-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The slabs approach the pyramid as they get thinner, and the coefficients 0.385, 0.338, … approach ⅓, so the volume is $\frac{1}{3}s^2h$.

GRADING NOTE REQUIRED WORK: Describe what the slabs approach and what the coefficients approach as n grows.

Any valid explanation. Essential: the slabs approximate the pyramid more closely as n grows; the coefficient trend toward ⅓; conclusion (an informal limit argument).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 1

Volume of a prism • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A prism has base area 36 and height 6. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q1/5

3064

QUESTION ID

S77-B1-Q1

TYPE

Short answer

CORRECT ANSWER

216

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cube has side 6. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q2/5

3065

QUESTION ID

S77-B1-Q2

TYPE

Short answer

CORRECT ANSWER

216

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

V =

A. Bh

B. ⅓Bh

C. 2Bh

D. B + h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q3/5

3066

QUESTION ID

S77-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Bh

GRADING NOTE Prism.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q3/5

True or false?

A cube is a prism.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q4/5

3067

QUESTION ID

S77-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Square base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q4/5

True or false?

A prism's volume depends on the number of its faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q5/5

3068

QUESTION ID
S77-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Base area and height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 2

Base area of a pyramid • Grade 6 • Geometry • 6.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A square base has side 10. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q1/5

3070

QUESTION ID

S77-B2-Q1

TYPE

Short answer

CORRECT ANSWER

100

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangular base has base 6 and height 4. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q2/5

3071

QUESTION ID

S77-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q2/5

The base of a square pyramid is a

A. triangle

B. square

C. rectangle that is not a square

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q3/5

3072

QUESTION ID

S77-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. square

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q3/5

True or false?

B in V = ⅓Bh means the area of the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q4/5

3073

QUESTION ID

S77-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q4/5

True or false?

B is the length of a base edge.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q5/5

3074

QUESTION ID

S77-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 4

Split a volume into equal parts • Grade 5 • Measurement & Data • 5.MD.C.5

ANSWER KEY Reduced copy of the question page, with its answer

A box of volume 60 cm³ is split into three equal parts. Volume of each?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q1/5

3076

QUESTION ID
S77-B4-Q1

TYPE
Short answer

CORRECT ANSWER
20 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q1/5

3076

ANSWER KEY Reduced copy of the question page, with its answer

Three congruent blocks fill a cube of volume 27. Volume of one block?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q2/5

3077

QUESTION ID

S77-B4-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q2/5

3077

ANSWER KEY Reduced copy of the question page, with its answer

A solid made of non-overlapping parts has volume

A. the largest part's volume

B. the sum of the parts' volumes

C. the product of the parts' volumes

D. the average volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q3/5

3078

QUESTION ID

S77-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. the sum of the parts' volumes

GRADING NOTE Additivity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q3/5

True or false?

If three equal pieces fill a prism, each piece has ⅓ of the prism's volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q4/5

3079

QUESTION ID

S77-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal shares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q4/5

True or false?

Cutting a solid into pieces changes its total volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q5

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q5/5

3080

QUESTION ID

S77-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volume is additive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q5

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q5/5

3080

SET 77 • HSG-GMD.A.1

FORWARD 1

Cavalieri's principle for pyramids • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid and a triangular pyramid both have base area 50 cm² and height 9 cm. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q1/5

3082

QUESTION ID

S77-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Equal: 150 cm³ each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An oblique pyramid has a square base with side 4 and vertical height 6. Find its volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q2/5

3083

QUESTION ID
S77-F1-Q2

TYPE
Short answer

CORRECT ANSWER
32

GRADING NOTE Any equivalent form. ⅓ · 16 · 6; use the vertical height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q2/5

3083

True or false?

Two pyramids with equal base areas and equal heights have equal volumes even if their apexes are in different positions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q3/5

3084

QUESTION ID

S77-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cavalieri.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q3/5

Why do two pyramids with equal base areas and equal heights have equal cross-section areas at every level?

A. Their bases have the same shape

B. Both cross-sections are scaled copies of the bases by the same factor

C. They have the same slant height

D. They have the same apex

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q4/5

3085

QUESTION ID
S77-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. Both cross-sections are scaled copies of the bases by the same factor

GRADING NOTE Each cross-section at height y is the base scaled by (h − y)/h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cross-section of a pyramid halfway up has what fraction of the base area?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q5/5

3086

QUESTION ID
S77-F1-Q5

TYPE
Short answer

CORRECT ANSWER
¼

GRADING NOTE Any equivalent form. Scale factor ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSG-GMD.A.1

FORWARD 2

Use the pyramid volume formula • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid has base side 9 m and height 10 m. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q1/5

3088

QUESTION ID
S77-F2-Q1

TYPE
Short answer

CORRECT ANSWER
270 m³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangular pyramid has a 6 in. by 4 in. base and height 7.5 in. Find its volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q2/5

3089

QUESTION ID
S77-F2-Q2

TYPE
Short answer

CORRECT ANSWER
60 in³

GRADING NOTE Any equivalent form. ⅓ · 24 · 7.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q2/5

3089

True or false?

A pyramid with the same base and height as a prism of volume 90 has volume 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q3/5

3090

QUESTION ID

S77-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has ⅓ of the volume: 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q3/5

3090

ANSWER KEY Reduced copy of the question page, with its answer

A pyramid with base area 48 and height 5 has volume

A. 120

B. 53

C. 240

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q4/5

3091

QUESTION ID

S77-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 80

GRADING NOTE $\frac{1}{3} \cdot 240$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A paperweight is a pyramid whose triangular base has base 6 cm and height 5 cm; the pyramid is 8 cm tall. Find its volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q5/5

3092

QUESTION ID
S77-F2-Q5

TYPE
Short answer

CORRECT ANSWER
40 cm³

GRADING NOTE Any equivalent form. B = 15; ⅓ · 15 · 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q5/5

3092

SET 104 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the volume of a cone

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	The cylinder volume argument	**HSG-GMD.A.1**
BACKWARD 3	The pyramid volume formula	**HSG-GMD.A.1**
FORWARD 1	Volumes of frustums (cones with the top cut off)	**HSG-GMD.A.3**
FORWARD 2	Use the cone volume formula	**HSG-GMD.A.3**

SET 104 • HSG-GMD.A.1

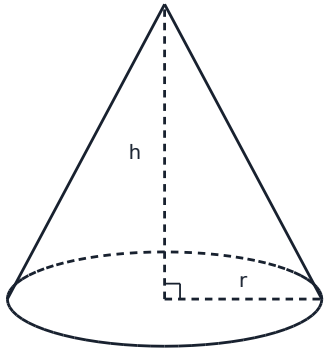
MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the volume of a cone is ⅓πr²h, using pyramids.

Work space



Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q1/5

3095

QUESTION ID

S104-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Pyramids with regular n-gon bases inscribed in the circle have V = ⅓Bh; as n grows, B approaches πr² and the pyramids approach the cone, so V = ⅓πr²h.

GRADING NOTE

REQUIRED WORK: Think of pyramids whose bases are regular polygons with more and more sides.

Any valid explanation. Essential: pyramid volume ⅓Bh; base polygons approach the circle (B → πr²); the pyramids approach the cone. Alternative: Cavalieri comparison with a pyramid of equal base area and height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has base radius r and height h. A square pyramid has base area πr² and the same height h. Use Cavalieri's principle to explain why the cone's volume is ⅓πr²h.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q2/5

3096

QUESTION ID

S104-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

At height y both cross-sections are copies of the bases scaled by (h − y)/h, so both have area πr²((h − y)/h)²; equal areas at every level and equal heights give equal volumes, ⅓πr²h.

GRADING NOTE REQUIRED WORK: Compare the cross-section areas of the two solids at the same height y.

Any valid explanation. Essential: the scale factor (h − y)/h for both solids; equal cross-section areas; Cavalieri's principle; the pyramid formula ⅓Bh with B = πr².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In an experiment, three cones full of water exactly fill a cylinder with the same base radius and height. Explain what this shows about the cone formula and why it is evidence, not a proof.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q3/5

3097

QUESTION ID

S104-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

One cone holds about ⅓ of πr²h, which supports V = ⅓πr²h; the experiment has measurement error and tests one cone, so it does not prove the formula.

GRADING NOTE REQUIRED WORK: Relate the cylinder volume πr²h to the three cone fillings, then discuss measurement.

Any valid explanation. Essential: the ⅓ relationship with the cylinder; evidence versus proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone with radius r and height h is approximated by n stacked disks of thickness h/n; the disk k steps from the top has radius kr/n. The total disk volume is $(\pi r^2 h/n^3) \times$ (the sum of the squares 1^2, 2^2, 3^2, and so on up to n^2), which equals $0.385\pi r^2 h$ for n = 10 and $0.338\pi r^2 h$ for n = 100. Explain how this supports $V = \frac{1}{3}\pi r^2 h$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q4/5

3098

QUESTION ID

S104-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

As the disks get thinner they fit the cone more closely, and the coefficients approach ⅓, so $V = \frac{1}{3}\pi r^2 h$.

GRADING NOTE REQUIRED WORK: Describe what the disks approach and what the coefficients approach.

Any valid explanation. Essential: the disks approximate the cone better as n grows; the trend toward ⅓; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has base radius r and height h. A plane parallel to the base cuts the cone at height y above the base. Explain why the cross-section is a circle of radius r(h − y)/h, and why this fact is what makes the cone's volume ⅓ of the cylinder's rather than ½.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q5/5

3099

QUESTION ID

S104-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The small cone above the cut is similar to the whole cone with ratio (h − y)/h, so the radius is r(h − y)/h. The areas shrink like the SQUARE of that ratio, $\pi r^2((h - y)/h)^2$, so most of the volume is near the base and the total is $\frac{1}{3}\pi r^2 h$, less than the ½ a linear shrink would suggest.

GRADING NOTE REQUIRED WORK: Use similar triangles in a vertical cross-section through the apex; compare how the cross-section areas shrink.

Any valid explanation. Essential: similar triangles give the radius; area scales by the square of the ratio; link to the ⅓ factor (via Cavalieri or slab sums).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q5/5

SET 104 • HSG-GMD.A.1

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle of radius 7?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q1/5

3101

QUESTION ID

S104-B1-Q1

TYPE

Short answer

CORRECT ANSWER

49π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle of diameter 8?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q2/5

3102

QUESTION ID

S104-B1-Q2

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value. r = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The base of a cone is a

A. triangle

B. sector

C. square

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q3/5

3103

QUESTION ID

S104-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. circle

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q3/5

True or false?

The base area of a cone with radius r is πr^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q4/5

3104

QUESTION ID

S104-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Circle area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The base area of a cone with radius 2 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q5/5

3105

QUESTION ID

S104-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q5/5

SET 104 • HSG-GMD.A.1

BACKWARD 2

The cylinder volume argument • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

A cylinder's volume is $\pi r^2 h$ because it is a stack of disks of area πr^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q1/5

3107

QUESTION ID

S104-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Set 76.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Cylinder radius 3, height 5: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q2/5

3108

QUESTION ID

S104-B2-Q2

TYPE

Short answer

CORRECT ANSWER

45π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q2/5

A cone fits inside a cylinder with the same base and height. The cone's volume is

A. ½ of the cylinder's

B. ⅓ of the cylinder's

C. ⅔ of the cylinder's

D. equal to the cylinder's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q3/5

3109

QUESTION ID

S104-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ⅓ of the cylinder's

GRADING NOTE Cone vs cylinder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder holds 60 cm³. A cone with the same base and height holds

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q4/5

3110

QUESTION ID

S104-B2-Q4

TYPE

Short answer

CORRECT ANSWER

20 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S104-B2-Q4**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q4/5

True or false?

A cone is a cylinder with the top cut off flat.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q5/5

3111

QUESTION ID

S104-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It comes to a point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q5/5

SET 104 • HSG-GMD.A.1

BACKWARD 3

The pyramid volume formula • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A pyramid has base area 30 and height 8. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q1/5

3113

QUESTION ID

S104-B3-Q1

TYPE

Short answer

CORRECT ANSWER

80

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q1/5

True or false?

V = ⅓Bh holds for every pyramid, whatever the shape of its base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q2/5

3114

QUESTION ID

S104-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Set 77.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q2/5

A pyramid and a prism have the same base and height. The pyramid's volume is

A. ⅓ of the prism's

B. ½ of the prism's

C. 3 times the prism's

D. equal to the prism's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q3/5

3115

QUESTION ID

S104-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ⅓ of the prism's

GRADING NOTE ⅓Bh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A pyramid has volume 40 and height 6. Base area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q4/5

3116

QUESTION ID

S104-B3-Q4

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form. 3 · 40 ÷ 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q4/5

True or false?

A pyramid has half the volume of the prism with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q5/5

3117

QUESTION ID

S104-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE One third.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q5/5

SET 104 • HSG-GMD.A.1

FORWARD 1

Volumes of frustums (cones with the top cut off) • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A cone has radius 6 and height 12. The top cone of height 4 is cut off by a plane parallel to the base. Find the exact volume of the remaining frustum.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q1/5

3119

QUESTION ID

S104-F1-Q1

TYPE

Short answer

CORRECT ANSWER

416π/3

GRADING NOTE Form required: exact value. Whole cone 144π; small cone radius 2, volume 16π/3; 144π − 16π/3 = 416π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cup is a frustum with bottom radius 3 cm, top radius 4 cm and height 6 cm. Extend it to a full cone and find the cup's volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q2/5

3120

QUESTION ID

S104-F1-Q2

TYPE

Short answer

CORRECT ANSWER

232 cm³

GRADING NOTE Rounding to the nearest whole number required. Full cone height 24 (radius 4); removed cone height 18 (radius 3); $\frac{1}{3}\pi(16 \cdot 24 - 9 \cdot 18) = 74\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A frustum's volume is the large cone's volume minus the small cone's volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q3/5

3121

QUESTION ID

S104-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtraction of non-overlapping parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone is cut halfway up its height, parallel to the base. The small top cone has what fraction of the whole cone's volume?

A. ⅓

B. ½

C. ⅛

D. ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q4/5

3122

QUESTION ID

S104-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. ⅛

GRADING NOTE $(\frac{1}{2})^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why cutting a cone halfway up leaves 7/8 of its volume in the bottom part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q5/5

3123

QUESTION ID

S104-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The top cone is similar with ratio ½, so its volume is (½)³ = ⅛ of the whole; 1 − ⅛ = 7/8 remains.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q5/5

SET 104 • HSG-GMD.A.1

FORWARD 2

Use the cone volume formula • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A cone has radius 3 in. and height 8 in. Find its volume to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q1/5

3125

QUESTION ID

S104-F2-Q1

TYPE

Short answer

CORRECT ANSWER

75.4 in³

GRADING NOTE Rounding to 0.1 required. 24π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S104-F2-Q1**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone-shaped cup has diameter 6 cm and height 9 cm. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q2/5

3126

QUESTION ID

S104-F2-Q2

TYPE

Short answer

CORRECT ANSWER

27π cm^3

GRADING NOTE Form required: exact value. r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q2/5

True or false?

A cone with the same base and height as a cylinder of volume 90 has volume 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q3/5

3127

QUESTION ID

S104-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has ⅓ of the volume: 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has base area 48 and height 5. Its volume is

A. 240

B. 53

C. 80

D. 120

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q4/5

3128

QUESTION ID

S104-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 80

GRADING NOTE $\frac{1}{3} \cdot 240$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A pile of sand is a cone 2 m tall with base radius 3 m. Find its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q5/5

3129

QUESTION ID
S104-F2-Q5

TYPE
Short answer

CORRECT ANSWER
18.8 m^3

GRADING NOTE Rounding to 0.1 required. 6π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q5/5

SET 78 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

Use volume formulas for cylinders, pyramids, and cones

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	Find a missing side with the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Evaluate a formula and round	**7.EE.B.3**
BACKWARD 4	Which solid uses ⅓?	**HSG-GMD.A.1**
FORWARD 1	Composite and hollow solids	**HSG-GMD.A.3**
FORWARD 2	Mass from volume and density	**HSG-MG.A.2**

SET 78 • HSG-GMD.A.3

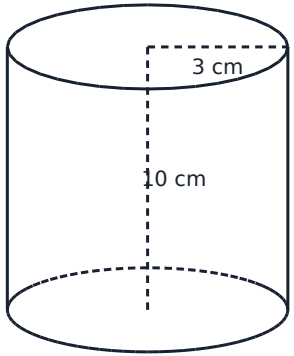
MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder has radius 3 cm and height 10 cm. Find its volume, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q1/5

3132

QUESTION ID

S78-M-Q1

TYPE

Short answer

CORRECT ANSWER

90π cm³ ≈ 282.7 cm³

GRADING NOTE Form required: both the exact value and rounding to 0.1, with units.

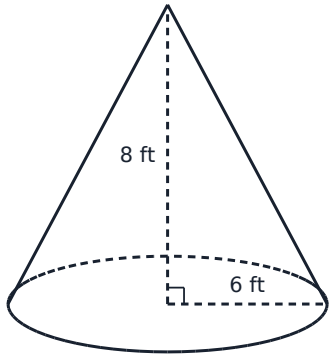
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has radius 6 ft and height 8 ft. Find its volume, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q2/5

3133

QUESTION ID

S78-M-Q2

TYPE

Short answer

CORRECT ANSWER

96π ft³ ≈ 301.6 ft³

GRADING NOTE Form required: both, with units. ⅓π · 36 · 8.

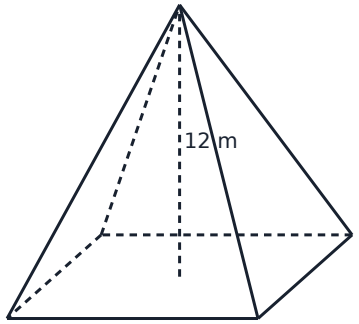
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid has base side 10 m and height 12 m. Find its volume.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q3/5

3134

QUESTION ID

S78-M-Q3

TYPE

Short answer

CORRECT ANSWER

400 m³

GRADING NOTE Any equivalent form, with units. ⅓ · 100 · 12.

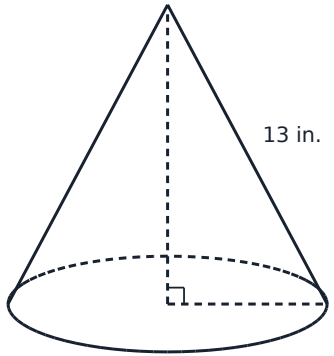
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has diameter 10 in. and slant height 13 in. Find its volume, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q4/5

3135

QUESTION ID

S78-M-Q4

TYPE

Short answer

CORRECT ANSWER

100π in^3 $\approx$ 314.2 in^3

GRADING NOTE Form required: both, with units. r = 5; height √(169 − 25) = 12; ⅓π · 25 · 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder and a cone have the same radius and height. The cylinder holds 600 mL. The cone holds

A. 300 mL

B. 200 mL

C. 600 mL

D. 1,800 mL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q5/5

3136

QUESTION ID

S78-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 200 mL

GRADING NOTE $\frac{1}{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle with radius 6?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q1/5

3138

QUESTION ID

S78-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle with diameter 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q2/5

3139

QUESTION ID

S78-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q2/5

A base of radius 3 has area

A. 9

B. 9π

C. 3π

D. 6π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q3/5

3140

QUESTION ID

S78-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9π

GRADING NOTE πr^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q3/5

True or false?

Squaring the radius comes before multiplying by π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q4/5

3141

QUESTION ID

S78-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Order of operations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q4/5

True or false?

πr^2 with $r = 3$ is $(3\pi)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q5/5

3142

QUESTION ID

S78-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9π, not $9\pi^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 2

Find a missing side with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 13 and one leg 5. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q1/5

3144

QUESTION ID

S78-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has radius 6 and slant height 10. Height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q2/5

3145

QUESTION ID

S78-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q2/5

Radius, height, and slant height of a cone form a

A. square

B. right triangle

C. equilateral triangle

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q3/5

3146

QUESTION ID

S78-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. right triangle

GRADING NOTE Height ⊥ base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q3/5

True or false?

The slant height is longer than the height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q4/5

3147

QUESTION ID

S78-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q4/5

True or false?

The slant height can be used as the height in $V = \frac{1}{3}\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q5/5

3148

QUESTION ID

S78-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use the perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 3

Evaluate a formula and round • Grade 7 • Expressions & Equations • 7.EE.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate ⅓ · 36 · 8.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q1/5

3150

QUESTION ID

S78-B3-Q1

TYPE

Short answer

CORRECT ANSWER

96

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Round 90π to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q2/5

3151

QUESTION ID

S78-B3-Q2

TYPE

Short answer

CORRECT ANSWER

282.7

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$96\pi \approx$

A. 301.6

B. 301.5

C. 3,016

D. 96.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q3/5

3152

QUESTION ID

S78-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 301.6

GRADING NOTE Use the π key.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q3/5

True or false?

Using 3.14 for π may change the last digit of a rounded answer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q4/5

3153

QUESTION ID

S78-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Use the π key.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{1}{3} \cdot 100 \cdot 12 = 1{,}200$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q5/5

3154

QUESTION ID

S78-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 4

Which solid uses ⅓? • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Pyramids and cones use V = ⅓Bh.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q1/5

3156

QUESTION ID

S78-B4-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pointed solids.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which has the formula V = Bh: a cone or a cylinder?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q2/5

3157

QUESTION ID

S78-B4-Q2

TYPE

Short answer

CORRECT ANSWER

Cylinder

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$V = \frac{1}{3}\pi r^2 h$ is the volume of a

A. cylinder

B. cone

C. prism

D. sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q3/5

3158

QUESTION ID

S78-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. cone

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q3/5

Why does a cone have ⅓ of the matching cylinder's volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q4/5

3159

QUESTION ID
S78-B4-Q4

TYPE
Short answer

CORRECT ANSWER
Three cones fill the cylinder (the informal argument).

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q4/5

True or false?

A pyramid uses V = Bh.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q5/5

3160

QUESTION ID

S78-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ⅓Bh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q5/5

SET 78 • HSG-GMD.A.3

FORWARD 1

Composite and hollow solids • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A silo is a cylinder (radius 4 m, height 10 m) topped by a cone (radius 4 m, height 3 m). Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q1/5

3162

QUESTION ID
S78-F1-Q1

TYPE
Short answer

CORRECT ANSWER
176π m^3

GRADING NOTE Form required: exact value. $160\pi + 16\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pipe is a cylinder of length 20 cm with outer radius 5 cm and inner radius 4 cm. Find the volume of the material, exact.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q2/5

3163

QUESTION ID
S78-F1-Q2

TYPE
Short answer

CORRECT ANSWER
180π cm^3

GRADING NOTE Form required: exact value. $\pi(25 - 16)(20)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q2/5

True or false?

The volume of a composite solid is the sum of the volumes of its non-overlapping parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q3/5

3164

QUESTION ID

S78-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Additivity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder (V = 300) has a cone with the same base and height carved out. Remaining volume?

A. 300

B. 200

C. 0

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q4/5

3165

QUESTION ID

S78-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 200

GRADING NOTE 300 − 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A pencil tip is a cone of radius 0.4 cm and height 1.5 cm on a cylinder of radius 0.4 cm and length 15 cm. Find the total volume to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q5/5

3166

QUESTION ID
S78-F1-Q5

TYPE
Short answer

CORRECT ANSWER
7.8 cm³

GRADING NOTE Rounding to 0.1 required. 2.4π + 0.08π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q5/5

SET 78 • HSG-GMD.A.3

FORWARD 2

Mass from volume and density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A steel cylinder has radius 2 cm and height 5 cm. Steel has density 7.85 g/cm³. Find the mass to the nearest gram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q1/5

3168

QUESTION ID
S78-F2-Q1

TYPE
Short answer

CORRECT ANSWER
493 g

GRADING NOTE Rounding to the nearest gram required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone of ice cream (radius 3 cm, height 10 cm) has density 0.55 g/cm³. Find the mass to the nearest gram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q2/5

3169

QUESTION ID
S78-F2-Q2

TYPE
Short answer

CORRECT ANSWER
52 g

GRADING NOTE Rounding to the nearest gram required. 30π cm³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q2/5

True or false?

Mass = volume ÷ density.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q3/5

3170

QUESTION ID

S78-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Mass = density × volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q3/5

A pyramid of volume 400 cm³ is made of material with density 2 g/cm³. Mass?

A. 200 g

B. 402 g

C. 1,600 g

D. 800 g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q4/5

3171

QUESTION ID

S78-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 800 g

GRADING NOTE 2 × 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A gold cube has volume 8 cm³ and mass 154.4 g. Find the density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q5/5

3172

QUESTION ID
S78-F2-Q5

TYPE
Short answer

CORRECT ANSWER
19.3 g/cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q5/5

SET 79 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

Use the volume formula for spheres, including composite solids

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Cubes of numbers	**6.EE.A.1**
BACKWARD 2	Radius from diameter	**7.G.B.4**
BACKWARD 3	Multiply fractions by numbers	**7.NS.A.2**
BACKWARD 4	Cylinder and cone volumes	**HSG-GMD.A.3**
FORWARD 1	Scaling spheres	**HSG-GMD.A.3**
FORWARD 2	Model real objects as spheres	**HSG-MG.A.1**

SET 79 • HSG-GMD.A.3

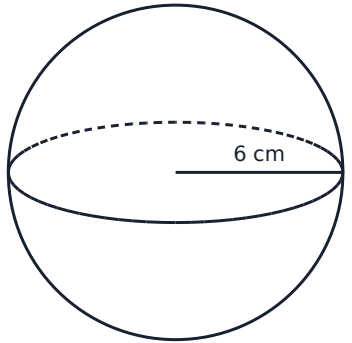
MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find the volume of a sphere with radius 6 cm, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q1/5

3175

QUESTION ID
S79-M-Q1

TYPE
Short answer

CORRECT ANSWER
288π cm³ ≈ 904.8 cm³

GRADING NOTE Form required: both, with units. 4/3 π · 216.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the volume of a ball with diameter 10 in., exact and to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q2/5

3176

QUESTION ID
S79-M-Q2

TYPE
Short answer

CORRECT ANSWER
500π/3 in³ ≈ 523.6 in³

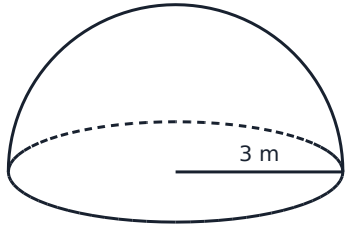
GRADING NOTE Form required: both, with units. r = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the volume of a hemisphere with radius 3 m, exact and to the nearest tenth.



Volume: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q3/5

3177

QUESTION ID

S79-M-Q3

TYPE

Short answer

CORRECT ANSWER

18π m³ ≈ 56.5 m³

GRADING NOTE Form required: both, with units. ½ · 4/3 π · 27.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A capsule is a cylinder of radius 3 cm and length 10 cm with a hemisphere on each end. Find its volume, exact and to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q4/5

3178

QUESTION ID

S79-M-Q4

TYPE

Short answer

CORRECT ANSWER

126π cm³ ≈ 395.8 cm³

GRADING NOTE Form required: both, with units. $90\pi + 36\pi$ (two hemispheres make one sphere).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

An ice cream cone (radius 2, height 9) is topped by a hemisphere of radius 2. The total exact volume is

A. 52π/3

B. 28π

C. 12π

D. 44π/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q5/5

3179

QUESTION ID
S79-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. 52π/3

GRADING NOTE 12π + 16π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 1

Cubes of numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

$6^3 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q1/5

3181

QUESTION ID

S79-B1-Q1

TYPE

Short answer

CORRECT ANSWER

216

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$5^3 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q2/5

3182

QUESTION ID

S79-B1-Q2

TYPE

Short answer

CORRECT ANSWER

125

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$3^3 =$

A. 9

B. 81

C. 6

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q3/5

3183

QUESTION ID

S79-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 27

GRADING NOTE $3 \cdot 3 \cdot 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$2^3 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q4/5

3184

QUESTION ID

S79-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cube.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$4^3 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q5/5

3185

QUESTION ID

S79-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 2

Radius from diameter • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A ball has diameter 10. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q1/5

3187

QUESTION ID

S79-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q1/5

True or false?

Using the diameter in $4/3\ \pi r^3$ makes the answer 8 times too large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q2/5

3188

QUESTION ID

S79-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q2/5

A sphere with diameter 12 has r^3 =

A. 216

B. 144

C. 36

D. 1,728

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q3/5

3189

QUESTION ID

S79-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 216

GRADING NOTE 6^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere has radius 2.5. Diameter?

Diameter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q4/5

3190

QUESTION ID

S79-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q4/5

True or false?

The radius of a sphere is twice its diameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q5/5

3191

QUESTION ID

S79-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 3

Multiply fractions by numbers • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$4/3 \cdot 216 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q1/5

3193

QUESTION ID

S79-B3-Q1

TYPE

Short answer

CORRECT ANSWER

288

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$\frac{1}{2} \cdot 4/3 \cdot 27 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q2/5

3194

QUESTION ID

S79-B3-Q2

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$4/3 \cdot 125 =$

A. 375/4

B. 500/3

C. 125/3

D. 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q3/5

3195

QUESTION ID

S79-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 500/3

GRADING NOTE Numerators.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$4/3 \cdot 27 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q4/5

3196

GRADING NOTE $27 \div 3 \times 4$.

QUESTION ID

S79-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q4/5

True or false?

$4/3 \cdot 6 = 24$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q5/5

3197

QUESTION ID

S79-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 4

Cylinder and cone volumes • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Cylinder radius 3, length 10: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q1/5

3199

QUESTION ID

S79-B4-Q1

TYPE

Short answer

CORRECT ANSWER

90π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Cone radius 2, height 9: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q2/5

3200

QUESTION ID
S79-B4-Q2

TYPE
Short answer

CORRECT ANSWER
12π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder and a cone with radius 3 and height 4 have volumes

A. 36π and 12π

B. 12π and 36π

C. 12π and 4π

D. 36π and 36π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q3/5

3201

QUESTION ID

S79-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 36π and 12π

GRADING NOTE ⅓.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q3/5

True or false?

Two hemispheres of radius r make one sphere of radius r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q4/5

3202

QUESTION ID

S79-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Halves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q4/5

True or false?

A composite volume is the product of the parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q5/5

3203

QUESTION ID

S79-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q5/5

SET 79 • HSG-GMD.A.3

FORWARD 1

Scaling spheres • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

The radius of a sphere is doubled. By what factor is the volume multiplied?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q1/5

3205

QUESTION ID

S79-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. 2^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q1/5

A sphere fits exactly inside a cube with side 6. Find the exact volume of the space in the cube outside the sphere.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q2/5

3206

QUESTION ID

S79-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$216 - 36\pi$

GRADING NOTE Form required: exact value. r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q2/5

True or false?

A sphere of radius 3 has 9 times the volume of a sphere of radius 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q3/5

3207

QUESTION ID

S79-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volume scales by $3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A balloon's radius grows from 10 cm to 15 cm. Its volume is multiplied by

A. 1.5

B. 2.25

C. 5

D. 3.375

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q4/5

3208

QUESTION ID

S79-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3.375

GRADING NOTE 1.5^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere and a cylinder have the same radius r, and the cylinder's height is 2r. What fraction of the cylinder's volume is the sphere's?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q5/5

3209

QUESTION ID

S79-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{3}$

GRADING NOTE Any equivalent form. $(4/3\ \pi r^3)/(2\pi r^3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q5/5

SET 79 • HSG-GMD.A.3

FORWARD 2

Model real objects as spheres • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Model Earth as a sphere with radius 6,371 km. Estimate its volume in cubic kilometers, in scientific notation to 3 significant figures.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q1/5

3211

QUESTION ID

S79-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.08×10^{12} km^3

GRADING NOTE Rounding to 3 significant figures required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A basketball has circumference 75 cm. Model it as a sphere and find its volume to the nearest 100 cm^3.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q2/5

3212

QUESTION ID
S79-F2-Q2

TYPE
Short answer

CORRECT ANSWER
7100 cm^3

GRADING NOTE Rounding to the nearest 100 required. r = 75/(2π) ≈ 11.94 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q2/5

True or false?

Modeling an orange as a sphere ignores its small bumps.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q3/5

3213

QUESTION ID

S79-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Modeling assumption.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q3/5

A scoop of ice cream is modeled as a sphere of radius 3 cm. Its volume is about

A. 28 cm³

B. 113 cm³

C. 339 cm³

D. 36 cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q4/5

3214

QUESTION ID

S79-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 113 cm³

GRADING NOTE 36π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give one reason a sphere is a good model for a ball bearing and one limitation of the model for a pumpkin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q5/5

3215

GRADING NOTE Any reasonable answer.

QUESTION ID

S79-F2-Q5

TYPE

Short answer

CORRECT ANSWER

A ball bearing is nearly perfectly round; a pumpkin has ridges, a flattened shape, and a stem, so the model is only approximate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q5/5

SET 80 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

Solve for an unknown dimension from a volume

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Solve a one-step equation with a coefficient	**HSA-REI.B.3**
BACKWARD 2	Square roots and cube roots	**8.EE.A.2**
BACKWARD 3	Rearrange a formula	**HSA-CED.A.4**
FORWARD 1	Find dimensions to meet a target volume	**HSG-GMD.A.3**
FORWARD 2	Design a container that meets constraints	**HSG-MG.A.3**

SET 80 • HSG-GMD.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder has volume 360π cm³ and radius 6 cm. Find its height.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q1/5

3218

QUESTION ID
S80-M-Q1

TYPE
Short answer

CORRECT ANSWER
10 cm

GRADING NOTE Any equivalent form, with units. $36\pi h = 360\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q1/5

A cone has volume 96π in³ and height 8 in. Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q2/5

3219

QUESTION ID

S80-M-Q2

TYPE

Short answer

CORRECT ANSWER

6 in.

GRADING NOTE Any equivalent form, with units. $\frac{1}{3}\pi r^2 \cdot 8 = 96\pi \rightarrow r^2 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere has volume 36π m³. Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q3/5

3220

QUESTION ID
S80-M-Q3

TYPE
Short answer

CORRECT ANSWER
3 m

GRADING NOTE Any equivalent form, with units. $r^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid has volume 256 ft³ and height 12 ft. Find the side of its base.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q4/5

3221

QUESTION ID
S80-M-Q4

TYPE
Short answer

CORRECT ANSWER
8 ft

GRADING NOTE Any equivalent form, with units. $\frac{1}{3}s^2 \cdot 12 = 256 \rightarrow s^2 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q4/5

A cylinder holds 500 cm³ and has height 10 cm. Its radius is closest to

A. 7.1 cm

B. 4.0 cm

C. 2.0 cm

D. 15.9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q5/5

3222

QUESTION ID

S80-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 4.0 cm

GRADING NOTE $r = \sqrt{(50/\pi)} \approx 3.99$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q5/5

SET 80 • HSG-GMD.A.3

BACKWARD 1

Solve a one-step equation with a coefficient • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve $36\pi h = 360\pi$.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q1/5

3224

QUESTION ID

S80-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ⅓x = 12.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q2/5

3225

QUESTION ID

S80-B1-Q2

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 4x = 108.

A. 25

B. 104

C. 432

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q3/5

3226

QUESTION ID

S80-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 27

GRADING NOTE Divide by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q3/5

True or false?

To solve $36\pi h = 360\pi$, dividing both sides by π is allowed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q4/5

3227

QUESTION ID

S80-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\pi \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q4/5

True or false?

To solve ⅓x = 12, divide by 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q5/5

3228

QUESTION ID

S80-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Multiply by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • HSG-GMD.A.3

BACKWARD 2

Square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $r^2 = 36$ for $r > 0$.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q1/5

3230

QUESTION ID

S80-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $r^3 = 27$.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q2/5

3231

QUESTION ID

S80-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sqrt[3]{125} =$

A. 25

B. 5

C. 41.7

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q3/5

3232

QUESTION ID

S80-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

GRADING NOTE $5^3 = 125$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{64} = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q4/5

3233

QUESTION ID

S80-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $8^2 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q4/5

True or false?

If $r^3 = 8$, then $r = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q5/5

3234

QUESTION ID

S80-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSG-GMD.A.3

BACKWARD 3

Rearrange a formula • High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $V = \pi r^2 h$ for h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q1/5

3236

QUESTION ID

S80-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$V/(\pi r^2)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B3-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve V = ⅓Bh for B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q2

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q2/5

3237

QUESTION ID

S80-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3V/h

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B3-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$V = 4/3\ \pi r^3$ solved for r^3 is

A. $4V/(3\pi)$

B. $V/(4\pi)$

C. $3V/(4\pi)$

D. $3\pi V/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q3/5

3238

QUESTION ID

S80-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3V/(4π)

GRADING NOTE Multiply by 3/(4π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q3/5

True or false?

Solving $V = \pi r^2 h$ for r needs a square root.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q4/5

3239

QUESTION ID

S80-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $r = \sqrt{(V/(\pi h))}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q4/5

True or false?

Solving V = Bh for h gives h = VB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q5/5

3240

QUESTION ID
S80-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE V/B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q5/5

SET 80 • HSG-GMD.A.3

FORWARD 1

Find dimensions to meet a target volume • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A drink can must hold at least 355 cm^3 and have radius 3.3 cm. What is the smallest height, in tenths of a centimeter, that meets the requirement?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q1/5

3242

QUESTION ID
S80-F1-Q1

TYPE
Short answer

CORRECT ANSWER
10.4 cm

GRADING NOTE Form required: a height in tenths that meets the constraint (round UP). $h = 355/(\pi \cdot 3.3^2) = 10.376$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A spherical tank must hold at least 1,000 m³. What is the smallest radius, in tenths of a meter, that meets the requirement?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q2/5

3243

QUESTION ID

S80-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6.3 m

GRADING NOTE Form required: a radius in tenths that meets the constraint (round UP). $r = \sqrt[3]{(750/\pi)} = 6.2035$; 6.2 m would hold only about 998.3 m³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q2/5

3243

True or false?

To double the volume of a cylinder while keeping its radius, double its height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q3/5

3244

QUESTION ID

S80-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Linear in h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q3/5

A cone has the same volume as a cylinder of equal radius. The cone's height is

A. 9 times the cylinder's

B. ⅓ of the cylinder's

C. 3 times the cylinder's

D. equal to the cylinder's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q4/5

3245

QUESTION ID

S80-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3 times the cylinder's

GRADING NOTE $\frac{1}{3}\pi r^2 h_1 = \pi r^2 h_2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cube has the same volume as a sphere of radius 3. Find the cube's edge to the nearest tenth.

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q5/5

3246

QUESTION ID

S80-F1-Q5

TYPE

Short answer

CORRECT ANSWER

4.8

GRADING NOTE Rounding to 0.1 required. $\sqrt[3]{(36\pi)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q5/5

3246

SET 80 • HSG-GMD.A.3

FORWARD 2

Design a container that meets constraints • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical vase must hold 2,000 cm³ and be 20 cm tall. Its radius must be at least 5 cm to fit a bouquet. Does the design work? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q1/5

3248

QUESTION ID

S80-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: r = √(2000/(20π)) ≈ 5.6 cm, which is at least 5 cm.

GRADING NOTE Form required: the conclusion with the computed radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A conical paper cup must hold at least 150 cm^3 and have radius 3.5 cm. What is the smallest height, in tenths of a centimeter, that meets the requirement?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q2/5

3249

QUESTION ID

S80-F2-Q2

TYPE

Short answer

CORRECT ANSWER

11.7 cm

GRADING NOTE Form required: a height in tenths that meets the constraint (round UP). $h = 450/(\pi \cdot 3.5^2) = 11.693$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q2/5

3249

True or false?

A container's volume requirement fixes each of its dimensions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q3/5

3250

QUESTION ID

S80-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It fixes only a relationship: many (r, h) pairs give the same volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q3/5

A cylinder must hold 100π cm³. Which (r, h) works?

A. (5, 4)

B. (5, 20)

C. (4, 5)

D. (10, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q4/5

3251

QUESTION ID

S80-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (5, 4)

GRADING NOTE $25 \cdot 4 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give two different cylinders (radius and height) that each hold 64π cm^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q5/5

3252

QUESTION ID
S80-F2-Q5

TYPE
Short answer

CORRECT ANSWER
For example r = 4, h = 4 and r = 2, h = 16.

GRADING NOTE Any two (r, h) pairs with $r^2h = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q5/5

Identify cross-sections of three-dimensional objects

MAIN	Geometry main question	**HSG-GMD.B.4**
BACKWARD 1	Cross-sections of right rectangular prisms and pyramids	**7.G.A.3**
BACKWARD 2	Properties of polygons	**5.G.B.4**
BACKWARD 3	Parts of solids	**6.G.A.4**
FORWARD 1	Areas of cross-sections	**HSG-GMD.B.4**
FORWARD 2	Use cross-section areas to compare volumes (Cavalieri)	**HSG-GMD.A.2**

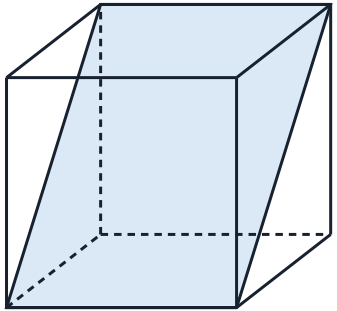
SET 81 • HSG-GMD.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A plane cuts the cube as shown. Name the shape of the cross-section.



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q1/5

3255

QUESTION ID

S81-M-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle (not a square)

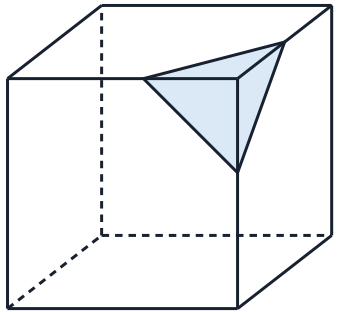
GRADING NOTE Any equivalent answer. Two sides are cube edges (s) and two are face diagonals (s√2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane cuts off one corner of the cube as shown. Name the shape of the cross-section.



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q2/5

3256

QUESTION ID

S81-M-Q2

TYPE

Short answer

CORRECT ANSWER

A triangle

GRADING NOTE Any equivalent answer. (Equilateral if the three cut points are equally far from the corner.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q2/5

A right circular cone is cut by a vertical plane through its apex. The cross-section is

A. a circle

B. a rectangle

C. a triangle (isosceles)

D. an ellipse

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q3/5

3257

QUESTION ID

S81-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. a triangle (isosceles)

GRADING NOTE Two slant edges and a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Name the cross-section of a cylinder cut by a plane (a) parallel to its base, (b) perpendicular to its base through its axis, and (c) slanted, cutting the curved side only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q4/5

3258

QUESTION ID

S81-M-Q4

TYPE

Short answer

CORRECT ANSWER

(a) circle (b) rectangle (c) ellipse

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere of radius 10 is cut by a plane 6 units from its center. Name the shape of the cross-section and find its radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q5/5

3259

QUESTION ID

S81-M-Q5

TYPE

Short answer

CORRECT ANSWER

A circle of radius 8

GRADING NOTE Form required: the shape and the radius. $\sqrt{(10^2 - 6^2)} = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 1

Cross-sections of right rectangular prisms and pyramids • Grade 7 • Geometry • 7.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A rectangular prism is cut parallel to its base. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q1/5

3261

QUESTION ID

S81-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle congruent to the base

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid is cut parallel to its base. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q2/5

3262

QUESTION ID

S81-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A smaller square

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q2/5

A square pyramid cut vertically through its apex and two opposite base edges' midpoints gives a

A. trapezoid

B. square

C. circle

D. triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q3/5

3263

QUESTION ID

S81-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. triangle

GRADING NOTE Through the apex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q3/5

True or false?

A vertical cut of a rectangular prism parallel to a side face gives a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q4/5

3264

QUESTION ID

S81-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Congruent to the face.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q4/5

True or false?

A cut of a cube parallel to a face can be a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q5/5

3265

QUESTION ID

S81-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 2

Properties of polygons • Grade 5 • Geometry • 5.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A quadrilateral with four right angles and unequal adjacent sides is a

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q1/5

3267

QUESTION ID

S81-B2-Q1

TYPE

Short answer

CORRECT ANSWER

rectangle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q1/5

True or false?

Every square is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q2/5

3268

QUESTION ID

S81-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Hierarchy.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q2/5

A polygon with six sides is a

A. hexagon

B. pentagon

C. octagon

D. heptagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q3/5

3269

QUESTION ID

S81-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. hexagon

GRADING NOTE Name.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle with two equal sides is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q4/5

3270

QUESTION ID

S81-B2-Q4

TYPE

Short answer

CORRECT ANSWER

isosceles

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A trapezoid has no parallel sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q5/5

3271

QUESTION ID

S81-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At least one pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 3

Parts of solids • Grade 6 • Geometry • 6.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

How many faces does a cube have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q1/5

3273

QUESTION ID

S81-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the shape of the base of a cone?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q2/5

3274

QUESTION ID

S81-B3-Q2

TYPE

Short answer

CORRECT ANSWER

A circle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q2/5

How many edges meet at each vertex of a cube?

A. 6

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q3/5

3275

QUESTION ID

S81-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE Corner.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q3/5

True or false?

A cylinder has two congruent circular bases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q4/5

3276

QUESTION ID

S81-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A sphere has one edge.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q5/5

3277

QUESTION ID

S81-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No edges.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q5/5

SET 81 • HSG-GMD.B.4

FORWARD 1

Areas of cross-sections • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A cube has edge 4. Find the area of the diagonal cross-section through two opposite edges.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q1/5

3279

QUESTION ID

S81-F1-Q1

TYPE

Short answer

CORRECT ANSWER

16√2

GRADING NOTE Form required: exact value. 4 × 4√2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has radius 6 and height 12. Find the area of the cross-section parallel to the base at height 4 (measured up from the base).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q2/5

3280

QUESTION ID

S81-F1-Q2

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value. Radius at that level = 6 · (12 − 4)/12 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q2/5

True or false?

A sphere of radius 5 cut 3 units from its center gives a circle of radius 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q3/5

3281

QUESTION ID

S81-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3-4-5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q3/5

A cube can have a cross-section that is a regular

A. octagon

B. pentagon

C. heptagon

D. hexagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q4/5

3282

QUESTION ID

S81-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. hexagon

GRADING NOTE Plane through the midpoints of six edges.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q4/5

A square pyramid has base side 10 and height 15. Find the exact area of the cross-section parallel to the base 5 units below the apex.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q5/5

3283

QUESTION ID

S81-F1-Q5

TYPE

Short answer

CORRECT ANSWER

100/9 (about 11.1)

GRADING NOTE Form required: exact value or rounding to 0.1. Side 10 · 5/15 = 10/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSG-GMD.B.4

FORWARD 2

Use cross-section areas to compare volumes (Cavalieri) • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Two solids have height 10 and every horizontal cross-section of each has area 7. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q1/5

3285

QUESTION ID

S81-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Equal: 70 each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A hemisphere has radius 5. Find the area of its cross-section 3 units above the base.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q2/5

3286

QUESTION ID

S81-F2-Q2

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value. Radius 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q2/5

True or false?

Cavalieri's principle can compare two solids of different heights.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q3/5

3287

QUESTION ID

S81-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solids must have equal heights.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A hemisphere of radius r rests on its flat base. At height y above the base plane ($0 \le y \le r$), its horizontal cross-section has area

A. $\pi(r^2 - y^2)$

B. $\pi(r - y)^2$

C. πr^2

D. πy^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q4/5

3288

QUESTION ID

S81-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $\pi(r^2 - y^2)$

GRADING NOTE Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q4/5

Explain why cross-section areas are useful for finding volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q5/5

3289

QUESTION ID

S81-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Volume can be built from thin slices; equal slice areas at each level give equal volumes.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q5/5

SET 82 • HSG-GMD.B.4 • GEOMETRIC MEASUREMENT AND DIMENSION

Identify three-dimensional objects generated by rotations of two-dimensional objects

MAIN	Geometry main question	**HSG-GMD.B.4**
BACKWARD 1	Rotation of a point about a center	**HSG-CO.A.4**
BACKWARD 2	Cross-sections perpendicular to the axis	**HSG-GMD.B.4**
BACKWARD 3	Names of solids	**6.G.A.4**
FORWARD 1	Volumes of solids of rotation	**HSG-GMD.B.4**
FORWARD 2	Design an object as a solid of rotation	**HSG-MG.A.3**

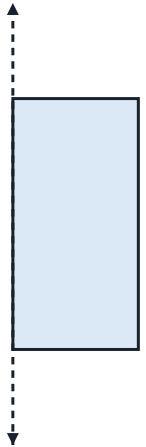
SET 82 • HSG-GMD.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The rectangle is rotated 360° about the dashed line. Name the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q1/5

3292

QUESTION ID

S82-M-Q1

TYPE

Short answer

CORRECT ANSWER

A cylinder

GRADING NOTE Any equivalent answer.

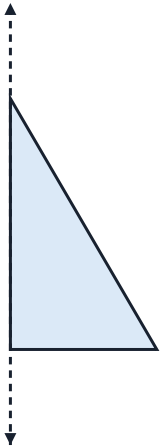
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The right triangle is rotated 360° about the dashed line. Name the solid formed.

Solid: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q2/5

3293

QUESTION ID

S82-M-Q2

TYPE

Short answer

CORRECT ANSWER

A cone

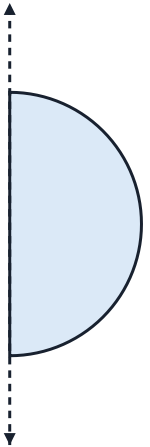
GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q2/5

3293

The semicircle is rotated 360° about the dashed line. Name the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q3/5

3294

QUESTION ID

S82-M-Q3

TYPE

Short answer

CORRECT ANSWER

A sphere

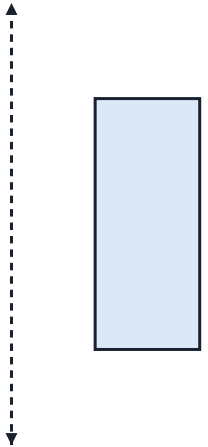
GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The rectangle is rotated 360° about the dashed line, which does not touch it. Describe the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q4/5

3295

QUESTION ID

S82-M-Q4

TYPE

Short answer

CORRECT ANSWER

A hollow cylinder (a cylinder with a cylindrical hole through the middle, like a pipe)

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle is rotated about its hypotenuse. The solid formed is

A. a sphere

B. one cone

C. two cones joined at their bases

D. a cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q5/5

3296

QUESTION ID

S82-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. two cones joined at their bases

GRADING NOTE Each leg sweeps a cone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 1

Rotation of a point about a center • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In a plane, rotating a point P (P different from the center O) 360° about O traces a circle with center O and radius OP.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q1/5

3298

QUESTION ID
S82-B1-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Every image is OP from O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q1/5

A point 3 cm from an axis is rotated about the axis. What does it trace?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q2/5

3299

QUESTION ID

S82-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A circle of radius 3 cm

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q2/5

A point ON the axis of rotation

A. traces a line

B. stays fixed

C. traces a circle

D. disappears

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q3/5

3300

QUESTION ID

S82-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. stays fixed

GRADING NOTE Distance 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q3/5

3300

ANSWER KEY Reduced copy of the question page, with its answer

A segment parallel to the axis, 2 units away, is rotated about it. What surface does it trace?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q4/5

3301

QUESTION ID

S82-B1-Q4

TYPE

Short answer

CORRECT ANSWER

The curved side of a cylinder

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q4/5

True or false?

Points farther from the axis trace smaller circles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q5/5

3302

QUESTION ID

S82-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Larger circles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 2

Cross-sections perpendicular to the axis • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

True or false?

A solid of rotation cut perpendicular to its axis has a circular cross-section (or a ring).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q1/5

3304

QUESTION ID

S82-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every point traces a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone is cut perpendicular to its axis. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q2/5

3305

QUESTION ID

S82-B2-Q2

TYPE

Short answer

CORRECT ANSWER

A circle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A pipe cut perpendicular to its axis gives a

A. rectangle

B. ring (annulus)

C. circle

D. triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q3/5

3306

QUESTION ID

S82-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ring (annulus)

GRADING NOTE Hole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere is cut through its center. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q4/5

3307

QUESTION ID

S82-B2-Q4

TYPE

Short answer

CORRECT ANSWER

A circle (a great circle)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q4/5

True or false?

A cylinder cut perpendicular to its axis gives a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q5/5

3308

QUESTION ID

S82-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 3

Names of solids • Grade 6 • Geometry • 6.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Name the solid with two parallel circular bases and a curved side.

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q1/5

3310

QUESTION ID

S82-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Cylinder

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Name the solid with one circular base and an apex.

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q2/5

3311

QUESTION ID

S82-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Cone

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q2/5

All points equidistant from a center in space form a

A. sphere

B. cone

C. circle

D. cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q3/5

3312

QUESTION ID

S82-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. sphere

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q3/5

True or false?

A cone has one flat face.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q4/5

3313

QUESTION ID

S82-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q4/5

True or false?

A sphere has a flat base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q5/5

3314

QUESTION ID

S82-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q5/5

SET 82 • HSG-GMD.B.4

FORWARD 1

Volumes of solids of rotation • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A 3-by-5 rectangle is rotated about its 5-unit side. Find the exact volume of the solid.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q1/5

3316

QUESTION ID
S82-F1-Q1

TYPE
Short answer

CORRECT ANSWER
45π

GRADING NOTE Form required: exact value. r = 3, h = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 3-by-5 rectangle is rotated about its 3-unit side. Find the exact volume of the solid.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q2/5

3317

QUESTION ID
S82-F1-Q2

TYPE
Short answer

CORRECT ANSWER
75π

GRADING NOTE Form required: exact value. r = 5, h = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q2/5

True or false?

Rotating one rectangle about two different sides can give solids with different volumes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q3/5

3318

QUESTION ID

S82-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A 3-by-5 rectangle gives 45π about its 5-unit side and 75π about its 3-unit side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle with legs 3 (on the axis) and 4 is rotated about the leg of length 3. Volume?

A. 48π

B. 9π

C. 12π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q4/5

3319

QUESTION ID

S82-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 16π

GRADING NOTE r = 4, h = 3: ⅓π · 16 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A semicircle of radius 3 is rotated about its diameter. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q5/5

3320

QUESTION ID
S82-F1-Q5

TYPE
Short answer

CORRECT ANSWER
36π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q5/5

SET 82 • HSG-GMD.B.4

FORWARD 2

Design an object as a solid of rotation • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A lathe turns a 2 cm by 10 cm rectangle about its 10 cm side to make a wooden dowel. Find the exact volume of wood.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q1/5

3322

QUESTION ID
S82-F2-Q1

TYPE
Short answer

CORRECT ANSWER
40π cm^3

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A designer wants a cone-shaped lampshade by rotating a right triangle about a 30 cm leg. What other leg length gives a base of diameter 40 cm?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q2/5

3323

QUESTION ID
S82-F2-Q2

TYPE
Short answer

CORRECT ANSWER
20 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q2/5

True or false?

A hemispherical bowl can be modeled by rotating a right triangle about one of its legs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q3/5

3324

QUESTION ID

S82-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That gives a cone; rotate a quarter-circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To make a ring-shaped washer by rotation, rotate a rectangle about a line in the rectangle's plane, parallel to one of its sides, that

A. passes through the rectangle's center

B. contains one side

C. lies outside the rectangle

D. cuts through the rectangle off-center

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q4/5

3325

QUESTION ID

S82-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. lies outside the rectangle

GRADING NOTE The gap between the axis and the rectangle becomes the hole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A vase is made by rotating a rectangle 6 cm wide and 20 cm tall about a line in the rectangle's plane, parallel to its long side and 2 cm from it, outside the rectangle. Find the exact volume of material.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q5/5

3326

QUESTION ID
S82-F2-Q5

TYPE
Short answer

CORRECT ANSWER
1,200π cm³

GRADING NOTE Form required: exact value. Outer radius 8, inner radius 2: π(64 − 4)(20).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q5/5

SET 83 • HSG-MG.A.1 • MODELING WITH GEOMETRY

Model objects with geometric shapes

MAIN	Geometry main question	**HSG-MG.A.1**
BACKWARD 1	Volume formulas	**HSG-GMD.A.3**
BACKWARD 2	Radius from a circumference	**7.G.B.4**
BACKWARD 3	Convert cubic units	**6.RP.A.3.d**
FORWARD 1	Judge a model against a measurement	**HSG-MG.A.1**
FORWARD 2	Use the model to estimate mass with density	**HSG-MG.A.2**

SET 83 • HSG-MG.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A tree trunk is about 6 m tall and has circumference 2.2 m. Model it as a cylinder and estimate its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q1/5

3329

QUESTION ID

S83-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

About 2.3 m³

GRADING NOTE REQUIRED WORK: Find the radius from the circumference, then use $V = \pi r^2 h$.

Rounding to 0.1 required. $r = 2.2/(2\pi) \approx 0.35$ m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pile of sand is shaped like a cone with base diameter 8 m and height 3 m. Estimate its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q2/5

3330

QUESTION ID

S83-M-Q2

TYPE

Short answer

CORRECT ANSWER

50.3 m^3

GRADING NOTE Rounding to 0.1 required. 16π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q2/5

Which shape is the best model for a soup can?

A. Cone

B. Cylinder

C. Square pyramid

D. Sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q3/5

3331

QUESTION ID

S83-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Cylinder

GRADING NOTE Two circular bases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A pool is modeled as a rectangular prism 25 m long, 10 m wide, and 1.8 m deep. How many liters of water does it hold? (1 m^3 = 1,000 L)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q4/5

3332

QUESTION ID

S83-M-Q4

TYPE

Short answer

CORRECT ANSWER

450,000 L

GRADING NOTE Any equivalent form. 450 m^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A pine tree is modeled as a cone. Name one measurement you would take and one way the model differs from the real tree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q5/5

3333

QUESTION ID
S83-M-Q5

TYPE
Short answer

CORRECT ANSWER
For example: height and the width at the bottom of the branches; the real tree has gaps, a trunk, and an uneven outline.

GRADING NOTE Any reasonable measurement and limitation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 1

Volume formulas • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Exact volume of a cylinder with r = 2 and h = 5?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q1/5

3335

QUESTION ID

S83-B1-Q1

TYPE

Short answer

CORRECT ANSWER

20π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact volume of a cone with r = 3 and h = 4?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q2/5

3336

QUESTION ID

S83-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a 2 × 3 × 4 prism?

A. 52

B. 9

C. 26

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q3/5

3337

QUESTION ID

S83-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 24

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$V = \frac{1}{3}\pi r^2 h$ is the cone formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q4/5

3338

QUESTION ID
S83-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q4/5

True or false?

A sphere's volume is $\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q5/5

3339

QUESTION ID

S83-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4/3\ \pi r^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 2

Radius from a circumference • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A circle has circumference 10π. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q1/5

3341

QUESTION ID

S83-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has circumference 2.2 m. Radius to the nearest hundredth?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q2/5

3342

QUESTION ID
S83-B2-Q2

TYPE
Short answer

CORRECT ANSWER
0.35 m

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

r =

A. C/(2π)

B. πC

C. 2πC

D. C/π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q3/5

3343

QUESTION ID

S83-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. C/(2π)

GRADING NOTE C = 2πr.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q3/5

True or false?

A tape measure around a trunk gives its circumference.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q4/5

3344

QUESTION ID
S83-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Distance around.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q4/5

True or false?

The radius is C/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q5/5

3345

QUESTION ID

S83-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE C/(2π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 3

Convert cubic units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

ANSWER KEY Reduced copy of the question page, with its answer

How many liters are in 2 m^3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q1/5

3347

QUESTION ID

S83-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2,000 L

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many cubic centimeters are in 1 L?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q2/5

3348

QUESTION ID

S83-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1,000 cm^3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

450 m^3 = ___ L

A. 45,000

B. 0.45

C. 450,000

D. 4,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q3/5

3349

QUESTION ID

S83-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 450,000

GRADING NOTE × 1,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1\ \text{mL} = 1\ \text{cm}^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q4/5

3350

QUESTION ID

S83-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1\ m^3 = 100\ L$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q5/5

3351

QUESTION ID

S83-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1,000 L.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q5/5

SET 83 • HSG-MG.A.1

FORWARD 1

Judge a model against a measurement • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A water bottle is modeled as a cylinder with radius 3.8 cm and height 17 cm. The label says 750 mL. Find the model volume to the nearest mL and the percent error.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q1/5

3353

QUESTION ID

S83-F1-Q1

TYPE

Short answer

CORRECT ANSWER

About 771 mL; about 2.8% too high

GRADING NOTE Form required: both (rounding to 1 mL and 0.1%). Percent error = (model − actual)/actual.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A silo is modeled as a cylinder (r = 3 m, h = 12 m) with a hemisphere on top. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q2/5

3354

QUESTION ID

S83-F1-Q2

TYPE

Short answer

CORRECT ANSWER

126π m³

GRADING NOTE Form required: exact value. 108π + 18π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q2/5

True or false?

A model that overestimates by 3% may still be useful for planning.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q3/5

3355

QUESTION ID

S83-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Judgment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A football (pointed ends) is best modeled by

A. two cones joined at their bases

B. a cube

C. a sphere

D. a cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q4/5

3356

QUESTION ID

S83-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. two cones joined at their bases

GRADING NOTE Pointed ends; other answers with reasons may be discussed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q4/5

Why might a cylinder model overestimate the volume of a bottle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q5/5

3357

QUESTION ID

S83-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The real bottle narrows at the neck and has thick walls, so it holds less than the full cylinder.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSG-MG.A.1

FORWARD 2

Use the model to estimate mass with density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A tree trunk modeled as a cylinder has volume about 2.3 m³. The wood has density 600 kg/m³. Estimate the trunk's mass to the nearest 10 kg.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q1/5

3359

QUESTION ID

S83-F2-Q1

TYPE

Short answer

CORRECT ANSWER

About 1,380 kg

GRADING NOTE Rounding to the nearest 10 kg required. 2.3 × 600 = 1,380 (using the unrounded volume 2.311 m³ gives 1390 kg, also accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sand pile modeled as a cone has volume 16π m³. The sand has density 1,600 kg/m³. Estimate its mass in metric tons (1 t = 1,000 kg), to the nearest ton.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q2/5

3360

QUESTION ID
S83-F2-Q2

TYPE
Short answer

CORRECT ANSWER
80 t

GRADING NOTE Rounding to the nearest ton required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q2/5

3360

True or false?

A more accurate volume model gives a more accurate mass estimate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q3/5

3361

QUESTION ID

S83-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Mass = density × volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A pool holds 450 m³ of water (1,000 kg/m³). Mass?

A. 450,000 kg

B. 450 kg

C. 45,000 kg

D. 4,500 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q4/5

3362

QUESTION ID

S83-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 450,000 kg

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q4/5

Explain how a geometric model and a density work together to estimate the mass of a log.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q5/5

3363

QUESTION ID
S83-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The model gives the volume; mass = density × volume.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q5/5

SET 84 • HSG-MG.A.2 • MODELING WITH GEOMETRY

Apply density based on area

MAIN	Geometry main question	**HSG-MG.A.2**
BACKWARD 1	Unit rates	**6.RP.A.3.b**
BACKWARD 2	Areas of rectangles and circles	**7.G.B.6**
BACKWARD 3	Compare ratios	**7.RP.A.2**
FORWARD 1	Convert area units in density problems	**HSG-MG.A.2**
FORWARD 2	Plan capacity from an area density	**HSG-MG.A.3**

SET 84 • HSG-MG.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A city has 840,000 people and an area of 120 square miles. Find its population density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q1/5

3366

QUESTION ID

S84-M-Q1

TYPE

Short answer

CORRECT ANSWER

7,000 people per square mile

GRADING NOTE Any equivalent form, with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q1/5

A circular forest region has radius 3 km and 4,500 trees. Find the tree density to the nearest tree per square kilometer.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q2/5

3367

QUESTION ID

S84-M-Q2

TYPE

Short answer

CORRECT ANSWER

159 trees per km²

GRADING NOTE Rounding to the nearest whole number required. Area $9\pi \approx 28.27$ km².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Town A has 12,000 people on 30 km². Town B has 9,000 people on 20 km². Which town is more densely populated? Show the densities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q3/5

3368

QUESTION ID

S84-M-Q3

TYPE

Short answer

CORRECT ANSWER

Town B (450 per km² vs 400 per km²)

GRADING NOTE Form required: the conclusion with both densities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A region has a population density of 250 people per km² and is a 4 km by 6 km rectangle. Estimate its population.

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q4/5

3369

QUESTION ID
S84-M-Q4

TYPE
Short answer

CORRECT ANSWER
6,000 people

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a unit of population density based on area?

A. people per mile

B. people per square mile

C. square miles per person per year

D. people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q5/5

3370

QUESTION ID

S84-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. people per square mile

GRADING NOTE Count ÷ area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 1

Unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

ANSWER KEY Reduced copy of the question page, with its answer

840 items in 12 boxes: items per box?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q1/5

3372

QUESTION ID

S84-B1-Q1

TYPE

Short answer

CORRECT ANSWER

70

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$45 for 9 m² of tile: dollars per square meter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q2/5

3373

QUESTION ID

S84-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$5 per m²

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Unit rate for 300 people on 4 km²?

A. 0.013 km² per person

B. 1,200 people per km²

C. 296 people per km²

D. 75 people per km²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q3/5

3374

QUESTION ID

S84-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 75 people per km²

GRADING NOTE 300 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q3/5

True or false?

A unit rate compares a quantity to one unit of another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q4/5

3375

QUESTION ID

S84-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q4/5

True or false?

600 people on 3 km² is 1,800 people per km².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q5/5

3376

QUESTION ID

S84-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 2

Areas of rectangles and circles • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Area of a 4 km by 6 km rectangle?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q1/5

3378

QUESTION ID

S84-B2-Q1

TYPE

Short answer

CORRECT ANSWER

24 km²

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle with radius 3 km?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q2/5

3379

QUESTION ID

S84-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9π km²

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A square region 5 mi on a side has area

A. 20 mi²

B. 10 mi²

C. 5 mi²

D. 25 mi²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q3/5

3380

QUESTION ID

S84-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 25 mi²

GRADING NOTE 5^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q3/5

3380

True or false?

Area units are square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q4/5

3381

QUESTION ID

S84-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two dimensions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q4/5

True or false?

A circle with radius 3 has area 6π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q5/5

3382

QUESTION ID

S84-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 3

Compare ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Which is greater: 12,000/30 or 9,000/20?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q1/5

3384

QUESTION ID

S84-B3-Q1

TYPE

Short answer

CORRECT ANSWER

9,000/20 (450 > 400)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q1/5

True or false?

To compare densities, compare the unit rates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q2/5

3385

QUESTION ID

S84-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which rate is greatest?

A. 80 per 4 km²

B. 50 per 2 km²

C. 60 per 3 km²

D. 20 per 1 km²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q3/5

3386

QUESTION ID

S84-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 50 per 2 km²

GRADING NOTE 25, 20, 20, 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A has 100 trees on 4 acres; B has 120 trees on 6 acres. Which is denser?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q4/5

3387

QUESTION ID

S84-B3-Q4

TYPE

Short answer

CORRECT ANSWER

A (25 per acre vs 20)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The town with more people is always denser.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q5/5

3388

QUESTION ID

S84-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Depends on area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q5/5

SET 84 • HSG-MG.A.2

FORWARD 1

Convert area units in density problems • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

1 square mile is about 2.59 km². Convert 7,000 people per square mile to people per km² (nearest whole number).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q1/5

3390

QUESTION ID
S84-F1-Q1

TYPE
Short answer

CORRECT ANSWER
2703 people per km²

GRADING NOTE Rounding to the nearest whole number required. Divide by 2.59.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q1/5

3390

ANSWER KEY Reduced copy of the question page, with its answer

A lawn of 400 m² needs grass seed at 35 g per m². How many kilograms of seed are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q2/5

3391

QUESTION ID

S84-F1-Q2

TYPE

Short answer

CORRECT ANSWER

14 kg

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q2/5

True or false?

A density in people per km² is a larger number than the same density in people per mi².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q3/5

3392

QUESTION ID

S84-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A km² is smaller, so fewer people per km².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A farm plants 2 trees per 10 m^2. How many trees on 1 hectare (10,000 m^2)?

A. 2,000

B. 200

C. 5,000

D. 20,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q4/5

3393

QUESTION ID

S84-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2,000

GRADING NOTE 0.2 per m^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q4/5

3393

ANSWER KEY Reduced copy of the question page, with its answer

A park of 3 hectares (1 ha = 10,000 m²) has 5,400 visitors. Visitors per 100 m²?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q5/5

3394

QUESTION ID

S84-F1-Q5

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Any equivalent form. 300 units of 100 m².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q5/5

3394

SET 84 • HSG-MG.A.2

FORWARD 2

Plan capacity from an area density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Safety rules allow 2 people per m^2 in a plaza. The plaza is a 30 m by 40 m rectangle. Maximum crowd?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q1/5

3396

QUESTION ID

S84-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2,400 people

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A festival expects 5,000 people and wants at most 1.5 people per m². What is the smallest whole number of square meters that meets the limit?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q2/5

3397

QUESTION ID

S84-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3,334 m²

GRADING NOTE Form required: a whole number of m² that meets the limit. 5,000 ÷ 1.5 = 3,333.3…, so round UP to 3,334 (3,333 m² would allow only 4,999.5 people).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q2/5

True or false?

When planning for a safety limit, round the capacity up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q3/5

3398

QUESTION ID

S84-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Round capacity down (and area up) to stay within the limit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q3/5

A circular dance floor of radius 5 m allows 3 people per m². Capacity is about

A. 235 people

B. 471 people

C. 75 people

D. 47 people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q4/5

3399

QUESTION ID

S84-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 235 people

GRADING NOTE $75\pi \approx 235.6$, round down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q4/5

3399

ANSWER KEY Reduced copy of the question page, with its answer

A circular dance floor of radius 5 m allows at most 3 people per m², and $3 \cdot 25\pi \approx 235.6$. Explain why the capacity is 235 people, not 236.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q5/5

3400

QUESTION ID

S84-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Rounding up would exceed the allowed density.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q5/5

SET 85 • HSG-MG.A.2 • MODELING WITH GEOMETRY

Apply density based on volume

MAIN	Geometry main question	**HSG-MG.A.2**
BACKWARD 1	Volume of prisms and cylinders	**HSG-GMD.A.3**
BACKWARD 2	Unit rates	**6.RP.A.3.b**
BACKWARD 3	Products of numbers in scientific notation	**8.EE.A.4**
FORWARD 1	Use density to identify materials and find volumes	**HSG-MG.A.2**
FORWARD 2	Design within a mass limit	**HSG-MG.A.3**

SET 85 • HSG-MG.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

An aluminum block measures 2 cm by 3 cm by 5 cm and has mass 81 g. Find its density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q1/5

3403

QUESTION ID
S85-M-Q1

TYPE
Short answer

CORRECT ANSWER
2.7 g/cm³

GRADING NOTE Any equivalent form, with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical tank (radius 0.5 m, height 2 m) is full of water (1,000 kg/m³). Find the mass of the water to the nearest kilogram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q2/5

3404

QUESTION ID

S85-M-Q2

TYPE

Short answer

CORRECT ANSWER

1571 kg

GRADING NOTE Rounding to the nearest kilogram required. 0.5π m³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A heater must supply 20 BTU per cubic foot of room. A room is 12 ft by 15 ft with an 8 ft ceiling. How many BTU are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q3/5

3405

QUESTION ID

S85-M-Q3

TYPE

Short answer

CORRECT ANSWER

28,800 BTU

GRADING NOTE Any equivalent form. 1,440 ft^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A sample has 3×10^6 bacteria per mL. How many bacteria are in 250 mL?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q4/5

3406

QUESTION ID
S85-M-Q4

TYPE
Short answer

CORRECT ANSWER
7.5×10^8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q4/5

Water has density 1 g/cm³. Which object floats in water?

A. A 1,000 g brick with volume 400 cm³

B. A 600 g block with volume 500 cm³

C. A 100 g ball with volume 50 cm³

D. A 400 g block with volume 500 cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q5/5

3407

QUESTION ID

S85-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. A 400 g block with volume 500 cm³

GRADING NOTE Density less than 1 g/cm³ (0.8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 1

Volume of prisms and cylinders • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a 2 × 3 × 5 cm block?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q1/5

3409

QUESTION ID

S85-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-B1-Q1**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact volume of a cylinder with r = 0.5 m, h = 2 m?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q2/5

3410

QUESTION ID

S85-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.5π m³

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q2/5

3410

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a 12 × 15 × 8 ft room?

A. 1,440 ft³

B. 180 ft³

C. 2,880 ft³

D. 35 ft³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q3/5

3411

QUESTION ID

S85-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1,440 ft³

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q3/5

True or false?

Volume is needed before a volume density can be used.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q4/5

3412

QUESTION ID
S85-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Density = amount ÷ volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A 1 m cube has volume 3 m³.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q5/5

3413

QUESTION ID

S85-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1 m³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 2

Unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

ANSWER KEY Reduced copy of the question page, with its answer

81 g in 30 cm³: grams per cm³?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q1/5

3415

QUESTION ID

S85-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2.7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

20 BTU per ft^3 for 100 ft^3: total?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q2/5

3416

QUESTION ID

S85-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2,000 BTU

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

500 g in 250 cm³ is

A. 250 g/cm³

B. 2 g/cm³

C. 0.5 g/cm³

D. 750 g/cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q3/5

3417

QUESTION ID

S85-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2 g/cm³

GRADING NOTE Divide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q3/5

True or false?

Amount = rate × quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q4/5

3418

QUESTION ID

S85-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rate use.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Density = volume ÷ mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q5/5

3419

QUESTION ID

S85-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Mass ÷ volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 3

Products of numbers in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$(3 \times 10^6) \times 250$ in scientific notation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q1/5

3421

QUESTION ID

S85-B3-Q1

TYPE

Short answer

CORRECT ANSWER

7.5×10^8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 2,400,000 in scientific notation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q2/5

3422

QUESTION ID

S85-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2.4×10^6

GRADING NOTE Form required: scientific notation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(2 \times 10^3)(4 \times 10^2) =$

A. 6×10^5

B. 8×10^5

C. 8×10^6

D. 8×10^1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q3/5

3423

QUESTION ID

S85-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8×10^5

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$10^3 \times 10^2 = 10^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q4/5

3424

QUESTION ID

S85-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Exponent rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$5 \times 10^2 \times 3 = 15 \times 10^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q5/5

3425

QUESTION ID

S85-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1.5×10^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q5/5

SET 85 • HSG-MG.A.2

FORWARD 1

Use density to identify materials and find volumes • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A ring has mass 9.65 g and volume 0.5 cm³. Gold has density 19.3 g/cm³. Could the ring be pure gold?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q1/5

3427

QUESTION ID

S85-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes:

9.65 ÷ 0.5 = 19.3 g/cm³

GRADING NOTE Form required: the conclusion with the density.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 1,350 g piece of aluminum (2.7 g/cm³) is melted into a cube. Find the edge of the cube, to the nearest tenth of a centimeter.

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q2/5

3428

QUESTION ID

S85-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7.9 cm

GRADING NOTE Rounding to 0.1 required. Volume 500 cm³; edge $\sqrt[3]{500}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q2/5

True or false?

Two objects with equal mass but different volumes have different densities.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q3/5

3429

QUESTION ID

S85-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Density = m/V.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q3/5

Oil (0.9 g/cm³) in a 2,000 cm³ jug has mass

A. 1,800 g

B. 2,000 g

C. 2,222 g

D. 1,100 g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q4/5

3430

QUESTION ID

S85-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1,800 g

GRADING NOTE 0.9 × 2,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q4/5

A balloon holds 0.02 m^3 of helium (0.18 kg/m^3). Mass of the helium in grams?

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q5/5

3431

QUESTION ID

S85-F1-Q5

TYPE

Short answer

CORRECT ANSWER

3.6 g

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q5/5

SET 85 • HSG-MG.A.2

FORWARD 2

Design within a mass limit • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A truck can carry 12,000 kg. Gravel has density 1,500 kg/m³. What is the largest volume of gravel it can carry?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q1/5

3433

QUESTION ID

S85-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8 m³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A shelf holds at most 40 kg. Books have density 0.8 g/cm³. How many cubic centimeters of books can it hold?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q2/5

3434

QUESTION ID
S85-F2-Q2

TYPE
Short answer

CORRECT ANSWER
50,000 cm³

GRADING NOTE Any equivalent form. 40,000 g ÷ 0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q2/5

True or false?

Mass limits become volume limits by multiplying by the density.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q3/5

3435

QUESTION ID

S85-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide by the density: V = m/d.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical water tank must weigh at most 3,140 kg when full (water 1,000 kg/m³, ignore the tank). With radius 1 m, its height can be at most about

A. 10 m

B. 3.14 m

C. 1 m

D. 0.32 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q4/5

3436

QUESTION ID

S85-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 m

GRADING NOTE 3.14 m³ ÷ π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A concrete block must be under 25 kg. Concrete is 2,400 kg/m³. Will a 20 cm × 20 cm × 30 cm block work?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q5/5

3437

QUESTION ID

S85-F2-Q5

TYPE

Short answer

CORRECT ANSWER

No: 0.012 m³ × 2,400 kg/m³ = 28.8 kg, which exceeds 25 kg.

GRADING NOTE Form required: the conclusion with the mass.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q5/5

SET 86 • HSG-MG.A.3 • MODELING WITH GEOMETRY

Design to satisfy physical constraints

MAIN	Geometry main question	**HSG-MG.A.3**
BACKWARD 1	Volume of a rectangular prism	**7.G.B.6**
BACKWARD 2	Use the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Ratios and rates	**7.RP.A.2**
BACKWARD 4	Inequalities as limits	**7.EE.B.4**
FORWARD 1	Satisfy several constraints at once	**HSG-MG.A.3**
FORWARD 2	Find the best design with a function graph	**HSF-IF.C.7.c**

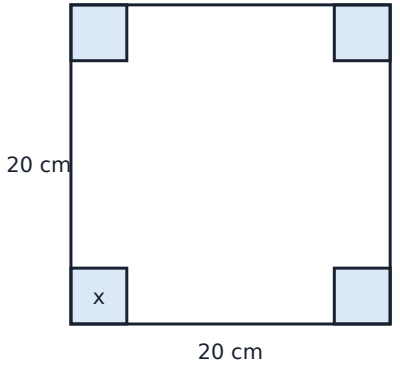
SET 86 • HSG-MG.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Compare the volumes for x = 2, 3, and 4. Which gives the largest box?



Not drawn to scale.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q1/5

3440

QUESTION ID

S86-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 2: 512 cm³; x = 3: 588 cm³; x = 4: 576 cm³; x = 3 is largest

GRADING NOTE Form required: the three volumes and the choice. $V = x(20 - 2x)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Accessibility rules allow a ramp slope of at most 1 : 12 (1 unit of rise per 12 units of run). A ramp must rise 30 in. What is the shortest horizontal run allowed, in feet?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q2/5

3441

QUESTION ID
S86-M-Q2

TYPE
Short answer

CORRECT ANSWER
30 ft

GRADING NOTE Any equivalent form. 30 × 12 = 360 in.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q2/5

A rectangular pen will use 40 m of fencing. Compare a 10 m by 10 m pen with a 5 m by 15 m pen. Which encloses more area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q3/5

3442

QUESTION ID

S86-M-Q3

TYPE

Short answer

CORRECT ANSWER

The 10 m by 10 m pen (100 m² vs 75 m²)

GRADING NOTE Form required: the choice with both areas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical tank must fit in a closet 2 m wide, 2 m deep, and 3 m tall. What is the largest possible volume, to the nearest tenth of a cubic meter?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q4/5

3443

QUESTION ID

S86-M-Q4

TYPE

Short answer

CORRECT ANSWER

9.4 m³

GRADING NOTE Rounding to 0.1 required. Radius 1 m, height 3 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A 13 ft ladder leans on a wall. For safety its base must be at least 5 ft from the wall. The highest point it can reach is

A. 18 ft

B. 12 ft

C. 13 ft

D. 8 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q5/5

3444

QUESTION ID

S86-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 12 ft

GRADING NOTE √(169 − 25).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 1

Volume of a rectangular prism • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a box 14 cm by 14 cm by 3 cm?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q1/5

3446

QUESTION ID

S86-B1-Q1

TYPE

Short answer

CORRECT ANSWER

588 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 20 cm side with 3 cm cut from each end leaves

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q2/5

3447

QUESTION ID

S86-B1-Q2

TYPE

Short answer

CORRECT ANSWER

14 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Volume of a box 12 by 12 by 4?

A. 28

B. 576

C. 48

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q3/5

3448

QUESTION ID

S86-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 576

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q3/5

True or false?

Cutting larger corners makes the box taller but its base smaller.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q4/5

3449

QUESTION ID

S86-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Trade-off.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q4/5

True or false?

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner. The box's base side is 20 − x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q5/5

3450

QUESTION ID
S86-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 20 − 2x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 2

Use the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

Hypotenuse 13, leg 5: other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q1/5

3452

QUESTION ID

S86-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Legs 30 in. and 360 in.: hypotenuse to the nearest inch?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q2/5

3453

QUESTION ID

S86-B2-Q2

TYPE

Short answer

CORRECT ANSWER

361 in.

GRADING NOTE Rounding to the nearest inch required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q2/5

A ladder, wall, and ground form a

A. right triangle

B. circle

C. square

D. equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q3/5

3454

QUESTION ID

S86-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. right triangle

GRADING NOTE Wall ⊥ ground.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q3/5

True or false?

A ladder leans against a vertical wall on level ground. The ladder is the hypotenuse of the right triangle formed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q4/5

3455

QUESTION ID

S86-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Opposite the right angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q4/5

True or false?

The ladder's height on the wall is 13 + 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q5/5

3456

QUESTION ID

S86-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use the theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 3

Ratios and rates • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A ratio of 1 : 12 for a rise of 2 in. gives what run?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q1/5

3458

QUESTION ID

S86-B3-Q1

TYPE

Short answer

CORRECT ANSWER

24 in.

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q1/5

Is a slope of 3 : 30 steeper than 1 : 12?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q2/5

3459

QUESTION ID

S86-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: 3 : 30 = 1 : 10, and 1/10 > 1/12.

GRADING NOTE Form required: yes/no with the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

360 in. =

A. 30 ft

B. 36 ft

C. 12 ft

D. 3.6 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q3/5

3460

GRADING NOTE ÷ 12.

QUESTION ID

S86-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 30 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q3/5

True or false?

A ratio of 1 : 12 means 1 inch up for every 12 inches across.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q4/5

3461

QUESTION ID

S86-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A 1 : 8 ramp is less steep than a 1 : 12 ramp.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q5/5

3462

QUESTION ID

S86-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Steeper.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 4

Inequalities as limits • Grade 7 • Expressions & Equations • 7.EE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The run r must satisfy r ≥ 360. Is r = 300 allowed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q1/5

3464

QUESTION ID

S86-B4-Q1

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A diameter d must satisfy d ≤ 2. Largest radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q2/5

3465

QUESTION ID

S86-B4-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

"At most 3 m" means

A. ≥ 3

B. < 3 only

C. ≤ 3

D. = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q3/5

3466

QUESTION ID

S86-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. ≤ 3

GRADING NOTE Inequality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

"At least 5 ft" allows exactly 5 ft.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q4/5

3467

QUESTION ID

S86-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ≥.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

"At most 12" excludes 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q5/5

3468

QUESTION ID

S86-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Includes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q5/5

SET 86 • HSG-MG.A.3

FORWARD 1

Satisfy several constraints at once • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Which whole-number values of x give a volume of at least 500 cm³ AND a height of at most 3 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q1/5

3470

QUESTION ID

S86-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 2 and x = 3

GRADING NOTE Form required: both values. Volumes 512 and 588; x = 4, 5 are too tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q1/5

A ramp must rise 2 ft and fit within 20 ft of horizontal space. Can it meet the 1 : 12 limit? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q2/5

3471

QUESTION ID

S86-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No: it needs 24 ft of run.

GRADING NOTE Form required: the conclusion with the run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q2/5

True or false?

A design that satisfies one constraint may fail another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q3/5

3472

QUESTION ID

S86-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Check all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangular garden needs area 48 m² and width at most 6 m. Which works?

A. 6 m wide by 8 m long

B. 4 m wide by 10 m long

C. 7 m wide by 7 m long

D. 8 m wide by 6 m long

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q4/5

3473

QUESTION ID

S86-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6 m wide by 8 m long

GRADING NOTE Width 6, area 48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder 3 m tall must hold at least 10 m^3 and stand on a 2 m by 2 m floor space. Is that possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q5/5

3474

QUESTION ID

S86-F1-Q5

TYPE

Short answer

CORRECT ANSWER

No: the largest is $r = 1$ m, giving $3\pi \approx 9.4\ m^3 < 10\ m^3$.

GRADING NOTE Form required: the conclusion with the volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q5/5

SET 86 • HSG-MG.A.3

FORWARD 2

Find the best design with a function graph • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Write its volume as a function V(x) and state the domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q1/5

3476

QUESTION ID

S86-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$V(x) = x(20 - 2x)^2$, $0 < x < 10$

GRADING NOTE Form required: the function and the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An open box made from a 20 cm by 20 cm sheet with x-cm corner squares cut out has volume $V(x) = x(20 - 2x)^2$. Use a graph or table to estimate the x that gives the largest volume, to the nearest tenth, and that volume to the nearest cm^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q2/5

3477

QUESTION ID

S86-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x \approx 3.3$; $V \approx 593\ cm^3$

GRADING NOTE Form required: both. (Exact: x = 10/3, V = 16,000/27 ≈ 592.6.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$V(x) = x(20 - 2x)^2$ has zeros at $x = 0$ and $x = 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q3/5

3478

QUESTION ID

S86-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The zeros are $x = 0$ and $x = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For an open box made from a 20 cm by 20 cm sheet, $V(x) = x(20 - 2x)^2$. The graph of V on $0 < x < 10$ rises and then falls. Its highest point gives

A. the cut that makes no box

B. the base area

C. the largest volume

D. the smallest volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q4/5

3479

QUESTION ID

S86-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. the largest volume

GRADING NOTE Maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S86-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For an open box made from a 20 cm by 20 cm sheet with x-cm corner cuts, $V(x) = x(20 - 2x)^2$ and V is largest near x = 3.3. Why is x = 5 not the best cut even though it is halfway between 0 and 10?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q5/5

3480

QUESTION ID
S86-F2-Q5

TYPE
Short answer

CORRECT ANSWER
V(5) = 500 < 593; the maximum of a cubic need not be at the middle.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q5/5

SET 87 • HSG-MG.A.3 • MODELING WITH GEOMETRY

Design to minimize cost

MAIN	Geometry main question	**HSG-MG.A.3**
BACKWARD 1	Surface area of prisms and cylinders	**7.G.B.6**
BACKWARD 4	Areas of surfaces with openings	**7.G.B.6**
BACKWARD 3	Unit prices and rounding up purchases	**7.EE.B.3**
FORWARD 1	Compare designs with equal volume for least material	**HSG-MG.A.3**
FORWARD 2	Decide between designs using expected cost	**HSS-MD.B.7**

SET 87 • HSG-MG.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A can with radius 4 cm and height 10 cm is made of metal costing \$0.002 per cm². Find the area of metal (top, bottom, and side) and the cost, to the nearest cent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q1/5

3483

QUESTION ID
S87-M-Q1

TYPE
Short answer

CORRECT ANSWER
112π cm² ≈ 351.9 cm²; cost \$0.70

GRADING NOTE Form required: the area and the cost. $2\pi r^2 + 2\pi rh = 32\pi + 80\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Can A has r = 4 cm, h = 10 cm. Can B has r = 5 cm, h = 6.4 cm. Show that they hold the same volume, then decide which uses less metal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q2/5

3484

QUESTION ID
S87-M-Q2

TYPE
Short answer

CORRECT ANSWER
Both hold 160π cm³; A uses 112π cm², B uses 114π cm², so A uses less metal.

GRADING NOTE Form required: both volumes, both areas, and the choice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A wall 12 m by 3 m has a window 1.5 m by 1.2 m. Paint covers 10 m² per liter and is sold only in whole liters at $8 each. Find the cost to paint the wall once.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q3/5

3485

QUESTION ID
S87-M-Q3

TYPE
Short answer

CORRECT ANSWER
$32

GRADING NOTE Any equivalent form. 34.2 m² needs 3.42 L, so buy 4 L.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two closed cardboard boxes each hold 600 cm³: box A is 10 cm by 10 cm by 6 cm and box B is 5 cm by 10 cm by 12 cm. Cardboard costs $0.01 per cm² (ignore thickness and overlaps). Find the cost of each box and choose the cheaper one.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q4/5

3486

QUESTION ID

S87-M-Q4

TYPE

Short answer

CORRECT ANSWER

A: 440 cm², $4.40; B: 460 cm², $4.60; box A is cheaper.

GRADING NOTE Form required: both surface areas, both costs, and the choice. A: 2(100 + 60 + 60); B: 2(50 + 60 + 120).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A 4 m by 3.5 m floor is covered with 0.5 m square tiles that cost $3.25 each and are sold individually. Plan A uses these tiles; plan B uses sheet flooring at $11 per m². Find the cost of each plan and choose the cheaper one.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q5/5

3487

QUESTION ID

S87-M-Q5

TYPE

Short answer

CORRECT ANSWER

A: 8 × 7 = 56 tiles, $182; B: 14 m² × $11 = $154; plan B is cheaper.

GRADING NOTE Form required: both costs and the choice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q5/5

SET 87 • HSG-MG.A.3

BACKWARD 1

Surface area of prisms and cylinders • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of the two circular ends of a can with radius 4?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q1/5

3489

QUESTION ID

S87-B1-Q1

TYPE

Short answer

CORRECT ANSWER

32π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of the side of a can with radius 4 and height 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q2/5

3490

QUESTION ID

S87-B1-Q2

TYPE

Short answer

CORRECT ANSWER

80π

GRADING NOTE Form required: exact value. 2πrh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q2/5

3490

ANSWER KEY Reduced copy of the question page, with its answer

Surface area of a 2 × 3 × 4 box?

A. 24

B. 48

C. 52

D. 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q3/5

3491

QUESTION ID

S87-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 52

GRADING NOTE 2(6 + 8 + 12).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q3/5

True or false?

The side of a cylinder unrolls into a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q4/5

3492

QUESTION ID

S87-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Net.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q4/5

True or false?

A cylinder's surface area is $\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q5/5

3493

QUESTION ID

S87-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q5/5

SET 87 • HSG-MG.A.3

BACKWARD 4

Areas of surfaces with openings • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A wall 12 m by 3 m has a 1.5 m by 1.2 m window. Area to paint?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q1/5

3495

QUESTION ID

S87-B4-Q1

TYPE

Short answer

CORRECT ANSWER

34.2 m²

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q1/5

3495

ANSWER KEY Reduced copy of the question page, with its answer

A 4 m by 3.5 m floor: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q2/5

3496

QUESTION ID

S87-B4-Q2

TYPE

Short answer

CORRECT ANSWER

14 m²

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q2/5

A 5 m by 2.4 m wall has a 0.9 m by 2 m door. Paintable area?

A. 10.2 m²

B. 12 m²

C. 1.8 m²

D. 13.8 m²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q3/5

3497

QUESTION ID

S87-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10.2 m²

GRADING NOTE 12 − 1.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q3/5

True or false?

Openings in a surface are subtracted from its area before buying material.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q4/5

3498

QUESTION ID

S87-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE No material is needed there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q4/5

True or false?

A floor of 14 m² needs 14 tiles that are 0.5 m on a side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q5/5

3499

QUESTION ID

S87-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each tile is 0.25 m², so 56 tiles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B4-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q5/5

3499

SET 87 • HSG-MG.A.3

BACKWARD 3

Unit prices and rounding up purchases • Grade 7 • Expressions & Equations • 7.EE.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Paint needed is 3.42 L, sold in whole liters. How many liters must be bought?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q1/5

3501

QUESTION ID

S87-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Metal costs $0.002 per cm². Cost of 350 cm²?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q2/5

3502

QUESTION ID

S87-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$0.70

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Fabric is $6 per yard, sold in whole yards. You need 4.2 yd. Cost?

A. $30

B. $25.20

C. $24

D. $26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q3/5

3503

QUESTION ID
S87-B3-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. $30

GRADING NOTE Buy 5 yd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

When items are sold in whole units, round the amount needed up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q4/5

3504

QUESTION ID

S87-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Enough material.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q4/5

3504

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

3.1 cans of paint can be bought as 3 cans.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q5/5

3505

QUESTION ID

S87-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Need 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q5/5

3505

SET 87 • HSG-MG.A.3

FORWARD 1

Compare designs with equal volume for least material • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A closed box with a square base must hold 1,000 cm^3 (ignore the thickness of the material). Compare the surface areas for base sides 8, 10, and 12 cm. Which uses the least material?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q1/5

3507

QUESTION ID

S87-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8 cm: 628 cm^2; 10 cm: 600 cm^2; 12 cm: about 621 cm^2; the 10 cm cube uses the least

GRADING NOTE Form required: the three areas and the choice. $h = 1000/s^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A closed cylinder must hold 1,000 cm^3. Compare its surface area (to the nearest cm^2) for radii 4 cm, 5.42 cm and 7 cm, and choose the radius that uses the least material.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q2/5

3508

QUESTION ID

S87-F1-Q2

TYPE

Short answer

CORRECT ANSWER

r = 4: 601 cm^2; r = 5.42: 554 cm^2; r = 7: 594 cm^2; r = 5.42 cm uses the least.

GRADING NOTE Form required: three areas and the choice. $h = 1000/(\pi r^2)$, so the area is $2\pi r^2 + 2000/r$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q2/5

True or false?

Among closed boxes with a square base and a fixed volume (ignoring material thickness), the cube has the least surface area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q3/5

3509

QUESTION ID
S87-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Classic result.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q3/5

For a closed cylinder with a fixed volume, the least surface area occurs when

A. the height equals the radius

B. the height is as large as possible

C. the radius is as large as possible

D. the height equals the diameter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q4/5

3510

QUESTION ID

S87-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. the height equals the diameter

GRADING NOTE h = 2r (r = 5.42 cm gives h ≈ 10.84 cm).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give one reason a manufacturer might NOT choose the least-material can shape.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q5/5

3511

QUESTION ID

S87-F1-Q5

TYPE

Short answer

CORRECT ANSWER

For example: it is hard to grip, does not fit standard shelves, or leaves too little label space.

GRADING NOTE Any reasonable reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q5/5

SET 87 • HSG-MG.A.3

FORWARD 2

Decide between designs using expected cost • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

Packaging A costs $1.20 per item and 2% of items break. Packaging B costs $0.90 and 8% break. A break costs $20. Find the expected cost per item for each and choose.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q1/5

3513

QUESTION ID
S87-F2-Q1

TYPE
Short answer

CORRECT ANSWER
A: $1.20 + 0.02 × $20 = $1.60; B: $0.90 + 0.08 × $20 = $2.50; choose A

GRADING NOTE Form required: both expected costs and the choice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The standard metal for a can never fails. A cheaper metal fails 5% of the time, and each failure costs $40 to replace. By how much must the cheaper metal's price per can be below the standard price for its expected cost per can to be strictly lower?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q2/5

3514

QUESTION ID
S87-F2-Q2

TYPE
Short answer

CORRECT ANSWER
By more than $2.00 (0.05 × $40 = $2.00 expected failure cost)

GRADING NOTE Form required: "more than $2.00"; exactly $2.00 cheaper gives equal expected costs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q2/5

True or false?

The cheapest material always gives the cheapest design.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q3/5

3515

QUESTION ID
S87-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Failures and replacement costs can make it more expensive overall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q3/5

Expected cost =

A. upfront cost × probability

B. upfront cost + probability of failure × cost of failure

C. cost of failure only

D. upfront cost − probability

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q4/5

3516

QUESTION ID
S87-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. upfront cost + probability of failure × cost of failure

GRADING NOTE Weighted cost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name one cost besides material that a can designer might consider.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q5/5

3517

QUESTION ID
S87-F2-Q5

TYPE
Short answer

CORRECT ANSWER
For example shipping weight, labor, breakage, or recycling.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q5/5

SET 105 • HSG-MG.A.3 • MODELING WITH GEOMETRY

Design typographic grid systems

MAIN	Geometry main question	**HSG-MG.A.3**
BACKWARD 1	Proportional reasoning with ratios	**7.RP.A.2**
BACKWARD 2	Operate with decimals	**6.NS.B.3**
BACKWARD 3	Factors of a whole number	**4.OA.B.4**
FORWARD 1	Choose grids that support several layouts	**HSG-MG.A.3**
FORWARD 2	Scale a grid layout to a new page size	**HSG-SRT.A.2**

SET 105 • HSG-MG.A.3

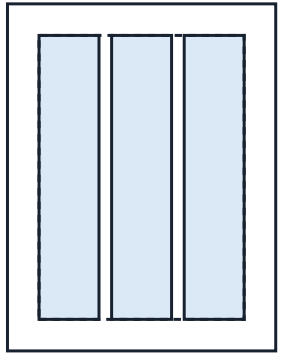
MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A page is 8.5 in. wide with 1 in. margins on each side. The text area has 3 equal columns separated by 2 gutters of 0.2 in. Find the width of each column to the nearest hundredth of an inch.

Width: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q1/5

3520

QUESTION ID
S105-M-Q1

TYPE
Short answer

CORRECT ANSWER
2.03 in.

GRADING NOTE Rounding to 0.01 required. (6.5 − 0.4) ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q1/5

3520

ANSWER KEY Reduced copy of the question page, with its answer

A poster grid uses the golden ratio (about 1.618 : 1) for height : width. A poster is 60 cm wide. Its height is about

A. 120 cm

B. 97.1 cm

C. 37.1 cm

D. 61.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q2/5

3521

QUESTION ID

S105-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 97.1 cm

GRADING NOTE 60 × 1.618.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A 12-column grid on a 6.5 in. text area has 11 gutters of 0.125 in. Find the width of each column to the nearest thousandth of an inch.

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q3/5

3522

QUESTION ID

S105-M-Q3

TYPE

Short answer

CORRECT ANSWER

0.427 in.

GRADING NOTE Rounding to 0.001 required. (6.5 − 1.375) ÷ 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q3/5

A web page 1,200 px wide uses a 12-column grid with 24 px gutters between columns and no outer margins. Find the column width and the width of an element that spans 4 columns (including the gutters inside it).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q4/5

3523

QUESTION ID

S105-M-Q4

TYPE

Short answer

CORRECT ANSWER

Column 78 px; 4-column element 384 px

GRADING NOTE Form required: both. (1,200 − 11 · 24) ÷ 12 = 78; 4 · 78 + 3 · 24 = 384.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A 6 in. by 9 in. page has 0.75 in. margins on all sides. A modular grid has 3 columns and 4 rows with 0.25 in. gutters in both directions. Find the width and height of each module to the nearest hundredth of an inch.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q5/5

3524

QUESTION ID
S105-M-Q5

TYPE
Short answer

CORRECT ANSWER
Width 1.33 in.; height 1.69 in.

GRADING NOTE Form required: both, rounded to 0.01. Text area 4.5 by 7.5; (4.5 − 2 · 0.25) ÷ 3 and (7.5 − 3 · 0.25) ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q5/5

SET 105 • HSG-MG.A.3

BACKWARD 1

Proportional reasoning with ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Height : width = 1.618 : 1 and width = 10. Height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q1/5

3526

QUESTION ID

S105-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16.18

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q1/5

Split 6.1 in. into 3 equal columns. Width of each (nearest hundredth)?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q2/5

3527

QUESTION ID

S105-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.03 in.

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A 2 : 3 rectangle 30 cm wide (width = 2 parts) is

A. 60 cm tall

B. 20 cm tall

C. 33 cm tall

D. 45 cm tall

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q3/5

3528

QUESTION ID

S105-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 45 cm tall

GRADING NOTE 15 per part.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q3/5

True or false?

Equal columns divide the available width equally.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q4/5

3529

QUESTION ID

S105-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A 1 : 2 ratio means the second is half the first.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q5/5

3530

QUESTION ID

S105-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q5/5

SET 105 • HSG-MG.A.3

BACKWARD 2

Operate with decimals • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY Reduced copy of the question page, with its answer

$8.5 - 2 \times 1 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q1/5

3532

QUESTION ID

S105-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$11 \times 0.125 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q2/5

3533

QUESTION ID

S105-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1.375

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

6.1 ÷ 3 to the nearest hundredth is

A. 2.3

B. 2.04

C. 2.03

D. 18.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q3/5

3534

QUESTION ID

S105-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2.03

GRADING NOTE 2.0333....

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$5.125 \div 12 \approx 0.427$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q4/5

3535

QUESTION ID

S105-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Division.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$0.2 + 0.2 = 0.04$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q5/5

3536

QUESTION ID

S105-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q5/5

SET 105 • HSG-MG.A.3

BACKWARD 3

Factors of a whole number • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

ANSWER KEY Reduced copy of the question page, with its answer

List all factors of 12.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q1/5

3538

QUESTION ID

S105-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1, 2, 3, 4, 6, 12

GRADING NOTE Form required: all six.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which numbers from 2 to 6 are factors of 10?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q2/5

3539

QUESTION ID

S105-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2 and 5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a factor of 12?

A. 8

B. 4

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q3/5

3540

QUESTION ID

S105-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE 12 ÷ 4 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q3/5

3540

True or false?

12 has more factors than 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q4/5

3541

QUESTION ID

S105-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Six factors vs four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

5 is a factor of 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q5/5

3542

QUESTION ID

S105-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 12 ÷ 5 is not whole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q5/5

3542

SET 105 • HSG-MG.A.3

FORWARD 1

Choose grids that support several layouts • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A 12-column grid can also be used as

A. 4 columns of 3, 3 of 4, or 2 of 6

B. 5 equal columns

C. 7 equal columns

D. only 12 columns

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q1/5

3544

QUESTION ID

S105-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. 4 columns of 3, 3 of 4, or 2 of 6

GRADING NOTE Divisors of 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Why do designers often choose 12-column grids?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q2/5

3545

QUESTION ID

S105-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12 divides evenly into 2, 3, 4, and 6, giving many layout options.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q2/5

True or false?

A 10-column grid can be split into 3 equal groups of whole columns.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q3/5

3546

QUESTION ID

S105-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3 is not a factor of 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A 12-column grid has 78 px columns and 24 px gutters. How wide is an element that spans 6 columns?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q4/5

3547

QUESTION ID

S105-F1-Q4

TYPE

Short answer

CORRECT ANSWER

588 px

GRADING NOTE Any equivalent form. 6 · 78 + 5 · 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A 6.5 in. text area must hold 4 columns, each 1.5 in. wide, with equal gutters between them. Find the exact gutter width.

Gutter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q5/5

3548

QUESTION ID

S105-F1-Q5

TYPE

Short answer

CORRECT ANSWER

⅙ in. ≈ 0.17 in.

GRADING NOTE Exact value or rounding to 0.01. (6.5 − 6) ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q5/5

SET 105 • HSG-MG.A.3

FORWARD 2

Scale a grid layout to a new page size • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

A poster layout is 60 cm wide and 97.1 cm tall. It is reduced to a flyer 24 cm wide with the same shape. Find the scale factor and the flyer's height.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q1/5

3550

QUESTION ID

S105-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Factor 0.4; height 38.84 cm

GRADING NOTE Form required: both. 24 ÷ 60; 97.1 × 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the poster above, a column is 15 cm wide and a gutter is 1.2 cm. Find their widths on the flyer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q2/5

3551

QUESTION ID

S105-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Column 6 cm; gutter 0.48 cm

GRADING NOTE Form required: both. Multiply by 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q2/5

True or false?

Scaling a layout by factor k multiplies the area of each module by k^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q3/5

3552

QUESTION ID

S105-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Similar figures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A layout is enlarged by factor 1.5. A 4 cm by 6 cm module becomes

A. 5.5 cm by 7.5 cm

B. 4 cm by 9 cm

C. 9 cm by 13.5 cm

D. 6 cm by 9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q4/5

3553

QUESTION ID
S105-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 6 cm by 9 cm

GRADING NOTE Multiply both by 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q4/5

Explain why a grid keeps its proportions when every measurement is multiplied by the same factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q5/5

3554

GRADING NOTE Any equivalent explanation.

QUESTION ID

S105-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Every ratio of two measurements is unchanged (k·a/k·b = a/b), so the scaled layout is similar to the original.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S105-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q5/5

SET 88 • HSG-SRT.D.9 • ADVANCED EXTENSION (+)

Derive and use the formula A = ½ab sin C

MAIN	Geometry main question	**HSG-SRT.D.9**
BACKWARD 1	The sine ratio in a right triangle	**HSG-SRT.C.6**
BACKWARD 2	Area of a triangle from base and height	**6.G.A.1**
BACKWARD 3	The included angle of two sides	**HSG-CO.B.8**
FORWARD 1	Solve for a side or an angle from an area	**HSG-SRT.D.9**
FORWARD 2	From equal areas to the Law of Sines	**HSG-SRT.D.10**

SET 88 • HSG-SRT.D.9

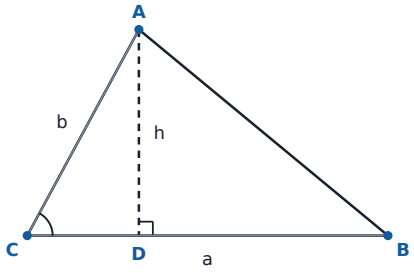
MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, the altitude from A meets line CB at D and has length h. Derive the formula Area = ½ab sin C. Your derivation must cover both cases: ∠C acute (D on side CB, as shown) and ∠C obtuse (D on line CB extended beyond C). (If ∠C is right, h = b and sin C = 1.)

Work space



Not drawn to scale.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q1/5

3557

QUESTION ID

S88-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Acute: h = b sin C. Obtuse: h = b sin(180° − C) = b sin C. In both cases Area = ½ah = ½ab sin C.

GRADING NOTE REQUIRED WORK: Express h using sine in right triangle ADC (acute case) or using the angle 180° − C (obtuse case), then use Area = ½ · base · height.

Any valid proof (paragraph, two-column, or flow). Essential: acute case sin C = h/b in right triangle ADC; obtuse case the right triangle ADC has angle 180° − C at C, so h = b sin(180° − C) and sin(180° − C) = sin C; Area = ½ · a · h in both cases. A derivation that covers only the acute case earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has sides a = 8 and b = 11 with included angle C = 40°. Find its area to the nearest tenth.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q2/5

3558

QUESTION ID

S88-M-Q2

TYPE

Short answer

CORRECT ANSWER

28.3

GRADING NOTE Rounding to 0.1 required. ½ · 8 · 11 · sin 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two sides of a triangle are 6 and 10, and the angle between them is 150°. Find the exact area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q3/5

3559

QUESTION ID

S88-M-Q3

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. sin 150° = ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram has sides 7 cm and 12 cm with an angle of 55° between them. Find its area to the nearest tenth.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q4/5

3560

QUESTION ID

S88-M-Q4

TYPE

Short answer

CORRECT ANSWER

68.8 cm²

GRADING NOTE Rounding to 0.1 required. Two congruent triangles: 7 · 12 · sin 55°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q4/5

Triangle PQR has PQ = 9, QR = 14 and an obtuse angle Q = 115°. Find its area to the nearest tenth, and explain why ½ab sin C still works when the included angle is obtuse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q5/5

3561

QUESTION ID

S88-M-Q5

TYPE

Short answer

CORRECT ANSWER

57.1; the altitude falls outside the triangle and has length 9 sin(180° − 115°) = 9 sin 115°, since sin(180° − Q) = sin Q.

GRADING NOTE Form required: the area (rounding to 0.1) and the explanation. ½ · 9 · 14 · sin 115°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 1

The sine ratio in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

In a right triangle, sin of an acute angle = ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q1/5

3563

QUESTION ID

S88-B1-Q1

TYPE

Short answer

CORRECT ANSWER

opposite ÷ hypotenuse

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse 10, angle 30°. Find the side opposite the angle.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q2/5

3564

QUESTION ID

S88-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If sin C = h/b, then h =

A. b + sin C

B. sin C/b

C. b/sin C

D. b sin C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q3/5

3565

QUESTION ID

S88-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. b sin C

GRADING NOTE Multiply by b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin 90° = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q4/5

3566

QUESTION ID

S88-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

sin C = adjacent ÷ hypotenuse

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q5/5

3567

QUESTION ID
S88-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE That is cos C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 2

Area of a triangle from base and height • Grade 6 • Geometry • 6.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Base 8, height 5: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q1/5

3569

QUESTION ID

S88-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Area = ½ · base · height

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q2/5

3570

QUESTION ID

S88-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q2/5

The height must be ___ to the base.

A. equal

B. adjacent

C. parallel

D. perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q3/5

3571

QUESTION ID

S88-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. perpendicular

GRADING NOTE Altitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Base a and height b sin C: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q4/5

3572

QUESTION ID

S88-B2-Q4

TYPE

Short answer

CORRECT ANSWER

½ab sin C

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A slanted side can be used as the height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q5/5

3573

QUESTION ID

S88-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 3

The included angle of two sides • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In triangle ABC, which angle is included between sides AB and AC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q1/5

3575

QUESTION ID

S88-B3-Q1

TYPE

Short answer

CORRECT ANSWER

∠A

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q1/5

True or false?

SAS uses the angle between the two given sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q2/5

3576

QUESTION ID

S88-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For sides a (BC) and b (CA), the included angle is

A. C

B. none

C. A

D. B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q3/5

3577

QUESTION ID

S88-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. C

GRADING NOTE Shared vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why must the angle be included for ½ab sin C?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q4/5

3578

QUESTION ID

S88-B3-Q4

TYPE

Short answer

CORRECT ANSWER

The altitude is built from side b and the angle between b and the base a.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Any angle of the triangle works in ½ab sin C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q5/5

3579

QUESTION ID

S88-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only the included one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q5/5

SET 88 • HSG-SRT.D.9

FORWARD 1

Solve for a side or an angle from an area • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

A triangle has sides 10 and 12 and area 30. Find all possible measures of the included angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q1/5

3581

QUESTION ID

S88-F1-Q1

TYPE

Short answer

CORRECT ANSWER

30° or 150°

GRADING NOTE Form required: both. sin C = ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q1/5

3581

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has area 50, one side 20, and an included angle of 45°. Find the other side to the nearest tenth.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q2/5

3582

QUESTION ID

S88-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7.1

GRADING NOTE Rounding to 0.1 required. 50 = ½ · 20 · b · sin 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q2/5

True or false?

Two different triangles can have the same two sides and the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q3/5

3583

QUESTION ID

S88-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE C and 180° − C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q3/5

The largest area of a triangle with sides 6 and 8 occurs when the included angle is

A. 60°

B. 180°

C. 90°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q4/5

3584

QUESTION ID

S88-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 90°

GRADING NOTE sin C = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of an equilateral triangle with side 6 using ½ab sin C (exact).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q5/5

3585

QUESTION ID
S88-F1-Q5

TYPE
Short answer

CORRECT ANSWER
9√3

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q5/5

SET 88 • HSG-SRT.D.9

FORWARD 2

From equal areas to the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

The area of triangle ABC can be written as ½bc sin A, ½ac sin B, or ½ab sin C. Use this to show that sin A/a = sin B/b = sin C/c.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q1/5

3587

QUESTION ID

S88-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

½bc sin A = ½ac sin B = ½ab sin C; dividing by ½abc gives sin A/a = sin B/b = sin C/c.

GRADING NOTE REQUIRED WORK: Set the three expressions equal and divide by ½abc.

Any valid proof (paragraph, two-column, or flow). Essential: three area expressions equal; division by ½abc; result.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use sin A/a = sin B/b: A = 30°, a = 5, b = 8. Find sin B.

sin B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q2/5

3588

QUESTION ID

S88-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q2/5

True or false?

The Law of Sines can be derived from the area formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q3/5

3589

QUESTION ID
S88-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE As above.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q3/5

Dividing ½bc sin A = ½ac sin B by ½abc gives

A. a sin A = b sin B

B. sin A/a = sin B/b

C. sin A = sin B

D. a/b = sin A sin B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q4/5

3590

QUESTION ID

S88-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. sin A/a = sin B/b

GRADING NOTE Cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why is it allowed to divide by ½abc?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q5/5

3591

QUESTION ID

S88-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Side lengths are positive, so ½abc ≠ 0.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q5/5

SET 89 • HSG-SRT.D.10 • ADVANCED EXTENSION (+)

Prove the Law of Sines

MAIN	Geometry main question	**HSG-SRT.D.10**
BACKWARD 1	Use trigonometry in right triangles	**HSG-SRT.C.8**
BACKWARD 2	The area formula ½ab sin C	**HSG-SRT.D.9**
BACKWARD 3	Proportions and cross-multiplying	**7.RP.A.2**
FORWARD 1	Use the Law of Sines (AAS and ASA)	**HSG-SRT.D.10**
FORWARD 2	Surveying with the Law of Sines	**HSG-SRT.D.11**

SET 89 • HSG-SRT.D.10

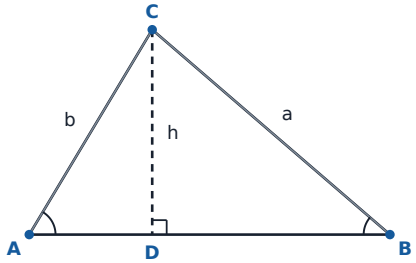
MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, CD is the altitude to AB. Prove that a/sin A = b/sin B.

Work space



Not drawn to scale.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q1/5

3594

QUESTION ID

S89-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

h = b sin A and h = a sin B, so b sin A = a sin B and a/sin A = b/sin B.

GRADING NOTE REQUIRED WORK: Write h two ways, using right triangles ADC and BDC.

Any valid proof (paragraph, two-column, or flow). Essential: sin A = h/b and sin B = h/a from the two right triangles; equate h; divide by sin A sin B. Repeating with another altitude gives c/sin C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has an obtuse angle at A, so the altitude from C meets line AB extended past A at D. Prove that a/sin A = b/sin B in this case.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q2/5

3595

QUESTION ID

S89-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

h = b sin(180° − A) = b sin A and h = a sin B, so a/sin A = b/sin B.

GRADING NOTE

REQUIRED WORK: Use the right triangle with angle 180° − A at A, and sin(180° − A) = sin A.

Any valid proof (paragraph, two-column, or flow). Essential: the exterior angle at A is 180° − A; h = b sin(180° − A); the identity sin(180° − A) = sin A; equate with h = a sin B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In acute triangle ABC, AE is the altitude from A to BC. Prove that b/sin B = c/sin C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q3/5

3596

QUESTION ID

S89-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

AE = c sin B and AE = b sin C, so c sin B = b sin C and b/sin B = c/sin C.

GRADING NOTE REQUIRED WORK: Write the altitude AE two ways, using right triangles ABE and ACE.

Any valid proof (paragraph, two-column, or flow). Essential: sin B = AE/c and sin C = AE/b from the two right triangles; equate; divide by sin B sin C. (If ∠B or ∠C is obtuse, use sin(180° − x) = sin x.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has a right angle at C. Prove that a/sin A = b/sin B = c/sin C in this case.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q4/5

3597

QUESTION ID

S89-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

sin A = a/c and sin B = b/c, so a/sin A = c = b/sin B; and c/sin C = c/1 = c.

GRADING NOTE REQUIRED WORK: Use the right-triangle definitions of sine and sin 90° = 1.

Any valid proof (paragraph, two-column, or flow). Essential: sin A = a/c, sin B = b/c (opposite over hypotenuse); each ratio equals c; sin 90° = 1 gives c/sin C = c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC with acute angle A is inscribed in a circle of radius R. Draw the diameter BD. Prove that a/sin A = 2R, and explain why this gives the Law of Sines.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q5/5

3598

QUESTION ID

S89-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

∠BDC ≅ ∠A (inscribed angles on arc BC); ∠BCD = 90°; sin D = BC/BD = a/(2R), so a/sin A = 2R. The same holds for b and c, so all three ratios equal 2R.

GRADING NOTE REQUIRED WORK: Compare ∠A with ∠D (same arc BC) and use right triangle BCD (angle in a semicircle).

Any valid proof (paragraph, two-column, or flow). Essential: inscribed angles on the same arc; angle in a semicircle; sine in right triangle BCD; each ratio equals 2R. (For obtuse A, ∠D = 180° − A and sin D = sin A.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q5/5

3598

SET 89 • HSG-SRT.D.10

BACKWARD 1

Use trigonometry in right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle: hypotenuse 12, angle 40°. Opposite side to the nearest tenth?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q1/5

3600

QUESTION ID

S89-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7.7

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-B1-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q1/5

True or false?

In an acute triangle, an altitude splits the triangle into two right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q2/5

3601

QUESTION ID

S89-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The foot lies on the opposite side; in an obtuse triangle, an altitude from an acute-angle vertex falls outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If sin B = h/a, then h =

A. h sin B

B. a/sin B

C. a sin B

D. sin B/a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q3/5

3602

QUESTION ID

S89-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. a sin B

GRADING NOTE Multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle: opposite 5, hypotenuse 10. Angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q4/5

3603

QUESTION ID

S89-B1-Q4

TYPE

Short answer

CORRECT ANSWER

30°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-B1-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q4/5

True or false?

sin is defined only for angles in right triangles, so it cannot be used after drawing an altitude.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q5/5

3604

QUESTION ID

S89-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The altitude creates right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q5/5

SET 89 • HSG-SRT.D.10

BACKWARD 2

The area formula ½ab sin C • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has b = 4, c = 5, and A = 90°. What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q1/5

3606

QUESTION ID

S89-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q1/5

True or false?

½bc sin A, ½ac sin B, and ½ab sin C all give the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S89-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q2/5

3607

QUESTION ID

S89-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S89-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Area of a triangle with sides 6, 6 and included 30°?

A. 3

B. 9

C. 36

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q3/5

3608

GRADING NOTE ½ · 36 · ½.

QUESTION ID

S89-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-B2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the area of triangle ABC using sides a and c.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q4/5

3609

QUESTION ID

S89-B2-Q4

TYPE

Short answer

CORRECT ANSWER

½ac sin B

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-B2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q4/5

True or false?

½ab sin C uses angle A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S89-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q5/5

3610

QUESTION ID

S89-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Angle C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S89-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q5/5

SET 89 • HSG-SRT.D.10

BACKWARD 3

Proportions and cross-multiplying • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve x/6 = 10/4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q1/5

3612

QUESTION ID

S89-B3-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q1/5

True or false?

If b sin A = a sin B, then a/sin A = b/sin B (when the sines are not 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q2/5

3613

QUESTION ID

S89-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Divide by sin A sin B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

From 3x = 4y, x/y =

A. 3/4

B. 4/3

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q3/5

3614

QUESTION ID

S89-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4/3

GRADING NOTE Divide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite p · q = r · s as a proportion with p and r on top.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q4/5

3615

QUESTION ID

S89-B3-Q4

TYPE

Short answer

CORRECT ANSWER

p/r = s/q (or p/s = r/q)

GRADING NOTE Any correct proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

a/b = c/d means a + d = b + c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q5/5

3616

QUESTION ID

S89-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ad = bc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q5/5

SET 89 • HSG-SRT.D.10

FORWARD 1

Use the Law of Sines (AAS and ASA) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, A = 40°, B = 60°, and a = 10. Find b to the nearest tenth.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q1/5

3618

QUESTION ID

S89-F1-Q1

TYPE

Short answer

CORRECT ANSWER

13.5

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q1/5

In triangle ABC, A = 40°, B = 60°, and a = 10. Find C, and find c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q2/5

3619

QUESTION ID

S89-F1-Q2

TYPE

Short answer

CORRECT ANSWER

C = 80°; c ≈ 15.3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q2/5

True or false?

Knowing two angles and any side determines the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q3/5

3620

QUESTION ID
S89-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE AAS/ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A = 30°, B = 45°, a = 6. b =

A. 3√2 ≈ 4.2

B. 6√2 ≈ 8.5

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q4/5

3621

QUESTION ID

S89-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6√2 ≈ 8.5

GRADING NOTE 6 sin 45°/sin 30°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q4/5

Why must the third angle be found before using the Law of Sines for c in an ASA problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q5/5

3622

QUESTION ID

S89-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The law pairs each side with its opposite angle, so C is needed for c.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q5/5

SET 89 • HSG-SRT.D.10

FORWARD 2

Surveying with the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Two observers 500 m apart sight a tree across a river. The angles at the observers (between the baseline and the tree) are 70° and 60°. How far is the tree from the observer at the 70° angle (nearest meter)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q1/5

3624

QUESTION ID
S89-F2-Q1

TYPE
Short answer

CORRECT ANSWER
565 m

GRADING NOTE Rounding to the nearest meter required. Third angle 50°; side opposite 60°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A surveyor measures a baseline of 200 m and angles of 52° and 71° to a peak. Find the distance from the 52° end to the peak (nearest meter).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q2/5

3625

QUESTION ID
S89-F2-Q2

TYPE
Short answer

CORRECT ANSWER
225 m

GRADING NOTE Rounding to the nearest meter required. Third angle 57°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q2/5

True or false?

Surveyors use a measured baseline and two angles to find distances they cannot walk.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q3/5

3626

QUESTION ID

S89-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Triangulation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q3/5

In a surveying triangle, the side opposite the largest angle is

A. equal to the others

B. the shortest

C. the longest

D. the baseline always

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q4/5

3627

QUESTION ID

S89-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. the longest

GRADING NOTE Law of Sines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q4/5

Explain why measuring angles is easier than measuring the distance across the river directly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q5/5

3628

QUESTION ID
S89-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The far side cannot be reached; angles can be measured from one bank.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q5/5

SET 90 • HSG-SRT.D.10 • ADVANCED EXTENSION (+)

Prove the Law of Cosines

MAIN	Geometry main question	**HSG-SRT.D.10**
BACKWARD 1	The Pythagorean Theorem	**8.G.B.7**
BACKWARD 2	The distance formula	**8.G.B.8**
BACKWARD 3	Sine and cosine in a right triangle	**HSG-SRT.C.6**
BACKWARD 4	Square a binomial	**HSA-APR.A.1**
FORWARD 1	Use the Law of Cosines (SAS and SSS)	**HSG-SRT.D.10**
FORWARD 2	Distances between moving objects	**HSG-SRT.D.11**

SET 90 • HSG-SRT.D.10

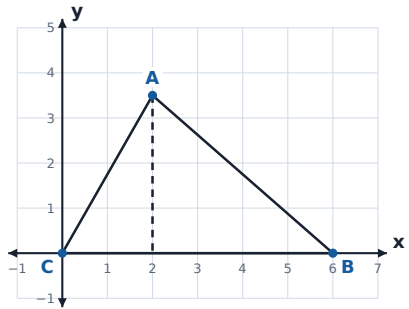
MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Place triangle ABC with C at the origin and B at (a, 0), so A = (b cos C, b sin C). Prove that $c^2 = a^2 + b^2 - 2ab \cos C$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q1/5

3631

QUESTION ID

S90-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$c^2 = (b \cos C - a)^2 + (b \sin C)^2 = a^2 + b^2 - 2ab \cos C$

GRADING NOTE REQUIRED WORK: Use the distance formula for AB and the identity $\sin^2 C + \cos^2 C = 1$.

Any valid proof (paragraph, two-column, or flow). Essential: A = (b cos C, b sin C) because CA = b; $c^2 = (b \cos C - a)^2 + b^2 \sin^2 C$; expand: $b^2\cos^2 C - 2ab \cos C + a^2 + b^2 \sin^2 C$; $\sin^2 C + \cos^2 C = 1$ gives $a^2 + b^2 - 2ab \cos C$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In acute triangle ABC, the altitude from A meets BC at D. Let CD = x. Use the two right triangles to prove $c^2 = a^2 + b^2 - 2ab \cos C$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q2/5

3632

QUESTION ID

S90-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$h^2 = b^2 - x^2 = c^2 - (a - x)^2$, so $c^2 = a^2 + b^2 - 2ax = a^2 + b^2 - 2ab \cos C$.

GRADING NOTE REQUIRED WORK: Write h^2 twice with the Pythagorean Theorem, then use $x = b \cos C$.

Any valid proof (paragraph, two-column, or flow). Essential: $BD = a - x$; $h^2 = b^2 - x^2$ and $h^2 = c^2 - (a - x)^2$; solve for c^2; substitute $x = b \cos C$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, ∠C is obtuse, so the altitude from A meets line BC extended beyond C at D. Let CD = x. Prove that $c^2 = a^2 + b^2 - 2ab\cos C$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q3/5

3633

QUESTION ID

S90-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$h^2 = b^2 - x^2 = c^2 - (a + x)^2$, so $c^2 = a^2 + b^2 + 2ax = a^2 + b^2 - 2ab\cos C$.

GRADING NOTE REQUIRED WORK: Write h^2 twice with the Pythagorean Theorem, using BD = a + x, then use $x = b\cos(180° - C) = -b\cos C$.

Any valid proof (paragraph, two-column, or flow). Essential: BD = a + x; two expressions for h^2; solve for c^2; $x = b\cos(180° - C)$ and $\cos(180° - C) = -\cos C$, so $2ax = -2ab\cos C$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the Law of Cosines to prove that in triangle ABC, if $c^2 > a^2 + b^2$, then $\angle C$ is obtuse.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q4/5

3634

QUESTION ID

S90-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

cos C = (a² + b² − c²)/(2ab) < 0, so C is between 90° and 180°: obtuse.

GRADING NOTE

REQUIRED WORK: Solve the law for cos C and decide its sign.

Any valid proof (paragraph, two-column, or flow). Essential: rearrangement; numerator negative and denominator positive; a negative cosine means an obtuse angle ($0° < C < 180°$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q4/5

Use the Law of Cosines to prove both directions: in triangle ABC, $\angle C = 90°$ if and only if $c^2 = a^2 + b^2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q5/5

3635

QUESTION ID

S90-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$C = 90°$ gives $c^2 = a^2 + b^2$; $c^2 = a^2 + b^2$ gives $\cos C = 0$, so $C = 90°$.

GRADING NOTE REQUIRED WORK: Substitute $\cos 90° = 0$ for one direction; for the converse solve for $\cos C$.

Any valid proof (paragraph, two-column, or flow). Essential: $\cos 90° = 0$; converse: $\cos C = 0$ with $0° < C < 180°$ forces $C = 90°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 1

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Legs 5 and 12: hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q1/5

3637

QUESTION ID

S90-B1-Q1

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse b, leg x: the other leg squared?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q2/5

3638

QUESTION ID

S90-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$b^2 - x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q2/5

In triangle ABC, the altitude from A to BC has length h and foot D, with CD = x and BC = a. In right triangle ABD, $h^2 + (a - x)^2 =$

A. a^2

B. c^2

C. x^2

D. b^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q3/5

3639

QUESTION ID

S90-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. c^2

GRADING NOTE Right triangle ABD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q3/5

True or false?

The Pythagorean Theorem applies to the right triangles made by an altitude.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q4/5

3640

QUESTION ID

S90-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Right angles at D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q4/5

True or false?

$a^2 + b^2 = c^2$ in every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q5/5

3641

QUESTION ID

S90-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Right triangles only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 2

The distance formula • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (0, 0) to (3, 4)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q1/5

3643

QUESTION ID

S90-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the squared distance from (p, q) to (a, 0).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q2/5

3644

QUESTION ID

S90-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(p - a)^2 + q^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q2/5

Squared distance from (1, 2) to (4, 6)?

A. 5

B. 25

C. 49

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q3/5

3645

QUESTION ID

S90-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 25

GRADING NOTE 9 + 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q3/5

True or false?

The distance formula comes from the Pythagorean Theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q4/5

3646

QUESTION ID

S90-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Legs are coordinate differences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q4/5

True or false?

Distance from (2, 0) to (5, 0) is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q5/5

3647

QUESTION ID

S90-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 3

Sine and cosine in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse b and angle C: adjacent side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q1/5

3649

QUESTION ID

S90-B3-Q1

TYPE

Short answer

CORRECT ANSWER

b cos C

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-B3-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hypotenuse b and angle C: opposite side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q2/5

3650

QUESTION ID

S90-B3-Q2

TYPE

Short answer

CORRECT ANSWER

b sin C

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-B3-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For a 3-4-5 triangle, $(\sin)^2 + (\cos)^2$ of an acute angle =

A. 25

B. 1

C. 12/25

D. 7/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q3/5

3651

QUESTION ID

S90-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1

GRADING NOTE 9/25 + 16/25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin^2 C + \cos^2 C = 1$ for acute angles C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q4/5

3652

QUESTION ID

S90-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pythagorean Theorem divided by the hypotenuse squared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

cos C = opposite ÷ hypotenuse

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q5/5

3653

QUESTION ID
S90-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Adjacent ÷ hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 4

Square a binomial • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(b \cos C - a)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q1/5

3655

QUESTION ID
S90-B4-Q1

TYPE
Short answer

CORRECT ANSWER
$b^2\cos^2 C - 2ab\cos C + a^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(a - x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q2/5

3656

QUESTION ID
S90-B4-Q2

TYPE
Short answer

CORRECT ANSWER
$a^2 - 2ax + x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(p - q)^2 =$

A. $p^2 - 2pq + q^2$

B. $p^2 - pq + q^2$

C. $p^2 - q^2$

D. $p^2 + q^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q3/5

3657

QUESTION ID
S90-B4-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. $p^2 - 2pq + q^2$

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a - x)^2 - x^2 = a^2 - 2ax$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q4/5

3658

QUESTION ID
S90-B4-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE x^2 cancels.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a - x)^2 = a^2 - x^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q5/5

3659

QUESTION ID
S90-B4-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Missing −2ax.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q5/5

SET 90 • HSG-SRT.D.10

FORWARD 1

Use the Law of Cosines (SAS and SSS) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, a = 7, b = 9, and C = 50°. Find c to the nearest tenth.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q1/5

3661

QUESTION ID
S90-F1-Q1

TYPE
Short answer

CORRECT ANSWER
7.0

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has sides 5, 7, and 8. Find the angle opposite the side of length 7.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q2/5

3662

QUESTION ID

S90-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form. cos = (25 + 64 − 49)/80 = ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q2/5

True or false?

The Law of Cosines can find an angle when all three sides are known.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q3/5

3663

QUESTION ID

S90-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE SSS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F1-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q3/5

Sides 3, 4, 6. The angle opposite 6 is

A. obtuse

B. acute

C. right

D. impossible to tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q4/5

3664

QUESTION ID

S90-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. obtuse

GRADING NOTE 36 > 9 + 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the largest angle of the triangle with sides 4, 5, and 6, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q5/5

3665

QUESTION ID

S90-F1-Q5

TYPE

Short answer

CORRECT ANSWER

82.8°

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q5/5

SET 90 • HSG-SRT.D.10

FORWARD 2

Distances between moving objects • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Two ships leave port at the same time on courses 70° apart. One travels 20 km and the other 30 km. How far apart are they, to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q1/5

3667

QUESTION ID

S90-F2-Q1

TYPE

Short answer

CORRECT ANSWER

29.8 km

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A golfer's ball is 150 yd from the tee; the hole is 180 yd from the tee; the angle between the two directions is 12°. How far is the ball from the hole (nearest yard)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q2/5

3668

QUESTION ID
S90-F2-Q2

TYPE
Short answer

CORRECT ANSWER
46 yd

GRADING NOTE Rounding to the nearest yard required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q2/5

True or false?

The Law of Cosines can be used only when the angle between the two known sides is a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q3/5

3669

QUESTION ID
S90-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It works for any included angle; with 90° it becomes the Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q3/5

Two hikers walk 3 km and 5 km from camp in directions 60° apart. Distance between them?

A. 8 km

B. √34 ≈ 5.8 km

C. 2 km

D. √19 ≈ 4.4 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q4/5

3670

QUESTION ID

S90-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. √19 ≈ 4.4 km

GRADING NOTE 9 + 25 − 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q4/5

Two ships leave port on courses 70° apart. Why can't the Pythagorean Theorem alone give the distance between them?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q5/5

3671

QUESTION ID
S90-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The angle between their courses is not 90°.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F2-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q5/5

SET 91 • HSG-SRT.D.11 • ADVANCED EXTENSION (+)

Apply the Law of Sines, including the ambiguous case

MAIN	Geometry main question	**HSG-SRT.D.11**
BACKWARD 1	The Law of Sines	**HSG-SRT.D.10**
BACKWARD 2	Inverse sine	**HSG-SRT.C.8**
BACKWARD 3	Triangle angle sum	**HSG-CO.C.10**
BACKWARD 4	Sine of supplementary angles	**HSG-SRT.D.9**
FORWARD 1	Solve both triangles in the ambiguous case	**HSG-SRT.D.11**
FORWARD 2	Bearings and navigation with the Law of Sines	**HSG-MG.A.1**

SET 91 • HSG-SRT.D.11

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Points A and B are 120 m apart on one bank of a river. A tree C is on the other bank. ∠A = 72° and ∠B = 63°. Find AC to the nearest tenth of a meter.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q1/5

3674

QUESTION ID

S91-M-Q1

TYPE

Short answer

CORRECT ANSWER

151.2 m

GRADING NOTE Rounding to 0.1 required. ∠C = 45°; AC is opposite ∠B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, A = 30°, a = 6, and b = 10. How many triangles are possible? Find all possible measures of B to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q2/5

3675

QUESTION ID

S91-M-Q2

TYPE

Short answer

CORRECT ANSWER

Two: B ≈ 56.4° or B ≈ 123.6°

GRADING NOTE Form required: the number and both angles. sin B = 10 sin 30°/6 ≈ 0.833.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, A = 40°, a = 12, and b = 10. How many triangles are possible? Find B to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q3/5

3676

QUESTION ID

S91-M-Q3

TYPE

Short answer

CORRECT ANSWER

One: B ≈ 32.4°

GRADING NOTE Form required: the number and the angle. The supplement 147.6° is too large (with A = 40° the sum exceeds 180°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q3/5

In triangle ABC, A = 50°, a = 5, and b = 10. How many triangles are possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q4/5

3677

QUESTION ID

S91-M-Q4

TYPE

Short answer

CORRECT ANSWER

None: sin B = 10 sin 50°/5 ≈ 1.53 > 1.

GRADING NOTE Form required: the number with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q4/5

A is acute and b sin A < a < b. How many triangles have these measurements?

A. Infinitely many

B. One

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q5/5

3678

QUESTION ID

S91-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Two

GRADING NOTE Side a can swing to two positions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 1

The Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Write the Law of Sines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q1/5

3680

QUESTION ID

S91-B1-Q1

TYPE

Short answer

CORRECT ANSWER

a/sin A = b/sin B = c/sin C

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A = 30°, B = 90°, a = 5. b =

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q2/5

3681

QUESTION ID

S91-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solving a/sin A = b/sin B for sin B gives

A. b sin A/a

B. a/(b sin A)

C. sin A

D. a sin A/b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q3/5

3682

QUESTION ID

S91-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. b sin A/a

GRADING NOTE Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q3/5

True or false?

Each side in the Law of Sines is paired with its opposite angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q4/5

3683

QUESTION ID

S91-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$a/\sin B = b/\sin A$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q5/5

3684

QUESTION ID

S91-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Pairs are opposite.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 2

Inverse sine • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

sin B = 0.5, B acute. B =

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q1/5

3686

QUESTION ID

S91-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

sin B = 0.8. Find the acute B to the nearest tenth.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q2/5

3687

QUESTION ID

S91-B2-Q2

TYPE

Short answer

CORRECT ANSWER

53.1°

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sin^{-1}(1) =$

A. 0°

B. 180°

C. 45°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q3/5

3688

QUESTION ID

S91-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 90°

GRADING NOTE Value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q3/5

True or false?

The calculator's $\sin^{-1}$ always gives an angle between $-90°$ and $90°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q4/5

3689

QUESTION ID

S91-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin^{-1}(1.2)$ is an angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q5/5

3690

QUESTION ID

S91-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Sine cannot exceed 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 3

Triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Two angles are 72° and 63°. Third angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q1/5

3692

QUESTION ID

S91-B3-Q1

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q1/5

True or false?

A triangle cannot have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q2/5

3693

QUESTION ID

S91-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Sum would exceed 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q2/5

3693

A = 40° and B = 147.6°. Is this a triangle?

A. Only if C is 0°

B. No: the sum exceeds 180°

C. Yes

D. Only if right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q3/5

3694

QUESTION ID

S91-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. No: the sum exceeds 180°

GRADING NOTE 187.6°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A = 30°, B = 123.6°. Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q4/5

3695

QUESTION ID

S91-B3-Q4

TYPE

Short answer

CORRECT ANSWER

26.4°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The angles of a triangle add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q5/5

3696

QUESTION ID

S91-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 4

Sine of supplementary angles • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin 150° = \sin 30°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q1/5

3698

QUESTION ID

S91-B4-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sin(180° - x) = \sin x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Name an obtuse angle with the same sine as 56.4°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q2/5

3699

QUESTION ID

S91-B4-Q2

TYPE

Short answer

CORRECT ANSWER

123.6°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B4-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

sin 120° equals

A. cos 120°

B. sin 60°

C. sin 30°

D. −sin 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q3/5

3700

QUESTION ID

S91-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. sin 60°

GRADING NOTE Supplement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B4-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q3/5

Why might sin B = 0.83 give two possible triangles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q4/5

3701

QUESTION ID

S91-B4-Q4

TYPE

Short answer

CORRECT ANSWER

Both B and 180° − B have that sine.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B4-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q4/5

True or false?

An obtuse angle has a negative sine.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q5/5

3702

QUESTION ID

S91-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B4-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q5/5

SET 91 • HSG-SRT.D.11

FORWARD 1

Solve both triangles in the ambiguous case • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

A = 30°, a = 6, b = 10. For the triangle with acute B, find C and c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q1/5

3704

QUESTION ID

S91-F1-Q1

TYPE

Short answer

CORRECT ANSWER

C ≈ 93.6°; c ≈ 12.0

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q1/5

In triangle ABC, A = 30°, a = 6, and b = 10, so B ≈ 56.4° or B ≈ 123.6°. For the triangle with obtuse B, find C and c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q2/5

3705

QUESTION ID

S91-F1-Q2

TYPE

Short answer

CORRECT ANSWER

C ≈ 26.4°; c ≈ 5.3

GRADING NOTE Form required: both. B ≈ 123.6°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q2/5

True or false?

In the ambiguous case the two triangles have different third sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q3/5

3706

QUESTION ID

S91-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Different shapes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q3/5

A = 30°, b = 10, and a = 5. How many triangles?

A. Three

B. Two

C. One (a right triangle)

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q4/5

3707

QUESTION ID

S91-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. One (a right triangle)

GRADING NOTE a = b sin A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A = 120°, a = 8, b = 10. How many triangles? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q5/5

3708

QUESTION ID

S91-F1-Q5

TYPE

Short answer

CORRECT ANSWER

None: with A obtuse, a must be the longest side, but a < b.

GRADING NOTE Form required: the number with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q5/5

SET 91 • HSG-SRT.D.11

FORWARD 2

Bearings and navigation with the Law of Sines • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A lighthouse is spotted from a ship at an angle of 35° to its course. After the ship sails 8 km farther, the angle is 70°. How far is the ship from the lighthouse at the second sighting (nearest tenth)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q1/5

3710

QUESTION ID

S91-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8.0 km

GRADING NOTE Rounding to 0.1 required. The triangle has angles 35°, 110°, 35°, so it is isosceles: 8.0 km.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q1/5

3710

ANSWER KEY Reduced copy of the question page, with its answer

A forest ranger at A sees a fire at an angle of 48° from the line to a second tower B, 10 km away. From B, the angle is 61°. How far is the fire from A (nearest tenth)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q2/5

3711

QUESTION ID

S91-F2-Q2

TYPE

Short answer

CORRECT ANSWER

9.3 km

GRADING NOTE Rounding to 0.1 required. Third angle 71°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q2/5

True or false?

Bearings and sightings are modeled as triangles with a known side and two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q3/5

3712

QUESTION ID

S91-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A ship sights a lighthouse at 35° to its course; after sailing 8 km farther along the course, the angle is 70°. In the triangle formed by the two ship positions and the lighthouse, the angle at the lighthouse is

A. 75°

B. 110°

C. 70°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q4/5

3713

QUESTION ID

S91-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 35°

GRADING NOTE 70° − 35° (exterior angle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q4/5

Name one reason a navigation model using a flat triangle might be inaccurate over very long distances.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q5/5

3714

QUESTION ID

S91-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Earth's surface is curved.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q5/5

SET 92 • HSG-SRT.D.11 • ADVANCED EXTENSION (+)

Apply the Law of Cosines, including resultant forces

MAIN	Geometry main question	**HSG-SRT.D.11**
BACKWARD 1	The Law of Cosines	**HSG-SRT.D.10**
BACKWARD 2	Inverse cosine	**HSG-SRT.C.8**
BACKWARD 3	Angles in a parallelogram	**HSG-CO.C.11**
FORWARD 1	Solve a triangle with both laws	**HSG-SRT.D.11**
FORWARD 2	Velocity vectors: airspeed and wind	**HSN-VM.A.3**

SET 92 • HSG-SRT.D.11

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

To find the length AB across a lake, a surveyor at C measures CA = 350 m, CB = 420 m, and ∠C = 68°. Find AB to the nearest tenth of a meter.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q1/5

3717

QUESTION ID
S92-M-Q1

TYPE
Short answer

CORRECT ANSWER
434.5 m

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two forces of 40 N and 60 N act on an object with an angle of 50° between them. Find the magnitude of the resultant force to the nearest tenth.

Resultant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q2/5

3718

QUESTION ID

S92-M-Q2

TYPE

Short answer

CORRECT ANSWER

91.0 N

GRADING NOTE Rounding to 0.1 required. In the force parallelogram the angle opposite the resultant is 130°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has sides 8, 10, and 13. Find its largest angle to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q3/5

3719

QUESTION ID

S92-M-Q3

TYPE

Short answer

CORRECT ANSWER

91.8°

GRADING NOTE Rounding to 0.1 required. Opposite 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-M-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane flies 200 km, turns 25° to the right, and flies 150 km. How far is it from its start, to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q4/5

3720

QUESTION ID

S92-M-Q4

TYPE

Short answer

CORRECT ANSWER

341.9 km

GRADING NOTE Rounding to 0.1 required. The angle between the legs inside the triangle is 155°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q4/5

Which law should be used first to solve a triangle given all three sides?

A. The Law of Sines

B. The Pythagorean Theorem

C. The angle sum

D. The Law of Cosines

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q5/5

3721

QUESTION ID

S92-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. The Law of Cosines

GRADING NOTE SSS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 1

The Law of Cosines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Write the Law of Cosines for side c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q1/5

3723

QUESTION ID

S92-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$c^2 = a^2 + b^2 - 2ab \cos C$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

a = 3, b = 4, C = 90°: c?

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q2/5

3724

QUESTION ID

S92-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-B1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q2/5

Solved for cos C:

A. $(a^2 + b^2 - c^2)/(2ab)$

B. $(c^2 - a^2 - b^2)/(2ab)$

C. $2ab/(a^2 + b^2 - c^2)$

D. $(a^2 + b^2 + c^2)/(2ab)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q3/5

3725

QUESTION ID

S92-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A.

$(a^2 + b^2 - c^2)/(2ab)$

GRADING NOTE Rearrange.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-B1-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q3/5

True or false?

The angle in the formula is the one opposite the side being found.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q4/5

3726

QUESTION ID

S92-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE C opposite c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$$c^2 = a^2 + b^2 + 2ab \cos C$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q5/5

3727

QUESTION ID

S92-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Minus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 2

Inverse cosine • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

cos C = 0.5. C =

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q1/5

3729

QUESTION ID

S92-B2-Q1

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

cos C = −0.03125. C to the nearest tenth?

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q2/5

3730

QUESTION ID

S92-B2-Q2

TYPE

Short answer

CORRECT ANSWER

91.8°

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A negative cosine means the angle is

A. acute

B. right

C. obtuse

D. zero

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q3/5

3731

QUESTION ID

S92-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. obtuse

GRADING NOTE Between 90° and 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q3/5

True or false?

$\cos^{-1}$ gives angles from 0° to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q4/5

3732

QUESTION ID

S92-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\cos^{-1}(0) = 0°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q5/5

3733

QUESTION ID
S92-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 3

Angles in a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

Consecutive angles of a parallelogram are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q1/5

3735

QUESTION ID

S92-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q1/5

3735

ANSWER KEY Reduced copy of the question page, with its answer

One angle of a parallelogram is 50°. The angle next to it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q2/5

3736

QUESTION ID

S92-B3-Q2

TYPE

Short answer

CORRECT ANSWER

130°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Forces at 50° form a parallelogram; the angle opposite the resultant in the triangle is

A. 130°

B. 25°

C. 50°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q3/5

3737

QUESTION ID

S92-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 130°

GRADING NOTE Supplement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane turns 25°. The angle inside the triangle at the turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q4/5

3738

QUESTION ID

S92-B3-Q4

TYPE

Short answer

CORRECT ANSWER

155°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q4/5

True or false?

The angle between two forces is the angle used in the Law of Cosines for the resultant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q5/5

3739

QUESTION ID

S92-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use its supplement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q5/5

3739

SET 92 • HSG-SRT.D.11

FORWARD 1

Solve a triangle with both laws • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, b = 12, c = 15, A = 40°. Find a to the nearest tenth.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q1/5

3741

QUESTION ID

S92-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9.7

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, b = 12, c = 15, and A = 40°. Find B to the nearest tenth of a degree (use the Law of Sines or Cosines).

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q2/5

3742

QUESTION ID

S92-F1-Q2

TYPE

Short answer

CORRECT ANSWER

53.0°

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q2/5

True or false?

After the Law of Cosines, it is safest to find the smaller remaining angle with the Law of Sines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q3/5

3743

QUESTION ID

S92-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Avoids the ambiguous obtuse case.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q3/5

Given two sides and the included angle, use first

A. $\sin^{-1}$

B. the Law of Sines

C. the Law of Cosines

D. the angle sum

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q4/5

3744

QUESTION ID

S92-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. the Law of Cosines

GRADING NOTE SAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use Heron's formula or ½ab sin C to find the area of the triangle with sides 8, 10, 13 (nearest tenth).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q5/5

3745

QUESTION ID

S92-F1-Q5

TYPE

Short answer

CORRECT ANSWER

40.0

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q5/5

SET 92 • HSG-SRT.D.11

FORWARD 2

Velocity vectors: airspeed and wind • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A plane flies due east at 300 km/h; a wind blows due north at 50 km/h. Find its ground speed (nearest tenth) and direction (degrees north of east, nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q1/5

3747

QUESTION ID

S92-F2-Q1

TYPE

Short answer

CORRECT ANSWER

304.1 km/h at 9.5° north of east

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q1/5

3747

ANSWER KEY Reduced copy of the question page, with its answer

A boat heads across a river at 8 km/h; the current is 3 km/h at 90° to its heading. Find its actual speed (nearest tenth).

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q2/5

3748

QUESTION ID

S92-F2-Q2

TYPE

Short answer

CORRECT ANSWER

8.5 km/h

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q2/5

True or false?

The resultant of two velocities is always found by adding their speeds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q3/5

3749

QUESTION ID

S92-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when they point the same way; otherwise use the parallelogram rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Velocities of 10 and 10 with 120° between them have a resultant of

A. 0

B. 17.3

C. 10

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q4/5

3750

QUESTION ID

S92-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

GRADING NOTE Law of Cosines with 60°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q4**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q4/5

3750

ANSWER KEY Reduced copy of the question page, with its answer

A plane's velocity is 400 km/h; a 60 km/h wind blows at 45° to its heading. Find the ground speed (nearest tenth).

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q5/5

3751

QUESTION ID

S92-F2-Q5

TYPE

Short answer

CORRECT ANSWER

444.5 km/h

GRADING NOTE Rounding to 0.1 required. Angle in the triangle 135°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q5/5

SET 93 • HSG-C.A.4 • ADVANCED EXTENSION (+)

Construct a tangent line from a point outside a circle

MAIN	Geometry main question	**HSG-C.A.4**
BACKWARD 1	Construct the midpoint of a segment	**HSG-CO.D.12**
BACKWARD 2	Inscribed angles on a diameter	**HSG-C.A.2**
BACKWARD 3	Tangent is perpendicular to the radius	**HSG-C.A.2**
FORWARD 1	Prove that the two tangent segments are congruent and compute…	**HSG-C.A.4**
FORWARD 2	Lines of sight to the horizon	**HSG-MG.A.1**

SET 93 • HSG-C.A.4

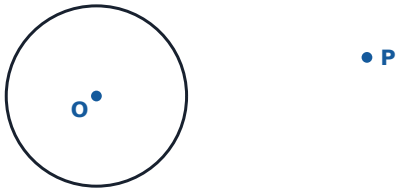
MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a line through P tangent to the circle with center O.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q1/5

3754

QUESTION ID

S93-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Draw OP; construct its midpoint M (perpendicular bisector); draw the circle with center M through O; it meets circle O at T_1 and T_2; draw PT_1 (and PT_2).

GRADING NOTE Grade the construction marks, not only the final figure: segment OP, the perpendicular bisector arcs locating M, the circle on diameter OP, and the tangent through a crossing point T.

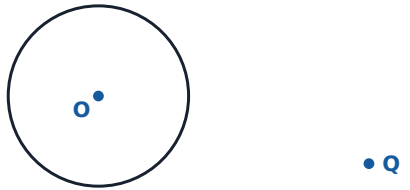
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct BOTH tangent lines from Q to the circle with center O.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q2/5

3755

QUESTION ID

S93-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same construction; both intersection points T_1 and T_2 give the two tangents QT_1 and QT_2.

GRADING NOTE Grade the construction marks, not only the final figure: all construction marks and both tangents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a line through R tangent to the circle with center O. Then write one sentence explaining why your line is tangent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q3/5

3756

QUESTION ID

S93-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Midpoint M of OR; circle with center M through O meets circle O at T; line RT. Tangent because ∠OTR is inscribed in a semicircle (90°), so RT ⊥ radius OT at T.

GRADING NOTE Grade the construction marks, not only the final figure: all construction marks for the midpoint and the helper circle, the tangent line, and the right-angle justification.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct both tangent lines from S to the circle with center O. Then, as evidence, set your compass to ST_1 and show that the same opening reaches T_2 from S.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q4/5

3757

QUESTION ID

S93-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Construct both tangents ST_1 and ST_2 with the circle on diameter OS; the compass check shows $ST_1 = ST_2$ (congruent tangent segments).

GRADING NOTE Grade the construction marks, not only the final figure: both tangents from the full construction and the compass check of equal tangent segments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct both tangent lines from P to the circle with center O. Then construct the bisector of the angle formed by the two tangents and show that it passes through O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q5/5

3758

QUESTION ID

S93-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Both tangents from the circle on diameter OP; bisecting $\angle T_1PT_2$ gives a ray through O (OP bisects the angle between the tangents).

GRADING NOTE Grade the construction marks, not only the final figure: both tangents, the angle bisector construction, and the observation that it passes through O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 1

Construct the midpoint of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Which construction finds the midpoint of OP?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q1/5

3760

QUESTION ID

S93-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of OP

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector crosses the segment at its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q2/5

3761

QUESTION ID

S93-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The perpendicular bisector construction uses arcs

A. of any radius

B. centered at the midpoint

C. of equal radius from both endpoints

D. from one endpoint only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q3/5

3762

QUESTION ID

S93-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. of equal radius from both endpoints

GRADING NOTE Equal radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To construct a tangent from a point P outside circle O, why is the midpoint of OP needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q4/5

3763

QUESTION ID

S93-B1-Q4

TYPE

Short answer

CORRECT ANSWER

It is the center of the circle with diameter OP.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The midpoint can be found by eye in a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q5/5

3764

QUESTION ID

S93-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It must be constructed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 2

Inscribed angles on a diameter • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

An angle inscribed in a semicircle is a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q1/5

3766

QUESTION ID

S93-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A point T is on the circle with diameter OP. m∠OTP = ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q2/5

3767

QUESTION ID

S93-B2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The intercepted arc of an angle inscribed in a semicircle is

A. 180°

B. 360°

C. 90°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q3/5

3768

QUESTION ID

S93-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 180°

GRADING NOTE Half of 180 is 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q3/5

To construct a tangent from a point P outside circle O, why is a circle with diameter OP drawn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q4/5

3769

QUESTION ID

S93-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Points on it make a right angle with O and P.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q4/5

True or false?

An inscribed angle on a diameter can be 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q5/5

3770

QUESTION ID

S93-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Always 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 3

Tangent is perpendicular to the radius • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

A line perpendicular to a radius at its endpoint on the circle is tangent to the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q1/5

3772

QUESTION ID

S93-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A tangent meets the radius at what angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q2/5

3773

QUESTION ID

S93-B3-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q2/5

If PT ⊥ OT and T is on circle O, then PT is

A. a secant

B. a diameter

C. a chord

D. tangent to the circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q3/5

3774

QUESTION ID

S93-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. tangent to the circle

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

OT = 5 and OP = 13 with PT tangent at T. PT = ?

PT = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q4/5

3775

QUESTION ID

S93-B3-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A tangent passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q5/5

3776

QUESTION ID

S93-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It touches at one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q5/5

SET 93 • HSG-C.A.4

FORWARD 1

Prove that the two tangent segments are congruent and compute lengths • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

ANSWER KEY Reduced copy of the question page, with its answer

PA and PB are tangent to circle O at A and B. Prove that PA = PB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q1/5

3778

QUESTION ID

S93-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

△OAP ≅ △OBP by HL, so PA = PB.

GRADING NOTE REQUIRED WORK: Use the right triangles OAP and OBP.

Any valid proof (paragraph, two-column, or flow). Essential: right angles at A and B (tangent ⊥ radius); OA = OB (radii); OP common hypotenuse; HL; CPCTC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has radius 5 and P is 13 from its center. Find the length of each tangent segment from P.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q2/5

3779

QUESTION ID

S93-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q2/5

True or false?

OP bisects the angle between the two tangents.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q3/5

3780

QUESTION ID
S93-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Congruent triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Tangents from P make a 60° angle. The angle AOB is

A. 30°

B. 120°

C. 60°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q4/5

3781

QUESTION ID

S93-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 120°

GRADING NOTE Quadrilateral with two right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Radius 6, tangent segment 8. Distance from P to the center?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q5/5

3782

QUESTION ID

S93-F1-Q5

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q5/5

SET 93 • HSG-C.A.4

FORWARD 2

Lines of sight to the horizon • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Model Earth as a sphere of radius 6,371 km and ignore atmospheric refraction (light travels in straight lines). A person's eyes are 2 m (0.002 km) above the sea. How far is the horizon (tangent distance), to the nearest tenth of a km?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q1/5

3784

QUESTION ID

S93-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5.0 km

GRADING NOTE Rounding to 0.1 required. $d^2 = (6371.002)^2 - 6371^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Model Earth as a sphere of radius 6,371 km and ignore atmospheric refraction. From the top of a 100 m (0.1 km) tower at sea level, how far is the horizon, to the nearest km?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q2/5

3785

QUESTION ID
S93-F2-Q2

TYPE
Short answer

CORRECT ANSWER
36 km

GRADING NOTE Rounding to the nearest km required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q2/5

3785

True or false?

The line of sight to the horizon passes through Earth's center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q3/5

3786

QUESTION ID

S93-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is tangent, perpendicular to the radius at the horizon point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q3/5

3786

ANSWER KEY Reduced copy of the question page, with its answer

Raising the eye height by a factor of 4 multiplies the horizon distance by about

A. 16

B. 8

C. 2

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q4/5

3787

QUESTION ID

S93-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE $d \approx \sqrt{(2Rh)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In the model of the distance to the horizon from a point above Earth, why is the right angle of the triangle at the horizon point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q5/5

3788

QUESTION ID

S93-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The tangent line of sight is perpendicular to the radius at the point of tangency.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q5/5

SET 94 • HSG-GPE.A.3 • ADVANCED EXTENSION (+)

Derive the equation of an ellipse from its foci

MAIN	Geometry main question	**HSG-GPE.A.3**
BACKWARD 1	The distance formula	**8.G.B.8**
BACKWARD 2	Derive a circle equation	**HSG-GPE.A.1**
BACKWARD 3	Expand squares	**HSA-APR.A.1**
FORWARD 1	Read foci and axes from an ellipse equation	**HSG-GPE.A.3**
FORWARD 2	Design with the reflective property of ellipses	**HSG-MG.A.3**

SET 94 • HSG-GPE.A.3

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

An ellipse is the set of points whose distances to the foci $(-3, 0)$ and $(3, 0)$ add to 10. Derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q1/5

3791

QUESTION ID

S94-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$$x^2/25 + y^2/16 = 1$$

GRADING NOTE REQUIRED WORK: Write the distance sum, isolate one radical, square twice, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{((x + 3)^2 + y^2)} + \sqrt{((x - 3)^2 + y^2)} = 10$; isolate and square; simplify to $3x - 25 = -5\sqrt{((x - 3)^2 + y^2)}$ (or equivalent); square again; $16x^2 + 25y^2 = 400$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (constant sum of distances to the foci), the equation of the ellipse with foci (0, −4) and (0, 4) whose distances to the foci add to 10.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q2/5

3792

QUESTION ID

S94-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$x^2/9 + y^2/25 = 1$

GRADING NOTE REQUIRED WORK: Write the distance sum, isolate one radical, square twice, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{(x^2 + (y + 4)^2)} + \sqrt{(x^2 + (y - 4)^2)} = 10$; isolate and square; $4y - 25 = -5\sqrt{(x^2 + (y - 4)^2)}$ (or equivalent); square again; $25x^2 + 9y^2 = 225$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (constant sum of distances to the foci), the equation of the ellipse with foci (±5, 0) and vertices (±13, 0).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q3/5

3793

QUESTION ID

S94-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$x^2/169 + y^2/144 = 1$

GRADING NOTE REQUIRED WORK: The vertices give the constant sum 2a = 26; then derive as before.

Any valid proof (paragraph, two-column, or flow). Essential: sum 26 (the vertex (13, 0) is 18 and 8 from the foci); $\sqrt{((x + 5)^2 + y^2)} + \sqrt{((x - 5)^2 + y^2)} = 26$; two squarings; $144x^2 + 169y^2 = 24{,}336$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (constant sum of distances to the foci), the equation of the ellipse with foci (1, 2) and (7, 2) whose distances to the foci add to 10. (You may translate the figure so the center is at the origin, derive, then translate back.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q4/5

3794

QUESTION ID

S94-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

$(x - 4)^2/25 + (y - 2)^2/16 = 1$

GRADING NOTE REQUIRED WORK: Find the center (4, 2); translate to foci (±3, 0); derive; replace x with x − 4 and y with y − 2.

Any valid proof (paragraph, two-column, or flow). Essential: center (4, 2); translated foci (±3, 0) with sum 10 give $x^2/25 + y^2/16 = 1$; translating back.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (constant sum of distances to the foci), the equation of the ellipse with foci $(-c, 0)$ and $(c, 0)$ whose distances add to $2a$ $(a > c > 0)$. Show that it can be written $x^2/a^2 + y^2/b^2 = 1$ with $b^2 = a^2 - c^2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q5/5

3795

QUESTION ID

S94-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$x^2/a^2 + y^2/b^2 = 1$, $b^2 = a^2 - c^2$

GRADING NOTE REQUIRED WORK: Isolate one radical, square twice, and divide by $a^2(a^2 - c^2)$.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{((x + c)^2 + y^2)} = 2a - \sqrt{((x - c)^2 + y^2)}$; square: $a\sqrt{((x - c)^2 + y^2)} = a^2 - cx$; square: $(a^2 - c^2)x^2 + a^2y^2 = a^2(a^2 - c^2)$; divide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 1

The distance formula • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (x, y) to (3, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q1/5

3797

QUESTION ID

S94-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt{((x - 3)^2 + y^2)}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q1/5

3797

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (0, 4) to (3, 0)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q2/5

3798

QUESTION ID

S94-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (5, 0) to (−3, 0)?

A. √34

B. 64

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q3/5

3799

QUESTION ID

S94-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8

GRADING NOTE Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The distance formula has a square root.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q4/5

3800

QUESTION ID

S94-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE √.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{(a^2 + b^2)} = a + b$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q5/5

3801

QUESTION ID

S94-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 2

Derive a circle equation • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Equation of the circle with center (0, 0) and radius 5?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q1/5

3803

QUESTION ID

S94-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + y^2 = 25$

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-B2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q1/5

True or false?

A circle is an ellipse whose two foci coincide.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q2/5

3804

QUESTION ID

S94-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Special case.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2/25 + y^2/25 = 1$ is

A. an ellipse with different axes

B. a parabola

C. a circle of radius 5

D. a hyperbola

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q3/5

3805

QUESTION ID

S94-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. a circle of radius 5

GRADING NOTE Equal denominators.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle is the set of points at a fixed distance from

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q4/5

3806

QUESTION ID

S94-B2-Q4

TYPE

Short answer

CORRECT ANSWER

one point (the center)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q4/5

True or false?

An ellipse has one focus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q5/5

3807

QUESTION ID

S94-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 3

Expand squares • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 3)^2 - (x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q1/5

3809

QUESTION ID
S94-B3-Q1

TYPE
Short answer

CORRECT ANSWER
12x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q2/5

3810

QUESTION ID
S94-B3-Q2

TYPE
Short answer

CORRECT ANSWER
$x^2 - 6x + 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(3x - 25)^2 =$

A. $9x^2 + 625$

B. $9x^2 - 75x + 625$

C. $3x^2 - 150x + 625$

D. $9x^2 - 150x + 625$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q3/5

3811

QUESTION ID
S94-B3-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. $9x^2 - 150x + 625$

GRADING NOTE Middle term $2 \cdot 3x \cdot 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a - b)^2 = a^2 - 2ab + b^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q4/5

3812

QUESTION ID
S94-B3-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x + 3)^2 = x^2 + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q5/5

3813

QUESTION ID
S94-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Missing 6x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q5/5

SET 94 • HSG-GPE.A.3

FORWARD 1

Read foci and axes from an ellipse equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find the foci of $x^2/25 + y^2/9 = 1$.

Foci: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q1/5

3815

QUESTION ID

S94-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(±4, 0)

GRADING NOTE Form required: both points. $c^2 = 25 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q1/5

3815

ANSWER KEY Reduced copy of the question page, with its answer

Find the vertices and foci of $x^2/16 + y^2/25 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q2/5

3816

QUESTION ID

S94-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Vertices (0, ±5); foci (0, ±3)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q2/5

3816

True or false?

For an ellipse, $c^2 = a^2 - b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q3/5

3817

QUESTION ID

S94-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which ellipse is closest to a circle?

A. $x^2/100 + y^2/9 = 1$

B. $x^2/25 + y^2/4 = 1$

C. $x^2/25 + y^2/24 = 1$

D. $x^2/25 + y^2/1 = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q4/5

3818

QUESTION ID

S94-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x^2/25 + y^2/24 = 1$

GRADING NOTE Nearly equal denominators.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the length of the major axis of $x^2/49 + y^2/36 = 1$.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q5/5

3819

QUESTION ID

S94-F1-Q5

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE Any equivalent form. 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q5/5

SET 94 • HSG-GPE.A.3

FORWARD 2

Design with the reflective property of ellipses • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A whispering gallery is an ellipse 40 m long and 16 m wide. How far from the center should two people stand (at the foci), to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q1/5

3821

QUESTION ID

S94-F2-Q1

TYPE

Short answer

CORRECT ANSWER

18.3 m

GRADING NOTE Rounding to 0.1 required. a = 20, b = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A medical device places a kidney stone at one focus of an ellipse $x^2/100 + y^2/64 = 1$ (cm). Where is the emitter placed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q2/5

3822

QUESTION ID

S94-F2-Q2

TYPE

Short answer

CORRECT ANSWER

At the other focus, 6 cm from the center: (−6, 0) if the stone is at (6, 0).

GRADING NOTE Any equivalent answer. c = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q2/5

True or false?

Sound from one focus of an elliptical room reflects back to the same focus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q3/5

3823

QUESTION ID

S94-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It reflects to the other focus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A garden ellipse is drawn with a 10 m rope tied to stakes 6 m apart. Its width (minor axis) is

A. 8 m

B. 10 m

C. 4 m

D. 6 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q4/5

3824

QUESTION ID

S94-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8 m

GRADING NOTE a = 5, c = 3, b = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q4/5

Explain how the string-and-stakes method uses the definition of an ellipse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q5/5

3825

QUESTION ID

S94-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The string length is the constant sum of the distances to the two stakes (foci).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q5/5

SET 95 • HSG-GPE.A.3 • ADVANCED EXTENSION (+)

Derive the equation of a hyperbola from its foci

MAIN	Geometry main question	**HSG-GPE.A.3**
BACKWARD 1	The distance formula	**8.G.B.8**
BACKWARD 2	Derive an ellipse equation	**HSG-GPE.A.3**
BACKWARD 3	Absolute value as distance	**6.NS.C.7**
FORWARD 1	Read foci and asymptotes from a hyperbola equation	**HSG-GPE.A.3**
FORWARD 2	Locate a position from a time difference	**HSG-MG.A.3**

SET 95 • HSG-GPE.A.3

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A hyperbola is the set of points whose distances to the foci (−5, 0) and (5, 0) differ by 6 (in absolute value). Derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q1/5

3828

QUESTION ID

S95-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x^2/9 - y^2/16 = 1$

GRADING NOTE REQUIRED WORK: Write the distance difference, isolate one radical, square twice, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $|\sqrt{((x + 5)^2 + y^2)} - \sqrt{((x - 5)^2 + y^2)}| = 6$; isolate and square; simplify to $5x - 9 = \pm 3\sqrt{((x - 5)^2 + y^2)}$ (or equivalent); square again; $16x^2 - 9y^2 = 144$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q1/5

Derive, from the definition (constant difference of distances to the foci), the equation of the hyperbola with foci (0, ±13) whose distances to the foci differ by 10 (in absolute value).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q2/5

3829

QUESTION ID

S95-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$$\frac{y^2}{25} - \frac{x^2}{144} = 1$$

GRADING NOTE

REQUIRED WORK: Write the distance difference, isolate one radical, square twice, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $|\sqrt{(x^2 + (y + 13)^2)} - \sqrt{(x^2 + (y - 13)^2)}| = 10$; two squarings; $144y^2 - 25x^2 = 3{,}600$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-M-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q2/5

Derive, from the definition (constant difference of distances to the foci), the equation of the hyperbola with foci (±10, 0) and vertices (±6, 0).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q3/5

3830

QUESTION ID

S95-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$x^2/36 - y^2/64 = 1$

GRADING NOTE REQUIRED WORK: The vertices give the constant difference 2a = 12; then derive.

Any valid proof (paragraph, two-column, or flow). Essential: difference 12 (from vertex (6, 0): 16 − 4); two squarings; $64x^2 - 36y^2 = 2{,}304$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (constant difference of distances to the foci), the equation of the hyperbola with foci $(-c, 0)$ and $(c, 0)$ whose distances differ by $2a$ $(0 < a < c)$. Show that it can be written $x^2/a^2 - y^2/b^2 = 1$ with $b^2 = c^2 - a^2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q4/5

3831

QUESTION ID

S95-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

$x^2/a^2 - y^2/b^2 = 1$, $b^2 = c^2 - a^2$

GRADING NOTE REQUIRED WORK: Isolate one radical, square twice, and divide by $a^2(c^2 - a^2)$.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{((x + c)^2 + y^2)} = \pm 2a + \sqrt{((x - c)^2 + y^2)}$; square: $cx - a^2 = \pm a\sqrt{((x - c)^2 + y^2)}$; square: $(c^2 - a^2)x^2 - a^2y^2 = a^2(c^2 - a^2)$; divide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (constant difference of distances to the foci), the equation of the hyperbola with foci (±5, 0) whose distances to the foci differ by 8. Then give its asymptotes.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q5/5

3832

QUESTION ID

S95-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$x^2/16 - y^2/9 = 1$; asymptotes $y = \pm(3/4)x$

GRADING NOTE REQUIRED WORK: Derive as before, then read b/a.

Any valid proof (paragraph, two-column, or flow). Essential: $|\sqrt{((x + 5)^2 + y^2)} - \sqrt{((x - 5)^2 + y^2)}| = 8$; two squarings; $9x^2 - 16y^2 = 144$; asymptotes from b/a = 3/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 1

The distance formula • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (x, y) to (−5, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q1/5

3834

QUESTION ID

S95-B1-Q1

TYPE

Short answer

CORRECT ANSWER

√((x + 5)² + y²)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (3, 0) to (5, 0) minus distance from (3, 0) to (−5, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q2/5

3835

QUESTION ID

S95-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−6 (difference 6 in absolute value)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (6, 0) to (−10, 0)?

A. 4

B. 60

C. 16

D. −16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q3/5

3836

QUESTION ID

S95-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 16

GRADING NOTE Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q3/5

True or false?

Distances are never negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q4/5

3837

QUESTION ID

S95-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q4/5

True or false?

Distance from (0, 13) to (0, −13) is 13.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q5/5

3838

QUESTION ID

S95-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 2

Derive an ellipse equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

True or false?

An ellipse uses a constant SUM of distances to the foci.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q1/5

3840

QUESTION ID

S95-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Ellipse with foci (±3, 0) and sum 10: equation?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q2/5

3841

QUESTION ID

S95-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2/25 + y^2/16 = 1$

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-B2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q2/5

For an ellipse, $b^2 =$

A. ac

B. $a^2 - c^2$

C. $a^2 + c^2$

D. $c^2 - a^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q3/5

3842

QUESTION ID

S95-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $a^2 - c^2$

GRADING NOTE Relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-B2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What changes in the definition to get a hyperbola?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q4/5

3843

QUESTION ID

S95-B2-Q4

TYPE

Short answer

CORRECT ANSWER

The difference (not the sum) of the distances is constant.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q4/5

True or false?

An ellipse equation has a minus sign between the squared terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q5/5

3844

QUESTION ID

S95-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Plus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 3

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7

ANSWER KEY Reduced copy of the question page, with its answer

$|4 - 10| =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q1/5

3846

QUESTION ID

S95-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$|10 - 4| =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q2/5

3847

QUESTION ID

S95-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q2/5

3847

ANSWER KEY Reduced copy of the question page, with its answer

|a − b| means

A. the size of the difference

B. a + b

C. a negative number

D. always a − b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q3/5

3848

QUESTION ID

S95-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the size of the difference

GRADING NOTE Distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$|x| = 6$ has two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q4/5

3849

QUESTION ID

S95-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ±6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q4/5

True or false?

$|-3| = -3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q5/5

3850

QUESTION ID

S95-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q5/5

SET 95 • HSG-GPE.A.3

FORWARD 1

Read foci and asymptotes from a hyperbola equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find the foci of $x^2/16 - y^2/9 = 1$.

Foci: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q1/5

3852

QUESTION ID

S95-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(±5, 0)

GRADING NOTE Form required: both points. $c^2 = 16 + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q1/5

3852

ANSWER KEY Reduced copy of the question page, with its answer

Find the vertices and asymptotes of $y^2/4 - x^2/9 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q2/5

3853

QUESTION ID

S95-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Vertices (0, ±2); asymptotes y = ±(2/3)x

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-F1-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q2/5

True or false?

For a hyperbola, $c^2 = a^2 + b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q3/5

3854

QUESTION ID

S95-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equation is a hyperbola?

A. $x^2/4 + y^2/9 = 1$

B. $y = x^2/4$

C. $x^2/4 - y^2/9 = 1$

D. $x^2 + y^2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q4/5

3855

QUESTION ID

S95-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x^2/4 - y^2/9 = 1$

GRADING NOTE Minus sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q4/5

Explain how to tell an ellipse from a hyperbola by its equation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q5/5

3856

QUESTION ID

S95-F1-Q5

TYPE

Short answer

CORRECT ANSWER

An ellipse has a plus sign between the squared terms; a hyperbola has a minus sign.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-F1-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q5/5

SET 95 • HSG-GPE.A.3

FORWARD 2

Locate a position from a time difference • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Two radio stations are 100 km apart at (±50, 0). A ship receives signals showing it is 60 km closer to one station than the other. Write the equation of the hyperbola it lies on.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q1/5

3858

QUESTION ID

S95-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2/900 - y^2/1600 = 1$

GRADING NOTE Form required: an equation. $2a = 60$, $c = 50$, $b^2 = 2500 - 900$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Sound travels 0.34 km/s. Two microphones 2 km apart hear a sound 3 s apart. What is the difference in distances to the source?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q2/5

3859

QUESTION ID

S95-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.02 km

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q2/5

True or false?

For one pair of stations, a single nonzero time difference (with the sign known, that is, which signal arrived first) places the ship on one branch of a hyperbola, not at a single point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q3/5

3860

QUESTION ID

S95-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A second, independent measurement is needed to fix the point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q3/5

Radio stations are at (−50, 0) and (50, 0) (km). A ship on the x-axis, to the right of the origin, is 60 km closer to the right station than to the left one. Its position is

A. (30, 0)

B. (40, 0)

C. (50, 0)

D. (60, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q4/5

3861

QUESTION ID

S95-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (30, 0)

GRADING NOTE Vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

One pair of stations and one nonzero signed time difference place a ship on one branch of a hyperbola. What extra measurement is needed to fix its position, and why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q5/5

3862

QUESTION ID

S95-F2-Q5

TYPE

Short answer

CORRECT ANSWER

A time difference from a second pair of stations (or another independent measurement such as a bearing); the second curve crosses the first branch, and the crossing point is the position.

GRADING NOTE Any equivalent answer naming an extra measurement and the intersection idea.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q5/5

SET 96 • HSG-GMD.A.2 • ADVANCED EXTENSION (+)

Use Cavalieri's principle to justify the volume of a sphere

MAIN	Geometry main question	**HSG-GMD.A.2**
BACKWARD 1	Cavalieri's principle for prisms and cylinders	**HSG-GMD.A.1**
BACKWARD 2	Cross-section radius with the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Volumes of cylinders and cones	**HSG-GMD.A.3**
BACKWARD 4	Area of a ring	**7.G.B.4**
FORWARD 1	Use Cavalieri's principle for other solids	**HSG-GMD.A.2**
FORWARD 2	Model a dome as a hemispherical shell	**HSG-MG.A.2**

SET 96 • HSG-GMD.A.2

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A hemisphere of radius r and a cylinder of radius r and height r with a cone removed (as shown) sit on the same plane. Use Cavalieri's principle to show that the hemisphere's volume is ⅔πr³, and conclude that a sphere has volume 4/3 πr³.

Work space

hemisphere, radius r cylinder minus cone

Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q1/5

3865

QUESTION ID

S96-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Both cross-sections have area π(r² − y²), so the volumes are equal: πr³ − ⅓πr³ = ⅔πr³; the sphere is two hemispheres: 4/3 πr³.

GRADING NOTE REQUIRED WORK: Compare the cross-section areas at height y.

Any valid explanation. Essential: hemisphere slice radius √(r² − y²) (Pythagorean), area π(r² − y²); the other slice is a ring with outer radius r and inner radius y (the cone radius at height y), area πr² − πy²; equal areas at every height and equal heights; Cavalieri; volume cylinder minus cone = ⅔πr³; double.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Let r = 6. A hemisphere of radius 6 rests on its flat base, and a cylinder of radius 6 and height 6 has a cone removed whose vertex is at the center of the bottom base and whose base is the top of the cylinder. Show that the two cross-sections have equal area at every height y ($0 \le y \le 6$), and use Cavalieri's principle to find the hemisphere's and the sphere's volumes.

Work space

hemisphere, radius r cylinder minus cone

Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q2/5

3866

QUESTION ID

S96-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Hemisphere slice $\pi(36 - y^2)$; ring $\pi(36 - y^2)$; equal at every y, equal heights, so V(hemisphere) = $216\pi - 72\pi = 144\pi$ and V(sphere) = 288π.

GRADING NOTE REQUIRED WORK: Write both slice areas in terms of y, compare, then compute the cylinder-minus-cone volume.

Any valid explanation. Essential: both areas as functions of y; Cavalieri's principle; 144π; doubled 288π = 4/3 π · 216.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere of radius r fits exactly inside a cylinder of radius r and height 2r. Use the Cavalieri result for the hemisphere to explain why the sphere has ⅔ of the cylinder's volume.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q3/5

3867

QUESTION ID

S96-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Each hemisphere has volume (cylinder of height r) − (cone) = $\pi r^3 - \frac{1}{3}\pi r^3 = \frac{2}{3}$ of πr^3; doubling, the sphere is ⅔ of the cylinder $2\pi r^3$: 4/3 πr^3.

GRADING NOTE REQUIRED WORK: Split the cylinder at the sphere's center into two cylinders of height r and compare each with a hemisphere.

Any valid explanation. Essential: two halves; each hemisphere is ⅔ of its half-cylinder by the Cavalieri comparison; ⅔ of the whole (Archimedes).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain, using Cavalieri's principle, why an oblique cylinder of radius r and vertical height h has the same volume as a right cylinder of radius r and height h.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q4/5

3868

QUESTION ID

S96-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

Every horizontal slice of both is a disk of area πr^2, and the heights are equal, so the volumes are equal: $\pi r^2 h$.

GRADING NOTE REQUIRED WORK: Compare cross-sections at every height.

Any valid explanation. Essential: equal slice areas at every level; equal heights; Cavalieri's principle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A student turns the removed cone upright (base on the bottom of the cylinder, vertex at the top) and compares the remaining solid with the hemisphere. Explain why Cavalieri's principle cannot be used for this comparison, by comparing the two cross-section areas at height y = 0.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q5/5

3869

QUESTION ID

S96-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

At y = 0 the hemisphere slice has area πr², but the cylinder slice is completely removed by the cone base (area πr² − πr² = 0); the areas differ, so the principle does not apply.

GRADING NOTE

REQUIRED WORK: Compute both slice areas at y = 0.

Any valid explanation. Essential: both areas at y = 0; unequal areas at some level; conclusion that the hypothesis of Cavalieri's principle fails.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 1

Cavalieri's principle for prisms and cylinders • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

Solids with equal heights and equal cross-section areas at every level have equal volumes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q1/5

3871

QUESTION ID

S96-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Principle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An oblique cylinder has radius 2 and height 5. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q2/5

3872

QUESTION ID

S96-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q2/5

A prism and a cylinder both have base area 30 and height 4. Volumes?

A. Cannot tell

B. 30 and 4

C. 120 and 40

D. Both 120

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q3/5

3873

QUESTION ID

S96-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Both 120

GRADING NOTE Cavalieri.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

State what must be equal for Cavalieri's principle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q4/5

3874

QUESTION ID

S96-B1-Q4

TYPE

Short answer

CORRECT ANSWER

The heights and the cross-section areas at every level.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q4/5

True or false?

Cavalieri's principle needs the cross-sections to be the same shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q5/5

3875

QUESTION ID
S96-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only equal areas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-B1-Q5**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 2

Cross-section radius with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A sphere of radius 5 is sliced 3 units from its center. Radius of the slice?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q1/5

3877

QUESTION ID

S96-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A hemisphere of radius r rests on its flat base; y is the height above the base plane (0 ≤ y ≤ r). Find the radius of the horizontal slice at height y.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q2/5

3878

QUESTION ID

S96-B2-Q2

TYPE

Short answer

CORRECT ANSWER

√(r² − y²)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q2/5

Radius 10, height 6: slice radius

A. 4

B. 8

C. 16

D. √136

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q3/5

3879

QUESTION ID

S96-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8

GRADING NOTE 100 − 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q3/5

True or false?

The slice radius, the height, and the sphere radius form a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q4/5

3880

QUESTION ID

S96-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Right angle at the slice center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q4/5

True or false?

A hemisphere of radius r rests on its flat base; y is the height above the base plane (0 ≤ y ≤ r). Its horizontal slices get bigger as y increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q5/5

3881

QUESTION ID

S96-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Smaller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 3

Volumes of cylinders and cones • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Cylinder r = 6, h = 6: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q1/5

3883

QUESTION ID

S96-B3-Q1

TYPE

Short answer

CORRECT ANSWER

216π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q1/5

3883

ANSWER KEY Reduced copy of the question page, with its answer

Cone r = 6, h = 6: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q2/5

3884

QUESTION ID
S96-B3-Q2

TYPE
Short answer

CORRECT ANSWER
72π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Cylinder minus cone (same r and h = r) =

A. $4/3\ \pi r^3$

B. πr^3

C. $\frac{2}{3}\pi r^3$

D. $\frac{1}{3}\pi r^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q3/5

3885

QUESTION ID

S96-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $\frac{2}{3}\pi r^3$

GRADING NOTE $\pi r^3 - \frac{1}{3}\pi r^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q3/5

True or false?

A cone is ⅓ of the cylinder with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q4/5

3886

QUESTION ID

S96-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\pi r^3 - \frac{1}{3}\pi r^3 = \frac{1}{3}\pi r^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q5/5

3887

QUESTION ID

S96-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\frac{2}{3}\pi r^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 4

Area of a ring • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Ring with outer radius 5 and inner radius 3: exact area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q1/5

3889

QUESTION ID

S96-B4-Q1

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q1/5

True or false?

Ring area = $\pi(R^2 - r^2)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q2/5

3890

QUESTION ID

S96-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q2/5

Ring with outer 4 and inner 1:

A. 3π

B. 15π

C. 9π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q3/5

3891

QUESTION ID

S96-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 15π

GRADING NOTE 16 − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Ring with outer r and inner y: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q4/5

3892

QUESTION ID

S96-B4-Q4

TYPE

Short answer

CORRECT ANSWER

$\pi(r^2 - y^2)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Ring area = $\pi(R - r)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q5/5

3893

QUESTION ID

S96-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\pi(R^2 - r^2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q5/5

SET 96 • HSG-GMD.A.2

FORWARD 1

Use Cavalieri's principle for other solids • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A twisted tower and a straight prism both have height 50 m and every horizontal slice is a 10 m by 10 m square. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q1/5

3895

QUESTION ID

S96-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Equal: 5,000 m³ each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-F1-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pyramid with base area 36 and height 10 and a cone with base area 36 and height 10: compare volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q2/5

3896

QUESTION ID

S96-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Equal: 120 each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q2/5

True or false?

Cavalieri's principle shows that every pyramid with base area B and height h has volume ⅓Bh, once one such pyramid is known.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q3/5

3897

QUESTION ID

S96-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal slices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q3/5

Two solids of height 8 have slice areas 5 and 5 at every level. Volumes?

A. 40 and 13.3

B. 5 each

C. Both 40

D. Cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q4/5

3898

QUESTION ID

S96-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Both 40

GRADING NOTE 5 × 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two solids have equal heights, but at one level their slice areas differ. Can Cavalieri's principle be used? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q5/5

3899

QUESTION ID

S96-F1-Q5

TYPE

Short answer

CORRECT ANSWER

No: the areas must be equal at every level.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q5/5

SET 96 • HSG-GMD.A.2

FORWARD 2

Model a dome as a hemispherical shell • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A concrete dome is a hemispherical shell with outer radius 10 m and inner radius 9.5 m. Find the volume of concrete to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q1/5

3901

QUESTION ID

S96-F2-Q1

TYPE

Short answer

CORRECT ANSWER

298.7 m³

GRADING NOTE Rounding to 0.1 required. $\frac{2}{3}\pi(10^3 - 9.5^3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A concrete dome is a hemispherical shell with outer radius 10 m and inner radius 9.5 m. Concrete has density 2.4 t/m³. Estimate the dome's mass to the nearest ton.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q2/5

3902

QUESTION ID

S96-F2-Q2

TYPE

Short answer

CORRECT ANSWER

717 t

GRADING NOTE Rounding to the nearest ton required. Volume $\frac{2}{3}\pi(10^3 - 9.5^3) \approx 298.71$ m³; × 2.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q2/5

True or false?

A shell's volume is the outer volume minus the inner volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q3/5

3903

QUESTION ID

S96-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q3/5

3903

ANSWER KEY Reduced copy of the question page, with its answer

A hemisphere of radius 3 has volume

A. 9π

B. 27π

C. 36π

D. 18π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q4/5

3904

QUESTION ID

S96-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 18π

GRADING NOTE ⅔π · 27.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A hemispherical shell has outer radius 10 m and inner radius 9.5 m. Why is its volume NOT $\frac{2}{3}\pi(10 - 9.5)^3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q5/5

3905

QUESTION ID
S96-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Volumes subtract, not radii: the shell is the difference of two hemispheres.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q5/5

3905